

Lecture Notes on Multidisciplinary Industrial Engineering
Series Editor: J. Paulo Davim

M. S. Shunmugam
M. Kanthababu *Editors*

Advances in Simulation, Product Design and Development

Proceedings of AIMTDR 2018

Lecture Notes on Multidisciplinary Industrial Engineering

Series Editor

J. Paulo Davim , Department of Mechanical Engineering, University of Aveiro, Aveiro, Portugal

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M. S. Shunmugam · M. Kanthababu
Editors

Advances in Simulation, Product Design and Development

Proceedings of AIMTDR 2018



Springer

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ISSN 2522-5022

ISSN 2522-5030 (electronic)

Lecture Notes on Multidisciplinary Industrial Engineering

ISBN 978-981-32-9486-8

ISBN 978-981-32-9487-5 (eBook)

<https://doi.org/10.1007/978-981-32-9487-5>

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Foreword

It gives us immense pleasure to present the Advances in Manufacturing Technology and Design—Proceedings of All India Manufacturing Technology, Design and Research (AIMTDR) Conference 2018.

We would like to express our deep gratitude to all the members of Organizing Committee of AIMTDR 2018 Conference and also to authors, reviewers, sponsors, volunteers, etc., for their wholehearted support and active participation. Our special thanks to Mr. P. Kaniappan, Managing Director, WABCO India Ltd, Chennai, who kindly agreed to act as President of National Advisory Committee (NAC) of the AIMTDR 2018 Conference. We also express our sincere thanks to Chairman Dr. S. Gowri, Honorary Professor, and Co-chairman Dr. P. Hariharan, Professor, Department of Manufacturing Engineering, Anna University, Chennai, for their wholehearted support. We would like to express our sincere thanks to Research Scholars Mr. K. R. Sunilkumar, Mr. U. Goutham, Mr. V. Mohankumar and Mr. R. Prabhu and also UG/PG students of the Department of Manufacturing Engineering, Anna University, for their contributions in the preparation of this volume.

High-quality papers have been selected after peer review by technical experts. We hope you find the papers included in the Proceedings of AIMTDR 2018 Conference are interesting and thought-provoking.

We also like to express our gratitude for the support provided by WABCO India Ltd., Chennai; Kistler Instruments India Pvt. Ltd., Chennai; AMETEK Instruments India Pvt. Ltd., Bengaluru; Central Manufacturing Technology Institute, Government of India, Bengaluru; Defence Research and Development Organisation, Government of India, New Delhi; and Ceeyes Engineering Industries Pvt Ltd., Trichy.

Finally, we would like to express our gratitude to the National Advisory Committee (NAC) members of AIMTDR 2018 Conference for providing the necessary guidance and support.

Guwahati, India

Uday Shanker Dixit
Vice-President
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Preface

All India Manufacturing Technology, Design and Research (AIMTDR) Conference is considered globally as one of the most prestigious conferences held once in two years. It was started in 1967 at national level at Jadavpur University, Kolkata, India, and achieved the international status in the year 2006. It was organized by various prestigious institutions such as Jadavpur University, IIT Bombay, IIT Madras, CMTI Bangalore, PSG iTech, IIT Kanpur, CMERI, IIT Delhi, NIT Warangal, IIT Kharagpur, BITS Ranchi, VIT Vellore, IIT Roorkee, Andhra University, IIT Guwahati and College of Engineering Pune.

The recent edition of the AIMTDR Conference, 7th International and 28th All India Manufacturing Technology, Design and Research (AIMTDR) Conference 2018, was jointly organized by the Departments of Manufacturing Engineering, Mechanical Engineering and Production Technology during 13–15 December 2018 at College of Engineering Guindy, Anna University, Chennai, India, with the theme ‘Make in India – Global Vision’. A major focus was given on recent developments and innovations in the field of manufacturing technology and design through keynote lectures. About 550 participants registered for the conference. During the conference, researchers from academia and industries presented their findings and exchanged ideas related to manufacturing technology and design.

Of the 750 papers received initially, 330 papers were finally selected after rigorous review process for publication in the Springer Proceedings. Selected papers from the conference are being published by Springer in the series Lecture Notes on Multidisciplinary Industrial Engineering in five volumes, namely **Volume 1**—Additive Manufacturing and Joining, **Volume 2**—Forming, Machining and Automation, **Volume 3**—Unconventional Machining and Composites, **Volume 4**—Micro and Nano Manufacturing and Surface Engineering and **Volume 5**—Simulation and Product Design and Development.

Chennai, India
May 2018

M. S. Shunmugam
M. Kanthababu

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About the Editors

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Part I

Simulation

Chapter 1

Some Investigations on Drilling of Aluminium Alloy from FEA-Based Simulation Using DEFORM-3D



R. Sreenivasulu and Ch. Srinivasa Rao

Abstract The production of holes is one of the most common operations among all the machining processes and is more complex than the other metal removal processes. During the drilling, burrs form on both the entry and exit side of the hole as a result of plastic deformation of the material. In order to investigate the burr height, finite element analysis (FEA)-based DEFORM-3D simulations are performed during drilling of aluminium 6061 and 7075 alloys. The influence of variable drill geometry and machining conditions on burr height, thrust force, stresses and strain rates apart from thermal aspects between the drill bit and work pieces are examined. Simulated results analyse the reduction in burr height which can be achieved using the selection of input parameters to attain multiple performance characteristics of output responses, will have a wide range of application prospects saving time and cost of post finishing operation of a drilled hole.

Keywords Thermal effect • Thrust force • Effective stress and strain rates • Burr height • FEA-based DEFORM-3D

1.1 Introduction

Manufacturing sector facing lot of problems due to abrupt changes in the design from day to day changes in taste of customers in their modern life in every aspect. Entrepreneurs who are entered newly in the competitive market were faced by a problem with capital investment incurred on development of design, data analysis followed by testing through experimentation. For conducting experimental tests, it takes more investment and wasting of time once it fails in testing. So, traditional

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_1

manufacturing support systems are not sufficient to meet the above-stated problems. Simulation, modelling and analysis help to fulfil the drawbacks of conventional systems by reducing experimental tests, flexibility in design of products according to customers satisfaction with less capital investment. This is possible only by adopting design software, controlled machining parameters using sensor-based technology with feedback system and optimizes the entire processes at every stage using advanced optimization techniques. Guo and Dornfeld stated in their work by controlling persuade of process parameters such as feed, depth of cut and tool geometry, then only optimization of output responses can be possible [1, 2]. The finite element modelling of machining was published by Strenkowski and Carroll [3], and a comprehensive review of general FEM code as applied towards machining has been reported by Marusich and Ortiz [4]. The author works reveal the detail improvements in the mathematical theory and how to apply them towards machining. DEFORM applies FEM theory in a user-friendly graphical user interface (GUI) that is very robust when compared to many custom FEM codes. DEFORM-3D software is a FEM-based system which works on simulation of manufacturing process, especially in designing and analysing the three-dimensional material flow in forming processes. The DEFORM-3D software club the automatic framework regenerator that can activate without human intervention, producing optimized network. The best choice of lattice system can be separated in the more accurate requisite areas; thus, the level of issue is compact and computation is noticeably enhanced [5]. However, this software is limited to guide template of turning and boring simulation in cutting aspects until now. Drilling and other kind of machining simulations have to be built up by individual. Hence, with the aid of this platform, its pre-processing unit is used to simulate the drilling operation with HSS twist drill and process the information data with the post-processing section [6]. In this paper, a FEM-based DEFORM-3D for drilling of aluminium alloys is first constructed, and then, simulations are drawn under different drilling speeds and feed rates in order to analyse and compare the influence of drilling parameters on drilling thrust force, torque, stresses, strain rates and temperature.

1.2 Modelling of Drilling Process

The work piece is modelled after assuming the material as perfectly plastic material where the material constitutive model of this deformable body is represented with Johnson–Cook material flow model. In the present analysis, Al 6061 and Al 7075 alloys are selected as work piece materials, and its properties are widely estimated in the literature. The parameters of Johnson–Cook model (JC model) obtained from the literature related to the same material during drilling operation under similar cutting conditions [7, 8]. JC model is one of the most extensively employed in a wide variety of manufacturing processes and engineering materials and a benchmark for comparison of different constitutive models. The original JC constitutive model, i.e. the combined form considering the strain, strain rate and temperature effect on flow

stress, can be mathematically expressed with the relation as presented in Eq. 1.1.

$$\bar{\sigma} = (A + B(\varepsilon)^n) \left(1 + C * \ln \left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right) \right) \left(1 - \left(\frac{T - T_0}{T_{\text{melt}} - T_0} \right)^m \right) \quad (1.1)$$

where $\bar{\sigma}$ is the flow stress, ε is the plastic strain, $\dot{\varepsilon}$ is the plastic strain rate, $\dot{\varepsilon}_0$ is the reference plastic strain rate (0.001 s^{-1}), T is the temperature of the work piece, T_{melt} is the melting temperature of the work piece material and T_0 be the room temperature (293 K); material constant A is the yield strength, B is the hardening modulus, C be the strain rate sensitivity, ' n ' is the strain-hardening exponent and ' m ' is the thermal softening exponent. Although a more realistic simulation model for the machining process should also take, the parameters for Al 6061 alloy are $A = 304.1 \text{ MPa}$, $B = 113.8 \text{ MPa}$, $n = 0.42$, $C = 0.002$, $m = 1.34$ and $T_{\text{melt}} = 785 \text{ K}$, and values for Al 7075 alloy are $A = 317.37 \text{ MPa}$, $B = 166.95 \text{ MPa}$, $C = 0.00736$, $n = 0.5091$ and $m = 1.5724$ and $T_{\text{melt}} = 900 \text{ K}$. The work piece is represented by a cylindrical model of 10 mm radius, where the twist drill bit is modelled as a rigid body, which rotates at the specified spindle speed. A fine mesh density is defined with an input size of 0.075 mm and size ratio 2 for work piece. Thermal boundary conditions are defined keeping in view that it will allow heat transfer from work piece to cutting tool. Heat transfer between the work piece and the tool is dependent on the pressure developed during machining.

1.2.1 Set the Characteristics of Tool and Work Piece Model

The geometric model of drill is imported from CATIA V5 R19 package, and then, the high-speed steel twist drill is set as adamant with respect to work piece. DEFORM-3D V11.0 can create a simple work piece model. In this software package, the adaptive meshing grid (AMG) technology is used to separate the grids. The framework of drill bit and work piece utilize the fixed type. The element size is set to 0.25 mm, the scale ratio of drill bit as set to 0.25 mm, the parent material web frame scale is set to 5, the side length is set to 0.25 as minimum and the web framework at the part of the parent material (Fig. 1.1).

Aluminium 6061 and 7075 alloy materials are chosen as work piece model includes 100,000 elements of polygon shape with 20,000 nodes. The bottom surface of the work piece is fixed in all directions. The twist drill bit is modelled as a rigid body using 30,000 (polygon) elements with 20,000 ports which move at the specified rotational speed. The work piece is represented by a cylindrical model of 10 mm diameter, where the cutting tool is modelled as a rigid body which moves at the specified cutting speed. A fine mesh density is defined with an input size of 0.075 mm and size ratio 2 for work piece. Thermal boundary conditions are defined keeping in view that it will allow heat transfer from work piece to the cutting tool.

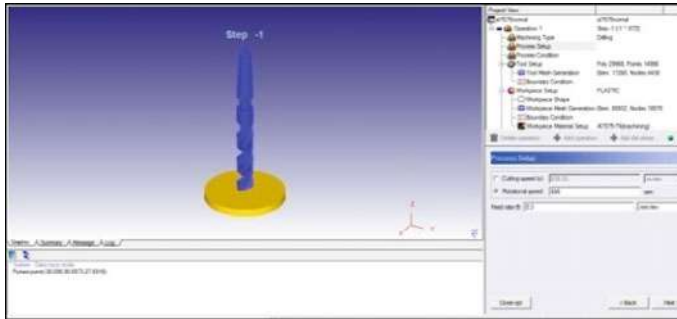


Fig. 1.1 FEA model of twist drill with work piece [9, 10]

Heat transfer between the work piece and the tool is dependent on the pressure developed during machining.

1.2.2 Boundary Conditions Setting to the Work Piece and Drill Bit

The initial setting of boundary conditions between the work piece especially at the bottom side and the drill bit along X , Y and Z directions assumed to be zero. Drill bit feeds along the Z -direction and rotates about Z -axis. Surrounding surfaces of parent work material and drill bit are set to heat dissipation into atmosphere, and the work piece dimensions' reparation is activated. When the networks are resealed, the volume of deformable body will be altered. The reparation of volumetric change is utilized to compensate the loss of volume of material. The reparation volume of material is relative to two segments of volume fractions, one is the factual volume of the parent work piece and the other is the equivalent network fraction of volumetric change.

1.3 Simulation Using DEFORM-3D

1.3.1 Set the Relations Between the Tool and the Objects

The relationship between master servant and workbench of DEFORM-3D software package is taken as the objects which are rigid (drill bit) are master parts and the distortable objects (parent work piece material) are servant parts. The heat transfer coefficient is set as $45 \text{ W/(m}^2 \text{ K)}$, coefficient of friction is taken as 0.6 constant throughout simulation by chosen shear friction and can be adopted in the simulation. However, the utmost spoil of the material arrived to a critical value to identify the

total steps meet then the simulation stops automatically. The atmosphere temperature is set to 293 K, and the heat transfer convection coefficient selected as 0.03 W/(m² K) from menu set-up. The SI system of units are considered for selection of magnitudes of numerical data required to run the simulation from menu set up.

1.4 Results and Discussion

Simulation of drilling of aluminium alloys (6061 and 7075 series) using DEFORM-3D V11.0 is applied and data as acquired, viz. stresses, strain rates, thrust force and torque.

1.4.1 Simulation Results of Al 6061 Alloy

Effective Stress and Strain Rates The effective stress and strain rates are acquired from the simulation at a simulation step of 2886 for Al 6061 alloy which is plotted in Figs. 1.3 and 1.4 with respect to time in steps. The obtained value of effective stress is 216 MPa and effective strain at the rate of 0.461 mm/mm. At the beginning of drilling operation the initial stress is more, once it attains yielding point then material plastically deformed, indicates that no noticeable influence on burr height by variation of input parameters.

Fig. 1.3 Effective stress variation in drilling [9, 10]

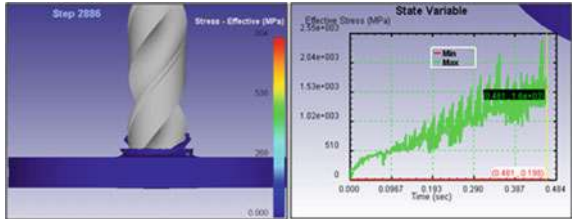
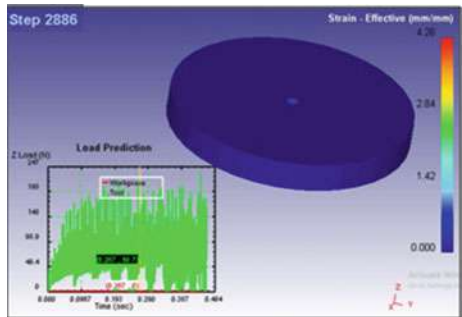


Fig. 1.4 Strain variation in drilling



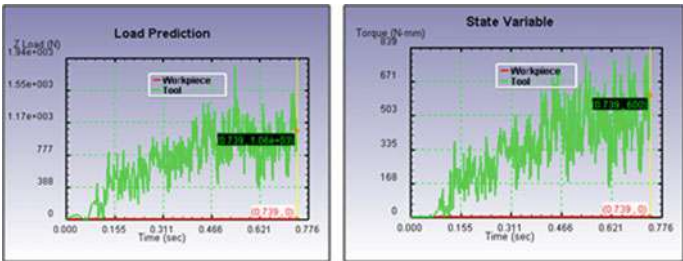


Fig. 1.5 Thrust and torque in drilling of Al 6061 alloy

Thrust and Torque During Drilling From Fig. 1.5, it is revealed that load (thrust force) prediction during drilling of aluminium 6061 alloy at the simulation step 2886 interval is obtained as the maximum of 319 N and maximum torque is obtained as 268 N mm at the same step interval. From the simulation, it is observed that burr height is minimum at minimum thrust load exerted between the drill bit and parent material. Also, little bit influence of point and clearance angles on drill geometry observed on burr height corresponding to thrust load.

Thermal Effect During Drilling At the step interval of 2886, generation of heat occurs between tool and work piece in the shear deformation zone, which leads to enhance the temperature in the range of 31–304 °C during drilling of Al 6061 alloy, depicted in Fig. 1.6. There is no significant effect of temperature on burr height by variation of input parameters while drilling, observed in the simulation steps.

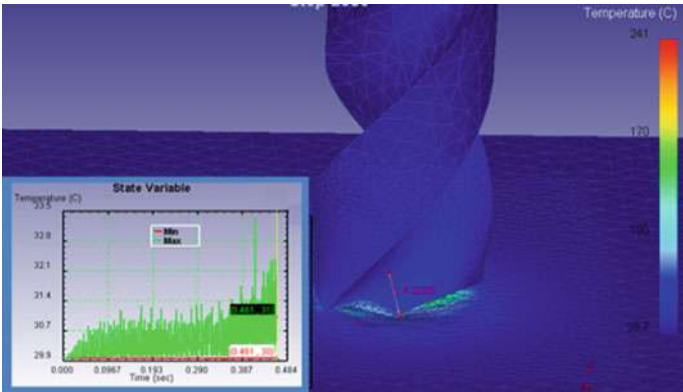


Fig. 1.6 Thermal variation in drilling of Al 6061 alloy

1.4.2 Simulation Results of Al 7075 Alloy

Effective Stress and Strain Distribution From the simulation, it reveals that burr height is increased for Al 7075 compared with Al 6061, and reason may be that the influence of composition elements in Al 7075 alloy. Stresses and strain rates are also proportionally increased. The effective stress and strain data acquired from the simulation based on finite element analysis using DEFORM-3D at a simulation step of 2981 for aluminium 7075 alloy and are captured an image shown in Figs. 1.7 and 1.8 with respect to time in seconds. The obtained value of effective stress is 378 MPa and effective strain at the rate of 0.347 mm/mm.

Thrust Force and Torque Prediction Figure 1.9 shows Al 7075 alloy load (thrust force) prediction of the drilling process at 2981 simulation step interval. The maximum load observed is 251 N, and maximum torque obtained is 143 N mm at 2981 step interval. Both thrust and torque are suddenly increased at the middle of the

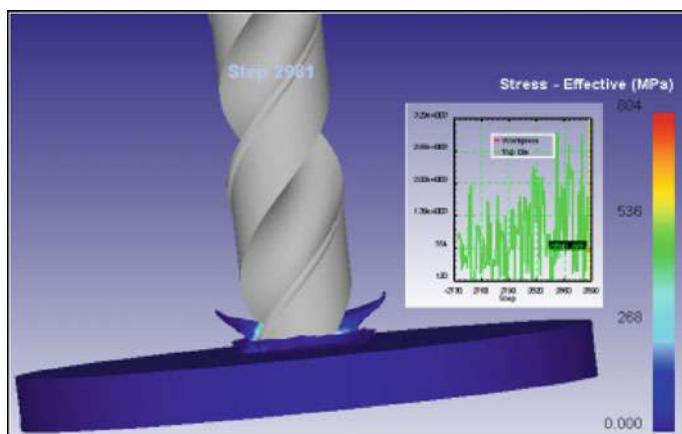


Fig. 1.7 Effective stress variation in drilling

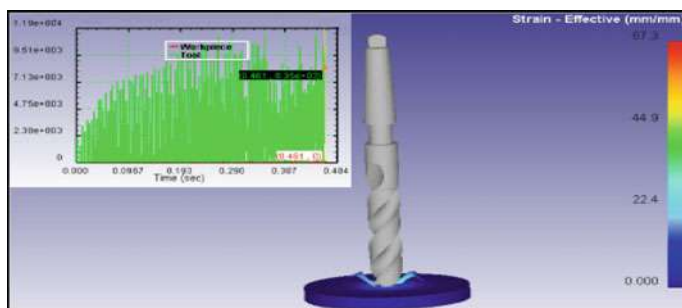


Fig. 1.8 Effective strain variation in drilling

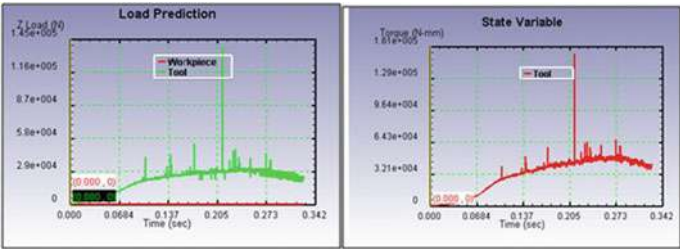
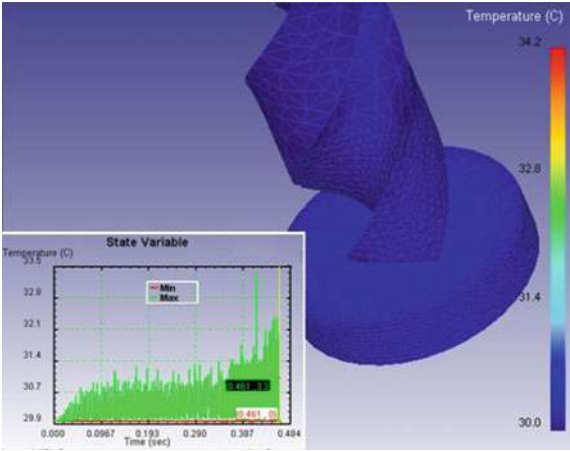


Fig. 1.9 Variation of thrust force and torque

Fig. 1.10 Temperature distribution during drilling of Al 7075 alloy



drilling operation that it is observed. The reason may be that moderate feed and spindle speed cause increase the load in between work and drill bit.

Temperature Variation At the step interval of 2981, generation of heat occurs between tool and work piece in the shear deformation zone, leads to enhance the temperature in the range of 33–358 °C during drilling of Al 7075 alloy, depicted in Fig. 1.10.

From simulation using DEFORM-3D, the burr size assessment is not possible to obtain directly from the software. So in the present work, burr height is measured in an alternate method using Java image processing program (ImageJ).

1.4.3 Burr Height Estimation Using ImageJ Software

ImageJ software can be able to found the area, perimeter and pixel value statistics of user-defined selections to measure distances and angles. It supports the standard

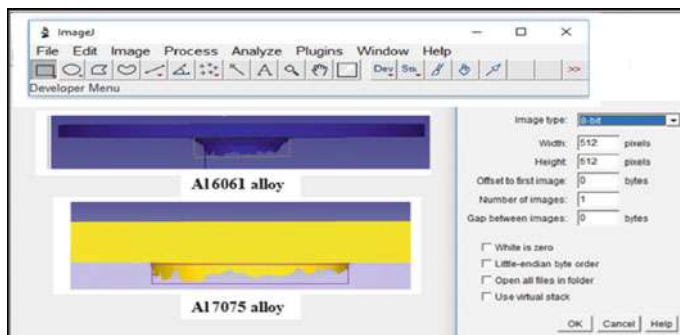


Fig. 1.11 Measurement of burr height using ImageJ

image processing functions such as contrast manipulation, sharpening, smoothing, edge detection and median filtering and also provides geometric transformations such as scaling, rotation and flips.

Image can be zooming out up to 32:1 and zoom in to 1:32 and analysis and processing functions are available at any magnification factor. The program supports any number of windows (images) simultaneously, limited only by available memory. The images of burrs for both the materials are calibrated initially with original pixel size and setting the scaling ratio from command window. By selecting the area to be measured on the burr images as a rectangle (shown in Fig. 1.11), then fix the global and local coordinates. The height of selected portion of rectangular area on the image is analysed to equivalent size of burr height with a chosen scale ratio.

1.5 Validation with Experimental Results

Experiments are conducted as per Taguchi L27 orthogonal array by considering the five parameters such as spindle speed (rpm), feed rate (mm/min), drill diameter (mm), point angle (degrees) and clearance angle (degrees) and three levels and measured the output responses such as burr height (mm), burr thickness (mm), thrust force (N) and torque (Nm) with well calibrated measuring apparatus, and the observations are tabulated in Table 1.1 [9, 10].

The developed simulations are compared with experimental results especially with significant parameters with output responses to optimize the burr size resulting in drilling of aluminium 2014 alloy. The simulation results are closely fitted with experimental values. Here, some of them are shown graphically especially with clearance angle variation which is the strong significant parameter to influence exit burr size (Figs. 1.12, 1.13, 1.14 and 1.15).

After validation of both results, it is observed that similar relationships are obtained, which reveal that good fitting.

Table 1.1 Output responses measured during the drilling of Al 2014 alloy

Exp No.	B _h (mm)	B _t (mm)	F _{th} (N)	M (N m)	Exp No.	B _h (mm)	B _t (mm)	F _{th} (N)	M (N m)
1	0.324	0.278	262	155	15	0.387	0.264	202	173
2	0.283	0.216	288	197	16	0.314	0.231	208	257
3	0.342	0.282	241	154	17	0.229	0.247	265	137
4	0.285	0.278	235	105	18	0.241	0.302	265	111
5	0.338	0.268	335	187	19	0.314	0.289	316	177
6	0.284	0.297	252	110	20	0.269	0.331	286	193
7	0.238	0.349	241	157	21	0.326	0.238	252	165
8	0.345	0.291	335	194	22	0.354	0.212	241	151
9	0.312	0.254	395	190	23	0.281	0.252	235	176
10	0.338	0.243	232	152	24	0.216	0.223	295	171
11	0.262	0.192	248	179	25	0.306	0.252	186	153
12	0.328	0.248	265	217	26	0.223	0.309	252	274
13	0.328	0.296	316	187	27	0.341	0.248	316	218
14	0.291	0.232	286	147	–				

Fig. 1.12 Variation of clearance with burr height

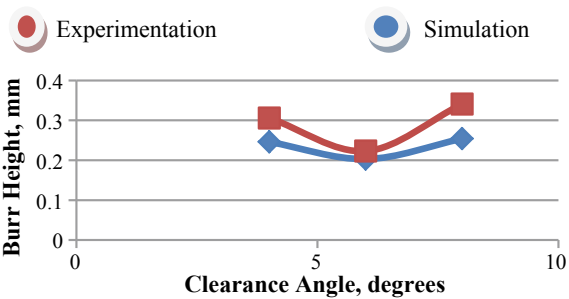


Fig. 1.13 Variation of clearance with burr thickness

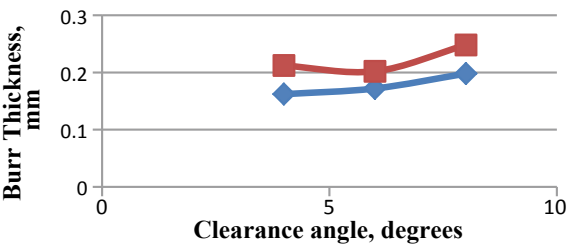


Fig. 1.14 Variation of clearance with thrust force

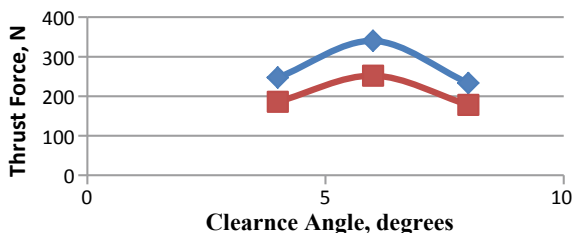
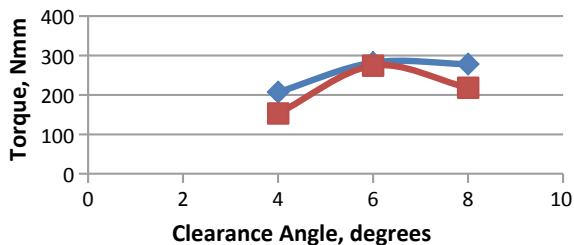


Fig. 1.15 Variation of clearance with torque



1.6 Conclusions

The following conclusions are drawn by comparing the simulation results with the experimental results of authors [9, 10].

The simulation results obtained from DEFORM-3D are also in good correlation at particular machining condition having the percentage variation of 1, 2.2 and 3.5% for burr height, thrust force and torque, respectively, for Al 6061 alloy, whereas for Al 7075 alloy 5.73, 3.7 and 6.2% occurred with the same output responses. The reason may be in the error occurred during the variation of shear strength, and Brinell hardness values lead to the cumulative error of higher values in case of Al 7075 alloy. The temperatures generated during drilling of aluminium 6061 and 7075 alloys with high-speed steel twist drill bits are simulated in DEFORM-3D at particular machining condition is correlating well measured with infrared pyrometer at the same machining condition, but it is observed that there is no significance of temperature on burr height. The reason may be that aluminium alloy is soft material and total heat generated at the contact of tool and work piece is spread over the top surface of the drilled hole. The effective stress and strain rates are obtained at a step interval of 2886 which is as 216 MPa and 0.461 mm/mm for aluminium 6061 alloy, respectively. In case of aluminium 7075 alloy, the obtained values at a step interval of 2981 are as 378 MPa and 0.347 mm/mm.

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Chapter 2

Self-Organizing Migrating Algorithm to Minimize Module Changes at Machine-Level in Reconfigurable Manufacturing



L. N. Pattanaik 

Abstract A reconfigurable manufacturing system (RMS) is designed at the outset with the capability of rapid adjustment of production capacity and functionality in response to fluctuations in product demand. This paper is presenting a model of RMS containing reconfigurable/modular machines assembled from sets of basic and auxiliary modules to exhibit two key characteristics: a defined range of functionality and scalable capacity. By suitable selection of modules, different operation capabilities with a varying degree of capacity can be developed. Products with alternative process plans and two discrete levels (low and high) of capacity requirements are considered for the modular machines. The objective of the work is to identify the best production sequence and respective process plans in order to minimize the total number of module changes while fulfilling the capacity constraint. Self-organizing migrating algorithm (SOMA) an evolutionary migration algorithm-based search is applied to find the near-optimal solution for the NP-hard combinatorial optimization problem. The approach is illustrated through a numerical problem along with computational results as applied to a hypothetical RMS model.

Keywords Reconfigurable machine tools · Reconfigurable manufacturing system · Modular machines · Basic and auxiliary modules · SOMA

2.1 Introduction

Unpredictable and dynamic market changes cause the design of manufacturing systems a challenging task in the present time. The existing conventional approaches are proving to be inefficient of several fronts. Some novel manufacturing philosophies are under developing stage to counter this. Reconfigurable manufacturing is a philosophy that strives for rapid changes in manufacturing system and its machines

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_2

by designing from the beginning. An RMS was defined by Koren et al. [1] as a manufacturing system which can quickly adjust its production capacity and functionality to counter dynamic demands in market or any regulatory needs. This is designed in the hardware and software components at the beginning in order to produce within a product family. The objective of an RMS is to provide the required amount of functionality and capacity within a short time. A distinguishing characteristic of RMS is that its physical configuration can be changed with time to provide the required functionality and capacity. These reconfigurations are carried out by either adding or removing full machines to or from the whole system, similarly axes/spindles to or from machine tools, varying process capabilities, controls, software, or machine structure to adjust production capacity, changing configuration of machine tools, changing the shop floor layout, material handling systems, etc., as described by Deif and ElMaraghy [2]. RMS is designed to possess a set of key characteristics like modularity, scalability, integrability, convertibility and diagnosability. Modularity implies that both software and hardware components are modularized, and a ‘plug-and-play’ is possible. Scalability means that the manufacturing system is scalable in the production quantity or capacity. Integrability is indicated by the design for ease of integrating system components and inclusion of updated technology [3].

A review of the literatures related to the various existing RMS models and methodologies are presented in Sect. 2.2. In Sect. 2.3, the proposed model is formulated and a new commonality measure among various operations is defined along with brief overview on meta-heuristic SOMA, fitness function and their implementations are discussed. An illustrative numerical case study and computational results are presented in Sect. 2.4. Finally, the conclusion and future directions of the research are presented in Sect. 2.5.

2.2 Relevant Literature

According to Koren et al. [1] definition, RMSs were assumed to be reconfigurable only within a particular product family. Xiaobo et al. [4] proposed a framework for a stochastic model of an RMS which involves three important issues: the optimal configurations in the design, the optimal selection policy in the utilization and the performance measure in the improvement of these systems.

RMS design aspects are classified into two main categories: system-level and machine-level issues [1]. At the system-level, a macroscopic view of the manufacturing system including material handling and machines is considered. At the machine-level, the deliberation is on reconfigurable and modular machines capable of delivering a range of functionality and capacity. Emergent methodologies, such as evolutionary soft computation, self-organization, machine learning, agent-based systems, are capable of finding robust and flexible solutions. Machine-level design issues include the modular or reconfigurable machine tool (RMT), task requirements of a RMT, kinematics constraints, dynamic stiffness and accuracy [5]. The

task requirements of a machine are represented by matrices of motion, and the screw theory is used to identify various components as proposed in Moon and Kota [6, 7].

ElMaraghy [8] divided the reconfiguration approach in manufacturing systems into hard or soft types. The hard type includes adding or removing of machines, machine modules and modifying the material handling systems. In the soft type, reconfiguration tasks like re-programming of RMTs, re-routing and scheduling and scaling up or down of operators. Pattanaik et al. [9] presented a methodology for machine cell formation in the presence of modular reconfigurable machines. A detailed review on the various reported works on RMS is compiled by Bi et al. [10]. Bensmaine et al. [11] presented a multi-objective genetic algorithm for the optimal selection of machines for reconfigurable manufacturing. Goyal et al. [12] proposed a measuring index for responsiveness of reconfigurable machine tools. Paolo [13] proposed a decision-making model for RMS based on Gale-Shapley theory. Khaled et al. [14] described the modular feature in RMS and the methodology to implement it during the design stage. Erik et al. [15] compared RMS performances through assessment of various reconfiguration schemes using lead times and resource utilizations. Yair [16] demonstrated a reconfigurable CNC machine tool is capable of performing three different operations, namely milling, grinding and polishing, by suitable reconfiguration of modules. In the present model, the RMS is assumed to be designed at the outset for different family of products with varying demands. The reconfigurability of the system can be achieved by modular machines which are capable of delivering different operational functions at scalable capacities. The objective of this work is to identify the best sequence of products for manufacturing and to select the process plans for each of them in order to minimize the total changes required in the machine modules.

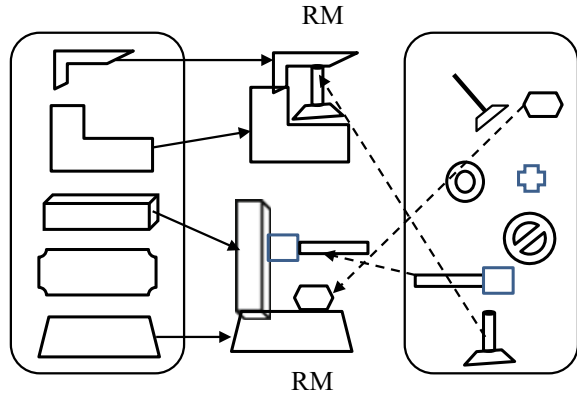
2.3 Model Framework

As mentioned in the earlier sections, the reconfigurable machines considered here are modular in nature and by combining a particular set of basic and auxiliary modules known operation functionality at a deterministic capacity level can be obtained. Two capacity levels: low and high are assumed here on the basis of the volume requirements of the products in the family. Although it is subjective to classify a demand as low or high, but the model can also include exact capacity requirements with a higher computational effort. Referring to Table 2.1, the sets of hypothetical basic and auxiliary modules required for different operations at different capacity levels on the RMTs are produced. The binary (1–0) entry in the table is used to represent the requirement of modules for operations. Each of the 1's indicating the corresponding module (column) is needed for the operation type along the row. The blank cells in the table contain 0's representing otherwise. The combinations of modules and operational capabilities for two RMTs are depicted as in Fig. 2.1. RMT1 is assembled using two basic modules like structural components (base, column, etc.) and one auxiliary module like motion drive with tool holder. Similarly, RMT2

Table 2.1 Module requirements and capacity of different operations on RMTs

Operations		Auxiliary modules													Basic modules						Capacity (<i>L</i>): low (<i>H</i>): high
		01	02	03	04	05	06	07	08	09	10	11	12	13	01	02	03	04	05	06	
RMT 1	O1 <i>L</i>		1		1	1		1		1				1		1		1	1		
	O2 <i>L</i>	1		1	1	1			1			1			1		1				
	O3 <i>L</i>	1		1			1		1		1			1	1		1		1		
	O4 <i>L</i>		1		1		1	1			1			1		1	1	1		1	
RMT 2	O4 <i>H</i>			1	1	1		1			1					1	1		1		
	O5 <i>L</i>		1		1		1		1				1		1		1				
	O3 <i>H</i>			1	1	1		1		1	1				1		1		1		
	O1 <i>H</i>	1	1			1	1	1				1	1			1	1		1		
RMT 3	O6 <i>L</i>		1	1	1		1				1	1				1	1	1			
	O2 <i>H</i>	1	1	1			1		1				1		1		1		1		
	O6 <i>H</i>			1	1			1		1			1		1		1	1			
	O5 <i>H</i>	1		1	1			1	1	1		1			1	1		1		1	

Fig. 2.1 Assembly of two RMTs using basic and auxiliary modules



is also assembled from different modules from the libraries of basic and auxiliary modules. Module change involves some assembly set-up tasks which consumes time, manpower and cost. The main objective of the present work is to minimize the total module changes, thereby minimizing the idle time and associated cost.

2.3.1 Formulation of Objective Function

A new commonality measure for operations with different capacities (low and high) is developed similar to the famous Jaccard's coefficient which considers the number of machine modules required at different levels of capacities. Commonality measure between operations O_i and O_j can be expressed as

$$S_{ij} = \frac{a}{(a + b + c)}, \quad 0 \leq S_{ij} \leq 1 \quad (2.1)$$

where

- a Number of machine modules common for both operations O_i and O_j
- b Number of machine modules required for O_i only
- c Number of machine modules required for O_j only.

For example, referring to Table 2.1,

S1L–1H (Commonality measure between O1—low capacity and O1—high capacity) = $\frac{5}{(5+4+5)} = 0.357$.

Similarly, S1L – 2L = $\frac{3}{(3+6+6)} = 0.2$, S2L – 2H = $\frac{4}{(4+5+6)} = 0.266$, S3L–3H = 0.467, S4L–4H = 0.357, S5L–5H = 0.312, S6L–6H = 0.384, etc.

As it can be seen from the above expression, a higher value of S_{ij} indicates more commonality of modules between the operations in the RMTs and vice-a-versa.

Adopting the given notations, two decision variables and one objective or fitness function for the problem can be formulated as follows:

Set of products (P): $P_1, P_2 \dots P_p$

Process plans: PP 1, PP 2 ... PP n

Set of operations: O1, O2 ... O m

$$X_{pn} = \begin{cases} 1, & \text{if process plan 'n' is used for the product 'p'} \\ 0, & \text{otherwise} \end{cases}$$

$$U_{mn} = \begin{cases} 1, & \text{if operation 'm' is required in process plan 'n'} \\ 0, & \text{otherwise} \end{cases}$$

$$F = \sum_{p=1}^{|P|} \sum_{n=1}^n \sum_{i=1}^m \sum_{j=1}^m S_{ij} \cdot X_{pn} \cdot U_{mn} + S_{ij} \text{ (transition)} \quad (2.2)$$

The above expression for objective or fitness function ' F ' adds the commonality measures among all possible pairs of operations in the selected process plans of the products. The term S_{ij} (transition) is adding the commonality measures among the last operations of a process plan and the first operations of the subsequent process plans as selected for a particular set of products.

2.3.2 Self-Organizing Migrating Algorithm (SOMA)

In order to find the best sequence for manufacturing the products of the family and to select a process plan for each product which can result in the minimization of total module changes, a SOMA-based search algorithm is implemented.

Zelinka and Lampinen [17] proposed a probabilistic evolutionary algorithm which was inspired from the behaviour of individuals living in a society to cooperate and organize among themselves. In this approach, an individual or a solution is considered as a leader based on the highest fitness it possesses. As in a society, the leader is likely to attract others individuals (solutions) towards it. The number of times these individuals will be moving towards the leader is termed as migration and is similar to the concept of generations in genetic algorithm. These movements of individuals towards the leader take place in predefined lengths of steps. Further, the length of the steps determines the precision achieved in the search process. When the length of the steps is small, the search becomes more precise or rigorous and vice-a-versa. However, with smaller steps, the probability to find a global minimum is more but at the cost of higher computational complexity. The direction of migration for the individuals towards leader is controlled by an application called perturbation. The random initial population can be generated using a random uniform distribution within the search space (Kumar et al. [18]).

```

    Input selected parameters  $[S_{ij}]$ ,  $[P]$  in matrix and
    constants  $n, m$ 
    Initialization step
        Select values of SOMA parameters,  $Q = 75$ ,
         $ITER = 3000$ ,  $N_{ex} = 10$ ,  $ST = 3$ ,  $N_{exf} = 20$ ,  $PRT = 0.1$ ,
        and  $PL = 1.3$ 
        Generation of initial random population of size
         $Q$ 
        Evaluation of fitness ( $F$ ) for each member
        Identify the leader and active
    Main loop
        Set the migration counter ( $migr. = 1$ )

        while ( $migr. < 3000$ ) //  $ITER = 3000$  (Stopping
        criteria)
            for  $Q = 1 : 75$ 
                Apply PRT to individuals
            end
             $migr. ++$ 
        end
        Find the optimal solution from current archive

```

Fig. 2.2 Pseudo-code for the SOMA

The pseudo-code (Fig. 2.2) and steps of SOMA are as follows:

- Step 1: Generate random initial population and evaluate fitness for all individuals
- Step 2: Arrange in descending order to select the best fitness individual as leader and worst as active
- Step 3: Generate PRT (degree of randomness during individual moves) vector for all individuals
- Step 4: New positions are created for active individual using equation for migration
- Step 5: The best position (leader) is selected, and it replaces the previous one by the new one
- Step 6: Stop if the termination criterion (maximum number of migrations) is achieved else go to step 2
- Step 7: The leader found in the last iteration is the optimal solution.

The migration of an individual is described by the following equation:

$$x_{i,j}^{ML+1} = x_{i,j, \text{start}}^{MK} + \left(x_{L,j}^{MK} - x_{i,j, \text{start}}^{MK} \right) * t * PRT_j \quad (2.3)$$

$x_{i,j}^{ML+1}$ is the next position of individual i
 $x_{i,j,\text{start}}^{MK}$ is the position of active individual
 $x_{L,j}^{MK}$ is the position of leader
 t takes value from 0–1 (migration step)
 PRT perturbation vector.

Some of the parameters selected for the single objective optimization using SOMA are as follows:

Size of the initial random population $Q = 75$
 Minimal size of the external archive $N_{\text{exf}} = 10$
 Maximum number of migration loops $\text{ITER} = 3000$
 Step numbers $\text{ST} = 3$
 Probability of perturbation $\text{PRT} = 0.1$
 Length of path $\text{PL} = 1.3$
 Maximum size of the external archive $5 \times N_{\text{exf}}$.

2.3.3 Coding and Decoding Schemes

To apply SOMA to an optimization or search problem, the decision variables are usually mapped and represented by a finite-length string of numerals or alphabets.

The representation scheme as permutation string adopted in the present paper is given here.

$$\left(\begin{array}{c} \text{Products} \\ 2 \ 1 \ 4 \ 6 \ 3 \ 5 \end{array} \right) \left(\begin{array}{c} \text{Process plans} \\ 1 \ 2 \ 3 \ 1 \ 2 \ 2 \end{array} \right) \Rightarrow [2 \ 1 \ 4 \ 6 \ 3 \ 5 \mid 1 \ 2 \ 3 \ 1 \ 2 \ 2] \text{ or } \left(\begin{array}{c} 2 \ 1 \ 4 \ 6 \ 3 \ 5 \\ 1 \ 2 \ 3 \ 1 \ 2 \ 2 \end{array} \right)$$

The first/top portion of the individual represents the sequence of products being manufactured while the last/bottom part shows the corresponding process plans for the products. For example, the sequence of products being selected for manufacturing is P_2, P_1, P_4, P_6, P_3 and P_5 . Similarly, the process plans selected for these products are PP 1, PP 2, PP 3, PP 1, PP 2 and PP 2, respectively.

2.3.4 Fitness Function

The fitness value of each potential solution is found using the commonality measures among the various operations as illustrated here for the given example.

For the individual $[2 \ 1 \ 4 \ 6 \ 3 \ 5 \mid 1 \ 2 \ 3 \ 1 \ 2 \ 2]$, referring to Table 2.2, PP 1 of product P_2 has operations O2, O4, O5 and O6. Using Eq. (2.2) for fitness or objective function, commonality measures among these operations can be added for low capacity

Table 2.2 Operation sequence and alternative process plans for the products

	<i>P1</i> low			<i>P2</i> low		<i>P3</i> high		<i>P4</i> low			<i>P5</i> high		<i>P6</i> high	
	1	2	3	1	2	1	2	1	2	3	1	2	1	2
O1						1	1	1	1				1	
O2	1	1	1	1	1			2		1		1		1
O3	2	2			2	2	2	3	2	2	1		2	
O4		3	2	2			3	4	3	3	2	2		2
O5	3		3	3	3	3			4	4		3	3	3
O6	4	4	4	4	4	4	4	5	5	5	3		4	4

requirement of product *P2*. Similarly, PP 2 of product *P1* of low demand/capacity contains operations O2, O3, O4 and O6, and lastly, PP 2 of product *P5* of high capacity requirement needs operations O2, O4 and O5. Apart from commonality measures among these operations, during the transition from one process plan of a product to another of a different product, the commonality measure between the last and first operations of those process plans has also to be added. The fitness value for the above-mentioned string of the solution can now be expressed as

$$F = \left(\sum S_{ij} \text{ for PP 1 of } P2 + \sum S_{ij} \text{ for PP 2 of } P1 + \sum S_{ij} \text{ for PP 3 of } P4 + \right. \\ \left. \sum S_{ij} \text{ for PP 1 of } P6 + \sum S_{ij} \text{ for PP 2 of } P3 + \sum S_{ij} \text{ for PP 2 of } P5 \right) \\ + S_{ij}(\text{transition}) \quad (2.4)$$

where

$S_{ij}(\text{transition}) = (S_{ij} \text{ between last operation of PP 1 of } P2 \text{ and first operation of PP 2 of } P1 + S_{ij} \text{ between last operation of PP 2 of } P1 \text{ and first operation of PP 3 of } P4 + \dots + S_{ij} \text{ between last operation of PP 2 of } P3 \text{ and first operation of PP 2 of } P5)$.

As can be seen from Table 2.2, the number of operation types is six (O1–O6) and therefore twelve operation types are possible (referring to Table 2.1) considering both low and high capacity levels for each operation. Then between all the pairs of operations, the commonality measure can be found using Eq. (2.1) and module requirements given in Table 2.1.

Some calculations on commonality measures are already discussed in Sect. 2.3.1. These commonality measures are used to evaluate a potential SOMA individual by finding its fitness value as given in Sect. 2.3.2. The representation of solutions or individuals is explained in Sect. 2.3.3 for six products.

Thereafter the SOMA can be applied with a well-defined stopping criteria, e.g. maximum number of migrations. The various steps of SOMA are coded using MATLAB® R2017b software with initial parameters as given in the previous section. The individual that is proved to be the leader after the stopping criteria is met is accepted as the optimal solution for the problem.

In the present case, the string [4 2 5 3 1 6| 2 2 1 1 3 1] is found at the end of the SOMA run which can be decoded as production sequence of P_4 , P_2 , P_5 , P_3 , P_1 and P_6 and corresponding process plans of PP 2, PP 2, PP 1, PP 1, PP 3 and PP 1, respectively. The output of the SOMA gives the best production sequence of the six products and their process plans which minimizes the total changes in module during the reconfiguration.

2.4 Computational Results

In order to find the best production sequence of products in the family and the corresponding process plans for the products which minimize the total changes in machine modules, an illustrative case study is presented here. The RMS is designed for a family of six products whose demands or production volumes are classified into either low or high capacity types. The products can be manufactured using two or three alternative process plans. The sequence of operations and capacity requirements of products are compiled as in Table 2.2.

2.5 Conclusions

The present paper addresses the issue of considering functionality and capacity at the machine-level during reconfigurable manufacturing. The reconfigurable machines are assumed to have multiple operation capabilities by combining different sets of basic and auxiliary modules. The RMS is designed at the outset for a particular group or family of products. These products with different levels of capacity requirements can be manufactured using alternative process plans. In order to select the best sequence of products for production and corresponding process plans which minimizes the total changes in modules, a SOMA-based optimization is performed. The output from the algorithm can be easily decoded to identify the best production sequence and process plans.

The present work considers only two levels of capacity requirements for simplicity, but the exact volume of products can be modelled to select the process plans and production sequence. Further, the cost and non-compatibility aspects (both hardware and software) among modules during reconfiguration are the other important factors to be considered. The future scope may also include a multi-objective formulation of conflicting responses like reconfiguration cost, production lead time, delivery priority.

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Chapter 3

Modelling and Simulation of Deep Drawing Process of Circular Cup on AL1200 Using Finite Element Analysis



Y. K. Sahu and M. K. Pradhan

Abstract Deep drawing process is the mostly used sheet forming process. This method is used in the automobile and aerospace industries and also used for the production of kitchen utensil and cold drink cans. In deep drawing, there are some factors which influence the process those factors are called process parameter of deep drawing. Blank holding force, friction, strain rate, thickness, blank shape, temperature, punch force and punch speed, etc. are the most important parameters. Wrinkles and spring back defects in the drawn component are highly undesirable defects. Wrinkling when insufficient holding force is applied on flange. The objective of this work is to analyze the wrinkling and spring back problem in the deep drawing of circular cup and to determine range of process parameters to minimize the wrinkling defect and spring back defect. A finite element model is developed for 3-D numerical simulation of a circular cup for blank material AL1200 forming process in finite element software ABAQUS 6.14 and ANSYS18.1/APDL. Properties and tool design parameters were used as input parameters for simulation. Wrinkling and spring back defect was observed in the simulated cup. From the FE simulation, we found out the minimum wrinkling occurs in the deep drawn cup at flange region when the Blank holding force is 1 KN and coefficient of friction is 0.02 and the minimum spring back effect occurs in the deep drawn at the BHF of 1 KN and friction of 0.01. Max punch pressure is needed to draw a component when punch velocity is 0.23 mm/s, friction 0.01 and BHP of 33 MPa. Defects like cracking, tearing and necking are not observed in the deep drawn components.

Keywords Deep drawing · Temperature · Coefficient of friction · Punch velocity and punch pressure · Blank holding force

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_3

3.1 Introduction

Deep drawing is most important method amongst the all sheet metal forming methods. This method is mainly used in the part of the car and aerospace industries and also used for the production of kitchen appliances. In this process, the metal blank is subjected to plastic deformation when the punch moves towards the die to get the desired shaped cup.

For the production of the defect-free cup and for the minimization of the cost of product, optimization of parameters is very important. Some important parameters are temperature, friction, blank holding force, punch force and punch speed and blank shape (Fig. 3.1).

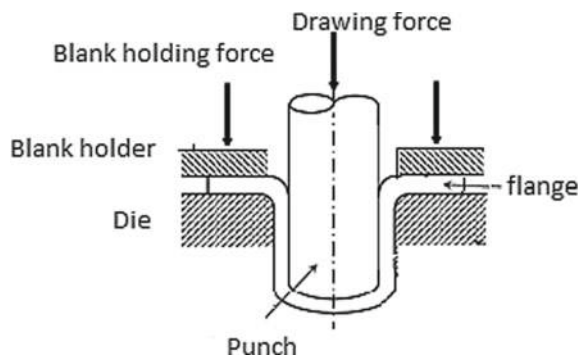
3.1.1 Defects in Deep Drawing Cup

A deep drawing process is intricate operation and more things can turn out badly during drawing operation, hence the number of defects generally occurs. Common defects are: wrinkling in the flange, tearing, earing and surface scratches (Fig. 3.2).

3.1.2 Literature Review

Yang [1] has investigated a lubrication/friction deep drawing model with finite element method to predict the friction coefficient and strain distribution. Simulation is done for combining an elastic-plastic of the deep drawing process with lubrication model. In the lubrication examination, the roughness of surface impact is incorporated by utilizing Wilson and Marsault's normal Reynolds equation. Numerical outcomes are compared with experimental outcomes for the film thickness and also strain distribution. The higher estimation of film oil brings the low friction and high

Fig. 3.1 Elements of the deep drawing process



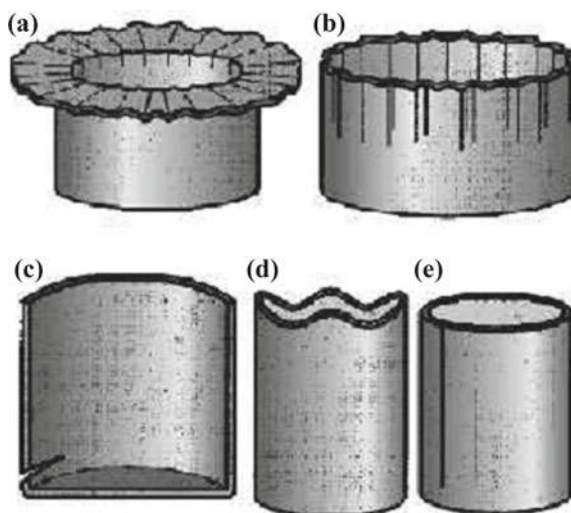
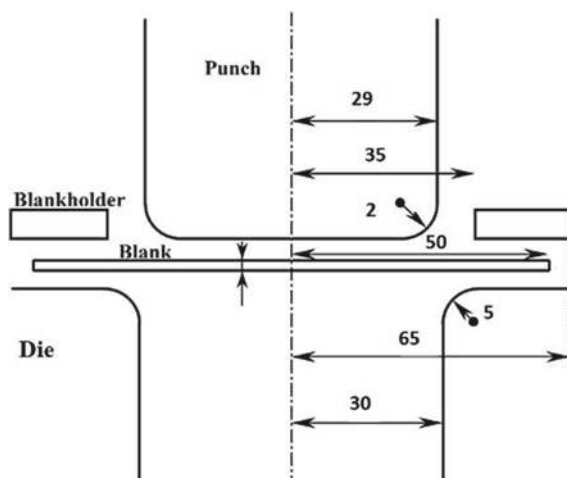


Fig. 3.2 Defects in deep drawing. **a** Wrinkling in the flange, **b** Wrinkling in the wall, **c** Tearing, **d** Earing, **e** Surface scratches

uniform strain distribution. Allen and Mahdavian [2] have built up a test set-up to see the aftereffects of grease on the development of die ring during deep drawing of steel blank. An experimental set-up was established to examine displacement, BHF, punch force and die hoop strain. Completely, lubrication oil is used between die and blank. Wall thickness and surface finish of the containers delivered during the experimental work were estimated to see any relationship with die expansion. Finite element model has been prepared for simulation at different lubrication environments and tooling on the technique. The experimental outcome based on die expansion beneath totally different lubrication condition shows a distinction between the levels of die hoop strain. This work also advocates that if a specific thickness lubricating oil film is given in deep drawing, an anticipated surface finish might be achieved on the final product. FEA models with a greasing up oil film thickness can replicate deep drawing more reasonable. Yamashita et al. [3] have researched the deep drawing by numerical investigation. Maslennikovs techniques are used in the numerical method. Numerical simulation is done by using explicit dynamic in DYNA3D. The impact of the parameters, frictional coefficient of the interface between the rubber ring and blank, for example, measurement and hardness of the rubber ring and mechanical properties of the sheet, on the sheet deformation is investigated. From the investigation, he found that for the production of deeper cup, harder rubber is used and for the shallow cup, thicker rubber ring is used. Demirci et al. [4] have simulated deep drawing for AL1050 by performing experiments and FEA by utilizing ANSYS/LS-DYNA, effects of the BHF on the thickness of cup drawn. This paper demonstrates that under consistent pressure the remains parts at a constant value drops drastically

close container corners and the outer edge of the cup flange. Blank material is considered anisotropic. From the analysis of the aluminium AL1050, they found that forces more than 10 MPa result in tearing of the cup at the wall portion. Abbadeni et al. [5] have investigated the hydro-mechanical deep drawing process. He proposed an analytical method for hydro-mechanical deep drawing; the flow of fluid is calculated from the Reynolds equation. The analytical simulation is done in the finite element software ABAQUS/Explicit. The numerical outputs are compared with the experimental data for the validation of the proposed model. After validation, the effect of BHF and the fluid pressure on the formability of cupping process for the blank material AA5086 aluminium alloy. Dao et al. [6] have examined the effect of thickness variation in trapezoid cup. By using the FEM with Taguchi method, he estimated the optimal values of the process parameters such as punch speed, chamber pressure and coefficient of friction. Brabie et al. [7] have examined the micro deep drawing having thickness variation from 0.05 to 0.20 mm. A mathematical model has been suggested to regulate and constrain the thickness variations on wall portion on the numerical and experimental simulation. Srinivas and Chennakesava Reddy [8] investigated the deep drawing process at an elevated temperature by finite element method, and Taguchi and ANOVA techniques are used. FE simulation is done to decide the significance of process parameters on the deep drawing of a round component. Process parameters like blank thickness, temperature, strain rate and friction coefficient. Result of FE simulation is compared to the experimental work for the validation of work. D-FORM software is used for finite element simulation. From the investigation, he found that the damage in cup occurs at high friction, temperature and strain rate and the wrinkling occurs less for the high value of friction temperature and strain rate. ANOVA, given that the parameters accepted at 90% confidence level. The percentage involvement shows that the temperature assists 61.74% of the variation, thickness parameter assists 24.67% of variations, temperature assists 61.74% of the variation, strain rate contributes 10.17% of variation and friction coefficient influences 2.91% of variation on the effective tensile stress. As the strain rate increases, the effective stress increases. Ayari et al. [9] have investigated deep drawing method to determine the effect on wrinkling by changing the process parameter thickness. FEM simulation is done for aluminium HFS and mild steel by software ABAQUS/Explicit. Choubey et al. [10] examined deep drawing of the conical cup. The thickness distribution of the cup was numerically examined to determine the optimal design. An experimental set-up for the validation of the simulation results for conical and cylindrical cup. The conical cups of 1.5 mm thick sheet with 2.7 limiting drawing ratio and 25 mm height were successfully obtained using an 18° die angle. The maximum punch pressure was 55 MPa in the conventional process and 31.82 MPa in the deep drawing process where in the blank holder was used.

Fig. 3.3 Tool geometry of cylindrical cup drawing



3.2 Research Methodology

3.2.1 Assumptions

It is assumed that the material is isotropic which means that it has similar properties in all directions. The mechanical interaction between all the contact surfaces is assumed to be the frictional contact.

3.2.2 Finite Element Analysis

Numerical simulation may be employed to replace the experimental trial and error process by a virtual trial and error process. Numerical simulation of deep drawing process employs the finite element method facilitating a more accurate determination of the effects of various process parameters on the deep drawing process. The accuracy of numerical simulation using finite element method greatly depends on exact numerical description of the tools, accurate description of material behaviour, contact behaviour and other process variables. The three main stages of any finite element analysis: pre-processing, simulating and post-processing (Fig. 3.3).

3.2.3 Modelling

Modelling in ANSYS This stage includes the creation of a geometric model it includes the following steps.

Define Blank Element Type The element used to simulate the blank is (SOLID185). SOLID185 Structural Solid is suitable for modelling general 3-D solid structures (Fig. 3.4).

Define Blank Material Properties Material properties of AL1200.

Define Contact Element The element used to represent the contact between the tool set (punch, die and blank holder) and the blank is [CONTA174 3-D Surface-to-Surface Contact Element (8 nodes)] (Fig. 3.5).

Generation of Geometric Model The axisymmetric FEM 3D model created for analysis is shown. The model generated using, key points, lines, area and volume. After completing the modelling of deep drawing assembly, next step is meshing

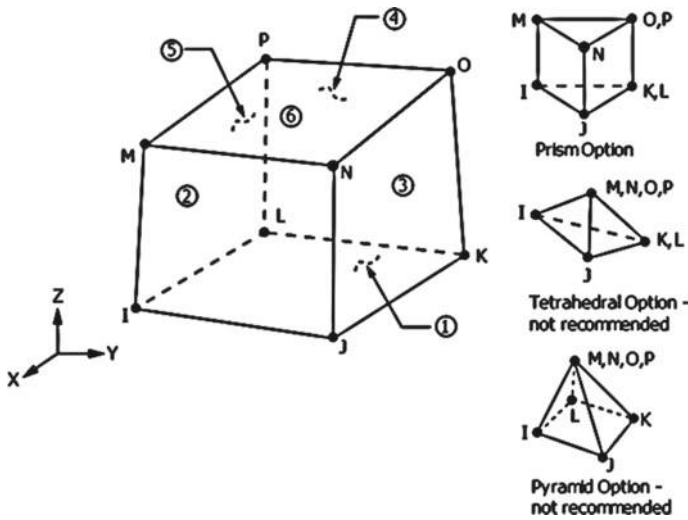


Fig. 3.4 SOLID185 3-D structural solid (ANSYS manual)

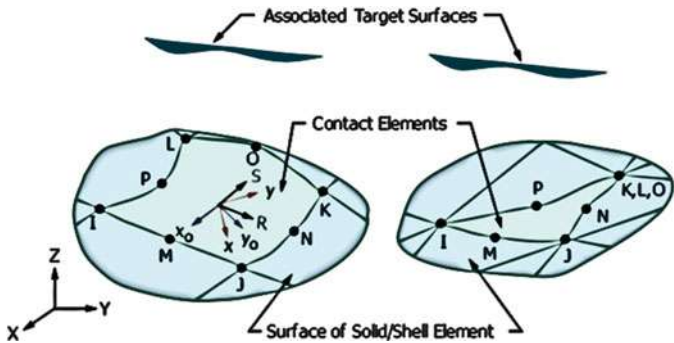


Fig. 3.5 CONTA174 3-D surface-to-surface contact element (8 nodes) (ANSYS manual)

where discretization has to be performed, the mesh created in this work the total node generated is 11,856 and total elements is 14,883, it is clear from the mesh geometry the node numbers and element numbers are almost six in digit which show that the mesh is very fine because the result accuracy depends on the mesh quality (Figs. 3.6 and 3.7).

Boundary Conditions Set the geometry to be axisymmetric around $y = 0$ (axis y - y) and set blank holder load and fixing the die at points (no displacement in both x and y directions). Punch moves downward to draw the blank inside the die to produce a cup; punch moves to total displacement of (35) mm with punch speed equal to 0.23–0.7 mm/s.

Fig. 3.6 3D model of deep drawing process in ANSYS

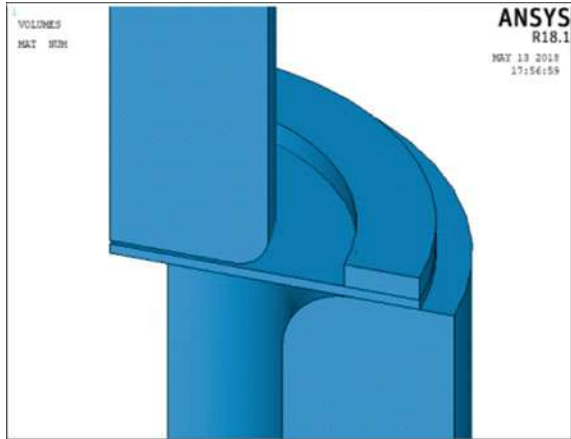
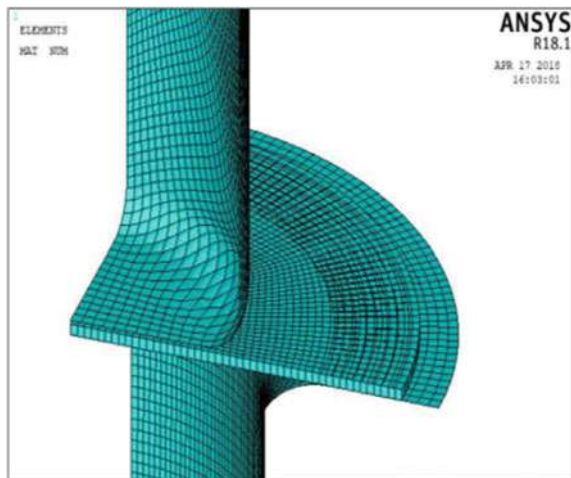


Fig. 3.7 3D meshed model of deep drawing process in ANSYS



3.2.4 Modelling in ABAQUS

All finite element models are created using ABAQUS/CAE pre-processor which are analyzed in this study and investigations of the punch, die and the blank holder are modelled as analytically rigid surfaces, whereas only the blank is defined as deformable body. The blank is meshed by the element C3D8R, a hexahedral element. And the punch and die surfaces are meshed with shell element of SFM3D4R type. And the blank holder is meshed with R3D4 element type.

Material Properties Aluminium of grade AL1200 is used in mechanical properties and chemical composition is in shown in Table 3.1 and 3.2 (Fig. 3.8).

Contact and Boundary Conditions The contact between the sheet and the punch, die and blank holder is upheld by a kinematic contact condition, using master-slave surface sets built up in the initial step of the arrangement. The surfaces of rigid body are defined as master surface and surface of deformable body is defined as slave surface. The friction of coefficient of all the contact surface is provided. The erosion

Table 3.1 Mechanical properties

S. No.	Property	Value (%)
1	Density	2.71 g/cm
2	Young’s modulus	65 GPa
3	Poisson ratio	0.3

Table 3.2 Chemical composition

S. No.	Element	Content (%)
1	Aluminium, Al	≥99
2	Silicon, Fe	≤1
3	Zinc, Zn	≤0.10
4	Copper, Cu	≤0.050
5	Titanium, Ti	≤0.050
6	Manganese, Mn	≤0.050

Fig. 3.8 3D model of deep drawing process in ABAQUS

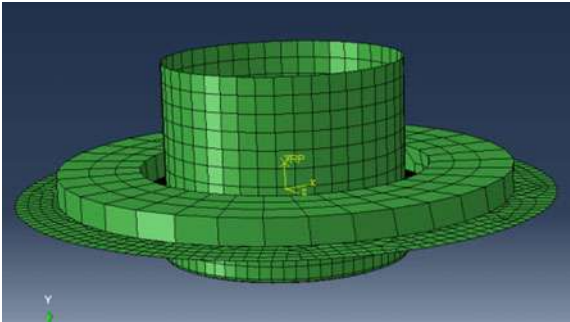


Table 3.3 Boundary condition used in ANSYS/APDL software

S. No.	BHF (MPa)	μ	Punch velocity (mm/s)
1	11	0.005	0.23
2	11	0.01	0.35
3	11	0.02	0.7
4	22	0.005	0.35
5	22	0.01	0.7
6	22	0.02	0.23
7	33	0.005	0.7
8	33	0.01	0.23
9	33	0.02	0.35

Table 3.4 Boundary condition used in ABAQUS software

S. No.	BHF (N)	μ
1	500	0.005
2	500	0.01
3	500	0.02
4	800	0.005
5	800	0.01
6	800	0.02
7	1000	0.005
8	1000	0.01
9	1000	0.02

between the contact surfaces. The boundary conditions are characterized for each step which characterizes the loading and unloading of force and defines the punch movement, blank displacement and degree of freedom (Tables 3.3 and 3.4).

Loading Conditions Blank holder is moved onto the blank with the prescribed displacement to establish the contact force is provided in the holder to hold the sheet; then, punch pushes the blank with the defined velocity into the die through a total distance of 17 mm, that is, the height of the cup (15 mm) plus the initial clearance (2 mm) between the punch and the top surface of the blank.

3.3 Results and Discussion

3.3.1 Results from Abaqus

The value of von Mises stress is a record used to determine the yielding of a ductile material, which is equivalent stress. If a value of von Mises surpasses the yielding

stress then yielding takes place though the value of principal stresses is lower than yield stress rather than checking of principal stress, von Mises stress is the better indicator to check plastic deformation. For cup drawing at 15 mm depth, it is seen that the greatest von Mises stress values are achieved at the die corner. In majority of the region in the flange, the average von Mises stress is greater than the stresses in punch nose region. Consequently, yielding is greater in the flange region. Whereas, in the wall of the drawn part, the von Mises stress has comparatively the lowest values just after the punch radius contact ends (Fig. 3.9 and Table 3.5).

Equivalent plastic strain gives the measure of lasting strain in a designing body. The equivalent plastic strain is figured from the segment plastic strain as characterized in the equivalent strain segment. Most of the basic engineering materials show a linear relationship between stress and strain up to that point is called proportional limit after that point stress-strain relation become nonlinear, but does not necessarily become inelastic (Figs. 3.10 and 3.11).

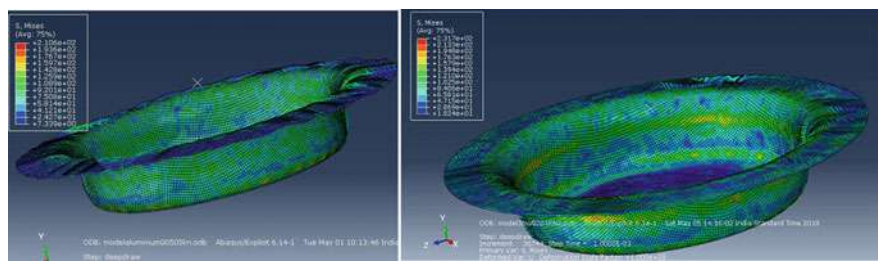


Fig. 3.9 Von-Mises stress distribution for cylindrical cup at 500 N BHF & 0.005 friction and at 1000 N BHF & 0.02 friction

Table 3.5 Results from ABAQUS software

S. No.	BHF (N)	Height of cup after punch force removed (mm)	Von-Mises stress (MPa)		Equivalent plastic strain	
			Max	Min	Max	Min
1	500	13.6	210.6	7.339	0.2704	0.009727
2	500	13.67	197.2	7.249	0.2718	0.009614
3	500	13.63	205.1	7.817	0.2703	0.009909
4	800	14.31	211.1	6.81	0.2707	0.009837
5	800	14.36	213	6.07	0.2728	0.0101
6	800	14.28	205	7.847	0.2678	0.01008
7	1000	14.55	217.9	4.24	0.2731	0.009372
8	1000	14.64	194	7.357	0.2778	0.009947
9	1000	14.64	231.7	10.24	0.2721	0.01036

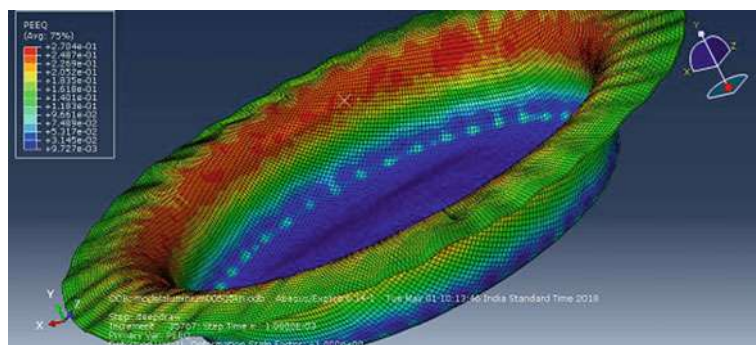
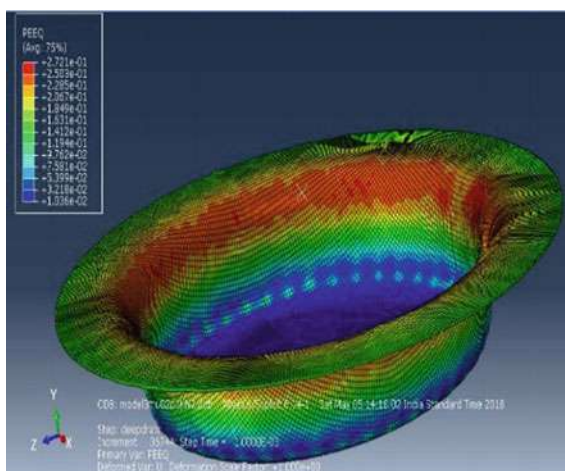


Fig. 3.10 EQPS cup at 500 N BHF and 0.005 friction

Fig. 3.11 EQPS cup at 1000 N BHF and 0.02 friction



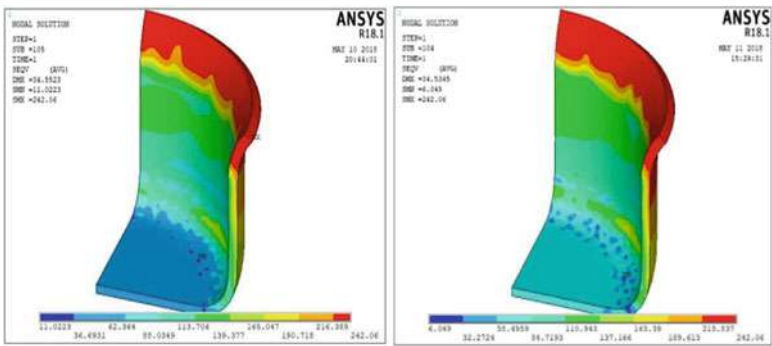
Plastic behaviour, characterized by non-recoverable strain begins when stress exceeds the value of yield stress. For the cylindrical cup drawing at a 15 mm cup depth, it is realized that the maximum equivalent plastic strain values are attained at the die corner.

3.3.2 Results from ANSYS Software

Stress

The first principal stress gives the estimation of stress that is normal to the plane where the shear stress is zero. It comprehends the max tensile stress prompted in the part because of the loading conditions. After performing finite element simulation for first

experiment, the maximum first principal stress obtain is 155.97 Mpa. The maximum second principal stress obtain is 67.99 Mpa. The third principal stress that is normal to the plane where the shear stress is zero. It comprehends the max compressive stress prompted in the part because of the loading conditions. The maximum third principal stress obtains 18.4768 Mpa.



Von-Mises stress is used to decide whether a given material will yield or crack. It is used the most of ductile material. The design will fail if the most extreme estimation of von Mises stress induced in the material is more than the strength of the material. The maximum von Mises stress obtain is 242.06 Mpa (Table 3.6).

Punch Pressure From the nine simulations for the three levels for three process parameters, parameters are blank holding force, velocity and friction. We found that the punch pressure drops rapidly after about 25 mm of punch displacement in the total displacement of 35 mm which indicates the onset of plastic instability in the part. The punch pressure increases by increasing BHF and friction (Figs. 3.12 and 3.13).

Table 3.6 Results from ANSYS software

S. No.	Von-Mises stress		Equivalent plastic strain		Punch	Pressure (MPa)
	Max	Min	Max	Min	Max	Min
1	242.06	11.02	0.606	0.024	174.76	19.42
2	242.06	15.41	0.605	0.0272	214.09	23.79
3	242.06	7.85	0.599	0.0374	283.66	31.52
4	242.06	6.96	0.608	0.0238	192.78	21.42
5	242.06	15.99	0.604	0.028	234.43	26.05
6	242.06	6.29	0.601	0.0395	297.84	33.09
7	242.06	7.17	0.608	0.0239	185.62	20.62
8	242.06	13.98	0.605	0.0271	207.95	23.11
9	242.06	6.05	0.600	0.039	264.96	29.44

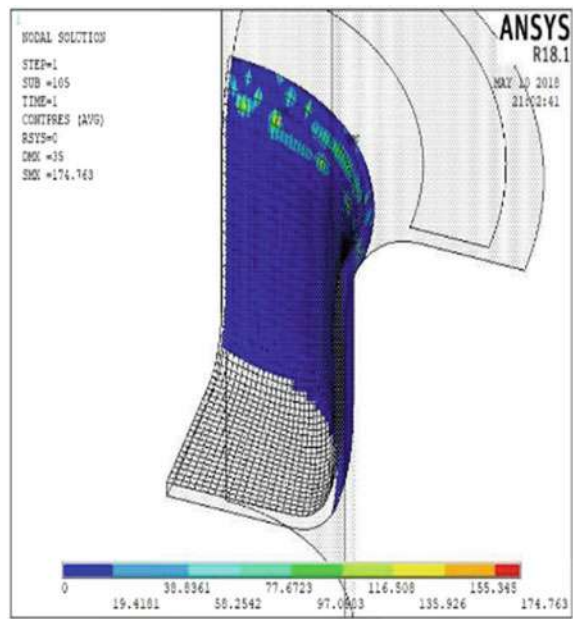


Fig. 3.12 Punch pressure at contact

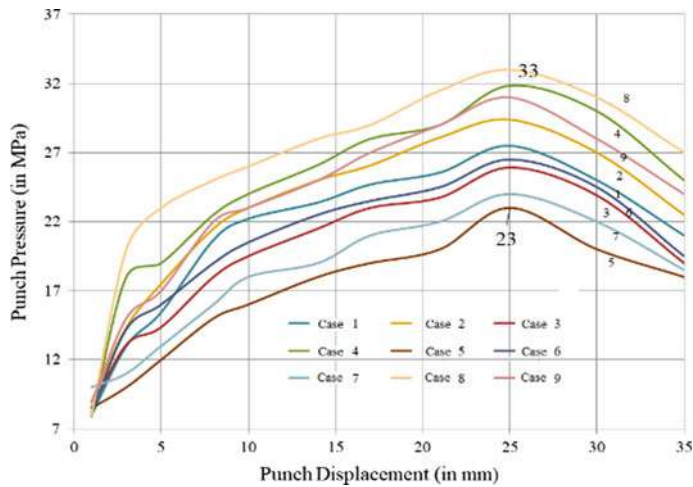


Fig. 3.13 Graph between punch pressure with punch displacement

3.4 Conclusions

The following important conclusions are obtained from the investigations deep drawing of aluminium alloy AL1200.

The minimum wrinkling occurs in the deep drawn cup at the flange region at the BHF of 1 KN and friction of 0.02.

Defects like cracking, tearing and necking are not observed in the deep drawn components of aluminium alloy AL1200.

The minimum spring back effect occurs in the deep drawn at the BHF of 1KN and friction of 0.01. Max punch pressure is needed when the coefficient of friction is 0.01 and BHP is of 1000 N.

The optimal parameter combination includes blank holder pressure 22 MPa, velocity of punch 0.35 mm/s and coefficient of friction 0.005.

Punch pressure is maximum at maximum blank holder pressure with 0.35 mm/s punch velocity.

Punch pressure was varied from 23 to 31 MPa for various experiments.

FEM simulation results showed that the maximum principal stress on deep drawn cup varies from 156.01 to 166.04 MPa and plastic strain varies from 0.59 to 0.61.

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Chapter 4

Numerical Investigation on Single Point Incremental Forming (SPIF) of Tailor Welded Blanks (TWBs)



Jeet Raut , Shalin Marathe and Harit Raval

Abstract TWB gives weight and cost-saving which are the ultimate desires of the automobile and shipbuilding industries. Despite having such tremendous advantages, the applications of TWBs have been limited due to the limitations of weld line movement and formability reduction compared to the base metal. In this research work, attempts have been made to study the forming behaviour of the tailor welded blanks (TWBs) during the single point incremental forming process. In this work, numerical simulation has been carried out using ABAQUS/Explicit. The TWBs are considered to be made from the friction stir welding (FSW) process. Different strategies of tool path movement have been adopted during the simulation of SPIF process and its effect on weld line movement has been studied. It has been found that the tool dragging effect plays an important role in weld line shift. As a conclusion, it was also observed that the initial position of the SPIF tool affects the weld line shift and stress–strain development as well.

Keywords Tailor welded blanks (TWBs) · ABAQUS/explicit · Single point incremental forming · Weld line movement

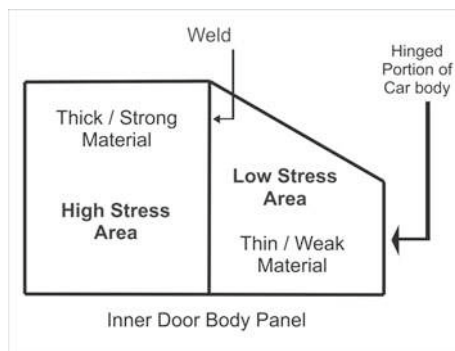
4.1 Introduction

In several industries like automotive and aerospace, the need for lightweight and cost-effective products with exceptional performance is essential for success. Tailor welded blanks (TWBs) offer an excellent means to meet these challenging and seemingly inconsistent demands without compromising the strength. It is a technology to join the similar/dissimilar material prior to any forming operation. Figure 4.1 represents the schematic of a typical example of a TWB application in the inner door panel. TWB developed from similar material having dissimilar gauges can lead to the cost as well as the overall weight saving of the component. The thick/strong

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_4

Fig. 4.1 Schematic representation of TWB application at inner door panel of a car body



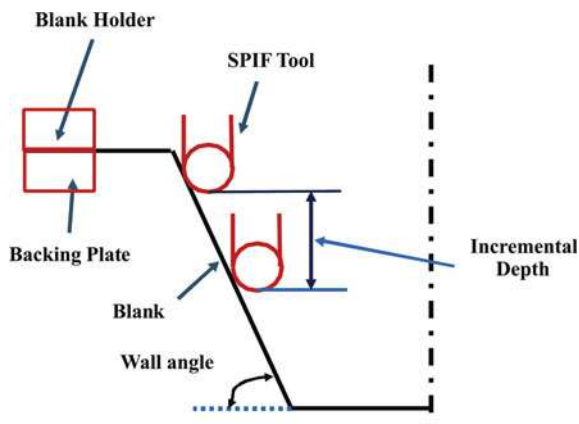
material can be placed where the stresses are generated more and accordingly the thin/weak material can be placed which will result in weight and cost-saving without compromising the strength of the joint.

In TWBs, the dissimilarity can be in the material's grade, thickness, strength or coating [1]. As mentioned, TWB is a cost-effective technology, General Motors estimated that a tailor-welded door inner panel saved the company \$ 4.9 million through the elimination of 14 dies, weld fixtures and check stands [1].

TWBs can be developed by using materials like aluminium [2] and its alloys, steel and its different grades as well. The selection of parent material to develop the TWB depends upon the joining process by which the joint will be made and the end application of the product or component. TWBs made from the steel are generally developed using the laser beam welding technology but for the aluminium material conventional welding processes are not suitable. The reason behind that is the aluminium material is low absorptive and highly reflective for the laser. For TWBs made from aluminium material, solid-state welding process like FSW is suitable.

Friction stir welding (FSW) is found to be giving excellent results regarding strength and the quality of the joint. The FSW process gives reasonably good quality of welds for particularly aluminium alloys [3, 4] which is difficult to achieve for the laser or any other conventional welding processes [5]. Despite such advantages of TWBs, there are some limitations to this technology such as formability reduction and weld line movement. Due to the differences in material as mentioned earlier, there will be the development of unequal stresses and strain that will result in the weld line movement towards the strong material during the conventional forming operations. It has been found that TWBs made of aluminium result in lower formability compared to that of the participating materials in it [5]. TWBs are related to the heterogeneous nature of the blank because of dissimilar materials and weld where the thinner/weaker material may deform preferentially and tear prematurely in conventional forming processes, which also results in weld line movement [5]. It has been also found that increase in thickness ratio of TWB materials causes a decrease in formability [6]. One of the reasons for the formability reduction and weld line movement can be the development of unbalanced forces during the conventional forming processes like deep drawing [5]. However, a lot of efforts are being done in order to develop and

Fig. 4.2 Schematic representation of SPIF process



modify the metal forming process and which have resulted in process like single point incremental forming (SPIF). In this process, localized deformation is produced on to the blank which is clamped in the blank holder (see Fig. 4.2).

Deformation is provided using a tool during the SPIF process and forming occurs due to the combined effect of stretching and bending [7, 8]. It has been also reported that the SPIF process results in better formability [7] for homogeneous blank compared to the other conventional forming processes [9]. Due to this advantage of SPIF process, it has been selected to study the forming behaviour of TWBs. The combination of SPIF and TWB technologies has not been much explored by many researchers yet.

So in this work, attempts have been made to investigate the formability of friction stir TWBs made of dissimilar material having same/different thickness during the SPIF process. Comparison of weld line movement for the different conditions is carried out.

4.2 Numerical Condition

In this present study, all the simulations were carried out using ABAQUS/explicit. This study has been divided into three sections. The detailed conditions of all the sections are reported in the upcoming sections. During the simulation in ABAQUS/explicit, individual sections were defined namely base metal-1, base metal-2 and weld metal, and accordingly, the material properties were assigned as per Table 4.1. In the present study, AA5052 H32 and AA5754 H22 are considered as strong and weak material, respectively, by comparing their respective yield strengths. It was also assumed that the materials mentioned in Table 4.1 obey the power law equation.

In the present work, square pyramid was developed from the TWB using the SPIF process. The blank size was considered as $125 * 125$ (mm²), and the step size or the

Table 4.1 Material properties consideration for simulation [10]

Properties	AA5754 H22	AA5052 H32	Weld Material
Density (g/m ³)	2.67	2.68	2.67
Young modulus (GPa)	68	70	69
Poisson ratio	0.33	0.33	0.33
Yield stress (MPa)	122	140	131
Strength coefficient (MPa)	252.04	258	255.37
Strain hardening index	0.112	0.125	0.1185

progressive downward deformation step was kept constant at 1.27 mm. The SPIF tool was considered to have the hemispherical end. The blank was considered as shell deformable-type body, and tool was considered as 3D analytical rigid body. Some other constant parameters are provided in Table 4.2

The distance of tool from the weld zone/line before forming process depends upon the dimensions of the target geometry to be produced. Considering the conditions mentioned in Table 4.2, the appropriate location of tool before forming process was selected.

Table 4.2 Parameters kept constant during simulation

Parameters	Values
Blank size (mm * mm)	125 * 125
Tool radius (mm)	6.35
Step size (mm)	1.27
Wall angle (degree)	45
Friction coefficient [11]	0.1
Draw depth (mm)	25.4
Tool type	3D analytic rigid shell
Mesh element	S4R

4.2.1 Section—I (Welded Blank of Same Thickness and Different Material)

In this section, the mentioned materials in Table 4.1 are considered to be joined using a friction stir welding process and both having 2 mm of thickness. One more simulation was carried out using same conditions for dissimilar thickness in which the thickness of AA5754 H22 and AA5052 H32 were considered as 2 and 1 mm, respectively. So, in this section effect of thickness ratio will be investigated on weld line movement. The assembly of tool and the TWB is indicated in Fig. 4.3. It can be observed that the tool movement will be in the XY plane, and the incremental depth will be in the Z (–ve)-direction.

During the simulation, the edges of the blank were fixed and all degrees of freedom (DOF) were fixed which is known the encastre boundary condition (see Fig. 4.4). The tool movement was provided by providing movement to the reference point of tool. In this section, the initial position of tool is at weak material (AA5754 H22) and it is moved towards the weld zone.

Fig. 4.3 Assembly of the tool and the TWB

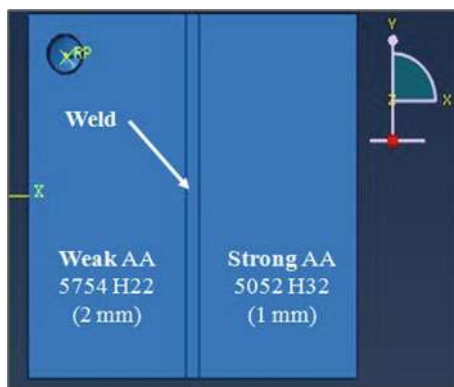


Fig. 4.4 Encastre boundary condition applied to the blank edges

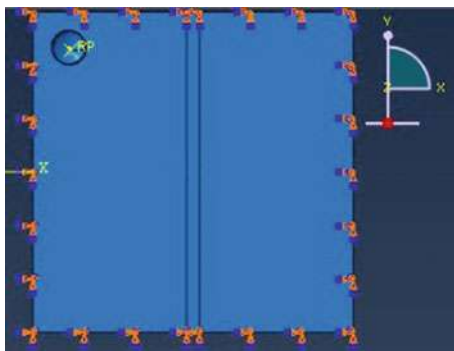
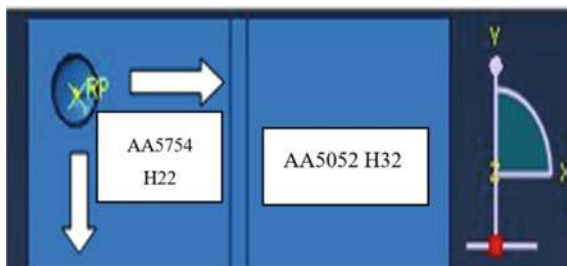


Fig. 4.5 Different tool movement direction from weak material for welded blank of different thickness and material



4.2.2 Section—II (Welded Blank of Different Thickness and Material)

In the present section, the TWB is considered to be made of dissimilar material and having different thickness. In this section (see Fig. 4.5), two different directions of tool are considered and compared. First, the tool is considered to move towards weld (+ve X-direction) and second that movement is considered parallel to weld (–ve Y-direction).

4.2.3 Section—III (Effect of Direction of Tool Movement)

In this section, TWBs are considered to be made of dissimilar material having similar thickness but the starting position of tool during forming is considered on weld (see Fig. 4.6). Two different conditions are considered for the tool movement from the initial position of the tool, in first condition, the tool will move towards strong (+ve X-direction) side and in second condition, tool will move towards weak material (–ve X-direction). The following section contains comparison of results for the earlier mentioned different simulation conditions.

Fig. 4.6 Two different conditions of tool movement



4.3 Result and Discussion

The basic mechanism responsible for the weld line movement depends upon the strength of the parent materials. During the forming process, same amount of load will be applied to both the parent materials of the blank but as one parent material having different properties than another; there will be the development of unequal stress–strain. The weak/thin material will expand and deform more compare to the strong/thick material.

4.3.1 Section—I (Welded Blank of Same Thickness and Different Material)

The results of weld line movement have been compared for this section as indicated in Fig. 4.7. In Fig. 4.7, the weld line shift towards negative and positive sides indicates shift of weld line towards weak and strong materials, respectively. It is noted that the weld line movement is more in the inclined wall section of the formed part compared to the centre part of the blank irrespective of the thickness ratio/difference. There are two peaks observed at 20–40 mm and 80–100 mm, it is due to the tool movement across the weld line. As the tool tries to drag the weld line in its direction of movement, the weld line shifts towards that direction and because of that such nature can be observed. It is also observed that, due to the thickness difference, the weak material

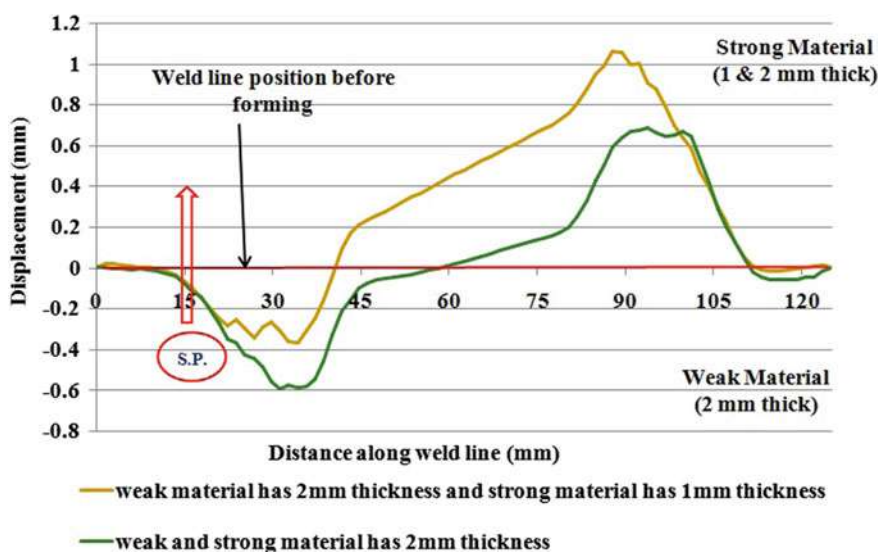


Fig. 4.7 Comparison of weld line movement for welded blank of same thickness and different material

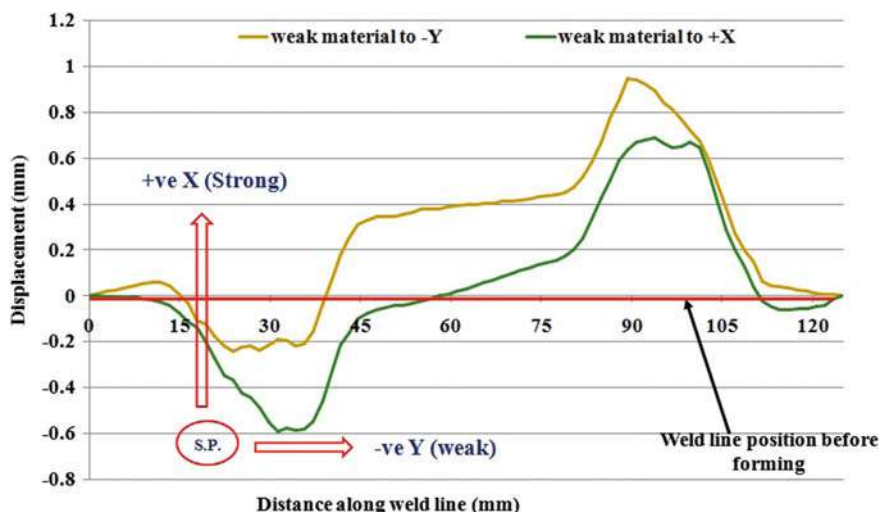


Fig. 4.8 Comparison of weld line movement for welded blank of different thickness and material

becomes weaker and due that the nature of the weld line changes as indicated in Fig. 4.7. The same figure also indicates that, as the thickness difference increases, the weld line shift increases. The increase in the difference of thickness of parent material will lead to in-homogeneity in the material of blank. This in-homogeneity leads to uneven distribution of forces and strains. That will result in the increase in the weld line shift during the forming process.

4.3.2 Section—II (Welded Blank of Different Thickness and Material)

For the conditions given in this section, again the peak value of the weld line movement is observed in the wall section of the square pyramid. From Fig. 4.8, it can be observed that the tool dragging effect plays the vital role in the weld line shifting. In Fig. 4.8, the weld line shift towards negative and positive sides indicates shift of weld line towards weak and strong material, respectively.

4.3.3 Section—III (Effect of Direction of Tool Movement)

The results of this section are in agreement with the same of the previous two sections regarding the peak weld line movement and weld line movement in the wall section of the square pyramid. Again, it must be noted that the tool dragging effect plays

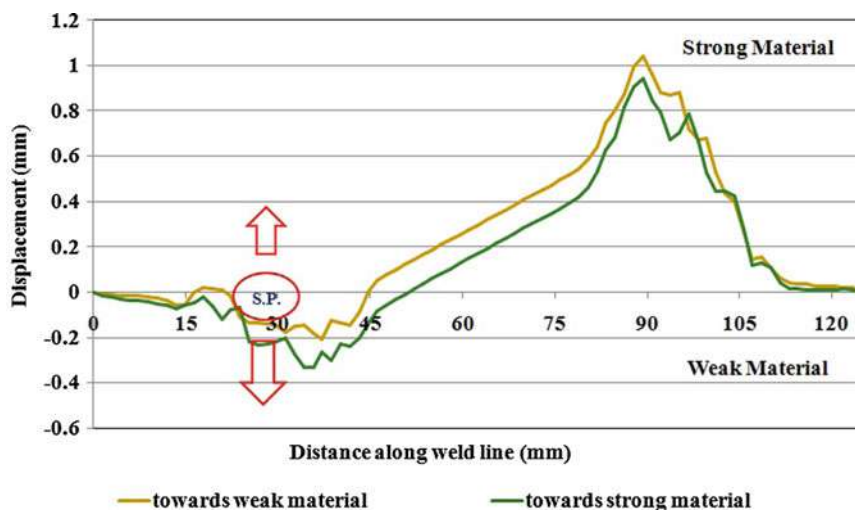


Fig. 4.9 Comparison of weld line movement for different direction of tool movement

the important role in the weld line shift. Since there will be no dragging effect in this case at 20–40 mm distance on weld line, the weld line shift is observed to be less at the mentioned region as shown in Fig. 4.9. In this figure, the weld line shift towards negative and positive sides indicates shift of weld line towards weak and strong materials, respectively.

Particularly for Section III, some additional results were extracted from the simulations regarding equivalent plastic strain (PEEQ) and von Mises stress. Figure 4.10 indicates the top view of the component formed with the help of SPIF process. It can be observed that the weld line/zone has become inclined due to inequality of stress–strain developed during the forming process.

Figure 4.11 and 4.12 indicate the von Mises stress and equivalent plastic strain developed during the SPIF of the TWBs, respectively. It can be clearly noted from Fig. 4.11 that the amount of stress developed in the weak/thin material is more compared to the same developed in the strong/thick material. Also, it should be noted that the maximum stresses are developed at the corner compared to the other part of the component being formed. When the tool reaches to a corner, tool stops and changes the direction of the movement and that result into the stress concentration effect leading to high stress and strain values.

Figure 4.12 indicates the equivalent plastic strain (PEEQ) developed in the part during SPIF process. From this figure, it can be noted that the maximum strain values are found to be at the corners. Added to that, a high magnitude of the strain is also observed at the weld line where tool moves in the downward direction by 1.27 mm after completing the planner movement. Due to downward movement of SPIF tool, more localized stretching and bending effects are produced at the weld line, and due to that it results in the high magnitude of the strain.

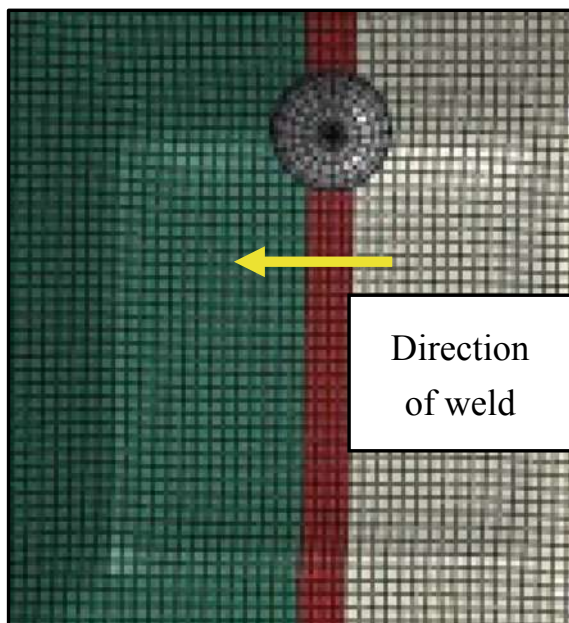


Fig. 4.10 Weld line movement observation

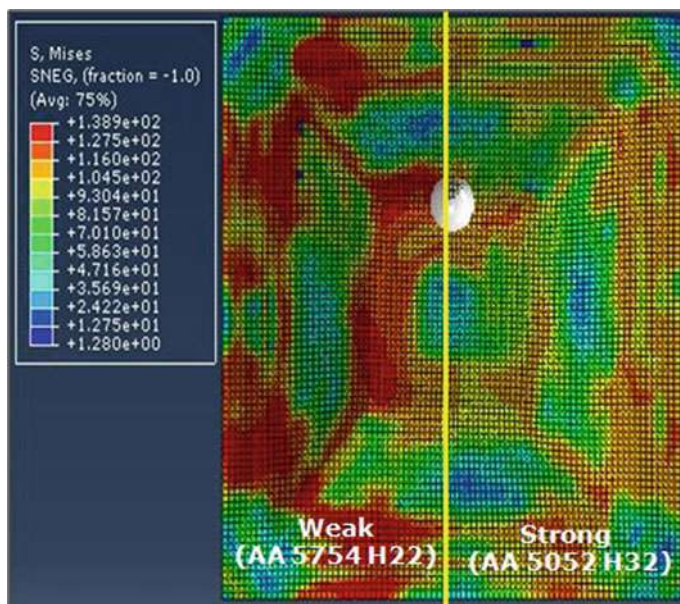


Fig. 4.11 Von Mises stress developed during the SPIF process of TWB for different direction of tool movement

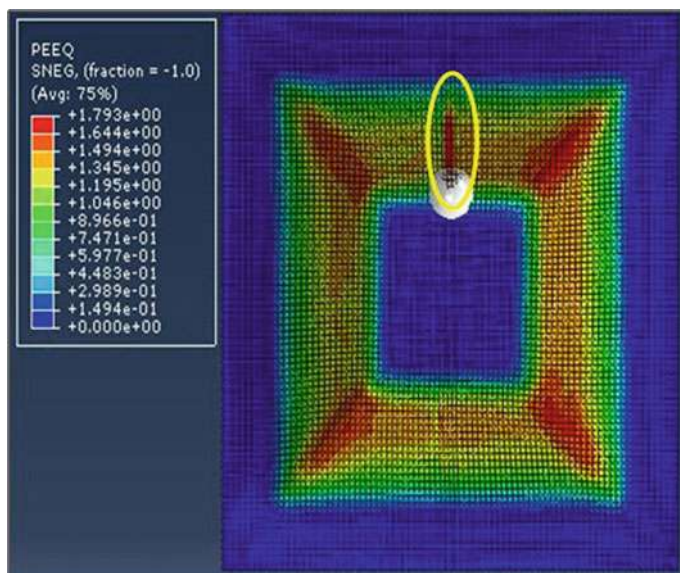


Fig. 4.12 Plastic equivalent strain (PEEQ) developed during the SPIF process of TWB for different direction of tool movement

4.4 Conclusions

In the present research work, numerical investigation on TWBs assumed to be made of FSW process was carried out. Combination of different material and thickness was investigated. From the simulation study carried out in the present work, following concluding points can be stated:

- Initial position of the forming tool plays a vital role in the weld line shift during the SPIF process, and therefore, it should be chosen accordingly.
- The thickness difference of the participating materials should be adopted very carefully because the weld line movement greatly depends upon it. More thickness difference leads to more in-homogeneity in the blank leading to uneven distribution of strain.
- Tool dragging effect plays an important role in shifting of weld line.
- Maximum weld line shift is observed in the wall section of the square pyramid.
- The weld line movement is least towards the starting point of deformation.
- The maximum value of von Mises stress and equivalent plastic strain (PEEQ) is found to be developed at the corners of the component being formed during the SPIF process.

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Chapter 5

Force and Thermal Variational Analysis by FE Approach on Dry Turning of Inconel 718



Bishal Das , Jibin T. Philip , Kore Mahesh and Basil Kuriachen

Abstract In the present work, dry turning process of Inconel 718 was simulated using DEFORM software. The Johnson–Cook fracture model was implemented in the FEM platform to develop and to analyse the 2D turning process model. The existing dire needs to reduce environmental pollution, and production costs have forced the manufacturers to limit the use of cutting fluids. In view of the above, the process was simulated at dry machining condition to investigate the influence of the two critical input factors: cutting speed (at 125, 300 and 475 m/min) and feed rate (at 0.05, 0.10 and 0.15 mm/rev). The variational analysis of the predicted machining responses, viz. cutting force, feed force, workpiece and tool interfacial temperature showed an incremental trend with time and cutting speed, respectively. The temperature simulation results were in agreeable concordance with the experimental results, from existing literature, with a maximum error of only 3.38%.

Keywords Dry turning · Inconel 718 · DEFORM · FEM · Simulation

5.1 Introduction

The highly disparate and hard to machine materials like Inconel 718, Ti6Al4V, and nitinol possesses excellent mechanical properties at elevated temperature [1]. They possess high strength and wear resistance with a wide range of applicability in the fields of aerospace, marine, biomedical, nuclear, etc. [2]. The high-temperature strength of Inconel 718 is due to the effects of the submicron gamma double prime precipitates, to a greater and the gamma prime precipitates, to a minimal extent, respectively. In addition, it possesses other advantageous material properties such as high corrosion resistance, high melting temperature, efficient thermal fatigue and shock resistance [3]. Researchers have identified that machining of nickel-based

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alloys is subjected to generation of surface cracks, plastic and metallurgical deformations, enhanced microhardness and induced residual stresses which can lead to significant loss and failure during production and manufacturing [4, 5].

Although a wide range of modern tools have been developed and used to improve the machining efficiency of the cutting process on Inconel 718, coated carbide tools are found to have an upper hand in terms of extended tool life and better surface roughness in the case of dry turning, in particular. The carbide tools made of cubic boron nitride (CBN) and polycrystalline cubic boron nitride (PCBN) present exceptional machining performance, despite having close competence from ceramic tools. But their comparatively high cost delimits their demand to some extent [3, 6]. The increasing demand for improved quality and productivity led to the development of ceramic tools. They are generally employed for high cutting speed operations on nickel-based alloys. Silicon nitride (Si_3N_4), a ceramic tool, was found to produce optimal machining efficiencies [7].

High-speed machining increases the efficiency of cutting by reducing the time taken to manufacture a product. The difficulty arises during the machining of the material to the desired shape and size due to its relatively high strength. The cooling effect of the lubricant within the cutting zone is not effective and not evident at high-speed machining. This leads to the development of dry machining. The advantages of dry machining are (i) non-pollution of environment, (ii) cost deduction due to eliminated coolants, (iii) reduced efforts for cleaning of worked surfaces and (iv) minimized disposal costs and energy consumption [8]. The practical approach for finding the experimental data involves high cost and may not be feasible at times.

The concomitant efficacy that can be achieved through the development of an efficient simulation routine in context of machining Inconel 718 or any such materials per se is countless. The efforts to test, validate, and optimize machining processes through rigorous experimental procedures are not practical enough. A few of the integrant factors that add up to the same include (i) excessive tool wear due to work hardening, (ii) the presence of microstructural carbide particles leading to abrasion, (iii) generation of high cutting temperatures at the rake face, (iv) diffusion wear resulting from material chemical affinity, (v) welding and adhesion initiating notching effect, (vi) low surface finish caused by vibrations generated by high cutting forces [8]. Such delimiting effects extensively increase the associated costs, material wastage, time required, manpower, etc., beyond economic and practical limits. Moreover, with the emergence of highly integrated and time-efficient simulation routines, which are made available through software's such as DEFORM (2D and 3D), ANSYS (APDL and Work Bench), MATLAB (R Series), COMSOL multiphysics directly proceeding for experimental analysis can only happen due to biased rationale or sheer negligence.

The finite element modelling (FEM) gives a detailed understanding of the cutting process by predicting the cutting forces and cutting temperatures during the machining process [9]. The modelling also forecasts the chip morphology which plays a significant role in determining the machining responses. Therefore, there is a strong need to develop a FEM model to predict the machining responses precisely. The cutting process can be modelled either in 2D or in 3D. Nevertheless, it should

Table 5.1 Input parameter setting [6, 7]

Parameter	Value
Initial temperature	20 °C
Cutting speed	125,300,475 (m/min)
Feed	0.05,0.10,0.15 (mm/rev)
Tool	Cubic Boron Nitride (CBN) and Silicon Nitride (Si_3N_4)

be considered that, 2D modelling requires only one-tenth of the time required in comparison with iteration using 3D models [10]. Therefore, the development of 2D model will be of great importance and advantage with regard to the subject context.

This paper focuses on the 2D turning process simulation of Inconel 718 and to investigate the effects of cutting speed and feed rate on the response parameters such as cutting forces and the cutting temperature. The accurate modelling and predictability of such responses will play a predominant part in determining the machinability of the material and thereby optimization of input parameters for maximum efficiency.

5.2 Process Parameters

A sophisticated dry turning process model was developed to perform high-speed machining of Inconel 718. The input parameter setting for the simulations is shown in Table 5.1. For the cutting process, only the cutting speed and the feed rate have been considered, as the simulations were carried out for 2D surface. The initial temperature of the workpiece and tool has been assumed to be 20 °C. A total of sixteen simulations have been performed to predict and evaluate the effect of the input factors on the machining responses.

5.3 Modelling Scheme

The entire modelling and simulation routines/procedures were developed on DEFORM 2D platform through FEM implementation. The values of the process parameters, viz. cutting speed and the feed rate, were inputted to the software through proper channel for the various levels considered, as shown in Table 5.3. The material model Inconel 718 was imported from the materials library, after proper geometric definitions and application of required material properties. For the simulation of force, the tool was created with a nose radius of value 0.4 mm, rake and clearance angle of -6 and 11 , respectively. CBN was used as the material for the tool. For the simulation of temperature, the tool was created with a nose radius of value 0.8 mm, rake and clearance angle of -6 and 6 , respectively. Silicon nitride (Si_3N_4) was used as the coating material for the tool.

The simulation model for turning process is shown in Fig. 5.1. The diameter of the workpiece was assigned with a value of 50 mm. The 2D model dimensions of the workpiece were specified with length and height values as 30 and 5 mm, respectively. Further, the tool and workpiece were meshed with 8847 and 3641 individual finite elements, respectively. The meshing procedure was executed with the enabled re-meshing criteria. As it can be a significant factor in obtaining accurate and precise outputs, due to its capability in repeated mesh generations, without any external aid or initiation, in case of any failure or improper meshing during the course of the process. Thereby, it ensures that the required nodal values are generated with fine meshing at the regions of ultimate interest and consideration of the user. The criteria avoid convergence and contact problem in long chips [2].

The Johnson–Cook model equation which governs the modelling scheme is shown in Eq. (5.1).

$$\sigma_{eq} = (A + B\epsilon^n) \left(1 + c \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right) \left(1 - ((T - T_{room}) / (T_m - T_{room}))^m \right) \quad (5.1)$$

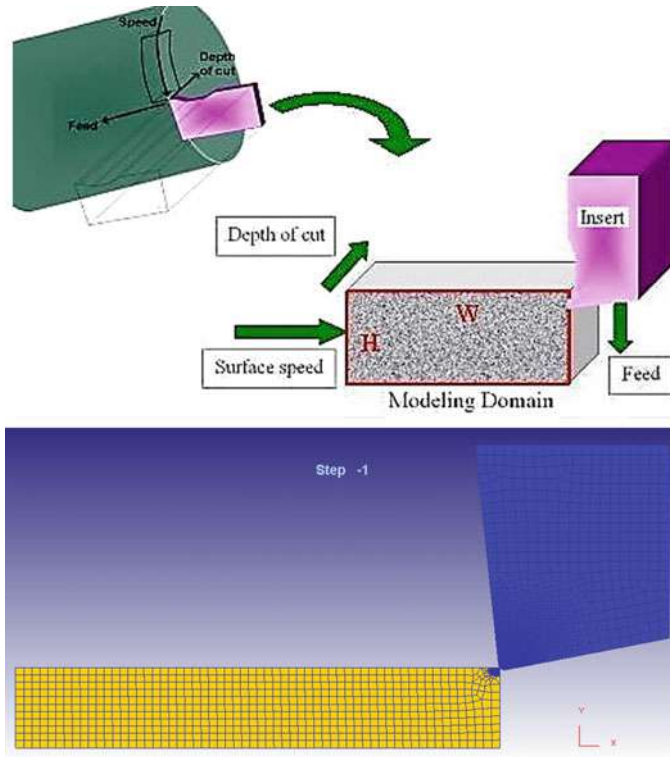


Fig. 5.1 Simulation model

The thermal and velocity boundary conditions were assigned to the tool and the workpiece. An initial temperature of 20 °C was assigned to both bodies. The cutting speed was given along the X -axis, and the Y -axis was kept fixed for the workpiece. Johnson–Cook material model was implemented because of its viscoplastic flowing regulation in the form of an exponential function which can describe the deformation localization during high-speed load. Experimental constants of Johnson–Cook model for Inconel 718 are A , B , c , n , m which possess values of 450 MPa, 1700 MPa, 0.017, 0.6522 and 1.3, respectively [11, 12].

5.4 Results and Discussion

The influence of the input factors of this research effort, viz. cutting speed and feed rate on the machining responses, has been discussed in detail in this section with significant plots, figures, tables and explanations. A total of nine simulations were conducted to verify the forces during machining and seven simulations for the validation of the temperature in the cutting zone. Further, simulation model values are validated with the experimental results. The machining simulation images at various time steps are as shown in Fig. 5.2.

The force simulation results predicted for cutting force and feed force alongside with experimental values extracted from the literature are noted in Table 5.2. Moreover, the plots depicting the variation of feed force and cutting force with feed rate for both simulation and experimental results [6] are shown in Figs. 5.3 and 5.4, respectively. The percentage of error between the actual and simulation values for cutting force and feed force was calculated. The former was observed to have a maximum error of 27.95%, while the latter, a value of 26.77%. The existence of the higher error% values, although below 30%, can be attributed to the exclusion of the depth of cut factor, which can be an influential factor in the case of cutting force and feed force. This variation in error can be minimized by extension of the existing 2D model to 3D, by implementing and incorporating the entire design to the DEFORM 3D software interface. Furthermore, the congruence in variational trends of forces for simulation and experimental results [6] can be clearly observed from Figs. 5.3 and 5.4, respectively, in spite of the aforementioned errors.

Force profile at cutting speed of 475 m/min and 0.15 mm/rev feed rate is shown in Fig. 5.5a, b. The profiles of cutting force and the feed force show that the force first increases to overcome the ultimate shear stress of the material to initiate the chip formation. Once the chip formation starts, the force experienced by the tool slightly reduces and maintains a nearly constant variational profile (remains at an average cutting force and feed force of 749.47 N and 362.38 N, respectively) with time. The underlying reason for this specific characteristic is that, further increase in force is non-essential for progressive chip removal, once the ultimate shear stress of the material has been exceeded initially.

The simulation data compared with that of the experimental temperature result are shown in Table 5.3. A rising trend in temperature with increased cutting speed

Fig. 5.2 Machining simulations

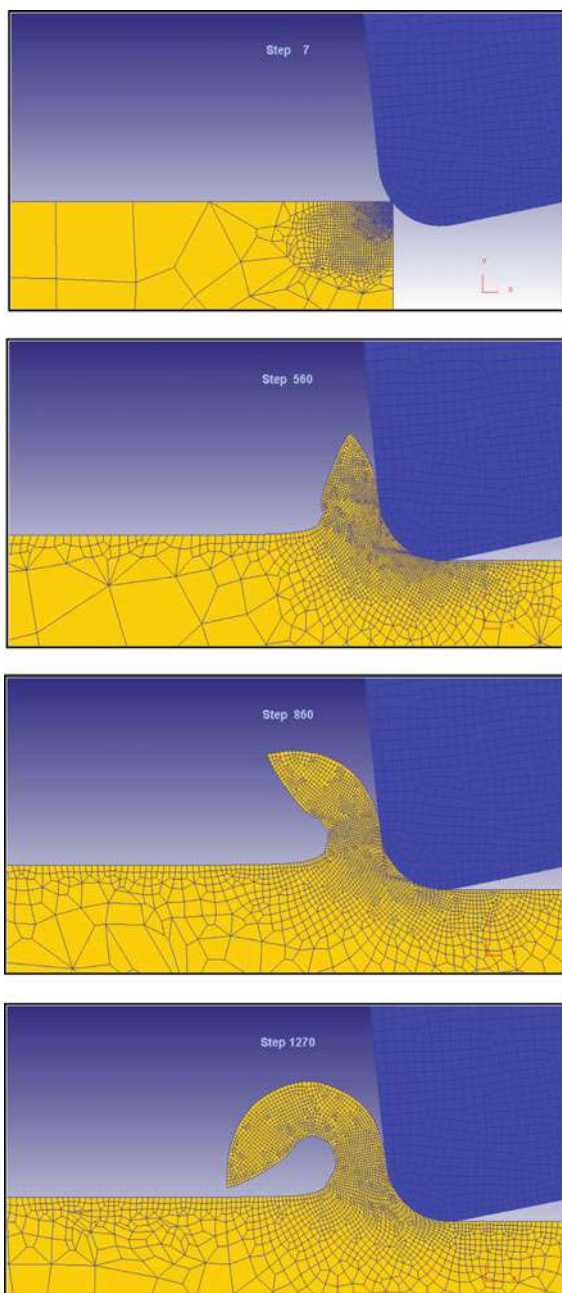


Table 5.2 Force results

Cutting Speed (m/min)	Feed rate (mm/rev)	Cutting Force (N)		% Error	Feed Force (N)		% Error
		Experiment [6]	Simulation		Experiment [6]	Simulation	
125	0.05	388.28	492.8	21.20	366.79	487.5	24.76
	0.10	492.75	579.7	14.99	382.41	456.3	16.19
	0.15	705.74	833.6	15.33	462.52	531.7	13.01
300	0.05	334.55	431.5	22.46	202.68	274.8	26.24
	0.10	342.37	475.2	27.95	241.75	318.6	24.12
	0.15	694.01	758.8	8.53	335.53	454.85	26.23
475	0.05	273.01	341.25	19.99	166.54	215.55	22.73
	0.10	390.23	525.5	25.74	231.99	316.8	26.77
	0.15	646.15	749.7	13.81	296.46	362.38	18.19

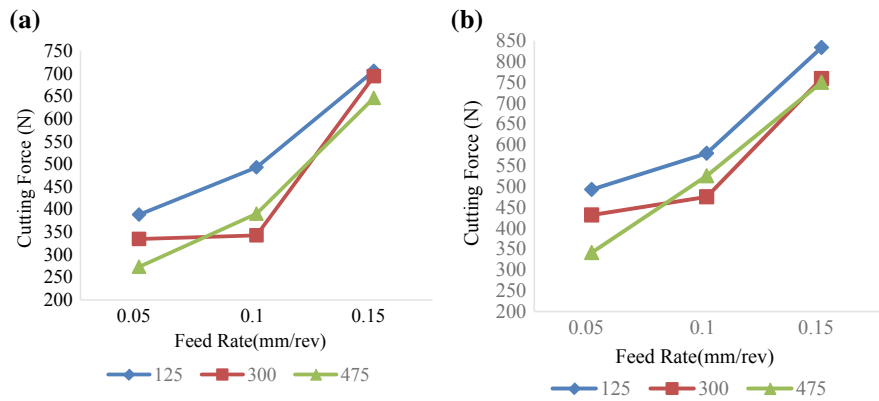


Fig. 5.3 Cutting force versus feed rate plot. **a** Experimental [6]. **b** Simulation

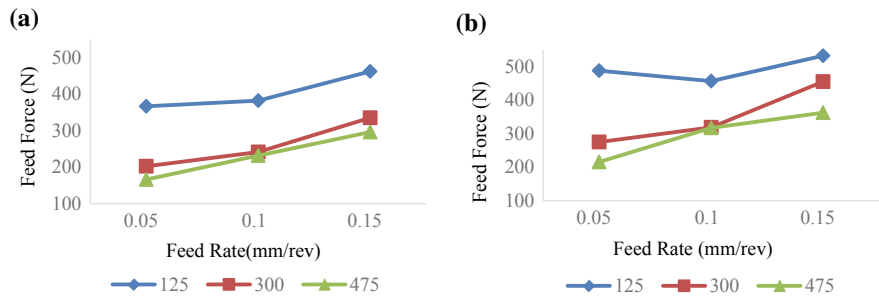
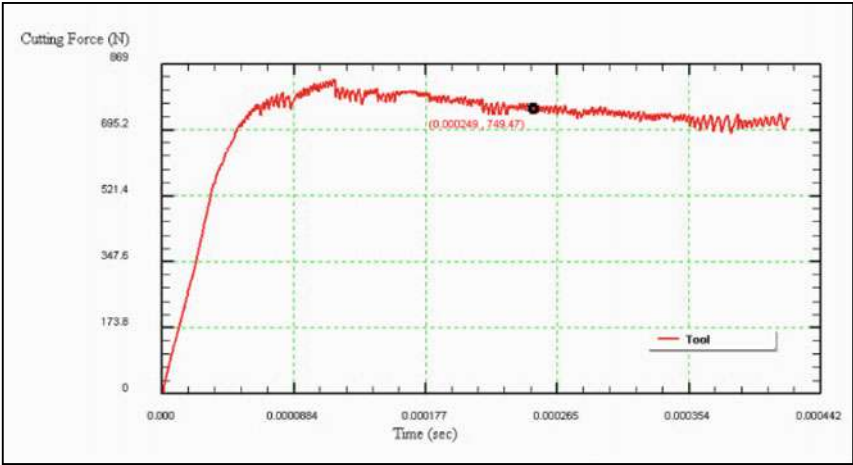
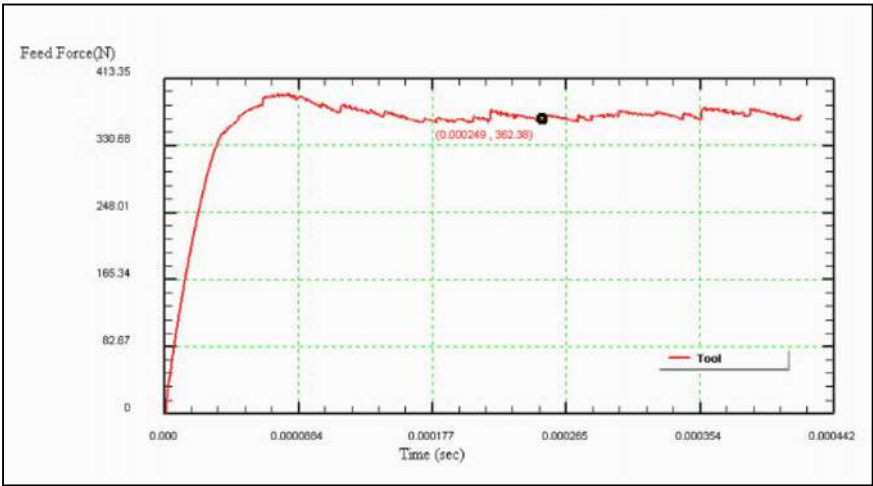


Fig. 5.4 Feed force versus feed rate plot. **a** Experimental [6]. **b** Simulation



(a) Cutting Force Profile



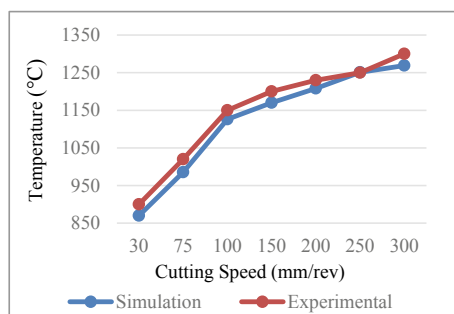
(b) Feed Force Profile

Fig. 5.5 Force profile at cutting speed of 475 m/min and 0.15 mm/rev feed rate

was observed from the temperature and cutting speed plot as shown in Fig. 5.6. This can be explained based on the occurrence of high frictional force that gets built-up at increased cutting speeds between the chips being removed and the cutting tool. This phenomenon positively contributes to the enhanced total energy of the system, complimenting for higher cutting temperatures at the tool chip interface [13]. The results have been verified with the experimental values, and maximum error is obtained to be 3.38%. Agreeable variation is seen between the experimental and simulated results. Such close congruence obtained will act as a solid foundation for

Table 5.3 Temperature results

Cutting speed (m/min)	Temperature		% Error
	Experimental [6]	Simulations	
30	900	870	3.33
75	1020	985.5	3.38
100	1150	1126	2.08
150	1200	1170	2.5
200	1230	1208	1.78
250	1250	1251	−0.08
300	1300	1269	2.38

Fig. 5.6 Temperature profile at feed rate of 0.19 mm/rev

proving the validity of the model developed, simulated, tested and presented with essential supplementary data, and the authors truly believe to have presented in this paper.

5.5 Conclusions

The FE approach to model, simulate and validate the turning process of Inconel 718 has been deliberately achieved through rigorous executorial routines on DEFORM 2D software platform. The variational analysis of the process responses, viz. cutting force and temperature, with the influential input factors such as cutting speed and feed rate have been inherently carried out as possible. The temperature simulation results presented a close compliance with the experimental results with negligible maximum error. Though higher maximum error values were obtained form force simulation, it is explainable by the fact that it can be due to the inability to consider the depth of cut factor. As the simulation was performed for 2D surface, and incorporation of more than 2 input factors is beyond the scope of this initiative.

The authors are in a progressive effort to extend this model to 3D, by utilization of DEFORM 3D software as the interface. The inclusion and consideration of other

influential factors, especially depth of cut, have been subjected to earnest consideration and inclusion to make the model more accurate and tangible.

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Chapter 6

Experimental Investigation and Finite Element Modelling of Electrical Discharge Machining Using Hollow Electrodes and Injection Flushing



Tony M. Shaju and G. L. Samuel

Abstract Electrical discharge machining (EDM) is a non-contact machining process in which rapid electric discharges are used to remove material from a workpiece by melting and vaporisation. The need for components with intricate and difficult to manufacture features have increased drastically over the past few years in different fields of application. The objective of this work was to carry out experimental investigations on the machining of EN31 steel alloy by varying input parameters like peak current and pulse-on-time, with different tool electrodes using the injection flushing configuration, to compare the effect of electrodes on material removal rate, tool wear rate and surface roughness in drilling annular holes on the workpiece. Finite element modelling of the process was done using COMSOL Multiphysics to calculate the temperature distribution and the volume of the craters formed during sparking. The results obtained from the model are then compared with the experimental data and were found that they are complementing each other. Copper is a better electrode material for machining EN31 alloy steel to obtain better material removal rate with least tool wear rate.

Keywords Electrical discharge machining · Injection flushing · Hollow electrodes · Material removal rate · COMSOL Multiphysics

6.1 Introduction

Technology is in the path of exponential growth with new and more innovations happening in various fields all around the world. This advancement in technology has to be assisted with modern capabilities for the manufacturing of the evolved products. There lies the significance of manufacturing technology in the development and progress of the world. In this scenario where competition is very tight, manufacturing a component is just not enough, but it has to be made within the

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_6

strict industrial specifications and at a higher production rate. Therefore, the terms efficiency and productivity assume higher importance compared to what was present a few decades ago. With the advent of new and improved materials, it has become more difficult to machine them to the desired shapes and size using the conventional manufacturing techniques. Electrical discharge machining (EDM) is one of the technologies widely used for machining of difficult to machine materials in the present-day industry. Two Russian scientists, Lazarenko and Lazarenko, developed EDM process in 1943 [1]. The major applications of EDM technology are in the field of tool-and-die making and in mould manufacturing for machining of tool steels and heat-treated steels. This technology is also used for processing advanced materials such as metal matrix composites, super-alloys and ceramic materials, for variety of applications. In EDM, electrode and workpiece do not make direct contact with each other during the machining process. This eliminates mechanical stresses, vibration and chatter that occur during conventional machining process [2]. The electrode is moved progressively towards the workpiece, till the gap is very small for the applied voltage, to ionise the dielectric fluid and help in developing a sparking phenomenon [3]. Any material that conducts electricity can be machined with EDM irrespective of their hardness [4].

6.2 Literature Review

In the study conducted by Yilmaz et al. [5], a comparative investigation of fast hole drilling of aerospace alloys using EDM was performed in order to explore the influence of electrode type and material. Pham et al. [6] studied the electrode wear behaviour of tube and rod electrodes, which are the two main kinds of electrodes for EDM drilling. Yan and Wang [7] investigated the machining characteristics of $\text{Al}_2\text{O}_3/6061\text{Al}$ composite using rotary electrical discharge machining with a copper tube electrode. Different flushing methods were studied and were found that flushing pressure and electrode rotation speed were having minor effects on MRR, TWR and Ra. Lonardo and Bruzzone [8] investigated the machining of Cr, Mo, V steel and observed the importance of electrode material, injection flushing and geometry of cutting on material removal rate, electrode wear and surface quality. Influence of dielectrics on the material removal rate was studied by Zhang et al. [9]. In the experiments conducted by Wong et al. [10], the influence of flushing on the efficiency and stability of machining conditions in EDM has been extensively investigated, including the effects of flushing configuration on the wear of the tool and the profile of the workpiece. Later on, the process of micro-EDM emerged and few people like Zahiruddin [11] studied their energy and removal efficiencies. Other significant experimental studies were also conducted to characterise material removal [12] and crater formation [13] during the EDM process. Dielectric medium and its properties also affect the performance of EDM. Zou et al. [14] studied the effect of using nitrogen plasma as the dielectric medium while machining brass H62 using tungsten electrode. They pointed out a significant improvement in discharge distance,

machining efficiency and surface quality. Apart from the dielectric medium tool, electrode material also has a predominant effect in EDM performance. Recently, Koyano et al. [15] investigated the effect of tool electrodes with high electrical resistivity (single-crystal silicon) in micro-electrical discharge machining of micro-rods. Their experimental results showed a reduction in the average diameter of discharge craters by 0.4 μm . Similar work has been carried out by Pilligrin et al. to find the effect of tool electrode materials in terms of tool wear rate, taper angle, overcut and surface roughness. From their experiments, it was observed that at higher discharge energy, MRR and TWR are more. The surface finish was observed to be good at low discharge energy. It is evident that EDM is one of the fast-evolving machining processes for machining of difficult to machine materials. The flushing efficiency plays a major role in the material removal rate as well as in the surface quality. As there are different flushing methods possible for the machining process, an attempt has been made in the present work to study the effectiveness of injection flushing technique during drilling hole using different electrodes in EDM.

6.3 Thermal Modelling of EDM Process

Finite element modelling of the EDM process was done with the help of COMSOL Multiphysics software version 4.3b. The crater shape which is developed as a result of the application of heat on to the surface of the workpiece was used to estimate the volume removed in a single crater. The following assumptions were made during the modelling. Workpiece material is assumed to be isotropic in nature and the model developed is axisymmetric about Z-axis. Radius of the heat source was kept constant with time. The thermal properties of the workpiece material are independent of the temperature induced on to the surface. The shape of the elements remains constant during thermal expansion which occurs due to the heating of the workpiece. The workpiece is not affected by any kind of stress that may occur during the EDM process. Conduction and convection are the only modes of heat transfer during the machining process. Any effect of radiation is neglected. Thermal analysis is transient, and the heat source has Gaussian distribution of heat flux incident on the workpiece surface. The heat transfer in a solid module of the COMSOL Multiphysics software is used to model the EDM process. The time-dependent mode of the solution module is employed to solve the problem. The governing equation used for solving the model is given in Eq. (6.1).

$$\rho C_p \frac{\delta T}{\delta t} + \rho C_p u \cdot \nabla T = \nabla \cdot (k \nabla T) + Q \quad (6.1)$$

The heat applied has a Gaussian distribution which is dependent on the distance of the spark from the axis of the workpiece, unlike the uniform distribution of heat flux that was applied in the initial works done by several researchers. The relation for Gaussian distributed heat flux is given in Eq. (6.2)

$$Q''(r) = \frac{4.57 F_c VI}{\pi R^2} e^{-4.5 \frac{r^2}{R^2}}$$

(6.2)

At the start of the EDM process ($t = 0$), the workpiece is immersed in the electrolyte, and the temperature of the whole domain is assumed to be at room temperature (T_0). The heat flux is applied on the top surface of the workpiece. The left-side vertical surface is taken as the axis of symmetry along which the 2D square can be rotated so that a cylindrical 3D object can be obtained. The other two surfaces, the bottom surface and the vertical surface on the right, are insulated thermally.

6.4 Experimental Details

The workpiece material used for conducting experiments in EDM is EN31 alloy steel. Major constituents of this alloy are iron, carbon and chromium. The mechanical properties of EN31 alloy steel are given in Table 6.1. Tool electrodes of two materials were used to conduct the experimental investigations. Copper and brass electrodes were used to machine the workpiece. Copper electrodes are known for their superior thermal conductivity. Relevant properties of copper and brass electrodes are shown in Table 6.2. The machine used to carry out the experimental investigations was Electronica Smart ZNC as shown in Fig. 6.1. The electrodes were having a geometry of hollow cylinders which was used to facilitate the flow of dielectric oil through the centre of the electrodes in order to implement the injection flushing configuration in the experiments. The electrodes were bought in the form of hollow geometry with

Table 6.1 Mechanical properties of EN31 alloy steel

Property	Value
Density	7810 kg/m ³
Melting point	1540 °C
Ultimate tensile strength	615 MPa
Yield strength	528 MPa
Elastic modulus	197 GPa
Poisson’s ratio	0.294
Specific heat	475 J/kg °C
Thermal conductivity	46.6 W/mK

Table 6.2 Properties of copper and brass electrodes

Properties	Copper	Brass
Thermal conductivity (W/m·K)	391	159
Melting point (°C)	1083	990
Electrical resistivity (ohm-cm)	1.69	4.7
Specific heat capacity (J/g·°C)	0.385	0.38

Fig. 6.1 Electronica Smart ZNC EDM machine

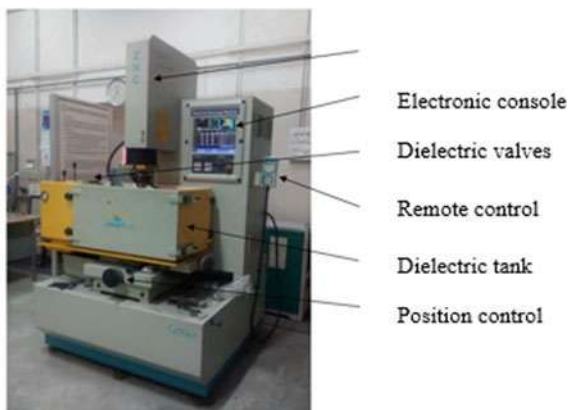
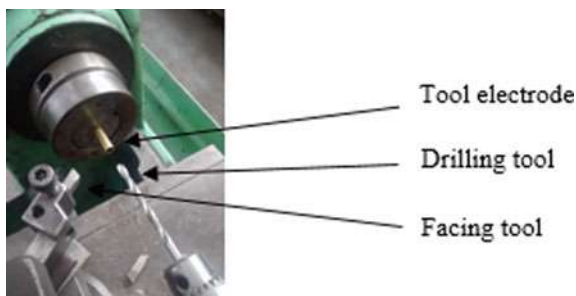


Fig. 6.2 One of the tool electrodes under preparation



a length of 300 mm and a diameter of 4 mm. Then, they were cut into small pieces using a hacksaw. After that, they were faced to get a very good finish and drilled to make the central hollow portion a perfect circle. After facing, the end surfaces were further finished by polishing using different grades of emery paper. Figure 6.2 shows one of the electrodes under preparation.

Electronica Smart ZNC has six valves to provide dielectric flushing into the system. Two of these valves are connected to flow dividers which have nozzles at the ends to provide jet flushing for clearing away the debris formed during the EDM process. Therefore, special tubes and connectors are attached to the dielectric supply system to provide an alternate route to facilitate the injection flushing configuration in the system, as shown in Fig. 6.3. Out of the major EDM parameters, the peak current and the pulse-on-time are most influential on the output of the machining process. So these two parameters were selected for preparing the design of experiments and conducting the experiments. Table 6.3 represents the experimental conditions in which the machining is carried out.



Fig. 6.3 Special tube connected to one of the six valves for injection flushing-type dielectric supply in the Electronica Smart ZNC EDM machine

Table 6.3 Experimental conditions for EDM machining

Parameter	Values
Sparkling voltage	60 V
Discharge current in steps	9, 12, 15 and 18 A
pulse-on-time in steps	50, 75, 100 and 150 μ s
Dielectric used	Commercial grade EDM oil
Flushing method used	Injection/pressure flushing
Flushing pressure	0.3 kg/cm ²
Workpiece hardness	58 HRC

6.5 Results and Discussion

6.5.1 Thermal Modelling of EDM

After running the simulation for the stipulated time, the required results were obtained under the result's tab in the model tree. The gap voltage is kept constant for the entire simulation. The value of heat flux is calculated by substituting the values of current, voltage and spark radius into the Gaussian heat distribution function. The spark radius is calculated with the help of an empirical formula which is dependent on the values of peak current and the pulse-on-time, as given in Eq. (6.3).

$$R = 2.04 \times 10^{-3} I^{0.43} T_{\text{on}}^{0.44} \quad (6.3)$$

Figure 6.4 shows the temperature distribution in the volume of the workpiece after the model is solved. It can be seen that the maximum temperature has formed at the centre point where the Gaussian heat distribution value is the maximum. Figure 6.5 represents the isothermal contour after the first spark and the subsequent spark that has been given to the workpiece. It denotes the line along which the temperature formed during the application of heat is at a value of 2770 °C. Figure 6.6 shows the variation of temperature with respect to the r-coordinate under different intervals of time for the first spark and for subsequent spark. Here, it is found that the maximum temperature for subsequent spark no longer occurs at the centre because the spark

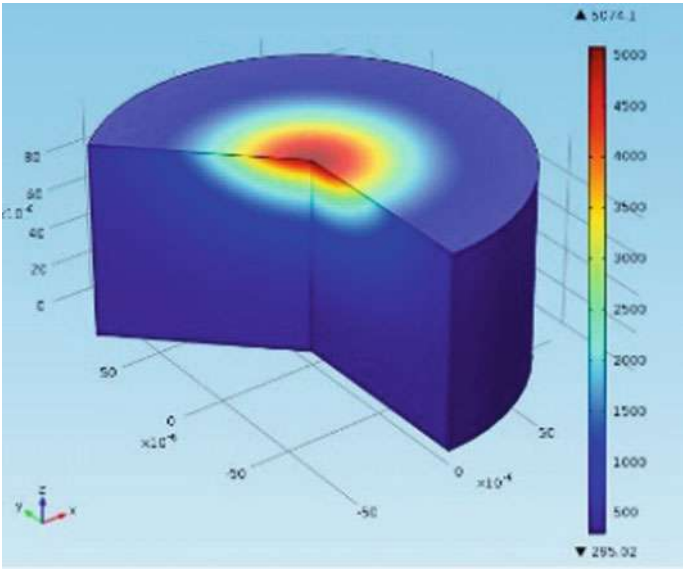


Fig. 6.4 Temperature distribution in Kelvin for 9A peak current and 50 μs pulse-on-time

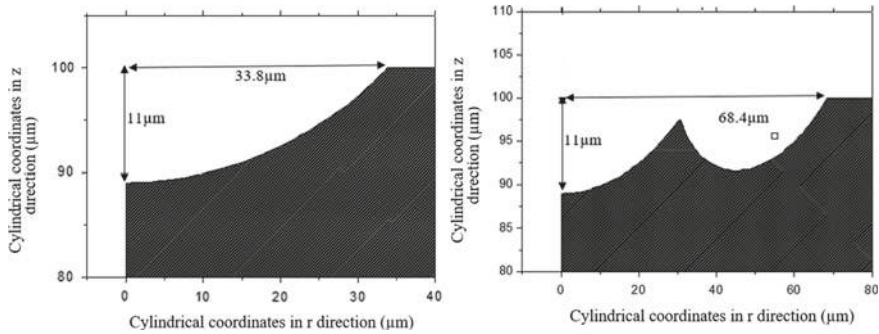


Fig. 6.5 Isothermal contour of melting point after the application of the first spark in the centre of the workpiece (left), and after the application of the subsequent spark

has occurred at a different point where the distance between the workpiece and the tool is minimum.

6.5.2 Machining with Hollow Copper and Brass Electrodes

Sixteen experiments each were conducted with hollow copper and brass electrodes by varying peak current and pulse-on-time in four steps. Graphs are plotted to find

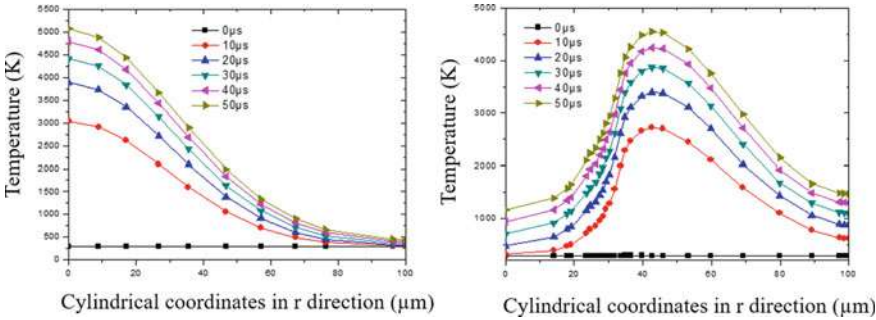


Fig. 6.6 Variation of temperature with respect to the r -coordinate under different time values for first spark (left) and for subsequent spark (right)

the effect of the input parameters on the outputs. The effect of peak current on MRR for different pulse-on-times for copper and brass electrodes is shown in Fig. 6.7. The MRR seems to be increasing steadily with an increase in peak current. This is due to the fact that the energy that has been discharged on to the workpiece increases with the current, as the current is directly proportional to discharge energy. This energy transfers more heat on to the workpiece which in turn melts away more material resulting in a higher material removal rate. This is in agreement with the Gaussian heat flux equation denoted earlier in the paper and with the findings of Govindan et al. [12]. It can also be observed that the values of MRR are comparatively low for brass electrodes with respect to copper electrodes.

Tool wear rate also shows the same trend as MRR, having an increasing tendency with an increase in peak current. This is because some part of the energy formed in the plasma channel is utilised to melt the tool electrode. As it is already mentioned that the discharged energy is proportional to the peak current applied; the energy transferred to the tool, though not as high as that transferred to the workpiece, also keeps increasing. Thus, a higher tool wear rate is obtained for higher peak current values. The effect of

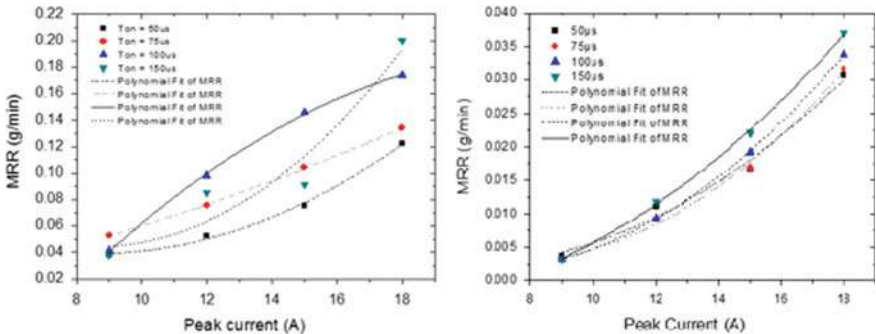


Fig. 6.7 Effect of peak current on material removal rate at different pulse-on-time values for copper electrodes (left) and brass electrodes (right)

peak current on tool wear rate for copper and brass electrodes at different pulse-on-time values is shown in Fig. 6.8. It can be observed that the magnitude of tool wear is also low for brass electrode compared to the copper electrode, but is not appreciably low.

Figure 6.9 shows the effect of peak current on surface roughness for both electrodes at different pulse-on-time values. Surface roughness tends to increase with an increase in peak current value. This trend occurs due to the bigger crater formed on the workpiece surface at higher currents. In simple mathematical terms, the crater formed can be modelled as hemispherical voids formed on the surface due to the bombardment of ions or electrons with high energy. Considering higher energy due to the high peak current, the diameter of the hemispherical crater formed is higher. This can be a direct indicator of a higher surface roughness value at higher peak current during machining.

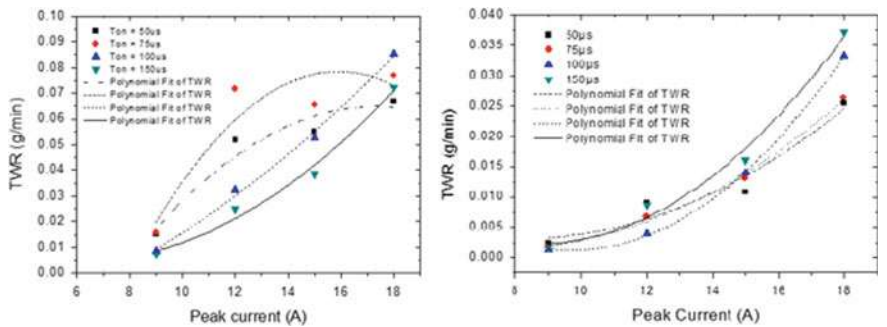


Fig. 6.8 Effect of peak current on tool wear rate at different pulse-on-time values for copper electrodes (left) and brass electrodes (right)

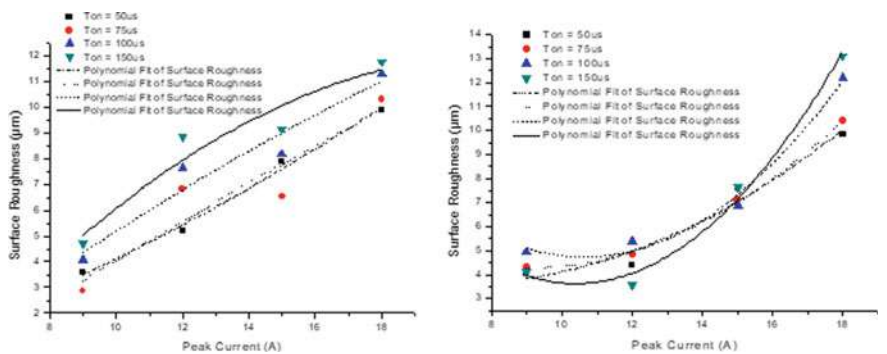


Fig. 6.9 Effect of peak current on Ra at different pulse-on-time values for copper electrodes (left) and brass electrodes (right)

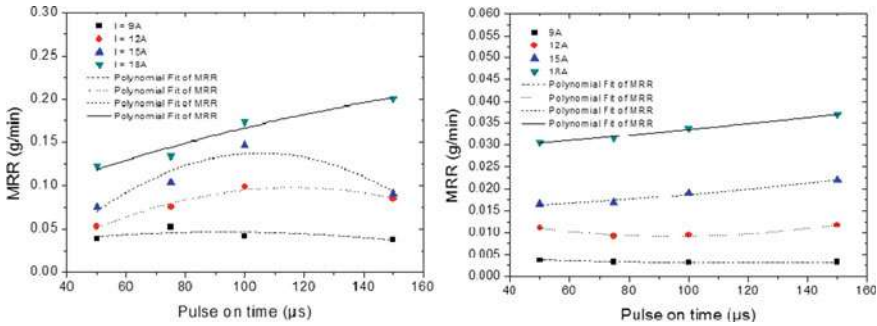


Fig. 6.10 Effect of pulse-on-time on MRR at different peak current values for copper electrodes (left) and brass electrodes (right)

The effect of pulse-on-time on material removal rate for copper and brass electrodes at different peak current values is shown in Fig. 6.10. With respect to pulse-on-time, MRR initially shows an increasing trend for copper electrodes, but gradually tends to decrease when very high pulse-on-time is given. This is due to the fact that the plasma channel radius which is dynamic in nature tends to increase with time, and thus, the energy which has been given to the workpiece tends to get more and more distributed along the workpiece and thus reduces the concentration of power on the required diameter. This also points to the fact that keeping on increasing discharge energy does not guarantee a higher removal rate during the machining operation. It is also observed that the material removal rate has not much been affected by the change in pulse-on-time for brass electrodes as compared to copper electrodes.

Tool wear rate increases initially with an increase in the pulse-on-time but later tends to decrease slightly for copper electrodes. The reason can be attributed to the same factors that cause a decreasing trend in MRR due to higher pulse-on-time values. For brass electrodes, tool wear rate also seems to remain almost constant for a change in pulse-on-time values in spite of the appreciable increase in tool wear rate for higher current values in the order of 18 A. The effect of pulse-on-time on tool wear rate for copper and brass electrodes at different peak current values is given in Fig. 6.11.

6.5.3 Comparison Between Copper and Brass Electrodes

Figure 6.12 compares the values of material removal rate and tool wear rate obtained using copper and brass electrodes for the machining of EN31 alloy steel. It is clear that copper has better material removal rate than brass. Even though the tool wear rate of copper is higher than that of brass, the difference is not appreciable as what we see in the material removal rate graph. This is because copper has low electrical resistivity, high melting point and high thermal conductivity compared to brass. So

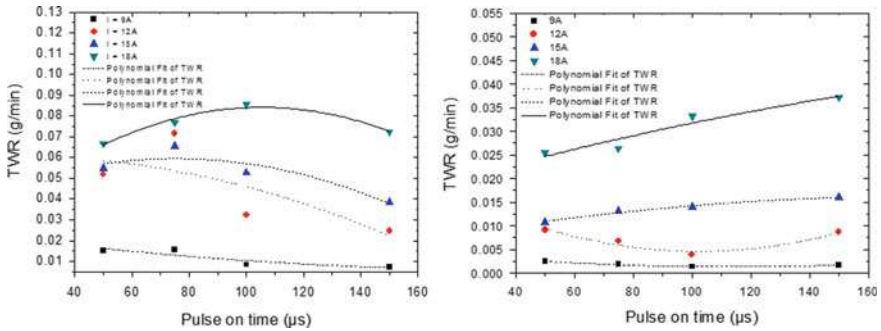


Fig. 6.11 Effect of pulse-on-time on tool wear rate at different peak current values for copper electrodes (left) and brass electrodes (right)

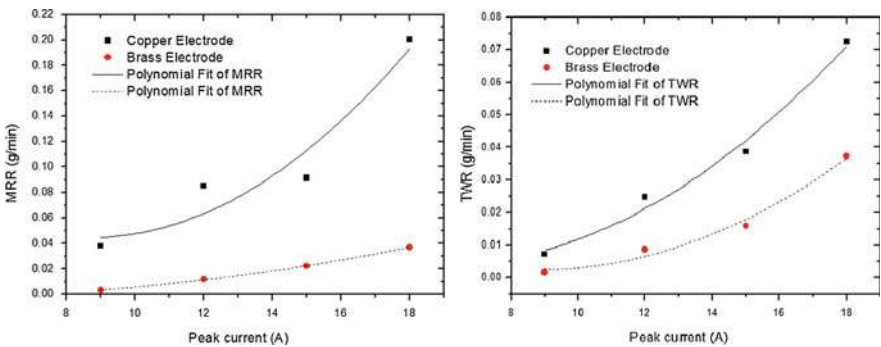


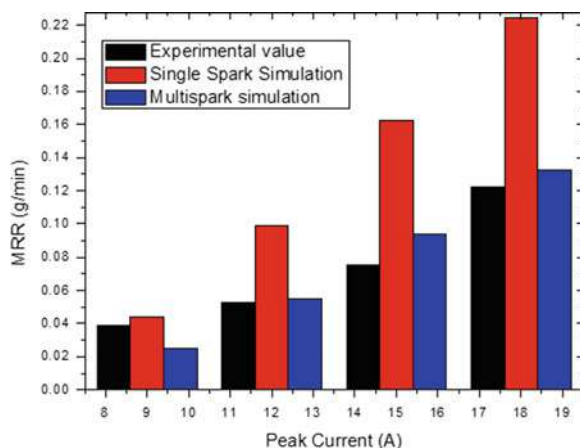
Fig. 6.12 Comparison between MRR (left) and TWR (right) for copper and brass electrodes

it can be concluded that copper is a better material to use for machining EN31 alloy steel to obtain better material removal rate with the least tool wear rate.

6.5.4 Comparison of Thermal Model with Experimental Results

The simulation for electrical discharge machining is done for a gap voltage of 60 V and a pulse-on-time of 50 μs with single spark and multi-spark, and the results are compared with experimental data. The comparison is shown in Fig. 6.13. It can be observed that the simulation results of MRR are higher than the experimental values. This can be attributed to the fact that the simulation is nearer to an ideal scenario, whereas the experimental material removal rate is affected by various external factors. Also, multi-spark simulation is closer to the experimental data than the single spark

Fig. 6.13 Comparison between experimental values, single spark simulation and multi-spark simulation



simulation because, in multi-spark simulation, the overlap of craters is also taken care of during the modelling of the problem.

6.6 Conclusions and Scope for Future Work

Finite element modelling of the EDM process was done with the help of COMSOL Multiphysics heat conduction in solid module to calculate the temperature distribution and the volume of the craters formed during sparking, thus calculating the MRR. Experimental investigations on the machining of EN31 steel alloy were done by varying the input parameters, and the results were compared with respect to the effect on material removal rate, tool wear rate and surface roughness in drilling annular holes on the workpiece. Material removal rate by using copper electrodes was found to be around 4.5 times than that of brass in the middle of the range of the experiments. This is because copper has low electrical resistivity, high melting point and high thermal conductivity compared to brass. Tool wear rate of copper is also high, and it is around 3.5 times that of brass electrodes. Although the tool wear rate is higher for copper, this can be compensated by the advantage of a high material removal rate. No appreciable difference was observed in the surface roughness measurements when the results from using the two different electrodes were compared. Therefore, copper is a better electrode than brass for machining shallow holes and fins on EN31 alloy steel with the help of injection flushing. The results obtained from the model were compared with the experimental data and were found that they are complementing each other. The variation between the experimental data and the simulation results may be due to the fact that the experimental data was obtained from real machining conditions and the simulation data had few assumptions taken.

An array of shallow holes are drilled on surfaces for weight reduction, and fabrication of micro-fins for better thermal management. Electrical discharge machining

carried out with hollow electrodes and injection flushing is a promising solution for making such features. As there are no much literature available on the use of such electrodes and flushing techniques, it was important to perform this experiment to identify the best tool electrode material and processing condition for obtaining the best machining results. In future, the effect of different flushing pressures can be studied to find the optimum pressure of the dielectric. The modelling can be extended for performing large number of sparks in a single simulation using the COMSOL with MATLAB interface. If the FEA simulation is repeated several times, there will be no pattern observed, as the next spark in simulation strikes at the location that is closest to the tool electrode, and once the simulation completes few initial runs, the spark will occur at random locations on the workpiece surface for the spark to act. It may be noted that EDM process is a stochastic process.

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Chapter 7

Experimental and Numerical Characterization of Residual Stresses in Tailor Welded Blanks After Springback



Vijay Gautam , D. Ravi Kumar and Subhajit Konar

Abstract Residual stresses are locked-in stresses in the material that is free of external forces and thermal gradients. These stresses self-equilibrate within the cross section of the material and can result in unexpected failure if not accounted for. A good knowledge of variation of the residual stresses and its distribution is of great importance for the accurate assessment and evaluation of fatigue life of cold-formed components. The present study is based on the experimental and numerical characterization of residual stresses in the tailor welded blanks of interstitial free steel with a thickness combination of 0.8 mm $\times$ 1.5 mm after springback using a new cos α technique using a portable X-ray device (μ -X360 residual stress analyser). The longitudinally welded specimens of tailor welded blanks are tested on V-bending set-up with three different punch profile radii, i.e. 10, 12.5 and 15 mm, and the residual stress is measured on inner and outer side of the tested samples. The effect of the punch profile radius on residual stress after springback is observed to be very significant in bending of tailor welded blanks. As the punch profile radius increases, residual stress decreases for a given thickness combination. The residual stress predicted by simulations agreed well with the experimental results for all punch radii except a few cases.

Keywords Residual stress · Fatigue life · Tailor welded blanks · Cos α technique · X-ray device · Rolling direction · Punch profile radius

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_7

7.1 Introduction

Residual stress or locked-in stress is defined as the stress possessed by a raw or processed material even after the removal of forces responsible for its occurrence. Residual stresses are self-equilibrating stresses and can result in unexpected failure if not accounted for [1]. These are of two types: tensile and compressive and can be developed in any stage of manufacturing processing. Tensile residual stresses are generally detrimental as they can promote crack initiation, stress corrosion and decrease in fatigue life and hence are avoided. Whereas, there are many methods to induce beneficial compressive residual stresses (CRS), like plastic deformation (shot-peening), temperature gradient (toughening of glass), metallurgical transformation in surface hardening to enhance the fatigue life of the component. For engineering materials which are plastically deformable, the residual stress directly adds to the magnitude of applied stress if both the stresses are of the same nature, i.e. either tensile or compressive. This addition of stresses accelerates the failure by plastic deformation and necessitates the pre-determination of stresses where failure may take place by plastic collapse. It is observed that the effect of the magnitude of left-over stress is small due to a lower magnitude of elastic misfit strains when compared with the plastic strains developed prior to failure [2]. In the cases, where the nature of residual stress is opposite to the applied stress, the net resultant stress is lower than the applied and delays the onset of failure by plastic deformation. A good knowledge of amplitude of the residual stresses and of its distribution is of great importance for the accurate assessment and evaluation of fatigue life of cold-formed components [3]. Considerable efforts are now being devoted by design engineers to develop the methods by which evaluation of the residual stress field can be accurately incorporated into design, especially in aerospace, nuclear, pressure vessels, automotive components and other engineering industries.

Rossi et al. [4] developed a numerical model to evaluate the distribution of residual stresses through thickness including the effects of coiling and uncoiling of high strength steel sheets before the cold bending of the corner. The elastic springback after flattening as well as after final shaping is also taken into account. In order to validate the model in predicting the residual stress distribution, the presented results are compared to the experimental measurements and numerical results.

Anand et al. [5] conducted experimental investigations on fatigue behaviour of tailor welded blanks of different thickness combination with different grades. It was concluded that fatigue limit of the uncoated TWB was found to be about 1.4 times higher than that of the galvanized TWBs. The fracture due to cyclic loading initiated in the thinner sheet of TWBs and stress concentration played a major role in the location of the fracture induced due to the difference in thicknesses of the TWBs. In the case of the TWBs with greater thickness difference, the fracture due to fatigue was observed to be located closer to the weld bead, whereas for TWBs with relatively lower thickness difference, the fatigue failure initiated always away from the weld bead. However, evaluation of residual stress was not carried out, but its contribution to fatigue failure should not be underestimated.

Martinson et al. [6] studied the comparison of the tangential residual stresses for resistance spot and laser spot welding and observed higher tensile residual stresses inside the weld ring than those found in the nugget of resistance spot weld. The weld region in laser spot found to be surrounded by a compressive region which is larger in magnitude than that of the resistance spot welding. Good agreement was seen between the results obtained by experiments and by simulated temperature field results using conical Gaussian heat source. It was also concluded that the residual stresses can be predicted qualitatively using a partially coupled simulation.

Gelfi et al. [7] proposed a new method to determine the magnitude and nature of residual stresses based on the analysis of a single Debye ring obtained by X-ray diffraction. Single Debye ring method is preferred for the evaluation of residual stresses in coatings and for the components with intricate geometry. Many examples were discussed and the results of residual stress obtained by this method compared with the results obtained by the conventional $d\text{-sin}^2 \psi$ method.

There are several destructive and non-destructive experimental techniques to evaluate/measure the residual stresses such as: wafer curvature, hole drilling, X-ray diffraction (XRD), neutron diffraction, ultrasonic method and electrical-magnetic methods [2, 8]. Among these techniques, the XRD and wafer curvature techniques are widely used for the determination of surface residual stresses [9, 10]. The evaluation of residual stresses using XRD- $\sin^2 \psi$ method depends mainly on the accuracy of determination of peak positions obtained in diffraction. Luo and Jones [9] conducted a research on accuracy of XRD- $\sin^2 \psi$ technique and evaluated four widely used methods of peak positioning in terms of the precision of the residual stress measurement, and measurements have been done on a number of different specimens including a thermal barrier coating sample deposited with electron beam evaporated ZrO_2 , several magnetron sputtered transition metal nitride coatings and shot-peened super-alloy components. The results obtained by the parabolic approach method produced the best linear regression [9].

The present study is based on the experimental and numerical characterization of residual stresses in the tailor welded blank with thickness combination of $0.8 \text{ mm} \times 1.5 \text{ mm}$, after springback using a new $\cos \alpha$ technique. The longitudinally welded specimens of tailor welded blanks are tested on V-bending set-up with three different punch profile radii, i.e. 10, 12.5 and 15 mm, and the residual stress is measured on the inner (compression) and outer (tension) side of the tested samples.

7.2 Experimental

7.2.1 Materials Used and Sample Preparation

Interstitial free (IF) steels of different thicknesses, i.e. 0.8 mm and 1.5 mm, are used to prepare tailor welded blanks (TWB) by an Nd-YAG laser welding process. The IF steels are predominantly used for automotive skin panels which require a

high degree of formability. The carbon content of this steel is lower than 0.004% by weight. To achieve the low carbon content, the processing of IF steel is done in vacuum degassing with the addition of titanium and niobium for the elimination of solutes in the final stage. In order to characterize the mechanical properties of the parent sheets, tensile test samples are prepared by laser cutting in three different orientations, i.e. parallel, inclined and transverse to the rolling direction. Similarly, TWB samples with thickness combination of 0.8 mm $\times$ 1.5 mm are prepared, and tensile samples are taken out from TWBs by keeping the weld line at 0°, 45° and 90° to the rolling direction.

7.2.2 Experimental Set-up for V-Bending

The experimental set-up for the determination of springback consists of a set of punches and dies of heat-treated D2 steel, designed to conduct the bending experiments of the samples by an included bend angle of 90°. In order to study the effects of punch profile radius on residual stresses, the punches are designed with three different profile radii of 10 mm, 12.5 mm and 15 mm with the three corresponding mating dies. The clearance in the die is kept equal to the sheet thickness to avoid any localized compression at the bottom of the die during the final stage of bending operation. The punch is designed to be secured to the movable cross-head while the die to the stationary wedge grip of a 50 kN UTM. To ensure plane strain condition in bending, bend specimens of parent sheets and TWBs with dimensions 150 mm $\times$ 25 mm are prepared by wire-cut EDM. In the bend samples of longitudinally welded TWBs, the weld line is always kept transverse to the bend axis, and the flat side of the bend sample is kept towards the punch.

7.2.3 Portable X-Ray Device to Measure Residual Stress

A portable X-ray device (μ -X360 residual stress analyser from Pulstec Industrial Co. Ltd.) mounted on a robust table is used to measure the residual stresses after springback in TWBs specimens. Various points are marked across the width of the specimen after bending along the bend axis on thicker and thinner sheets both on compression and tension side. Detector gun of the stress analyser is inclined at an angle of 35° for the IF steel (ferrite) to get the accurate results.

Specimen is placed on the adjustable table exposing the surface to be assessed for the residual stress establishing a red spot of laser (approximately 2 mm) on the mark where the X-ray beam will incident. The setting of the specimen is achieved by focusing a green checkbox which shows the specified area under inspection by sensors. A dedicated μ -360 residual stress measurement system software version 2.5.6.0 is used to make the measurements. Value of the residual stress with certain deviation is obtained from the data acquisition system. Similar procedure is repeated

for all the points marked on the specimen both on compression and tension side of the specimens.

7.3 Stress Measurement Using $\sin^2 \psi$ and $\cos \alpha$ Techniques

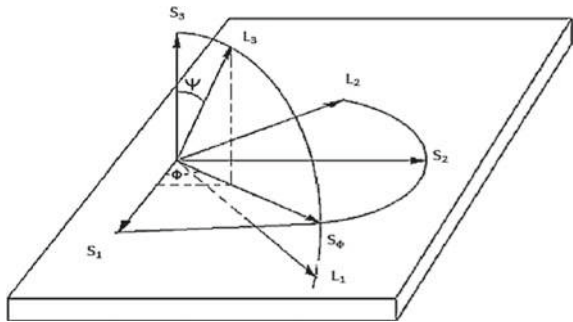
In conventional $\sin^2 \psi$ technique, the variation of lattice interplanar distance is detected by changing the angle for a large number of times for different angular measurements of incident X-rays. When the stress is applied, X-ray incident angle ψ is large, compressive stress changes the strains and results in smaller interplanar distance d , and consequently, the changed angle θ increases $n\lambda = 2d \sin\theta$. The $\sin^2 \psi$ technique calculates the residual stresses existing in the surface layer of polycrystalline materials by assuming a plane-stress state.

The measurement depends on the ability to evaluate fundamental atomic plane spacing, d_0 , and the change interplanar distance d resulted due to deformation. The measured lattice spacing is expressed as $d\psi\phi$ with respect to the measured lattice spacing at various diffractometer angles, ψ and ϕ as shown in Fig. 7.1. This translation from the diffractometer space to the sample space is fundamental to the derivation of both the $\sin^2 \psi$ and $\cos \alpha$ methods. By relating this measured lattice spacing to the strain components in the sample space, the measured lattice spacing may be represented from the diffraction vector n , through Einstein notation as $\varepsilon_{\varphi\psi} = n_i n_j \varepsilon_{ij}$. This strain is used in Hooke's law to yield equation as given below:

$$n = \begin{bmatrix} \sin \psi \cos \phi \\ \sin \psi \sin \phi \\ \cos \psi \end{bmatrix} \quad (7.1)$$

$$\varepsilon_n = n_1^2 \varepsilon_{11} + n_2^2 \varepsilon_{22} + n_3^2 \varepsilon_{33} + 2n_1 n_2 \varepsilon_{12} + 2n_1 n_3 \varepsilon_{13} + 2n_2 n_3 \varepsilon_{23} \quad (7.2)$$

Fig. 7.1 Coordinate system



$$\varepsilon_{\phi\psi} = \frac{(d_{\phi\psi} - d_o)}{d_o} = \frac{1 + \nu}{E} (\sigma_{11} \cos^2 \vartheta + \sigma_{12} \sin^2 \vartheta + \sigma_{22} \sin^2 \vartheta) \sin^2 \psi - \frac{\nu}{E} (\sigma_{11} + \sigma_{22}) \quad (7.3)$$

Equation (7.3) is commonly used as fundamental equation for $\sin^2 \psi$ technique.

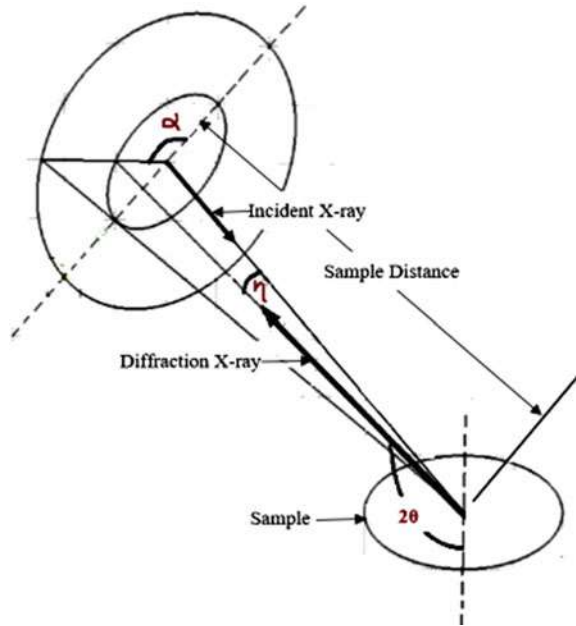
The $\cos \alpha$ method utilizes the Debye ring collected with a single measurement using a 2D detector (Fig. 7.2), as described by Sasaki et al. [11] and others. The translation from the diffractometer space to the sample space is inherently more complex due to the 2D planar geometry of the measurement. The sample space axes (φ , ψ , ω) are followed, along with the critical α , η and 2θ angles on the Debye ring as shown in Figs. 7.2 and 7.3.

The translation matrix is given as

$$n = \begin{bmatrix} \cos \eta \sin \psi_o + \sin \eta \cos \psi_o \cos \alpha \\ \cos \eta \sin \psi_o \sin \vartheta_o + \sin \eta \cos \psi_o \sin \psi_o \cos \alpha + \sin \eta \cos \vartheta_o \sin \alpha \\ \cos \eta \cos \psi_o - \sin \eta \sin \psi_o \cos \alpha \end{bmatrix} \quad (7.4)$$

where n can be written in the form of the expression for the translation of strain, $\varepsilon_\alpha = n_i n_j \varepsilon_{ij}$, which can be inserted into Hooke's law to form as given below:

Fig. 7.2 Schematic representation of $\cos \alpha$ geometry using a 2D detector [11]



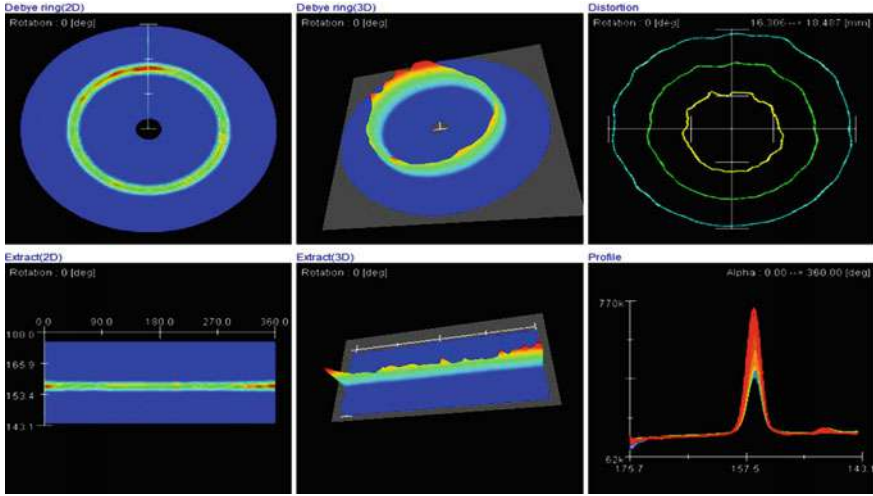


Fig. 7.3 Debye ring mapping to evaluate residual stresses in the bend samples using $\cos \alpha$ method

$$\varepsilon_{\alpha} = \frac{1 + \nu}{E} n_i n_j \varepsilon_{ij} - \frac{\nu}{E} \sigma_{kk} \quad (7.5)$$

Defining the two parameters, a_1 and a_2 allows the linear determination of the stresses σ_{11} and σ_{22} if both angles φ_0 and ψ_0 have been chosen and kept constant for the data collection.

$$a_1 = \frac{1}{2} [(\varepsilon_{\alpha} - \varepsilon_{\pi+\alpha}) + (\varepsilon_{-\alpha} - \varepsilon_{\pi-\alpha})] \quad (7.6)$$

$$a_2 = \frac{1}{2} [(\varepsilon_{\alpha} - \varepsilon_{\pi+\alpha}) - (\varepsilon_{-\alpha} - \varepsilon_{\pi-\alpha})] \quad (7.7)$$

$$\sigma_x = -\frac{E}{1 + \nu} \cdot \frac{1}{\sin^2 \eta} \cdot \frac{1}{\sin^2 \psi_0} \cdot \left(\frac{\partial a_1}{(\partial \cos \alpha)} \right) \quad (7.8)$$

Equation (7.8) gives the residual stresses.

7.4 Modelling and Simulations

In the present study, ABAQUS software is used to simulate both loading (bending) and unloading (springback) conditions. The punch and the die are modelled as analytical rigid and the blank as deformable. Shell planar elements (S4R) are selected for the blank and are placed on the die with the flat surface facing towards the punch. The TWB blank is modelled with 3902 number of elements and 4079 number of nodes.

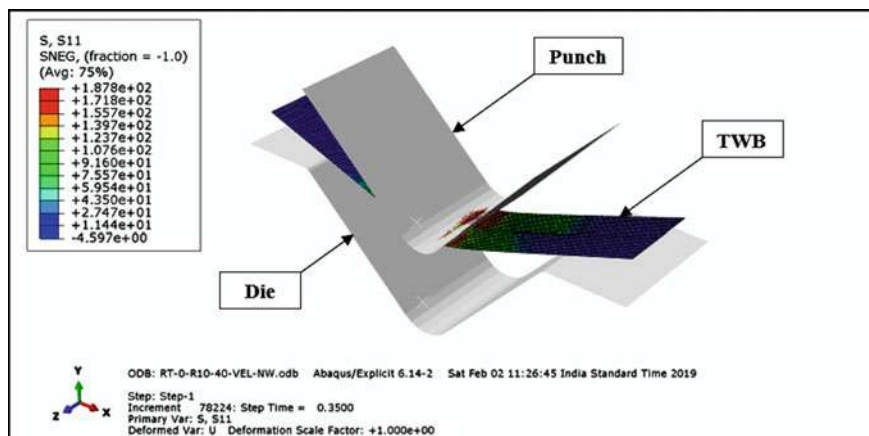


Fig. 7.4 Assembly of tools in bending simulations

The total number of variables in the model is 24,474 only. The surface-to-surface contact properties between the punch and blank surface and between die and blank surface are assigned using penalty contact method as this preferred for pinched contact of shells and rigid bodies. The coefficient of friction is assigned as 0.125 between blank-die and 0.05 between blank-punch interfaces. The die is fixed and the punch is allowed to move equal to the die depth by assigning movable boundary condition to the punch as shown in Fig. 7.4.

Hill's anisotropic yield criterion is used as a material model for deformable blank. The true stress-true strain data obtained from uniaxial tension test with isotropic strain hardening exponent and strength coefficient are used in the material model for TWBs. Although studies are restricted to the determination of longitudinal stress after bending and springback, the weld zone is not modelled in the TWBs. A 5-point integration method using Simpson's rule is adopted to determine the longitudinal stress variation through the thickness of the blank.

In order to determine the residual stress after springback, the simulations are performed by removing the punch and the die from the FEA model. The blank is assigned initial history data of the bending simulation containing a history of loading. A central node is assigned with zero-velocity boundary condition so that the blank does not move from its initial position, but both the arms of the blank are free to move in response to the removal of the constraint. The coordinates of the blank before and after springback are captured, and the springback is determined as the difference in the bend angle before and after springback. The longitudinal stress variation located in the deformed zone across the width of the specimen and along the bend axis is also captured.

7.5 Results and Discussion

7.5.1 Tensile Properties of Parent Sheets and TWBs

The tensile properties of the IF steels of different thicknesses and in different orientations with respect to the rolling direction are shown in Figs. 7.5 and 7.6. Elastic modulus of the steel is lower in thinner sheet than the thicker sheet and varies from 196 to 199 GPa, whereas in thicker sheet, it varies from 200 to 205 GPa. The yield strength and ultimate tensile strength of the specimens oriented at 45° to the rolling direction are higher than the other orientations in both the thicknesses. The average percentage elongation of the thicker sheet is higher than the thinner sheet and is

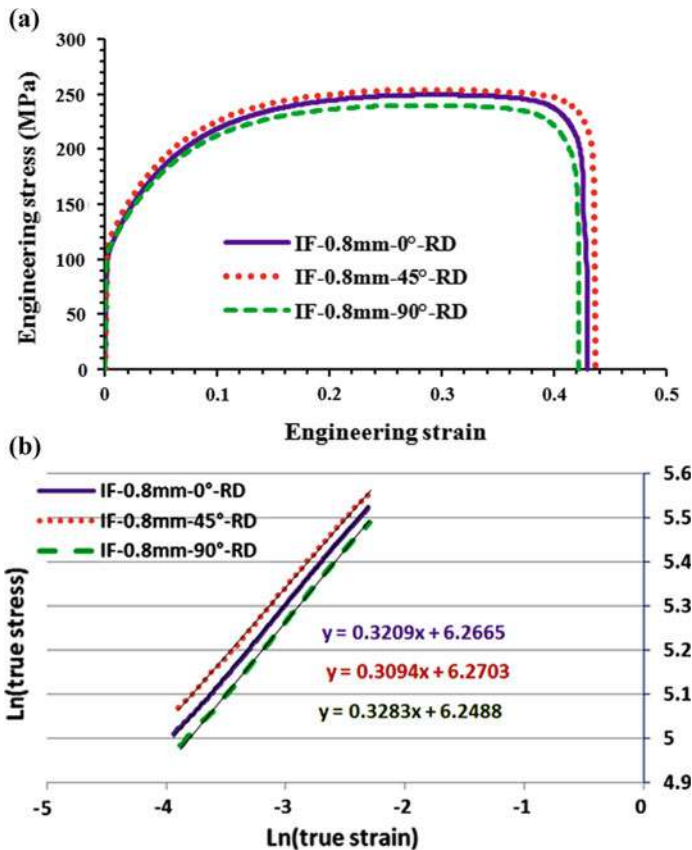


Fig. 7.5 Tensile properties of interstitial free steel: **a** Engineering stress–strain plots of tensile testing of 0.8-mm-thick sheet specimens oriented at 0°, 45° and 90° w.r.t. rolling direction and **b** in true stress versus in true strain of the same plots to determine strain hardening exponent and strength coefficients

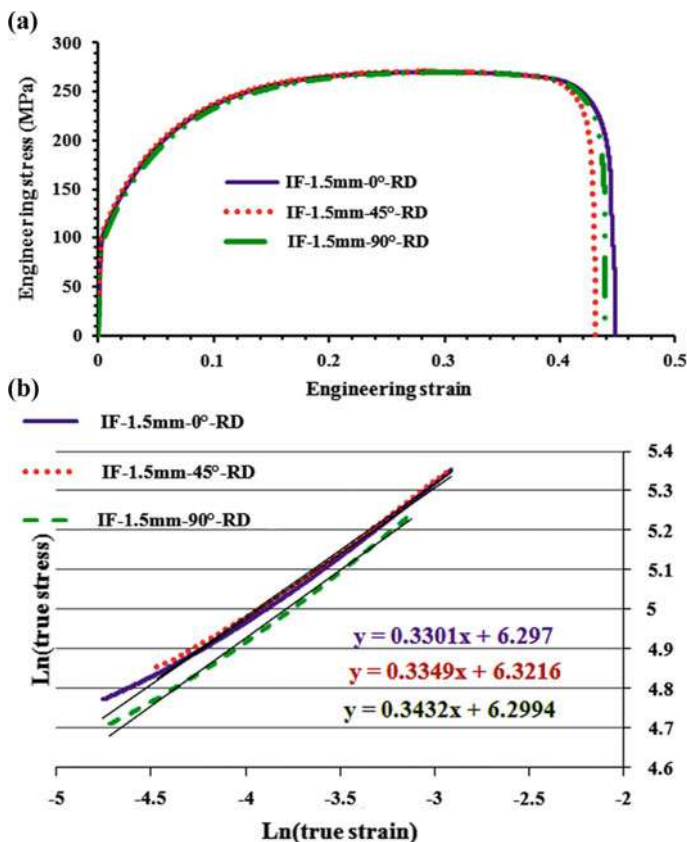


Fig. 7.6 Tensile properties of interstitial free steel: **a** Engineering stress–strain plots of tensile testing of 1.5-mm-thick sheet specimens oriented at 0°, 45° and 90° w.r.t. rolling direction and **b** in true stress versus in true strain of the same plots to determine strain hardening exponent and strength coefficients

approximately 43%. In case of both the thicknesses, the average strain hardening exponent is approximately 0.33, which indicates excellent stretchability with uniform deformation. The average plastic strain ratio of the thicker sheet is lower than the thinner sheet which indicates the better drawability with a thinner sheet. The normal anisotropy, a measure of deep drawability, of the thicker sheet is higher than the thinner sheet and is observed to be 1.92 [12].

The results of tensile tests performed on specimens of longitudinally welded TWBs are summarized in Table 7.1. It is observed that yield strength and tensile strength of TWBs are much higher than the parent sheets, but the average elongation is limited to 38% only indicating reduced ductility. The value of strain hardening exponent of the TWB specimens has also been reduced to 0.27 on an average basis. The higher values of tensile strength and lower values of strain hardening exponent

Table 7.1 Tensile properties of TWBs of IF steel

Thickness combination (mm × mm)	Weld orientation w.r.t. RD (°)	Yield stress (MPa)	Ultimate tensile strength (MPa)	% Elongation	Strain hardening exponent <i>n</i>	Strength coefficient (MPa) <i>K</i>
1.5 × 0.8	0	158.00	262.5	36.7	0.272	486.91
	45	159.40	280.8	34.9	0.282	523.20
	90	157.80	259.4	43.7	0.261	485.31

and elongation can be attributed to the weld zone. The weld zone influences the overall mechanical properties of TWBs [12].

7.5.2 Distribution of Longitudinal Stress

The variation of longitudinal stress across the width and contour plots obtained from FE simulations of bending and springback of TWBs of IF steel with thickness combination of 0.8 mm × 1.5 mm with punch profile radius of 15 mm is shown in Fig. 7.7.

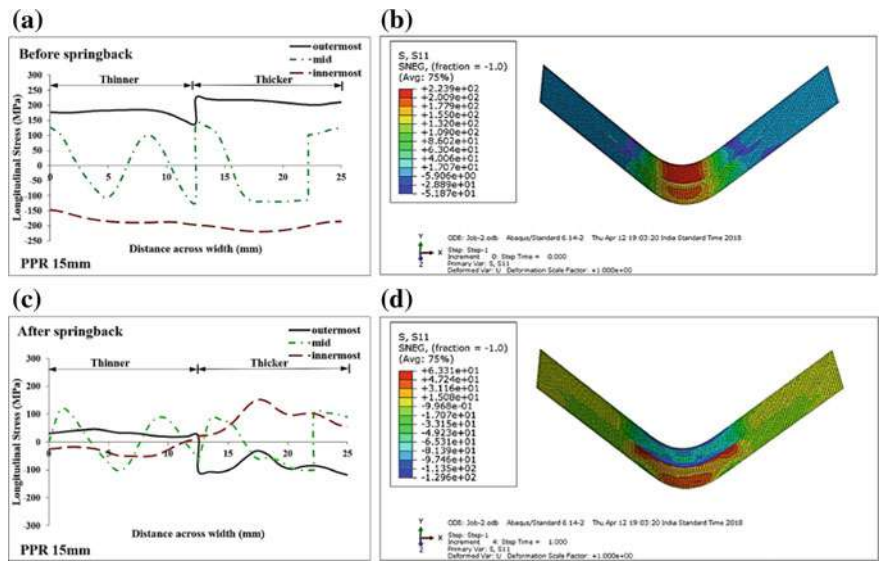


Fig. 7.7 a Variation of longitudinal stress at different points through thickness before and c after springback, b contours of longitudinal stress before springback and d after springback for a punch profile radius of 15 mm

As shown in Fig. 7.7a, the variation of longitudinal stress before springback is almost constant in the thinner and the thicker sheets, but higher values of longitudinal stress is observed in both the outermost (tension side) and innermost layers (compression side) in the thicker sheet than the thinner sheet, and a sudden drop in the values of stress is seen at the junction of thinner and thicker areas. The sudden increase in the longitudinal stress at the junction of thinner and thicker sheets on the outermost layer is due to the step in the two sheets resulting in localized stress concentration. The flat side of the TWB which is the punch side or the innermost layer of the bent sheet shows gradual increase in the stress values towards the thicker side. The higher values of stress in the thicker sheet can be attributed to the higher tensile strength of the sheet and higher plastic deformation due to bending for a given punch profile radius. The stresses are tensile on the outer side and compressive on the inner side, but the stress value in the mid-layer are elastic as these are well below the yield strength of IF steel. The similar results are observed for punch profile radii of 10 mm and 12.5 mm.

It is also observed that as the profile radius increases, the stress values decrease significantly during bending of the specimen. A stress value of 248 MPa for punch profile radius of 10 mm reduces to 232 MPa for 12.5 mm and further reduces to 224 MPa for punch profile radius of 15 mm (Fig. 7.7b). The die depth is slightly higher with lower punch profile radius, and therefore, the higher coining effect appears to have occurred in this case due to localized compressive stresses between the punch and the die at the end of bending. Since the longitudinal stress is higher in the case of lower punch profile radius, the fraction of thickness on either side of a neutral surface where stresses are lower than the yield stress would be lower, which in turn reduces the elastic core. In the case of higher punch profile radius, the coining force is lower due to which the elastic core is higher and therefore springback is higher.

As shown in Fig. 7.7c, it is observed that the variation of residual stress after springback depicts an increasing and decreasing trend across the width of thinner and thicker sheet, both on tension and compression side. It is also observed that after springback, the residual stress on thicker side is compressive as the case of a normal sheet but the stress on the thinner side is still tensile. This may be attributed to the combined effects of thicker and thinner portions on resultant springback of TWB. The thinner side was constrained from undergoing complete springback by the thicker side due to higher stiffness and lower springback, and hence, the residual stress in the thinner sheet is tensile. Similar results of variation of residual stresses are observed with punch profile radii of 10 and 12.5 mm. In all the cases, the variation of the residual stress in the mid-layer remains same after springback and is observed with increasing and decreasing trend as seen before springback.

A comparison of residual stress values in the outermost and innermost surface layers obtained from simulations after springback and experimental technique using $\cos \alpha$ method for different punch profile radii is given in Table 7.2.

As discussed, the values of residual stress obtained from FE simulations, on an average basis across the width of the specimen are in good agreement with the experimental values both in thinner and thicker sheet in a given thickness combination of $0.8 \text{ mm} \times 1.5 \text{ mm}$ for all punch profile radii. The residual stress in the thinner sheet

Table 7.2 Comparison of experimental and predicted results

Punch profile radius (mm)	Location	Residual stress (MPa)			
		Thinner		Thicker	
		FEA	EXP.	FEA	EXP.
10	Outermost layer	49.98	56	−99.44	−75
	Innermost layer	−38.1	−46	98.49	83
12.5	Outermost layer	40.23	36	−94.62	−70
	Innermost layer	−36.4	−21	93.53	31
15	Outermost layer	25	16	−87.3	−46
	Innermost layer	−33.6	−12	25.7	20

is tensile in the outermost layer and compressive in the innermost layer, but the thicker sheet is able to rearrange its stress distribution to compressive on outermost layer and tensile on innermost layer. There is a large variation in the residual stress values obtained by FEA when compared with the experimental results in the innermost layer of thicker sheet with punch profile radius of 12.5 mm. This variation could be attributed to the fact that the values are averaged over the entire width of thinner and thicker sheets although the trend of the stress in innermost layer is increasing and decreasing in both the sheets. Moreover, the experimental values are determined at three different places on thinner and thicker sheet and do not indicate the complete stress mapping on the entire width due to the limitation of the measuring machine.

As the profile radius increases, it is observed that residual stress value decreases in the outermost and innermost layer in the cases of both thicker and thinner sheets, and the similar results are observed in experimental values.

7.6 Conclusions

The present study is based on the experimental and numerical characterization of residual stresses in the tailor welded blanks, before and after springback. With reference to the above studies and results, the following conclusions can be drawn:

1. Higher values of longitudinal stress before springback obtained from bending simulations are observed in both outermost (tension side) and innermost layers (compression side) in the thicker sheet than the thinner sheet due to higher plastic deformation in thicker sheet.
2. The sudden increase in the values of longitudinal stress obtained by bending simulations at the junction of thinner and thicker sheets on the outermost layer is due to the step in the longitudinally welded TWB resulting in localized stress concentration, whereas the flat side of the TWB which is the punch side or the innermost layer of the bent sheet shows gradual increase in the stress values towards the thicker side.

3. As the profile radius increases, the longitudinal stress values obtained from FE simulations decreases significantly during bending of the specimen.
4. The effect of the punch profile radius on residual stress after springback is observed to be very significant in TWBs. As the punch profile radius increases, residual stress decreases for a given thickness combination. The residual stress in thicker sheet is higher than the thinner sheet for all punch profile radii.
5. It is observed that after springback, the residual stress on thicker side is compressive as the case of a normal sheet, but the stress on thinner side is still tensile. This may be attributed to the combined effects of thicker and thinner portions on resultant springback of TWB.
6. Residual stress predicted by FE simulations agreed well with the experimental results for all punch radii except a few cases.

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Chapter 8

Prediction of Cutting Forces in Micro-milling of P-20 Steel by TiAlN-Coated WC Tool: An Analytical Approach



P. Sahoo , T. Pratap and K. Patra

Abstract Micro-milling is one of the emerging tool-based micro-manufacturing processes to fabricate miniaturized features on different materials. Still, there is a need of enhancement in tooling performance as the tool wear and breakage significantly affect the produced feature quality. Hard coating of the tool is a significant approach to improve its tooling performance by reducing the tool wear and tool breakage. However, the coating can increase the edge radius which increases the cutting forces. Therefore, the prior prediction of cutting forces is very much essential to enhance the tooling performance. We proposed an analytical approach for prediction of cutting forces by combing FEM simulation for orthogonal cutting followed by mechanistic modelling by the consideration of tool run out, minimum chip thickness and elastic recovery for TiAlN-coated tool. Finally, adequacy of the model is verified by experiments, and good agreement between predicted and experimental results is observed.

Keywords Micro-milling · Cutting forces · TiAlN-coated tool · Analytical modelling · FEM

8.1 Nomenclature

dF_t, dF_n : Force components in tangential and normal direction (N), respectively	K_{te}, K_{ne} : Ploughing/rubbing coefficient in tangential and normal direction, respectively (N/mm)
K_{tc}, k_{nc} : Shearing coefficient in tangential and normal direction, respectively (N/mm ²)	θ : Immersion or rotational angle of the cutter (°)
f_t : Feed (mm/tooth)	δ : Lag angle (°)
η : Helix angle (°)	w : Angular velocity (radian/sec)

(continued)

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_8

(continued)

r : Cutter radius (mm)	E : Elastic recovery rate
N : Number of teeth	ε : Plastic strain
Z : Axial depth of cut (mm)	$\dot{\varepsilon}$: Strain rate
$h(\theta)$: Instantaneous UCT (mm)	$\dot{\varepsilon}_0$: Reference strain rate

8.2 Introduction

Extensive applications of micro-features in aerospace, microelectronics and biomedical fields spur the development of several micro-fabrication processes. Among various micro-fabrication processes, micro-milling is one of the facile manufacturing approaches due to its enhanced characteristics such as higher material removal rate, applicability to wide range of engineering materials, capability towards producing three-dimensional complex and precise micro-features [1, 2]. Usually, it is considered that micro-milling tool (Fig. 8.1a) is just the downscaling of conventional milling cutter. However, due to this scaling effect, chip formation process encounters severe problem as edge radius of the cutting tool comes in the range of uncut chip thickness (Fig. 8.1b). This scaling effect is also commonly known as size effect [3]. Because of the size effect, chip formation is not possible when uncut chip thickness is smaller than minimum chip thickness. This leads to rubbing and ploughing followed by elastic recovery rather than shearing of workpiece material. In addition, various undesired phenomena come into picture like higher force generation, worse surface quality, burr formation, tool wear and tool breakage [1, 4]. Again, edge radius of the micro-tool is comparable to the grain size of the workpiece material, and sometimes feed and depth of cut are in the range of the grain size, so the chip formation is not isotropic and homogeneous like macro-milling process [3]. As a consequence,

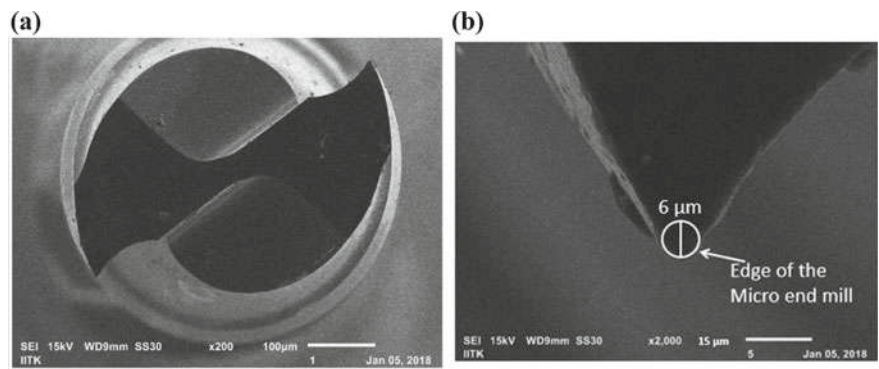


Fig. 8.1 a Top face of micro-end mill and b enlarged view of the edge

the performance of micro-end mill becomes more critical compared to conventional scale end mill cutter.

Tool wear and tool breakage are major impediments in micro-scale milling [5]. Especially these phenomena are prone to be more critical during machining of hard materials such as hardened tool steel, titanium alloy and nickel-based super alloy. Among such hard materials, P-20 steel which is a type of tool steel has enormous applications in micro-mould and die preparation due to superior characteristics such as wear resistant, higher toughness and hardness [6].

To meet the desired surface quality criteria, minimization of burr formation and tool wear and enhancement of the micro-tool performances, hard coating is generally implemented on the tool surface [4]. Aramcharoen et al. [6] reported the effect of ultra-thin coating of TiN, TiCN, TiAlN, CrN and CrTiAlN on tungsten carbide (WC) micro-end mill cutter for machining of hardened tool steel. Their research concluded that all coated tool enhanced the performance in comparison to uncoated tool in terms of edge chipping and flank wear. TiAlN-coated tool was established as the flagship among all coated tools. Thepsonthi and Ozel [4] examined the effect of CBN coating in micro-end mill cutter for machining of Ti-6Al-4V by both FEM simulation and experimental method. Their study revealed the improved characteristics of CBN-coated tool in terms of burr formation, tool wear, surface roughness and temperature generation. Furthermore, effects of various types of coated tools such as diamond-like carbon (DLC), AlTiN, AlCrN and TiAlN+ AlCrN in micro-milling of Inconel-718 have been performed by Uzun et al. [5]. Their study showed that coated tools outperformed the uncoated tool by reducing surface roughness, burr formation, flank wear, edge chipping and edge rounding. The above literature concludes the improvement in performance of the coated tool for machining of hard and difficult to machine material. However, tool wear and tool breakage can be reduced by coating but cannot be eliminated completely due to increase in cutting force due to increase in edge radius by application of coating material. So prediction of cutting forces is much requisite to control the surface quality and tool failure by optimizing the machining parameters in advance. Many researchers showcase their work in modelling of cutting forces; some of them are described in this section with brevity.

Bao and Tansel [7] reported the analytical modelling for the prediction of cutting forces in micro-milling by differentiating the total cutting zone into three parts and added size effect with tool run out in their model. Jun et al. [8] proposed a cutting force model by incorporation of effective rake angle, ploughing, rubbing and elastic recovery. They developed two separate mechanistic models for shearing dominant region and ploughing dominant region, respectively. Lu et al. [9] proposed an analytical cutting force model by incorporating flank wear during micro-milling. Srinivasa and Shunmugam [10] reported purely analytical model for force prediction in micro-milling by considering material strengthening effect, edge radius effect and basics of oblique cutting. Further, with recent advances in hardware and software, an Arbitrary Lagrangian–Eulerian (ALE) formulation is used by most of the software packages such as ABAQUS, ANSYS and DEFORM 3D for modelling material removal processes. Jin and Altintas [11] reported the fine element method (FEM)-based simulation for calibration of cutting coefficient in turning of brass. Further, they

implemented the cutting coefficient results to find the force value in micro-milling by aggregating the effect of tool run out and helix angle. FEM simulation for the prediction of cutting forces, temperature and stress with their experimental validation for micro-milling of Ti-6Al-4V has been presented by Uzun et al. [12]. However, they did not consider the effect of minimum chip thickness, tool run out, elastic recovery, etc., which are very significant parameters associated in micro-milling.

The aforesaid literature survey unveils the effect of coating in micro-milling and the necessity of analytical model. However, all the analytical cutting force models for micro-milling reported in literature are valid for uncoated carbide tool, and these may not work properly for coated tool as properties of the tool changes from substrate material due to coating. Further, most of the models extract the cutting coefficients from cutting calibration experiment which are also not cost-effective. Again, few literatures are available those extracted the cutting coefficient from FEM simulation.

Therefore, by considering the aforementioned scopes and to bridge such gap, the current study proposes an analytical model for the prediction of cutting forces in micro-milling for TiAlN-coated tool. The analytical model has been established by extracting the force coefficients from thermomechanical-based FEM simulation. The cutting coefficients obtained by FEM have been used in mechanistic model by aggregating the effects of tool run out, minimum chip thickness and elastic recovery. Finally, the adequacy of the proposed model has been taken care by validating it with experimental data.

8.3 Modified Undeformed Chip Thickness (UCT) Algorithm

The undeformed chip thickness model for a conventional end mill cutter can be expressed as [7].

$$h(\theta) = f_t \sin(\theta) \quad (8.1)$$

where $h(\theta)$ is the instantaneous undeformed chip thickness (mm), f_t is feed per tooth (mm/tooth) and θ is the rotational angle of the tool ($^\circ$). However, the effect of tool run out comes into picture which creates unbalance of the undeformed chip thickness for different teeth of the milling cutter.

As shown in Fig. 8.2, values of undeformed chip thickness generated by two teeth are not same for same positional angle. Due to lower feed value and tool diameter as compared to macro-end milling, run out effect cannot be neglected. The undeformed chip thickness model by considering tool run out can be expressed as [9]

$$AA' = h(\theta) = r + L \sin \sin \left(\text{wt} - \frac{2\pi i}{N} - \delta + \lambda \right)$$

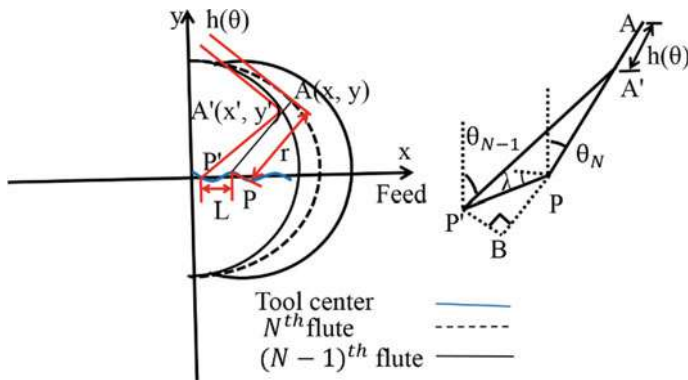


Fig. 8.2 Tool centre and tool flute trajectory of the micro-mill cutter due to tool run out

$$- \left(r^2 - L^2 \cos^2 \left(\omega t - \frac{2\pi i}{N} - \delta + \lambda \right) \right)^{0.5} \quad (8.2)$$

where r is radius of cutter (mm), L is the distance between the centre of the tool for two consecutive passes (mm), ω is the angular velocity of the cutter (radian/second), δ is the lag angle due to helical flute ($^\circ$), N is the number of flute and $i = 0, 1, 2, \dots, (N-1)$.

Further, size effect has a dominating effect during chip formation in micro-cutting. When undeformed chip thickness is below a critical value (minimum chip thickness), proper chip formation is not possible; only some part of the workpiece plastically deformed and rest elastically recovered after passing of the tool [8]. Therefore, effect of minimum chip thickness and elastic recovery should be added to make the undeformed chip thickness algorithm more reliable and realistic, which can be expressed as

$$h(\theta)_{\text{actual}} = (1 - E)h(\theta), \text{ when } h(\theta) \leq h_{\min} \quad (8.3a)$$

$$h(\theta)_{\text{actual}} = h(\theta), \text{ when } h(\theta) > h_{\min} \quad (8.3b)$$

where E is the elastic recovery rate and $h_{\min}$ is the minimum undeformed chip thickness which can be considered from the literature [8] and [13], respectively.

8.4 Analytical Cutting Force Model

8.4.1 Mechanistic Model for Prediction of Cutting Forces

The geometrical and Cartesian coordinate presentation of an end milling process has been shown in Fig. 8.3. Here x -axis represents the feed direction, y -axis represents the transverse to feed direction, and z -axis represents the axial direction. During the modelling of cutting forces, the micro-end mill has been divided into number of elements in axial direction, and each element acts as an oblique cutting tool (Fig. 8.3b). The forces which act on a discrete element can be represented as [10].

$$dF_t(\theta) = (K_{tc}h(\theta) + K_{te})dZ \quad (8.4a)$$

$$dF_n(\theta) = (K_{nc}h(\theta) + K_{ne})dZ \quad (8.4b)$$

where dF_t and dF_n are the elemental forces in tangential and normal directions, respectively. K_{tc} and K_{nc} are cutting coefficients; and K_{te} and K_{ne} are rubbing coefficients in tangential and normal directions, respectively. The elemental axial depth of cut (dZ) can be calculated as

$$dZ = \frac{rd\theta}{\tan \eta \tan \theta},$$

where η is the helix angle.

However, the cutting forces can be measured only in feed, transverse and axial direction during experiment. So, the Cartesian coordinate can be converted into global coordinate such as

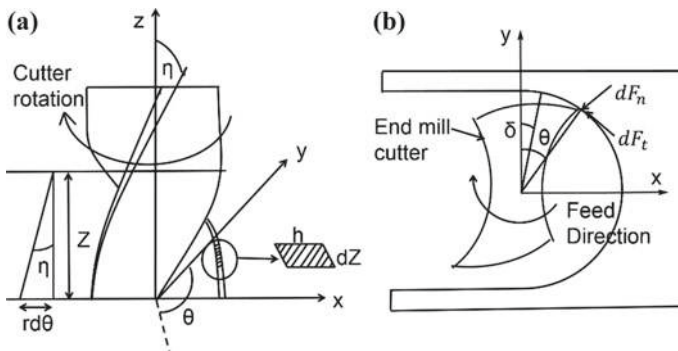


Fig. 8.3 Geometrical and coordinate presentation of end mill process

$$\begin{pmatrix} dF_x(\theta) \\ dF_y(\theta) \\ dF_z(\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} dF_t(\theta) \\ dF_n(\theta) \\ dF_a(\theta) \end{pmatrix} \quad (8.5)$$

Further, in micro-end milling process multi-tooth engagement occurs due to helical shape of the tool [10]. Therefore, by considering the assumptions proposed by Srinivasa and Shunmugam [10] and by integrating Eq. (8.5) within the limit of entry and exit angle of the flute, the total forces acting on the tool can be expressed as

$$F_x(\theta) = \frac{r}{\tan \eta} \sum_{i=0}^{i=N-1} \int_{\theta_{\text{exit}}}^{\theta_{\text{entry}}} [K_{tc}h(\theta) \cos \theta + K_{te} \cos \theta + K_{nc}h(\theta) \sin \theta + K_{ne} \sin \theta] \quad (8.6a)$$

$$F_y(\theta) = \frac{r}{\tan \eta} \sum_{i=0}^{i=N-1} \int_{\theta_{\text{exit}}}^{\theta_{\text{entry}}} [-K_{tc}h(\theta) \sin \theta - K_{te} \sin \theta + K_{nc}h(\theta) \cos \theta + K_{ne} \cos \theta] \quad (8.6b)$$

$$F_z(\theta) = \frac{r}{\tan \eta} \sum_{i=0}^{i=N-1} \int_{\theta_{\text{exit}}}^{\theta_{\text{entry}}} [-K_{ac}h(\theta) - K_{ae}] \quad (8.6c)$$

8.4.2 FEM Simulation for Force Coefficient Calculation

The cutting coefficients have been evaluated through finite element simulation for orthogonal machining as shown in Fig. 8.4 by DEFORM 3D (version 10.1) software. Five layers of TiAlN coating of 0.2 μm each having total coating thickness of 1 μm have been applied on the tool as shown by coating window in Fig. 8.4. During simulation, workpiece and tool have been considered as viscoplastic and rigid body, respectively. To justify the effect of edge radius, the edge radius of the tool (both coated and uncoated) has been taken as 3 μm as per the original dimension of the micro-end mill cutter (Fig. 8.1b).

Johnson–Cook material model [10] as given Eq. (8.7) has been used for the simulation (where first bracket represents strain hardening, second bracket represents strain rate effect, and third bracket indicates the flow softening). The values of the constant terms for the parametric equation have been given in Table 8.1. The mesh structures have been considered as tetrahedral for both tool and workpiece. The total number of mesh elements for workpiece is taken as 31,996 having minimum size 1 μm , and tool has been meshed by 5266 number of elements having mesh size 0.5 μm . Remeshing technique has been applied to avoid separation and distortion of

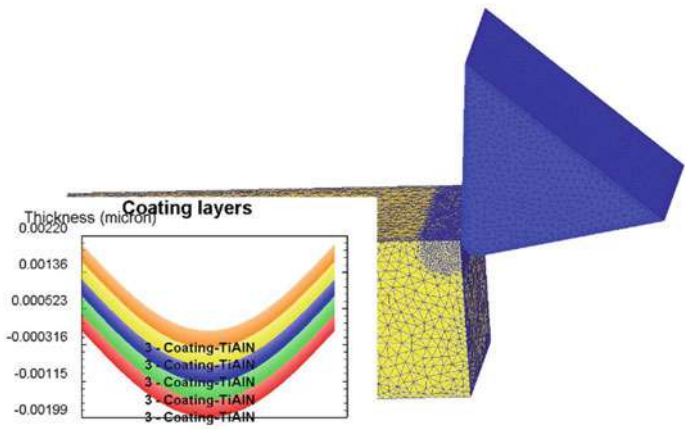


Fig. 8.4 Coating window showing the application of coating on the tool

Table 8.1 Johnson–Cook parameters [14]

A (MPa)	B (MPa)	n	c	m	T _{room} (°C)	T _{melting} (°C)
908.54	321.39	0.278	0.028	1.18	20	1487

the tool. Tool has been assumed to be rapidly heated having conductance coefficient in the order of $10^7 \text{ NS}^{-1} \text{ mm}^{-1} \text{ C}^{-1}$. The heat transfer between tool and workpiece with environment is associated with the convection coefficient of $0.02 \text{ NS}^{-1} \text{ mm}^{-1} \text{ C}^{-1}$. The remaining thermomechanical properties for P-20 Steel, WC/Co and TiAlN are shown in Table 8.2.

$$\sigma_{\text{flow}} = \left(A + B\varepsilon^n \right) \left(1 + c \ln \ln \frac{\dot{\varepsilon}}{\varepsilon_0} \right) \left(1 - \left(\frac{T - T_{\text{room}}}{T_{\text{melting}} - T_{\text{room}}} \right)^m \right) \tag{8.7}$$

The contact between tool and workpiece has been chosen to be hybrid relationship in DEFORM 3D software. Hybrid friction model has been chosen for assignment of the inter-object relationship. The constant for shearing friction has been taken as 0.9, and coefficient of Coulomb friction has been taken as 0.4 for TiAlN coating and 0.6 for uncoated WC/Co tool, respectively [4]. Friction factor between chip and

Table 8.2 Thermomechanical properties [4, 14]

	P-20 Steel	WC/Co	TiAlN
Thermal conductivity (N/S C^{-1})	28.4	55	$0.008T + 11.95$
Heat capacity ($\text{N/mm}^2 \text{ C}^{-1}$)	4.396	$0.0005T + 2.07$	$0.0003T + 0.57$
Thermal expansion coefficient (C^{-1})	12.8×10^{-6}	4.7×10^{-6}	9.4×10^{-6}
Young's modulus (MPa)	211,000	5.6×10^5	6×10^5

workpiece has been taken as 0.2. Simulation for orthogonal machining has been carried out at feed rates of 2, 4, 6 and 8 $\mu\text{m/tooth}$ and a constant depth of cut and cutting speed as 30 μm and 523.59 mm/sec, respectively.

The specific cutting force values (RMS of force values/depth of cut) obtained by simulation for both coated and uncoated tool are shown in Fig. 8.5. The figure shows higher magnitude of cutting forces both in tangential and normal directions for uncoated tool compared to those of coated tool. Further, induced temperature by uncoated tool is higher than that by coated tool as shown in Fig. 8.6. The higher values of cutting force and temperature of the uncoated tool are due the difference in thermomechanical properties and higher friction coefficient as compared to the coated tool. Finally, the force coefficients have been calculated by linearly fitting the specific cutting forces obtained in both tangential and normal directions (Table 8.3).

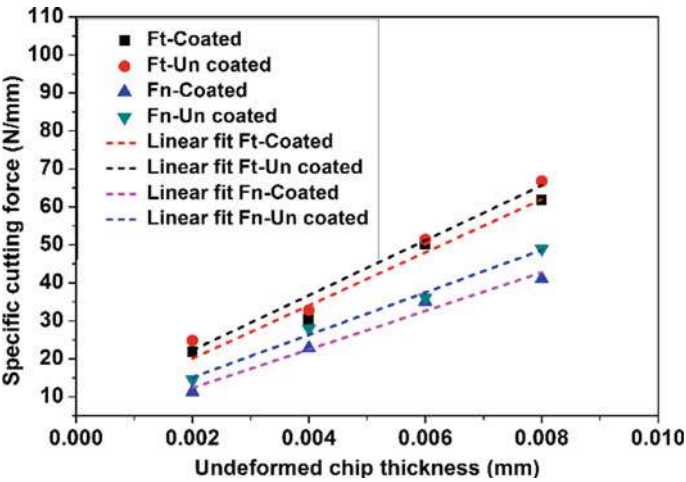


Fig. 8.5 Cutting coefficient calculation by linear fitting of simulated cutting forces

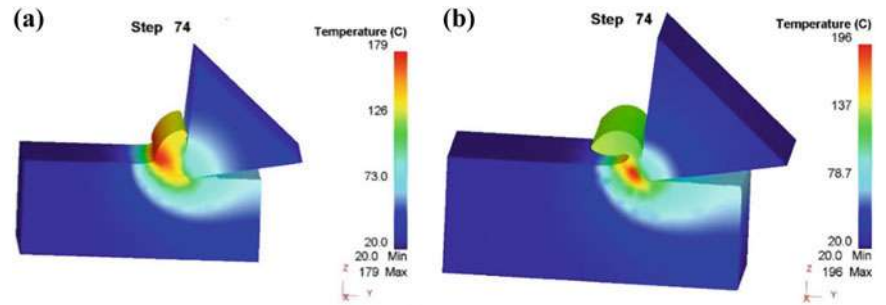


Fig. 8.6 Simulated temperature distribution using DEFORM 3D **a** TiAlN-coated tool and **b** uncoated tool

Table 8.3 Force coefficients

	Coated tool	Uncoated tool
$K_{tc} \left(\frac{N}{mm^2} \right)$	6985.6	7227
$K_{te} \left(\frac{N}{mm} \right)$	6.1	7.815
$K_{nc} \left(\frac{N}{mm^2} \right)$	5060	5586
$K_{ne} \left(\frac{N}{mm} \right)$	2.25	3.95

Cutting coefficients have been evaluated as function of undeformed chip thickness and rubbing coefficient as y-intercept by considering Eq. (8.4a, b).

8.5 Experimentation

A 3-axis hybrid micro-machining centre (Mikrotools, DT-110) has been used for the micro-milling experiments as shown in Fig. 8.7. During each experiment, a new TiAlN micro-tool having 500 μm diameter, 30° helix angle, 10° rake angle, 5° clearance angle and 3 μm edge radius has been used. All the experiments have been carried out at spindle speed 20,000 rpm, depth of cut of 30 μm and three level of feed rate, viz. 1, 2, 3 μm/ tooth. Cutting force data has been acquired by a tri-axial mini dynamometer (Kistler, 9256C2) and a data acquisition system (Kistler, 5697) by considering sampling frequency as 10,000 Hz. Further, to avoid the unwanted noises, a low-pass filter of 50 Hz has been set in the data acquisition system. Cutting tool run out value has been measured by laser displacement sensor (RF-60,360, Riftek), and run out angle has been obtained by the model proposed by Sahoo and Patra [15, 16].

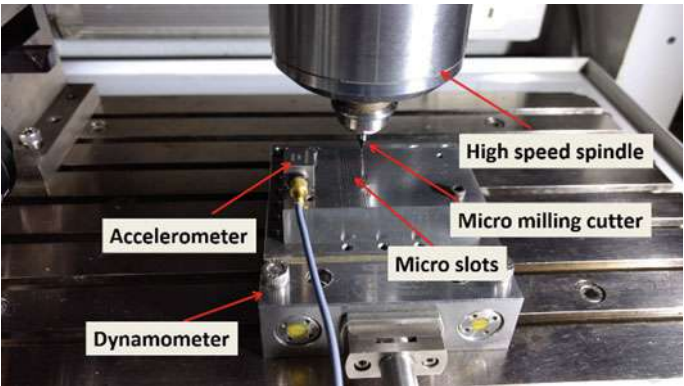


Fig. 8.7 Experimental set-up

8.6 Results and Discussion

The comparison of predicted force values with experimental results is depicted in Figs. 8.8, 8.9 and 8.10. From Figs. 8.9 and 8.10, it can be revealed that the trends

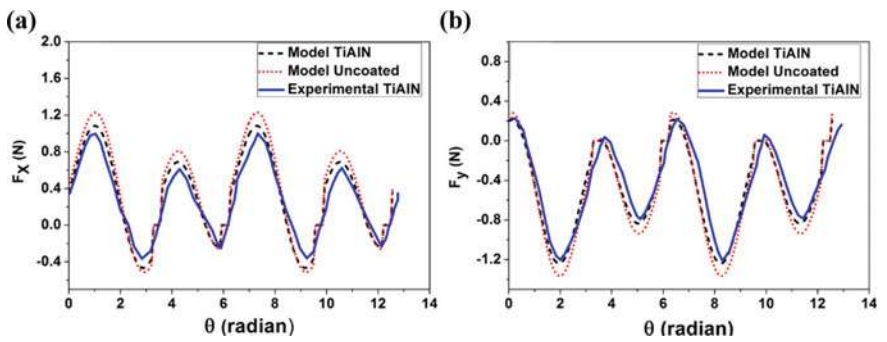


Fig. 8.8 Cutting force at feed rate $3 \mu\text{m/tooth}$; **a** x-direction and **b** y-direction

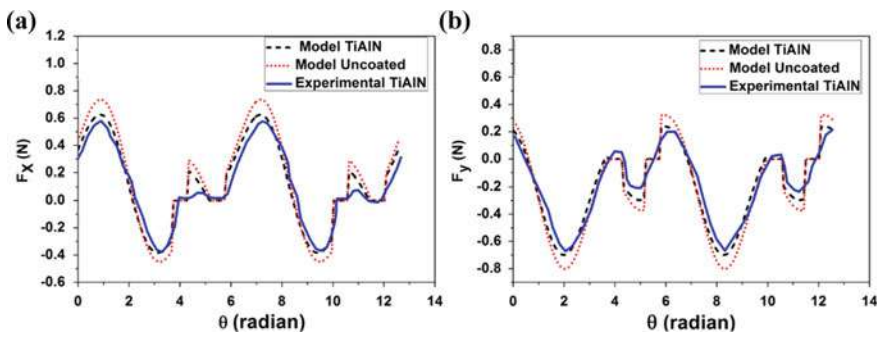


Fig. 8.9 Cutting force at feed rate $1 \mu\text{m/tooth}$; **a** x-direction and **b** y-direction

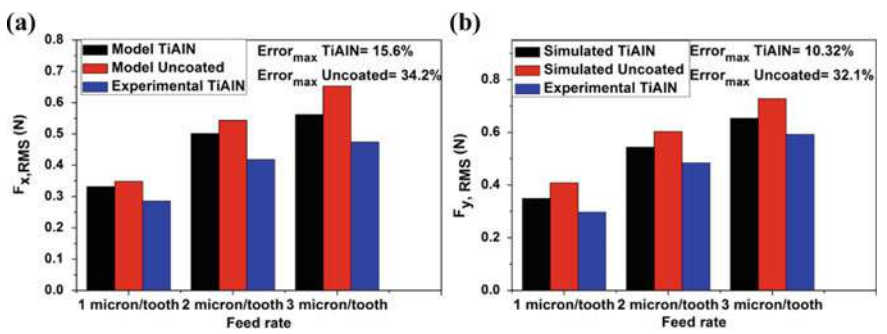


Fig. 8.10 Comparison of RMS values of cutting forces; **a** x-direction and **b** y-direction

of cutting forces generation by the proposed model are very much similar with experimental results. The cutting force values predicted for TiAlN-coated tools are closer to the experimental forces than those of the uncoated tools. Further, the effects of tool run out and elastic recovery rate are observed significantly at lower feed rate as shown in Fig. 8.9. This is due to more ploughing and rubbing at lower feed rate where the effects of minimum chip thickness and the workpiece material spring back after passing of the tool are more. The RMS values of cutting forces obtained by both model and experimentation are shown in Fig. 8.10. The figure shows a good correlation between the predicted force values obtained for TiAlN-coated tool with experimental results as compared to those of uncoated tool. The maximum error obtained by TiAlN-coated tool with experiment is only 15.6 and 10.32% in x and y directions, respectively, whereas the maximum error in x -direction is 34.2% and in y -direction is 32.1% in case of uncoated tool.

8.7 Conclusion

This paper presented an analytical model for prediction of cutting forces for TiAlN-coated tool in micro-milling by extracting the force coefficients by FEM simulation instead of the experimental data. The proposed model conveyed its adequacy for prediction of cutting forces in both higher and lower feed values with consideration of tool run out and elastic recovery rate in chip thickness model. In addition, consideration of coating effect during coefficient calculation was proved to be viable as prediction of cutting forces by TiAlN coating model was more closer to the experiment as compared the uncoated tool. Moreover, the improvement in the performance of the model has been observed by reduction in prediction error by considering coated tool as per the real application during experiment instead of uncoated tool.

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Chapter 9

Effect of Mechanical Constraints on Thermo-Mechanical Behaviour of Laser-Welded Dissimilar Joints



Bikash Kumar , Rachit Nawani and Swarup Bag

Abstract Titanium and its alloy exhibit eminent properties such as low density, creep and corrosion resistance, which attribute miniature applications in medical industry. Joining of dissimilar material poses challenge due to great difference in their thermal and mechanical properties. Residual stress is an important cogitation for the component integrity and life assessment of welded joints where its magnitude arises up to yield strength. The present study involves finite element-based modelling of dissimilar welding (Ti–SS) to examine the thermo-mechanical behaviour of welded joints. The temperature profiles are validated with experimental data. In thermo-mechanical analysis, the mechanical constraint plays an important role which substitutes the practical welding condition. Hence, the influence of different restraint conditions on the residual stress and distortion are analysed in the present work. No significant difference is found in magnitude and trend of residual stresses for different boundary condition. However, remarkable variation is observed in distortion analysis for different conditions.

Keywords Dissimilar welding · Finite element model · Mechanical constraints · Thermo-mechanical analysis

9.1 Introduction

Dissimilar joints are accomplished in order to abduct privilege of certain features of each component to strengthen the potential of a product. Mostly, two dissimilar materials required to join when one acquires characteristics essential for operation of device but has unavoidable disadvantage like more cost or dangerous. Titanium and its alloy possess an eminent combination of mechanical properties, creep and corrosion resistance which has led significant miniature application in various industries like nuclear, biomedical, etc. Due to owning very high cost, it is important to

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_9

join it with material which is economical (like steel alloy) and can attribute similar properties at some extent and can produce improved properties even after joining. It is reported that characteristics such as precise, localised or concentrated beam, high welding velocity, narrow heat-affected zone (HAZ), low residual stress generation have given some success in the area of dissimilar joining by using high energy beam process [1]. Despite the capability of dissimilar joining by laser, non-uniform alloying concentration in the fusion zone (FZ) and variation in thermal expansion properties lead to low joint strength and possible crack formation [2]. Joining of thin sheets susceptible to some major defects such as burn-through, buckling, warping, oxide formation, etc. [3].

A number of numerical and experimental inspections of laser-welded dissimilar joints have been revealed by researchers. Dissimilar welding of steel–copper, steel–nickel, steel–aluminium and Ti–SS has been already studied in literature. The effects of phase transformation and intermetallic compounds on the mechanical properties and microstructural morphology have been studied for dissimilar joint of SS 304 and Ti-6Al-4V at different pressure. Grain growth and interfacial morphology have been studied by EDAX and XRD analysis [4]. Tomashchuk et al. studied the influence of operational parameters on the thermo-mechanical stability and local phase composition of electron beam welds between titanium alloy and stainless steel with foil (copper) as an intermediate layer [5]. It is found that two types of morphologies can be achieved depending upon the beam offset from the centre line. When beam is offset towards the titanium alloy, a large amount of brittle TiFe_2 phase is formed which can be reduced by offsetting the beam towards the stainless steel side. Temperature profile for thin sheet of both Ti-6Al-4V and Al alloy dissimilar joint has been reported experimentally and numerically by using fibre laser process [6]. The primary thermo-physical properties such as thermal conductivity, thermal expansion coefficient, absorptivity, specific heat capacity, density and melting temperature play important role in laser welding of dissimilar material. Metal of higher thermal conductivity like stainless steel (304) as compared to Ti-6Al-4V dissipates energy promptly, making it difficult to maintain stable weld pool. Non-uniform heat conduction in the joint comprising two metals with widely differing thermal conductivity can result misaligned welded structure [7].

Not only these factors but also air gaps, misalignment of weldments and inappropriate fixturing are major sources of weld defect or in progress of residual stress in the structure. The fusion welding characterised by heating and cooling phenomena or non-uniform expansion and shrinkage of different zones simultaneously leads to residual stress and associated distortion in welded structure. The stress exists even after removal of load due to external agents within the material is known as residual stress. Ranjbarnodeh et al. developed a 3-D thermal model for dissimilar steel alloy joint for evaluation of temperature distribution and bead geometry in absence of pool convection. It is reported that due to difference in thermal conductivity, maximum temperature is shifted towards low carbon steel [8]. Author has studied the behaviour of residual stress in post-welded heat-treated condition for multi-pass welding of ferritic and austenitic steel pipe numerically and validated it with experimental result [9]. Support vector regression (SVR) optimised model-based finite element method

has been developed to predict the residual stress for maintaining the performance and reliability of dissimilar joint of steel alloy [10]. Taran et al. investigated the residual stress distribution between pipe joining made of stainless steel and titanium alloy [11]. The residual stress in a SS–Ti adapter by using the neutron-diffraction method is reported as well as measurement of axial, radial and hoop strain components is carried out. Thermal-elastic-plastic finite element-based 3-D model has been developed for different grade of steel alloy dissimilar joints [12]. It is revealed that the residual stresses in longitudinal direction due to welding increase with the increasing yield stress of steel.

A few literatures have been found corresponding to experimental and numerical investigation of Ti–SS dissimilar joints. However, there is scarcity of literature which focuses on effect of constraints on thermo-mechanical behaviour of dissimilar thin plate welded joint numerically. The present study includes temperature distribution, weld bead estimation study for dissimilar joint of thin plate of Ti-alloy and AISI 304 using numerical model by the application of ABAQUS commercial software package and validated it with experimental result as reported in literature. Based on this validated model, three different thermo-mechanical models have been developed with various boundary conditions to examine its influences on residual stress and distortion distribution.

In the present study, firstly, 3-D thermal model has been developed for the heat input of 65 J/mm. Three-dimensional eight-noded linear brick element is taken for modelling. 300 K is taken as ambient temperature and stored in predefined field for thermal analysis. DFLUX subroutine is implemented for the movement of laser heat source in a Gaussian fashion. Thereafter, time–temperature history exhibited by the thermal cycle has been recorded. These nodal temperature outputs are taken as predefined field input for mechanical analysis. Afterwards, sequential coupled thermo-mechanical model for dissimilar weld joint has been established for different boundary conditions. Model which is used in thermal analysis is also taken for mechanical analysis. Same meshing is applied for both models; however, diffusive heat transfer element is used for thermal model and 3-D stress element is taken for mechanical model. Furthermore, residual stress and distortion in different direction for different boundary condition have been analysed at the middle of the plate.

9.2 Thermal Analysis

A 3-D finite element-based thermal model is developed for the investigation of temperature field evolved from the temporal and spatial movement of heat flux during laser welding of dissimilar material using commercial software package ABAQUS. The temperature-dependent properties like specific heat, conductivity and static value of latent heat of fusion and density are assumed for simulation. The thermo-physical and mechanical properties are considered from literature [13, 14]. Full plate geometry is taken for analysis where half-plate represents solution domain of Ti-alloy and another part signifies SS304. The 3-D governing heat conduction equation based on

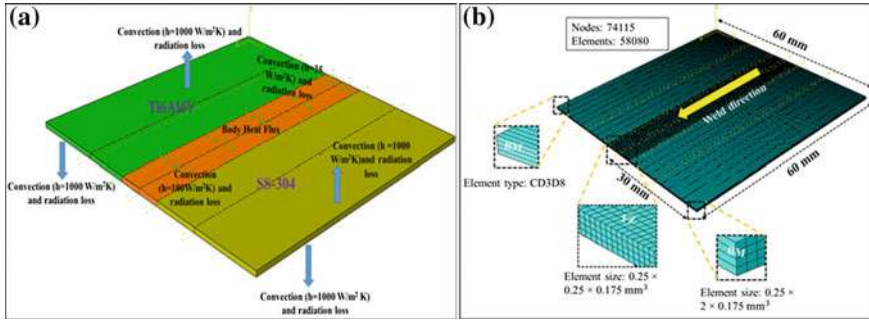


Fig. 9.1 **a** Thermal boundary condition on solution domain; **b** Mesh specification for dissimilar plate model

energy conservation is given by Eq. (9.1).

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + Q = \rho C \left(\frac{\partial T}{\partial t} - v \frac{\partial T}{\partial x} \right) \quad (9.1)$$

where ρ is density of material (kg/m^3), C is specific heat (J/kg K), k is the thermal conductivity (W/m K), Q is rate of heat generation in the domain. T is temperature (K), v is the welding velocity (m/s), t is time (s) and x, y, z are the coordinates in the reference system. Different boundary conditions or thermal constrained associated with fixture (to avoid misalignment during welding) and heat losses due to unavoidable conduction, convection and radiation phenomena are expressed as (Fig. 9.1a).

Dirichlet-type thermal restraint, where the preliminary temperature for full plate solution domain is assumed as ambient temperature, i.e.

At time

$$t = 0, T(x, y, z, 0) = T_i = 300 \text{ K.}$$

Neumann boundary condition: A heat loss at different surfaces due to conduction, convection and radiation is given by Eq. (9.2).

$$k \frac{\partial T}{\partial n} - q_g + \varepsilon_r \sigma (T^4 - T_i^4) + h_c (T - T_i) = 0 \quad (9.2)$$

where T_i is initial temperature, k is isotropic thermal conductivity of the material, q_g is Gaussian distributed heat flux impinges on substrate, ε_r is the emissivity, σ denoted as Stefan–Boltzmann constant ($5.67 \times 10^{-8} \text{ Jm}^{-2} \text{ s}^{-1} \text{ K}^{-4}$), h_c is the convective heat transfer coefficient between workpiece and ambient surrounding.

The geometrical configuration of welded joint with optimised non-uniform mesh specification in FZ and far away surface for both the material is shown in Fig. 9.1b. For the spatial and temporal movement of laser beam on the substrate, double-ellipsoidal heat source model is considered for the numerical simulation [15].

9.3 Structural Analysis

The nodal time–temperature data extracted from the thermal analysis has been employed as input thermal loading for the sequential coupled thermo-mechanical analysis. Similar type of finite element-based mesh is employed here as used in thermal modelling. Due to absence of information regarding phase transformation of dissimilar material (Ti-alloy and steel alloy), phase transformation effect is neglected in this analysis. The material is considered to follow von Mises yield criterion and associated flow rule for incorporation of rate-independent plasticity. In mechanical analysis, the governing equation based on principle of equilibrium in the form of tensor is given by Eqs. (9.3) and (9.4).

$$\sigma_{ij,j} + b_i = 0 \quad (9.3)$$

$$\sigma_{ij} = \sigma_{ji} \quad (9.4)$$

where b_i is body force vector and σ_{ij} denotes stress tensor. Correspondingly, constitutive equations for thermo-elasto-plastic model are considered. A stress–strain relation is expressed as Eq. (9.5).

$$[d\sigma] = ([C^{\text{el}}] + [C^{\text{pl}}])\{d\varepsilon\} + [K^{\text{th}}]\{dT\} \quad (9.5)$$

where $[C^{\text{el}}]$, $[C^{\text{pl}}]$, $[K^{\text{th}}]$ are elastic, plastic and thermal stiffness matrix. The relationship between thermal stress and strains based on von Mises yield criterion is given by Eq. (9.6).

$$\varepsilon_{ij} = \frac{1 + \mu}{E}\sigma_{ij} - \frac{\nu}{E}\sigma_{ij}\delta_{ij} + \lambda S_{ij} + [\alpha(T - T_i)\delta_{ij}] \quad (9.6)$$

where E , α , λ , S_{ij} and δ_{ij} are modulus of elasticity, coefficient of thermal expansion, plastic flow factor, deviatoric stress component and Kronecker delta notation, respectively.

Several factors are responsible for the generation of residual stress and distortion in welded structure. On those, mechanical boundary condition plays a remarkable role in prediction of distortion in welded joints. Mostly, thermal and mechanical responses are susceptible towards different types of restraints during joining of similar or dissimilar thin sheets. Researcher reported that application of fixture or different constraints alters the temperature distribution within plate and can decrease depth of penetration by 21%. Significant variation in the magnitude of longitudinal and transverse residual stresses has been found due variation of mechanical constraints [16]. Three different types of restrain or boundary conditions employed for the prediction of residual stress and deformation in the present study are shown in Fig. 9.2.

In Fig. 9.2, Case 1 shows that displacement in transverse (Y -axis) and thickness (Z -axis) direction (V_1 , W_1) is restricted at the corner of Ti-alloy plate whereas all

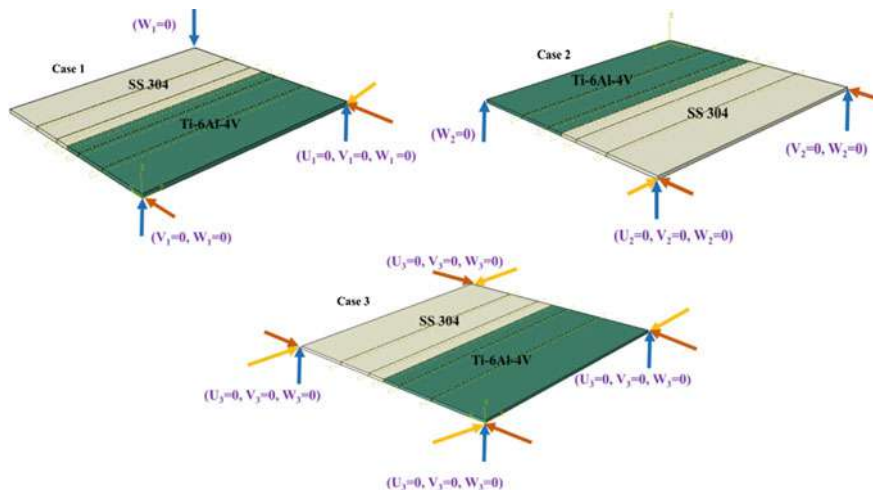


Fig. 9.2 Three different boundary conditions employed in numerical analysis

degrees of freedom (U_1, V_1, W_1) are seized for corner opposite to local coordinate system or end edge of Ti-alloy. Displacement in thickness (Z-axis) direction (W_1) only restricted at end edge of steel alloy. For Case 2, the only difference is that whatever restriction was applied to Ti and steel plate side is reversed to steel and Ti-plate side, respectively. However, in Case 3, all degrees of freedom (U_3, V_3, W_3) are seized for all corners of plates.

9.4 Results and Discussion

The simulation of temperature, distortion and residual stress field is performed corresponding to the welding velocity of 4.2 mm/s and current of 11 A for the heat input of 65 J/mm [13]. The result obtained from the thermal model, i.e. welds bead geometry and temperature profile for both the part of materials are validated with experimental output [13, 17]. A comparison of literature-based result which is regarded as experimental result and numerical weld bead shape for the laser-welded dissimilar joint of dual-phase Ti-alloy and SS 304 at specified process condition is shown in Fig. 9.3. Macro-structural features of numerically developed bead geometry are of similar pattern as predicted by experimental result [18]. Mushy zone which is characterised by solidus (Ti-alloy ~1877 K; SS ~1673 K) and liquidus temperature (Ti-alloy~1933 K; SS ~1723 K), heat-affected zone (HAZ) which exhibits solid-state phase transformation and fusion zone (FZ) is represented by different isotherm contour. Red colour represents FZ attained maximum temperature ~2400 K followed by mushy zone (yellow band) and HAZ (region between yellow and blue). The attained maximum temperature is well below the boiling temperature of both base materials

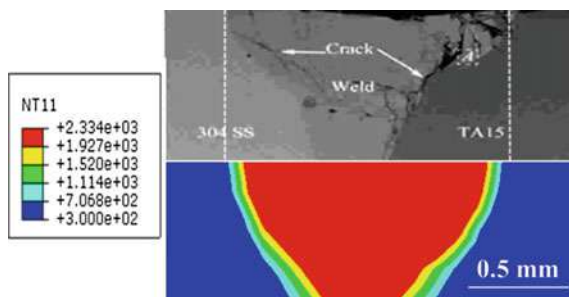


Fig. 9.3 Comparison of experimental [18] and numerical weld bead shape for dissimilar joint

(i.e. < 3200 K). Therefore, It can conclude that conduction mode of laser welding is predominant over keyhole for the considered welding condition in the present investigation [19].

Figure 9.4 depicts the experimental and numerical interpretation of temperature profile for steel alloy (Fig. 9.4a) and Ti-alloy (Fig. 9.4b) in HAZ region. As it is too complex to evaluate the temperature in fusion zone experimentally due to involvement of melting phenomenon during welding, HAZ temperature range is employed for validation. Maximum temperature of ~ 1200 K and ~ 1320 K is observed for stainless steel and Ti-alloy, respectively. The heat-affected zone temperature is well below of fusion zone temperature; this is due to the fact that less amount of heat conducted far away from the weld line which results lower peak temperature. It is clearly visible that the agreement between experimental and numerical result is quite reasonable for steel alloy in cooling cycle and more satisfactory in case of Ti-alloy.

It is clearly visible in Fig. 9.5 that temperature attained by Ti-alloy is higher than that of steel alloy in different zone. In this range of temperature, thermal diffusivity

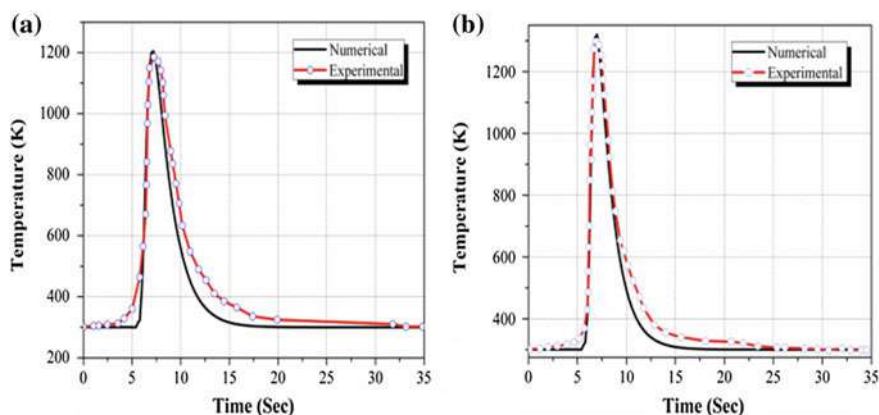


Fig. 9.4 Comparison of literature-based extracted result (i.e. written as experimental in the figure) and numerical thermal profile of **a** SS 304 [17]; **b** Ti-6Al-4V [13] in HAZ

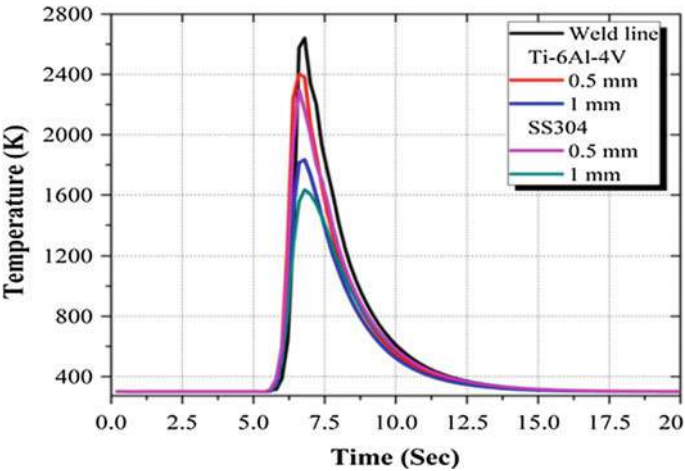


Fig. 9.5 Temperature distribution across the weld

of Ti6Al4V is higher than that of SS-304 which causes more diffusion of heat on Ti6Al4V side resembles comparatively higher peak temperature.

Figure 9.6a represents the deflection in transverse (Y -axis) direction (U_2) is opposite in nature for both the material for Case 1. For Ti-alloy, it is around ~ 0.013 mm in negative Y -direction and ~ 0.05 mm in positive Y -direction for SS 304. This is due to the fact that different mechanical constraints are imposed on both ends of the plate. Figure 9.6b depicts the out of the plane deflection (U_3) which is much more symmetric in nature for dissimilar material. The maximum value of deflection is ~ 0.35 mm in negative Z -direction near the welding interface. The deflection (i.e.

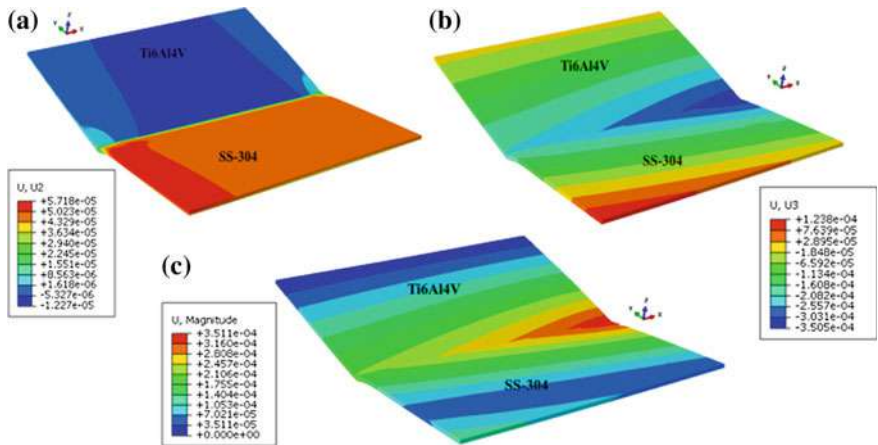


Fig. 9.6 Contour plot of longitudinal (U_1), transverse (U_2) and total (U) deflection for Case 1

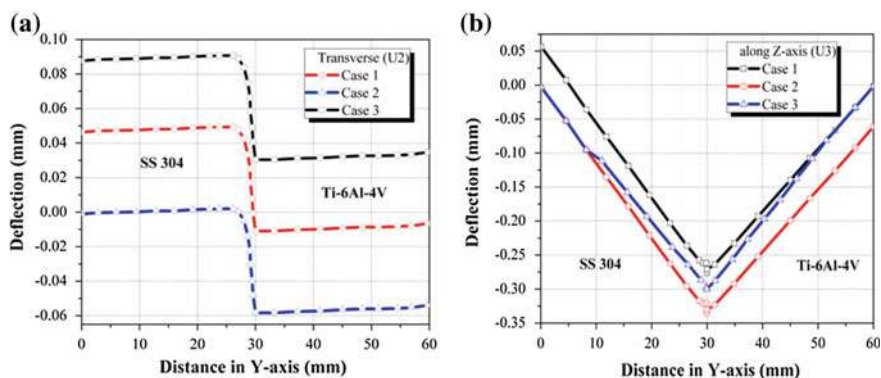


Fig. 9.7 Variation of deflection in **a** transverse, **b** thickness direction with distance in Y-axis for different Cases

U_3) decreases linearly at a distance away from the weld line. It is clearly seen in Fig. 9.6c that the magnitude of total deflection (U) is symmetric in nature for the two materials as U_3 . It is reported that the deflection in Z-direction is important deciding factor for the nature of total deflection in thin sheet welding. Similarly, it is found that the magnitude of out of plane deflection (U_3) is much larger than the other two components of deflection, i.e. U_1 , U_2 . The maximum value of total deflection (U) is ~ 0.35 mm, which is around 50% of the thickness of the plate. Moreover, the nature of deflection is sagging type for both the materials in Z-direction.

Figure 9.7a shows the graphical representation of variation of transverse deflection (i.e. U_2) with distance in Y-axis for the three restraint conditions. It shows that the trend of the curves is found to be similar in nature but the values differ for the three Cases. For Case 1, the deflection on SS304 plate is much larger than Ti-alloy due to the presence of lesser constraints which allows it to deform freely. The magnitude of transverse deflection for Ti-alloy reaches ~ 0.01 mm while around ~ 0.05 mm for SS 304. However, U_2 attains 0.06 mm for Ti-alloy whereas ~ 0.01 mm for SS304 in Case 2. Here, SS 304 plate is in more strictly restraint condition which resembles lower deflection. It is noted that both the plates show comparable deflection, i.e. ~ 0.04 mm and ~ 0.07 mm for Ti-alloy and SS 304, respectively in Case 3. Figure 9.7b shows that the out of plane deflection (i.e. U_3) is symmetrically distributed on both sides of welding line. It is observed that deflection is decreasing as the distance away from the welding line is increasing. The value of the maximum deflection is having values of ~ 0.26 mm, ~ 0.33 mm and ~ 0.28 mm for the Cases 1, 2 and 3, respectively at the weld line. The nature of deflection is of sagging type for all the three cases. The maximum value of resultant (total) deflection is ~ 0.27 mm, ~ 0.37 mm and ~ 0.29 mm for the three cases, respectively. Welding is characterised by highly non-uniform temperature distribution corresponding to different heat input. So, the stresses induced during welding are highly heterogeneous in nature. The two major components namely the longitudinal stress and transverse stress for three boundary

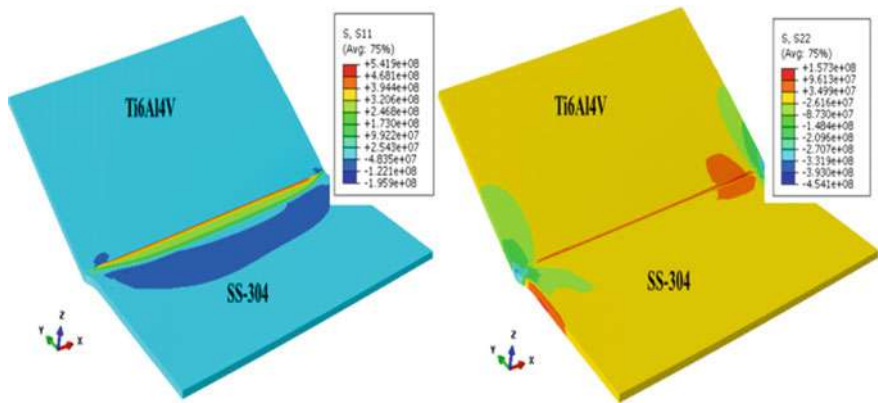


Fig. 9.8 Contour plot of longitudinal and transverse stresses for Case 1

conditions are studied here. Figure 9.8 signifies the contour plot of distribution of longitudinal and transverse stresses for Case 1.

Figure 9.9a shows the variation in longitudinal stress (i.e. S11) along the Y-axis for all cases. It reveals that the distribution of longitudinal stress is asymmetric across the weld centre line because of different mechanical properties of two materials. For SS 304 side, the longitudinal stress is tensile in nature having value of ~185.7 MPa, ~188 MPa and ~191.4 MPa whereas for Ti-alloy, it is compressive in nature having comparatively lower magnitude of ~28.4 MPa, ~28.5 and ~29.6 MPa for all three cases, respectively, in fusion zone. As we move away from the welding line, there is sudden drop in the value of stress and converges towards zero value at the end due to considerably lower temperature.

Figure 9.9b displays the distribution of transverse residual stress (i.e. S22) along the Y-direction in different boundary conditions. It can be seen that the transverse

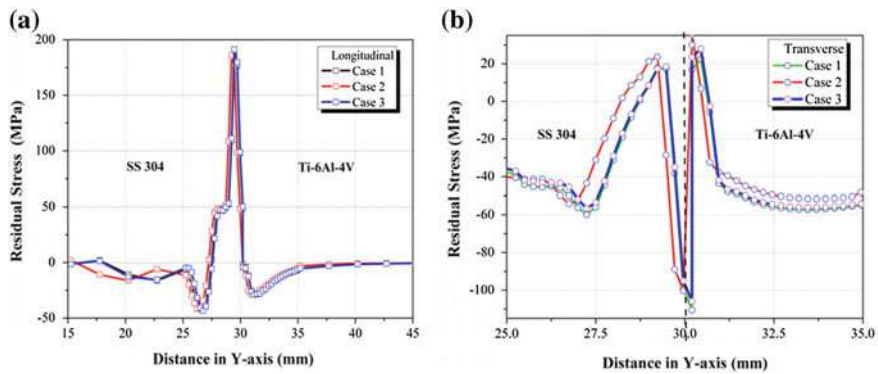


Fig. 9.9 Distribution of residual stress in **a** longitudinal (S11); **b** transverse direction (S22) across the weld line

stresses are compressive in nature in the FZ and converted to tensile in the HAZ. Further away, the nature again turns compressive but with a lesser magnitude. In the FZ, the maximum value of compressive stress reaches ~110 MPa, ~100 MPa and ~95 MPa while ~57 MPa, ~51 MPa and ~56 MPa of tensile stress in HAZ, respectively. As the present study is dealing with thin plates, the normal component of the residual stress is negligible. Also, the shear components of the stress tensors are negligible in magnitude and will have least impact over the distortion pattern of the weld plates. It is concluded that longitudinal component has predominant influence in the distortion and failure of material as compared to transverse stresses.

In the present work, mechanical model has not been validated with the experimental result. It may possible that some discrepancy would be there in predicted result. Therefore, it is essential to validate numerical output with experimental result for different boundary conditions. During mechanical analysis, fluid flow phenomena and effect of phase transformation are not considered. It would be better if these phenomena were included for the estimation of residual stress and distortion because these parameters also play important role in generation of residual stress and distortion.

9.5 Conclusions

In the present investigation, sequentially coupled thermo-mechanical analysis of dissimilar Ti-6Al-4V and stainless steel 304 weld joint has been performed to investigate the temperature field, residual stresses distribution as well as distortion of welded structure. The following conclusions have been drawn from the present work.

- Full depth of penetration has been observed for the heat input of 65 J/mm as mentioned in experimental result. Due to involvement of lower temperature (< boiling temperature), vaporisation phenomena have not perceived here.
- Numerical simulation-based macrograph has shown bead geometry of similar nature as reported in literature. The maximum temperature attained in Ti-plate (2400 K) is comparatively higher than that obtained in SS304 (2300 K).
- The longitudinal residual stresses are quite asymmetric across the welding line. On the SS 304 side, it is tensile in nature while compressive on Ti-alloy side but their magnitude is slightly different for the three restraint conditions.
- The transverse residual stresses are lower in magnitude than their longitudinal counterparts in FZ while their values are even smaller in HAZ. Moreover, they are compressive in FZ while tensile in nature in HAZ for both the materials.
- The distortion pattern obtained with the three boundary conditions shows a similar trend across the transverse axis but has different magnitudes in both y and z directions.
- The out of the plane distortion is the major one of the three components and the magnitude of deflection is around 0.35 mm which is around 50% of the thickness of the plates.

- It is also found that the value of longitudinal stress increases as the number of constraints increases whereas the magnitude of transverse residual stress decreases with the increases in constraint.

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Chapter 10

Thermal Modeling and Simulation of Crater Generation on Wire Electrode During Wire EDM Operation



Sanghamitra Das and Shrikrishna N. Joshi

Abstract In the present research, the wire cross section is considered for analysis to evaluate the peak temperature obtained by the wire during machining. The crater shape and crater area on the wire have been estimated from the temperature profile which can be used to evaluate the wire erosion rate. Gaussian heat flux is considered in the model as it gives better results compared to other approaches as available in literature. In the present work, we have considered the latent heat of melting of brass wire as it is a very important factor in the thermal modeling of EDM because latent heat signifies the consumption of the amount of supplied heat in the phase change of the wire material. The developed model successfully predicted the temperature profiles across the wire cross section. It will be useful in the prediction of wire rupture during adverse process conditions.

Keywords Wire EDM · Wire electrode life · Wire rupture · Finite element method · Thermal modeling

10.1 Introduction

Wire electric discharge machining (WEDM) is an unconventional machining process that uses spark discharges to remove material from both the electrodes. The wire electrode is a small but very important part of a WEDM machine. However, wire breakage is the major concern for a wire EDM process as it hampers the overall productivity of the system. Frequent occurrence of wire rupture drastically reduces the efficiency and accuracy of the wire EDM operation. The wire is subjected to combined thermal and mechanical load during machining. Thermal load characterized by the peak temperature obtained in the wire electrode is one of the major causes of wire rupture which causes the wire to erode thereby losing its mechanical strength [1].

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_10

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There are a number of reports on the numerical modeling of wire electrode in a WEDM process to predict and prevent the wire rupture phenomenon to ensure better machining productivity. Literature shows that many researchers have developed thermal models as well as structural models of wire electrode using finite difference method and finite element method to predict wire breakage. Dekeyser et al. [2] first developed a preliminary thermal model to predict the wire rupture phenomenon during wire EDM process using finite difference method. Banerjee et al. [3, 4] also developed a finite difference model to estimate the thermal loads on the wire electrode along the wire axis as well as along the wire cross section for varying values of input parameters, viz. input power, pulse on time, wire velocity, and wire diameter. The results showed that the temperature increased in the zone of discharge channel with increased power input. The effect of wire velocity was found to be negligible on temperature distribution and reduced diameter of wire lead to greater thermal load, thus posing serious threat to the wire. A simple finite element model (FEM) was developed to predict the thermal distribution in the wire for increased wire velocity and reduction in heat transfer coefficient. The input parameters were optimized to prevent wire rupture [1]. Banerjee and Prasad [5] further proposed a one-dimensional explicit finite difference thermal model for estimating the transient temperature distribution along the length of the wire during wire EDM machining under the conditions of randomly located spatial sparks with and without the formation of clusters. Yang et al. [6] incorporated the moving heat source characteristics into the heat transfer model of wire electrode and simulated the temperature field in a single pulse discharge using finite element method-based software, ANSYS. Murphy and Lin [7] developed a combined structural–thermal model to describe the vibration and stability characteristics of an EDM wire and to determine how various parameters (such as the transport speed, the tension force, and the discharge energy) influence the machining accuracy. Han et al. [8] discussed the coupled thermomechanical analysis, and both the three-dimensional temperature and the stress distributions in the micro-wire are determined. The tension of the micro-wire is optimized in accordance with the discharge energy input into the wire electrode in order to improve machining accuracy and prevent wire breakage. Das and Joshi [9] incorporated plasma features, moving heat source characteristics, multi-spark phenomenon and wire vibrational effect into the wire model to predict the cathode erosion rate in micro-wire EDM process. Further, Tomura and Kunieda [10] justified the mechanism of electromagnetic force applied to the wire electrode during wire EDM process by developing a two-dimensional finite element model for electromagnetic field analysis by taking into account electromagnetic induction. In recent years, coupled multi-physics model (thermal model, structural model, electromagnetic model) were developed to control the vibration of wire electrode and to study the effect of process parameters in cutting thin plate process [11].

In the present paper, a two-dimensional nonlinear transient thermal model of the wire electrode is developed to predict the peak temperature in the wire after a single discharge for varying levels of input parameters, viz. voltage, current, and pulse on time. The crater geometry and crater area can be evaluated from the temperature distribution which gives the amount of wire eroded after a single discharge. Use

of optimal levels of input parameters shall prevent the failure of the wire due to excessive thermal loads, thus obtaining a better machining productivity.

10.2 Development of Thermal Model

In EDM, intense localized heating due to spark discharges at the narrowest gap between the tool and work piece melts and vaporizes the material from both the electrodes. In the present work, a preliminary two-dimensional transient thermal model for a single discharge is developed for the wire electrode using finite element-based software, ANSYS. In this model, the temperature profile and wire erosion after a single discharge are obtained on circular cross section of wire electrode. This model shall pave the way to estimate the total thermal load in the wire.

10.2.1 Assumptions

The thermal model is developed under the following assumptions:

- The material of the wire is homogeneous and isotropic.
- The model is developed for a single spark.
- The thermal properties like thermal conductivity are dependent on temperature.
- Transient analysis is considered.
- Heat flux is assumed to be Gaussian distributed [12].
- Joule heating and cross-vibration effects of the wire are neglected.
- Plasma flushing efficiency is considered to be 100%.

10.2.2 Governing Equation

Heat flow through the wire is governed by a two-dimensional transient heat conduction equation for the wire cross section.

$$\frac{\partial}{\partial x} \left(k \frac{\partial \theta}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial \theta}{\partial y} \right) = \rho c_p \frac{\partial \theta}{\partial t} \quad (10.1)$$

where $\theta = T - T_\infty$ (T is temperature and T_∞ is ambient temperature), k is thermal conductivity (W/mK), ρ is density (kg/m³), c_p is specific heat (J/kg K), and t is time (s).

10.2.3 Boundary Conditions

For the thermal modeling of the spark formation at the wire electrode in WEDM process, conduction is considered as the primary mode of heat transfer between the plasma channel and the electrodes.

Figure 10.1 shows the interaction between the plasma channel and the wire electrode during the discharge phenomenon. Figure 10.2 shows the cross-sectional view of the wire electrode considered for thermal analysis. The heat flux is distributed over the wire region A-B (zone 1). On the remaining part of the wire (zone 2), convection between the wire surface and the dielectric was considered as the boundary condition.

Mathematically, these boundary conditions can be described as follows:

For zone 1:

$$k \frac{\partial \theta}{\partial r} = Q(r) \quad \text{for } r = R_w \quad (10.2)$$

Fig. 10.1 Interaction between the discharge channel and wire electrode

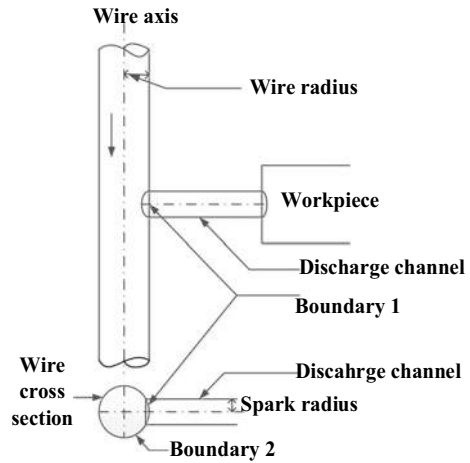
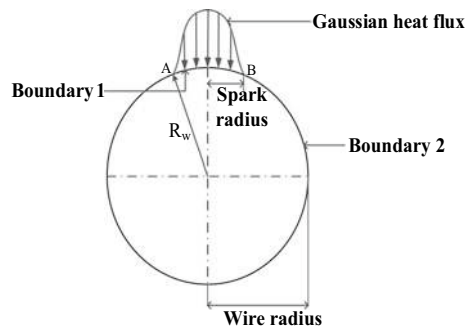


Fig. 10.2 Two-dimensional thermal model of the wire cross section



where $Q(r)$ is the heat flux applied and R_w is the radius of the wire.

For zone 2:

$$k \frac{\partial \theta}{\partial r} = h\theta \quad \text{for } r = R_w \quad (10.3)$$

where h is the convective heat transfer coefficient ($\text{W/m}^2 \text{ K}$).

The initial temperature of the wire at time $t = 0$ is assumed to be at room temperature of 300 K.

10.2.4 Spark Radius

The radius of the spark during discharge duration is a very important factor in developing a thermal model for wire EDM process. Many researchers have proposed different approaches for spark radius equations. Ikai and Hashiguchi [13] have derived a semi-empirical equation of spark radius termed as “equivalent heat input radius” which is a function of discharge current, I (A) and discharge on-time, t_{on} (μs) (Eq. 10.4).

$$R = (2.04 \times 10^{-3}) I^{0.43} t_{\text{on}}^{0.44} (\mu\text{m}) \quad (10.4)$$

In the present work, this approach has been used to calculate the spark radius as it gives more realistic results compared to other approaches.

10.2.5 Heat Flux on Wire Electrode

Different approaches of heat flux were used by a number of researchers in the numerical modeling of electrical discharge machining process. Research shows that most thermal models have considered uniformly distributed cylindrical heat flux for the spark. In this work, the approach of Gaussian distribution of heat flux as suggested by Joshi and Pande [14] is used. The Gaussian heat flux equation is:

$$Q(r) = \frac{4.57 F_c V I}{\pi R^2} \exp\left(-4.5 \frac{r^2}{R^2}\right) \quad (10.5)$$

where r is taken along x -axis, F_c is fraction of total EDM spark power going to the wire electrode (cathode); V is discharge voltage (V); I is discharge current (A) and R is spark radius. Energy distribution factor (F_c) is a very important factor in the heat flux equation because it gives the fraction of total energy absorbed by the wire electrode. The discharge energy is distributed between the wire electrode and the work piece, and the rest of the heat is carried away by the dielectric. Various values of F_c were proposed in the literature. DiBitonto et al. [15] gathered data over

a long period of time at different operating conditions and by tuning F_c , suggested that the energy distribution factor for cathode should be chosen as 0.183 for good correlation between the analytical and experimental results. In the present model, we have chosen this value of F_c to calculate the heat flux and to see its effect on the wire material removal as the wire acts as the cathode for a wire EDM process.

10.2.6 Convective Heat Transfer Coefficient

The heat transfer phenomenon in wire EDM process is quite complex; therefore, calculating the convective heat transfer coefficient (h) is quite difficult. Most of the researchers have assumed a constant value of h . The convective heat transfer coefficient is assumed to have a constant value of 10,000 W/m² K referring to Jennes et al. [2]. Banerjee et al. [3] have also used this value of h , and the results were quite satisfactory.

10.2.7 Solution Methodology

The main aim of this model is to predict the temperature field and the wire erosion rate in the cross section of the wire electrode during wire EDM process for a single discharge. The governing equation (Eq. 10.1) along with the boundary conditions was solved by finite element method (FEM) to compute the temperature profile at the end of a single discharge. The process model was solved by using ANSYSTM with Parametric Design Language (APDL), a FEM solver. A two-dimensional continuum of the wire cross section was considered for analysis. Material properties like thermal conductivity are assumed to be temperature dependent. For meshing, a thermal solid element with four nodes and single degree of freedom (temperature) at each node (PLANE 55) was used for this model. The mesh was refined at the spark location to get better results. The transient heat transfer problem was solved by applying the heat flux at the spark location. Convection heat transfer is applied at the remaining part of the wire periphery where heat flux is not applied.

In the present work, a two-dimensional geometry (cross section) of brass wire with a diameter of 0.25 mm is considered for the analysis. The properties of brass wire are listed in Table 10.1.

In the present model, we have considered the latent heat of melting of wire as it is a very important factor in the thermal modeling of EDM because latent heat signifies the consumption of the amount of supplied heat in the phase change of the work/tool material. The effective specific heat c_{peff} is calculated by using Eq. 10.6.

$$c_{\text{peff}} = c_p + \frac{L_H}{\Delta T} \quad (10.6)$$

Table 10.1 Properties of brass wire

Composition	65% Copper, 35% Zinc
Density (kg/m ³) (at 20 °C)	8470
Temperature-dependent thermal conductivity (W/mK)	50.208 (73 K), 125.52 (293 K), 142.256 (473 K)
Specific heat (J/kg K) (at 293 K)	380
Melting temperature (K)	1193
Latent heat of melting (KJ/kg)	168

Table 10.2 Process parameters levels

Factors	Level 1	Level 2	Level 3
Voltage (V)	20	30	40
Current (A)	16	24	32
Pulse on time (μs)	3.2	6.4	12.8

where c_p is the specific heat of the wire material, L_H is the latent heat of melting, and ΔT is the temperature difference between melting point temperature of wire material and room temperature. After the temperature profile is obtained, the elements that attain a temperature above the melting point of brass wire are eliminated from the model geometry to obtain the crater geometry. In the present model, three input parameters, viz. gap voltage, current, and pulse on time, are varied at different levels. The selected levels for each process parameters are shown in Table 10.2.

10.3 Results and Discussion

In order to investigate the effects of these process parameters on temperature and wire erosion, a total of $3^3 = 27$ experiments (3 factors varied at 3 levels) have been carried out. The values of peak temperature obtained and the crater area on the wire after a single discharge at different sets of process parameters are listed in Table 10.3. Figures 10.3 and 10.4 show the temperature profile obtained after a single discharge and the crater obtained on the wire cross section for a process condition: voltage (V) = 20 V, current (I) = 16 A, pulse on time (t_{on}) = 3.2 μs. The peak temperature achieved by the wire is around 8000 K. The crater depth predicted in this case is about 13 μm which is quite in agreement with the measured value of 19 μm as available in the literature [4]. The crater area obtained is 566.1 μm² after removing the elements from the model geometry that have attained the temperature above the melting point (1193 K) of wire material, i.e., brass. Figure 10.5 depicts the increase of temperature with the time and shows the peak temperature obtained by the wire at the end of the pulse duration.

Table 10.3 Values of peak temperatures and crater areas obtained at different process conditions

Voltage (V)	Current (A)	Pulse on time (μ s)	Peak temp (K)	Crater area (μ m ²)
20	16	3.2	7939.7	566.1
20	24	3.2	8894.21	693.7
20	32	3.2	9786.35	792.7
20	16	6.4	6346.18	911.1
20	24	6.4	7070.1	1121
20	32	6.4	7599.59	1298
20	16	12.8	4948.7	1416
20	24	12.8	5521.56	1779
20	32	12.8	5942.42	2082
30	16	3.2	12147.1	736
30	24	3.2	13530.3	873.7
30	32	3.2	14179.5	1005
30	16	6.4	9371.5	1212
30	24	6.4	10,457	1455
30	32	6.4	11244.8	1657
30	16	12.8	7273.11	1959
30	24	12.8	7936.63	2378
30	32	12.8	8762.62	2743
40	16	3.2	16091.5	858.7
40	24	3.2	17936.9	1007
40	32	3.2	18809.3	1163
40	16	6.4	12394.5	1438
40	24	6.4	13840.2	1702
40	32	6.4	14898.9	1923
40	16	12.8	9596.68	2372
40	24	12.8	10741.8	2840
40	32	12.8	11583.3	3227

10.3.1 Effect of Process Parameters on the Peak Temperature Obtained by the Wire

Figures 10.6, 10.7, and 10.8 show the effect of process parameters (voltage, current, and pulse on time) on the temperature achieved by the wire during a single discharge phenomenon.

The peak temperature obtained in the wire electrode after a single discharge increases with the increase in voltage and current. The reason behind this is due to the increase of power with the increase in voltage and current, and the temperature in the wire increases. However, the peak temperature obtained in the wire electrode

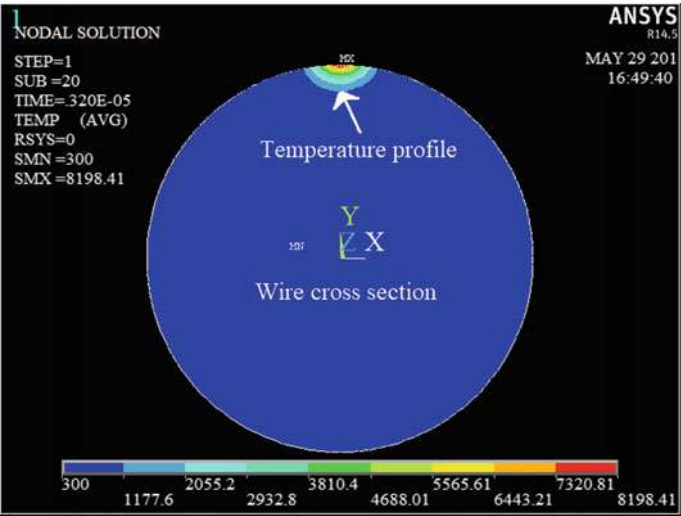


Fig. 10.3 Temperature profile for a single discharge ($V = 20\text{ V}$, $I = 16\text{ A}$, $t_{\text{on}} = 3.2\text{ }\mu\text{s}$)

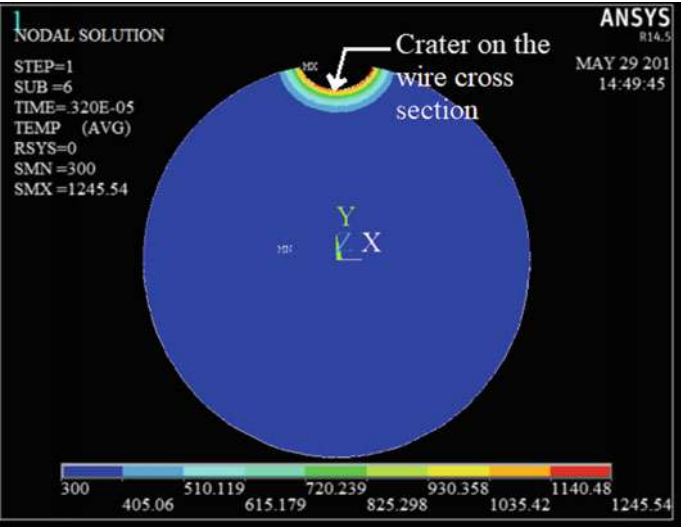


Fig. 10.4 Crater profile on the wire cross section ($V = 20\text{ V}$, $I = 16\text{ A}$, $t_{\text{on}} = 3.2\text{ }\mu\text{s}$)

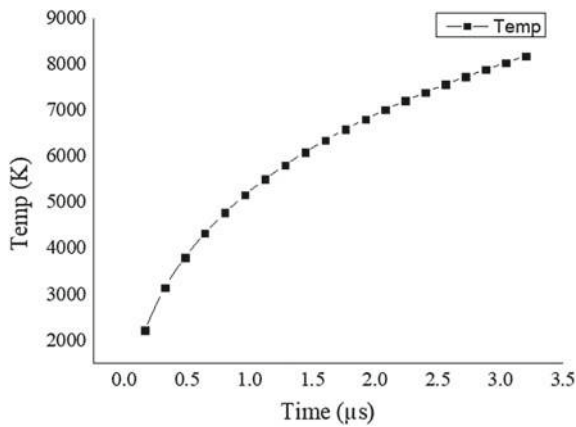


Fig. 10.5 Time versus temperature plot for a single discharge

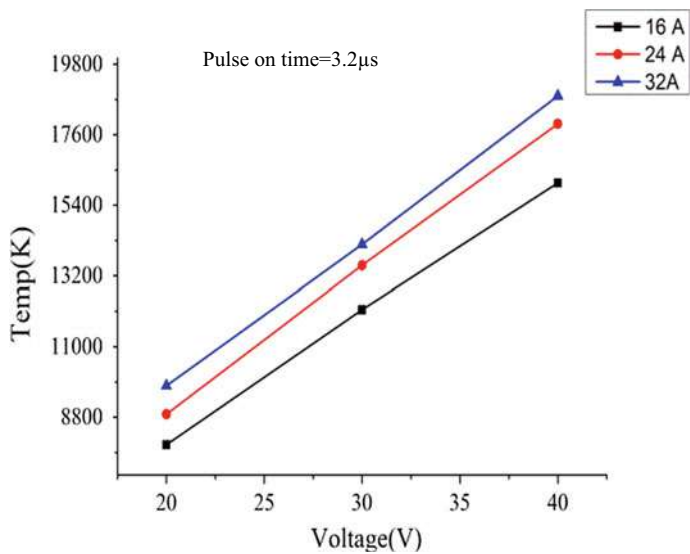


Fig. 10.6 Voltage versus temperature at constant pulse on time

after a single discharge decreases with increase in pulse on time. This may be because the spark radius over which the heat flux is applied increases due to increase in pulse on time; thus, the heat is dissipated in a larger area of the wire cross section, thus reducing the peak temperature obtained in the wire. These trends of variations were found to be in agreement with the available literature [3].

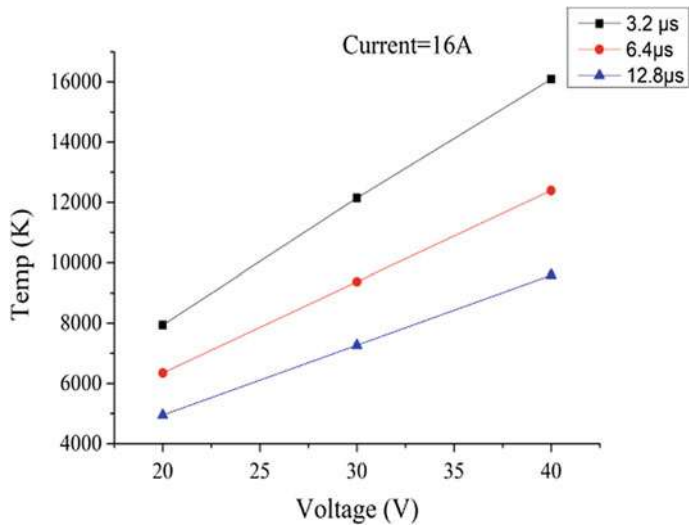


Fig. 10.7 Voltage versus temperature at constant current

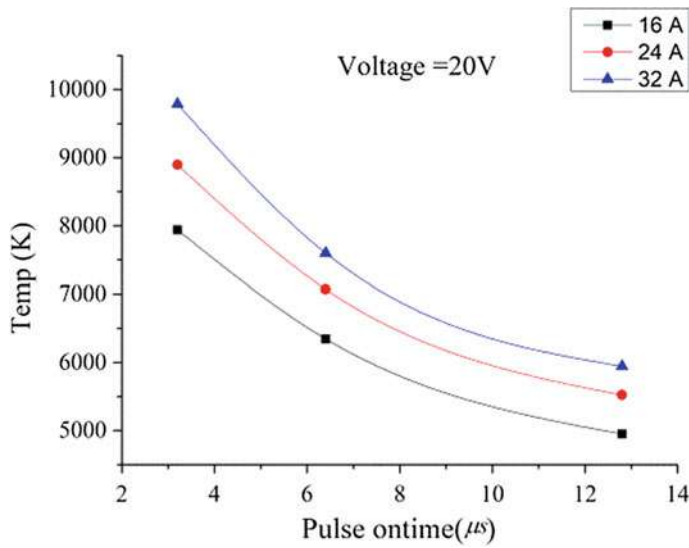


Fig. 10.8 Pulse on time versus temp at constant voltage

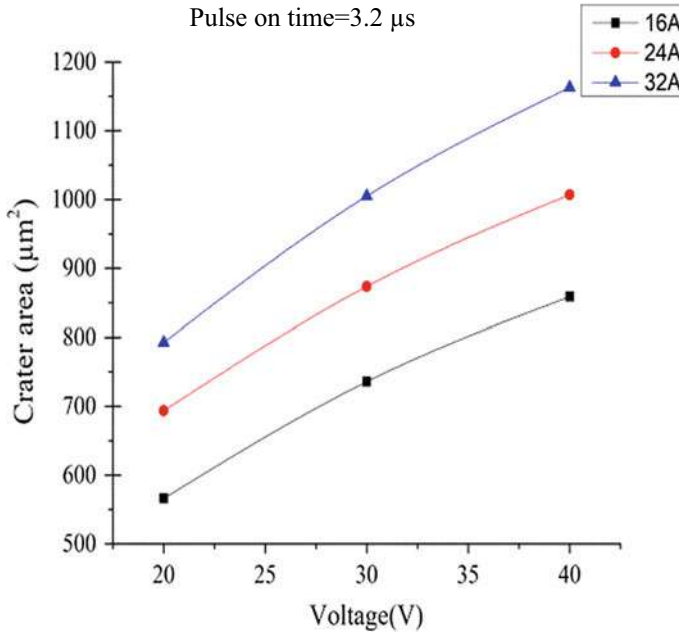


Fig. 10.9 Voltage versus crater area at constant pulse on time

10.3.2 Effect of Process Parameters on the Wire Erosion Rate

To study the wire erosion rate during machining, the crater area obtained in the wire electrode after a single discharge is plotted for varying levels of input parameters. Figures 10.9, 10.10, and 10.11 show the effect of these parameters on the crater area obtained on the wire cross section.

The crater area in the wire electrode after a single discharge phenomenon during WEDM machining increases with increase in voltage and current due to increase in power. The crater area after a single discharge in the wire electrode also increases with increase in pulse on time. This is because the crater volume and the width of heat-affected zone increases with input power and discharge duration [4]. Thus, the wire erosion rate increases with increasing values of voltage, current, and pulse on time, thus reducing the load bearing strength of the wire.

10.4 Conclusions

A two-dimensional nonlinear transient heat transfer model of the wire electrode is developed using finite element method. The model predicts the peak temperature obtained by the wire and the crater shape after a single discharge phenomenon at

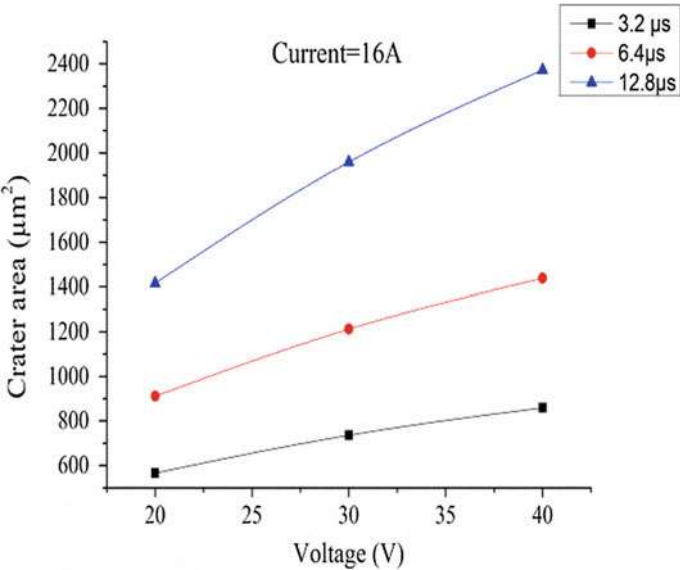


Fig. 10.10 Voltage versus crater area at constant current

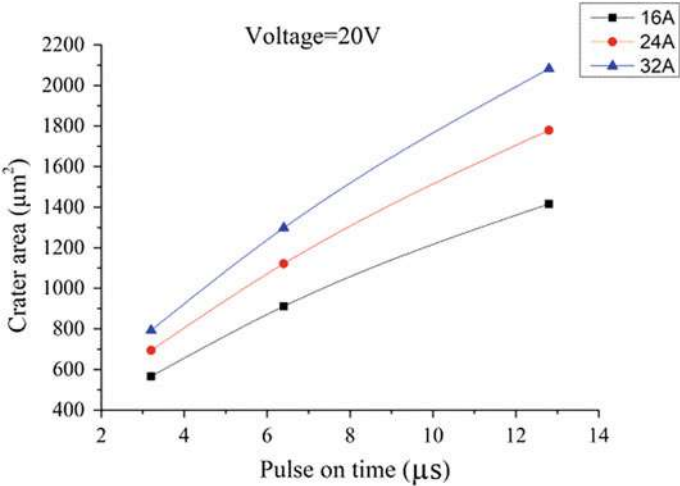


Fig. 10.11 Pulse on time versus crater area at constant voltage

different levels of input parameters. The crater area gives the amount of wire material eroded when the plasma flushing efficiency is considered as 100%. It is observed that the peak temperature and crater area on the wire cross section vary at varying levels of process parameters, viz. voltage, current, and pulse on time. Optimization of these input parameters is essential to predict and minimize wire rupture risk. The results of the present work are summarized as follows:

- The peak temperature achieved by the wire after a single discharge increases with voltage and current due to increase of input power. However, temperature decreases with increase in pulse on time.
- The crater area obtained on the wire increases with increasing values of voltage, current, and pulse on time, thus increasing the wire erosion rate, thereby reducing the load bearing strength of the wire.

The two-dimensional nonlinear transient thermal model developed in the present paper shall pave the way to estimate the total thermal load in the wire during machining and to evaluate the wire erosion rate at different input parameters, thus predicting wire breakage.

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Chapter 11

Optimal Vendor-Managed Inventory Models for Single-Vendor Multiple-Retailer Supply Chains



Narayan C. Nayak and Amar C. Mohanty

Abstract This research developed an analytical approach for a single-vendor multiple retailer's vendor managed inventory system. In this paper, various factors like lead time, probability of producing defective items when process goes out of control, shipment size, and safety stock have been considered to modify the models available in earlier literature. Numerical models have been developed that helps in optimizing the total cost that may incur due to various factors at both vendor and retailer levels. Analysis has been carried out for single-vendor and multiple (four) retailers. The total cost that has been optimized is the summation of total vendor cost and total retailer cost. The condition under which each approaches may be preferred has been thoroughly discussed. Significant factors that influence the total cost of the system have been identified by means of various plots.

Keywords Supply chain · Vendor-managed inventory · Retailer · Safety stock · Economic order quantity · Optimization

11.1 Introduction

A supply chain (SC) is the network involved in the creation and sale of a product, starting from delivery of source materials from the supplier to the manufacturer and finally delivering to the end user. While improving SC performance by adopting the plans and the objectives of individual enterprises, supply chain coordination (SCC) plays a very important role. In case of distribution settings, SCC focuses on inventory management as well as in decision making [11]. SC performance is improved by vendor-managed inventory (VMI). VMI, due to its versatile and negotiable nature,

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_11

is a very flexible tool that can be modified as and when it is required in a particular situation. Traditional ordering process is changed in VMI where collaboration between suppliers and their customers (e.g., distributor, retailer, or product end user) takes place. Walmart and many other large retailer houses found VMI as one of the successful business models. It prevents overflowing warehouses or shortages at manufacturing level as well as costly labor, purchasing, and accounting. Businesses maintain a proper inventory level in VMI, and optimized inventory leads to easy access and fast processing with reduced labor costs. It reduces inventory-related costs throughout the supply chain and keeps inventory level low. While shifting the responsibility of replenishing and managing inventory to vendors, it helps organizations to reduce the inventory-associated costs [14].

Good managerial ability for integrating and coordinating the intricate network of business relationships among supply chain partners leads to success of a firm. Enhancing the operational efficiency, profitability, and competitive position of a firm, and its supply chain partners are the main objectives of VMI. The partnership model of VMI between the supply chain partners enhances the profitability of each of the partners [13]. An important aspect in the supply chain management is to establish co-operation among the SC partners.

Yao et al. [15] developed an analytical model that explores the effect of SC parameters (ordering cost and carrying cost) on inventory cost savings that is to be realized from VMI. Collaborative initiatives of VMI realize the distribution of savings between buyers and suppliers affecting the cost. Almehdawe and Mantin [2] considered a supply chain where a single capacitated manufacturer and multiple retailers exist. Stackelberg game VMI framework modeled by them function under two scenarios: (i) traditional approach followed considering manufacturer as the leader and (ii) dominant player of the supply chain was fixed as one of the retailers. Pasandideh et al. [10] developed an EOQ model that was more applicable to real-world production and inventory control problems. The model considered several products in which shortages were backordered, there was an upper bound on the total number of orders, and the storage had limited capacity. Dye [5] considered a deterministic EOQ model with generalized demand, deterioration, and unit purchase cost functions under two levels of trade credit policy. Optimal values of selling prices, replenishment number, and replenishment scheme that maximize the total profit over the finite planning horizon have been found out. Ben-Daya et al. [4] developed a consignment (CS) and vendor-managed inventory policy for a single-vendor and multiple-buyer supply chain with known demand. Hariga et al. [7] considered a supply chain where retailers specify their maximum allowed inventory levels in which a single vendor manages multiple-retailer stocks under a VMI contract. Hoque [8] considered some realistic factors and developed two single-vendor multi-buyer integrated inventory models. The production flow is synchronized by transferring the lot with equal-and/or-unequal-sized batches (sub-lots), where the largest unequal-sized batch was considered equal to that of the equal-sized batches. Mateen and Chatterjee [9] developed analytical models for a single-vendor multiple-retailer system coordinated through VMI through various approaches. Belalia and Ghaiti [3] have given important managerial guidelines for VMI strategy in an autocorrelated demand environment. AIDurgam et al. [1] studied

a single-vendor-single-manufacturer joint economic lot size problem under stochastic demand. Sahu and Nayak [12] developed a new methodology for performance evaluation of SC through a case study at an aluminum industry. Gani and Dharik [6] developed a framework for a single-vendor multi-buyer supply chain considering a CS and VMI policy.

Based on the economic order quantity (which is the optimal order size obtained when minimized the retailer cost function), the supplier receives orders from the retailer, in case of decentralized supply chain (VMI does not exist). The case of VMI is different than that of decentralized type in the sense that inventory for the supplier will manage both entities where the supplier has full access to the buyer demand information. The supplier order quantity is assumed to be an integer multiple of the buyer in case of normal inventory model. In order to minimize system's total cost, the supplier has to decide on his optimal quantity, the buyer optimal quantity and the replenishment frequency.

This research provides some useful implications for organizations while adopting VMI. Extended mathematical models have been developed in this work, for single-vendor multi-retailer centralized VMI system. Models have been validated using the secondary data from Mateen and Chatterjee [9], and the inferences are listed out. Total cost has been calculated considering all the cost components in the system. Plots have been drawn for relating various important factors. Importance is given on the entire research process, and particularly, the concepts and theories upon which the research work carried out.

11.2 Problem Statement

In this research, a system with a single-vendor and multiple retailers has been considered. The vendor supplies the manufactured product to the retailer according to its demand. The demand faced by retailer is deterministic in nature. The EOQ model has been modified in this paper to show the effect of few other parameters like lead time, safety factors, and probability of producing defective products in case the process goes out of control. Even though these parameters have very less effect on the total cost of the system, still these factors cannot be completely ignored. Total cost at the vendor site and the retailer site has been analyzed through plots at both the vendor and retailer levels. The model developed in this research has been compared with existing standard models (derived for economic order quantity where parameters such as lead time, safety factors, and probability factor have not been considered).

11.3 Assumptions

The total production rate by the vendor assumed higher than the total demand faced by the retailers. Therefore, there is no backordering. That means the vendors will always

have certain amount of inventory during the shipment of products for a given length of time. Vendor has sufficient capacity to meet all the buyers' demand. Assumptions considered for developing models are as follows:

- Under the VMI model, vendor bears the ordering costs and is responsible for the replenishment decisions. Thus, their respective transportation cost and holding cost are borne by the retailer site. Further, it assumed that the ordering cost would remain the same even if it were borne by the vendor.
- The lead time is considered to be constant for each individual delivery of the products during a period of time.
- While producing part/component, the process may go out of control with a given probability each time and subsequently another unit/part is produced. The process is assumed to be in control during beginning of the process. Once system goes out of control, and process produces defective parts until corrective action is initiated.

11.4 Notations

In this paper, following notations have been used to record the centralized single-vendor multiple-retailer VMI system. The notations are used to describe all the records in all the three models.

P_s	Production setup cost (Rs. per setup)
O_i	Ordering cost for the i th retailer (Rs. per order; borne by the vendor under VMI)
H_i	Holding cost of the retailer (Rs. per unit per year)
H_v	Holding cost for the vendor (Rs. per unit per year)
T_i	Transportation cost from the vendor to the i th retailer (Rs. per order)
D	Total demand rate across all the retailers (units per year)
P	Production rate for the vendor (units per year)
R	Replenishment cycle for the i th retailer when operating independently (years)
q_i	Shipment size of j th retailer
P_r	Probability that the vendors production process can go out of control
L	Length of lead time for retailer
σ	Variance factor
n	Number of shipments
Q	Order quantity of the retailer
r_w	Vendors' unit rework cost per defective item
K	Safety factor
α_{ij}	Stock of inventories after fulfilling the demand D_{ij} by the i th retailer in j th shipment
r	Number of retailers

11.5 Model Development

Three VMI models have been developed capturing different assumptions under which the system can function, with different cost implication associated with each of them. Transportation cost, ordering and setup cost, and rework cost associated with each model are same. In all the three models, rework cost has been incorporated by considering its probability factor that the process can go out of control. Further, when the process goes out of control, the entire system produces defective items and for those defective items, vendor has to bear some rework cost.

Mathematical models for transportation cost, ordering and setup cost, and rework cost have been formulated, and the total cost has been calculated by adding different costs associated with the VMI system (transportation cost, ordering and setup cost, rework cost, vendor and retailer inventory handling cost, etc.).

Setup and Ordering Cost for the Vendor. Let R be the length of the optimal replenishment cycle for n shipments delivered to each retailer.

$$\text{The total setup and ordering cost for the vendor} = \frac{P_s + n \sum_{i=1}^r O_i}{R} \quad (11.1)$$

$$\text{Transportation cost} = \frac{n\beta \sum_{i=1 \rightarrow r} T_i}{R} \quad (11.2)$$

$$\text{Rework cost} = \frac{r_w n D Q P_r}{2} \quad (11.3)$$

Total Cost for VMI System. Total cost of the system = Setup and Ordering Cost + Vendor Inventory Handling Cost + Retailer Inventory Handling Cost + Transportation Cost + Rework Cost.

The vendor and retailer inventory handling cost have been formulated while considering the VMI mathematical models as given below.

11.5.1 VMI Model-1

In this model (Fig. 11.1), the vendor sends shipments to each retailer cyclically. Y-axis (dependent variable) shows the amount of inventory in this figure. As per the literature model [4], extra items produced by the supplier have been kept under retailers' inventory while supplying in equal shipment size during its every single delivery, irrespective of demand. Thus, no retailers may handle such large amount of inventory. Many of the retailers want to keep the stock as they need, and they do not want to spend extra for their unnecessary inventory. Thus, the literature model needs to be modified as there exist unequal shipments. The suppliers supply in equal replenishment time of the retailers' according to the consumers' demand. In the current research, before the inventory in retailer site is finished, the next shipment is

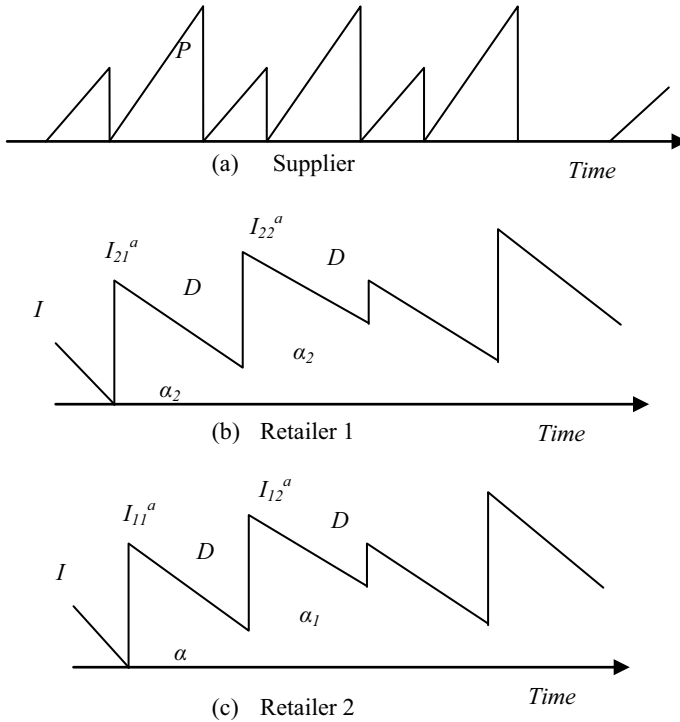


Fig. 11.1 Rectified inventory profile under VMI Model-1

given to the retailer. Hence, inventory in retailer's site increases after each shipment, and thus, safety stock is not required. While considering the above assumptions, empirical relations among its variables have been developed and are discussed below.

Vendor Inventory Handling Cost. The total area under the curve for the vendor corresponding to the single shipment to all the retailers (in time R/n) can be written as the sum of areas of all the triangles and can be written as:

$$\begin{aligned}
 \text{Vendor's inventory cost} &= \sum_{i=1}^r \left[\frac{1}{2} \left(\frac{D_i R}{n} \right) \left(\frac{D_i R}{nP} \right) \right] \\
 &= \frac{H_v}{\left(\frac{R}{n} \right)} \sum_{i=1}^r \left[\frac{1}{2} \left(\frac{D_i R}{n} \right) \left(\frac{D_i R}{nP} \right) \right] = \frac{H_v R}{2nP} \sum_{i=1}^r D_i^2 \quad (11.4a)
 \end{aligned}$$

Retailer Inventory Handling Cost. Let the inventory level of i th retailer after the receipt of j th shipment be denoted as I_{ij}^a and the stock of inventory after fulfillment of the demand D_{ij} from I_{ij}^a inventory may be denoted as α_{ij} . Now the inventory holding cost for r -retailers is the sum of area under the curve of retailer's inventory profile for n shipments which can be written as:

$$\begin{aligned}
\text{Area} &= L(I_{10}^a - \alpha_{10}) - \frac{1}{2}D_{10}L + L(I_{11}^a - \alpha_{11}) - \frac{1}{2}D_{11}L \\
&\quad + L(I_{12}^a - \alpha_{12}) - \frac{1}{2}D_{12}L + L(I_{20}^a - \alpha_{20}) - \frac{1}{2}D_{20}L \\
&\quad + L(I_{21}^a - \alpha_{21}) - \frac{1}{2}D_{21}L + L(I_{22}^a - \alpha_{22}) - \frac{1}{2}D_{22}L + \dots \\
&= L \left(\sum_{j=0}^n I_{ij}^a - \sum_{j=0}^n \alpha_{ij} \right) - \frac{1}{2}L \cdot \sum_{j=0}^n D_{ij}
\end{aligned}$$

Taking $\left(\sum_{j=0}^n I_{ij}^a\right)$ as the total production of vendor over R replenish time is $P * R$ (Here $R = 1$ for 1 year) and $\sum_{j=0 \rightarrow n}^{i=1 \rightarrow r} D_{ij}$ can be reduced to $\sum_{i=1 \rightarrow r} D_i$, where D_i is the total demand of the item faced by i th retailer. Now the area can be reduced to

$$L \left(p - \sum_{\substack{i=1 \rightarrow r \\ j=0 \rightarrow n}} \alpha_{ij} \right) - \frac{1}{2}L \cdot \sum_{i=1 \rightarrow r} D_i$$

Hence, the total holding cost for all the retailers over R replenish time is

$$L \times \left[\sum_{i=1 \rightarrow r} H_i \right] \left[\left(p - \sum_{\substack{i=1 \rightarrow r \\ j=0 \rightarrow n}} \alpha_{ij} \right) - \frac{1}{2} \sum_{i=1 \rightarrow r} D_i \right] \div R \quad (11.4b)$$

Total Cost for VMI System. Total cost of the system = Setup and Ordering Cost + Vendor Inventory Handling Cost + Retailer Inventory Handling Cost + Transportation Cost + Rework Cost.

11.5.2 VMI Model-2

In this model (Fig. 11.2), the vendor delivers product to the retailers in increasing sub-batch sizes, where y-axis shows about quantity. As per the literature model, proposed by Gani and Dharik [6] and Hariga et al. [7], the vendor sends shipment to the retailer when the existing stock of the retailer is finished. Therefore, risk of inventory stock-outs arises during the time of demand uncertainty. While modeling in this research, lead time is considered, and thus, some safety stock is maintained. In addition, probability factor has been considered here in order to mitigate the errors of total system cost.

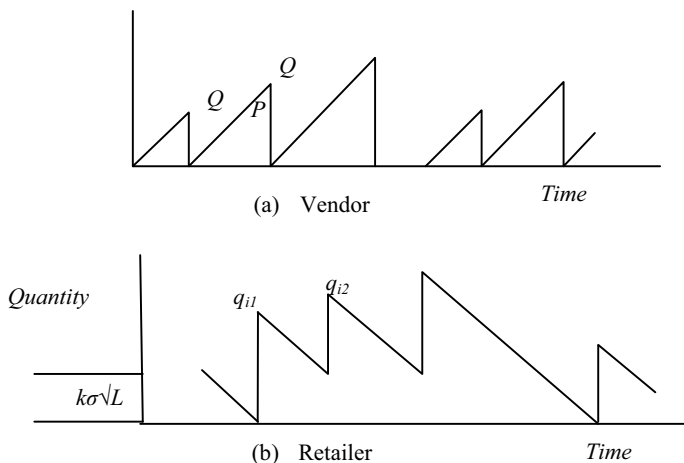


Fig. 11.2 Rectified inventory profile under VMI Model-2

Vendor Inventory Handling Cost. Since Time required to produce Q_{k+1} = Time to consume Q_k = Time to consume q_{ik}

$$\text{i.e., } \frac{Q_{k+1}}{P} = \frac{Q_k}{D} = \frac{q_{ik}}{D_i} \quad (11.5)$$

If $Q = DR$ is the total production batch size, we can write: $Q = Q_1 + Q_2 + Q_3 + Q_4 + \dots + Q_n$.

Also in this model,

$$Q_{k+1} = xQ_k \quad \text{and} \quad Q = Q_1 \left[\frac{x^n - 1}{x} \right] \quad (11.6)$$

$$\begin{aligned} \text{Average annual inventory cost at the vendor} &= \frac{D}{Q} \left[\frac{Q_1^2}{2P} + \frac{Q_2^2}{2P} + \frac{Q_3^2}{2P} + \dots + \frac{Q_n^2}{2P} \right] \\ &= \frac{H_v R D^2}{2P} \left[\frac{x^{2n} - 1}{x^2 - 1} \right] \left[\frac{x - 1}{x^n - 1} \right]^2 \end{aligned} \quad (11.7a)$$

Retailer Inventory Handling Cost. Average inventory for i th retailer will be the area under the inventory profile multiplied by (D/Q) .

$$\begin{aligned} \text{Average Inventory for } i\text{th retailer} &= \left\{ \frac{1}{2} \left[\frac{q_{i1}^2}{D_i} + \frac{q_{i2}^2}{D_i} + \dots + \frac{q_{in-1}^2}{D_i} \right] \right. \\ &\quad \left. + k\sigma\sqrt{L} \left[\frac{q_{i1}^2}{D_i} + \frac{q_{i2}^2}{D_i} + \dots + \frac{q_{in-1}^2}{D_i} \right] \right\} \end{aligned}$$

$$+ \frac{1}{2D_i} \left[q_{in} + k\sigma\sqrt{L} \right]^2 \left\} \frac{D}{Q}$$

Simplifying using Eqs. (11.5) and (11.6), we get area

$$\begin{aligned} &= \left\{ \frac{D_i Q^2}{2D} \left[\frac{x-1}{x^n-1} \right]^2 \left[\frac{x^{2n}-1}{x^2-1} \right] \right. \\ &\quad \left. + k\sigma\sqrt{L} \frac{Q}{D} \left[\frac{x-1}{x^n-1} \right] [1+x+x^2+\dots+x^{n-1}] \right. \\ &\quad \left. + \frac{1}{2D_i} \left[\frac{x^{n-1} Q D_i}{D} \left[\frac{x-1}{x^n-1} \right] + k\sigma\sqrt{L} \right]^2 \right\} \frac{D}{Q} \end{aligned}$$

Thus, total holding cost

$$\begin{aligned} &= \left\{ \frac{D_i Q^2}{2D} \left[\frac{x-1}{x^n-1} \right]^2 \left[\frac{x^{2n}-1}{x^2-1} \right] \right. \\ &\quad \left. + k\sigma\sqrt{L} \frac{Q}{D} \left[\frac{x-1}{x^n-1} \right] [1+x+x^2+\dots+x^{n-1}] \right. \\ &\quad \left. + \frac{1}{2D_i} \left[\frac{x^{n-1} Q D_i}{D} \left[\frac{x-1}{x^n-1} \right] + k\sigma\sqrt{L} \right]^2 \right\} \frac{D H_i}{Q} \end{aligned} \quad (11.7b)$$

Total Cost for VMI System. Total cost of the system = Setup and Ordering Cost + Vendor Inventory Handling Cost + Retailer Inventory Handling Cost + Transportation Cost + Rework Cost.

11.5.3 VMI Model-3

In this model (Fig. 11.3), it assumed that the vendor replenishes all the retailers at the same time. The delivery batch sizes synchronized in such a way that the time taken to fully consume the delivered sub-batch is same for all the retailers. In order to ensure this, the batch sizes have to be allocated in the ratio of the demand rate faced by the respective retailers. The said model has been adapted from Mateen and Chatterjee [9] and considered it for single-vendor multi-retailer supply chain. Moreover, probability factor has been considered in our model to make the system cost error free. Since the retailers already have high inventory with them, safety stock has not been taken into account.

In this case, a similar approach for VMI models has been taken as available in the literature; the following relations are obtained.

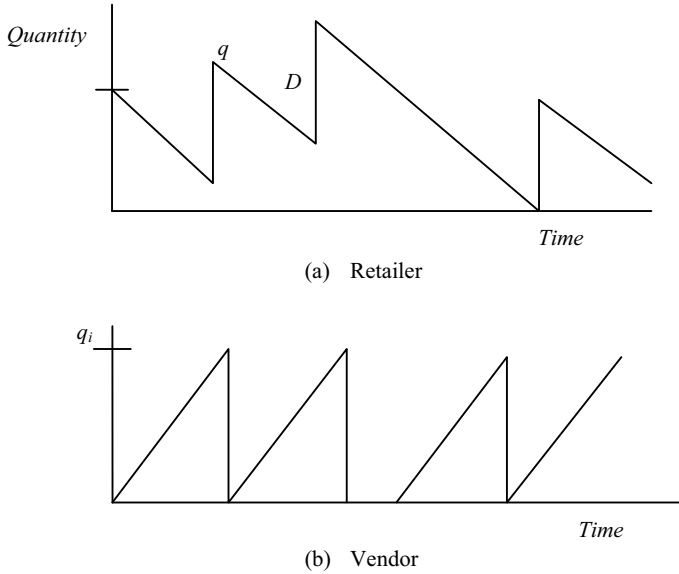


Fig. 11.3 Inventory profile under VMI Model-3

Vendor Inventory Handling Cost. The area under the inventory curve (Fig. 11.3b) = $\frac{nQ}{2} \left(\frac{Q}{P} \right)$.

Since $nQ = DT$,

$$\text{Total holding cost incurred by the vendor} = \frac{H_v R D^2}{2nP} \quad (11.8a)$$

Retailer Inventory Handling Cost. Let the production batch size is $Q (= \sum_{i=1}^r q_i)$, where q_i is the quantity delivered to the i th retailer. While the i th retailer consumed the first shipment, the second sub-batch arrives at $\left(\frac{Q}{P} \right)$. Therefore, its consumption will not be started till the first batch is consumed at $\left(\frac{q_i}{D_i} \right)$. In the mean time, third lot would arrive at $2 \left(\frac{Q}{P} \right)$. Thus, area under the inventory curve can be written as (Fig. 11.3a):

$$= n \left(\frac{1}{2} q_i \frac{q_i}{D_i} \right) + \left(\frac{q_i}{D_i} - \frac{Q}{P} \right) q_i (1 + 2 + 3 + \dots + n - 1)$$

Since $R = \frac{nQ}{D}$ and $q_i = \frac{D_i Q}{D}$

$$\text{The average inventory cost per year} = \sum_{i=1}^r \frac{D_i H_i}{2} \left[\frac{R}{n} + (n-1) \left(\frac{1}{D} - \frac{1}{P} \right) \frac{DR}{n} \right] \quad (11.8b)$$

Total Cost for VMI System. Total cost of the system = Setup and Ordering Cost + Vendor Inventory Handling Cost + Retailer Inventory Handling Cost + Transportation Cost + Rework Cost.

11.6 Numerical Analysis

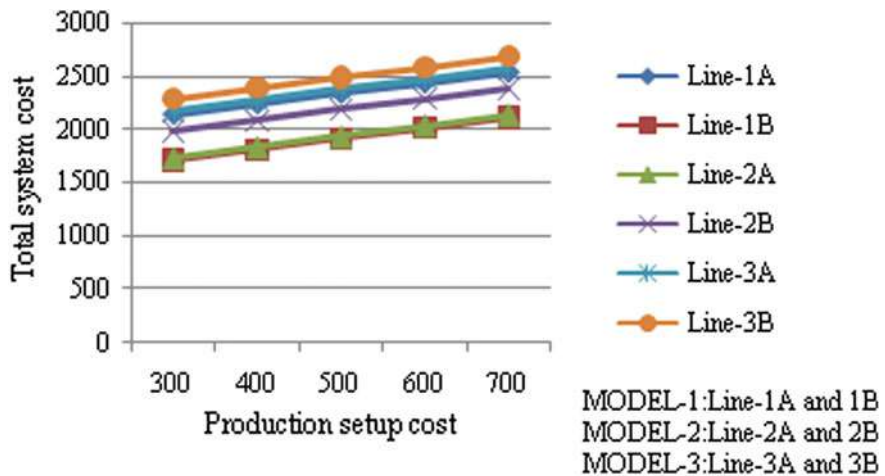
A numerical example from previous literature (Table 11.1) has been taken for in depth understanding of how various cost components affect the total benefits, which can be derived from a VMI relationship. In this connection, a system with one vendor and four retailers has been considered.

11.6.1 Impact of Change in Production Setup Cost

The change in the total system cost increases with an increase in the production setup cost (Fig. 11.4). It is to be noted that in case of Model-1 (Line-1B) in which vendor shipment size is dependent of retailers' demand and all retailers place order independently. It has also been observed that with the increase of production setup cost, the total system cost is less, with unequal shipment size in comparison to the VMI model with equal shipment size. The system cost is least in case of modified VMI system of Model-1 (Line-1B) and highest in modified VMI Model-3 (Line-3B).

Table 11.1 Data for the VMI system

$L = 15$ days $= 0.041$ yr	Retailer 1	Retailer 2	Retailer 3	Retailer 4
$P = 1500$	$O_1 = 13$	$O_2 = 15$	$O_3 = 10$	$O_4 = 12$
$R = 1$ year				
$n = 4$	$I_1^a = 200$	$I_2^a = 200$	$I_3^a = 120$	$I_4^a = 120$
$H_v = 4$	$H_1 = 6$	$H_2 = 7$	$H_3 = 7$	$H_4 = 5$
$P_s = 500$				
$Q = 50$	$T_1 = 18$	$T_2 = 12$	$T_3 = 14$	$T_4 = 12$
$\beta = 0.6$				
$r_w = 15$	$D_1 = 200$	$D_2 = 80$	$D_3 = 120$	$D_4 = 350$
$\sigma = 52, K = 1.51$	$\alpha_1 = 0$	$\alpha_2 = 120$	$\alpha_3 = 120$	$\alpha_4 = 0$
$P_r = 0.0001$				



1A, 2A, 3A presents corresponding literature models
1B, 2B, 3B presents corresponding models developed in the literature

Fig. 11.4 Effect of change in production setup cost on total system cost

In the above situation, delivery batch sizes are synchronized in such a way that the time taken to fully consume the delivered sub-batch is same for all the retailers.

11.6.2 Impact of Change in Vendor Holding Cost

The change in total system cost increases with the change in the vendor holding cost (Fig. 11.5). The total system cost for the modified Model-1 (Line-1B) found to be lower than those obtained from the rest of the models. It has been observed that initially total system cost in literature Model-2 (Line-2A, in which retailer replenish time is unequal with in hand safety stock) gives lower cost as compared to all VMI models.

11.6.3 Impact of Change in Production Rate

As shown in Fig. 11.6, the total system cost across all the models increases with an increase in the vendor production rate. Further, the total system cost is lowest in modified Model-1 (Line-1B), and it has lowest production setup cost and vendor holding cost.

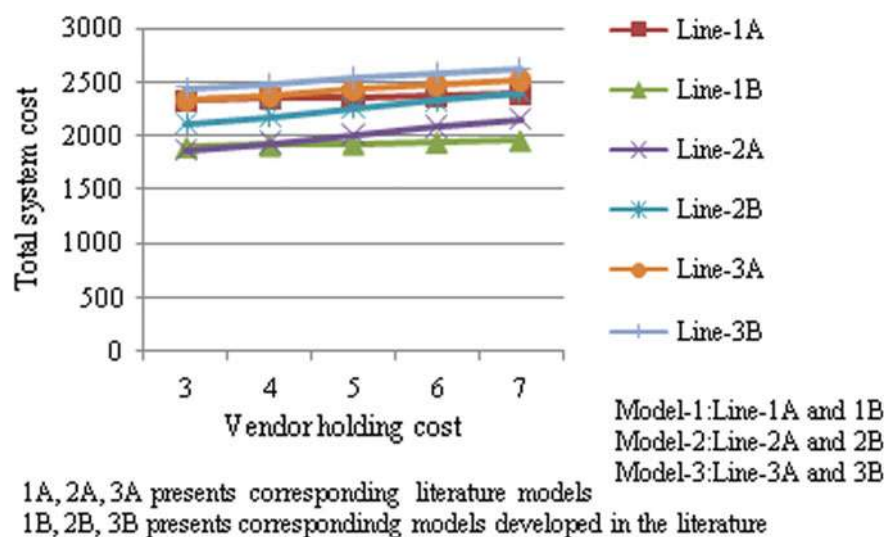


Fig. 11.5 Effect of change in vendor holding cost on total system cost

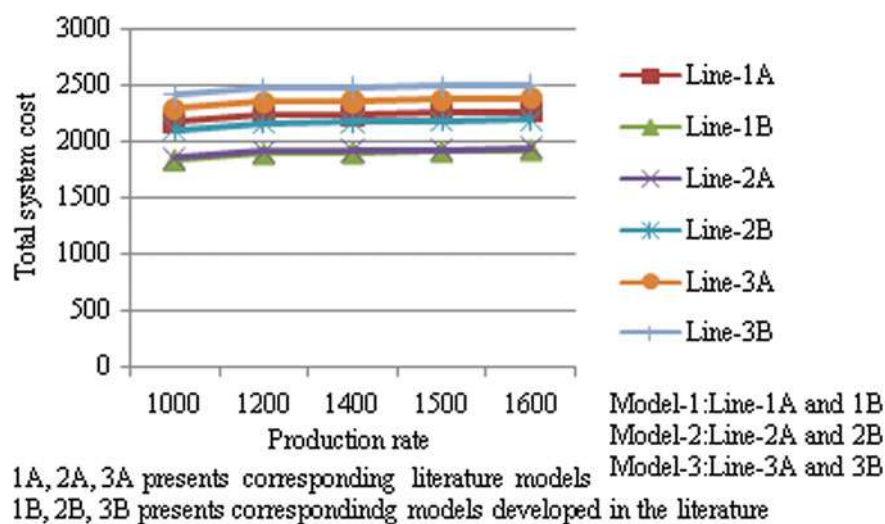


Fig. 11.6 Effect of change in production rate on total system cost

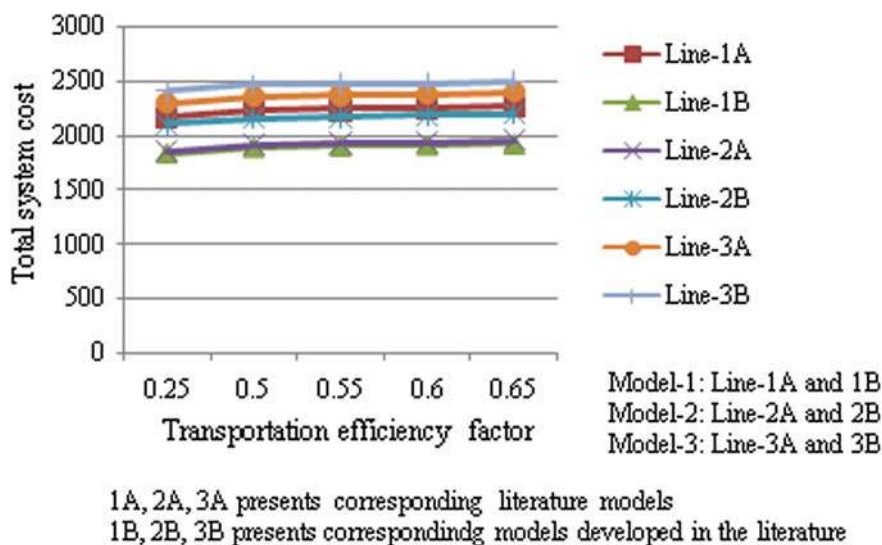


Fig. 11.7 Effect of change in transportation efficiency on total system cost

11.6.4 Impact of Change in Transportation Efficiency Factor

The impact of change in transportation efficiency upon the total system cost is shown in Fig. 11.7. As the transportation efficiency factor decreases, the total system cost goes down. Therefore, by grouping together the retailer deliveries, more benefits can be derived. For the same value of β , pattern of literature Model-2 (Line-2A) and modified Model-1 (Line-1B) is same. The impact upon the above factor becomes more pronounced as the transportation cost increase.

11.7 Discussions

In supply chain management, there exists a shared understanding of the roles of different supply chain partners. The proposed partnership between the supply chain partners enhances the profitability of each of the partners. Establishing co-operation among the SC partners is important in the supply chain management.

Numerical analysis has been carried out to find out the relationships between various important parameters. Total cost in VMI system increases due to the factors like production rate, vendor holding cost, and transportation efficiency. Further, the above factors are influenced by shipment size, safety stock, lead time, and the probability of process going out of control. Selecting the right replenishment policy for the SC system also plays a vital role in the supply chain management. It is also observed that the optimization of transportation cost provides large benefit to the VMI system.

It is also worth mentioning that if lead time increases, then the effect on the total cost increases. Hence, it may be noted that in order to increase the profit, the vendor should try to keep the lead time less (i.e., the time between the placement of orders and delivery of the products should be minimized as far as possible). Therefore, the process of supply chain would be more effective.

As far as the safety factor is concerned, with the increase in factor of safety, the overall cost of VMI system increases slightly. It depends upon the vendor to take optimal value of factor of safety for the product, keeping in mind the failure of the products.

Sometimes, it may so happen that the production process goes out control after functioning a long span of time. The above leads to the production of defective products which may incur losses to the entire VMI system. The period of operation of any particular machine should optimally specify to avoid the situation of getting defective products, which may lead to increase the total cost of VMI system.

11.8 Conclusions

The objectives of this research work, in continuation with previous work, have been successfully achieved. The development and analysis of various parameters in three different optimal models for cost reduction and error findings in the VMI system with single-vendor and multiple retailers are carried out. This research adds various approaches to SC coordination through VMI. This research considered transportation cost and rework cost along with safety stock while modeling. Model-1 has been modified into unequal shipment size to avoid extra inventory handling cost, thus significantly affecting the performance of VMI model.

Various factors such as production setup cost, holding cost, production rate, and transportation efficiency factor which influence the total cost of the system have been analyzed graphically. From this analytical research work, it has been found that changing the shipment size and lead time of the retailers can reduce the total system cost. Further, it is very important to regulate replenish time and shipment size. We have carefully utilized the standard economic order quantity model for this purpose as it can be seen in the developed models. In this project work, rework cost, probability of the process goes out of control, and safety stocks have been added to make our model more accurate than its literature model. However, it has been seen that by adding these factors, there is minor increment of total system cost that can affect neither the vendor nor the retailer to bear that increased cost. From this, we found that the proposed models are more beneficial than that of literature VMI models. The condition under which each approaches may be preferred has been discussed. The factors that influence the total cost of the system have been identified by means of various plots.

In this research, it has been assumed that all the buyers' demand will be met by the vendor those having sufficient capacity. Thus, investigation may be carried out where the vendor has scarce capacity resources (the issues of capacity allocation

and demand shortages may also to be considered). The sequencing of the shipment may be considered as another research issue. The above situations may be studied in a stochastic demand environment also. The functioning of the system may be affected due to vendor's capacity flexibility and thus needs to be investigated. It may be crucial in determining the actual benefits considering relative bargaining power of the parties involved in the supply chain. All the above issues may be taken up in detail in future.

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Chapter 12

Simulation of Torsional–Axial Chatter Vibrations in Indexable Drilling for Noise Generated



Pavan Joshi , Mahesh Todkar , B. S. Suresh and Ravi Halasur

Abstract Nowadays, indexable drills are the most commonly used drills for short hole-making operations because of their high performance and economic usage. Although there is much advancement in the indexable drill designs, they make high-pitched noise during the drilling operation which is unpleasant, cause poor surface finish, and may damage the tool too. Regenerative chatter vibrations are the main cause for this high-pitched noise, and a study conducted showed that the coupling between the axial and angular deflections in these drills causes the chatter. Thus, a numerical simulation model of the chatter occurring in these kinds of drills is done by considering torsional–axial vibrations. The model is used to predict the dominant frequency of the drill in which it is working, deflections of the drills under vibrations, and the variation in the forces because of these vibrations. Finally, the spectrum obtained from the noise generated, during the drilling operation, is compared with the spectrum obtained from the simulation, which shows that the numerical simulation is giving the agreeable results.

Keywords Indexable drills · Regenerative chatter · Torsional–axial vibrations · Cutting forces · Time domain

12.1 Introduction

Hole-making process raised its level to a higher degree with the introduction of indexable drills. These contain indexable inserts, separate flute designs for central and peripheral inserts for better chip evacuation. With these advancements, also there is high-pitched noise during operation, which is troublesome and causes chatter marks on the surface and lowering the life of tool. Thus, a study needs to be carried out on vibrations causing this noise in indexable drill.

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_12

Vibrations in machine tool can be classified in many numbers of ways. As observed in the machine tool system, vibrations can be classified as free vibrations, forced vibrations, and self-excited vibrations [1].

Free and forced vibrations can be isolated by using the anti-vibration mounts. The self-excited vibrations which are caused because of the mechanism itself affect the drilling operation and cause the chatter marks. The concept of helical flute design helps in better chip evacuation in drilling. These flutes cause coupling between angular and axial deformations in the drill structure. The modal analysis shows that the third mode shape of the drill is the torsional–axial mode which causes the coupling. Torsional–axial coupling causes the change in the force which in turn changes the chip thickness, thus again causes the change in force. Hence making a cycle of change in cutting force and chip thickness, this is regenerative in nature [2]. These regenerative vibrations cause the high-pitched noise in the current design of the indexable drills. This paper attempts to simulate such noise generated in drilling operation due to torsional–axial vibrations in indexable drills.

Mathematical modeling of such vibrations will give a better understanding of the chatter and noise in drilling operation. Some investigations have been carried out in this regard, by considering the geometrically defined cutting edges. The mechanics between tool edge and the work are modeled to predict the friction and normal forces. After that, a combined static and dynamic chip thickness is modeled as a function of tool geometry. Cutting force is generated by considering orientation of cutting edges and the kinematics of the operation which are applied to structural dynamics of machine tool and chatter stability and are predicted by simulation in the time domain [3, 4].

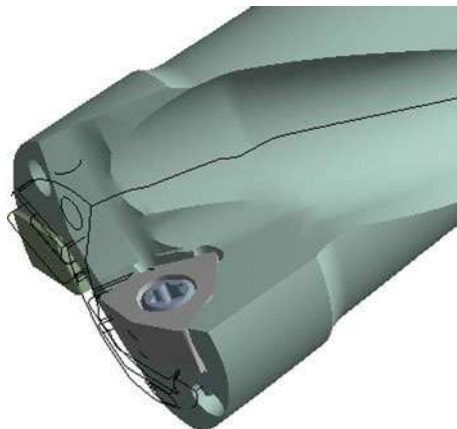
Most of the models consider the position of the cutting edges to predict the chatter occurring and show that the drilling operation is mostly affected by torsional–axial vibrations, and in the current indexable drill models, the dominant frequency is slightly below the natural frequency of the drill body [2]. During operation, sometimes the drill may rotate in backward direction due to torsional vibrations. Consideration of this in the model gives the better prediction of the regenerative chatter vibrations in the simulation model [5, 6].

Time is the most fundamental thing in any of the machining industry whether its small scale or large scale. Thus in this paper, a simulation model is built using the MATLAB code so as to reduce the consumption of time for analysis; here, the variations in the forces because of the backward rotations are considered and noise generated due to torsional–axial deflections is simulated.

12.2 Mathematical Model

In this work, indexable drill considered has two inserts: one central insert which cuts the center part of the hole and the other peripheral insert which cuts the remaining part of the hole. Because of the asymmetries in the helical tool path for the chip evacuation in the central and peripheral insert, chip loads cause the vibrations in

Fig. 12.1 Deformation of the drill in the torsional–axial mode



the axial–torsional direction during operation. Because of these fluctuations, the cutting forces vary which in turn vary the chip thickness. Thus, this phenomenon is regenerative. Based on this, a simulation model is proposed consisting of the following main steps

- Force equation in terms of chip thickness is found by linear regression.
- System parameters: mass, stiffness, and damping constant are found from FE analysis using ANSYS.
- Simulation is done by considering the above parameters.

Effects caused by the machine tool, workpiece, and bending of the tool are not considered here, as the drill operation is highly dominated by the torsional–axial vibrations. This causes the deformation in the angular and axial direction of the drill axis. As the bending is neglected, it is assumed that drill has two degrees of freedom: one is in the axial direction of the drill axis and other in the angular direction of the drill axis. Figure 12.1 is showing the deformation of the drill in torsional–axial mode.

12.2.1 Formulation of Force Equations

Determination of force equation in terms of chip thickness requires force model, from which axial force and torque are determined for different feed rates. A linear relation between forces and feed rate is assumed, and a regression of the force and feed rate is developed to get the force equations in terms of the feed rate.

An indexable drill, whose specification given in Table 12.1, is considered for determination of forces and torque. From the force model for a cutting speed of 200 m/min at feed rates ranging from f_1 mm/rev to f_5 mm/rev, force and torque are determined. Then the regression plots are developed as shown in Figs. 12.2 and 12.3.

Table 12.1 Product details

Drill body	DFSP180R3WB25M
Center insert	SPG060304RXX
Peripheral insert	DFT030304XX

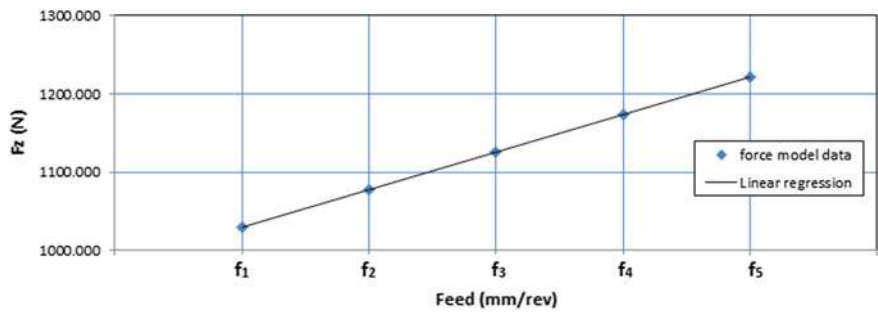


Fig. 12.2 Linear regression of axial force

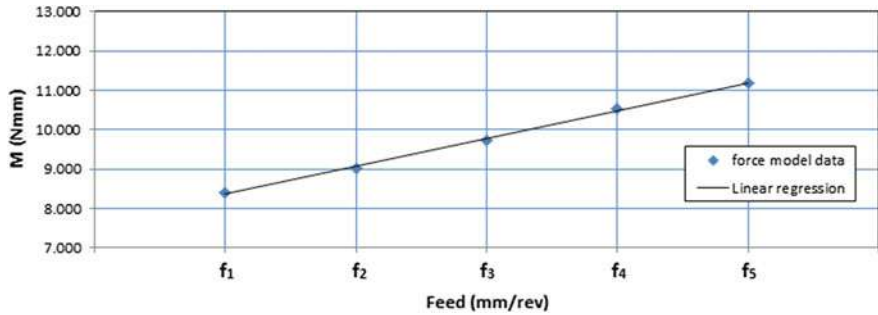


Fig. 12.3 Linear regression of the torque

Figures 12.2 and 12.3 show linear dependency between feed rate and the forces. From these, the following equations of force and torque can be written as in Eqs. (12.1) and (12.2).

$$F_z = 4798 f_n + 645.77 \tag{12.1}$$

$$M_z = 70.532 f_n + 2.7122 \tag{12.2}$$

Here F_z is an axial force in N, M_z is the torque in Nm, and f_n is the feed rate in mm/rev. In this work, it is assumed that the feed rate is equal to chip thickness so that the force and torque can be determined from the two equations for the entire time domain. Thus, the two equations can be written in terms of the chip thickness as follows:

$$F_z = 4798 h + 645.77 \quad (12.3)$$

$$M_z = 70.532 h + 2.7122 \quad (12.4)$$

12.2.2 Determination of Modal Parameters

For simulation, the mathematical modeling of the system needs to be identified. For this system, equation of motion can be written as [6],

$$m \frac{d^2 u}{dt^2} + c \frac{du}{dt} + ku = F \quad (12.5)$$

where m , c , and k are modal parameters of the system, F is the force acting on the system, and u is the response of the system. As two degrees of freedom are considered, the loading and deflection relationship can be described by 4 frequency response functions as in Table 12.2.

Here, F_z is the axial force, M_z is the torque, $H_\theta M_z$, $H_\theta F_z$, $H_z M_z$, and $H_z F_z$ are frequency response functions corresponding to the loading conditions in the corresponding direction. For example, in $H_z F_z$ shows ' F_z ' the axial force is applied and the ' Z ' response is observed in the axial direction. These frequency response functions are found from FE analysis using ANSYS. The material properties required for determining these FRF's are given in Table 12.3.

Initially, modal analysis is performed on the model to get the exact range of the dominant frequency. In this case, the dominant mode is torsional–axial mode. Thus, harmonic analysis is performed by concentrating on this mode and the frequency response functions are obtained. From these FRFs, the dynamic parameters are determined. The dynamic parameters are as shown in Table 12.4.

Table 12.2 List of FRF's

Load	Axial motion	Angular motion
F_z	$H_z F_z$	$H_\theta F_z$
M_z	$H_z M_z$	$H_\theta M_z$

Table 12.3 Material properties

Material	Young's modulus	Poisson's ratio	Density
Tool steel	207 GPa	0.27	7810 kg/m ³

Table 12.4 Dynamic parameters

Modal parameters (FRF)	Mass	Damping	Stiffness
$H_z F_z$	$-8.1 \times 10^{-2} \text{ kg}$	$-1.56 \times 10^2 \text{ N/s}$	$-2.58 \times 10^8 \text{ N/m}$
$H_z M_z$	$9.61 \times 10^{-5} \text{ kg m}$	$1.84 \times 10^{-1} \text{ Nm/s}$	$3.05 \times 10^5 \text{ Nm/m}$
$H_\theta F_z$	$9.61 \times 10^{-5} \text{ kg m/rad}$	$1.84 \times 10^{-1} \text{ N s/rad}$	$3.05 \times 10^5 \text{ N/rad}$
$H_\theta M_z$	$-1.3300 \times 10^{-7} \text{ kg m/rad}$	$-2.5500 \times 10^{-4} \text{ Nm s/rad}$	$-4.22 \times 10^2 \text{ Nm/rad}$

12.2.3 Simulation Model

Now a simulation model is developed for vibrations caused by the regenerative chatter vibrations. While drilling the drill rotates and moves in the axial direction corresponding to the workpiece [6]. These angular and rigid-body motions of the drill are given by Eqs. (12.6) and (12.7).

$$\Delta \theta^{\text{rigid}} = \omega \cdot \Delta t \quad (12.6)$$

$$\Delta Z^{\text{rigid}} = \frac{f_n}{2\pi} \cdot \Delta \theta^{\text{rigid}} \quad (12.7)$$

Here, $\Delta \theta^{\text{rigid}}$ is the rigid angular motion, ω is the angular speed in rad/s, Δt is the discrete-time step chosen in such a way that $1/\Delta t$ should be as high as 21 times the natural frequency of the dominant mode. ΔZ^{rigid} is the rigid axial motion of the drill in mm, and f_n is the feed rate in mm/rev. The angular speed is determined by Eqs. (12.8) and (12.9).

$$\omega = \frac{2\pi N}{60} \quad (12.8)$$

$$N = \frac{Vc}{\pi D} \quad (12.9)$$

Vc is the cutting speed in m/min, D is the diameter of the drill in m, and N is the rotational speed of spindle. For every time step, the deflections are calculated by solving the mathematical model equation and these deflections are used to obtain the axial and angular position of the drill at n th time step as in Eq. (12.10),

$$\begin{pmatrix} Z \\ \theta \end{pmatrix} = \begin{pmatrix} n \Delta z^{\text{rigid}} + \sum_{j=1}^2 u_{1j} \\ n \Delta \theta^{\text{rigid}} + \sum_{j=1}^2 u_{2j} \end{pmatrix} \quad (12.10)$$

Simulation is done by considering chip thickness equal to feed rate initially [5]. After that, the chip thickness is calculated after each iteration and it is used to obtain the forces from the regression force Eqs. (12.3) and (12.4). These forces are loaded on to the dynamic system to obtain the deflections. The responses in terms of the deflections are calculated by Euler method. These responses are added to the rigid body motion of the drill so as to obtain the new position of the cutting edges. Here rigid body motions are motions caused by spindle rotation and feed, in axial and angular directions. The new position of the edges is compared with the previous cuts to obtain the new chip thickness. The new chip thickness is obtained from the following equation after each iteration [6].

$$h(n) = Z(n) - \max Z |(\theta = \theta_n \text{ to } \theta_{(n-360)})| \quad (12.11)$$

Here, $h(n)$ is new chip thickness at n iteration, $Z(n)$ is current position of the cutting edge, θ_n is current angular position, $\theta(n - 360)$ is angular position in the previous revolution. Here the current position of the drill is subtracted from the previous maximum position of the drill during the previous revolution ($\theta = \theta_n$ to $\theta(n - 360)$). The chip thickness is used to calculate the forces and torque. This process continues until the end of the simulation.

Angular vibrations cause the backward rotation of the drill. The cumulative relative rotation of the drill depends on the rigid body motions and the angular (torsional) deflections. In some time intervals, the amount of backward rotation is more than the forward rotation of the drill. This causes the drill to rotate in the backward direction; during this backward rotation, the drill edge does not cut the material, and it just rubs the cut surface [5]. Thus, during these conditions variation of forces need to be simulated.

The assumption made is that the cutting edge touches the surface by its flank face in the backward rotation and actually does not cut the material. As the forces are dependent on the chip thickness, in this case, it is not contributing to the force generation. The first term in the force equation is not contributing during these kinds of rotations. From these, the following approximations can be made to the force equations.

$$F_z = \begin{cases} 4798(h) + 645.77, & h(n) > 0 \text{ and } \Delta\theta' + \omega \geq 0 \\ 645.77, & h(n) > 0 \text{ and } \Delta\theta' + \omega < 0 \\ 0, & h(n) \leq 0 \end{cases} \quad (12.12)$$

$$M_z = \begin{cases} 70.53(h) + 2.7122, & h(n) > 0 \text{ and } \Delta\theta' + \omega \geq 0 \\ 2.7122, & h(n) > 0 \text{ and } \Delta\theta' + \omega < 0 \\ 0, & h(n) \leq 0 \end{cases} \quad (12.13)$$

In this case when drill rotates backward, the flank face can only rub the recently cut surface, which causes torque in opposite direction of the rotation. The torque is calculated by Eq. (12.4) considering the chip thickness to be equal to zero at that moment. Therefore, a constant torque of 2.7122 Nm can be observed. When the

drill is rotating backward, thrust force is assumed to be obtained from Eq. (12.3) considering chip thickness to be equal to zero. Thus, a constant force of 645.77 N is observed.

For determination of the forced responses from the mathematical model, Euler integration approach is used which is an iterative process for solving the second-order differential equations,

Equation (12.3) can be rewritten as Eq. (12.14),

$$\frac{d^2u}{dt^2} = \frac{F - c \frac{du}{dt} - ku}{m} \quad (12.14)$$

Initially considering $u = 0$ and $\frac{du}{dt} = 0$, and discrete-time step (Δt) is chosen. From Eq. (12.14), $\frac{d^2u}{dt^2}$ is found, which is used for further iterations. New velocity of the system and position for the next time step is found from Eqs. (12.15) and (12.16).

$$\frac{du}{dt}(\text{new}) = \frac{du}{dt}(\text{old}) + \frac{d^2u}{dt^2}(\Delta t) \quad (12.15)$$

$$u(\text{new}) = u(\text{old}) + \frac{du}{dt}(\text{new})(\Delta t) \quad (12.16)$$

These equations are used in the chatter simulation for determining the new position and velocities. The simulation flow is as shown in Fig. 12.4.

12.3 Simulation Results and Experimental Validation

Using the above model a simulation¹ is done by considering the feed rate of 0.08 mm/rev at a cutting speed of 200 m/min on the same drill as described earlier. Deflections in axial and angular direction are simulated which are as shown in Figs. 12.5 and 12.6. Effects of the backward rotations on force and torque can be observed in Figs. 12.7 and 12.8.

Experimentation is carried out by considering the same parameters, the feed rate of 0.08 mm/rev, cutting speed of 200 m/min on the same tool described in the previous section. Here the noise generated is taken as the experimental validation data. The noise generated is measured using the microphone, and then the normalization is done by considering the mean value, and the spectrum of measured noise is obtained as shown in Fig. 12.10.

Comparing the spectrum of simulated torque is shown in Fig. 12.9, and the noise-generated spectrum is shown in Fig. 12.10; the dominant frequency is observed to be 7968 Hz from simulation. Also, the dominant frequency of 7988.82 Hz is

¹MATLAB is used for Simulation.

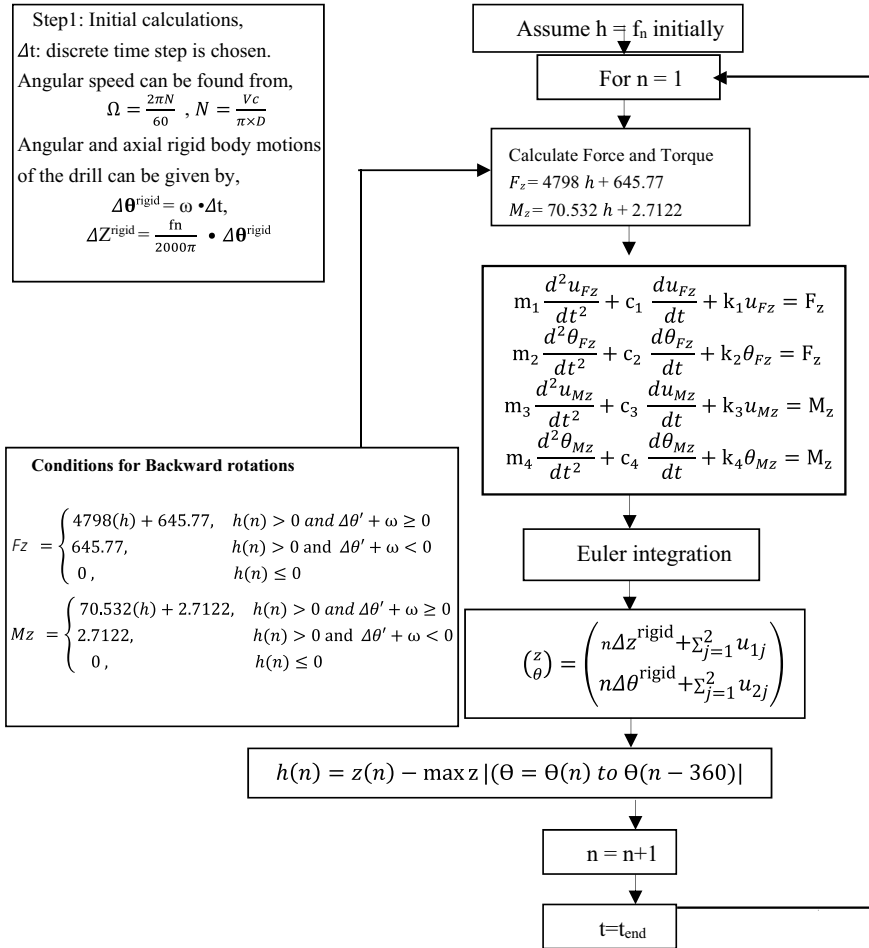


Fig. 12.4 Simulation flowchart

observed using noise spectrum. This means the simulation model is showing the good agreement with the noise generated.

12.4 Conclusions

A simulation based on the torsional–axial chatter vibrations in indexable drilling is done by considering the forces during the backward rotation of the drill. The simulated results are showing the good agreement with the noise spectrum. From this simulation, the forces varying during backward rotations of the drill and the

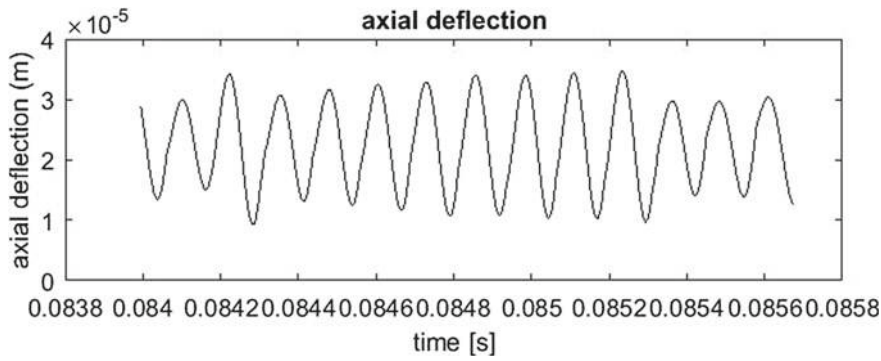


Fig. 12.5 Axial vibrations due to axial force and torque

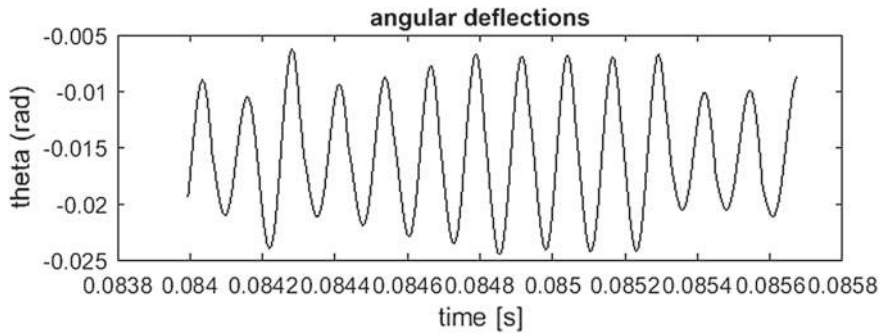


Fig. 12.6 Angular vibrations due to axial force and torque

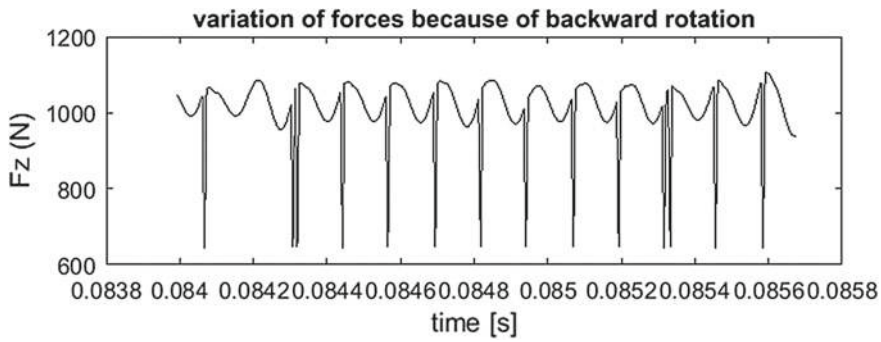


Fig. 12.7 Variation of forces due to backward rotations of the drill

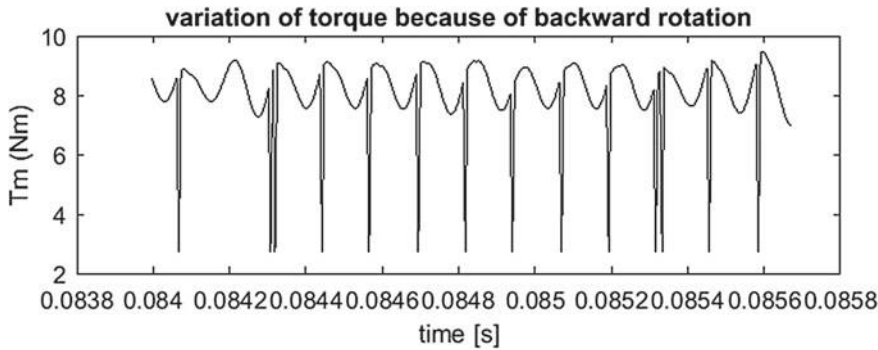


Fig. 12.8 Variation of torque due to backward rotations of the drill

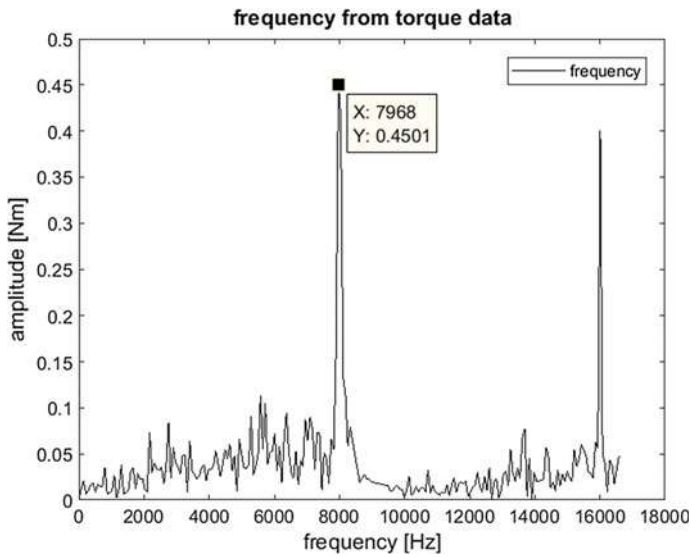


Fig. 12.9 Spectrum obtained from the simulated torque

deflections because of these forces can also be determined. A designer can appropriately be able to choose the drill model, and this may reduce the number of physical prototyping.

As changing the tool, geometry will change the modal parameters and the cutting forces; the results obtained from the simulation will vary if the geometry of the tool is changed. Thus, automation in this field for determination of the modal parameters due to changing geometries in tool is recommended.

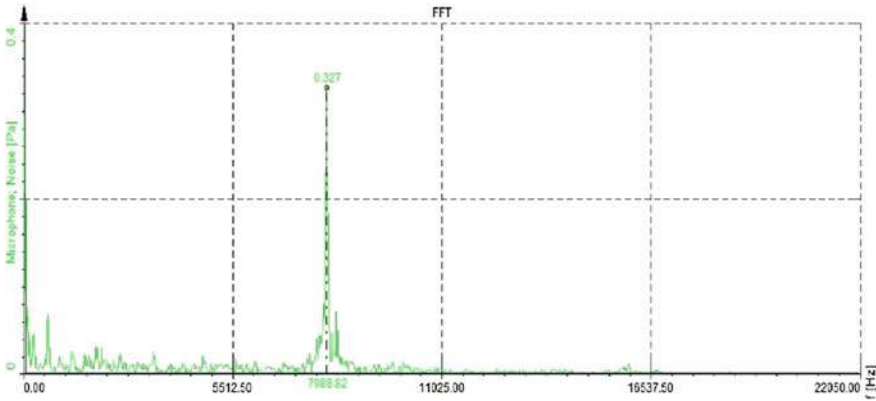


Fig. 12.10 Spectrum obtained from the noise generated

Acknowledgements The investigation was carried out in Kennametal India Ltd. in cooperation with BMSCE, Bengaluru. Supports from both organizations are highly appreciated. Guidance from Guruprasad Sunkad and Hariharan for this work is acknowledged.

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Chapter 13

Finite Element Analysis of Sheet Thickness and Force Variation in AA6063 During Single Point Incremental Forming



Saurabh Rai , Hreetabh Kishore , Harish Kumar Nirala
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Abstract The single point incremental forming (SPIF) using aluminum alloy-based sheets is widely used in automobile and aerospace industries due to its high strength to weight ratio. SPIF is one of the evolving manufacturing processes due to its potential for die-less forming of metallic sheets. The maximum allowable formability of AA-6063 is limited to the elongation ranging from 12 to 30%. This process generally uses hemispherical end-shaped forming tool which traces the generated CNC code path to acquire the desired shape. In this study, a conical geometry is formed through experiments and simulations using SPIF. This study presents a comparative finite element analysis (FEA) between implicit- and explicit-based computational techniques for SPIF using Abaqus®. In this study, output responses include sheet thickness variation, Von-Mises stress distribution, fracture limit curve and solver time for each computational method. Implicit computational method proves its advantages over explicit for accuracy.

Keywords SPIF · FEA · Tool path · Explicit and implicit computational method · ABAQUS

13.1 Introduction

Incremental forming process is a die-less manufacturing process in which free-formed shapes are made without using dedicated tooling. It is a sheet metal forming process with a high potential economic payoff for rapid prototyping applications and small quantity production. Schematic representation of SPIF process is shown in Fig. 13.1. Essential components of the process are; (i) sheet metal blank, (ii) blank holder, (iii) backing plate, and (iv) forming tool [1].

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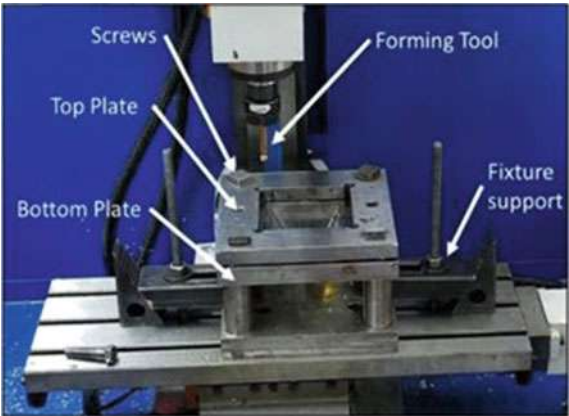
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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_13

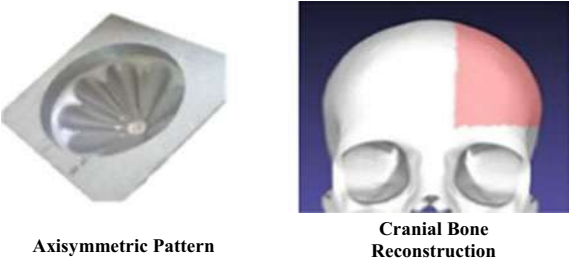
Fig. 13.1 Pictorial view of SPIF



The blank holder is utilized for clamping and holding the sheet in a fixed location during SPIF process. The backing plate supports the sheet, and it avoids the undesired bending of the sheet along its periphery. The tool generally used is of hemispherical dome-shaped, which continuously deforms the sheet to a desired component configuration, and the respective NC tool path is designed using the respective MATLAB code. As tool follows a designated path, it incrementally deforms the workpiece into the desired shape. To perform SPIF process, no specialized machine is required; one can use a conventional three-axis milling machine. Incremental forming led to higher formability as compared to conventional forming process and has been significantly employed for the manufacturing of biomedical parts. These parts have been used in the surgery of patients during cranioplasty [2]. Figure 13.2 shows some applications of SPIF. Some of the products include safety helmets, cranial plate, automobiles, etc. (Fig. 13.2).

The SPIF process having unique feature of subtracting the use of die block which is placed on the back surface of the sheet. Recent investigations of SPIF process are concerned with the applications of product formed and the process capability [3]. As formability of the process is defined by using four major parameters [4]: (i) percentage thickness reduction of the sheet (ii) step-depth (iii) speed (both rotational

Fig. 13.2 Applications of SPIF process [10]



and feed rate), and (iv) radius of the forming tool. The influence of the first parameter is commonly explained using the sine law. The second parameter explains about formability that decreases with increasing the step size. Results reported by Ham and Jeswiet [5] indicated that step size itself does not have a significant effect on the formability. The speed of the forming tool also influences formability because of its direct effect on the frictional conditions at the tool–sheet interface. The smaller tool radius with better formability results in higher productivity due to the smaller strains concentration at the region of deformation on the sheet using the forming tool. Likewise in the conventional stamping, larger forming tool results larger distribution of strains over a larger region makes the process similar to it. Jeswiet et al. [5], Fratini et al. [6] and Allwood et al. [7], and other authors improved the process performance regarding various industrial and research aspects. This could be done by the characterization of the products and the optimal process parameters. Different studies have been performed regarding thinning phenomena of sheet. Singh et al. [3] experimented and numerically simulated incremental forming process and reported the results of sheet thickness forming force, etc. by performing experiments on aluminum sheet AA6063 using three-axis CNC machine at four different forming angles 30°, 40°, 60°, and 75°. They studied sheet thinning variation along with the formed depth. Many authors have suggested for explicit solver method in order to have better results. In the present study, variation in thickness with the depth of truncated cone, using two different solver methods in numerical simulation is studied. The comparison of the results of two different solver methods is done with the actual experimental data [3].

Recently, researchers analyzed the effective formability limits by utilizing the different damage models and various fracture criterion combined to improve the process capability [1]. A peer review on the increased formability to compare the conventional stamping and deep drawing operations using numerical simulation has already been reported. It was done by the utilization of fracture forming limit diagrams before the fracture in the formed sheet, instead of earlier forming limit diagrams which is based on the necking of the sheet [3].

13.2 Materials and Methodology

Aluminum alloy AA6063 is a standard strength alloy generally referred to as an architectural alloy. It is usually used in intricate extrusions of workpiece. It has a good surface finish; high corrosion resistance is suited to welding and can be easily anodized. Most commonly alloys available as T6 temper and T4 show excellent formability. These materials are used for architectural applications, shop fittings, irrigation tubing, balustrading, window frames, extrusions, and doors [1]. AA6063-T4 is used in our work. The chemical composition of AA6063-T4 is presented in Table 13.1. The mechanical properties of AA6063-T4 are given in Table 13.2.

This work presents a closed-form analysis, by modeling the SPIF process and the experimental data is used to simulate numerically to obtained results which is not

Table 13.1 Chemical composition of AA6063-T4

Elements	Si	Fe	Cu	Mg	Cr	Zn	Ti	Al
Weight%	0.5	0.35	0.1	0.9	0.1	0.1	0.1	Rem

Table 13.2 Mechanical properties of AA6063-T4

Properties	Density	Melting point	Poisson's ratio	Young's modulus
Values	2.7 gm/cc	600 °C	0.33	70 GPa

possible by experimental study. The model is based on the membrane analysis with bidirectional, in-plane contact friction, and is focused on the extreme two different methods of analysis used in SPIF. This process deals with the plasticity of material, and the deformation causes due to the generated forces between tool and sheet that becomes highly nonlinear. The application of load is time-dependent and it goes on increasing as the forming depth of conical geometry increases. Usually, different solver algorithm shows different results and time duration depends on step width which may be either minimum or maximum. The material undergoes severe deformation when the plasticity increases above 50% [8].

Simulations are performed by many researchers using Abaqus explicit method to analyze SPIF process. From the Abaqus® manual, it becomes clear that explicit method is applicable to large, nonlinear, quasi-static analyses [9]. For SPIF, both explicit and implicit computational methods are equally capable of providing the desired solution. Both techniques take care of severe nonlinearity due to excessive distortion and fluctuating load.

While solving using explicit method, the equation is in the form

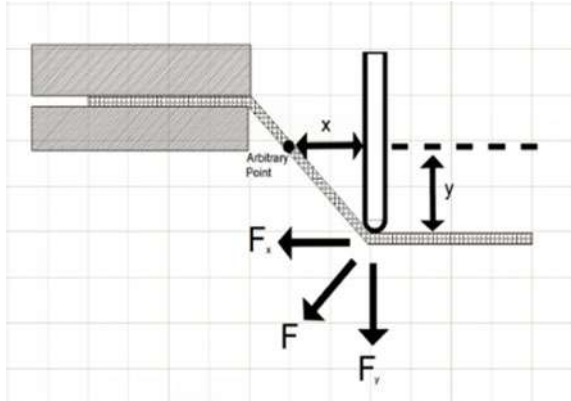
$$KU = F \quad (13.1)$$

Here, K depicts the stiffness of the material, U is the displacement of the node, and F is the force acting on the node. Calculation is performed for every node with respect to unbalanced force.

$$F' = F_{\text{ext}} - F_{\text{int}} \quad (13.2)$$

where F' is unbalanced force, F_{ext} is force applied by the tool, and F_{int} is reaction force generated from the sheet. Implicit solver solves F' to reach near to zero and updates the values of K after each iteration [8]. This way of solving the forming problem makes it more compatible to deal with plastic deformation. But the main problem with this solver is the convergence of the result, and it is also time-consuming in terms of computational time. The mathematical modeling for incremental forming can be done by assuming the cantilever beam fixed at one end [9]. The tool applies load from one end and another end of a sheet is fixed by restricting all the degree of freedoms. The bending moment stress at all the point can be calculated by using the

Fig. 13.4 Force interaction between Tool and deformed sheet



Eq. (13.3)

$$M = F_x Y + F_y X \quad (13.3)$$

where F_x represents the horizontal component of force, F_y is a vertical component and X is horizontal distance from point and Y is vertical distance of point from the tool. Further, simple bending equation is utilized to calculate the stress over each. Due to loading, we get elastic as well as plastic deformation by using Eqs. (13.4–13.5) Fig. 13.4 shows the force interaction between sheet and rigid tool.

Some basic equations involved in understanding of plasticity of elastoplastic model that is utilized by any FEA software to calculate the strain at each point during the simulation are presented in Eqs. (13.4)–(13.9).

$$[d\epsilon] = \begin{bmatrix} d\epsilon_{xx} & d\epsilon_{xy} & d\epsilon_{xz} \\ d\epsilon_{yx} & d\epsilon_{yy} & d\epsilon_{yz} \\ d\epsilon_{zx} & d\epsilon_{zy} & d\epsilon_{zz} \end{bmatrix} \quad (13.4)$$

where $d\epsilon$ denote strain tensor, this is the basic representation of strain in three directions, to calculate total strain; the incremental strain is calculated for each node by adding the incremental strain in each iteration. This equation is implemented by the solver to calculate the strain.

In this simulation, initial displacement is given to the tool so that indirectly the value of the U or the displacement of the node is known; further, it is also required to calculate incremental strain that will be present in the sheet when tool starts to move.

Equation (13.5) denotes the general form of incremental strain in terms of i and j (where $i, j = 1, 2, 3$). The subscripts 1, 2, and 3 represent the parameters in three mutually perpendicular directions.

The plasticity in the material is governed by Eq. (13.5). The strain is denoted by:

$$d\varepsilon_{ij} = \frac{1}{2} \frac{\partial(du_i)}{\partial x_j} + \frac{\partial(du_j)}{\partial x_i} = \frac{1}{2} (du_{i,j} + du_{j,i}) \quad (13.5)$$

Differentiating Eq. (13.5) with respect to time will give Eq. (13.6) which is called the strain-rate velocity relation, which is important for incremental strain phenomena. Elastic–plastic material governing equations are given as the strain-rate velocity relation:

$$\varepsilon_{ij} = \frac{1}{2} (v_{i,j} + v_{j,i}) \quad (13.6)$$

Due to strain hardening, the value of stress increases and the total plastic strain is given by Eq. (13.7):

$$\sigma_{eq} = \sigma_Y + K \left(\dot{\varepsilon}_{eq}^p \right)^n, \quad \varepsilon_{eq}^p = \int \dot{\varepsilon}_{eq}^p dt \quad (13.7)$$

In elastic region is given by Eq. (13.8):

$$\dot{\varepsilon}_{kk} = \frac{\sigma_{kk}}{3K}, \quad \dot{\varepsilon}_{ij} = \frac{1}{2} \dot{\sigma}_{ij} \quad (13.8)$$

Equilibrium Eq. (13.9), ρ is density, b_i is body force

$$\rho \frac{\partial v_i}{\partial t} + v_{i,j} v_j = \rho b_i + \sigma_{ij,j} \quad (13.9)$$

These are the basic equations which were used in understanding the simulation and providing some analytical understanding of the results.

13.2.1 Simulation Procedure

Incremental forming simulations have been performed using implicit and explicit solver methods. The results obtained were analyzed and compared for both techniques. To perform the simulation, a tool path is generated for frustum of cone using MATLAB. Large number of points along the tool path has been generated in order to enhance the accuracy of the process. Furthermore, the model of the sheet is made using 3D deformable shell having a radius of 45 mm and rigid hemispherical dome-shaped tool of radius 3 mm.

13.2.2 Boundary Condition

The material properties of AA6063 were assigned to the sheet. The analysis is done using a commercial software Abaqus®. The tool was defined as analytic rigid. The motion of the tool is given by three-directional amplitude. The amplitude is defined by fixed time/frequency method. Computational time increases with smaller mesh element size and also convergence difficulty in implicit computational technique. The problem with higher time step is that computational time gets increased. Table 13.3 gives information about the simulation parameters for the sheet.

The sheet is defined as a deformable shell with thickness 1 mm. The sheet thickness parameter is only defined for shell element. The fixed displacement boundary condition is given to the circumference of the sheet by restricting all the degrees of freedom. The tool motion is given by defining amplitude and the three-axis coordinated points. The values of the path were assigned to the tool tip by defining a reference point on the tip of the tool.

The tool path has a conical shape with 75° inclined forming angle, 35 mm radius of first revolution, and a height of 13 mm as shown in Fig. 13.5.

The generated tool path is imported in amplitude section of Abaqus®. The tool path is given to tool tip which is taken as reference point in the simulation. The master–slave surface interaction criteria are given to the tool and sheet as provided

Table 13.3 Simulation parameters

Parameters	Values
Nodes	1600
Amplitude (fixed)	0.01
Time period (explicit, implicit method)	48.7
Cone angle	75°
Radius of the cone	35 mm
Height	13 mm

Fig. 13.5 Tool path (frustum of cone)

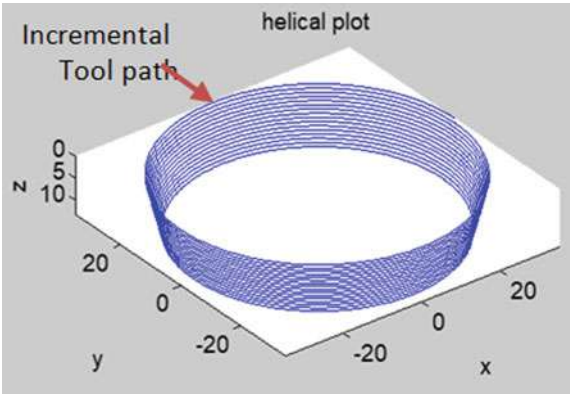
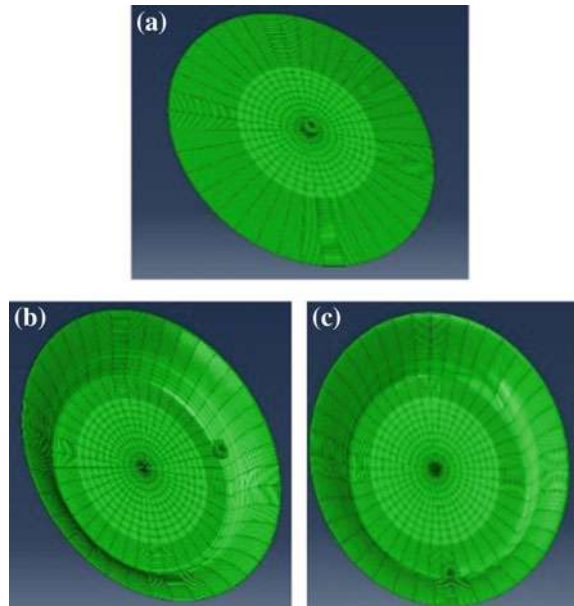


Fig. 13.6 **a** Undeformed sheet, **b** SPIF using implicit solver and **c** SPIF using explicit solver



by taking sheet surface as slave and tool tip as master and interaction properties are tangential behavior with friction coefficient value of 0.25. Mesh type used in this study comprises of a four-noded doubly curved thick shell. The position of tool on the plate after and before deformation is shown in Fig. 13.6a, b.

In this paper, comparison of the results has been shown with the actual experimental results [3]. The deformed elements using explicit and implicit method are depicted in Fig. 13.6b, c which has a significant variation in results as elaborated in the next section.

13.3 Results and Discussion

Experimental results available from literature are being used as reference to validate the study. This study has considered the thinning and strain as main focus of study [3]. Outputs and particular parametric study of each solver method is reported in this section.

13.3.1 Thickness Variation

Thickness of the formed structure in SPIF is primarily a function of forming angle “ θ ” and is given by cosine law.

$$t_f = t_i \cos \emptyset \quad (13.10)$$

where t_f is the final formed thickness, t_i is the initial thickness, and $\emptyset$ is the forming angle. Therefore, influence of forming angle (75°) for a conical profile on the section thickness is evaluated. Rest of the parameters kept constant [1]. From the results, it can be observed that the section thickness is comparatively nonuniform, using explicit solver method with a decreasing trend along the forming depth, unlike uniform thickness predicted by cosine law. A sharp reduction in section thickness after a certain depth is also observed owing to transition from bending zone to actual stretching in forming. Product quality in SPIF mainly affected by thickness variation in the deformed parts. Continuous deformation of the sheet metal in formed component may lead to fracture if it reaches a threshold value. FEA helps to simulate the process to predict the thickness reduction to avoid fracture. The results from both methods are in close agreement with the experiments, though having variations owing to the shearing effect and consequent material removal in incremental forming which have not been considered in finite element modeling. Nonuniform thickness distribution is attributed mainly to change in deformation mechanics of both solvers from plane strain to biaxial thinning with increasing forming depth in incremental forming. The results obtained from implicit solver prove to be better than explicit solver in terms of desired accuracy. The contour plot of thickness using explicit method and implicit method in Fig. 13.7a, b shows the variation of thickness to be in the range of 0.9–0.3 mm (implicit method) and 0.9–0.1 mm (explicit method) for a sheet having initial thickness of 1 mm. Figure 13.7c shows the thickness variation of both implicit and explicit solver methods their comparison with experimental data.

13.3.2 Stress Distribution

The yield stress of material under plane stress condition depends on the strength of sheet material, which is defined in terms of flow stress. The value of flow stress depends on the amount of plastic deformation during process. This value changes with change in the deformation of sheet, and more the deformation force more strain hardening is observed. The Von-Mises theory is most suitable for considering yielding in material while undergoing plastic behavior. Considering all other conditions exactly same [7], implicit solver technique gives better results than explicit method in terms of thickness distribution and uniform stress distribution. In a particular case study, it has been found that implicit solver method takes 42 mins for the completion of simulation run whereas explicit took 12 mins. Here, Von-Mises stress contour plot has been used to show the stress distribution in the formed component. Figure 13.8a, b depicts the Von-Mises stress distribution of two solver methods.

A forming limit diagram (FLD) sometimes also termed as fracture limit diagram utilized to predict forming behavior of sheet metal. This diagram shows graphically the description of material failure, etc. Forming limit diagram of the thinner sheet has been considered as the failure limit and is plotted between major strain versus

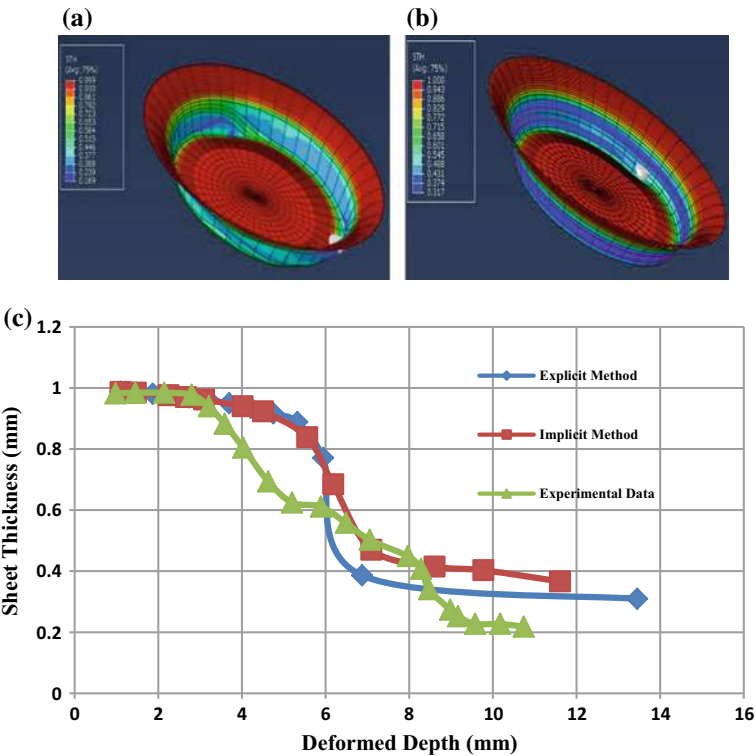


Fig. 13.7 **a** Simulation result of thickness variation using explicit method, **b** simulation result of thickness variation using implicit method, **c** sheet thickness versus forming depth

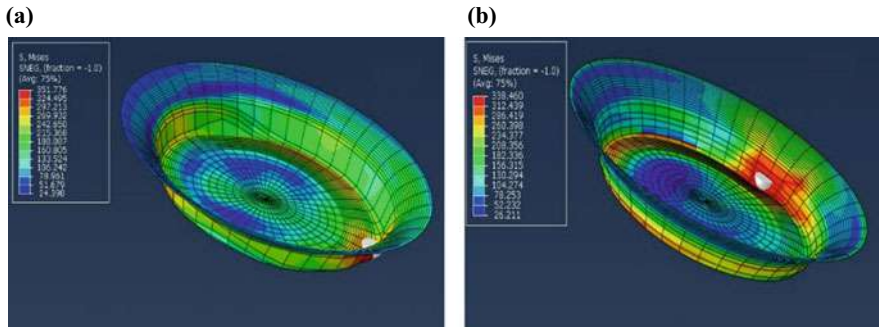


Fig. 13.8 **a** Explicit contour plot, **b** implicit contour plot

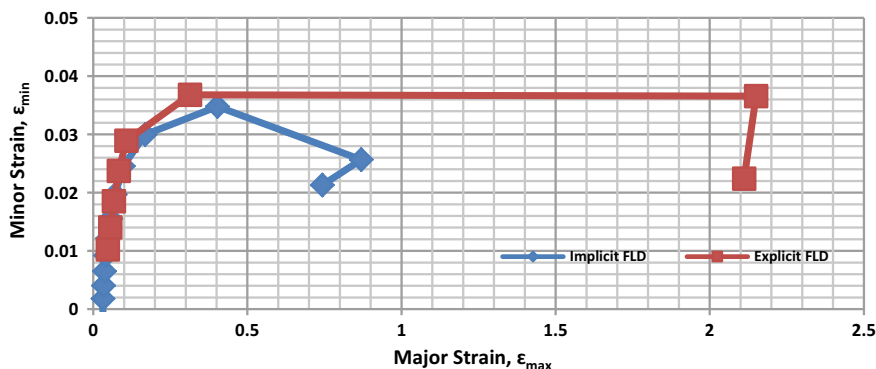


Fig. 13.9 Forming limit diagram of both methods

minor strain. FLD is used as failure criteria to determine failure by stretching the sheet by a dome-shaped tool and then observing the strain.

In incremental forming, the value of FLD is generally much higher than conventional forming process. FLD is measured by measuring strains on part produced by incremental forming. Present work shows forming limit plot which has been evaluated numerically. The comparison of FLD values for explicit and implicit solver technique is shown in Fig. 13.9. It is found that in explicit, the stretching is not comparatively uniformly distributed as it comprises of higher strain, but using implicit method, the strain distribution was found to be uniform. Implicit method of SPIF proves more accurate and results are closer to the actual data.

13.4 Conclusions

Implicit solver is better than explicit solver, but still, researcher goes with explicit solver to simulate this process as explicit solver requires less computational time. The present study is focused on the finite element analysis of SPIF process for AA-6063-T4 sheet. Results obtained from numerical analysis help in predicting the thickness distribution, plastic strain, and stress distribution in the formed component. This study focuses on change in results due to solver technique. The obtained thickness distribution is found approximately the same and uniform in implicit method and uneven in explicit method as compared to experimental results. It is also found that mesh distortion was uneven in explicit having higher strain value, and even in implicit solver method. In implicit method, the mesh distortion reflects even distribution of thickness and strain throughout the deformation. The simulation time required for implicit solver method is more than the explicit solver method. It was 10 min for explicit and 15 min for implicit. So, depending on the accuracy and higher formability, implicit solver method is highly recommended for precision but the time required in performing the computation is more. Further, study on process optimization and

mesh refinement by considering more parameters such as tool force, residual force, etc. is required for better understanding.

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Chapter 14

Analysis and Prediction of Electrical Discharge Coating Using Artificial Neural Network (ANN)



R. Tyagi , S. Kumar , V. Kumar , S. Mohanty , A. K. Das
and A. Mandal

Abstract Surface modification through electric discharge coating (EDC), a common feature of EDM machine, was done with the use of green compact electrode at negative polarity that builds a layer on the workpiece. Green compact sintered electrodes were prepared from the mixture made up of tungsten (WS_2) and copper (Cu) powder in different proportions. In this study, effect of input experimental parameters (duty factor, peak current, and powder mixing ratio) on output parameters (tool wear rate, mass transfer rate, microhardness, and coating thickness) was observed. From FESEM and EDS results, a good coating feature was detected on the top coating with coating material presence. The artificial neural network was applied for prediction of output parameters response. The experimental results and predicted results using the artificial neural network (ANN) showed good agreement. There was a good agreement observed in regression and performance plot between actual experimental results and ANN predicted results.

Keywords EDC · Mild steel · Green compact electrode · ANN · Microhardness · Coating thickness

14.1 Introduction

Mild steel is the cheapest and most common type of steel used for many applications. Because of some its properties such as good hardness and strength, cheap and easy accessibility, it has a wide range of application in making machine parts likewise nuts and bolts, automobile and ship body parts, magnets, pipes, knives, etc. It is required to protect it against corrosion and wear by using some types of paints, oil, or grease. These methods are costly and time-consuming and are not reliable. Hence, we need a method which can provide better resistance to corrosion, wear, and friction. With the help of electro-discharge coating (EDC), these problems can be rectified and by

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_14

applying solid lubricant coating on mild steel. EDC can be used to improve surface properties like hardness, resistance to wear, and corrosion of different engineering materials, with negligible change on the original characteristics of the base material. During the process of EDC, due to electric sparks and formation of plasma, channel tool materials get melted and get deposited over the surface, and this deposited material gets solidified by rapid quenching (cooling medium) in dielectric fluid.

In EDC technique, requirement of highly complicated equipments (vacuum apparatus) is not necessary. Also, hard and wear-resistant layers of dissimilar material composition can be easily produced on the workpiece surface. Coating layers are produced in different parts of the workpiece, and control in coating thickness is easily possible. EDC process is also used in aerospace applications and automotive industries such as coating of cutting tools, alloying of dies, and coating of roll surface in a rolling mill. Electro-discharge texturing in rolls increases roll hardness and its performance whereas reduces friction coefficient.

Solid lubricants or dry lubricants are used as an additive in lubricating oil or greases, as main ingredient in antifriction coatings or as free-flowing powder. The solid lubricants provide efficient lubrication, minimizing friction and wear under extreme operating conditions. Solid lubricants can be used where lubricating oil or grease fails to reduce friction and wear like high pressure, high temperature, reciprocating motion, ceramics, etc. WS_2 is one of the best lubricants in the world. It can work in high temperature and high pressure also. The coefficient of friction of WS_2 is very low (0.03–0.07). Several researchers used EDC process to modify the workpiece surface as discussed below.

14.2 Literature Review

Shunmugam et al. [1] performed experiment by using P/M compact electrode of mixture of WC (40%) and iron (60%). Improvement in abrasion resistance was observed as 20–60% and 20–50% reduction in cutting force. Simão et al. [2] done surface modification and electrical discharge texturing (EDT) on hardened Sendzimir rolls. The green compact electrode was made of TiC/WC/Co and WC/Co. In the study, it was found that when the sintering temperature and compacting pressure were increased, then the electrical, mechanical, microstructural, physical, and thermal characteristics of electrode were changed resulting in greater texturing performance. Patowari et al. [3] carried out surface modification by green compact of W–Cu and thereafter prepared ANN model for comparing the experimental result with predicted value. Kumar et al. [4] coupled ANN process with Taguchi to predict the surface roughness of ED machined titanium alloys which showed the good agreement of predicted and experimental values. Chakraborty et al. [5] modified Al alloy surface using EDC with SiC–Cu compact electrode. They established that MDR, surface roughness, and TWR increase with increasing current and pulse duration and decrease with compaction load. Tyagi et al. [6] performed electrical discharge coating of solid lubricant on mild steel surface. There was an increase in wear resistance and a

decrease in hardness from 180 HV to 44.11 HV was achieved due to solid lubricant deposition. Hwang et al. deposited the multilayer coating using titanium and graphite electrode that exhibit remarkable wear resistance [7].

From the literature work, it can be analyzed that EDC coating was performed from the decades but limited research was reported on the analysis and prediction of deposition of solid lubricant coating through this process. Very few research work was carried out on EDC of WS_2 and Cu by applying artificial neural network (ANN) approach in order to predict the coating behavior with respect to various variable input parameters. The purpose of present work is to develop a solid lubricant WS_2 -Cu coating over the mild steel by electric discharge process using a compact electrode prepared with WS_2 and Cu powder. The ANN predicted response was compared with actual experimental parameters settings.

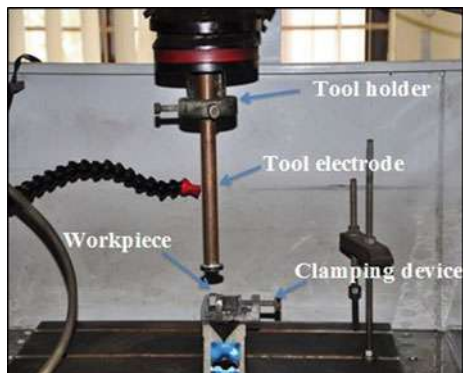
14.3 Methodology

14.3.1 Experimental Setup

Electrical discharge coating is done on electrical discharge machine (EDM). For the experiment, Sparkonix ZNC/ENC 35 die-sink EDM is used. The image of die-sink EDM with its parts is shown in Fig. 14.1. The machine consists of work tank, main controller with control panel, X and Y axis control wheel, servo head, dielectric filled in tank below work tank and a pump. Servo head of EDM machine provides automatic adjustment to maintain the gap/spark gap in between the tool electrode and the substrate throughout the EDC process. The servo head can be raised or lowered by the buttons provided in the control panel. The process parameters like voltage, peak current (I_p), pulse on time, pulse off time, machining time duty factor (τ) can be fed and programed in the control panel. The place of tool can be altered physically by movement of X-Y stage. The setup of EDC is prepared inside the machine



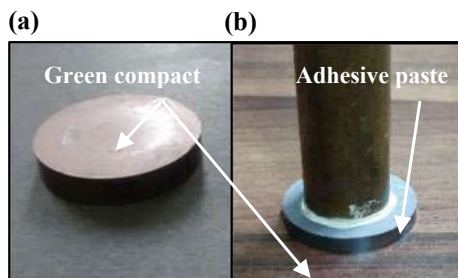
Fig. 14.1 Sparkonix ZNC/ENC 35 die-sink EDM

Fig. 14.2 Setup for EDM

work tank which consists of magnetic vice, clamping vice, mild steel workpiece ($10 \times 10 \times 4$ mm), tool holder which holds the tool and is attached to servo head as shown in Fig. 14.2.

14.3.2 Preparation of Compacted Tool Electrode by Powder Mixture of WS_2 and Cu

Electrodes prepared for EDC process were prepared using a mixture of tungsten disulfide and copper at different proportions, which are 40:60/50:50/60:40 weight% composition. Weight% composition 40:60 implies 6 gm of WS_2 and 9 gm of Cu, composition 50:50 implies 7.5 gm of WS_2 and 7.5 gm of Cu, and composition 60:40 implies 9 gm of MoS_2 and 6 gm of Cu. Here, copper serves as a binding material and improves the conductivity of tool. For uniform mixing, each composition was mixed for 1 h in mortar and pestle, After mixing, powders were compacted in hot mounting press by selecting the parameters such as compaction pressure 200 kg/cm^2 , sintering temperature 130°C , 10 min heating time, 5 min cooling time, compaction die diameter of 15 mm, and thickness of 3 mm. Figure 14.3a, b Electrodes composed

**Fig. 14.3** a Green compact electrodes and tool extension, b setup for EDC

of two parts; one part is tool extension made of copper for appropriate holding the electrode and other one is powder metallurgical compacted pellet which acts as an electrode. The tool extension and compacted pellet part then joined together using electrically conductive silver paste and from electrode part, during experimentation material gets erode and transferred over the work surface throughout the process.

14.3.3 Working Principle of EDC

In EDC, the tool electrode is used at positive polarity while the workpiece is used to negative polarity. The dielectric used can be DI water, kerosene, or any other EDM oil. When a high-ampere pulsed voltage is applied across the gap between the two electrodes, the dielectric fluid presents between the gap breaks down resulting in the liberation of electrons, which in the presence of electric field, collides with the molecules of dielectric fluid causing more electrons to be liberated from the dielectric's molecule. The process multiplies and an avalanche of ions and electrons is emitted which drops the resistance of dielectric and ionizes it. Due to this, electrical energy is discharged in the gap creating a spark whose temperature is very high. And because of this high temperature (around 10,000 °C) spark, there is melting of some material from both the electrodes. As the pulsed voltage is in off condition (Toff), there is some time for cooling the molten material over the surface of workpiece which results in coating on the workpiece. The carbon particles are also disintegrated from dielectric fluid composed of hydrocarbon oil and resulting into hard carbide on the substrate surface. All the experiments have been repeated three times and average of all the value has been taken to plot the results. In this study, tool wear rate (TWR) and material removal rate (MTR) were calculated by the change in initial and final weight of tool and workpiece for each experiment. Further, Vicker hardness tester and optical microscope helped in measuring the microhardness (MH) and thickness of coating (CT), respectively.

14.4 Artificial Neural Network (ANN) Solution and Methodology

ANN can be defined as a flexible modeling tool which exhibits the ability to resolve the problems associated with nonlinear process with the help of software computing technique by studying the mapping of input and output parameters. With the help of multilayered algorithm, tool wear rate (TWR), mass transfer rate (MTR), microhardness, and coating thickness can be validated. In MATLAB software, neural network toolbox is used for ANN validation. In order to train the network, multilayer feedforward network is employed along with backpropagation algorithm. Neural Networks

give a large number of various possibilities which can be applied to change the algorithms. The backpropagation algorithm is most used algorithm as it gives acceptable results. It also allows input, output, and target. A layer neural network composed three input neurons, four output neurons, and one hidden layer of twenty neurons for prediction. The absolute error presented in Table 14.2 has been obtained by the subtracting the predicted value from the experimental value.

$$\text{Absolute error} = \text{Experimental value} - \text{Predicted value}$$

A total of 36 experiments were performed with two repetitions to check the variation in output parameters owing to variable level of the three input process parameters. Out of 36 experiments, 27 has been taken for training and other nine was used for testing. Thus, total 36 experiments were successfully carried out for this analysis. The experimental settings and experimental result along with ANN predicted results and % relative error (in between experimental results and ANN predicted data) for all the 36 observations are shown in Tables 14.1 and 14.2. Table 14.3 shows the type of parameters chooses for ANN in MATLAB. The data produced was favored for testing additionally for training of ANN using feedforward backpropagation (FFBP) algorithm. The neural network algorithm comprises of with three input neurons, four output neurons, and two hidden layers of fifty neurons for prediction (Fig. 14.4). The neural network applied to the given study is feedforward backpropagation network (FFBP). FFBP algorithm reflects a decent technique in order to learn and train the ANN technique. ANN technique can automatically regulate its weights and threshold values throughout training in such a way that the change in between the target and sampled outputs is kept low. The adjustments are calculated using propagation algorithm. The performance plot after training presented the best results. Out of the all validation performance of the network, best value obtained was 17.4207 at epoch 60 and model is completed in 1000 epochs (Fig. 14.6). The % relative errors of actual experimental results and ANN predicted results for TWR, MTR, coating thickness (CT), and microhardness are given in Tables 14.1 and 14.2. In addition, for analyzing the ability of the neural network, linear regression in between response of network and experimental target value was executed. For the present case, the entire output data was put through for training, validation, and testing to perform the regression analysis. The obtained regression results are been presented separately for the output, which is shown in Fig. 14.5. The values of correlation coefficient (R) for training were 0.99454, for validation 0.99092, for testing 0.98267, and for overall 0.99181 in simulation. ANN model depicts a good agreement in between the experimental results (Table 14.1) and the ANN predicted results (Table 14.2) through FFBP neural network. Figure 14.6 illustrates the convergence or gradient of mean square error (MSE) for output parameter with the number of epochs during training of the particular network.

The graph of performance of the results obtained after training has been shown in Fig. 14.6 which shows the best results. Best validation performance of the network is 17.4207 at epoch 60 and model is completed in 1000 epochs. Tables 14.1 and 14.2 show the percentage relative errors of experimental and ANN predicted results.

Table 14.1 Experimental settings with actual experimental results [6]

Ex. No.	Input parameters				Output parameters		
	WS ₂ :Cu	I_p (A)	τ (%)	TWR (g/min)	MTR (g/min)	CT (mm)	MH (0.2 kg load)
1	0.66	4	30	−0.83	−0.5	0.26	51.39
2	0.66	4	50	0.9	0.16	0.29	63.27
3	0.66	4	70	0.72	0.13	0.3	58.86
4	0.66	4	90	0.54	0.9	0.31	52.11
5	0.66	7	30	−0.5	−0.2	0.3	78.95
6	0.66	7	50	1.27	0.19	0.32	67.25
7	0.66	7	70	1.2	0.15	0.29	72.33
8	0.66	7	90	1.2	0.1	0.31	73.29
9	0.66	10	30	−0.3	−0.1	0.4	100.1
10	0.66	10	50	1.53	0.22	0.51	70.66
11	0.66	10	70	1.45	0.13	0.49	75.58
12	0.66	10	90	1.3	0.11	0.48	75.8
13	1	4	30	1.7	0.22	0.28	90.44
14	1	4	50	2.12	0.25	0.35	64.37
15	1	4	70	1.92	0.2	0.34	85.85
16	1	4	90	1.54	0.19	0.33	92.1
17	1	7	30	2.14	0.24	0.4	65.12
18	1	7	50	2.39	0.28	0.55	62.11
19	1	7	70	1.99	0.23	0.4	69.81
20	1	7	90	1.45	0.2	0.36	72.84
21	1	10	30	2.4	0.26	0.53	54.33
22	1	10	50	2.69	0.3	0.57	46.24
23	1	10	70	2.54	0.29	0.56	70.46
24	1	10	90	2.49	0.26	0.51	51.55
25	1.5	4	30	2.89	0.28	0.34	76.64
26	1.5	4	50	2.9	0.3	0.37	54.11
27	1.5	4	70	2.82	0.23	0.26	55.15
28	1.5	4	90	2.75	0.21	0.22	81.92
29	1.5	7	30	2.99	0.32	0.57	75.12
30	1.5	7	50	3.22	0.35	0.61	52.99
31	1.5	7	70	2.92	0.29	0.59	54.92
32	1.5	7	90	2.9	0.25	0.58	72.64
33	1.5	10	30	3	0.38	0.58	72.39
34	1.5	10	50	3.58	0.4	0.66	44.11
35	1.5	10	70	3.42	0.37	0.65	51.56
36	1.5	10	90	3.2	0.34	0.6	57.55

Table 14.2 ANN predicted results and absolute error

Ex. No.	ANN predicted value				Absolute error			
	TWR (g/min)	MTR (g/min)	CT (mm)	MH (0.2 kg load)	TWR (g/min)	MTR (g/min)	CT (mm)	MH (0.2 kg load)
1	-0.51758	-0.40412	0.33388	58.5367	0.31242	-0.09588	-0.07388	-7.1467
2	0.12018	-0.4682	0.35492	50.3303	0.77982	0.6282	-0.064923	12.9397
3	0.11691	-0.4440	0.55827	55.9053	0.60309	0.57408	-0.25827	2.9547
4	1.2963	-0.4282	0.60068	57.5794	-0.75626	1.3282	-0.29068	-5.4694
5	-0.82247	0.40332	0.23542	80.7	0.32247	-0.60332	0.064584	-1.75
6	-0.7673	-0.1622	0.26703	67.2516	2.0373	0.3522	0.052973	-0.00155
7	-0.11312	-0.3841	0.29824	75.876	1.3131	0.5341	-0.008243	-3.546
8	2.9824	-0.47759	0.34907	75.9798	-1.7824	0.57759	-0.039073	-2.6898
9	-0.82439	0.74897	0.37884	88.0857	0.52439	-0.84897	0.021155	12.0143
10	-0.8017	0.28218	0.38892	75.5965	2.3317	-0.062176	0.12108	-4.9365
11	-0.66517	0.59403	0.45379	77.8351	2.1152	-0.46403	0.036209	-2.2551
12	2.6517	-0.25153	0.53588	56.043	-1.3517	0.36153	-0.055878	19.757
13	-0.74323	-0.48591	0.57911	79.0467	2.4432	0.70591	-0.29911	11.3933
14	1.2859	-0.4964	0.50638	63.5946	0.8341	0.7464	-0.15638	0.77543
15	3.0338	-0.49763	0.618	69.4384	-1.1138	0.69763	-0.278	16.4116
16	3.3758	-0.49856	0.64311	85.589	-1.8358	0.68856	-0.31311	6.511
17	-0.82333	-0.21852	0.44379	74.4387	2.9633	0.45852	-0.043793	-9.3187
18	-0.44493	-0.4933	0.43349	58.6493	2.8349	0.7733	0.11651	3.4607
19	1.7663	-0.49443	0.56929	78.7793	0.22371	0.72443	-0.16929	-8.9693
20	3.2217	-0.47703	0.57751	73.0092	-1.7717	0.67703	-0.21751	-0.16922

(continued)

Table 14.2 (continued)

Ex. No.	ANN predicted value				Absolute error			
	TWR (g/min)	MTR (g/min)	CT (mm)	MH (0.2 kg load)	TWR (g/min)	MTR (g/min)	CT (mm)	MH (0.2 kg load)
21	-0.82484	0.63543	0.58822	61.5365	3.2248	-0.37543	-0.058215	-7.2065
22	-0.60931	0.15323	0.46136	47.3185	3.2993	0.14677	0.10864	-1.0785
23	0.95176	-0.12575	0.5862	53.2703	1.5882	0.41575	-0.026203	17.1897
24	2.5328	-0.22382	0.57924	54.7765	-0.042836	0.48382	-0.069239	-3.2265
25	0.47247	-0.23663	0.64983	81.1472	2.4175	0.51663	-0.30983	-4.5072
26	2.8082	-0.4435	0.64916	52.1995	0.091833	0.74359	-0.27916	1.9105
27	3.3992	-0.45943	0.63579	56.598	-0.57922	0.68943	-0.37579	-1.448
28	3.4729	-0.47746	0.55249	79.5865	-0.72294	0.68746	-0.33249	2.3335
29	-0.23131	-0.40128	0.6481	71.4562	3.2213	0.72128	-0.078101	3.6638
30	1.6772	-0.48328	0.64192	48.4994	1.5428	0.83328	-0.031924	4.4906
31	1.633	-0.4787	0.54511	56.8116	1.287	0.76877	0.044885	-1.8916
32	2.3651	-0.47355	0.56404	73.1579	0.53495	0.72355	0.015964	-0.51794
33	-0.6156	0.19477	0.61498	69.815	3.6156	0.18523	-0.034977	2.575
34	-0.27576	-0.16206	0.43894	50.5478	3.8558	0.56206	0.22106	-6.4378
35	1.3417	-0.39569	0.60157	47.0329	2.0783	0.76569	0.048427	4.5271
36	3.2067	-0.4804	0.6087	53.0718	-0.006669	0.82043	-0.008701	4.4782

Table 14.3 Types of parameters

Name	Type
Network type	Feedforward backpropagation
Number of hidden layers	1
Transfer function used	LOGSIS
Training function	TRAINGDX
Learning function used	LEARNGDM
Performed function	MSE 12
Number of neurons	50
Number of epoch	1000
Learning factor	0

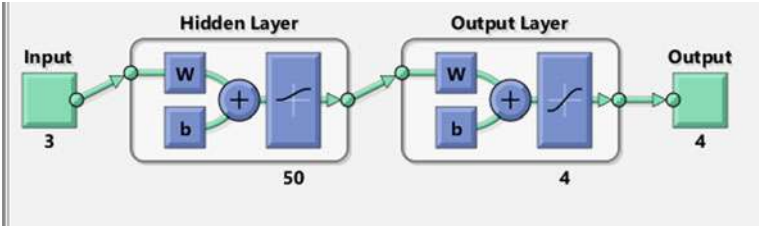


Fig. 14.4 A neural network with three input neurons, four output neurons, and two hidden layers of fifty neurons

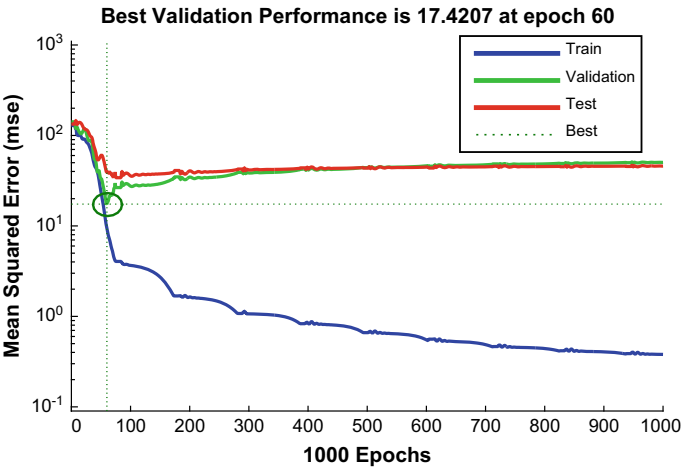


Fig. 14.5 Performance plot

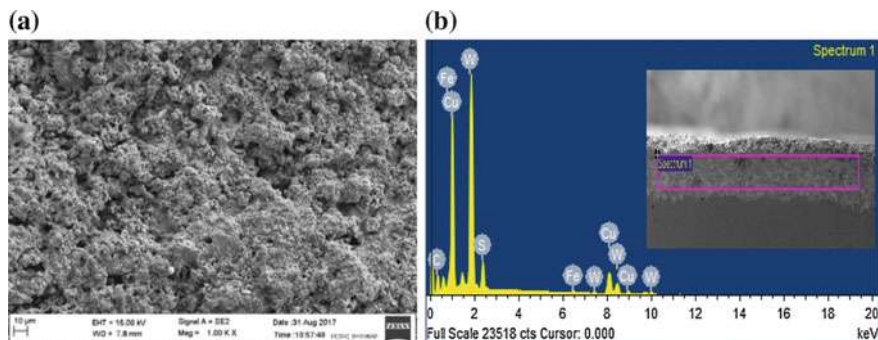


Fig. 14.6 **a** FESEM of top coating surface, **b** cross-section EDX

Hence, ANN model represents a good agreement between the achieved experimental and predicted results by the FFBP neural network.

14.5 FESEM and EDS Analysis

Field emission scanning electron microscopy (FESEM) depicted the good coating surface by showing the very few pores at top (Fig. 14.7a) with negligible gap at the interface of cross-section (Fig. 14.7b). Energy dispersive spectroscopy (EDS) results confirm the coating deposition through tool by showing the coating material or element W, S, and Cu along with other metallic elements at the cross-section (Fig. 14.7b). The variation in FESEM images captured at the cross-sectional area of coating was also examined with change in peak current as shown in Fig. 14.7a–c. It can be inferred that the coating thickness increases when the peak current increases and maximum thickness was achieved at 10 A peak current. There was good bonding observed between the coating and workpiece interface. Although, formation of pores increases with increase in current values, while nonuniform coating was observed at low current values. Hence, it can be said that the better microstructure with uniform coating was obtained at 10 A peak current. There was good bonding observed between the coating and workpiece interface. Although, formation of pores increases with increase in current values, while nonuniform coating was observed at low current values. Hence, it can be said that the better microstructure with uniform coating was obtained at 10 A peak current.

14.6 Conclusions

Electro-discharge coating of $\text{WS}_2 + \text{Cu}$ powder was successfully performed onto the mild steel substrate. Copper was used as a binding material due to poor electrical

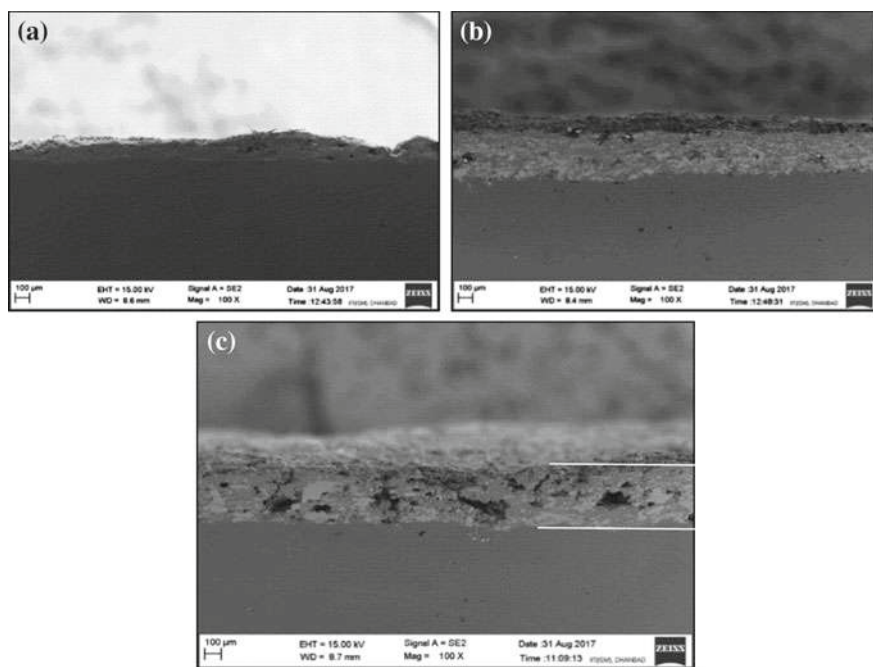


Fig. 14.7 FE SEM images of coating cross-section for the peak current, **a** 4 A, **b** 7 A, **c** 10 A

properties of WS_2 . Also, dry nature of WS_2 powder makes it difficult to be used alone for preparing its powder compact. Hence, various other types of solid lubricants with proper additive can also be used to prepare lubricant surfaces. However, EDC allows the use of wide variety of material for making green compact electrode. ED coating also facilitates the use of different shapes of electrodes for coating complex shapes parts. A good coating quality without pores and voids was observed onto the top coating surface. EDS spectrum showed the principal coating elements present at the cross-section. Further, FESEM images taken at coating cross-section depicted increases in coating thickness with increase of peak current which proved that peak current is the major factor to achieve higher coating height. Both the experimental and predicted data confirm the validity of ANN. ANN also helped in obtaining the parameter required for achieving better surface quality. Hence, it showed the better consistency between predicted and experimental values which confirmed the existence of this model.

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Chapter 15

Machining Performance Prediction for Zirconia Toughened Alumina Insert in Machining of High Carbon Steel Using Computational Approach



Subhrojyoti Mazumder and N. Mandal

Abstract This work aims to develop a finite element model for zirconia toughened alumina cutting insert to predict the cutting performances in machining of AISI 1095 steel using an implicit Lagrangian computational method by means of commercially available Deform 3D machining software package. Different cutting forces associated with the turning operation, temperature distribution at the tool tip as well as workpiece deformation zones, induced stress and strain rate at the workpiece shearing regimes are evaluated using this FE model. Material removal rate is also calculated using this computational approach. This computational technique has been found as a suitable approach to predict the cutting performances of the modelled zirconia toughened alumina cutting insert turning against the high carbon steel.

Keywords Finite element simulation · ZTA insert · AISI 1095 · High carbon steel

15.1 Introduction

Implementation of computational approach into the machining process has proved to be a faultless way to predict the performances of cutting inserts while turning against different material levels [1, 2]. Sometime, it is quite difficult to assess the performance of newly developed cutting tool employed in hard turning since it comprises of many complex conditions during cutting operation. With the advancement of technology, inducement of finite element simulation into the cutting operation has paved a new horizon to analyze the critical circumstances coming out due to material deformation. On the other hand, application of such computational method greatly reduces the material waste and unnecessary man power which are of course a great concern for any industry. Soft computational tool like Deform 3D can be used as a prime tool to predict the performance of newly proposed cutting inserts before undergoing any level of production. Cutting tool wear, temperature distribution at the tool, different

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_15

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stages of forces can readily be evaluated using such finite elemental method [3–5]. Beside that estimation of induced stress and strain rate at the deformation zones is the significant aspect in order to design any cutting insert while machining a specified job. Deform computational package helps to solve these parameters effortlessly [6]. The objective of this paper is to develop a FE model primarily to predict all these performances of a proposed triangular type ZTA cutting insert while turning AISI 1095 high carbon steel.

15.2 Development of Finite Element Model

15.2.1 Simulation Hypothesis

The simulation process is carried out with the help of commercially available Deform 3D V11.2 software package. Three-dimensional orthogonal cutting is performed by considering implicit Lagrangian computational approach for predicting the cutting performances of the ZTA tool. Few assumptions are made in order to simplify the process since high-speed machining is associated with many complex phenomena [7]. The simulation process follows a constitutive mechanical property. The cutting insert is considered as rigid while the workpiece is considered to be plastic. Thus, the elastic deformation can be neglected. Both the insert and workpiece are assumed to be isotropic in nature.

15.2.2 Materials Modelling

15.2.2.1 Modelling of the Cutting Insert

In this present study, a finite element model is developed to execute the cutting performances of the ZTA cutting insert. The insert primitive is designed according to the standard TNMA432 which is predefined in the tool library. The schematic of the insert is given in Fig. 15.1. The detail specifications for the cutting insert are given in Table 15.1. Thermo-mechanical material properties for the ZTA insert are depicted in Table 15.2.

15.2.2.2 Workpiece Selection

Deform 3D software package comprises of a series of machining steels. In this study, AISI 1095 high carbon machining steel is chosen from the workpiece library by following to the Oxley's flow stress model which is given in Eq. (15.1).

Fig. 15.1 Schematic of the TNMA 432 cutting insert

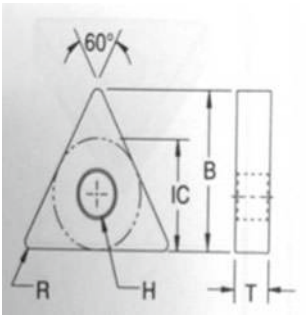


Table 15.1 Specification of the cutting insert

SI no.	Parameters	Values (mm)
1	Inscribed circle diameter (IC)	12.7
2	Inner hole diameter (H)	5.16
3	Corner radius (R)	0.79
4	Thickness (T)	4.76
5	Edge length (B)	18.26

Table 15.2 Thermo-mechanical properties of the ceramic cutting tool

Density “ ρ ” (kg/m ³)	Hardness (HRc)	Elastic modulus “ E ” (GPa)	Poison ratio “ ν ”	Thermal conductivity “ k ” (W/mK)	Specific heat “ C_P ” (J/kgK)	Thermal expansion coefficient (10 ⁻⁶ /K)
4100	80	310	0.26	21	819	8.1

$$\bar{\sigma} = \bar{\sigma}(\bar{\epsilon}, \dot{\bar{\epsilon}}, T) \tag{15.1}$$

where $\bar{\sigma}$ = flow stress, $\bar{\epsilon}$ = equivalent plastic strain, $\dot{\bar{\epsilon}}$ = strain rate and T = temperature.

The selected AISI 1095 steel is considered for machining with the help of the triangular type modelled insert. The material properties and composition are given in Tables 15.3 and 15.4 [8], respectively.

Table 15.3 Thermo-mechanical properties for the workpiece AISI 1095

Density (Kg/m ³)	Hardness (HRc)	Young’s modulus (GPa)	Poison’s ratio	Coefficient of thermal expansion (10 ⁻⁶ /K)	Thermal conductivity (W/mK)	Heat capacity (J/KgK)
7850	13	206	0.3	12	49.8	461

Table 15.4 Chemical composition of the workpiece AISI 1095

Component	Content (%)
Iron (Fe)	98.38–98.80
Carbon (C)	0.9–1.03
Phosphorous (P)	0.04
Sulphur (S)	0.05
Manganese (Mn)	0.3–0.5

15.2.3 Mesh Generation

Deform 3D follows an adaptive mesh generation technique. Relative mesh size of 50,000 is considered for the insert, and absolute mesh size is considered for the workpiece which is 25% of the feed rate. The mesh type is tetrahedral for both the cases. The size ratios are 4 and 7 for insert and workpiece, respectively.

15.2.4 Boundary Condition

Boundary conditions play an important role to develop an appropriate FE model. The boundary conditions at the tool-workpiece contact are 0.6 shear friction factors, 0.02 convection coefficient and 1305 kW/m²K as the global heat transfer coefficient. Starting environmental temperature is considered as 20 °C. The global heat transfer coefficient is calculated using Eq. (15.2) [9, 10].

$$h = 442 - 2.36 V_c - 7950 f + 0.0276 V_c^2 + 40,600 f^2 \quad (15.2)$$

where V_c = cutting speed and f = feed rate. Figure 15.2 shows the isometric view of the tool-workpiece configuration where the directions of cutting, feed and depth of cut are +Y, −X and +Z, respectively.

15.2.5 Simulation Process Setup

Before performing the FE analysis, all the simulation process should be defined appropriately. The cutting conditions, material properties, boundary conditions, operating environment conditions, viz. environment temperature, convection coefficient and heat transfer coefficient are to be given in the pre-processor. CTJNR2020K-16 tool holder is considered to hold the triangular type TNMA 432 cutting insert. The machining parameters, all boundary conditions, workpiece and tool type and tool holder details are depicted in Table 15.5.

Fig. 15.2 Tool-workpiece configuration at the starting point

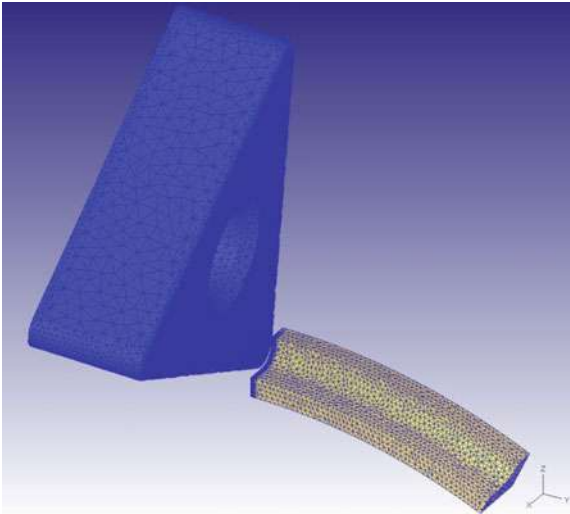


Table 15.5 Simulation conditions

Sl no.	Conditions	Values
1	Shear friction factor	0.6
2	Coefficient of heat transfer	1305 kW/m ² K
3	Convection coefficient	0.02
4	Environment temperature	20 °C
5	Insert mesh generation	No. of elements: 44,384, nodes: 10,270
6	Workpiece mesh generation	No. of elements: 37,366, nodes: 8534
7	Workpiece	AISI 1095
8	Workpiece shape	Curved
9	Workpiece dimension	Diameter: 50 mm
10	Cutting tool	ZTA insert
11	Tool dimension	12.7 × 12.7 × 4.76 mm, 0.8 mm nose radius
12	Tool holder specification	CTJNR2020K-16 Side rake angle: −6°, back rake angle: −6°, side cutting edge angle: −3°
13	Number of simulation steps	1000
14	Step increment to save	25
15	Arc angle to cut	25°
16	Lubricating condition	Dry
17	Cutting conditions	Cutting speed (V_c) = 200 m/min Depth of cut (d) = 1 mm Feed rate (f) = 0.14 mm/rev

15.3 Results and Discussion

15.3.1 Assessment of the Different Forces Associated During Turning

It is important to select the shear friction factor properly in order to get the appropriate values of the forces at the tool-chip interface. A shear friction factor of 0.6 shows optimum result while dry cutting condition is performed. The schematic of the cutting force, axial thrust force and radial force is given in Fig. 15.3. The mean values for all the three forces have been reported in Table 15.6. It is found that the approximate cutting force is 611 N at the tool tip. Fine mesh size provides less fluctuation in the forces generated during the cutting performances. Figure 15.4 shows the good agreement for selected mesh size and friction factor since the fluctuation in the forces at the tool tip is significantly low.

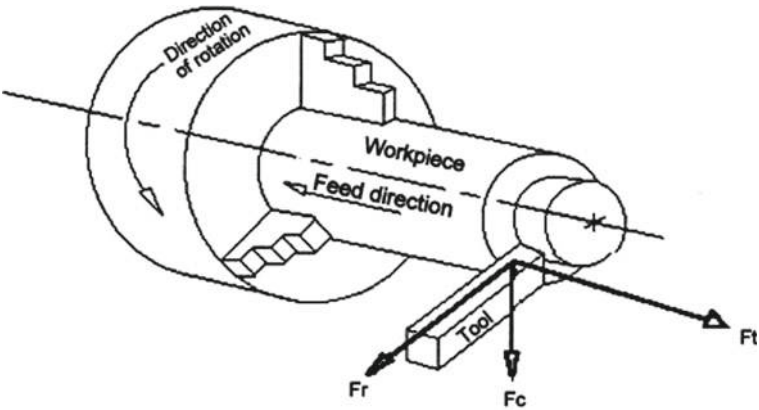


Fig. 15.3 Different forces associated with turning operation

Table 15.6 Different forces associated with the cutting operation

Force	Cutting force, F_c (N)	Axial thrust force, F_t (N)	Radial thrust force, F_r (N)
Value	611	384	444

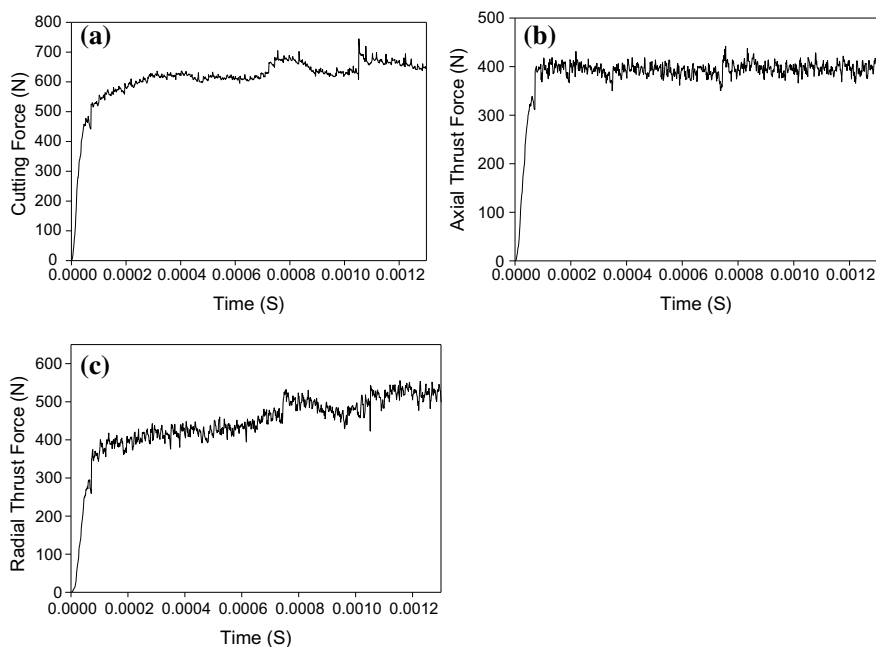


Fig. 15.4 **a** Cutting force, **b** axial thrust force and **c** radial thrust force distribution as a function of time

15.3.2 Temperature Distribution Prediction

15.3.2.1 Temperature Distribution at the Tool Tip

FE analysis is a prime tool to evaluate the temperature regimes in the tool face. Figure 15.5 shows the temperature distribution at the rake face of the cutting insert. It is found that maximum temperature is rose up to $\sim 1060^\circ\text{C}$ at the tool-chip interface.

15.3.2.2 Temperature Distribution at the Shear Zones

It is a challenging practice to measure the temperature at the workpiece deformation zones during the process of machining. With the aid of FE modelling, the shear zone temperature distribution can be predicted readily. It is clearly shown in the Fig. 15.6 that how the temperature is increasing as the primary shear zone shifting towards the secondary zone. During the turning operation, a major portion of heat is being transferred via chip flow. A total 20 number of different nodal points are selected for evaluating the temperature, stress and strain rate at the primary and secondary deformation zones. The average values are reported in Table 15.7. The

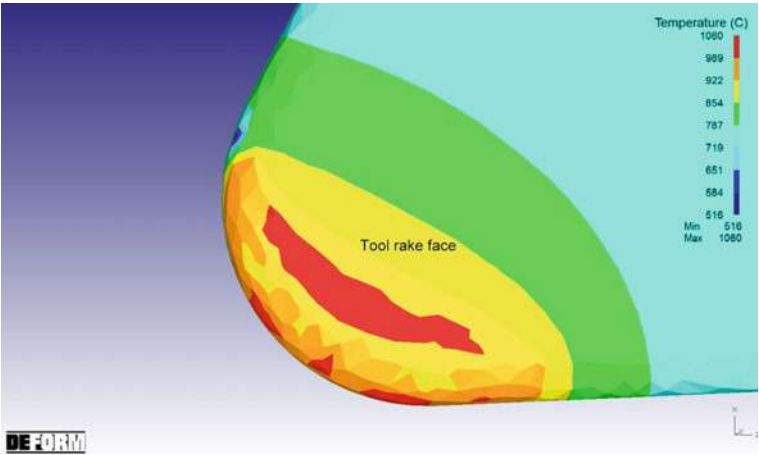


Fig. 15.5 Temperature distribution profile at the tool rake face

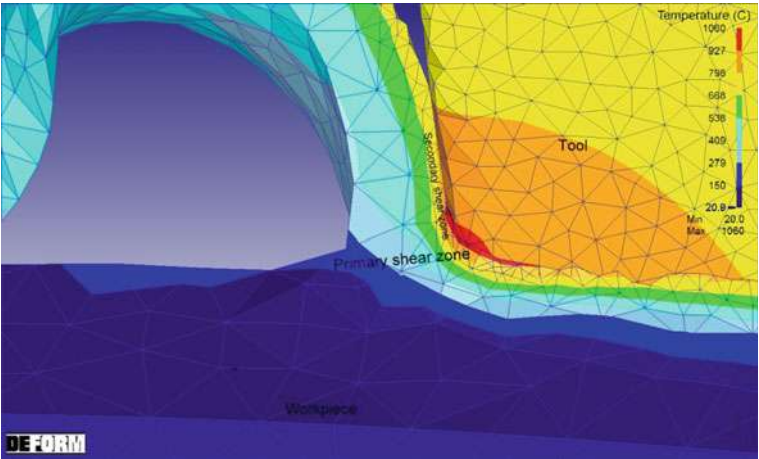


Fig. 15.6 Temperature distribution at the deformation zones

Table 15.7 Average effective stress, strain rate and temperature at the deformation zone

	Effective stress (MPa)	Effective strain rate (S^{-1})	Temperature ($^{\circ}C$)
Primary shear zone	914	12,091	568
Secondary shear zone	500	5804	772

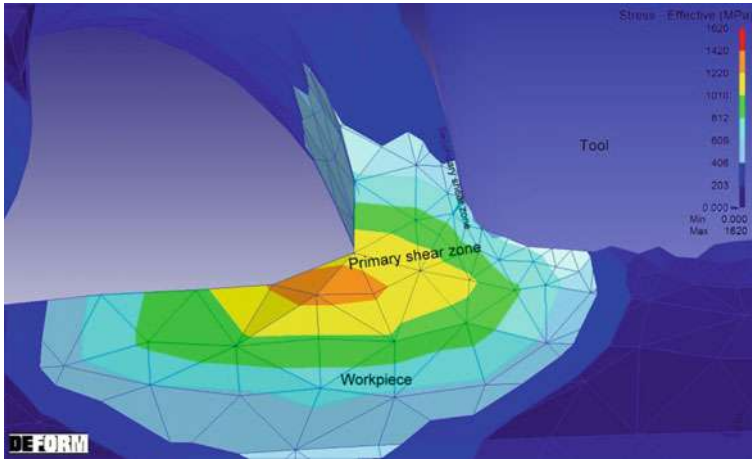


Fig. 15.7 Effective stress distribution at the deformation zones

temperatures at the primary and secondary shear zones are found to be ~ 568 and ~ 772 $^{\circ}\text{C}$, respectively.

15.3.3 Prediction of the Effective Stress and Strain Rate Distribution at the Shearing Zones

The average value of effective stress and strain rate is calculated at both the primary and secondary shear zones by picking the nodal points as stated earlier. The values are reported in Table 15.7. Figures 15.7 and 15.8 show the stress and strain rate distribution at the deformation zone as a function of time after solving 1000 steps. Both the stress and strain rate levels at the primary deformation zone (~ 914 MPa, $\sim 12,091$ S^{-1}) are much higher than the secondary deformation zone (~ 500 MPa, ~ 5804 S^{-1}).

15.3.4 Estimation of Material Removing Rate

Material removing rate (MRR) is an important factor in order to estimate the quality and productivity of the machining operations. Deform 3D is a substantial tool to evaluate the MRR at different level of cutting conditions [11]. The MRR during the machining of AISI 1095 steel is given in Fig. 15.9. An approximate value of

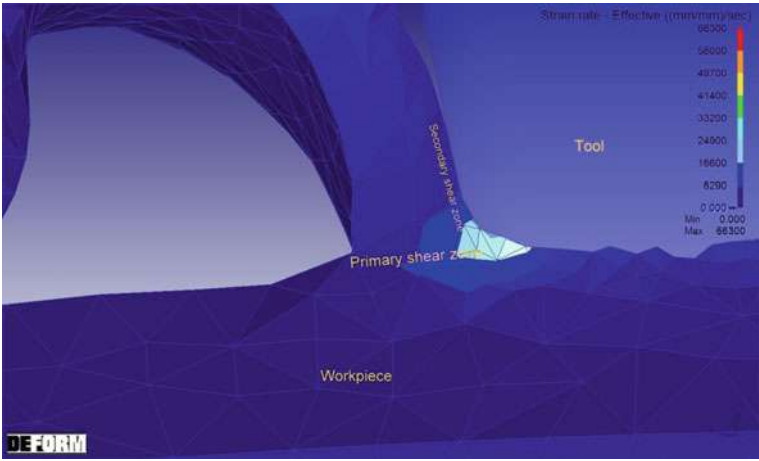
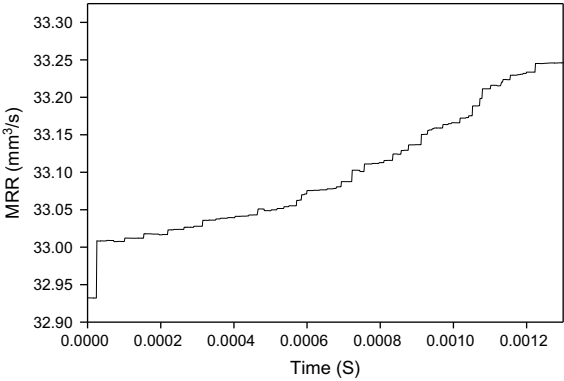


Fig. 15.8 Effective strain rate distribution at the deformation zones

Fig. 15.9 Variation in material removing rate as a function of time



33 mm³/s is found as the MRR at the specified cutting conditions ($V_c = 200$ m/min, $d = 1$ mm and $f = 0.14$ mm/rev) after solving 1000 steps.

15.4 Conclusion

A three-dimensional finite element model has been established for the proposed triangular type zirconia toughened alumina cutting insert in machining of AISI 1095 steel. The following conclusions have been drawn:

- a. FE analysis is found as a suitable approach to predict the cutting performances for the proposed cutting insert.

- b. The tangential cutting force at the tool tip is ~ 611 N, while the cutting conditions are $V_c = 200$ m/min, $d = 1$ mm and $f = 0.14$ mm/rev.
- c. 1060°C is the maximum level of temperature that is been encountered at the tool tip.
- d. The average effective stress and strain rate at the primary deformation zone is calculated as 914 MPa and $12,091\text{ S}^{-1}$ respectively.
- e. The material removing rate is found to be $\sim 33\text{ mm}^3/\text{s}$ at the same specified cutting conditions.

Acknowledgements The authors would like to express their sincere acknowledgment for supporting the work towards Nanomission project (SR/NM/NT-1062/2015), Department of Science and Technology (DST), Govt. of India.

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Chapter 16

FEM Approach to Predict Three Jaw Chuck Stiffness and Its Effect on Gripping Force for High Speed Turning and Experimental Verification



K. S. Karthik , Aslam Pasha Taj and S. R. Chandramouli

Abstract Higher safety norms and precision machining has pushed machine tool manufacturers to build high-speed machines with reliable work-holding devices. With advances in bearing manufacturing techniques and easy availability of precision roller bearing, hydrostatic bearings, active magnetic bearings and efforts are reduced to manufacture high speed and high precision spindle, leaving workpiece clamping subsystem as the weakest link. Popular work-holding device in lathe is a power-operated three-jaw chuck because of its self-centering properties. However, the problem with power-operated three jaw is the loss of gripping force at high speeds due to large centrifugal forces that act on three-jaw chuck. This loss in gripping force makes the machine operation detrimental in terms of safety of operator/machine as well as the accuracy of machined components due to loss of stiffness at the work side. It will be advantageous if the stiffness behavior of the work holding is known for the operating range. Though the supplier provides speed versus gripping force plot, this not sufficient as the measurement is done for ideal work holding diameter. The speed versus gripping force plots varies for different static gripping forces as well as for different holding diameters as the stiffness varies due to jaw positioning for different diameters. This paper proposes finite element method to predict the loss in gripping force/stiffness due to high spindle speed for various holding diameters. The finite element results are verified with experimental results.

Keywords High speed turning · Three jaw chuck · Finite element analysis of chuck · Gripping force

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_16

16.1 Introduction

In recent times, demand for precision manufacturing is very high; the components machined should conform to closer dimensional, positional accuracies and form tolerances in order to have an edge over competitor's products. Modern mechanical engineering is advancing in high-speed cutting technology. High speeds up to 30 m/s with high durable cutting tools are required, in order to meet these demands innovative components like high-frequency motor spindle, thermal error compensation, linear drive motor, etc., have been greatly developed. Machine tool spindle plays an evident role in machining, its rigidity, rotational accuracy and its critical speeds are crucial for producing precision components [1].

Apart from spindle, work holding devices are equally contributing to the accuracy of component produced. Clamping forces on cylindrical surfaces are the main contributing factors for the resulting accuracy because it induces an elastic deformation on the workpiece leading to triangularity form error after declamping. Hence measuring and controlling the gripping force, especially for the thin-walled workpiece, is very important [2].

Power-operated chucks are commonly used work holding devices. Figure 16.1 consists of master jaws, which are wedge-driven for clamping and jaws mounted on a serrated face of master jaws using tenon and bolt, these serrations avoid slipping of jaws at higher speeds. Failure of a clamping system causing inadequate clamping can be hazardous for machine and operator. Hence analytically predicting loss of clamping force is of high importance. Externally, clamped jaws are prone to slip due to centrifugal force acting on jaws. These forces increase with radius of rotating mass w.r.t axis of rotation and square times the rotational speed. The gripping force loss should be less than two-third of the initial static gripping forces [3]. At high

Fig. 16.1 Cross section of hydraulic power chuck

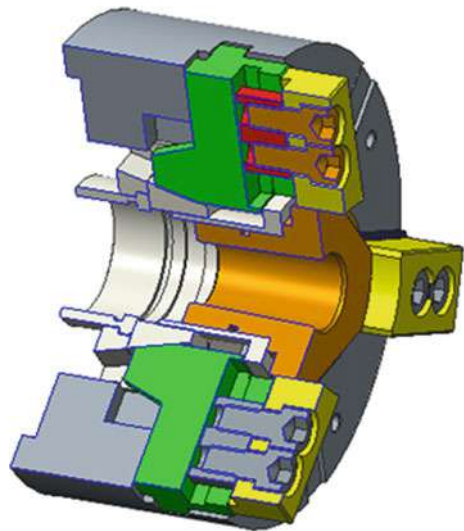
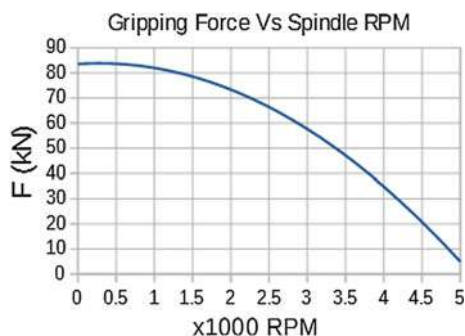


Fig. 16.2 Gripping force versus spindle rpm



rotational speed, power chucks must clamp workpieces reliably. The main factor limiting the rotational speed of power chucks is the strength failure of chuck parts and insufficient gripping force, owing to centrifugal gripping force of chuck jaws. Therefore, the insufficient static gripping force and dynamic loss of gripping force are of main concern.

There is no straightforward solution for this due to complex characteristics such as contact condition between chuck-jaw and workpiece-jaw. Contact condition and gripping force has huge effect during initiation of chatter vibration during machining. It has been observed that the gripping force decreases with increase in fluctuation of chucking, especially, this trend appears remarkably in the presence of chatter vibration [4].

The decay of gripping force is a quadratic function of spindle speed rpm, and affected by configuration, mass and elastic properties of chuck and workpiece [5], a typical plot experimental plot of gripping force versus spindle speed is as shown in Fig. 16.2.

The gripping force loss not only varies with static gripping force value, i.e., gripping force at 0 RPM but also diameter of the workpiece. The reason being, change in contact area between serrations of master jaw and jaw. The loss is due to the elastic deformations of master jaw, tenon, bolts and jaw.

Typically, in order to compensate for the loss of centrifugal forces, practices like adding counter weights to increase the clamping force during rotation, use of lighter materials for the jaws like aluminum are adopted [6–8].

Previous researchers worked extensively on establishing measuring technique [9] and finite element analysis technique, to measure and predict gripping force loss at various speeds [10]. The main objective of finite element analysis is to study the variation of clamping force at different holding diameters and at various rotational speeds. In present paper, finite element model is prepared and results are compared with experiments.

Fig. 16.3 Experimental setup



16.2 Determination of Gripping Force by Experiments

16.2.1 Experimental Setup

Figure 16.3 shows typical experimental setup for gripping force loss measurement, 500 mm diameter power chuck is considered for experiments with work holding diameter of 105 mm. The gripping force testing equipment consists of (i) load cell with adapter of various diameter to measure force exerted by the jaws (ii) tachometer to measure spindle speed mounted on independent stand (iii) hand-held instrument to view the force and speed details. The signals from load cell are transmitted to the hand-held instrument wirelessly.

16.2.2 Testing Procedure

Testing procedure carried out to emphasize on the effect of different chucking pressures and different rotational speeds (tested up to 2000 rpm with the intervals of 100 rpm).

Step 1: Jaws adjusted to accommodate diameter of 105 mm, clamp load cell in chuck

Table 16.1 Results of experiments conducted for gripping force loss for three different chucking pressures

Trail no.	Chucking pressure (bar)	Static gripping force (kN)	Dynamic gripping force (kN @2000 rpm)	Loss in griping force (kN)
1	30	116	89.6	26.4
2	20	81.9	55.5	26.4
3	15	57.3	30.3	27

- Step 2: Readings recorded at static condition (0 RPM) for different chucking pressures
- Step 3: Spindle rotated up to 2000 rpm in intervals of 100 RPM and recorded readings at each interval.

Table 16.1 shows three sets of readings for 30, 20 and 15 bar chucking pressures for holding diameter of 105 mm.

16.3 Determination of Gripping Force by Finite Element Analysis

16.3.1 Geometry

Figure 16.4 shows assembly of 500 mm diameter chuck considered for analysis. CAD model imported in.stp format. The unwanted features such as holes, fillets and chamfers are removed for better control of mesh. For in-depth analysis to predict loss in gripping force, part of the chuck body, master jaw, jaw, tenon and two bolts (which connects master jaw and jaw through tenon) are considered.

Fig. 16.4 Full model chuck

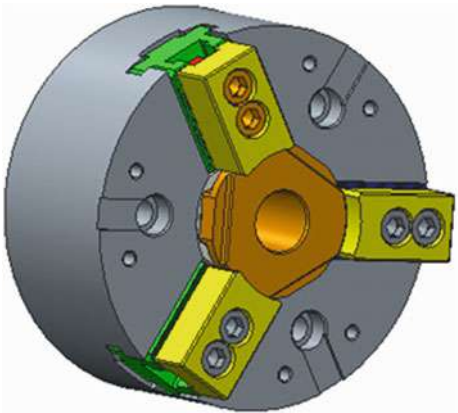
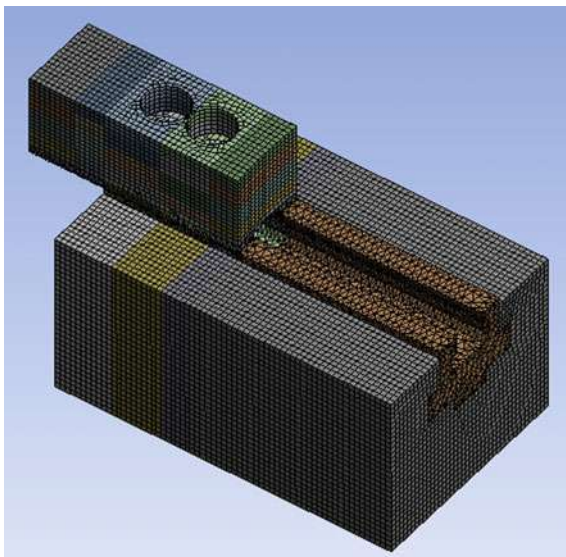


Fig. 16.5 Meshed model of axis symmetric model taken for analysis



16.3.2 Mesh Generation

For detailed analysis, regions of importance such as interaction between serrations of master jaw and jaw, tenon and bolt are meshed using controlled features. All other regions are meshed with default meshing option (Fig. 16.5).

16.3.3 Load Steps

Loads in finite element applied are in three steps to capture the force reactions between master jaw and top jaws for bolt pretension, power-actuated clamping and rotational speed.

- Step 1: Bolt pretension: M20 tenon bolts for tightening and engaging of jaw in the serrations applied in load step 1. Applied pretension = 103 kN per bolt (Fig. 16.6).
- Step 2: In addition to bolt pretension, hydraulic clamping force of 115 kN is applied at the surfaces in contact with workpiece. This force is equivalent to force exerted by chucking cylinder at 30 bar pressure when tested practically on machine. Refer Table 16.1 (Fig. 16.7).
- Step 3: In addition to bolt pretension and hydraulic clamping force, centrifugal force applied by rotating chuck assembly at speed of 2000 rpm (Fig. 16.8).

Fig. 16.6 Load step 1: bolt pretension

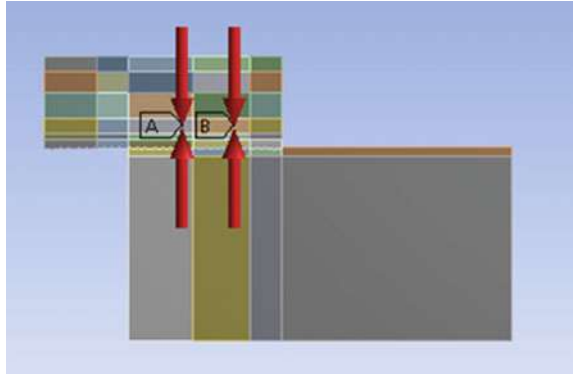
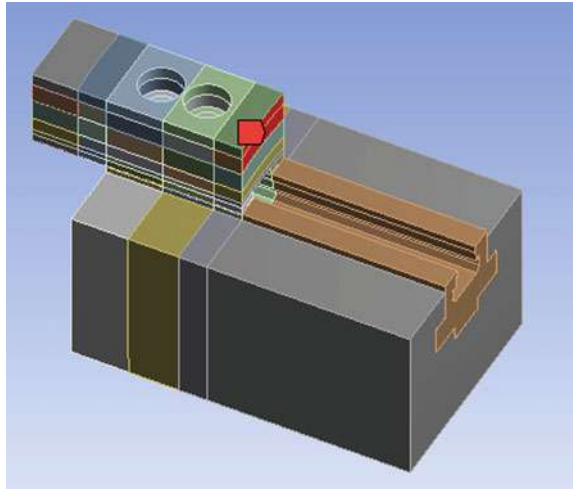


Fig. 16.7 Load step 2: power-actuated clamping



16.3.4 Finite Element Analysis Results

Finite element analysis carried out for two extreme positions of the jaw. For each position, analysis is carried out three load steps as explained in Sect 16.3.3.

Position 1: Holding diameter 300 mm. For this holding diameter, tenon is located at extreme back end. Half of the jaw is unsupported at this position. Here, only 35 serrations are in contact. Refer Fig. 16.9a.

Position 2: Holding diameter 105 mm. Tenon is located at middle of the chuck, and jaw is fully supported by the master jaw. Serrations are identified by giving numbers as shown in Fig. 16.9b. Here, around 90 serrations are in contact. Finite element results are as tabulated in Table 16.3 for 2000 rpm (Table 16.2).

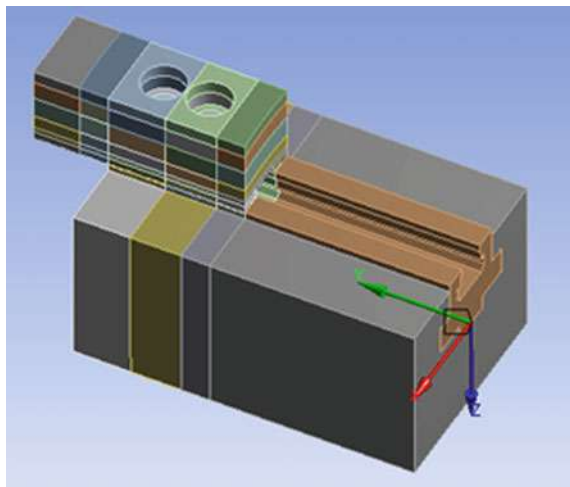


Fig. 16.8 Load step 3: rotational speed about z-axis

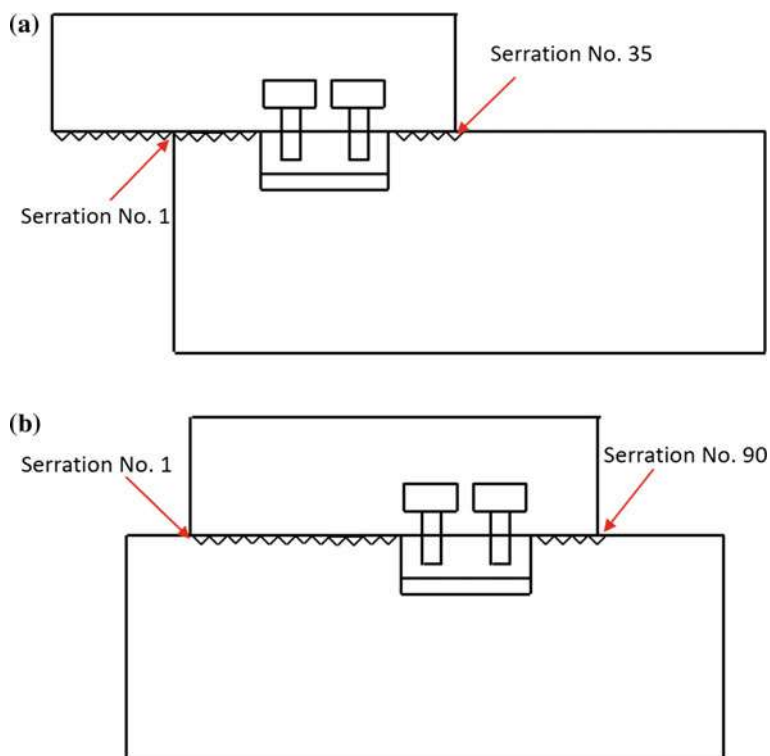


Fig. 16.9 **a** Serration number considered for analysis position 1. **b** Serration number considered for analysis position 2

Table 16.2 Comparison of finite element results for position 1 and position 2

Force reaction (kN)	Position 1 (Dia. 300 mm)	Position 2 (Dia. 105 mm)
Static clamping	70.98	68.04
Dynamic clamping at 2000 rpm	84.25	77.24
Gripping force loss per jaw	13.27	9.20
Total gripping force loss for three jaws	39.81	27.60

16.4 Results and Discussion

16.4.1 Observation

Co-ordinate system considered for post-processing is as shown in Fig. 16.10a, b. Jaw serrations are not equally loaded for the entire length Fig. 16.10a. The teeth in front loose contact due to elastic deformations of jaws when clamped statically, these deformations further increase in dynamic condition, i.e., when rotated at higher rpm's due to outward centrifugal force.

Loss of serration contact is substantially more in position 1 compared to position 2. This loss of serration contact is directly proportional to loss of clamping forces. Clamping forces are calculated by addition of force reactions on serrations of master jaws in Fz direction. Force reaction, Fz plotted against serration number for three forces, i.e., bolt tension, static clamping and rotational speed. It can be observed from Fig. 16.11 that tenon and bolt ensure positive butting of the serrations hence distributes load due to static bolt tension uniformly over the length.

From Fig. 16.11, it can be clearly seen that only few serrations get loaded after applying static clamping force and rotational speed. The maximum static reaction force is experienced at serration no. 9 (4.19 kN) and minimum at serration no. 35, this is due to loss of contact between jaw and master jaw see Fig. 16.10a. Similarly, when rotated at 600 rpm, the centrifugal force reaction on serration is maximum at serration no. 9 (4.38kN) and minimum at serration no. 35. Difference of 0.19 kN is due to outward centrifugal force experienced.

Figure 16.12 shows the force reaction Fz on serrations for bolt pretension and different holding diameters. For 300 mm diameter holding, only few serrations are loaded, i.e., 0–15 serrations and serration no. 15–30 are loaded with minimum force, whereas for 105 mm holding diameter all the serrations are uniformly loaded (refer ploy. Dia 105, trend line of 3rd order polynomial). This is because half of the jaw is unsupported for 300 mm holding diameter hence experiencing bending loads in tenon and bolts, which deforms elastically; due to this deformation, there is a loss of contact between master jaw and the jaw.

Fig. 16.10 **a** Lifting of jaw when clamped for position 1.
b Lifting of jaw when clamped for position 2

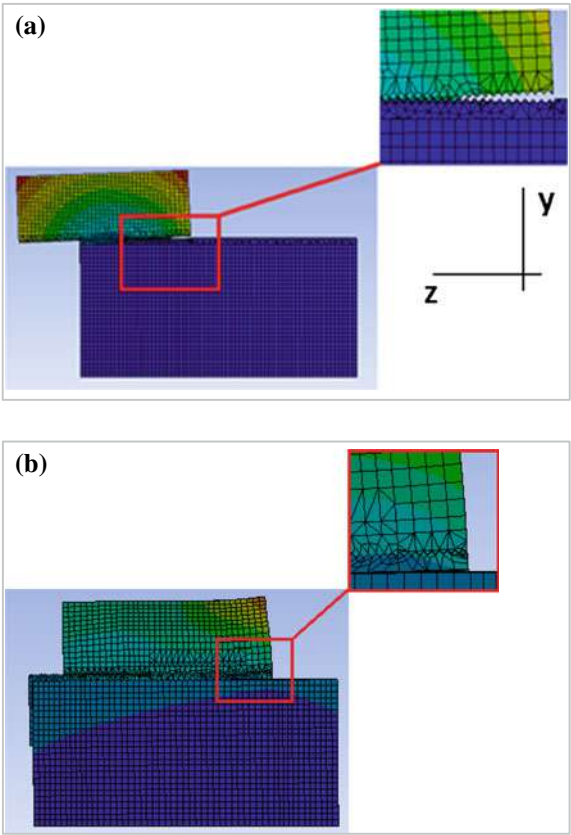
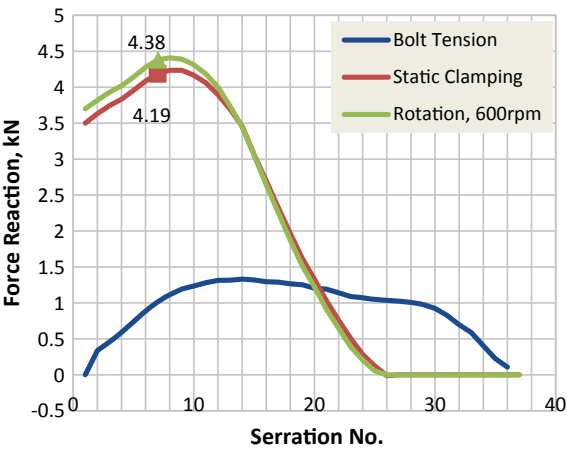


Fig. 16.11 Force reaction Fz versus serration no for position 1



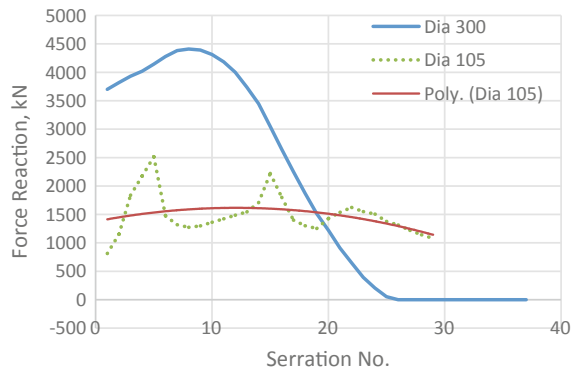


Fig. 16.12 Force reaction F_z for bolt pretension versus serration no. for position 1 and position 2

Figure 16.13 explains loss of gripping force per jaw for speed ranging from 0 to 2000 rpm. It can be clearly seen that the gripping loss not only depends on speed but also holding diameter.

Table 16.3 shows comparison of results between finite element analysis and experiments for 105 mm holding diameter with 5% deviations.

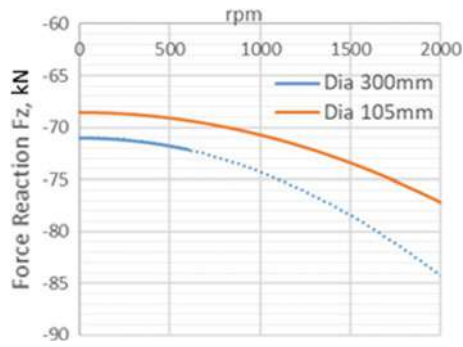


Fig. 16.13 Gripping force loss

Table 16.3 Comparison of finite element method and experimental at 2000 rpm for 105 mm holding diameter

Sl. no.	Method	Predicated loss (kN)
1	Experiments	26.40
2	Finite element method	27.60

16.5 Conclusion

In this present work, gripping force loss of a three-jaw chuck is calculated using finite element analysis, and FEA model is validated through experimentation. This paper mainly emphasizes on the effect of holding diameter and rotational speed on gripping force loss. Larger holding diameter and higher rotational speed lead to higher gripping force losses. Finite element results and experiments results for 105 mm holding diameter are in good agreement with only 5% deviations. Finite element model of chuck assembly is accurate enough and can be used to calculate gripping force loss for the combinations of different holding diameters and rotational speeds.

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Chapter 17

Experimental Investigation and Numerical Analysis of Thermal Fields and Residual Stresses in Multi-pass GTA Welding of AA 6061T6 Plates



Narender Kumar and H. Chelladurai

Abstract Welding is one of the most widely used materials joining processes in the industries. Plates of different thicknesses used for the fabrication of components can be welded using multi-pass welding, depending upon the applications. However, residual stresses are induced in the welded joints due to the rapid heating and cooling, which leads to inhomogeneous distribution of dimensional changes and consequently the failure of welded joint occurs. This manuscript aims to predict temperature distribution and residual stresses during multi-pass butt joint on gas tungsten arc welding (GTAW) of aluminum alloy (AA) 6061T6 weldments. Transient thermal analysis and mechanical stress contour in three dimensions have been estimated considering three modes of heat transfer, i.e., conduction, convection, and radiation. Temperature-dependent properties such as thermal conductivity, heat capacity, yield stress, elastic modulus, and thermal expansion are employed in the welding simulations. The experimental results of temperature distribution in AA 6061T6 weldments are validated using ANSYS 18.1.

Keywords Multi-pass welding · GTAW · Numerical simulation · AA 6061T6 plates

17.1 Introduction

Welding is one of the most used joining processes in aerospace industries, ship industries, automotive industries, etc. During different stages of welding, i.e., heating and cooling, differential weld thermal cycle is experienced by weld metal which has the consequences of the residual stresses in welded joints in region close to the fusion boundary, i.e., heat-affected zone causes post-weld deformation of the welded

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_17

structure. Residual stresses induced in weldments are mainly due to non-uniform distributions of plastic and thermal strains. As temperature of the plate increases, the yield strength of the material decreases, and thermally induced residual stresses increase.

Plates of different thicknesses are used for the fabrication of components, depending upon the applications. Multi-pass welding is adopted when plate thickness exceeds the limitation of two-pass welding technique.

Different authors work on the prediction of thermal cycles and residual stresses of different materials. So, this paper focuses on the analysis of temperature distribution and residual stresses experimentally and numerically during multi-pass GTA welding of AA 6061T6 having 6 mm thickness. Fluke thermal camera 560 is used to measure the temperature distribution, and numerical analysis is done on ANSYS 18.1 workbench software.

Manurung et al. [1] investigated the effect of welding sequence induced angular distortion and thermal stress distribution on multi-pass weld for low manganese–carbon steel. Authors carried out different welding experiments to validate the results. The first welding sequence was carried out from inside to outside and the second sequence from outside to inside for combined butt and T-joint. According to the simulation, there was not much difference in two sequences. The angular distortion for welding sequence one was less than that for welding sequence two.

Bajpei et al. [2] performed experimental analysis on temperature distribution, longitudinal and transverse residual stresses, and distortions in joining two thin dissimilar aluminum alloys AA5052 and AA6061 plates. Three-dimensional thermo-mechanical finite element model was used to determine transient temperature, residual stresses, and distortions. X-ray diffraction machine and coordinate measuring machine were used to validate the simulation results. Table 17.1 shows properties of AA 6061T6 grade aluminum at different temperatures.

Jiang et al. [3] studied multi-pass gas welding for Al 5083 alloy plate with thickness of 30 mm. The main aim of this study was to study the effect of the thermal cycle on metallurgical characteristics and mechanical properties of the material being welded. It was observed that multiple thermal cycles contributed an increase of precipitate particles which resulted in the strengthening of the welded joint. Zubairuddin et al. [4] did the numerical simulation for multi-pass welding of grade 91 steel. Study of multi-pass GTA welding of 6 mm thick grade 91 steel plate was carried out using three different models which include a 2D model, a 3D coarse meshed model, and a 3D fine-meshed model. It was observed from both simulations and numerical study that preheating of grade 91 steel plate up to 200° helps in reduction of distortion. It has been cited in various literatures [5–8] that there are limited data available in the area of temperature distribution during multipass welding of plates along with the residual stress analysis. The novelty of this experiment is that it will guide in enhancing the strength of the structure by performing certain post weld heat treatments.

Table 17.1 Temperature-dependent thermo-mechanical properties of AA6061

Temperature (°C)	Density (kg/m ³)	Specific heat (KJ/kg °C)	Therm. conductivity (W/m °C)	Coefficient of thermal expansion (μ/°C)	Young's modulus (GPa)
25	2700	0.896	167	22	68.9
37.8	2685	0.92	170	23.45	68.54
93.3	2685	0.978	177	24.61	66.19
148.9	2667	1.004	184	25.67	63.09
204.4	2657	1.028	192	26.6	59.16
260	2657	1.052	201	27.56	53.99
315.6	2630	1.078	207	28.53	47.48
371.1	2620	1.104	217	29.57	40.34
426.7	2602	1.133	223	30.71	31.72

Bajpei et al. [2]

17.2 Materials and Experimental Methods

17.2.1 Welding Conditions

Two aluminum alloy 6061T6 plates each of size 150 mm × 100 mm × 6 mm are butt welded using manual GTA welding. For filling the material in the weld bead zone, ER-4043 filler wire having diameter 1.2 mm is used. For shielding purpose, argon gas with 99.99% purity and flow rate 12 L/min is used. To remove dust, grease, and oil contaminants, the plates are cleaned with acetone. The chemical composition of alloy and filler material is listed in Table 17.2.

On the basis of trial experiments and the literature review, the following welding conditions and welding parameters are used for the experimentation which is enlisted in Table 17.3.

The base metal sheets of the required size are cut on the shearing machine, and a single V-groove butt joint with groove angle 60°, root face 1.5 mm, and root gap 1.5 mm is prepared on shaper machine for better penetration for multi-pass weldments as shown in Fig. 17.1. Time lapses between two consecutive passes are around 2 min in experiments and simulation parts.

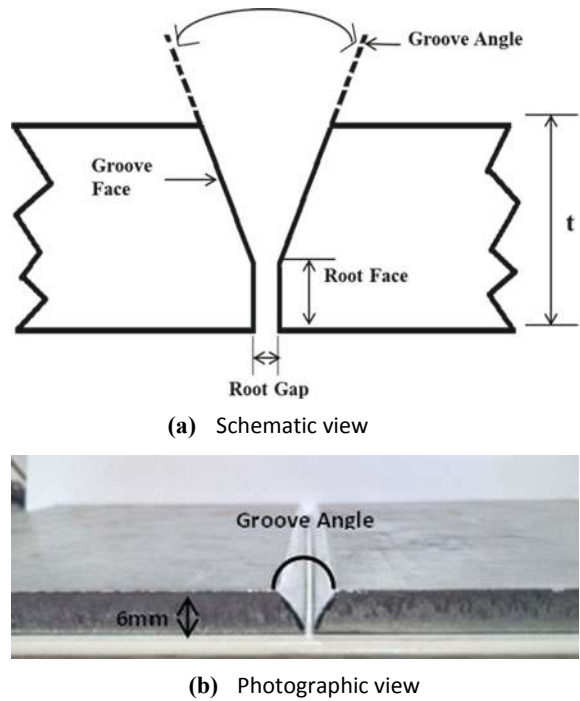
Table 17.2 Temperature-dependent thermo-mechanical properties of AA6061

Elements	Mg	Si	Cu	Cr	Mn	Zn	Ti	Al
AA6061T6	0.085	0.68	0.22	0.06	0.32	0.07	0.05	98.52
ER-4043	0.05	4.80	0.17	0.05	0.24	0.05	0.05	94.59

Table 17.3 Welding parameters used in analysis

Number of passes	3
Welding voltages (V)	24
Welding current (A)	170
Weld speed (mm/s)	2.5
Power of arc (W)	3060
Argon flow rate (L/min)	12

Fig. 17.1 Configuration of V-groove butt joint with groove angle 60°



17.2.2 Temperature Measurement

For determining the temperature fields, an infrared Fluke thermal camera 560 is used as shown in Fig. 17.2. An infrared thermal camera also called thermographic camera is a device which makes images using IR (infrared radiation) similar to a common camera which makes images using visible light. As in the visible light camera, wavelength range is 400–700 nm, and in infrared cameras, range is up to 14,000 nm. In this study, emissivity of aluminum alloy is set as 0.95 and temperature range up to 1500 °C.



Fig. 17.2 Fluke thermal camera 560 (<http://www.fluke.com>)

17.3 Finite Element Analysis

To develop a model for the GTAW process, different process parameters such as the number of steps and sub-steps, the weld speed, deposition of filler material, and material nonlinearities have been considered. Following assumptions with regard to the finite element, thermal-structural analysis is considered:

1. Both convection effects due to argon and air, along with radiation phenomenon have been considered.
2. Finite element death and birth procedure is used for simulating the filler metal deposition during the welding process.
3. Temperature-dependent properties have been considered for analysis as shown in Table 17.1.

17.3.1 Thermal Analysis

In thermal analysis, the finite element formulation based on following partial differential equation is used for transient heat conduction analysis in a three-dimensional body

$$\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q}_g = \rho c_p \frac{\partial T}{\partial \tau} \quad (17.1)$$

where $T(x, y, z, \tau)$ is the temperature; x , y , and z represent the welding direction in longitudinal, transverse, and thickness direction; τ is time; ρ is the density; c_p is specific heat; and k_x , k_y , k_z are the thermal conductivities in x -, y -, and z -direction, respectively.

The heat-flow density for convection (q_c) in the environment is given by Newton's heat transfer law, as follows:

$$q_c = \rho(T)h_c(T - T_0) \quad (17.2)$$

where T is temperature of surface of the plate, T_0 is ambient temperature, and h_c is the coefficient of convective heat transfer.

The heat-flow density for radiation q_r is given by Stefan–Boltzmann law, as follows:

$$q_r = \varepsilon_r \sigma_r (T^4 - T_0^4) \quad (17.3)$$

where ε_r is the emissivity of material surface whose value is considered as 0.3, and σ_r is the Stefan–Boltzmann constant whose value is $5.67 \times 10^{-8} \text{ W/m}^2$.

17.3.2 Residual Stress Analysis

For analysis of residual stresses during multi-pass GTA welding process, thermo-elastic-plastic model based on von Mises yield criteria is adopted. As the properties of a material depend upon the temperature history, the resulting stresses and strains depend upon the path. The equivalent stress is given as follows:

$$\sigma_v = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \quad (17.4)$$

The resulting stresses and strains are calculated by employing incremental stress–strain relationship as follows:

$$d\dot{\varepsilon}_{ij}^T = [\{d\dot{\varepsilon}_{ij}^E\} + \{d\dot{\varepsilon}_{ij}^P\} + \{d\dot{\varepsilon}_{ij}^{TH}\}] \quad (17.5)$$

where $d\dot{\varepsilon}_{ij}^E$ is elastic strain, $d\dot{\varepsilon}_{ij}^P$ is plastic strain, and $d\dot{\varepsilon}_{ij}^{TH}$ is thermal strain.

17.4 Results and Discussion

17.4.1 Thermal Variations

The transient temperature cycles in plates during welding are extracted with the help of Fluke thermal camera at various points to validate with the simulation results. The higher value of temperature is obtained near the welding torch. As the measuring point moves away from welding torch and center of weld line, the temperature value

Table 17.4 Comparison of results of temperature distribution

Distance from center line (mm)	Temperature (°C) (From simulation)	Temperature (°C) (From experiment)
5	406	397.5
13.5	205	193.2
21.5	162	177.89

Fig. 17.3 Temperature distribution at fusion zone after third passes by thermal camera



decreases as shown in Table 17.4 which shows the comparison between the experimental value and numerical value at various points measured from the center line of the weld bead.

Figure 17.3 shows the transient temperature distribution in plates as GTAW heat source travels along weld direction. The peak temperature during the third pass reaches 751.5 °C during experimentation which indicates melting of both the plates.

Temperature distribution by numerical analysis is calculated in ANSYS 18.1 software as shown in Fig. 17.4. The maximum temperature value reaches 785 °C in fusion zone after the third pass of welding shows the proper fusion of material.

Temperature variation with respect to time at various points at the middle of the plate at distances 2, 5, 13.5, and 21.5 mm is calculated by simulation and found that the temperature value increases with time and maximum value reaches when torch crosses the point location. The three peaks shown in Fig. 17.5 indicate the maximum temperature during three passes of GTA welding process.

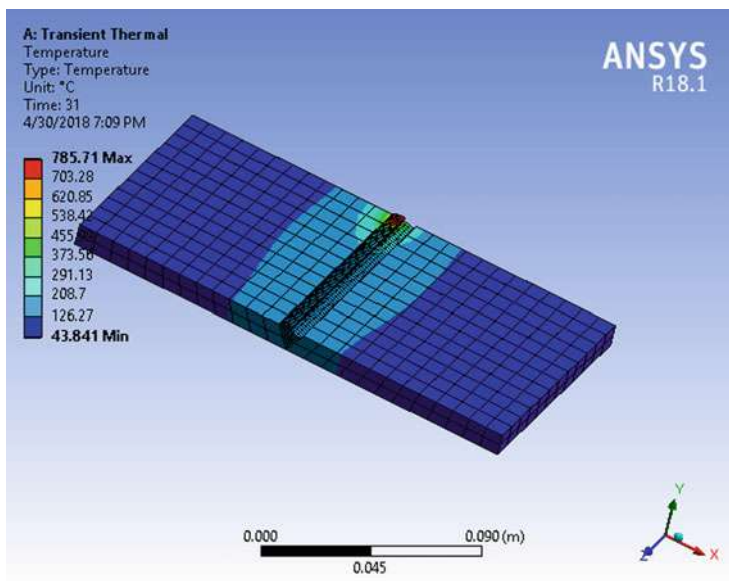


Fig. 17.4 Temperature distribution at fusion zone after third passes by simulation

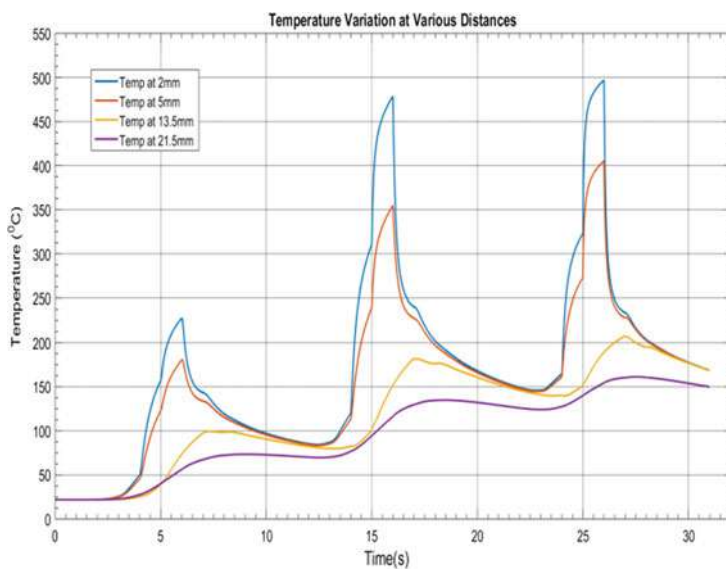


Fig. 17.5 Temperature variation with respect to time

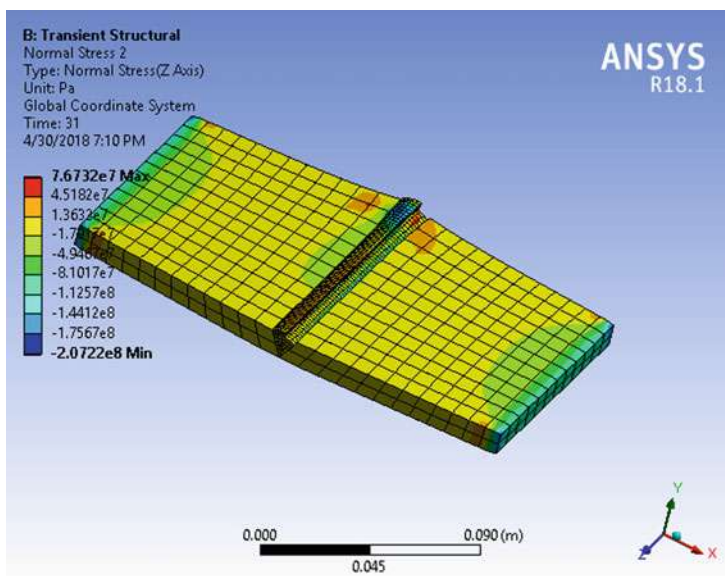


Fig. 17.6 Longitudinal stresses in z-direction

17.4.2 Residual Stresses

Residual stresses are highly sensitive to transient temperature distribution, which is related to total heat applied and heat distribution patterns within the weld domain. Hence, for determination of realistic temperature profile, careful and accurate thermal analysis is required. After performing the transient thermal analysis, the obtained results are imported in order to do transient structure analysis. From results, it is observed that tensile residual stresses were produced near the weld zone, and their magnitude decreases when moving away from the weld zone. High tensile stresses are induced in the weld vicinity because the contraction of weld metal is restricted by parent material and due to the presence of clamps. Also, it is found that the magnitude of longitudinal stress is higher than transverse stress as shown in Figs. 17.6 and 17.7.

17.5 Conclusions

The present research article emphasis on the study of the temperature distribution and residual stresses by GTA welding. The temperature distribution after GTA welding on the sample AA6061T6 is measured using the Fluke thermal camera 560. The numerical simulation is carried out using ANSYS 18.1 software and results are validated with the experiment. The average deviation between experimental and

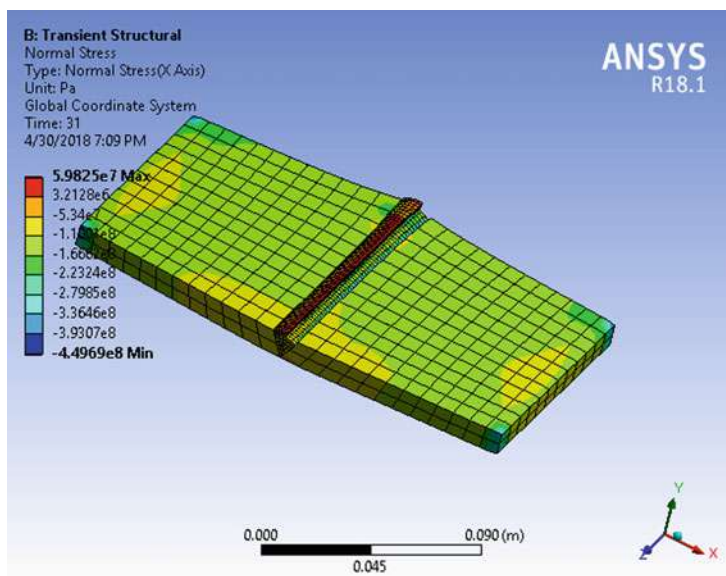


Fig. 17.7 Transverse stresses in x -direction

numerical result lies within a range of 5–15%. It is estimated that the maximum value of the temperature measured at 5 mm from center line of weld during experiment and simulation is 397 °C and 406 °C respectively. The maximum values of longitudinal and transverse stresses during GTA welding are around 76.63 MPa and 59.8 MPa.

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Chapter 18

Effect of Johnson–Cook Material Model Constants on Predicted Chip Morphology and Forces in FE Simulations of Machining Operation for 93% WHA Alloy



Chithajalu Kiran Sagar, Amrita Priyadarshini and Amit Kumar Gupta

Abstract Tungsten heavy alloys (WHAs) with W content 90–95% possess a good combination of high tensile strength as well as high density, thus finding wide applications as counterweights and ballast, radiation shielding, ballistic penetrators, vibration-damped tooling and sporting goods. However, these properties make machining of WHAs to desired dimensions and finish very difficult. A proper understanding of the mechanism of chip formation during machining is surely required that helps in finding the right combination of cutting parameters for achieving higher productivity and better finish. Finite element (FE) simulations help understand the chip formation mechanism with minimum number of experiments. The basic purpose of the current work is to develop an FE model by taking into account three different sets of JC model constants and compare the predicted output variables with experimental machining tests available in the literature.

Keywords Johnson–Cook model · Machining · Cutting forces · Chip morphology · Finite element modelling

18.1 Introduction

Tungsten heavy alloy (WHA) is one of the best choices for use in radiation shielding, counterweights and ballast, ballistic penetrators, vibration-damped tooling, etc. WHAs have 90–98 weight per cent tungsten that majorly contributes for high density. High density in WHAs makes it unique and most suitable for various aerospace applications where compact, precisely positioned concentrations of mass are critical for stability and smooth operation of flight controls and engines. But the properties such as density, high elastic modulus and toughness that make the alloy unique can

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_18

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pose machining challenges. To overcome such challenges, there is a need for proper understanding of the mechanism of chip formation during machining. This would help in finding the right combination of cutting parameters for higher productivity. Finite element method (FEM) is one such approach that helps in reducing the number of experimental tests considerably which are nevertheless required both for better understanding and optimization of the process, thus saving much of the material, money, time and effort. However, depicting the material behaviour of material undergoing plastic deformation accurately during machining process is very crucial for successfully simulating machining processes. In typical machining processes, the work material is subjected to shearing and severe plastic deformations, wherein the strain may reach as high as 200% and strain rates around $10^6/s$ or higher in the primary shear deformation zone [1], thus making it difficult to determine the flow stresses of work material under equivalent conditions. Various material models are available that take plasticity into consideration and characterize the material flow stresses in terms of strain, strain rate and temperature with varying degrees of accuracy [2]. A comparison was done for the three models, namely Johnson–Cook (JC) model, modified JC model and Arrhenius model to study how accurately these models can predict temperature and stress values. These constitutive models were found for a specific steel alloy by taking experimental true strain-true stress data into consideration [3]. Similarly, the influence of different material models, namely JC model, El-Magd's model and Koppka's model was analysed for two different steel alloys in FE modelling of cutting forces [4]. Out of all, JC model has gained much of the popularity over the years owing to its robustness. This model defines the plastic behaviour as a function of strain, strain rate and temperature. JC model is equally capable of simulating segmented and discontinuous chips that are generally expected while machining hardened steel, titanium and other tough alloys, in addition to continuous chips [5]. However, every model has its own limitations, and hence, JC model may also give unsatisfactory results under certain circumstances.

In addition, to select right kind of material model, proper choice of the material model constants is equally crucial in order to predict forces, temperatures, chip morphology, etc., with a reasonable accuracy [6]. Umbrello et al. [7] investigated the effect of five sets of JC constants obtained from literature on the modelling of cutting forces and residual stresses during machining of AISI 316L and implemented in a FE model. Burley et al. [8] presented a methodology that combined ballistic impact test with FE modelling. Strain-rate sensitivity parameter for plastic deformation of work materials was evaluated using impact test, and FE modelling was performed. The predicted outcomes (residual indent shapes and displacement-time plots) were thoroughly compared with experimental ones, and the right value of the parameter was found by looking for the maximized value of a 'goodness of fit' parameter. Sobolev and Radchenko [9] compared the results obtained using the flow stress description given by the JC model with that of a tabulated function by simulating two drop tests, one with an impact on stainless steel cask lid and the other on the high-strength cast iron cask bottom using LS-Dyna. Shrot and Baker [10] developed an inverse approach to recalculate JC model constants under high strain rates and temperatures using Levenberg–Marquardt search algorithm based

on the chip morphology and cutting forces obtained during machining tests. Rahimi et al. [11] obtained static constants of JC model by combining experimental tests and FE simulations with optimization techniques by taking geometric parameters that can be measured easily to define the objective function. In few of the recent works, authors identified the various sets of JC constants from the literature for Ti6Al4V alloy and introduced the same in FE orthogonal cutting model based on different formulations such as Lagrangian, Arbitrary Lagrangian–Eulerian (ALE) and Coupled Eulerian–Lagrangian (CEL), and their results were compared at predetermined cutting conditions [12, 13].

Machining of WHAs into desired dimensions and finish for aerospace and defence applications is not only crucial but also challenging. However, not much of the works are available, so far, that focus on the basic understanding of machining behaviour of WHAs, both experimentally and numerically. Hence, the main objective of the present work is to develop a robust 2D FE model to simulate orthogonal cutting of one of the commonly used WHAs, i.e. with 93%W. The basic intention is to explore the applicability of three sets of JC constants in the developed FE model by comparing the obtained predicted results (chip morphology and forces) with those obtained from experimental studies. A set is finally recommended which would be capable of accurate estimation of the cutting force, feed force and chip thickness values.

18.2 Methodology

The Split Hopkinson pressure bar (SHPB) experimental data were taken from Lee et al. [14] and utilized to determine JC model constants under wide range of strain rates and temperatures for 93% WHA. Experimental stress-strain curves obtained from the mentioned literature are converted to true stress-strain curves, and data points are taken in plasticity region till ultimate tensile strength (UTS). These data points are then used for determining the JC model constants, and this approach was considered as M1. Next step was to fine-tune the calculated JC model constants obtained from M1 using GA (genetic algorithm) based optimization technique. The new constants were determined, and this approach was considered as M2. In addition, another set of JC constants was considered, namely M3, which had been directly taken from the literature, wherein Rohr and Nahme [15] had derived JC model constants for 93% WHA at varied strain rates and temperatures. The values of JC constants obtained from M1, M2 and M3 have already been presented in [16] wherein a comparison of all three models has been made by calculating absolute error and coefficient of correlation. The current work is basically an extension of [16] which aims at demonstrating the applicability of predicted values of JC constants for 93% WHA in FE modelling of machining process. The FE results predicted from all the three models are compared to an experimental reference [17] under equivalent cutting conditions.

18.3 FE Simulation Procedure

18.3.1 *Model Features and Boundary Conditions*

FE model was developed using Abaqus/Explicit (version 6.14) to simulate the chip formation in orthogonal cutting. Since machining operation involves both plastic deformation and heat generation, coupled temperature-displacement (explicit) analysis module was used to predict stresses, strains and forces. The explicit dynamics analysis takes care of large deformations and complicated contact conditions, as observed in case of metal cutting, very efficiently as compared implicit one. Adaptive meshing technique, or in other words, Arbitrary Lagrangian–Eulerian (ALE) was used in conjunction with dynamic explicit analysis as this controls mesh distortion effectively which is much expected in the case of machining simulations. The geometric model of the developed 2D FE model consists of a section of perfectly sharp cutting tool and workpiece in the form of rectangular block. The cutting tool is considered to be perfectly sharp based on the fact that the effect of tool edge radius hardly plays any role once a steady state is reached in cutting. Such assumption has been taken for the simplicity. The orthogonal cutting model is based on plane strain conditions where uncut chip thickness is taken as feed (along Y -axis) and depth of cut is defined as plane strain depth (along Z -axis) while creating the geometric model. The geometrical angles of the cutting are as follows: principle cutting edge angle = 90° , rake angle = 0° and clearance angle = 7° .

Both the workpiece and the tool were considered as deformable bodies and discretized with four-node bilinear plane strain quadrilateral, displacement with hourglass control (CPE4R) type elements. Note that CPE4R elements are also compatible with coupled temperature-displacement dynamic explicit analysis module and have been taken to keep the analysis faster. The total number of elements on workpiece is 8000 and on cutting tool is 698. As boundary conditions, the cutting tool movement is constricted in vertical direction by fixing the top edge in Y -axis and velocity equal to the cutting speed (in m/s) given in negative X -direction. The bottom of the workpiece block is fixed in Y -axis, and left edge is fixed in X -axis. The tool and the workpiece were kept initially at the room temperature. The model features of the FE model with meshing are shown in Fig. 18.1.

18.3.2 *Material Properties*

Table 18.1 lists the material properties of both the workpiece and cutting tool used in the simulation of the chip formation process [18, 19]. The chemical composition of the workpiece material is as follows: 93W–4.9Ni–1.4Fe–0.7Co.

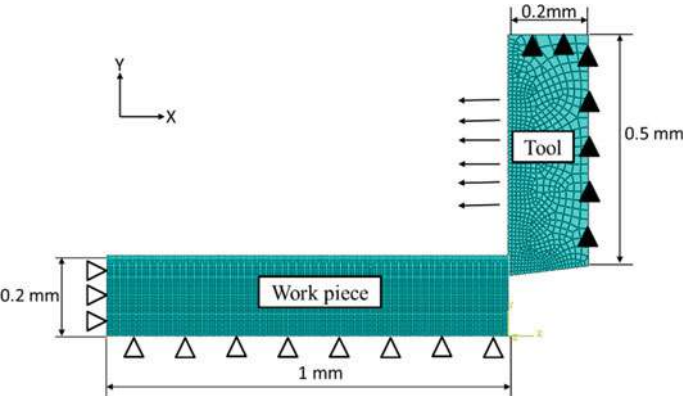


Fig. 18.1 Model features and meshing

Table 18.1 Material properties of cutting tool and workpiece

Parameter	Workpiece (93% WHA) [17, 18]	Tool (Tungsten carbide) [19]
Thermal conductivity, k (W/m °C)	60	50
Density, ρ (Kg/m ³)	17,760	11,900
Young's modulus, E (N/m ²)	362×10^9	534×10^9
Poisson's ratio, ν	0.27	0.22
Specific heat, C_p (J/Kg °C)	134	400
Expansion coefficient, α (m/m °C)	4.2×10^{-6}	

18.3.3 Material Model

JC model is the most robust model and is often used as the benchmark for comparison of different other models. JC model is not restricted to model the continuous chip formation only and is capable of simulating segmented and discontinuous chips as well. This material model defines the flow stress as a function of strain, strain rate and temperature such that it not only considers the strain rates over a large range but also temperature changes due to thermal softening by large plastic deformation. The workpiece material used in the present work, i.e. 93% W. WHA is expected to produce discontinuous chips and attain higher cutting temperatures while machining. Hence, JC model is one of the appropriate models to be considered for FE simulations for machining WHAs.

According to Johnson–Cook constitutive material model, flow stress of the work-piece is described as follows:

Table 18.2 JC model constants for 93% WHA [16]

Model	A MPa	B MPa	n	c	m	Δ %	R
M1	1103	707.5	0.22	0.11	1.4	8.72	0.96
M2	1103	875.5	0.35	0.09	1.9	2.93	0.97
M3 [15]	1197	580	0.05	0.02	1.9	16.5	0.92

$$\sigma = (A + B\varepsilon^n)(1 + C \ln \dot{\varepsilon}^*)(1 - T^{*m}) \quad (18.1)$$

where σ is the equivalent stress, A is the initial yield stress (MPa), B is the hardening modulus (MPa), n is the work-hardening exponent, C is the strain rate dependency coefficient, m is the thermal softening coefficient, ε is the equivalent plastic strain, $\dot{\varepsilon}$ is the plastic strain rate, $\dot{\varepsilon}_{\text{ref}}$ is the reference strain rate, T_{ref} is room temperature and T_m is the melting temperature.

The present study considers three sets of Johnson–Cook material constants, namely M1, M2 and M3. The detailed stepwise procedure for M1 and M2 has been documented in [16]. For the sets M1 and M2, the true stress-strain was plotted for the strain rates of 4000, 2500, 1600/s and temperatures of 298, 573, 773, 973, 1173, 1373 K. M1 is the calculated value, while M2 is the optimized value using genetic algorithm. For the set M3, JC constants were determined for the strain rates of 2×10^{-1} , 1×10^{-2} , 8×10^{-4} , 6×10^{-5} , 1000, 10,000/s and temperatures of 300 and 827 K [15]. The values of M1, M2 and M3 as well as the corresponding absolute error percentage (Δ) and coefficient of correlation (R) are listed in Table 18.2.

A ductile failure model, namely JC damage model is also incorporated along with the material model as material property input for workpiece material for simulating discontinuous chips as expected in case of machining 93%WHA alloy. In addition, the ELEMENT DELETION = YES module of the software has been used to delete the elements that fail. It is to be noted that no damage model has been considered for tool material as the main objective of the work is to predict forces and right kind of chip morphology in machining operation. JC damage criterion is of the following form:

$$\varepsilon_f = [D_1 + D_2 \exp(D_3 \sigma^*)] \times [1 + D_4 \ln \dot{\varepsilon}^p] x \left[1 + D_5 \left(\frac{T - T_{\text{room}}}{T_{\text{melt}} - T_{\text{room}}} \right) \right] \quad (18.1)$$

where D_1 – D_5 are constants determined by trial and error method in FE machining model, and the values of which are listed in Table 18.3.

Table 18.3 JC damage model constants

D_1	D_2	D_3	D_4	D_5
0.208	0.005	−1.48	0.35	0.5

18.3.4 Friction Model and Contact Conditions

For defining contact between tool and workpiece, master-slave contact pair has been used such that master surface is rake surface of the cutting tool, and slave surface is the workpiece region that would come as chip. A kinematic contact algorithm has been used to impose contact constraints in the chip tool interface. This method is based upon conservation of momentum between the contacting bodies. Frictional conditions are modelled using Coulomb's friction law, and heat generation due to friction is considered by defining a conversion factor of 0.9.

18.4 Results and Discussions

18.4.1 Experimental Validation

An experimental investigation carried out on machining of 93%WHA using carbide cutting tools by Nandam et al. [17] has been utilized as the reference data for validation of the FE simulation results. The cutting parameters considered for machining 93% WHA alloy were as follows: cutting speed = 105 m/min, feed = 0.05 mm/rev and depth of cut = 0.3 mm. Short discontinuous chips were found while machining. Table 18.4 lists the experimentally measured values as well as the predicted values of cutting forces and chip thickness using M1, M2 and M3.

Figure 18.2 shows the error percentage of F_c and t_c simulation results of M1, M2 and M3 with respect to the experimental results under similar cutting conditions. It is observed that model M2 proved better over other two models in predicting cutting force giving an error as low as 2%. This is in accordance with the Δ (%) and R as listed in Table 18.2.

Experimental values of chip thickness are determined using Optical Profile Projector under $10 \times$ magnification [19]. While predicted chip thickness values were determined by measuring the perpendicular distance between two nodes in the direction of chip thickness. It is observed that chip thickness values are consistently lower for all the three models when compared with that of the experimental ones. The probable reason for mismatch could be the error in measuring the chip thickness accurately

Table 18.4 Predicted values of output variables

Model	Output variable	
	F_c (N)	t_c (mm)
M1	35.2	0.05
M2	34.3	0.06
M3	28.4	0.06
Exp [17]	33.5	0.12

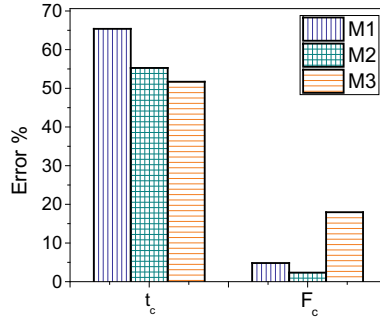


Fig. 18.2 Error percentage with respect to experimental values of t_c and F_c under similar cutting conditions for M1, M2 and M3

either in the case of experiments or simulations. The chips obtained experimentally or predicted numerically are discontinuous and very short, thus making it challenging to measure the chip thickness precisely.

In 2D orthogonal cutting, side cutting edge angle (SCEA) is 0° (or principal cutting edge angle is 90°). This makes uncut chip thickness (t) equals to feed (f). The equation is as follows:

$$t = f \times \cos(\text{SCEA}) \quad (20.3)$$

But in 3D oblique turning, side cutting angle is greater than 0° , and as the side cutting edge angle increases, uncut chip thickness decreases. This phenomenon may lower the cutting forces to some extent. Hence, in 3D oblique turning, forces will be slightly lower than that of the orthogonal cutting under similar machining conditions. Similar rationality has been used in the present work for comparing the forces predicted using 2D FE model with that of the actual oblique turning operation. Comparable observations were found in the present work, where forces predicted by M1 and M2 are found to be slightly higher than the experimental ones. However, lower forces are observed for M3. This can be attributed to that fact that values of JC constants, namely B and n are considerably lower as compared to the other two sets (Fig. 18.2).

18.4.2 Effect of Cutting Parameters on Predicted Results

FE simulations were carried under varied cutting conditions to check whether the developed models are capable of capturing the change in output variables as the cutting velocity or feed changes.

Chip Morphology. Figure 18.3 shows the chip morphology predicted by M1, M2 and M3 at three different cutting speeds for constant values of feed (0.05 mm/rev)

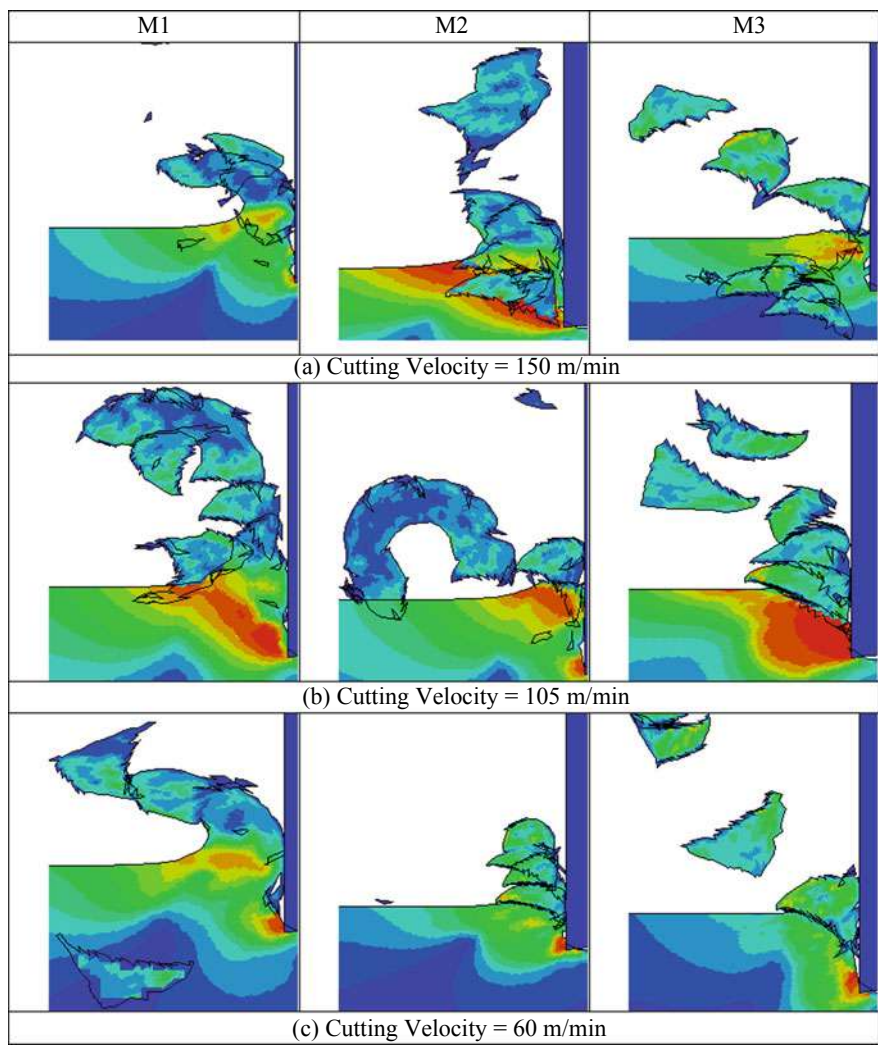


Fig. 18.3 Predicted chip morphology under variable cutting speeds, constant feed and depth of cut for M1, M2 and M3

and depth of cut (0.3 mm). It is observed that all the models are capable of simulating discontinuous chips which is expected while machining 93% WHA. Model M3 produces well-defined fragments as compared to that of other two models. However, it could not capture any changes in the chip morphology with the increase in cutting velocity. While, in case of M2, it is found that fragmentation of chips becomes prominent as the cutting speed increases. Though M1 and M2 could replicate discontinuous chips, slight fine-tuning of meshing may enhance the chip quality further.

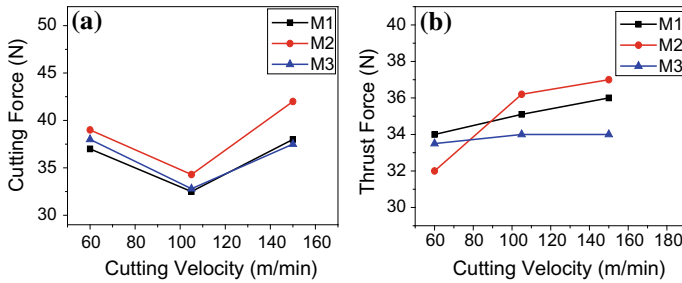


Fig. 18.4 Effect of cutting velocity on **a** cutting force and **b** thrust force

Cutting Forces and Chip Thickness. Figure 18.4 shows the predicted cutting forces and thrust forces for M1, M2 and M3 with the change in cutting speed.

In Fig. 18.4a, all the three models follow the same trend with M3, predicting slightly higher values of cutting forces over the entire range of cutting speed, while values predicted by M1 and M2 closely matches. It is observed as the cutting speed increases from 60 m/min to 105 m/min, cutting force decreases which is a common observation during machining process. But, when the cutting speed increases further, there is a sharp increase in the values of cutting forces. This may be attributed to sudden increase in the tool wear with the increased cutting speed. The trend observed in case of thrust force (see Fig. 18.4b) is different as compared to cutting force. As the cutting speed increases, thrust force increases for all the three cases. This again could be attributed to the gradual increase of flank wear, and it is known that thrust force is more susceptible to flank wear compared to cutting force.

Figure 18.5a, b shows the effect of feed on cutting force and thrust force, respectively. As it is known with the increase in feed, forces should increase, and all the models could replicate this phenomenon fairly well.

Chip Thickness and Shear Angle. The effect of cutting speed and feed on chip thickness was studied and presented in Fig. 18.6. There is not much change in the chip thickness with the increase in cutting speed for all the three cases. Whereas, all three models effectively show increase in the chip thickness values as the feed

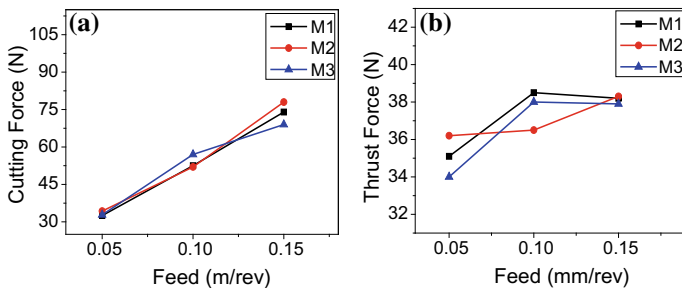


Fig. 18.5 Effect of feed on **a** cutting force and **b** thrust force

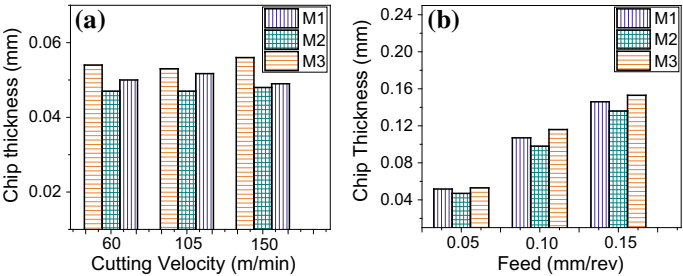


Fig. 18.6 Effect of **a** cutting velocity and **b** feed on chip thickness

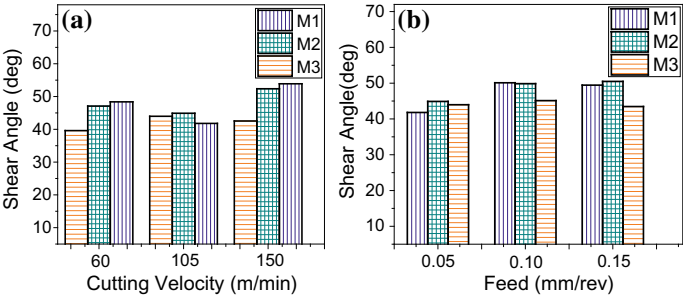


Fig. 18.7 Effect of **a** cutting velocity and **b** feed on shear angle

increases which is expected. Similarly, shear angle is measured under different cutting conditions and presented in Fig. 18.7. It is difficult to observe a trend, but in general, it can be stated that as the cutting speed increases, shear angle increases in general. This is expected because shear angle increases as the chips become more discontinuous and fragmented. Some deviations are found at cutting speed of 105 m/min for M2 and M3. This deviance could be because of the approximation in the measurement of chip thickness of irregularly shaped discontinuous chips. The trend obtained for shear angle closely matches with the chip morphology presented in Fig. 18.7. Similarly, an overall increase in the shear angle is observed especially for models M1 and M2 with the increase in feed.

18.5 Conclusions

Three sets of JC model constants were used as inputs for FE simulations of machining process of 93% WHA. All the three models could predict various machining outputs such as cutting force, chip thickness and shear angle values fairly well. Cutting forces predicted using M2 showed minimum deviation followed by M1 and M3. FE models using M3 could simulate smoother chips with prominent fragmentation over

the entire range of cutting speed. It is observed that model M2 is a fair compromise in terms of chip morphology, forces and chip thickness over the entire range of cutting speed and feed and hence can be used for carrying out further simulations which are essentially required for optimization of machining 93% WHA alloy.

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Chapter 19

Numerical Simulation of Heat Transfer and Fluid Flow in Co-axial Laser Cladding of Ti6Al4V Alloys



Vijay Mandal , Shashank Sharma and J. Ramkumar

Abstract In this article, a 2D FEM model is built to simulate the heat transfer and fluid flow in the laser cladding process of the Ti6Al4V alloy. Physical phenomena such as melt pool generation, mass addition due to powder flow, Marangoni convection, and re-solidification of the melt pool have been incorporated in the developed model. The governing equations pertaining to mass, momentum, and energy were solved in a Lagrangian moving frame to predict temperature and velocity field along with geometrical dimensions of the deposited clad. The temperature and temperature gradients were calculated at “14” located points in three different directions, to scrutinize the thermal behavior of the melt pool. Further, the influence of driving forces such as Marangoni force and thermal buoyancy force was analyzed. The prediction of microstructure evolution was based on the estimation of the temperature gradient, cooling rate, and solidification rate in the fusion zone.

Keywords Laser cladding · Heat transfer and fluid flow · Solidification rate

19.1 Introduction

Titanium and its alloys have become essential for structural applications in aerospace, electronics, and bio-medical industries, due to its lightweight and high strength to weight ratio [1, 2]. The surface properties such as corrosion resistance, thermal resistance, and tribological properties profoundly affect the functionality of any component. The major drawback of Ti alloys is relatively high wear rate [3]. Enhancement

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_19

of surface properties with the help of surface modification techniques such as electroplating, chemical coating, anodic oxidation process, hot dipping, thermal spraying, and vapor deposition is extensively used to overcome the drawbacks mentioned above. However, these processes pose several limitations (i.e., the additional medium is required, toxic, pyrophoric, corrosive, etc.) [4]. One such method that can alleviate the impediments offered by the aforementioned techniques is laser cladding process. Laser has the capability to accumulate high energy on a material which is temporally and spatially confined. The large thermal gradients and high cooling rates transform the grain structure rendering excellent wear-resistant surface. The direct metal deposition through laser is a process of free-form powder deposition by means of the heat source with a high thermal gradient, and the net shape of structure can be obtained directly from the metal powder [5].

Therefore, in laser cladding, continuous efforts have been put forth by the researchers all around the world to improve the various process; for instance, Peyre et al. [6] developed a thermal model which predicts morphology and thermal behavior during multi-layer laser cladding using COMSOL Multiphysics. They reported that the numerical approach is in good covenant with the experimental results such as powder temperature with radius, melt pool geometry at different processing parameters (i.e., scanning speed, laser intensity, and radius of powder delivery), and spatial temperature variation with respect to time. In a similar approach, Kong and Kovacevic [7] conducted experiments and validated with the numerical simulation data. The influence of clad geometry on the input process parameters, i.e., laser power (P), scanning speed (V), and the mass flow rate ($\dot{m}$) was studied. They reported that clad height increases with an increase in laser power (P) and powder $\dot{m}$. However, dilution decreases with an increase in V and powder feed rate. Bedenko et al. [8] investigate an experimental and theoretical model in the laser cladding process. They reported that characteristic size of the bead (i.e., height (H) and width (W) of clad) decreases with increase in V , however, increases with an increase in P and $\dot{m}$. The scanning speed (V) plays a vital role in fluid flow in the melt pool. At higher V , due to less interaction time, Marangoni convection is less significant, and the powder particles impinging the melt pool surfaces govern the fluid flow at this stage. However, at low scanning speed, fluid convection due to Marangoni effect is dominant. Kumar and Roy [9] scrutinized the effect of Marangoni convection on microstructural evolution. They reported finer microstructures at the bottom of the clad parts for a positive value of Marangoni number (Ma), whereas for a negative value of Ma , the finer microstructures were observed at the top surface of the melt pool. They also noted that the influence of microstructure does not largely depend on V . Moreover, the microstructure plays a vital role in the surface properties of the clad part. For instance, Gan et al. [10] simulated the heat transfer, fluid flow, and multi-component mass transport of alloy. They predicted the solidification parameters such as solidification rate (R), cooling rate (T_t), and temperature gradient (G). They reported that the temperature gradient in the periphery is higher (1352 K/mm); however, at the center of the melt pool, G was comparatively lower (650 k/mm). They also reported that the microstructure changes from planner front to equiaxed dendrite from bottom to top of the melt pool.

Therefore, to better understand the process dynamics, a simplified two-dimensional numerical model is constructed. The present study takes into account heat transfer, fluid flow, and transient powder addition on the substrate to scrutinize the levels of temperature, temperature gradients, and Marangoni convection induced velocity profiles during laser cladding. Further, the dimensional analysis was studied to comprehend the importance of heat transfer by diffusion and advection, and roles of driving forces such as Marangoni force and thermal buoyancy force. For the microstructure prediction, G , Tt , and R have been discussed.

19.2 Model Implementation and Assumptions

To understand the physical process occurring in laser cladding, a 2D model considering heat transfer, melt flow dynamics along with analysis of solidification parameters such as thermal gradient and solidification rate has been formulated. In this model, there are following assumptions to be made: (1) Incompressible, laminar fluid flow in the melt region is taken into consideration. (2) The thermo-physical properties of the work materials are temperature dependent. (3) The buoyancy effect in the melt region of this model is taken into account using Boussinesq approximation. (4) The shrinkage effect during solidification is not considered in this model. (5) The vaporization effect is also not considered.

19.2.1 Governing Equations

Mass Conservation: The governing continuity equation is defined by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0 \quad (19.1)$$

Energy Conservation: The governing energy conservation equation is defined by

$$\frac{\partial(\rho C_p T)}{\partial t} + \vec{u} \cdot \nabla(\rho C_p T) = \nabla \cdot (K \nabla T) \quad (19.2)$$

During the phase change, values of ρ , C_p , and k were calculated by the following equations:

$$\rho = \theta \rho_{\text{phase 1}} + (1 - \theta) \rho_{\text{phase 2}} \quad (19.3)$$

$$C_p = \frac{1}{\rho} (\theta \rho_{\text{phase 1}} C_{p,1} + (1 - \theta) \rho_{\text{phase 2}} C_{p,2}) + L \frac{\partial \alpha_m}{\partial T} \quad (19.4)$$

$$k = \theta k_{\text{phase 1}} + (1 - \theta) k_{\text{phase 2}} \quad (19.5)$$

In Eqs. (19.3)–(19.5), phase 1 symbolizes the solid phase, i.e., solid metal, phase 2 symbolizes the liquid phase, i.e., molten metal, and θ is a linear function, lies between 0 and 1.

Momentum Conservation: The momentum conservation equation is defined by

$$\frac{\partial(\rho \vec{u})}{\partial t} + \vec{u} \cdot \nabla(\rho \vec{u}) = -\nabla p + \nabla \cdot (\mu(\nabla \vec{u} + (\nabla \vec{u})^T)) + \vec{F}_c + \vec{F}_d \quad (19.6)$$

This equation is implemented in the entire computational domain including melted and solid metal region. For dissipating velocity field in the solid region, the viscosity in the solid region is taken to be a very high value (~ 1000). The source term $\vec{F}_c$ is used in the mushy zone (or porous media). The term $\vec{F}_d$ can be defined as

$$\vec{F}_d = -\frac{C(1 - f_l)^2}{b + f_l^3} \vec{u} \quad (19.7)$$

Here, $\vec{F}_d$ represents the momentum sink term for the mushy zone as indicated by the Carman–Kozeny equation in porous media. The constant term “ C ” represents the morphology constant, and very large value is taken, i.e., 2×10^7 [10]. Another constant term b represents very small, i.e., 10^{-5} to avoid the source term, and $\vec{F}_d$ is coming to be infinity. In this equation, the term f_l represents the liquid fraction depends on solidus temperature (T_s) and liquidus temperature (T_l) and is defined as

$$f_l = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_l - T_s} & T_s < T < T_l \\ 1 & T > T_l \end{cases} \quad (19.8)$$

The source term $\vec{F}_c$ in Eq. (19.6) is defined by

$$\vec{F}_c = \rho_{\text{liquid}} \vec{g} \beta_T (T - T_{\text{ref}}) \quad (19.9)$$

In Eq. (19.9), ρ_{liquid} is density of the liquid, and β_T and T_{ref} are listed in Table 19.1.

Marangoni Convection: The above equation defines the forces induced due to Marangoni convection at the interface (or free surface).

$$\mu \left(\frac{\partial u}{\partial y} \right) = - \left(\frac{\partial \gamma}{\partial T} \right) \left(\frac{\partial T}{\partial x} \right) \quad (19.10)$$

In this Eq. (19.10), μ and γ represent the viscosity and surface tension, respectively.

Table 19.1 Thermo-physical properties of Ti6Al4V

Property	Symbol	Value	Ref.
Liquidus temperature	T_l	1923 K	[11]
Solidus temperature	T	1873 K	[11]
Ambient temperature	T_{ref}	293.15 K	
Density	ρ	4420 kg/m ³	
Thermal conductivity	k	W/mK	[12]
Specific heat capacity	c_p	J/(kg K)	[12]
Convection coefficient	h	50 W/m ² K	
Emissivity	ε	0.4	[11]
Dynamic viscosity	μ	4×10^{-3} kg/(m s)	[11]
Absorptivity	A	0.4	[11]
Stefan–Boltzmann constant	σ	5.67×10^{-8} W/m ² K ⁴	[11]
Thermocapillary coefficient	$\partial\gamma/\partial T$	-2.7×10^{-4} N/mK	[11]
Thermal expansion coefficient	β_T	2×10^{-4} 1/K	[11]

Boundary Conditions: The time-dependent Gaussian heat source as shown in Fig. 19.1 is estimated by

$$q = \frac{2AP}{\pi R^2} e^{\frac{-x^2}{R^2}} e^{\frac{-2(vt-R)^2}{R^2}}$$

(19.11)

where A is absorptivity of Ti6Al4V. At top of the surface, energy balance equation is defined by

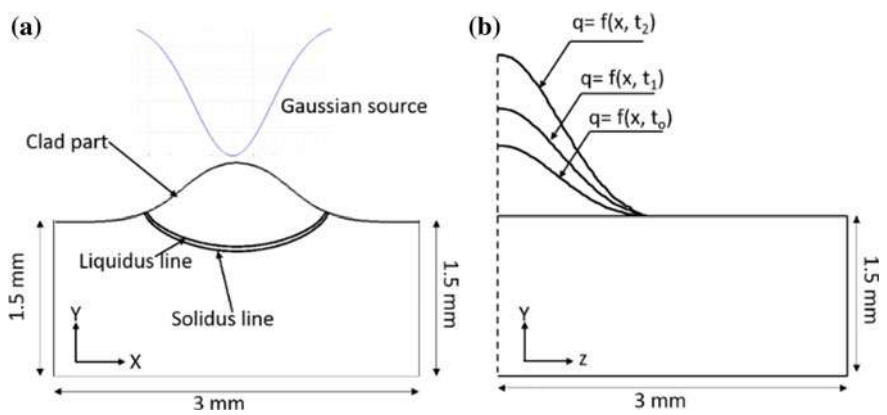


Fig. 19.1 **a** Schematic diagram of the laser cladding process and **b** variation of laser intensity with respect to time

$$k \frac{\partial T}{\partial n} = q - h(T - T_{\infty}) - \varepsilon \sigma (T^4 - T_{\infty}^4) \quad (19.12)$$

In this Eq. (19.12), 1st, 2nd, and 3rd terms on the right-hand side are incoming heat energy, heat loss due to convection, and radiation heat loss to the ambient, respectively. In Eq. (19.12), the terms h , ε , and σ are listed in Table 19.1. Force balance at the top-most surface is the following boundary conditions.

$$\left[\mu \left(\vec{\nabla} u + \left(\vec{\nabla} u \right)^T \right) \right] \vec{n} = \sigma k \nabla \phi \vec{i} + \frac{\partial \sigma}{\partial T} \frac{\partial T}{\partial x} \vec{j} \quad (19.13)$$

Above boundary condition mainly defines the melt pool dynamics in laser cladding. The first term describes towards the effect of surface tension, i.e., minimizing energy distribution by changing the shape of its surface, whereas the second term accounts for Marangoni convection (Eq. 19.10) which is responsible for distribution of molten metal according to the distribution of temperature in the computational domain.

As for rest of three boundaries, balance heat flux at the surfaces of the following boundary equation

$$-K \nabla T = -h(T - T_{\text{ref}}) \quad (19.14)$$

No slip at the rest of the surfaces of the resulting boundary equation

$$u = 0 \quad (19.15)$$

To predict the surface profile during laser cladding process, a Lagrangian mesh is used.

Mass addition: In this model, it is assumed that the mass addition is Gaussian in nature, and also the velocity of the heat source and mass addition are same. The term u_p describes the moving velocity owing to the mass addition. It can be evaluated by the following equation:

$$u_p = \frac{\dot{m} \varepsilon_p}{\rho \pi r_p^2} \exp\left(\frac{-x^2}{r_p^2}\right) f(t) \quad (19.16)$$

In this Eq. (19.16), ρ = density of powder, and $\dot{m}$, ε_p , and r_p are listed in Table 19.2. The term $f(t)$ is defined as

$$f(t) = \begin{cases} 1 & 0 < t < 130 \text{ ms} \\ 0 & t > 130 \text{ ms} \end{cases} \quad (17)$$

Table 19.2 Parameters used for simulation

Parameters	Symbol	Value
Laser power	P	400 W
Scanning speed	V	400 mm/min
Beam radius	R	0.65 mm
Mass flow rate	$\dot{m}$	1.5 g/min
Mass flow radius	r_p	0.65 mm
Efficiency of powder catchment	ε_p	0.6

19.2.2 Material and Parameters Used for Simulation

In this study, Ti6Al4V was used for both powder and substrate materials. The thermo-physical properties of Ti6Al4V alloys are listed in Table 19.1. Thermal conductivity and specific heat capacity are important characteristics of the materials, and these two parameters depend on the temperature [12]. The processing parameters for the study of laser cladding of Ti6Al4V are shown in Table 19.2.

19.3 Results and Discussion

19.3.1 Study of Thermal Cycle at Various Location in the Melt Pool

In Fig. 19.2a, different points (5–17) were located in the computational domain. These points are located in three different directions such as horizontal, vertical, and 45° from the horizontal. The located point “10” positions at the midpoint of the melt pool. In this model, surface deformation has been coupled with heat transfer and fluid flow. Normal Gaussian distribution velocity is acted at the top of the surface, due to this velocity, the top surface of the computational domain will move upward with time. The located points are mapped into the moving frame, that is with the direction of deformation, to capture the levels of temperature and temperature gradient with transient powder addition. The rate of deposition depends on mass flow rate and interaction time. With the increase in the value of mass flow rate and interaction time, the deposition rate increases, and it leads to an increase in the dimension of the clad part. The intensity of the laser source at the center is comparatively higher with respect to radial direction. Due to change in its intensity, the temperature at a different location in the melt pool varies. The intensity of the heat source is increased from $t = 0$ ms to $t = 130$ ms and then decreases from $t = 130$ ms to $t = 400$ ms. The variation of the temperature of all the located points periodically increases up to 130 ms, while the temperature variation exponentially decreases from 130 ms to 400 ms as shown in Fig. 19.2a. At the located point “10,” the maximum temperature

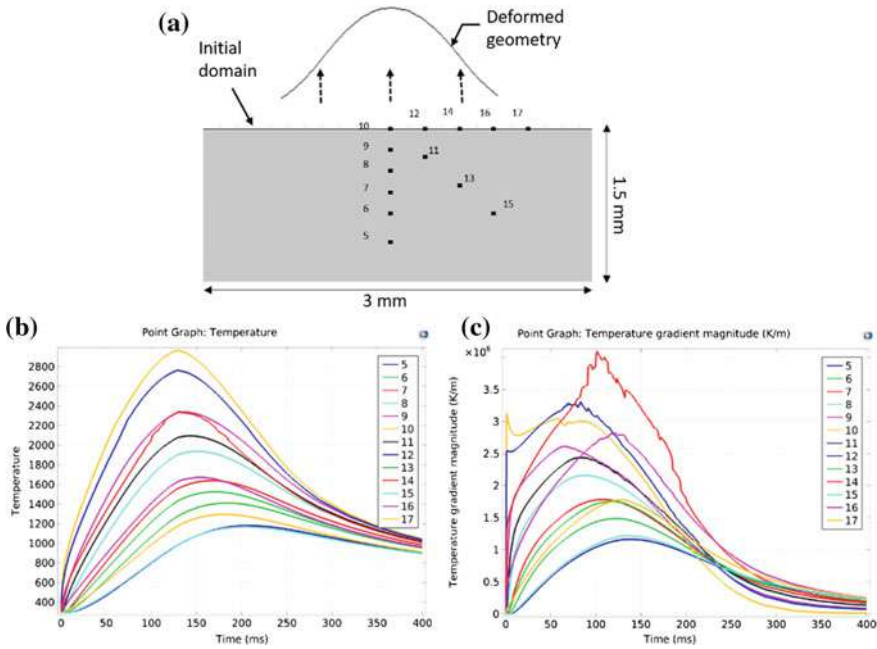


Fig. 19.2 a Schematic diagram of computational domain depicting deformed geometry, b temperature variation with time, and c magnitude of the temperature gradient with time

was found to have a value of 2966 K which is less than the vaporization temperature [13] of Ti6Al4V alloys. However, the points “5” and “15” depict comparatively lower temperature and reach up to 890 K.

In this model, the magnitude of the temperature gradient (G) at the located points is also predicted as shown in Fig. 19.2c. From the above figure, it can be observed that up to 38.8 ms, the maximum value of G is noticed at the point 10. However, the maximum of G was observed in the range of 38.8–77 ms, at the designated point 12. In the time period from 77 ms to 233.5 ms, the magnitude of the maximum temperature gradient was seen at the located point 14. It is pertinent to note that after reaching the maximum value of G at the located point 10, it is suddenly decreasing because of convection loss. Thus from the results, it can be concluded that the G attains their maximum value along the edge of the melt pool, due to a drastic change in temperature from its center. From the above results, it can be concluded that G varies from point to point with time. Also from the predicted model, the value of G at any located point is not the global maxima at all the time period. The aforementioned variation in G can be attributed to the presence of strong Marangoni currents present in the melt pool, discussed in later section.

19.3.2 Study of Heat Transfer and Hydrodynamics Fluid Flow in Clad Melt Pool

In this section, the study of temperature distribution, velocity field, and geometry of the melt pool has been demonstrated. Figure 19.3a–h represents the temperature distribution and evolution of geometry of the clad melt pool during the heating period at 30, 90, 110, and 130 ms and during the cooling period at 140, 170, 200, and 220 ms. Similarly, the velocity field distribution during heating and cooling period at different time steps as shown in Fig. 19.4a–h. Note that the laser intensity increases from 0 ms to 130 ms and decreases from 130 ms to 400 ms, according to the interaction time function discussed in Eq. (19.11) to incorporate the moving heat source condition. The spatial temperature and velocity profiles of fluid flow in the clad melt pool of cladding are indicated by color map and velocity vector arrows, respectively. The two temperature contours at the bottom are representing the solidus and liquidus

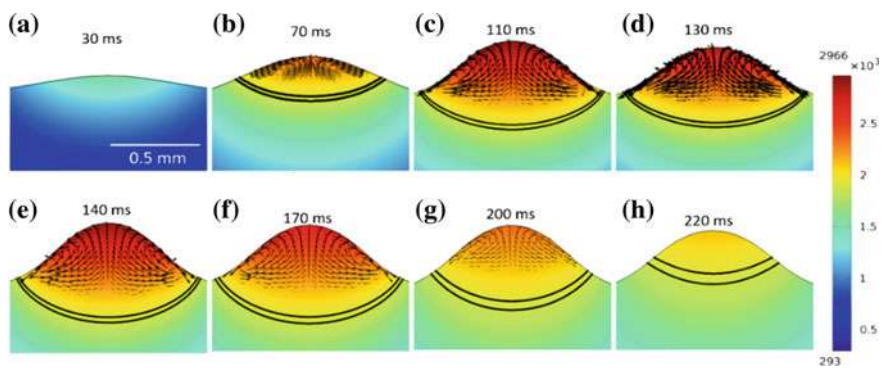


Fig. 19.3 Temperature distribution during heating a–d and cooling e–h, temperature in K

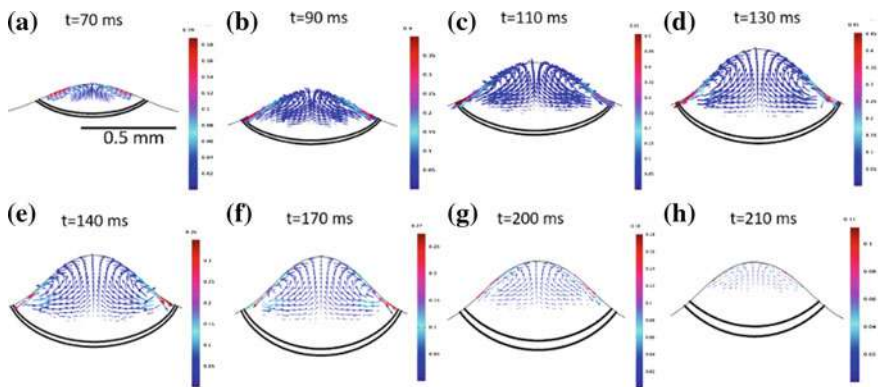


Fig. 19.4 Velocity field distribution during heating a–d and cooling e–h, velocity in (m/s)

temperature. In this figure, it can be seen that the dimension of the melt pool (i.e., width and height) continuously grows because of the addition of material and increase in temperature. The above phenomena are consistent till the intensity of laser starts to decrease at that instant the powder addition condition also ceases to act. Therefore, the melt pool and dimension of the clad increase up to 130 ms, and after 130 ms, the dimension of the clad remains constant, while the melt pool dimension decreases continuously.

The fluid flow velocity is an essential characteristic of the laser cladding process. At the initial stages of the time period the fluid flow, Marangoni convection is relatively low and flows due to the material deposition, and thermal buoyancy prevails. At later stages, after 70 ms, velocity of fluid flow is relatively higher (>0.4 m/s) because of strong Marangoni convection acting at the surface of the melt pool. The maximum value of temperature in the clad melt pool was found to be 2966 K at 130 ms, while the maximum magnitude of velocity was achieved 0.51 m/s at 110 ms as shown in Fig. 19.4e. Also, observed from the above Fig. 19.4a–h, the magnitude of velocity in the clad melt pool at $t = 110$ ms is 0.51 m/s and at $t = 130$ ms is 0.45 m/s, while the intensity of the beam is higher at 130 ms as compared to $t = 110$ ms because of persistent melt pool convection owing to the presence of Marangoni effects. Moreover, the melt pool velocity decreases after 130 ms because of an abrupt drop of the incoming heat flux, resulting in the decrease of G .

19.3.3 Analysis of the Non-dimensional Number in the Clad Melt Pool

The three non-dimensional numbers such as Marangoni number (Ma), Peclet number (Pe), and Grashof number (Gr) were analyzed to comprehend the hydrodynamics performance of fluid flow in the clad melt pool. In this model, the fluid flow in the melt pool due to thermal buoyancy and Marangoni force was considered. The Marangoni

number is defined as $Ma = \frac{\rho l_c \Delta T \left| \frac{\partial \gamma}{\partial T} \right|}{\mu^2}$ where l_c is the characteristics length, ΔT is the difference of temperature at the center of the melt pool and solidus temperature, and ρ , $\frac{\partial \gamma}{\partial T}$, and μ are listed in Table 19.1. Moreover, the Grashof is defined as $Gr = \frac{\rho^2 \beta g \Delta T l_d^3}{\mu^2}$ where g is termed as acceleration due to gravity, and l_d is the characteristics length (1/8th of width) of the clad melt pool. The another non-dimensional quantity is Peclet number (Pe) which is defined as $Pe = \frac{u_m \rho C_p l_c}{k}$ where u_m is the mean velocity of the melt pool. Each of three non-dimensional numbers plays a vital role in the clad melt pool. The formulated non-dimensional number (Rf) is the ratio of Ma to Gr. The non-dimensional number Rf, having much higher value than the unity as shown in Table 19.3, leads to fluid flow in the clad melt pool mainly signifies the dominance of the Marangoni convection over the buoyancy force in the melt pool. In this model, the calculated value of non-dimensional number Pe is considerably

Table 19.3 Magnitude value of various dimensionless number at $t = 130$ ms

Power	Ma	Gr	Pe	Rf = Ma/Gr
400 W	40,762.068	0.5785	27	70,461.65

higher than unity, demonstrates the prominent effect of heat transfer by advection, while the heat transfer by diffusion has a minor role.

19.3.4 Analysis of Melt Pool Geometry

The geometry of the melt pool [i.e., width (W) and height (H)] of clad was computed and analyzed for the different time period. Figure 19.5 shows the variation of clad width and height with respect to time. As shown in the figure, up to 40 ms, both H and W are found to be zero because $T_{\max}$ is less than T_{liquidus} for this period of time. After 40 ms, both H and W gradually increase with time, until initiation of solidification. Note that the solidification is not started from 130 ms, despite laser getting switched off after 130 ms.

From this model, it is observed that the predicted H and W start decreasing from 150 ms. In the period from 130 to 150 ms, H and W increase in a short range, while the laser is not an active mode because of stagnation of temperature and high thermal conductivity reaches their $T_{\max}$. From the above figure, it can be concluded that at the given process parameters (Table 19.2), the value of W is always higher (approximate double) than that of H , and just after switching off the laser source, it is not necessary that H and W suddenly start to decrease.

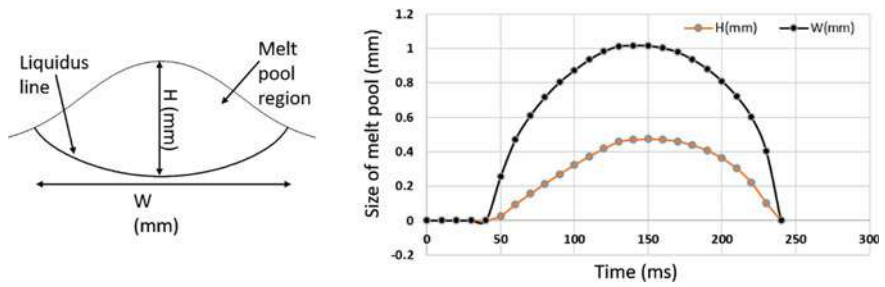


Fig. 19.5 Variation of melt pool height and width with time

19.3.5 Study of Solidification in the Clad Melt Pool

The microstructure formation in the rapid solidification plays a vital role in the surface properties for the various industrial applications. In order to comprehend the rapid solidification processes in the clad melt pool, the solidification parameters such as Tt , G , and R were analyzed for the prediction of the microstructure. Mathematically, R can be expressed as $R = Tt/G$ where $Tt = dT/dt$ (K/s) and $G = dT/dx$ or dT/dy (K/m). Here, T , t , x , and y represent the temperature, time, and coordinates in millimeters, respectively [14]. The chemical composition of alloys and undercooling plays a substantial role in the prediction of microstructure during solidification in the clad melt pool. In this model, to simplify the calculations, the undercooling effect is not considered. Therefore, the solidification parameters are estimated by only heat transfer and fluid flow in the clad melt pool. In this study, the solidification parameters Tt , G , and R to be calculated in the 14 different locations “a–n” in the clad melt pool are as shown in Fig. 19.6a. Figure 19.6b represents the variation of cooling rate at different positions in the clad melt pool. The value of the magnitude of the (Tt) along Y -axis (at the center of the melt pool) varies with respect to every point. From the above figure, the cooling rate at the central point is lower in comparison with the top and bottom point of the clad melt pool which is well agreed with Gan et al. [15], and variation of magnitude of Tt is significantly higher along the Y -direction in comparison with X -axis (central line). The maximum value of cooling rate is located at the point “n” at the right edge of the melt pool. The magnitude of G along Y -direction varies from top to bottom of the clad melt. The value near the top surface

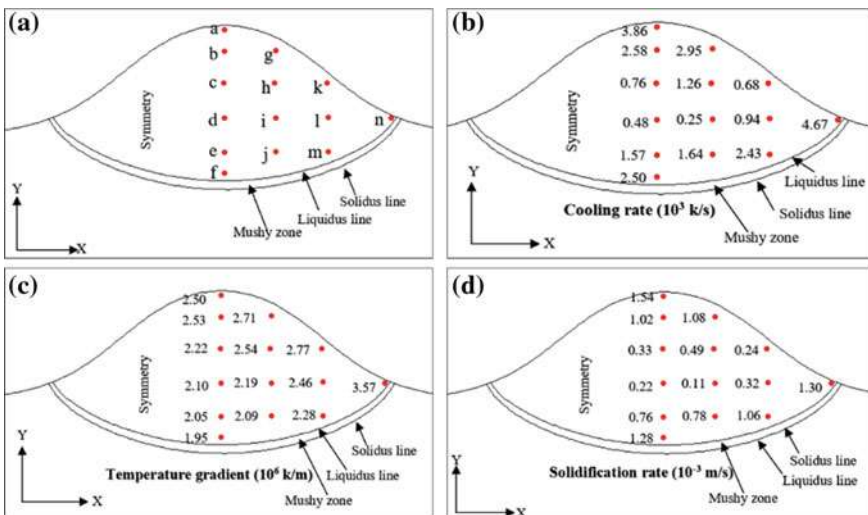


Fig. 19.6 a Representation of calculation points in the melt pool, b cooling rate, c temperature gradient, and d solidification rate in the clad melt pool

is higher and decreases along the bottom of the surface. The variation of the G along X -axis is higher in comparison with the Y -axis.

As shown in Fig. 19.6c, the solidification rate along the Y -axis first decreases and then increases from top to bottom of the melt pool. The R along Y -direction (in central line) is higher in comparison with X -direction (central line) because of energy density along Y -direction is higher with respect to the X -direction. Therefore, the values of G , Tt , and R obtained from the above study depict a clear understanding of the thermal map in laser cladding process of Ti alloy. Further, with the help of these values, the state of microstructure evolution can be predicted, as with time, the G will decrease, and the R increases leading to sharp drop in G/R value which ultimately dictates the transition of columnar to equiaxed microstructures in the solidified zone. The developed thermos-fluidic model can be employed as the initial condition for the microstructure prediction model.

19.4 Conclusions

A 2D finite element model (FEM) is developed to simulate the heat transfer and fluid flow in the cladding process of the Ti6Al4V alloy. The following conclusions have been drawn from the developed model. The fluid flow in the clad melt is predominantly governed by Marangoni convection, while the thermal buoyancy force plays a minor role. Heat transfer through conduction mode is dominant only in the initial stage of the melt pool, while the heat transfer through advection mode is dominant in the later stages. The maximum value of temperature ($T_{\max}$) in the melt pool was found to be 2966 K at 130 ms, while the maximum magnitude of velocity was achieved 0.51 m/s at 110 ms. The geometry of the melt pool was successfully evaluated, and it was found that both H and W vary in a parabolic nature; further, it has been observed that just after switching off the laser, it is not necessary that height (H) and width (W) suddenly start to decrease. The proposed thermal model can act as a basis for microstructure prediction tool, as it correctly captures the variation in solidification parameters Tt , G , and R due to the dominating effect of Marangoni convection. In the future, a coupled model will be developed to predict microstructures simultaneously.

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Chapter 20

FEA of Electrical Discharge Machining on the Particle Metal Matrix Composite



K. Benarji , Y. Ravi Kumar and S. Kanmani Subbu

Abstract Electrical discharge machining (EDM) is one of the nonconventional machining processes suitable for machining of metal matrix composites (MMC). In this work, the heat transfer model has been adopted to predict temperature distribution within selected MMC and considered other machining conditions also such as temperature-dependent thermal properties, Gaussian heat flux and plasma radius. The MMC (Al–4Cu–6Si+10%SiC) is used as a workpiece material in this present investigation. The FEA finite element analysis (FEA) axisymmetric model was generated to simulate the MMC using FEA package multi-physics and predicted Material Removal Rate (MRR). Also, the results have been validated with experimental results.

Keywords Metal matrix composites · Electrical discharge machining · Material removal rate

20.1 Introduction

Metal matrix composites (MMC), nanomaterials, ceramics and superalloys that are under the category of difficult to cut materials [1]. MMC is the combination of metallic properties with ceramic properties which results in high compressive and shear strength and high service temperature. The presence of hard and brittle reinforcement in MMC enters into the category of difficult to cut materials. However, they are used for different applications such as aerospace, defense, marine, and automotive industries because of their high specific modulus, thermal stability, strength to weight ratio, corrosion resistance, and hardness [2]. Therefore, machining such materials with high dimensional accuracy and low surface roughness has been challenging

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_20

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task using the conventional machining process. EDM is one of the unconventional machining processes, extensively preferred for machining of hard materials. In the process, the material removal is taken place by single or repeating sparks between the surface of anode and cathode surface in the presence of dielectric medium [3]. FEA model is the best tool to predict the thermo physics involved during EDM of composite material. Little research has been done on the EDM of Particle Metal Matrix Composites (PMMC) using FEA model analysis. Vishwakarma et al. [4] investigated the FEA modeling of material removal rate (MRR) of Al6063/SiC composites using EDM. The influence of single and multi-spark EDM on composites in terms of MRR and temperature distribution has been studied. In addition, the comparative study has been carried out between the simulated and experimental results referred [5] in terms of material removal rate. Arya et al. [6] have studied thermal stress analysis along with MRR mechanism using ANSYS multi-Physics Mechanical APDL during EDM of composites. The influence of voltage, current, and pulse duration on the temperature distribution and residual stresses induced in the workpiece have been studied by Avinash and Mohan [7] and observed that the compressive and tensile stresses at the crater surface lead to crack initiation after multiple discharges. The effect of different input parameters such as pulse on/off time, shape and size of heat source, and phase change on the thermal behavior and material removal mechanism have been analyzed for the powder mixed EDM. It reported that shape and size of the crater were less than the EDM under the similar conditions [8].

The FEA of die-sink EDM has been performed on the crater shape and MRR using the input parameters such as discharge voltage, discharge duration, duty cycle, and discharge current. Reported that, crater shape and MRR obtained by the developed model was close approximate to experimental results at the given different input parameters [9].

In this present investigation, COMSOL 5.3 multi-physics has been selected for predicting the material removal rate and residual stresses during EDM of metal matrix composites. The extremely fine mesh chosen to examine the output results such as MRR and stresses induced while EDM of metal matrix composite. Influence of current and voltage on MRR mechanism during machining of composite material (Al–4Cu–6Si+10%SiC) also investigated.

20.2 Thermal Model of EDM

The following assumptions have been made for FEA modeling to simplify problem analysis.

- An axisymmetric model has been considered for simulating given composite material.
- The shape of reinforcement in matrix is spherical.
- The composition of workpiece material is homogeneous.
- The mode of heat transfer in workpiece is conduction only.

- The initial temperature is set to room temperature in single discharge analysis.
- The flushing efficiency is considered 100%.
- The spark energy is partly conducted to workpiece remaining energy transfer taken place by convection and radiation.
- The Gaussian distribution of heat flux is selected for heat source which is applied on the surface of the workpiece.
- The effect of recast layer is not considered.

Three-dimensional transient heat conduction equation of cylindrical coordinate system for axisymmetric model is given by [10]

$$\frac{1}{r} \frac{\partial}{\partial r} \left(Kr \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(Kz \frac{\partial T}{\partial z} \right) = \rho Cp \left(\frac{\partial T}{\partial t} \right) \quad (20.1)$$

where K is thermal conductivity of the workpiece, ρ is density, Cp is specific heat, t is time, T is temperature, and r and Z are cylindrical coordinates of the workpiece.

20.2.1 Boundary and Initial Conditions

The minor cylindrical portion of workpiece is considered as domain FABCDE and is subjected to Gaussian heat source as shown in Fig. 20.1. In this analysis, ABCD is taken as axisymmetric model and boundaries 1, 2, 3, and 4 are mentioned as shown in figure. Here, boundary 1 is subjected to Gaussian heat distribution and convective heat transfer in the presence of kerosene dielectric medium. Boundaries 2, 3, and 4 are assumed to be no heat flux.

20.2.2 Heat Source

Yeo et al. [11] have been applied heat source in the form of disk and point, whereas typically several researchers [4, 12–14] have been considered the Gaussian heat flux distribution which is considered to be approximate to the given heat input. The maximum heat flux is located at the center of a spark and gradually decreasing with radius (r) as shown in Fig. 20.2. The equation of heat flux at any given radius r is given by [4]

$$Q(r) = \frac{4.45 PVI}{\pi R^2} \exp \left(-4.5 \left(\frac{r}{R} \right)^2 \right) \quad (20.2)$$

where P is fraction of heat input to workpiece, V is voltage, I is current, and R is radius of spark. In this present investigation, percentage of heat input transfer to

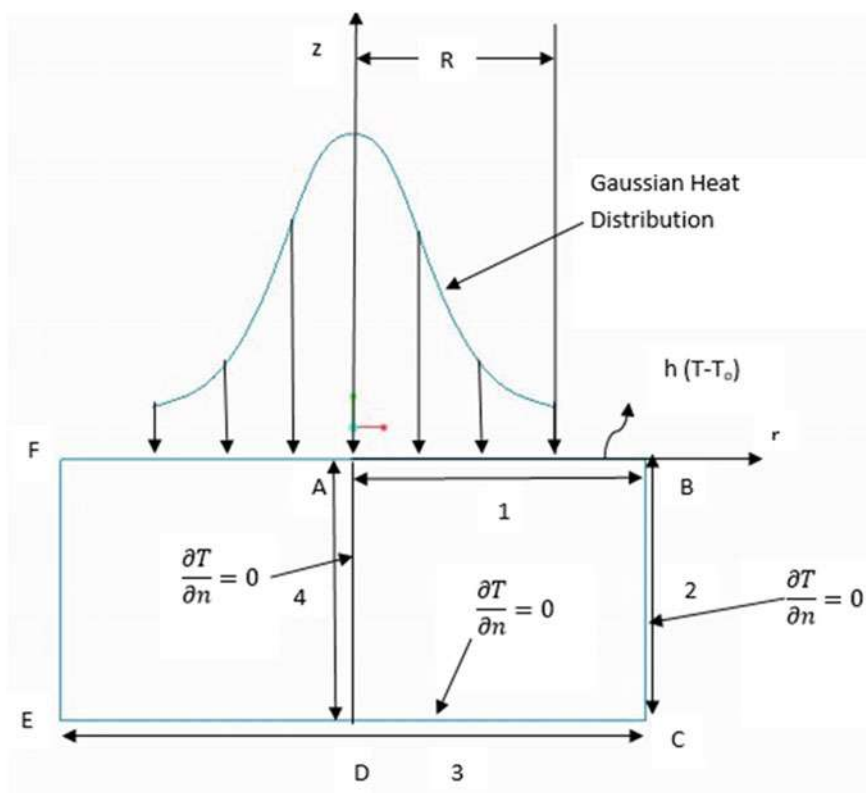


Fig. 20.1 Schematic view of the EDM process simulation. Boundary conditions are when $t > 0$, $Kz \frac{\partial T}{\partial z} = Q(r)$, when $R < r$ for boundary 1, $Kz \frac{\partial T}{\partial z} = h_f(T - T_o)$, when $R \geq r$ for boundary 1, $Kz \frac{\partial T}{\partial n} = 0$, at boundary 2, 3 & 4. Here, h_f is convective heat transfer coefficient

workpiece is 0.08 [4] and no reliable model has been derived for obtaining plasma channel radius so that taken as the value of radius $120 \mu\text{m}$ [6].

20.2.3 Modeling Procedure

Thermal model analysis of EDM of composite material is very difficult. Therefore, powerful tool is required to analyze the given composite material. The three-dimensional composite model has been designed with the dimension of $450 \mu\text{m} \times 200 \mu\text{m} \times 30 \mu\text{m}$ along with spherical reinforcement with diameter of $30 \mu\text{m}$ as per 10% volume fraction of reinforcement.

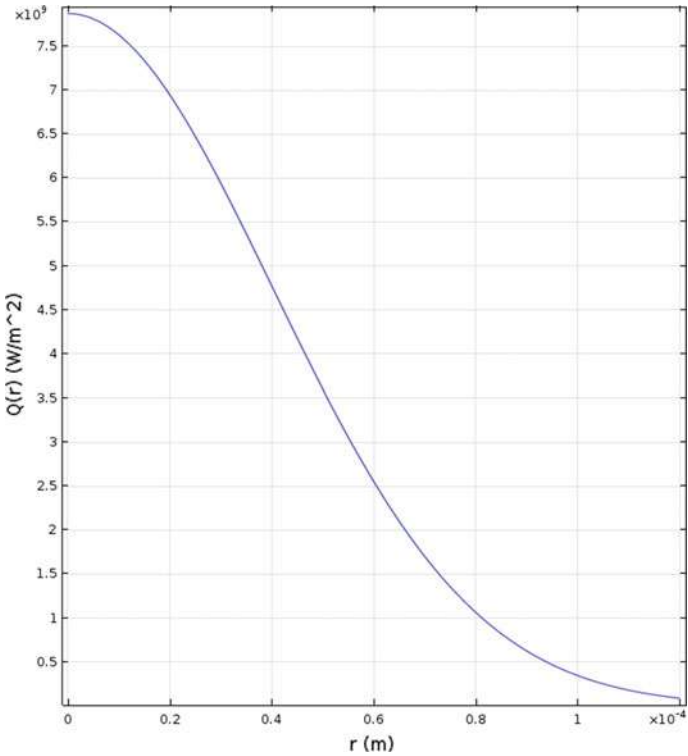


Fig. 20.2 Gaussian heat distribution with plasma channel radius for an axisymmetric model

20.2.4 Spark Radius

The model has been implemented by the Dibitonto [15] for the spark radius which is the function of discharge time and power and Erden integral equation [16] which was modification of Dibitonto equation as shown in below equation.

$$R(t) = Z P^m t^n$$

where $Z = \frac{L}{lm+0.5N}$, $m = M + 0.5n$, z, m, n are constants and L, M, N are experimental coefficients. However, no realistic model has been developed for the plasma channel radius. Since spark radius considered in this present study was 120 μm .

20.2.5 Material Removal Rate

The material removal criteria are based on the temperature distribution inside the workpiece. The removal is initiated when the material reaches its melting temperature (T_m) and above. MRR is calculated based on the crater morphology which is assumed as spherical dome shape.

$$\text{MRR} = \frac{C_v \times \text{No. of pulses}}{\text{Machining time}} \tag{20.3}$$

$$C_v = \frac{1}{6}\pi h(3r^2 + h^2) \tag{20.4}$$

The spherical dome volume (C_v) is calculated using the formula. Where r is radius of crater volume and h is depth of crater as shown Fig. 20.3 (Tables 20.1, 20.2 and 20.3).

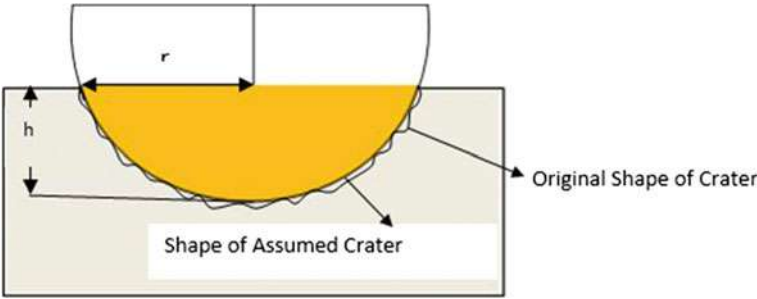


Fig. 20.3 Crater volume

Table 20.1 Thermal properties of Al4Cu6Si at room temperature

Thermal conductivity	173 W/m K
Density	2700 kg/m ³
Specific heat	890 J/kg K
Melting temperature	950 K

Table 20.2 Thermal properties of SiC at room temperature

Thermal conductivity	120 W/m K
Density	3200 kg/m ³
Specific heat	750 J/kg K
Melting temperature	3003 K

Table 20.3 Process parameters used in EDM of composite material

Fraction of heat input to the workpiece (p)	0.08
Voltage (V)	40,60
Current (I)	20, 30, 40 A
Pulse discharge	200 μ s
Convective heat transfer coefficient (h)	10,000 W/m ² K
Initial temperature (T_0)	295 K
Spark radius (R)	120 μ m

20.3 Results and Discussion

The basic composite (Al–6Cu–4si+10%SiC) model has been created as shown in Fig. 20.4. The maximum temperature can be observed at the center of the spark as shown in Fig. 20.5a, and it implies that temperature distribution in composite material indicated that the applied heat source is Gaussian distribution. The process parameters have been tabulated as per experimental results.

The temperature distribution in MMC is portrayed like isothermal bowl shape surface curves while EDM of composite materials as shown in Fig. 20.5b. The mechanism by which the material removal is taken place in EDM process is melting and evaporation. Therefore, the material removal was opted by filtering the composite material when temperature reaches its melting point and above.

However, the reinforcement particles which are exposed to heat source having higher melting point than matrix material, so particles are excavated without being melted as shown in Fig. 20.6a. The similar results were observed by previous researchers [4, 6] using ANSYS software (Fig. 20.7).

The increase in MRR was observed with the rise in voltage and current, and similar observation was made by the previous researches [8, 9, 17]. However, the influence of current was more significant on the MRR rather than applied voltage as shown in Figs. 20.8 and 20.9. Firstly, it was ascribed to the generation of electrons

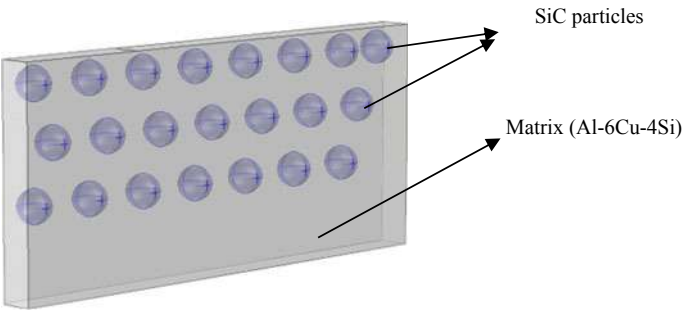


Fig. 20.4 Modeling of composite material (Al–6Cu–4si+10%SiC)

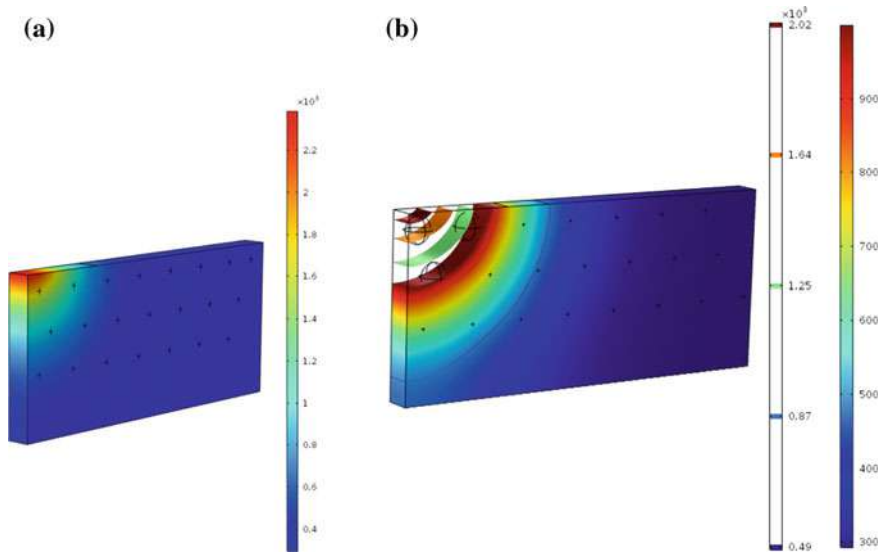


Fig. 20.5 Temperature distribution at the end of the single pulse at $V = 40$ V, $I = 20$ A, and $T_{on} = 200 \mu s$

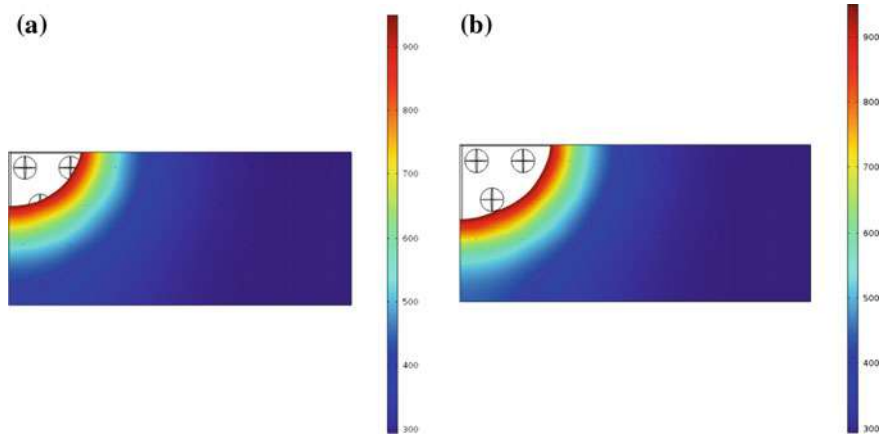


Fig. 20.6 EDM modeling for material removal rate at the end of single pulse at **a** $V = 40$ V, $I = 20$ A **b** $V = 40$ V, $I = 30$ A

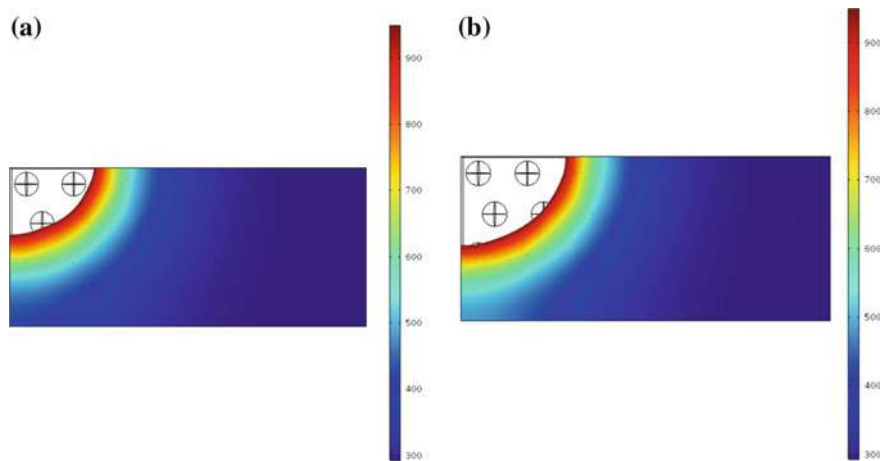


Fig. 20.7 EDM modeling for material removal rate at the end of single pulse at **a** $V = 50\text{ V}$, $I = 20\text{ A}$ **b** $V = 50\text{ V}$, $I = 30\text{ A}$

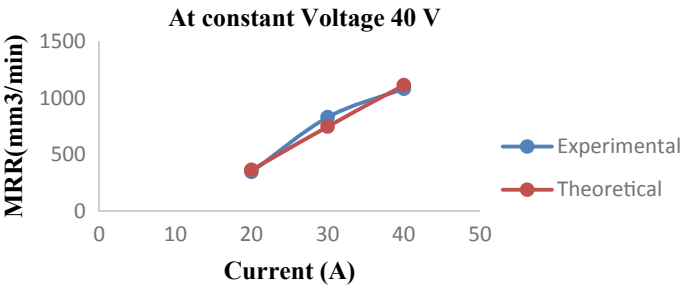


Fig. 20.8 Material removal rate of composite material during EDM at constant voltage = 40 V

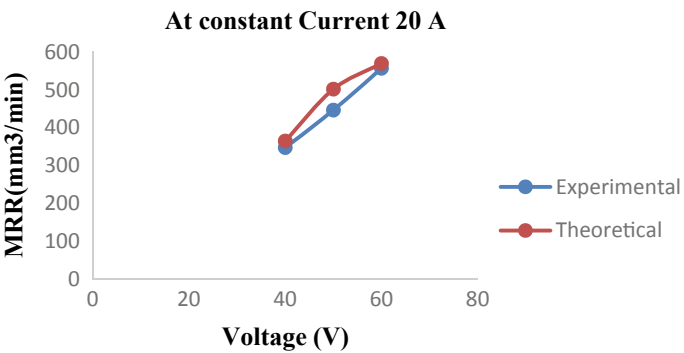


Fig. 20.9 Material removal rate of composite material during EDM at constant current = 20 A

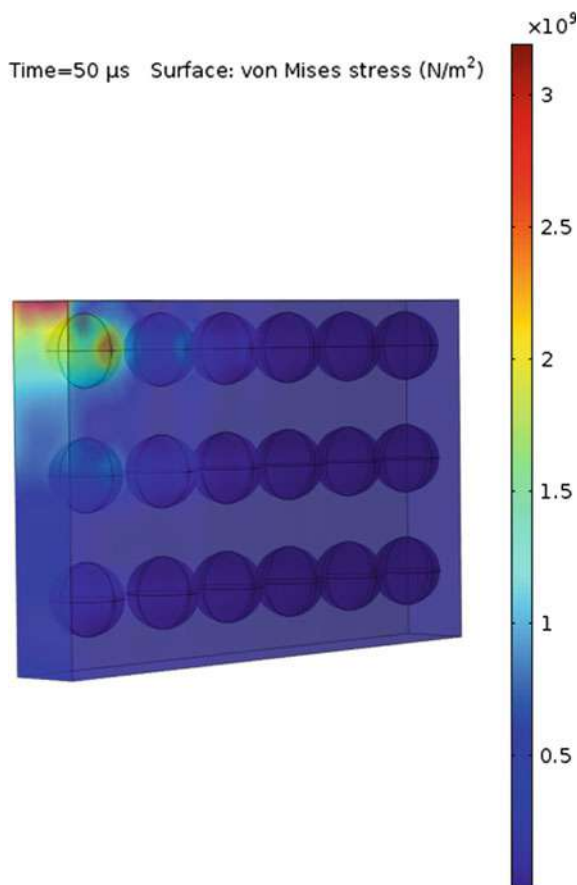


Fig. 20.10 Generation of thermal stresses in EDM of MMC

with high kinetic energy and moving towards workpiece resulting in the greater amount of heat energy at the workpiece. Secondly, the increase in plasma radius with rise of current and voltage might have ascribed to enhance the dimensions of the crater which directly enhance the MRR. The similar results were identified by previous researches [8]. Therefore, the increasing trend was observed from the results obtained by FEA modeling and the experiment as depicted in Figs. 20.8 and 20.9. Apart from this, the generation of thermal stresses was investigated during EDM of MMC. The maximum stresses can be observed at the near-surface of the crater and inside the reinforcement particle. It could be attributed to different coefficient thermal expansions of matrix and reinforcement material, consequently thermal distortion in melt pool that leads to crack formation inside the crater volume. In addition, the stresses induced in SiC particle exceed the limit of yield stress consequently it leads to fracture in reinforcement particle, similar results were observed by researchers experimentally [6] (Figs. 20.10, 20.11).

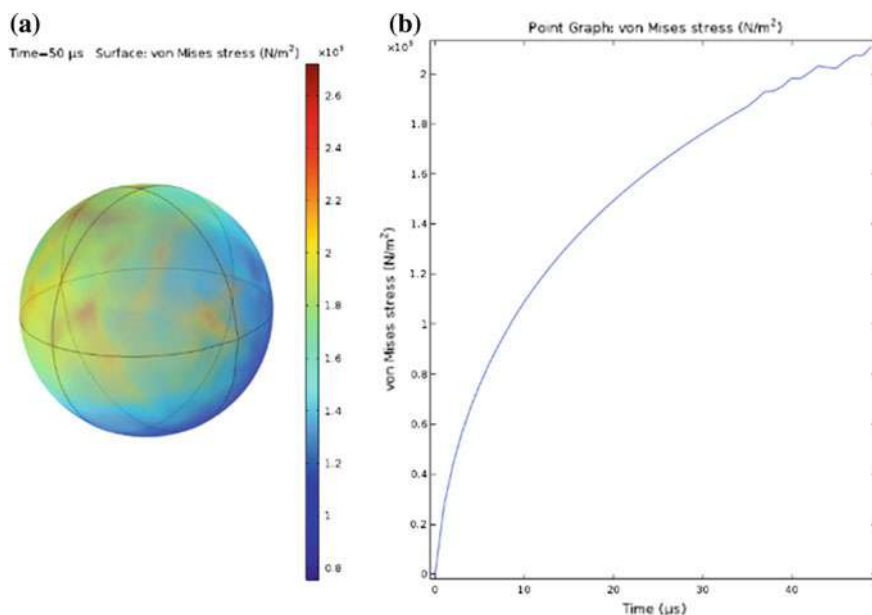


Fig. 20.11 a Thermal stresses distribution in reinforcement particle b increase in thermal stresses with time duration up to 50 μs at voltage (V) = 20 V and current (I) = 10 A

20.4 Conclusions

The finite analysis of EDM on metal matrix composite was evaluated in terms of temperature distribution, crater volume, and residual stresses. The influence of current and voltage gap on the MRR investigated during EDM of MMC.

The isothermal bowl shape surfaces noticed when temperature distributed in composite material at different machining conditions. The increase in material removal rate observed with current as well as voltage gap. However, the rise in current was more significant towards improving the MRR rather than applied voltage. The minimum percentage of error obtained between experimental and simulated results is 5.01%. The maximum limit of stress induced in the reinforce particle observed as 2 Gpa, which exceeds the yield strength of SiC particles, consequently initiates cracking.

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Chapter 21

Development and Analysis of a Discrete Particle Swarm Optimisation for Bi-criteria Scheduling of a Flow Shop with Sequence-Dependent Setup Time



V. Anjana , R. Sridharan and P. N. Ram Kumar

Abstract Most studies in flow shop scheduling neglect the setup times or consider the setup times along with the processing times. However, in industries that manufacture paint, textiles, ceramic tiles, etc., the setup times are significant and are sequence dependent. This paper addresses the problem of scheduling a flow shop operating in a sequence-dependent setup time (SDST) environment considering the objectives, namely minimisation of makespan and mean tardiness. The evolutionary method of discrete particle swarm optimisation (DPSO) based on weighted approach is developed and applied to SDST benchmark problems of flow shop scheduling. The efficacy of the metaheuristic is compared with that of a hybrid genetic algorithm, and it is observed that on an average, the proposed DPSO provides an improvement of 7.8, 22.3 and 11.3% in the values of mean ideal distance, computational time and diversification matrix, respectively. For most problems, the proposed DPSO performs superior to the hybrid genetic algorithm.

Keywords Permutation flow shop · Sequence-dependent setup time · Discrete particle swarm optimisation · Hybrid genetic algorithm

21.1 Introduction

Flow shop is a shop floor configuration where all the jobs share the same sequence of processing. Every job has a deterministic span of time for completing its operation in each machine known as the processing time [1]. For processing a job, setup activities needed to be performed on each machine and the time incurred for doing these preparations is known as the setup time. Setup includes activities such as obtaining tools, cleaning the machines, setting the machines, fixing and removing jobs and returning

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_21

tools. [2]. From the literature, it is observed that most studies in flow shops neglect the setup time or consider the setup time along with the processing time. However, in real-life situations, the presence of setup times cannot be neglected, especially in industries that manufacture paint, tiles, pharmaceuticals, automobiles, drugs and cosmetics, where the setup times are sequence dependent. In a sequence-dependent setup time (SDST) environment, the setup times of jobs depend on the current job to be processed and also on the previous job that has already been processed [3].

The studies on flow shops focus on optimising a single objective such as minimisation of makespan, total flow time, tardiness and number of setups [1, 3–5]. In real-life situations, a decision-maker has to deal with the optimisation of more than one objective. The presence of setup times and the attainment of multiple objectives increase the complexity of the scheduling problem. The solution to the multi-objective problem is obtained as a set of Pareto-optimal solutions or non-dominated solutions. A solution is said to be non-dominating, if it is not dominated by any other solutions of the multi-objective optimisation problem. Each solution in the Pareto-front has a better value for any one of the objectives and is a solution to the problem [6]. The decision-maker has to select a solution from the Pareto-front depending on the importance of the objective to be achieved. Researchers and practitioners adopt the weighted and non-weighted approach for solving the multi-objective problems. In a weighted approach, weights are attached to the objectives such that the multi-objective problem is converted to an equivalent single-objective problem. Hence, the present study focuses on the development of discrete particle swarm optimisation (DPSO) metaheuristic based on the weighted approach for solving the multi-objective SDST flow shop scheduling problem. The efficacy of the proposed metaheuristic is compared with the hybrid genetic algorithm.

The weighted approach has been adopted by many researchers like Rajendran and Ziegler [7], Eren and Guner [8], Eren [9] and Dhingra and Chandna [10] for solving the SDST flow shops with multiple objectives. Rajendran and Ziegler develop heuristics for scheduling an SDST flow shop with the objective of minimising weighted flow time and weighted tardiness. The researchers compare the performance of their heuristic with an existing heuristic, random search procedure and a greedy local search. The performance analysis reveals the better performance of the proposed heuristic. Eren and Guner address the scheduling problem in an SDST flow shop with the objective of minimising the weighted sum of total completion time and total tardiness. The authors develop an integer programming model for solving problems of up to 12 jobs. For solving larger-size problems, special heuristic algorithms and Tabu search are developed. The results from the computational studies indicate that the algorithms are effective for solving problems up to 1000 jobs. Eren considers the scheduling of a two-machine SDST flow shop with the objective of minimising four criteria. An integer programming model and Tabu search are developed for solving the multi-objective problem of up to 1000 jobs. Dhingra and Chandna develop a hybrid genetic algorithm minimising the weighted sum of total weighted squared tardiness and makespan of an SDST flow shop.

It is observed from the literature that the works related to SDST flow shops with multiple objectives are less in number. Further, no works have been reported with

discrete particle swarm optimisation (DPSO) based on weighted approach for solving an SDST flow shop scheduling problem with multiple objectives. Thus, the objectives of the present study are as follows.

- Development of a metaheuristic based on discrete particle approach for scheduling an SDST permutation flow shop with the objective of minimising makespan and mean tardiness.
- Determination of the best set of parameters of the proposed metaheuristic.
- Experimentation of the metaheuristic using benchmark problems.
- Comparison of the proposed metaheuristic with a hybrid genetic algorithm.

The rest of the paper is organised as follows. The problem definition and the assumptions related to the study are presented in Sect. 21.2. Section 21.3 provides the detailed description of the proposed metaheuristic. Section 21.4 presents the method of determining the best set of parameters of the proposed metaheuristic. The experimentation details are described in Sect. 21.5. Section 21.6 presents the analysis of the results, and Sect. 21.7 provides the conclusion.

21.2 Problem Definition

The present study addresses the scheduling problem of an n job $\times$ m machine SDST flow shop with the objective of minimising makespan and mean tardiness. The assumptions made in the study are as follows.

- All the jobs are available for processing at time zero.
- The processing times of operations of jobs are known in advance.
- Setup times for operations are considered explicitly from processing time and are sequence dependent.
- Each machine can process only one job at a time.
- No pre-emption is allowed.
- The machines are continuously available, that is no breakdown of machines.

Notations

n	Number of jobs to be scheduled
m	Number of machines in the flow shop
p_{ji}	Processing time of job j on machine i
d_j	Due date of job j
C_j	Completion time of job j
s_{ijk}	Setup time on i th machine if job j is preceded by job k
σ	Ordered set of jobs already scheduled, out of n jobs; partial sequence
$q(\sigma, i)$	Completion time of partial sequence σ on machine i (i.e. the release time of machine i after processing all jobs in partial sequence σ)
$q(\sigma_j, i)$	Completion time of job j on machine i , when the job is appended to partial sequence σ

$f_1(x)$	Makespan of sequence x
$f_2(x)$	Mean tardiness of sequence x
w_1	Weight assigned to makespan
w_2	Weight assigned to mean tardiness.

The objective function for an SDST flow shop scheduling problem is expressed as follows:

$$\text{Min } f(x) = w_1 \times f_1(x) + w_2 \times f_2(x) \quad (21.1)$$

Since a sequence-dependent flow shop is considered, the recursive equation for the completion time of job j on machine i is determined using Eq. 21.2.

$$q(\sigma_j, i) = \max\{q(\sigma, i) + s_{ijk}, q(\sigma_j, i - 1)\} + p_{ji}, \quad (21.2)$$

where job k precedes job j ; $q(\sigma_j, i)$ is the completion time of job j on machine i , when the job is appended to partial sequence σ ; $(q(\sigma, i) + s_{ijk})$ denotes the sum of the completion time of processing of job k on machine i and the setup time for job j on machine i ; $q(\sigma_j, i - 1)$ is the completion time of the immediately preceding operation of job j on the previous machine; and p_{ji} is the processing time of job j on machine i .

The flow time of job j , C_j , is given by

$$C_j = q(\sigma_j, m), \quad (21.3)$$

where $q(\sigma_j, m)$ is the completion time of the last operation of job j on machine m .

The makespan for a sequence of jobs is given by

$$f_1 = \max(C_j, \quad j = 1, 2, \dots, n) \quad (21.4)$$

The tardiness of a job is given by

$$t_j = \max\{0, (C_j - d_j)\} \quad (21.5)$$

The mean tardiness of a sequence of jobs is given by

$$f_2 = \frac{\sum_{j=1}^n t_j}{n} \quad (21.6)$$

21.3 Discrete Particle Swarm Optimisation

Particle swarm optimisation (PSO) developed by Kennedy and Eberhart [11] for optimising continuous linear functions mimics the social behaviour of birds gathering their food. PSO optimises a problem by having a population of candidate solutions and moving these particles around in the search space over the particle's position and velocity. The continuous PSO is not sufficient to solve the real-life problems with discrete problem features. Thus, the developers of PSO modified it to address the discrete problem, namely flow shop scheduling [12]. Discrete particle swarm optimisation (DPSO) is the discrete version of particle swarm optimisation. The difference between PSO and DPSO occurs in the representation of the solution. When PSO is applied for solving discrete optimisation problems (scheduling problems), the solution representation of PSO is modified to represent the discrete solutions [13]. In the present study, a variant of DPSO based on the weighted approach is developed for scheduling an SDST flow shop with the objective of minimising makespan and mean tardiness. The proposed metaheuristic is described in detail in Sect. 21.3.1.

21.3.1 The Proposed DPSO Metaheuristic

In DPSO, the initial population is considered as the swarm and each solution in the swarm is termed as the particle. The initial swarm for the present research is generated using the NEH heuristic [14] and the pair-wise interchange method. The generation of the initial swarm is followed by the computation of the objective function values of the particles in the swarm. Each particle in the swarm is represented as $X_1, X_2, X_3, \dots, X_N$, where N denotes the number of particles in the swarm. The personal best matrix corresponds to the objective function values of each particle in swarm. The objective function values are determined as the weighted sum of the objective function values. The lowest value among the personal best values is considered as the global best. Once the personal best matrix is formed, the position of the particles is updated. In PSO, every move of the particle to the next position is influenced by its own previous position, the position of the neighbouring particles and the particle in the leading position. The position of the particles is obtained from the velocity components. The position update is performed by two types of crossover and a mutation operation. The types of crossover involved are social crossover and cognition crossover. The previous position, the position of the neighbouring particles and the particle in the leading position are given by the mutation operation, the cognition velocity component and the social velocity component, respectively. The three components are determined using the following relations.

The position update equation consists of three components:

$$\lambda_i^t = w \otimes F_1(X_i^{t-1}) \quad (21.7)$$

$$\delta_i^t = c_1 \otimes F_2(\lambda_i^t, p_i^{t-1}) \quad (21.8)$$

$$\mu_i^t = c_2 \otimes F_3(\delta_i^t, g_i^{t-1}) \quad (21.9)$$

The first component given by Eq. 21.7 represents the velocity of the particle. In Eq. 21.7, F_1 represents the mutation operator with a probability of w . A uniform random number r is generated between 0 and 1. If r is less than w , then the mutation operator is applied to generate a perturbed permutation of the particle, otherwise the particle is kept without any change. The second component obtained using Eq. 21.8 represents the cognition part of the particle. F_2 in Eq. 21.8 represents the cognition crossover operator with a crossover probability c_1 . Here, λ_i^t and p_i^{t-1} are the two parents for crossover where λ_i^t is the particle obtained from mutation and p_i^{t-1} is the particle in the personal best matrix. The occurrence of this crossover operation depends on the random number generated. The third or social component is provided by Eq. 21.9 where F_3 and c_2 represent the crossover operator and social crossover probability, respectively. The parents for crossover are δ_i^t and g_i^{t-1} where δ_i^t is the particle obtained from the cognition crossover and g_i^{t-1} is the global best solution. The crossover operation depends on the random number generated. The objective function values of the velocity components are determined. Since the problem is of minimisation type, the least value among the three components is considered and the position is updated. A local search is performed on these solutions, which increases the diversity of the solutions. A non-dominant sorting procedure is applied to the solutions obtained from the local search, and the set of non-dominated solutions are updated in each generation. The solutions obtained from the local search become the swarm for the next generation. The procedure is repeated until it reaches the specified termination criteria.

21.3.2 Hybrid Genetic Algorithm

The hybrid genetic algorithm (HGA) is developed by combining the evolutionary method of genetic algorithm with a local search method. The initial population is generated using NEH and the pair-wise interchange method. The population is then subjected to genetic operators of selection, crossover and mutation. A local search is applied to the solutions obtained from mutation. The set of Pareto-optimal solutions is obtained by applying the non-dominant sorting algorithm to the offspring of mutation. The procedure is repeated for a specified number of generations. The hybrid GA is applied to the benchmark problems with the best set of parameters obtained from Taguchi's robust design and the utility index concept.

21.4 Parameter Configuration of DPSO

The parameters of the proposed DPSO include type of mutation, probability of mutation, type of cognition crossover, probability of cognition crossover, type of social crossover, probability of social crossover and the swarm size. The parameters of the DPSO metaheuristic are determined using Taguchi's robust design in combination with the concept of utility index [15, 16]. The parameters and their different levels are shown in Table 21.1. The L18 orthogonal array is selected, and the objective function values are determined for each trial. Once the objective function values are computed, the average response value of each objective function for each factor level is determined. The average values of the objectives for each factor level are presented in Table 21.2. The average response value corresponding to each level of the parameters is computed from the objective function values and is shown in Table 21.3.

The preference number for each objective function is obtained using Eq. 21.10.

$$P_b = Z \log \frac{y_b}{y'_b}, \tag{21.10}$$

where y_b is the value of the objective b , y'_b is the maximum or minimum acceptable value of the objective and Z is a constant. The value of Z has to be determined for computing the preference number for each objective. It is assumed that at optimum, value of the objective $P_b = 9$, and hence, the value of Z is computed as follows.

$$\text{At optimum, } y'_b \text{ of objective } b, P_b = 9; Z = \frac{9}{\log \frac{y_b^*}{y_b}}, \tag{21.11}$$

where y_b^* is the predicted optimal value of attribute b .

The predicted optimal value of makespan = 1651.87 + 1645.39 + 1643.81 + 1638.61 + 1645.72 + 1640.81 + 1642.64 - 3 × 1655.48 = 6838.3.

Table 21.1 Parameters and their levels

Sl. No.	Parameter	Code	Level		
			1	2	3
1	Mutation type	A	Shift	Swap	–
2	Mutation probability	B	0.1	0.2	0.3
3	Type of cognition	C	Single point	Two point	PMX
4	Cognition probability	D	0.7	0.8	0.9
5	Type of social crossover	E	Single point	Two point	PMX
6	Social crossover probability	F	0.7	0.8	0.9
7	Swarm size	G	20	30	50

Table 21.2 Average response value by factor levels

Sl. No.	Parameter				Probability of cognition crossover	Type of cognition crossover	Probability of cognition crossover	Type of social crossover	Probability of social crossover	Swarm size	Average makespan	Average mean tardiness
	Type of mutation	Probability of mutation	Type of cognition crossover	Probability of cognition crossover								
1	Shift	0.1	Single point	0.7	Single point	0.7	Single point	0.7	20	1663.67	308.23	
2	Shift	0.1	Two point	0.8	Two point	0.8	Two point	0.8	30	1667.67	296.77	
3	Shift	0.1	PMX	0.9	PMX	0.9	PMX	0.9	50	1682.50	288.40	
4	Shift	0.2	Single point	0.7	Single point	0.7	Two point	0.8	50	1654.67	289.00	
5	Shift	0.2	Two point	0.8	Two point	0.8	PMX	0.9	20	1659.33	292.80	
6	Shift	0.2	PMX	0.9	PMX	0.9	Single point	0.7	30	1628.50	291.15	
7	Shift	0.3	Single point	0.8	Single point	0.8	Single point	0.9	30	1669.67	299.63	
8	Shift	0.3	Two point	0.9	Two point	0.9	Two point	0.7	50	1682.50	289.10	
9	Shift	0.3	PMX	0.7	PMX	0.7	PMX	0.8	20	1623.33	307.60	
10	Swap	0.1	Single point	0.9	Single point	0.9	PMX	0.8	30	1685.00	299.30	
11	Swap	0.1	Two point	0.7	Two point	0.7	Single point	0.9	50	1685.00	292.65	
12	Swap	0.1	PMX	0.8	PMX	0.8	Two point	0.7	20	1646.00	301.53	
13	Swap	0.2	Single point	0.8	Single point	0.8	PMX	0.7	50	1654.67	301.97	
14	Swap	0.2	Two point	0.9	Two point	0.9	Single point	0.8	20	1663.67	308.20	
15	Swap	0.2	PMX	0.7	PMX	0.7	Two point	0.9	30	1635.50	315.68	
16	Swap	0.3	Single point	0.9	Single point	0.9	Two point	0.9	20	1680.50	285.35	
17	Swap	0.3	Two point	0.7	Two point	0.7	PMX	0.7	30	1569.50	281.05	
18	Swap	0.3	PMX	0.8	PMX	0.8	Single point	0.8	50	1647.00	299.13	

The predicted optimal value of mean tardiness = $295.85 + 293.64 + 293.43 + 251.64 + 295.19 + 253.50 + 293.38 - 3 \times 292.88 = 1098$.

The preference number is determined for makespan and mean tardiness from the predicted optimal values using Eq. 21.10. The utility index is computed using Eq. 21.12.

$$U_d = \sum_{b=1}^l a_b P_b, \tag{21.12}$$

where a_b is the weight assigned to the objective b , P_b denotes the preference number of objective b , l is the number of objectives and d is the experiment number. The preference number and the utility index corresponding to each experiment of the orthogonal array are shown in Table 21.4. The average utility index corresponding to each level of parameters is determined and is provided in Table 21.5. From Table 21.5, it is observed that parameter B , that is the mutation probability, has the highest range, and hence, it is the influencing factor on the performance characteristics of the algorithm. The order of importance of the parameters on the performance of the algorithm can be listed as follows: mutation probability, probability of social

Table 21.4 Preference number and utility index

Experiment No.	Preference number		Utility index
	Makespan	Mean tardiness	
1	0.082	0.180	0.131
2	0.066	0.455	0.260
3	0.010	0.662	0.336
4	0.116	0.647	0.381
5	0.098	0.552	0.325
6	0.218	0.593	0.406
7	0.059	0.385	0.222
8	0.010	0.644	0.327
9	0.239	0.195	0.217
10	0.000	0.393	0.197
11	0.000	0.556	0.278
12	0.150	0.339	0.245
13	0.116	0.329	0.223
14	0.082	0.181	0.131
15	0.191	0.007	0.099
16	0.017	0.739	0.378
17	0.455	0.849	0.652
18	0.146	0.397	0.272

Table 21.5 Average utility value for each level of parameters

Parameter	Utility value			Max–Min (range)
	Level 1	Level 2	Level 3	
A	0.289	0.275	–	0.015
B	0.241	0.261	0.344	0.103
C	0.255	0.329	0.262	0.074
D	0.293	0.258	0.296	0.038
E	0.240	0.282	0.325	0.085
F	0.330	0.243	0.325	0.087
G	0.238	0.306	0.303	0.068

crossover, type of social crossover, type of cognition crossover, swarm size, probability of cognition crossover and type of mutation. The different levels of parameters are plotted with the average utility index, and the parameter with the highest utility value is selected as the best parameter. The utility index value for each parameter at different levels is shown in Fig. 21.1. The parameter values with the highest utility values are A1 B3 C2 D3 E3 F1 G2.

The best set of parameters of the proposed DPSO is obtained from Taguchi’s method, and the concept of utility value is shown in Table 21.6.

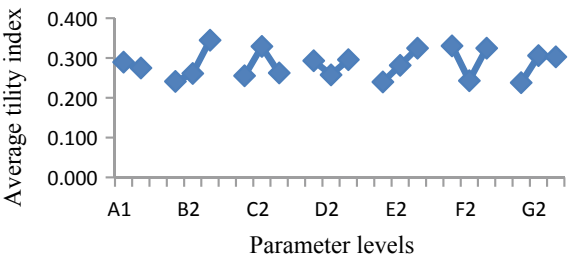


Fig. 21.1 Utility index value for each parameter at various levels

Table 21.6 Best set of parameters of DPSO

Sl. No.	Code	Parameter	The best setting level
1	A	Mutation type	Shift
2	B	Mutation probability	0.3
3	C	Type of cognition crossover	Two point
4	D	Probability of cognition crossover	0.9
5	E	Type of social crossover	PMX
6	F	Probability of social crossover	0.7
7	G	Swarm size	30

21.5 Experimentation

Experimentation of the proposed DPSO is carried out on the SDST benchmark problems of flow shop scheduling. The study is conducted on 20 jobs, 50 jobs and 100 jobs with the machine size as 5, 10 and 20. The method for generating the setup times and the due dates required for the study are provided in the following subsection. The algorithms are applied with the best set of parameters determined using Taguchi's orthogonal array and utility index as described in the preceding section. The DPSO metaheuristic terminates after examining 1000 solutions for 20 jobs and 50 jobs, whereas the termination occurs after examining 1500 solutions for 100 jobs. All the problem instances are solved using MATLAB software on a desktop computer that runs on an Intel Core Processor with 3 GHz RAM.

21.5.1 Data Generation for the Problems for Computational Studies

In the present study, the setup times of jobs are considered explicitly. Hence, the setup times of jobs are generated using the setup time level concept. The setup time level is expressed as the ratio of maximum setup time to the maximum processing time. The setup time for the jobs is expressed using the following relation.

$$\text{Setup time level} = \frac{\max s_{ijk}}{\max p_{ijk}} \times 100$$

for all $i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$; $k = 1, 2, \dots, n$, (21.13)

where p_{ijk} is the maximum time element of the processing time matrix and s_{ijk} is the maximum time element of the setup time matrix. The setup time level is assumed to be 25%, and hence, the setup time of jobs is generated in a uniform distribution in the interval between 1 and 25. The processing times are obtained from the benchmark problems of flow shop scheduling [17]. The due dates of jobs required for the study are generated using the method of total work content. The due date of a job is expressed as follows.

$$\begin{aligned} \text{Due date of a job} &= \text{arrival time} + r \\ &\quad \times (\text{processing time of the job} + \text{setup time of the job}), \end{aligned}$$

(21.14)

where r is the allowance factor and it is set equal to 2.

$$\begin{aligned} \text{Setup time of a job} \\ &= \text{number of operations of the job} \times \text{average setup time of an operation} \end{aligned}$$

(21.15)

21.6 Results and Discussion

The Pareto-optimal solutions are determined for the nine problem sizes using the DPSO metaheuristic. The efficacy of the proposed metaheuristic is compared with the hybrid genetic algorithm based on the performance measures such as mean ideal distance (MID), computational time, diversification matrix (DM), average objective values and minimum objective values. The obtained Pareto-optimal solutions for the nine SDST benchmark problem sizes are shown in Table 21.7. It is observed from the values in Table 21.7 that for most of the problem sizes, the values of makespan and mean tardiness obtained from DPSO are better than the hybrid genetic algorithm. In the 20 job $\times$ 20 machine problem, both the metaheuristics provide mean tardiness values as zero for one of the solutions in the Pareto-optimal set. However, a better makespan value is obtained from DPSO for the zero mean tardiness value.

21.6.1 Performance Analysis of the Proposed Metaheuristic Based on the Mean Ideal Distance, Computational Time and the Diversification Matrix

The MID values, computational time and DM values of the proposed metaheuristic are shown in Table 21.8. From Table 21.8, it is observed that for all the problem

Table 21.7 MID, computational time and the diversification matrix for the SDST benchmark problems

Sl. No.	Problem size $n \times m$	Mean ideal distance		Computational time (s)		Diversification matrix	
		Hybrid genetic algorithm	DPSO	Hybrid genetic algorithm	DPSO	Hybrid genetic algorithm	DPSO
1	20 $\times$ 5	1612.24	1634.97	3.12	0.83	22.74	95.13
2	20 $\times$ 10	2059.68	1891.51	3.27	0.81	47.58	41.23
3	20 $\times$ 20	2834.00	2426.01	3.61	1.46	174.03	408.08
4	50 $\times$ 5	3747.80	3695.62	6.37	14.97	61.94	198.46
5	50 $\times$ 10	4160.81	3735.86	6.96	15.85	58.33	105.89
6	50 $\times$ 20	4917.66	3940.43	13.53	15.86	79.96	165.13
7	100 $\times$ 5	7421.30	7398.17	24.13	1.93	25.00	55.25
8	100 $\times$ 10	7460.86	7136.28	24.15	1.77	61.19	93.02
9	100 $\times$ 20	8863.25	7654.33	35.75	2.08	48.71	48.96

n —Number of jobs; m —number of machines
The values are provided in bold to show the better performance of the discrete particle swarm optimisation

Table 21.8 Minimum and average makespan values for the SDST benchmark problems

Problem size $n \times m$	Hybrid genetic algorithm		DPSO	
	Minimum	Average	Minimum	Average
20×5	1581	1589.33	1563	1609.50
20×10	2042	2053.33	1871	1890.00
20×20	2740	2834.00	2222	2426.00
50×5	3534	3552.20	3432	3508.50
50×10	4016	4027.25	3619	3664.00
50×20	4839	4869.25	3881	3929.50
100×5	6893	6895.50	6852	6879.00
100×10	7049	7067.67	6745	6791.50
100×20	8527	8551.00	7431	7454.50

n —Number of jobs; m —number of machines
The values are provided in bold to show the better performance of the discrete particle swarm optimisation

sizes except the $20 \text{ job} \times 5 \text{ machine}$ problem instance, the MID values are lower for the DPSO metaheuristic. Lower MID values indicate better performance of the metaheuristic. Hence, it is evident from the MID values that the DPSO metaheuristic performs better than the hybrid genetic algorithm. When the computational time is considered, the time taken for the DPSO metaheuristic to provide the Pareto-optimal solutions is lower for most of the problem sizes when compared to the hybrid genetic algorithm. The DM values also reveal the superior performance of DPSO metaheuristic to hybrid genetic algorithm. The higher values of DM indicate the better performance of the metaheuristic. Thus, in terms of MID values, computational time and DM values, the DPSO metaheuristic outperforms the hybrid genetic algorithm.

21.6.2 Performance Analysis Based on the Average and Minimum Objective Function Values

The average values and the minimum values of makespan and mean tardiness obtained for the proposed algorithms are shown in Tables 21.9 and 21.10, respectively. From Table 21.10, it is observed that the minimum value of makespan is obtained from the DPSO metaheuristic for all the problem instances. Further, the average makespan provided by the DPSO metaheuristic has lower values for all the problem sizes except the $20 \text{ job} \times 5 \text{ machine}$ problem. In that problem instance, though the average makespan value is better for HGA, the minimum makespan is provided by the DPSO metaheuristic. Similarly, the minimum and average values of mean tardiness have better values for the DPSO metaheuristic except the $20 \text{ job} \times$

Table 21.9 Minimum and average mean tardiness values for the SDST benchmark problems

Problem size $n \times m$	Hybrid genetic algorithm		DPSO	
	Minimum	Average	Minimum	Average
20×5	265.20	270.73	276.90	286.90
20×10	148.30	160.57	67.10	75.10
20×20	0.00	1.47	0.00	4.15
50×5	1175.50	1194.78	1100.70	1158.90
50×10	1021.80	1045.55	700.40	728.30
50×20	673.10	688.14	227.20	290.48
100×5	2731.40	2743.65	2716.70	2722.55
100×10	2372.50	2390.02	2190.40	2191.30
100×20	2327.70	2331.85	1730.70	1737.55

n —Number of jobs; m —number of machines

The values are provided in bold to show the better performance of the discrete particle swarm optimisation

Table 21.10 Pareto-optimal solutions for the SDST benchmark problems

Sl. No.	Problem size $n \times m$	Hybrid genetic algorithm		DPSO	
		Makespan	Mean tardiness	Makespan	Mean tardiness
1	20×5	1599	265.20	1563	296.9
		1588	267.90	1656	276.9
		1581	279.10		
2	20×10	2073	148.30	1909	67.1
		2045	149.00	1871	83.1
		2042	184.40		
3	20×20	2914	0.00	2630	0
		2848	1.40	2222	8.3
		2740	3.00		
4	50×5	3569	1175.50	3590	1100.7
		3557	1181.30	3557	1113.7
		3554	1193.30	3432	1220.8
		3547	1197.20	3455	1200.4
		3534	1226.60		
5	50×10	4041	1021.80	3709	700.4
		4030	1035.10	3619	756.2

(continued)

Table 21.10 (continued)

Sl. No.	Problem size $n \times m$	Hybrid genetic algorithm		DPSO	
		Makespan	Mean tardiness	Makespan	Mean tardiness
6	50×20	4022	1050.80		
		4016	1074.50		
		4901	673.10	3881	323.7
		4888	677.00	4015	227.2
		4884	679.80	3902	310
		4880	682.00	3920	301
		4876	683.20		
		4875	683.50		
		4870	685.40		
		4867	689.40		
		4854	691.30		
		4849	694.60		
		4848	694.80		
		4839	723.60		
7	100×5	6898	2731.40	6852	2716.7
		6893	2755.90	6906	2728.4
8	100×10	7091	2372.50	6745	2192.2
		7075	2374.30	6838	2190.4
		7070	2382.30		
		7065	2394.00		
		7056	2400.00		
		7049	2417.00		
9	100×20	8575	2327.70	7431	1744.4
		8527	2336.00	7478	1730.7

n —Number of jobs; m —number of machines

5 machine case. Hence, it is evident from the results that the DPSO metaheuristic performs better when compared to the hybrid genetic algorithm.

21.7 Conclusions

The present study proposes a DPSO for solving the bi-objective scheduling problem of an SDST flow shop. Computational studies using the SDST benchmark problems reveal that on an average, the proposed DPSO provides an improvement of 7.8, 22.3

and 11.3% when compared with HGA for the measures such as MID values, computational time and diversification matrix, respectively. In the present study, continuous availability of the machines is assumed. However, in real-life situations, we may encounter breakdown and repair of machines. Thus, the work can be extended by integrating appropriate scheduling and maintenance policies. Other methods for generating the initial population and due dates can be examined. Performance measures other than makespan and mean tardiness can also be considered.

Acknowledgements The authors express their sincere thanks to the reviewers for their suggestions which helped in improving the initial version of the paper.

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Chapter 22

A MATLAB-Based Application to Solve Vehicle Routing Problem Using GA



Nikki Rathore , P. K. Jain and M. Parida

Abstract Application of vehicle routing problem in real-life logistics operations is a need of today's world, and this paper focuses on developing a vehicle routing problem for the delivery and pickup of products from multiple depot to the graphically scattered customers. The proposed model can be used in real-life applications of various logistic operations where there is a need to determine the optimized location of warehouse for setup so that the demand of customers is fully satisfied. To do so, a genetic algorithm-based solution methodology is proposed to solve the above-stated problem. The proposed algorithm is tested on generated data based on real-life scenarios. The experiments show that the proposed algorithm successfully finds the potential locations for warehouse setup based on the demand and location of customers for minimum transportation cost. The presented approach can provide good solutions to a large-scale problem generally found in real life.

Keywords VRP · Multi-depot · Time windows · Genetic algorithm · Real life · Uttarakhand

22.1 Introduction

Routing of vehicle is an important part of any transportation strategy and can give economic advantage [5] and for such reason, the researchers have shown a great interest in vehicle routing problem (VRP). Till date many variants of VRP are introduced in the literature such as dynamic VRP (DVRP) [11, 9], capacitated VRP (CVRP) [18], VRP with pickup and delivery (VRPPD) [6], and VRP with time windows (VRPTW) [15, 10]. Of all the variants of vehicle routing, capacitated vehicle routing (CVRP) [7] serves as the basis among all the classical transportation optimization

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_22

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models. Its aim is to produce a set of routes with minimum cost from central depot to graphically distributed customers such that the demand of all is satisfied without violating the capacity constraints of vehicle. If the pickup demand along with the delivery demand are added to model, the VRP will become pickup and delivery problem [16] which is the generalization of CVRP.

To gain the economic advantage and to be more competitive in nature, many small- and medium-scale companies come together as a single firm. As a result of which, a need to manage multiple depots along with operating costs arises, and faced with customer's requests that they will not accept any kind of service if the vehicle visits them beyond the time windows [3]. If the firm wants to enter in a new market, the main motive is to locate the optimum location so that the total transportation cost can be minimized. Some of the researchers have worked on this aspect such as Özyurt and Aksen [12] and developed a solution methodology where the optimum location of depots is found out along with the feasible routes to serve the customers from previously proposed depots; Alaia et al. [1] have suggested a new genetic algorithm-based solution to find the new depot locations so that the pickup and delivery demand of customers can be satisfied with minimum travel cost; Shen and Chen [13] have proposed a dual-state particle swarm-based methodology where in first stage the optimized locations of depots are identified and then feasible routes are formed for servicing customers; Liu and Lin [8] have developed a hybrid heuristic methodology which is a combination of simulated annealing and tabu search to solve the depot-location allocation routing and inventory problem.

To deal with time windows and customer preferences, various models have been proposed in the literature so far: Balakrishnan [2] proposed several exact heuristics to solve VRP with semi-soft time windows where the penalties are considered only on late arrival of vehicle while the early arrival is acceptable; Taillard et al. [14] and Chiang and Russell [4] proposed heuristics based on tabu search.

In this research, the authors have considered a vehicle routing problem where delivery to graphically scattered customers is given by multiple depots whose locations are to be allocated on the basis of customers demand and their time windows. The problem in hand deals with soft time windows where the violation of time windows is acceptable with some penalty. According to vehicle routing taxonomy, this problem can be referred to as the multiple depot pickup and delivery vehicle routing problem with soft time windows. To the best of author's knowledge, no solution methodology is proposed so far to solve such problem.

The outline of rest of the paper is as follows: in Sect. 22.2, the mathematical formulation of the problem is illustrated and presented; in Sect. 22.3, a new solution methodology based on GA is presented and results and experimental analysis are explained in Sect. 22.4; finally, in Sect. 22.5, conclusion of the proposed work and future scope is given.

22.2 Problem Description and Formulation

The vehicle routing problem studied in this paper involves locating a set of low-cost feasible routes from multiple depots to graphically distributed customers having delivery and pickup demand. Each customer has their own time windows and service will be given in that time duration only, failing to which a penalty cost should be incurred into the total transportation cost borne by the supplier. The formulated problem includes optimization of total transportation cost. The optimization function includes the total transportation cost and penalty associated with time windows violation.

The problem can be modeled on a complete directed graph $G: (V, A)$ in which V represents the set of nodes representing customers and depots, and A is the set of arcs connecting two points from i to j which denotes a route, where $i, j \in V$. Following notations are used in the formulation:

$G: (V, A)$	Directed graph
V	Set of nodes
X_i and Y_i	Geographical coordinates of node $i \forall i \in V$
$V_C: \{1, \dots, N\}$	Set of customers
$V_D: \{N + 1, \dots, N + M\}$	Set of depots
$V_C \cup V_D = V$	Union of two represents total node set
$V_C \cap V_D = \phi$	Null set
A	Set of arcs
C_{ij}	Distance between node i and $j \forall i, j \in V$
D_j	Delivery demand by customer node $j \in V_C$
P_j	Pickup demand of customer node $j \in V_C$
V_H	Available vehicle
VH	Total no. of vehicles
Q_v	Total capacity of a vehicle $v \in V_H$
$M = \max \left\{ \sum_{j \in V_C} (D_j + P_j), \sum_{i \in V} \sum_{\substack{j \in V_C \\ i \neq j}} C_{ij} \right\}$	Sufficiently large no.
Λ	Positive coefficient to convert time units to cost
t_D	Vehicle start time from depot
SP_v	Speed of vehicle $v \in V_H$
t_{ij}	Time to travel from node i to $j \forall i, j \in V$
a_{iv}	Arrival time of vehicle $v \in V_H$ at vertex $i \in V$
$[a_i, b_i]$	Time window for each customer $\forall i \in V_C$
a_i	Earliest start time to service customer $i \in V_C$
b_i	Latest finish time at customer $i \in V_C$

w_{iv}	Wait time of vehicle $v \in V_H$ at customer location $i \in V_C$
p_{iv}	Tardiness of delivery to customer location $i \in V_C$ by vehicle $v \in V_H$
λ_{1i}	Penalty for early arrival at demand/customer node $i \in V_C$
λ_{2i}	Penalty for lateness at demand/customer node $i, i \in V_C$
s_{iv}	Time when service at $i \in V_C$ ends
$s_i = f_k(D_i + P_i)$	Service time (for loading and unloading activities) for each customer $i \in V_C$
T_L	Maximum route duration length.

Decision Variables

$x_{ijv} = \begin{cases} 1, & \text{if vehicle } v \in V_H \text{ traverse edge } i, j \\ 0, & \text{otherwise} \end{cases}$	Binary decision variable indicating whether a vehicle travels on a given edge in the solution
y_{ijv}	Total pickup load by vehicle $v \in V_H$ while travelling arc i, j
z_{ijv}	Total delivered load by vehicle $v \in V_H$ while travelling arc i, j
S	Variable used in subtour elimination.

Given all the above-stated sets, parameters, and variables, the vehicle routing problem for the multiple depots with picked up and delivered load with customer time windows is formulated as:

Objective Function:

Minimize

$$\left[\Lambda \left\{ \sum_{v \in V_H} \sum_{i \in V_D} \sum_{j \in V_C} C_{ij} x_{ijv} \right\} + \left\{ \sum_{v \in V_H} \sum_{j \in V_C} (\lambda_{1j} E(w_{jv}) + \lambda_{2j} E(p_{jv})) \right\} \right] \quad (22.1)$$

Subjected to:

$$\sum_{v \in V_H} \sum_{i \in V} x_{ijv} \geq 1 \quad \forall j \in V_C \quad (22.2)$$

$$\sum_{i \in V_D} \sum_{j \in V_C} x_{ijv} - \sum_{i \in V_C} \sum_{j \in V_D} x_{ijv} = 0 \quad \forall v \in V_H \quad (22.3)$$

$$\sum_{i \in V} x_{ipv} - \sum_{j \in V} x_{p j v} = 0 \quad \begin{matrix} \forall p \in V, \\ v \in V_H, i \neq j \end{matrix} \quad (22.4)$$

$$\sum_{\substack{i \in V_C \\ j \in V_D}} \sum_{v \in V_H} y_{ijv} = \sum_{i \in V_C} P_i \quad (22.5)$$

$$\sum_{\substack{i \in V_D \\ j \in V_C}} \sum_{v \in V_H} z_{ijv} = \sum_{j \in V_C} D_j \quad (22.6)$$

$$y_{ijv} + z_{jiv} \leq Q_v \quad (22.7)$$

$$\sum_{\substack{i \in V_C \\ j \in V_C}} \sum_{v \in V_H} y_{ijv} = P_i \quad \forall i \neq j \quad (22.8)$$

$$\sum_{\substack{i \in V_C \\ j \in V_C}} \sum_{v \in V_H} z_{ijv} = D_j \quad \forall i \neq j \quad (22.9)$$

$$\sum_{j \in V_C} x_{jiv} D_i \geq z_{iv} \quad \forall v \in V_H, i \in V_C \quad (22.10)$$

$$\sum_{v \in V_H} \sum_{j \in V_C} x_{jiv} \geq \left[\frac{P_i + D_j}{Q_v} \right] \quad \forall i \in V_C \quad (22.11)$$

$$\sum_{i \in V_D} \sum_{v \in V_H} x_{iv} \leq V_H \quad (22.12)$$

$$a_{iv} = t_D + \sum_{i \in V_C, j \in V} t_{ji} + \sum_{j \in V_C} s_j \quad (22.13)$$

$$C_{ij} = \sqrt{(X_j - X_i)^2 + (Y_j - Y_i)^2} \quad \forall i, j \in V \quad (22.14)$$

$$t_{ij} = C_{ij} / SP_v \quad \forall v \in V_H, l \in V, j \in V, i \neq j \quad (22.15)$$

$$w_{iv} = \max\{a_i - a_{iv}, 0\} \quad \forall i \in V_C, v \in V_H \quad (22.16)$$

$$p_{iv} = \max\{s_{iv} - b_i, 0\} \quad \forall i \in V_C, v \in V_H \quad (22.17)$$

$$\sum_{i \in S} \sum_{j \in S \setminus \{i\}} x_{ijv} \leq |S| - 1 \quad \forall S \in V_C, |S| \geq 2, v \in V_H \quad (22.18)$$

$$x_{ijv} \in \{0, 1\} \quad \forall i \in V, j \in V, v \in V_H \quad (22.19)$$

$$y_{ijv} \geq 0 \quad \forall i \in V, j \in V, v \in V_H \quad (22.20)$$

$$z_{ijv} \geq 0 \quad \forall i \in V, j \in V, v \in V_H \quad (22.21)$$

Equation 22.2 gives the objective function which minimizes the total transportation costs including total penalty due to time window violation. The minimization function is subjected to some constraints which are given as: Eq. 22.3 guarantees that each customer must be visited at least once so that no customer will be left unserved among all the routes. Equation 22.4 states that each vehicle should start and end its trip at the same depot. Equation 22.5 describes the flow of path of the vehicle. Equations 22.6 and 22.7 give the flow equations for picked up and delivered loads, respectively. Equation 22.8 is a capacity constraint. Equations 22.9 and 22.10 ensure that the demand of each customer node is fulfilled. Equation 22.11 guaranteed that at any point i , the load delivered should not be greater than the demand of that point i . Equation 22.12 imposes constraint on number of vehicles used. Equation 22.13 calculates the vehicle arrival time at node i . Equation 22.14 calculates the distance between two nodes. Equation 22.15 used to estimate the travel time between node i and j . Equations 22.16 and 22.17 compute the wait time and tardiness of service at customer node i . Equation 22.18 gives subtour elimination constraint. Equations 22.19, 22.20, and 22.21 are the definition constraints on decision variables.

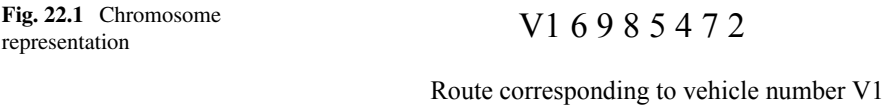
22.3 Solution Approach

Genetic algorithm (GA) is extensively used in the previous works to solve the vehicle routing problem and its variants [17]. In this paper, the authors also propose a new modified GA to solve the multiple depot pickup and delivery VRP with time windows. The proposed solution approach consists of three steps: (1) set up the warehouses, (2) identify the feasible routes from warehouse to customers, and (3) minimize the total cost. The proposed GA is used at three places in the whole algorithm, once for setting up warehouses to find the optimum location with respect of routes. After the setting up of warehouses, second step is to pass the customer locations into GA to find possible routes in-between warehouse and customers and third step is to calculate minimum cost route by passing the demand and time window of each customer into the GA which is the main focus of our algorithm. The objective function is also used to parameterize the fitness function for GA which will give the optimized minimum cost route. A graphical user interface is developed for the above-mentioned step to provide the visual effect of each operation and to easily edit the input based on user's requirements.

22.3.1 Genetic Algorithm

Genetic algorithm is a meta-heuristic inspired by the natural selection and evolution of organisms. GAs are used to generate high-quality solutions to optimization and search problems using bio-inspired genetic operators such as selection, crossover, and mutation. A population of N chromosomes which represents cluster of routes is

Fig. 22.1 Chromosome representation



initialized and then these are subjected to an evolutionary process until the stopping criteria is met. In the proposed GA, an alternating edge crossover (AEX) is used. In our method, each chromosome represents feasible vehicle routes satisfying customer demand for minimum possible cost of travel and penalty imposed. The length of chromosome is changeable and will depend on the number of consumers served by the single depot which will vary according to capacity and time window constraints. Each cluster is an ordered subset of vehicle and customer nodes as shown in Fig. 22.1. The vehicle number acts as a delimiter which indicates the start or end of a current route in a specified chromosome.

Initialization. To produce the initial population, the authors have used random permutations of N customers per depot. The routing technique takes into significance that the vehicle capacity constraint is not desecrated before adding a demand node to the existing route. A fresh route is initialized every time a new demand node is encountered that cannot be joined to the present route due to limitations of the model. This procedure is repeated till all customers/demand nodes have been allocated.

Fitness Evaluation. The fitness value of each chromosome is determined by using the objective function as a fitness function because of variability of the problem.

Selection. Tournament selection strategy is used for the selection of parents. It is a fitness-based selection scheme in which m individuals are chosen at random from the population set and the finest individuals on the basis of their fitness value are selected out of these to become a parent. Figure 22.2 shows the example of selection criteria.

Crossover. To solve the problem, we have used the crossover operator named as alternating edges crossover (AEX). The AEX operator interprets a chromosome as

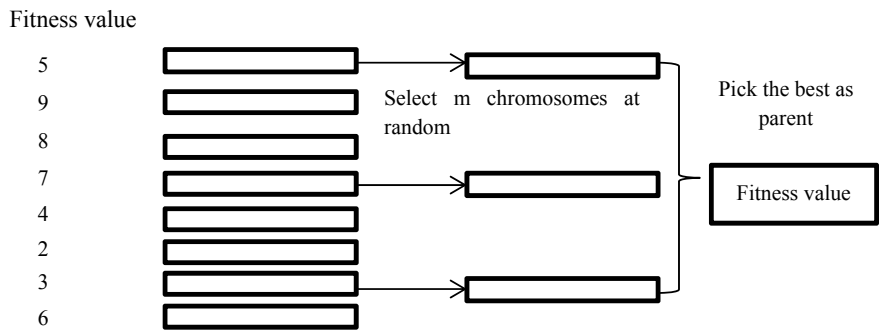


Fig. 22.2 Tournament selection

a directed cycle of arcs. The child cycle is formed by choosing in alternation arcs from the first and from the second parent, with further random selections in case of infeasibility. For instance, let the two parents be:

$$P1 = (5 \ 1 \ 7 \ 8 \ 4 \ 9 \ 6 \ 2 \ 3)$$

$$P2 = (3 \ 6 \ 2 \ 5 \ 1 \ 9 \ 8 \ 4 \ 7)$$

The process initiates by picking the arc $5 \rightarrow 1$ from $P1$ as the first arc. Consequently, the child is initialized as

$$C = (5 \ 1 \ _ \ _ \ _ \ _ \ _ \ _)$$

Next, the arc from $P2$ is added that is next to 1, i.e., $1 \rightarrow 9$. Hence, the child turns out to be as

$$C = (5 \ 1 \ 9 \ _ \ _ \ _ \ _ \ _)$$

Next, the arc which is directed outward from 9 in $P1$ is added, etc. Subsequently repeating the few steps, the partly formed child is produced as:

$$C = (5 \ 1 \ 9 \ 6 \ 2 \ 3 \ _ \ _ \ _)$$

The next arc going out from 3 should be selected from $P2$, but this adoption is infeasible because it would close the circle too early. To prevent this state, from the unvisited vertexes, one node is selected at random, for example 7. Thus, the child becomes

$$C = (5 \ 1 \ 9 \ 6 \ 2 \ 3 \ 7 \ _ \ _)$$

After this step, the conventional procedure can be continued again by selecting the arc $7 \rightarrow 8$ from $P1$, and then $8 \rightarrow 4$ from $P2$. The finalized child appears as follows:

$$C = (5 \ 1 \ 9 \ 6 \ 2 \ 3 \ 7 \ 8 \ 4)$$

Mutation. Swap mutation is used here to produce offspring. It can be done by selecting any two customers from random two routes and exchange their positions. We have applied swap mutation between the routes of same vector and in between the routes of two different routes.

Obtain the offspring using tournament selection and then perform crossover and mutation. Stop the algorithm when stopping criteria is met and report the feasible number of routes obtained.

To effectively cover all the customers at minimum possible cost, a proper setup of warehouses is very important. The proposed Model effectively covers all the customers minimizing the operational cost.

22.4 Experimental Setup and Results

The execution of proposed algorithm for multi-depot pickup and delivery VRP with customer time windows is implemented using MATLAB R2014b. All the experiments were done on a 64-bit operating system equipped with Intel (R) Core (TM) i5-5200U running at 2.20 GHz. A data set is generated based on author's knowledge

of real-life scenarios having real-time assumptions and conditions to depict a real-life problem. The algorithm is tested on the generated data set and the results are presented accordingly.

Following assumptions have been made before the generation of real-time data set:

1. A data set of 100 customers with their demand and time window is considered in Uttarakhand, India, and its nearby states.
2. The potential locations of warehouses are considered on the border of Uttarakhand, India. The state covers the total area of 53,483 km².
3. Five vehicles in each warehouse.
4. A vehicle can cover the maximum of 200 km of area in a single route.
5. All vehicles are connected with GPS-based tracking device to track the location of vehicle at any point of time.

For solving the problem, the entire geographical area is divided into 15 zones so that identifying the customer could become easy and the customers are served zonewise.

Figure 22.3 shows graphical user interface (GUI) designed to solve the problem. Once we click the setup button, warehouse locations proposed by our GA on the basis of customer's locations are identified as shown in Fig. 22.4. The working of GA for the first step is shown by Fig. 22.5. After the initial setup, the demand of each customer along with their time windows is provided to the algorithm which can be done using read data. Once the input is given to the algorithm, click on start to get started the process of finding minimum cost route to serve the customers. 100 customer's locations with their respective pickup and delivery demands along with the time windows and service time are provided as an input to solve the problem.

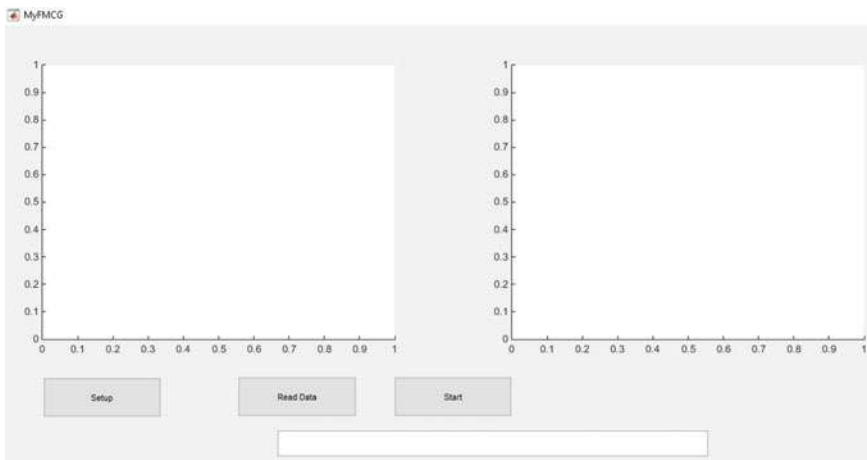


Fig. 22.3 Graphical user interface

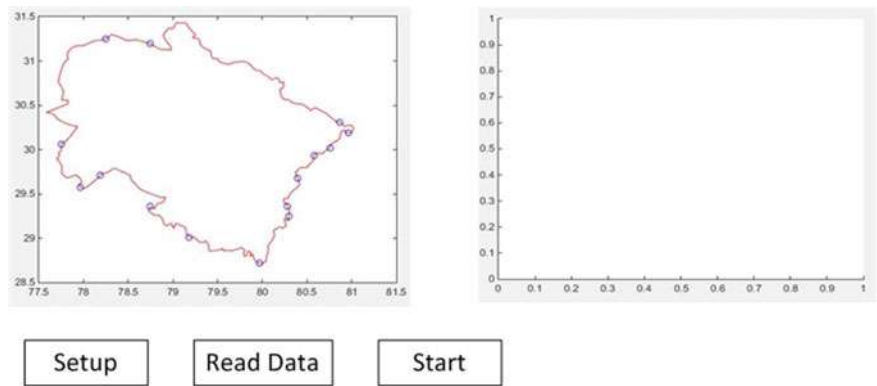


Fig. 22.4 GUI indicating the location of warehouse (left) and customer (right)

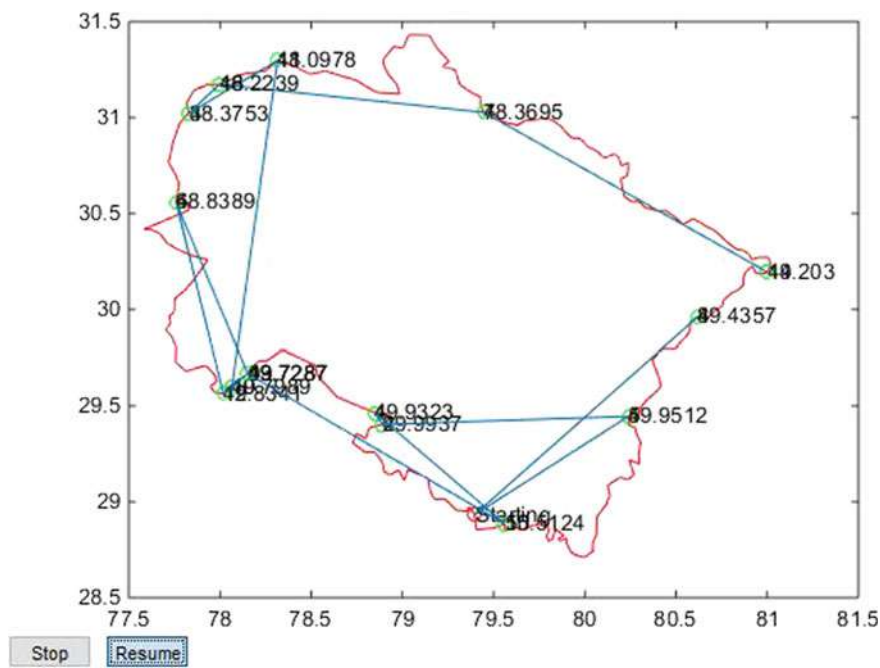


Fig. 22.5 Working of GA for identification of optimum locations of warehouses

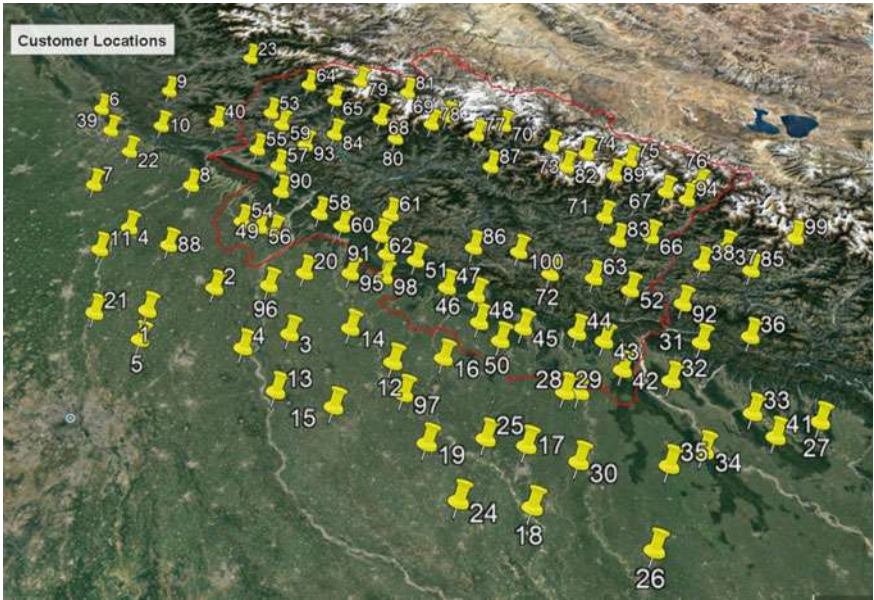


Fig. 22.6 Customer locations around border of Uttarakhand, India

Figure 22.6 shows the geographical locations of 100 customers around the border of Uttarakhand state of India within the 100 km range as the authors have put total route coverage constraint on vehicle which states that a single vehicle can cover a maximum of 200 km distance in a single route.

Once the simulation run is over, many routes connecting the customers to the proposed warehouse locations are obtained with minimized transportation cost. In Fig. 22.7, different optimized routes originating from different warehouses and connecting various numbers of customers are shown. It shows that each customer is part of a route; hence, the constraint which states that demand of each is fulfilled is satisfied.

The obtained results show that the total optimized distance travelled by the vehicles is found to be 6328.07 km and numbers of vehicle used are 34 out of 75 (5 vehicles at each of the 15 warehouses) which are taken initially and rest of the vehicles remain unutilized. It indicates that initial assignment of vehicles was over-estimated well above the actual requirement. Our algorithm minimizes the total cost as well as number of vehicles used successfully. Table 22.1 indicates various characteristics of each route, i.e., the number customer served, total number of vehicles used, and total distance covered.

From Table 22.1, it has been found that the average number of customers served per warehouse is 7. Warehouse number 10 and 13 serves the maximum number of customers, i.e., 10. The visitation order of 10 customers served from warehouse number 13 is shown in Fig. 22.8. The total optimized distance of all the vehicles

travelled from warehouse number 13 is 393.1 km collectively. It is also observed that each vehicle travelled a distance of approximately 186.11 km which is very close to our total route length limit of 200 km. The results obtained are competitive in nature and shows that we can apply our algorithm to solve the problem of large instances which are generally faced in real life.

22.5 Conclusions and Future Work

In this paper, a new solution methodology based on genetic algorithm is developed to solve the multi-depot pickup and delivery vehicle routing problem with customer time windows. A mathematical model is also developed for the proposed problem and the objective function is used as fitness function due to the complicated nature of the problem. The developed model can be applied to problems where there is a need to allocate new locations to the warehouses/depots in an area in order to satisfy the demands of customers with the objective to minimize the transportation. Some assumptions have been taken while formulation of the model such as each depot has fixed number of vehicles which will start and end the route at the same depot after servicing the customers. Each vehicle has a predefined capacity and sum of customer requests fulfilled by the vehicle will not increase its maximum capacity. Such type of scenarios is seen when company wants to enter in a new market. The implementation of problem in hand is done using MATLAB. The algorithm developed can successfully solve the problem with 100 customers and 15 warehouses. Results shows that the developed algorithm is running optimally for the proposed problem. For future work, we propose to enrich this problem by taking other factors such as fixed, variable, loading, and unloading cost into optimization function and develop an application which continuously updates the inventory level at each warehouse to make the model more realistic and integrate self-learning algorithm into our GA for pattern recognition.

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Chapter 23

On Modeling the Thermal Behavior of Single and Quad Laser Melting of Powdered Nickel Alloy



Hemnath Anandan Kumar  and Senthilkumaran Kumaraguru

Abstract Selective laser melting is an additive manufacturing process that uses high laser power beam to melt the powders and fuse together to form three-dimensional parts from CAD model. Inconel 625 is a nickel-based alloy that is widely used in aerospace, chemical, nuclear reactors, and marine applications. As these applications need high service temperatures and corrosion resistance properties, the quality of the parts fabricated should be taken into consideration while fabrication of parts. The formation of the temperature gradient is critical as it affects the stability and dimension of a molten pool, which in turn affects the surface finish and densification of the parts. However, for producing a quality part from selective laser melting, understanding the thermal behavior under laser melting is necessary, when subjected to different process settings. In this paper, using a thermal analysis was used to study the melting by selective laser melting. The different process settings chosen for analysis include laser power and scan speed using constant energy density model. The Gaussian model has been adopted for symmetrical distribution of laser irradiance across the beam. The simulation for temperature analysis was carried out using commercial FEM software for single and quad laser configurations. The temperature profiles were observed at several nodes by varying the process parameter and the temperature distribution during the fabrication was predicted.

Keywords Selective laser melting · Inconel 625 · Additive manufacturing · Modeling · Simulation

23.1 Introduction

Selective laser melting (SLM) is one of the successful commercial additive manufacturing (AM) technique in which a scanned laser beam melts the material locally in a powder bed using a sliced 3D CAD model data. In the SLM process, initially,

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_23

a powder delivery system lays a thin layer of metal powder on a build platform and spread using a recoater. After spreading, the powder layer is selectively melted by a laser source based on a prescribed hatching pattern. Once the layer scanning is completed, the build platform descends down by a distance equal to the layer thickness (20–50 μm in case of SLM process) and a fresh layer of powder is spread across the part bed. This cycle of recoating and scanning is carried out until the parts are completely built. SLM process is commonly carried out in a chamber shielded by inert gas environment. SLM has the capability to produce fully functional parts right from metal powders without any post processing operation. But, melting and rapid solidification events have significant effects on the part quality and precision of the products. SLM comprises of several physical phenomena like fluid flow, heat and mass transfer, phase transformation, absorption and scattering of laser radiation, evaporation of material, and chemical reactions. As there is a prevalence of very high operating temperature, it will significantly affect various types of heat and mass transfer and also momentum transfer that occur during the process. During the SLM of metals, prevalence of huge thermal gradients result in increment of residual stresses and hence the deformation. Sometimes, very high residual stresses may also cause formation of crack in the final product. SLM products face a serious issue of thermal distortion which can be rectified by rigorous research. Therefore, understanding the influence of process parameters and process mechanisms is trivial for producing quality parts from SLM.

23.2 Background

Though additive manufacturing technology facilitates design complexity and flexibility, production of parts with desired mechanical properties is quite difficult. In the metal additive manufacturing process, there will be a prevalence of high temperature due to the high energy density laser. This will definitely lead to unavoidable defects in the fabricated parts. Hence, prediction of such thermal history is necessary to avoid such defects and optimize the process. Several attempts have been made by different researchers to study the influence of fluid flow and heat transfer mechanisms happening in metal AM process adopting several numerical method techniques and finite element methods. Zeng et al. [1] reviewed various analytical and numerical methods for solving thermal problems in additive manufacturing. They have taken a case study and analyzed temperature variation with respect to time, scan speed, and laser power. A 3D FE model was done by Dai and Shaw [2] to investigate the temperature distribution using a moving laser source for processing multi-material components. In this work, they have not only analyzed the residual stresses generated, but also the distortion of the fabricated product. Later, a three-dimensional model was also developed by Roberts et al. to study about the temperature field during the SLM process and justified the developed model by experimentation methods. But the main disadvantage of using such model is larger time consumption [3]. Rather than developing a full part, a single layer simulation was carried out by Patil et al.

and the temperature distribution was studied using titanium powder [4]. They have highlighted that laser power, beam diameter, and hatch spacing have a great influence in determining the temperature distribution. A FEM based on a numerical model was proposed by Kolossov et al. [5] to determine the temperature field on the top surface of powder layer during SLS process in which the specific heat as a function of temperature was considered.

A three-dimensional FEM model was proposed by Dong et al. [6] for selective laser sintering (SLS) process to analyzing temperature profiles in which they considered the impact of process parameters such as scan speed and laser power. They observed that when the laser power is high, temperature and density increase, whereas if the scan speed increases, both temperature and density decrease. Fu and Guo [7] did a 3D FE simulation on SLM of Ti6Al4V alloy and predicted the melt pool dimensions and validated experimentally. Li and Gu [8] also studied the thermal behavior of selective laser melted titanium alloy. Temperature-dependent physical properties were considered and the heat flux was used as Gaussian moving heat source for analysis. They made some conclusions like as the scanning speed increases, the temperature gradient of the melt pool also increases. Hussein et al. [9] investigated about temperature distribution, molten pool analysis, and stress analysis of SLM process in SS 316L. In this work, the authors tried to study the stress distribution and temperature fields in SS 316L built without support structures in SLM process by a 3D FE-based simulation. It was observed that at higher scan speed, the predicted melt pool length increases while both depth and width of the melt pool decreases. Criales et al. [10] proposed a two-dimensional method for thermal and molten pool analysis for Inconel 625 in SLM process. In this study, effect of various process settings on peak temperature and melt pool dimensions was investigated. More predictions would have been carried out if it is a 3D model.

The temperature distribution studies presented in the past research shows that there is a need for accurate temperature simulations that can be useful to predict densification and melt pool geometry and part quality. It is also found that very scanty works were done on the thermal history simulation of Inconel 625 powder. In this work, we attempt to understand the influence of scan speed and laser power on the temperature evolution and further use the temperature results to understand the nature of thermal gradients induced by laser exposures at multiple points on the powder bed surface. The process parameters chosen for modeling are listed out in Table 23.1.

Table 23.1 Process parameter levels and settings

	Process parameters	
	Laser power (P)	Scan speed (v)
	W	mm/s
Set 1	195	800
Set 2	170	697
Set 3	145	594
Set 4	120	492

From the past research, we know that there is lack of accurate thermal simulations to predict the geometry of the melt pool and part quality.

23.3 Modeling

Nickel-based alloys have been recently employed in the aerospace and defense applications. The Nickel alloys have good corrosion resistance and it has higher service temperature compared to ferrous alloys. In this study, Inconel 625 in powder form is considered and its composition, physical properties, and temperature-dependent thermal properties are listed in Tables 23.2, 23.3, and 23.4, respectively.

The model chosen for the study is a rectangular-shaped layer of powder bed with dimensions (3×1 mm) with a thickness of $50 \mu\text{m}$ as shown in Fig. 23.1. The process chamber is maintained at a preheated temperature (around 80°C). The laser is moving along X -direction and the chosen process parameters are scan speed and laser power.

23.3.1 Assumptions

1. The laser beam serves a heat flux which is assumed to have a Gaussian distribution and applied directly on the top of the powder layer.
2. The conductivity of the porous powder layer is modeled using effective thermal conductivity and the simulated layer is considered to be completely sintered at the melting temperature.
3. The material is considered to be isotropic with respect to the thermal properties.
4. The absorptivity of the powder is assumed to be that of the solid material.

23.3.2 Heat Transfer Modeling

The problem formulation for heat transfer analysis is given below. The energy balance equation and the initial and boundary conditions are:

$$\rho c \frac{\partial T}{\partial t} = k \left[\frac{\partial}{\partial x} \left(\frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial T}{\partial z} \right) \right] \quad (23.1)$$

where ρ is the density of the material in kg/m^3 , c is specific heat capacity in J/kg.K , T is the temperature of powder system in K , t is interaction time in s , and k is thermal conductivity of material in W/m.K .

Table 23.2 Chemical elements and composition of Inconel 625 [11]

Elements	Ni	Cr	Mo	Nb	Fe	C	Si	Al	Ti	Mn	S
%	61	21.5	9	3.6	4	0.05	0.20	0.20	0.20	0.20	0.001

Table 23.3 Physical properties of Inconel 625

Property (units)	Value
Density (kg/m^3)	8440
Modulus of elasticity (Gpa)	207.5
Yield strength (Mpa)	448
Ultimate tensile strength (Mpa)	862
Melting range ($^{\circ}\text{C}$)	1290–1350

Table 23.4 Temperature-dependent thermal properties of Inconel 625

Temperature (K)	Specific heat (J/kg.K)	Thermal conductivity (W/m.K)
293	–	–
373	496	12.4
473	521	14.2
573	538	16
673	555	17.7
773	573	19.3
873	620	21.5
973	654	26.8
1073	663	26.8
1173	677	26.7
1273	684	28.2
1373	695	29.6
1473	705	–

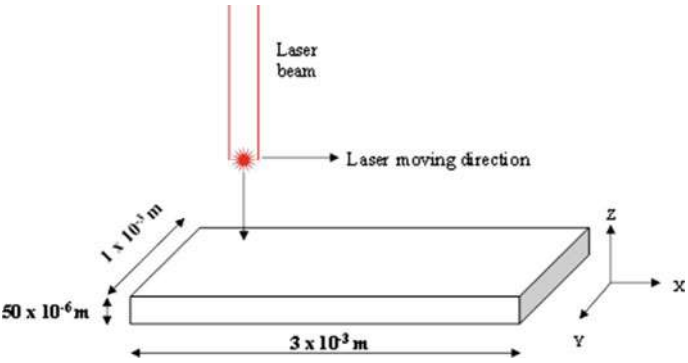


Fig. 23.1 The physical model of the powder bed

The initial conditions are

$$T(x, y, z, t) = T_0; \text{ at } t = 0 \text{ s} \quad (23.2)$$

where T_0 = preheated temperature.

The natural boundary condition is given by

$$k \frac{\partial T}{\partial n} - q' + q_c = 0, (x, y, z) \in S \quad (23.3)$$

where S is the surfaces attached to imposed heat flux (top surface of the layer, which is exposed to convection), q_c is heat convection, q' is input heat flux and n is normal vector of S .

$$q_c = h(T - T_a) \quad (23.4)$$

where h is the heat convection coefficient in $\text{W/m}^2\text{K}$.

The heat flux q' is modeled using the symmetrical distribution of laser irradiance across the beam. The laser beam is assumed to be distributed as Gaussian distribution and the equation for distribution is

$$q' = \frac{2AP}{\pi R^2} \exp\left(\frac{-2((x - vt)^2 + y^2)}{R^2}\right) \quad (23.5)$$

where P is the laser power in W , R is the effective laser beam radius at which the energy density is reduced to $1/e^2$ at the center of the laser spot, v is the scanning speed in m/s , and A is the absorptivity of the powder (Table 23.5).

In this work, along with the single line scanning of the laser to study about the thermal behavior, an attempt has been made to study about the effect of temperature in quad laser process. In this process, four lasers will be moving simultaneously to enhance the process speed and faster production rate. The dimension of the entire

Table 23.5 Inputs to FEM solver

Element	Solid 70
Material properties	Thermal conductivity, density, specific heat
Model extents	$3 \times 2 \times 1 \text{ mm}$
Meshing	Fine meshing
Heat flux input function and variables	$q' = \frac{2AP}{\pi R^2} \exp\left(\frac{-2((x-vt)^2+y^2)}{R^2}\right)$ Variables: x, y, t , and all others are constant
Boundary conditions	Initial conditions (on all nodes), heat flux (on laser moving surface), heat convection (on all surfaces)
Analysis type	Transient (full)
Load step options	Stepped loading with 25 substeps

bed is found to be 400×400 mm. The quad laser system is a new technology introduced by a two commercial AM machine manufacturer. In common powder bed fusion systems, there will be a single laser source, which will be utilized to process the powder materials, whereas in quad laser system, there are four dedicated laser sources to process the materials. The entire powder bed 400×400 mm is divided into four quadrants, each having a dimension of 200×200 mm. Each laser source is allowed to process the materials present within that particular quadrant and has an overlap of 50 mm with the neighboring quadrants. The significant advantages of this system are large build platform and higher build rate which will result in improved productivity with reduced time consumption. The schematic of the quad laser system is shown below.

The temperature distribution prevailing in the powder bed of quad laser process is assumed to vary from the edges to the center in a range of $5\text{--}6^\circ\text{C}$. To study the effect of temperature on different quadrants, the entire powder bed was divided into four equal quadrants of equal dimensions 200×200 mm with a layer thickness of $50\text{ }\mu\text{m}$ and laser source was moved over it. The assumed boundary conditions for this quad laser process are same as that of the previous single line scanning simulation.

23.4 Results and Discussions

A commercial FE solver is employed to obtain the solution to the problem formulated. After meshing, the numbers of elements in the model were 8000 and the no. of nodes were 16,482. The mesh size used was $50\text{ }\mu\text{m}$. The temperature fields at different laser powers and velocities are shown in Figs. 23.2a–d, 23.3 and 23.4.

23.4.1 *Effect of Process Parameters on Single Line Scan*

The simulations of temperature distribution for two different nodes (namely A and B) are shown in Figs. 23.5 and 23.6, respectively. In Fig. 23.5, the laser movement is along X-direction and there is a sudden increase in the temperature for all process parameters set. This is because of the spontaneous influence of laser exposure on the powder bed where the preheated powder bed at 353 K rapidly increases to high temperature. Within some short period, the temperature reaches a high temperature and from the graph, it is visible that the temperature profile descends down as the laser moves away from the particular node. It means that the powder gets melted and cools down rapidly as the laser source moves away. It can also be observed that the temperature increases with higher laser power. It is also seen that as the energy density is higher, the peak temperature will also be higher.

In Fig. 23.6, the graph represents the temperature profile followed at node B for a different set of parameters. Here, for a particular time, there is a constant linear profile which is the initial condition 353 K. This node is far away from the laser

Fig. 23.2 Temperature profiles at **a** 120 W power and 492 mm/s velocity; **b** 140 W power and 594 mm/s velocity; **c** 170 W power and 697 mm/s velocity; and **d** 195 W power and 800 mm/s velocity

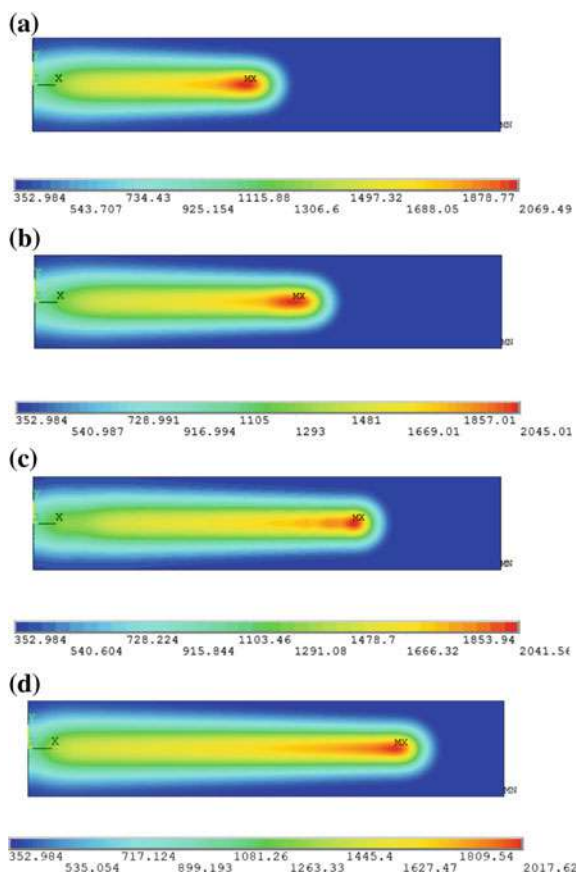
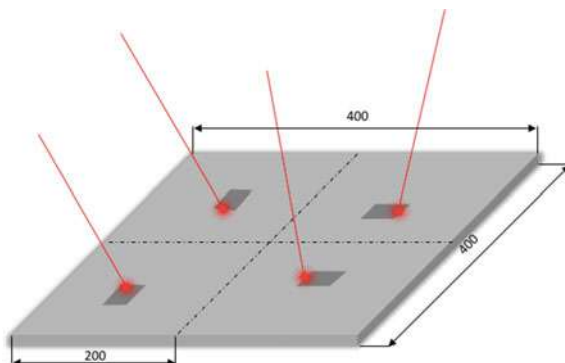


Fig. 23.3 Schematic of quad laser powder bed



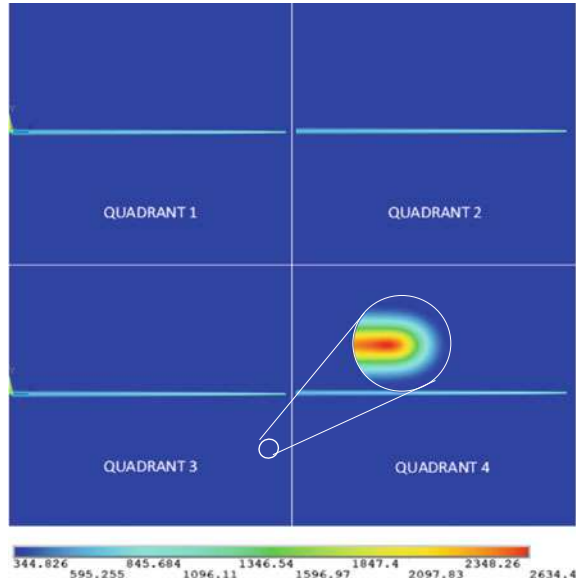


Fig. 23.4 The temperature profile for laser movement with 195 W and 800 mm/s in four different quadrants for a quad laser bed

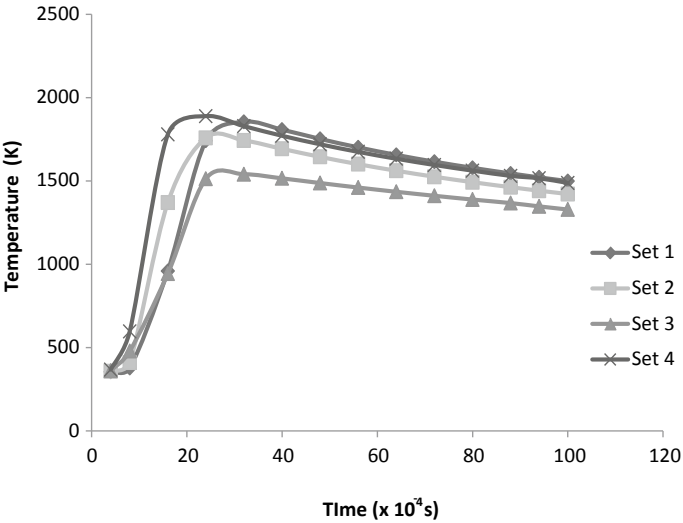


Fig. 23.5 The temperature at node A for different time steps for different process parameter sets

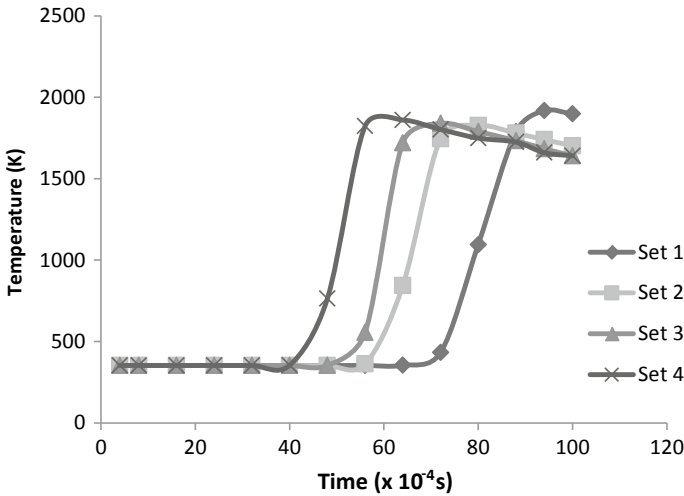


Fig. 23.6 The temperature at node B for different time steps for different process parameter sets

source at the beginning of the simulation. When the node comes in contact with the laser source, it follows a similar profile as of the previously selected node A. If the temperature gradient is plotted, we can observe that for different scan length, temperature gradient varies accordingly.

Williams and Deckard [12] found that the delay times have a significant effect on the SLS processes and also found that the temperature gradient decreases with increased scan length which means the delay time is higher. When the scan speed is low, the peak temperature is attained at very lesser time delays, whereas if the scan speed is higher, it is vice versa.

23.4.2 Effect of Laser Movement in Quad Laser Bed

The laser movement on four different quadrants of the quad laser bed is shown in Fig. 23.4. Similar boundary conditions were given as input. The temperature distribution was studied by selecting three different nodes in the midline along which the laser passes through. The temperature recorded at three different nodes 1, 2, and 3 are plotted in Fig. 23.7. There is a sudden increase in temperature which implies the point of contact of the node with the laser source and as the time progresses, the laser moves away from the respective node and temperature gradually decreases. On comparing with the previous model, the temperature fall is different because in quad laser simulation, there is a huge surface area which facilitates heat dissipation at a faster rate than the small model. On observing into the quad laser simulation, there was not much effect on the variation of temperature when the laser movement occurs as per the observation from the simulation results. Hence, when there is a

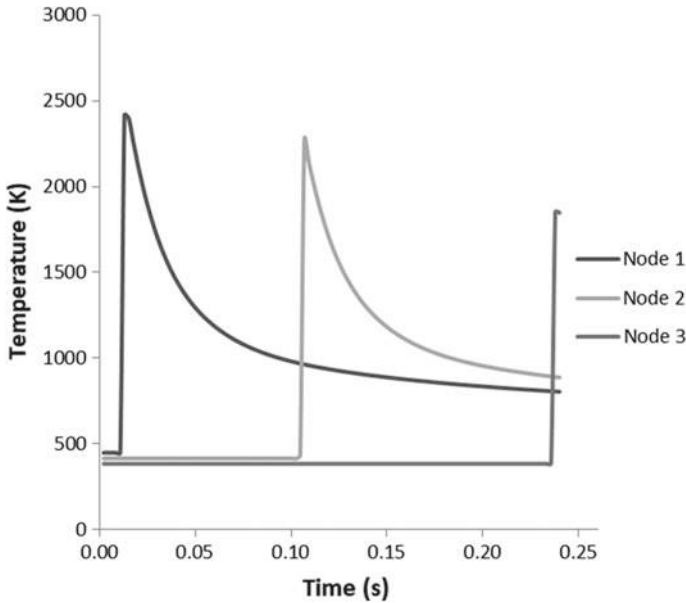


Fig. 23.7 The temperature at different nodes for different time steps in a quad laser bed

large temperature gradient occurs within the quad laser bed, it may have an effect on the increase in the temperature during laser movement.

23.5 Conclusions

In this work, the thermal behavior of Inconel 625 was analyzed using the process parameters like laser power and scan speed. A 3D simulation model was developed for predicting temperature behavior of Inconel 625 powder bed at various process settings. The peak temperature was observed around 2000 K in all cases. The cooling rates are uniform in all the profiles. Using those temperature profiles, prediction of melt pool dimensions and velocity profiles was studied. Along with that an attempt to simulate quad laser has been carried out. This helped the authors to know more about the temperature distribution during a simple laser melting and quad laser melting process. In future, experimental validation of the results will be carried out with SLM setup.

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Chapter 24

Numerical Analysis of Cutting Modes in High-Speed Machining of Aluminum Alloys with PCD and CBN Tool Inserts



I. Sri Phani Sushma and G. L. Samuel

Abstract In manufacturing industries, high-speed machining of aluminum alloys is highly recommended for achieving better productivity in terms of cutting force reduction and improved surface finish. Even though an overwhelming number of process parameters affect the high-speed machining operations, tool material is considered to be the most predominant factor in determining the machining performance. Hence, in the present work, experimental and simulation analyses are carried out for understanding the effect of different tool materials in high-speed machining of aluminum alloy. Formation of dead metal zone is taken as the fundamental criterion for analyzing the discrepancy in cutting forces, and the same is discussed in detail in the present paper.

Keywords High-speed machining · Aluminum alloys · Dead metal zone

24.1 Introduction

Machining is the most critical unit in any manufacturing industry, because of the complexity in metal cutting mechanisms which alters the tool geometry under extreme machining conditions. This will be even more challenging while machining aluminum alloys at higher speeds due to severe built-up edge formation. High-speed machining has gained wide attention in many industries due to its capability in improving the machined product quality generating lower cutting forces and improved surface finish. The productivity of any machining processes can be enhanced only by limiting the down time. Proper benchmarking of all machining units, selection of optimum cutting parameters, and appropriate integration of

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_24

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advanced machining techniques can eliminate the bottlenecks in the manufacturing line, which ultimately brings higher productivity with minimum utilization of energy resources. In this aspect, high-speed machining is always advantageous for manufacturing discrete metal parts, due to the lower power consumption which eventually results in higher productivity. However, the knowledge of tool edge geometry and cutting conditions is highly significant as it significantly affects the performance of high-speed cutting. Kalyan and Samuel [1] reported the effectiveness of PCD tools in minimizing cutting forces while machining aluminum at high cutting speeds owing to the thermal softening effect and the reduction in the formation of dead metal zone (DMZ).

Advanced machining techniques such as hard turning, hard milling, and micromechanical machining are reported to have the uncut chip thickness in the same dimensions of tool edge; this highly demands the usage of cutting edges which can withstand the increase in mechanical and thermal stresses. Numerous designs for cutting edges were developed which includes chamfered, chamfer hone, double chamfered honed, etc. [2]. Among these, chamfered tools are reported to be successful in various turning operations as a result of strength improvement of cutting edge by the formation of a stable trapped material which can be termed as “dead metal zone.” Various researchers have reported the underlying mechanisms and the factors responsible for the improvement in cutting performance while using prepared edge tools, which includes analytical [3], computational [4, 5], and experimental [6, 7] procedures. The prime and foremost importance in all investigations is to understand the influence of DMZ in machining performance in terms of cutting forces and surface roughness. Generally, under all machining processes, a tertiary shear zone will form at the contact regime between tool and workpiece which eventually increases the cutting forces. Mayer and Stauffer [8] studied the same, analytically by slip-line field model and also by performing experiments with negative rake curvilinear PCD tools.

Although significant works are available in the literature on analyzing the cutting mode mechanism under high-speed machining, limited works are reported in case of aluminum alloys. Hence, in the present paper, high-speed machining of aluminum alloys has been studied both experimentally and numerically using different tool materials specifically PCD and CBN. Experiments are carried out on Al 6061 at high speeds with low feed and low depth of cuts. In addition, a numerical model is developed using the finite element software, DEFORM 3D for validating the observed results. All the comparisons are carried out based on dead metal zone formation along with the slip-line field model for better understanding of the alterations in tool–chip and tool–workpiece interfaces.

24.2 Numerical Model

In the present work, a numerical model has been developed using DEFORM 3D software to analyze the effect of various process parameters on cutting forces during orthogonal cutting operation. DEFORM 3D is a finite element simulation software

based on an updated Lagrangian analysis method. An adaptive re-meshing method has been employed for modeling the chip formation. The following section explains the details of the modeling procedure used in the current study.

(a) **Modeling of Cutting Tool**

The geometrical characteristics of the cutting tools including the tool material are selected from the software tool library as shown in Fig. 24.1. Where “ R ” represents the nose radius and is selected as 0.20 mm for the current study. In the finite element domain, the tool has been defined as a rigid four-node tetrahedral element.

(b) **Modeling of Workpiece**

In the present model, the workpiece is considered to be viscoplastic material. For representing the flow stress of the material, a modified Johnson–Cook model with strain gradient plasticity has been used for the analysis. The cutting process in the model was simulated by keeping the bottom portion of workpiece stationary and the tool moving in the cutting direction as shown in Fig. 24.2.

(c) **Post-Analysis–Effective Strain, Tool Wear Interface Temperature, Cutting Forces, and DMZ**

The effect of cutting parameters during tool–work material interaction has been studied based on the variation in effective strain, tool wear interface temperature, cutting forces, and the formation of dead metal zone. The present study mainly focuses on explaining the effect of tool material on cutting performance

Fig. 24.1 Defining tool nomenclature ($L1 = 1.0$, $L2 = 1.0$, $B = 5^\circ$, $C = 5^\circ$, $R = 0.2$)

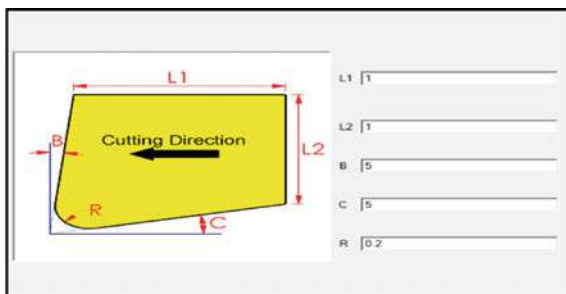
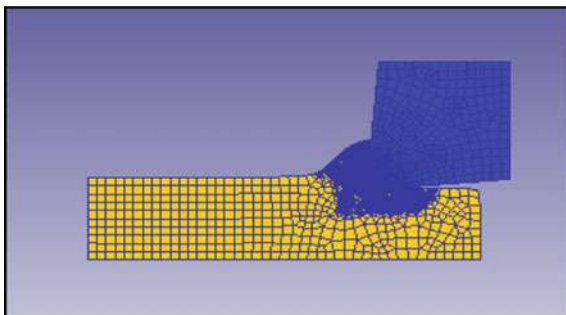


Fig. 24.2 Tool–workpiece interaction



in terms of DMZ formation under various cutting conditions along with the generation of cutting forces. Figures 24.3 and 24.4 represent the details of the analysis of force components (cutting force and thrust force) during tool–work material interaction. The same procedure has been repeated for various cutting conditions, and the corresponding generated forces are tabulated for further analysis.

The analysis of DMZ formation is executed as mentioned in Fig. 24.5, where the edges of the DMZ are joined to form a triangle and the evaluation of the same is carried out. The formation of the dead metal zone is estimated by using Heron’s formula, where the area of the formed triangle can be expressed as,

$$A = \sqrt{s(s - a)(s - b)(s - c)} \tag{23.1}$$

where $s = (a + b + c)/2$ is the semi-perimeter of the triangle $a \leq b \leq c$.

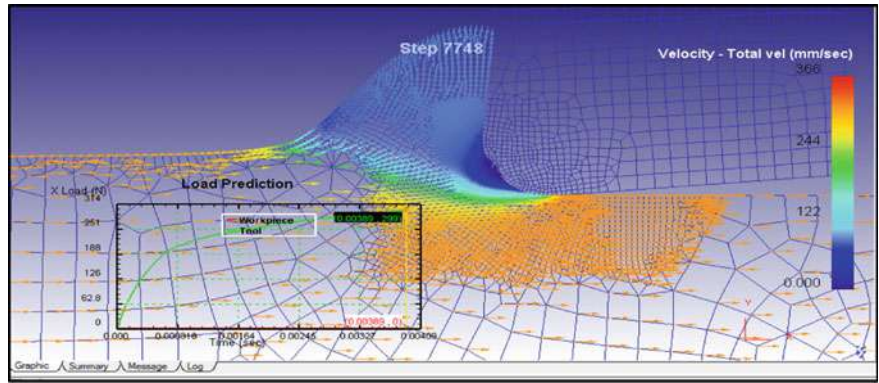


Fig. 24.3 Variation in cutting force

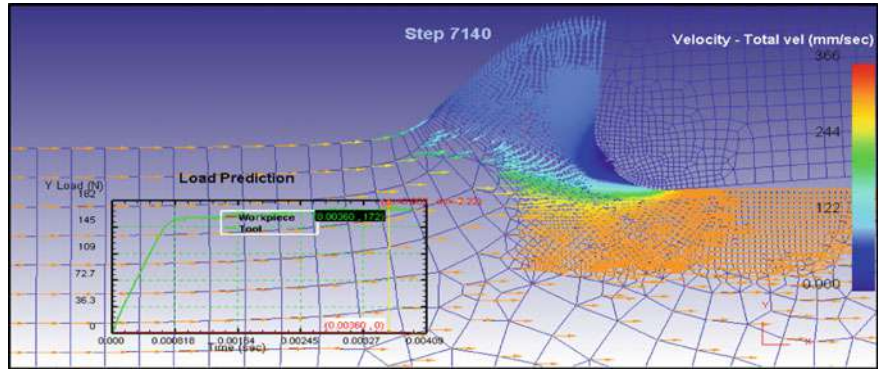


Fig. 24.4 Variation in thrust force

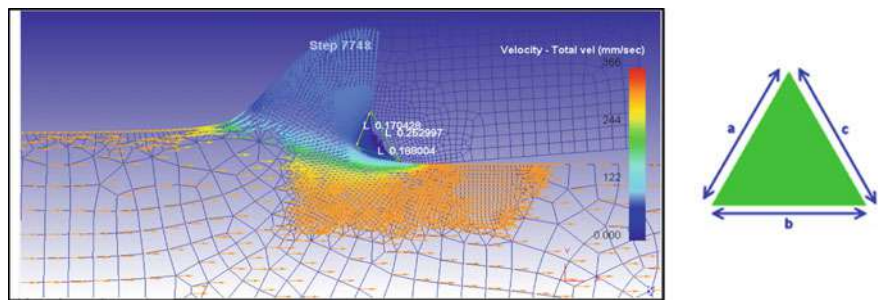


Fig. 24.5 Analysis of DMZ formation

24.3 Experimental Details

In the present work, turning experiments were carried out to investigate the effect of PCD and CBN inserted tool while performing high-speed machining on aluminum alloys. Material properties of the PCD tool insert are given in Table 24.1. Table 24.2 gives the details of the tool and tool holder. The experiments were performed with cutting speeds ranging from 300 to 785 m/min by maintaining a constant feed of 0.1 mm/rev and 0.5 mm depth of cut. Various levels in experiments are provided in Table 24.3. The variations in cutting forces under all experiments were analyzed with respect to dead metal zone formation as it will affect the surface finish of the component. For better understanding and validation, results are compared with the simulation results.

24.4 Results and Discussion

The performance evaluation of PCD and CBN inserts in high-speed machining on aluminum alloy was executed by comparing the various cutting forces generated at different cutting speeds. All the force details were recorded separately for the comparative study. Figure 24.6 depicts the surface image of the PCD and CBN tool

Table 24.1 Material properties of inserts

S. No.	Property	PCD	CBN
1	Young’s modulus (E)	850 GPa	648 GPa
2	Poisson ratio (ν)	0.086	0.174
3	Coefficient of thermal expansion (α)	0.00000015 C ⁻¹	0.00000038 C ⁻¹
4	Thermal conductivity (K)	540 W/mK	100 W/mK
5	Heat capacity (C)	750 kJ/kgK	750 kJ/kgK
6	Emissivity (ϵ)	0.95	0.95

Table 24.2 Specifications of insert and tool holder

S. No.	Specification	Specification details	
1	Insert	CNMG 120404- HM	TNML120404- HM
2	Material	Polycrystalline diamond	Cubic Boron Nitride
3	Included angle	80°	80°
4	Clearance angle	7°	7°
5	Edge length	9 mm	9 mm
6	Nose radius	0.8 mm	0.8 mm
7	Insert thickness	4 mm	4 mm
8	Shank cross section	13 × 3	13 × 3
9	Back rake angle	−5°	−5°
10	Side rake angle	−5°	−5°
11	Side cutting angle	0°	0°
12	Tool Inclination angle	0°	0°

Table 24.3 Details of experimental levels

Iteration No.	Feed mm/rev	Speed m/min	Depth of cut, mm
1	0.1	314	0.1
2	0.14	314	0.2
3	0.18	314	0.3
4	0.1	565	0.2
5	0.14	565	0.3
6	0.18	565	0.1
7	0.1	785	0.3
8	0.18	785	0.2
9	0.14	785	0.1

captured using a Zeiss optical microscope after performing cuts at a higher cutting speed of 785 m/min. Due to the increase in cutting temperature (tool–chip interface) at higher speeds, the tool surface has been observed to be highly damaged. In the metal cutting process, a large percentage of heat generated during cutting is carried away by the chips which slide along the rake face of the tool. This can result in built-up edge formation, and its magnitude depends on the temperature increment at the cutting zone, which highly depends on the machining parameters and mechanical properties of tool and work material. As aluminum is a highly soft material, the rise

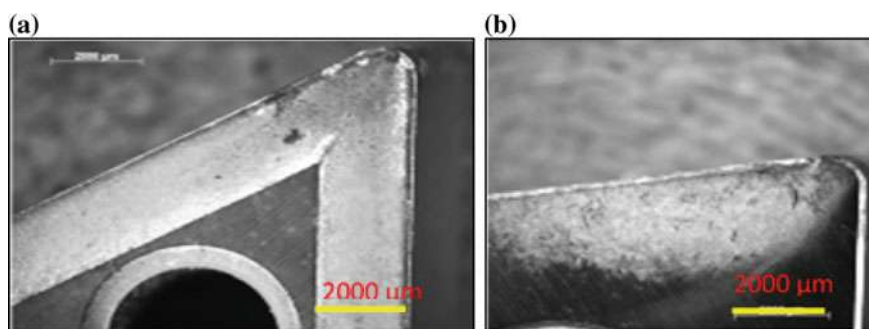


Fig. 24.6 Topographic images of the inserts at 785 m/min cutting speed for **a** PCD and **b** CBN cutting tool inserts

in tool temperature resulted in high adherence of work material on tool surface which eventually deteriorated the surface integrity of both the cutting inserts.

Investigation on the surface integrity of tool revealed the formation of higher surface defects which includes surface cracks, material smearing, micro-pits, adherence of work material, etc. All these factors will deteriorate the tool life and will result in the rapid propagation of tool wear while machining at higher speeds. PCD tools were found to be less affected from the surface and subsurface defects leading to better machinability even at higher speeds. Machining at higher speeds will result in thermal softening leading to the adherence of material on the tool rake face. This will also bring variation in chip flow velocity. As the material softening occurs at higher speeds, chip flow will get obstructed due to the sticking effect on the tool rake face. This will result in the increase of sliding friction in the cutting regime of chip–tool interface leading to chip welding on the tool surface, which causes the formation of built-up edge with machining time. Further, machining using the tool with the built-up edge will lead to higher cutting forces, whose magnitude will greatly depend on the area of dead metal zone formed at the shear area. In the present work, the above-mentioned effect during machining aluminum at various speeds was analyzed both experimentally and by performing finite element simulation. Rise in cutting force, plowing force, shear force, and thrust force were taken as the major factors for evaluation. Tables 24.4 and 24.5 give the details of various forces developed using PCD and CBN inserts. A combination of nine trials was used for analyzing the tool performance. Figure 24.7 represents the variation in various forces while machining using PCD insert at different cutting speed.

Machining at higher speeds resulted in lower cutting force due to the thermal softening of the material which makes the material removal easier. But further increment in cutting speed had an adverse effect which results in the formation of material adherence on the tool surface leading to higher cutting forces with time. From the force data, it was observed that shear force is less in magnitude when compared with all other cutting forces due to lower specific cutting energy.

Table 24.4 Comparison of experimental and simulated cutting forces for PCD inserts

No.	Cutting force (N)		Thrust force (N)		Shear force (N)		Plowing force (N)	
	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>
1	290	202	259	103	83	126	306	101
2	290	299	262	172	105	63	286	282
3	285	391	274	244	115	49	280	412
4	329	236	296	158	98	34	345	250
5	361	316	311	235	140	119	336	275
6	391	308	297	137	151	203	344	134
7	426	255	420	222	139	134	459	204
8	439	305	418	180	172	64	434	290
9	473	243	407	115	154	157	470	112

E- Experimental, *M*- Modeling

Table 24.5 Comparison of experimental and simulated cutting forces for CBN inserts

No.	Cutting force (N)		Thrust force (N)		Shear force (N)		Plowing force (N)	
	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>	<i>E</i>	<i>M</i>
1	281	202	255	103	81	126	299	101
2	294	299	286	172	96	63	313	282
3	298	391	329	244	83	49	361	412
4	307	236	221	158	131	34	285	250
5	315	316	286	235	156	119	290	275
6	358	308	294	137	124	203	138	134
7	383	255	294	222	145	134	338	204
8	384	305	307	180	160	64	331	290
9	419	243	288	115	229	157	280	112

E- Experimental, *M*- Modeling

Simulation results show a higher degree of fluctuation in cutting force and thrust force as resulting from the formation of tool adherence while increasing speed. This can be further understood from the metal cutting principle, as the cutting speed increases, the material softening occurs, resulting in tool built-up edge, further developing the tool wear due to adhesion and deformation mechanisms occurring at the microscopic level and its magnitude can be used as the index for evaluating the fluctuations in the cutting forces. Figure 24.8 represents the variation in cutting forces with respect to cutting speed while machining aluminum alloy with CBN insert at various cutting speeds.

From the plot, it can be understood that at higher speed, shear force increased, resulting from the improvement in friction factor in the cutting regime. An increase in friction at the tool–chip interface will lead to increment in the shear area leading

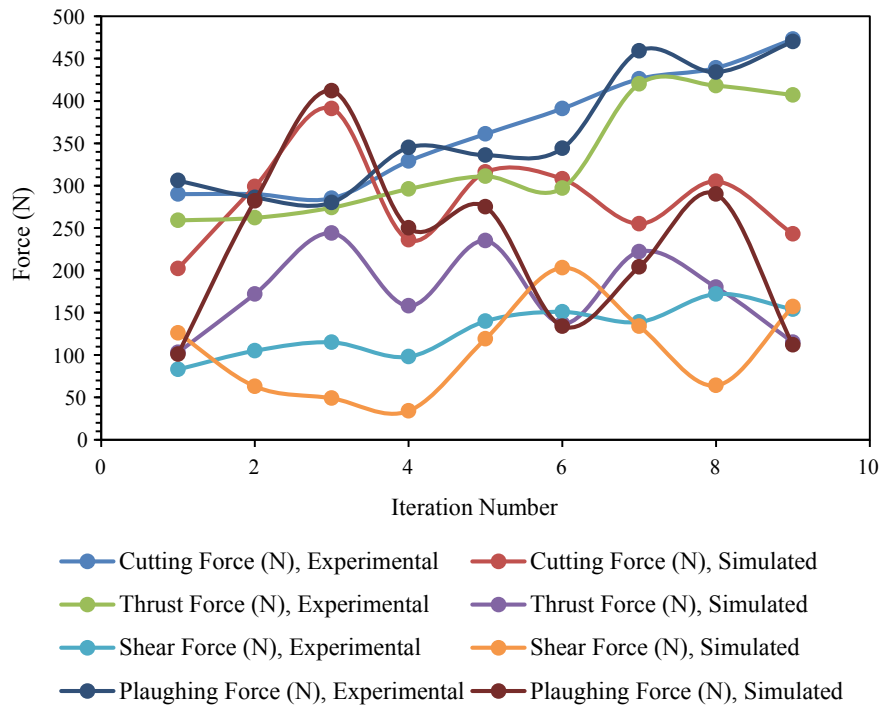


Fig. 24.7 Comparison of experimental results with simulated data of PCD inserts

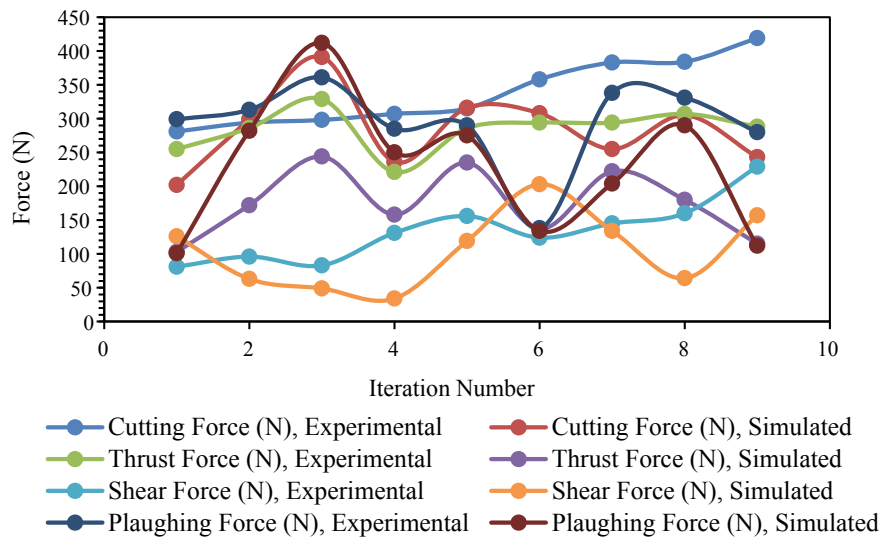


Fig. 24.8 Comparison of experimental results with simulated data of CBN inserts

to higher cutting energy for the material removal. As the material got softened at a higher cutting speed, the cutting force decreases up to a critical speed beyond which increase in cutting speed will not affect the cutting force, beyond which the force will increase due to the formation of material adhesion. The variation in roughness and dead metal zone values while performing machining by PCD and CBN tool are represented in Figs. 24.9 and 24.10. From the plot, it can be understood that as cutting speed increases dead metal zone shows an increasing trend which causes the roughness to vary rapidly at higher speeds.

For better understanding, a comparative study on the cutting forces obtained from simulation and experiments was carried out with DMZ as shown in Fig. 24.11. From the plots, it can be understood that, as the DMZ increases, there is a significant rise in thrust force. This justifies the formation of tool adhesion making the tool to wear out rapidly which in turn will result in higher cutting temperature which is not advisable for accomplishing better machinability.

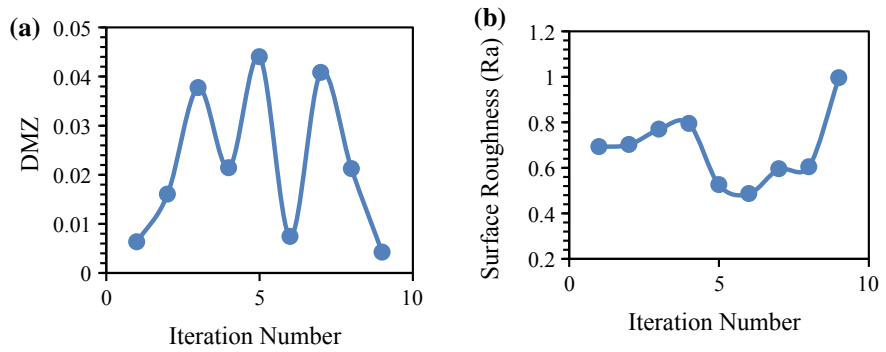


Fig. 24.9 **a** Variation of DMZ while using PCD inserts and **b** graph shows the surface roughness obtained by experimentation while machining at various cutting speeds of PCD inserts

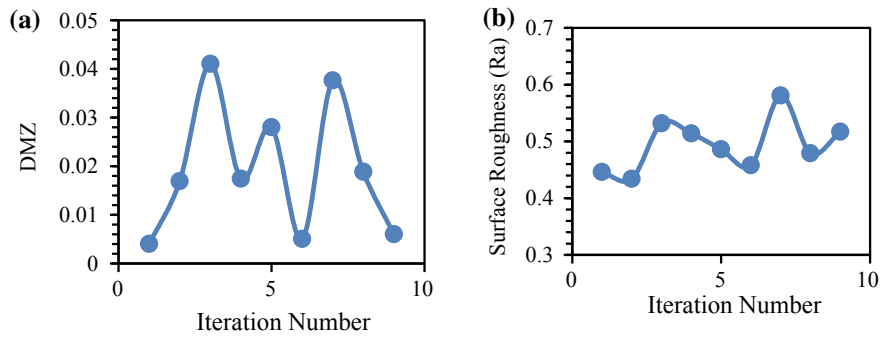
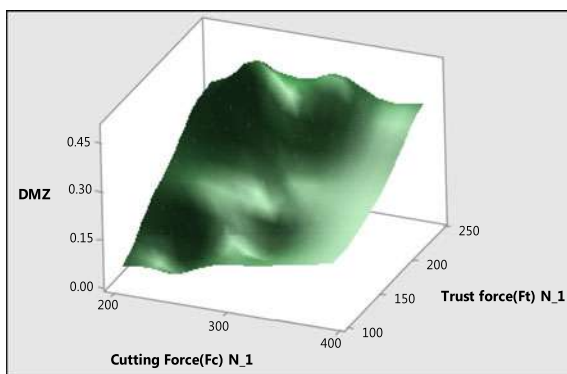


Fig. 24.10 **a** Graph shows DMZ simulation results of CBN inserts and **b** graph shows the surface roughness obtained by experimentation while machining at various cutting speeds of CBN inserts

Fig. 24.11 Graph shows comparison of cutting forces obtained and 3D surface plot



Further investigations were carried out for analyzing the variation in cutting forces while using CBN inserts, both experimentally and by simulation. The present study for CBN tool showed a similar result as that of PCD tool where the thrust force and cutting force rise with respect to cutting speed and a huge reduction in shear force is observed as the cutting speed increased from lower to higher values. An increase in variation of the dead metal zone with respect to plowing force and cutting force was observed in the simulation study. DMZ was found to be reduced while performing machining at higher speeds and higher fluctuation in surface roughness was found due to the tool built-up. The formation of DMZ and its influence in cutting force and roughness was studied using simulation analysis in DEFORM 3D software. Simulations were repeated for various cutting speeds, and all the results were recorded and investigated separately for further analysis. Figure 24.12 shows the formation of DMZ at various speeds while using PCD and CBN tool, which depicted the formation of the lower dead metal zone in case of PCD tools than CBN tools. Hence, PCD tool inserts are highly recommended for the better machinability of aluminum alloys at higher speeds.

24.5 Conclusions

In the present work, an investigation was carried out on the variation in cutting forces while machining aluminum alloy at higher cutting speeds. A numerical model using DEFORM 3D software was also used for understanding the machining performance by simulating the same under similar cutting conditions. Both the experimental and simulation analyses were performed separately for PCD and CBN cutting tool. The machinability of cutting tools was evaluated based on the fluctuations in cutting forces generated while performing high-speed machining. Cutting force seems to be following a similar trend for both the tools, though the performance of the PCD tool was observed to be better due to lower cutting forces and dead metal zone. Forces recorded while performing experiments showed a little increase than the simulated

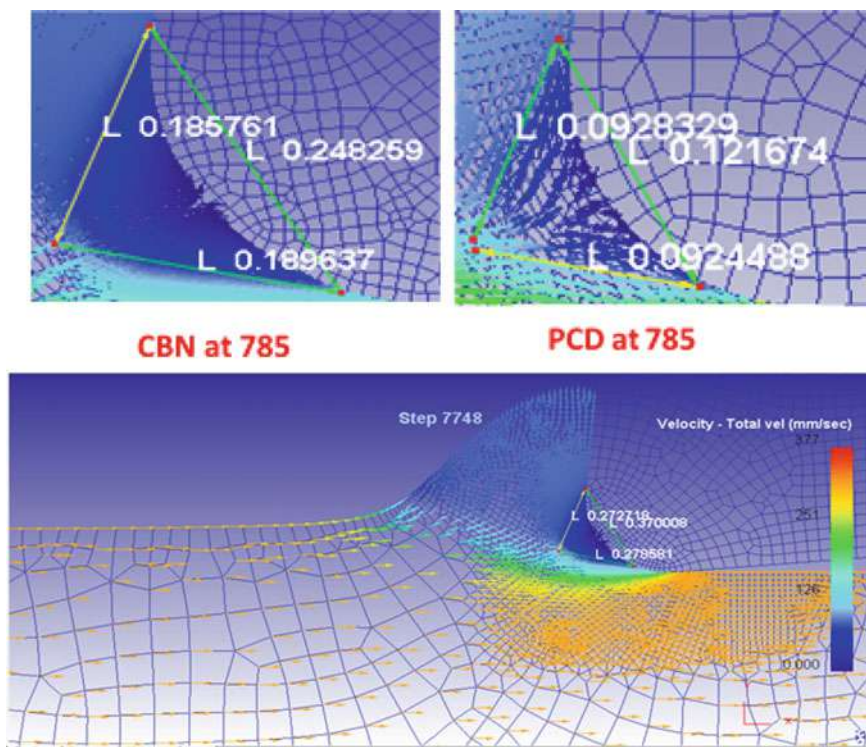


Fig. 24.12 Simulation-based estimation of DMZ at various cutting speeds

force, which can be attributed to the variation in the occurrence of dead metal zone and also the frictional effect at the sliding regime, which was not considered in the simulation analysis. The findings from the present work based on analysis of the force variations and the formation of “dead metal zone” with respect to tool material demonstrated the importance of various cutting modes in high-speed machining of aluminum alloy for better productivity.

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Chapter 25

Design of Row-based Machine Layout—A Case Study



Chandanam Srinivas , Ravela Naveen and Bijjam Ramgopal Reddy

Abstract The layout problem of machines is the determination of the relative location of machines in the available space to minimize the total material handling cost. Machine layout problems are cumbersome and are non-polynomial in nature. Generally, metaheuristics give closer to optimal solution but not precise solution. Since machine layout problems are complex it is, therefore, necessary to obtain the solution by more than one technique like genetic algorithm and ant colony algorithms. The objective of the present study is a case study of Lakshmi Engineering Workshop is taken and optimum arrangement of machines which yield minimum total transportation cost is found out by applying genetic algorithm and ant colony algorithm techniques in a multi-row machine layout. By adopting present method, the cost for the optimum layout decreased by 38% when compared with the existing layout cost.

Keywords Machine layout · Genetic algorithm · Ant colony algorithm

25.1 Introduction

The machine layout is an accomplishment of maximum efficiency by arranging the resources like machines in the given space. In the case of large machine layout, layout of machines plays a greater role in production time and cost. In machine layouts, it was found out that nearly half of the production costs are due to transportation of workpieces [1]. Chiang and Kouvelis [2] report that material handling constitutes 30–70% of the production costs. Hence it is paramount to contemplate about design of machine layout in the early stage itself and design it in a comprehensive manner. Genetic algorithm is a complex search optimization technique through a space ill-defined possibilities and yet done randomly in a structured manner [3]. It involves parallel evaluation of feasible solutions in a search space. In engineering

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_25

field, GA's has many diverse applications and has outperformed many traditional methods. Another technique by name ant colony optimization operates on the inspiration of ants uses a probabilistic technique for solving complex engineering problems. It helps in identifying the optimal solution among good solutions [4]. There are few evidences in the literature, where, more than one metaheuristics are used to determine the solution to a machine layout problem. Also, much work has not been reported in the literature on the design of multi-row machine layouts.

25.1.1 Problem of Designing the Machine Layout

Layout problems are known to be complex and are generally NP-hard. If the problems are not solvable in polynomial time, then it is called NP-hard problem. The layout design is a combinatorial optimization problem that arises frequently in real-life applications. Arrangement of facilities in an efficient manner in a single row and finding the optimum cost is called single-row layout problem [5]. Based on the placement of machines, considering the frequencies between each pair of machines and cost of transport, transportation cost is found out [6]. In multi-row layout, machines are arranged in a straight line in parallel rows. Thus, multi-row is an extension of single-row layout in which machines are located in parallel rows [7]. The objective is to minimize the sum of the products of cost of transport, frequency between the machines and the center-center distance between each pair of machines.

25.2 The Machine Layout Design

The most suitable form of arrangement of the machines in the machine layout is in a single or in multiple rows. Prior to solving the problem, the following assumptions are made [8]:

- All machines are of rectangular shape.
- All machines are operated in the center of that space.
- The available surface of machines is rectangular in shape.
- The available surface of machines is limited along width.

For such manner of solving the problem, it is necessary to know the dimensions of machines and the minimum allowable distances transport quantities and transport costs between all the pairs of machines. Further, it is necessary to know the transport quantities, transportation cost between the individual machines during a certain time period. We also need to know the width of transport (w), the greatest length of the row (a) and the width of the row (r).

25.3 Genetic Algorithm

GA's are a new approach to solving complex problems such as determination of machine layout; they can be defined as metaheuristic-based systems. GA's became known through the work of John Holland in the 1960s. The GA's contain the elements of the methods of blind search for the solution and of directed and stochastic search. Initially, the algorithm makes a quest for the solution in the entire search space and later by crossover it searches in surrounding of promising solutions. So GA's employed random, yet directed search for locating the globally optimal solution [9]. The typical steps required to implement GA's are encoding of feasible solutions into organisms using a representation method, evaluation of fitness function, selection strategy, setting of GA's parameters and criteria to terminate the process.

25.4 Ant Colony Algorithm

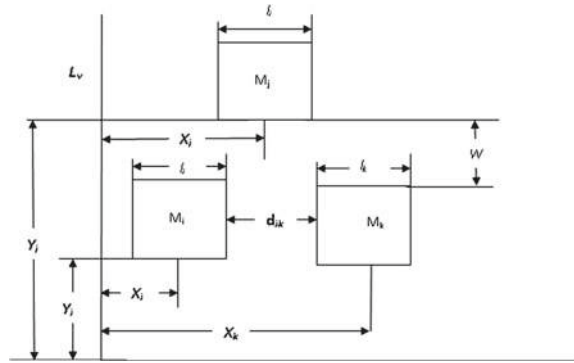
NP-hard optimization problems are solved efficiently by a recent technique by name ant colony algorithm [3]. Based on the behavior of real ant colonies, an algorithm called ant colony optimization is developed which was efficiently applied to NP-hard combinatorial optimization problems. Based on the experience of ant called as pheromone trails, information is modified while solving a problem. A description of the procedure followed in the ACO algorithms is given as follows [4]:

- Step 1: Initialisation of the parameters and pheromone trails
- Step 2: Construction of a complete solution for each ant
- Step 3: Local pheromone trail update
- Step 4: Improvement of each solution to its local optimum
- Step 5: Global pheromone trail update
- Step 6: If termination condition reached then stop otherwise go to step 2.

25.5 Mathematical Model

The machines are arranged along well-defined rows because in most of the cases the separation between rows can be predetermined according to the type of the material handling system used [3]. Multirow layout and its parameters and decision variables are illustrated in Fig. 25.1.

Fig. 25.1 Illustration of parameters and decision variables for multi-row problem



25.5.1 Objective Function

$$Z = \min \sum_{i=1}^{n-1} \sum_{j=i+1}^n C_{ij} F_{ij} L_{ij} \quad (25.1)$$

$$\text{s.t. } |x_i - x_j| Z_{ik} Z_{jk} \geq \frac{(l_i + l_j)}{2} + d_{ij}$$

$$\sum_{k=1}^m y_i = w(k-1) Z_{ik} \quad i = 1 \text{ to } n$$

$$\sum_{k=1}^m Z_{ik} = 1 \quad i = 1 \text{ to } n$$

$$\sum_{i=1}^n Z_{ik} < n \quad k = 1 \text{ to } m$$

$$x_i, y_i \geq 0 \quad i = 1 \text{ to } n$$

$$Z_{ik} = 0, l_i = 1 \text{ to } n \quad k = 1 \text{ to } m$$

n is the number of machines; m is the number of rows;

w is the separation between two adjacent rows;

l_i is the length of the machine i ;

d_{ij} is the distance between machines i and j ;

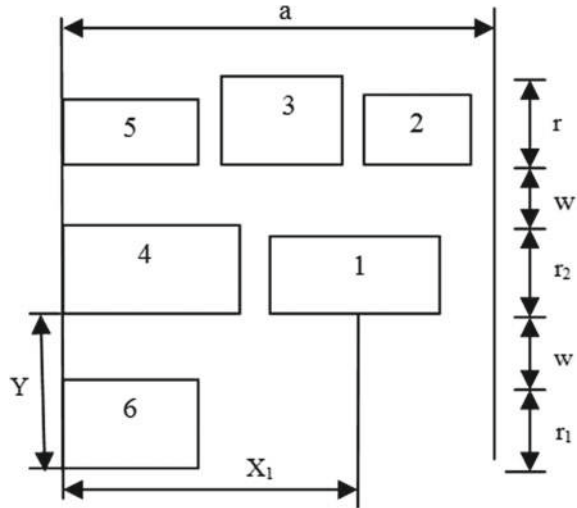
f_{ij} is the frequency of trips between machines i and j ;

c_{ij} is the transport cost per unit distance travelled between machines i and j ;

x_i is the distance between the center of machine i and the vertical reference line L_v ;

Y_i is the distance between the center of machine i and the horizontal reference line L_h (Fig. 25.2).

Fig. 25.2 Representation of arrangement



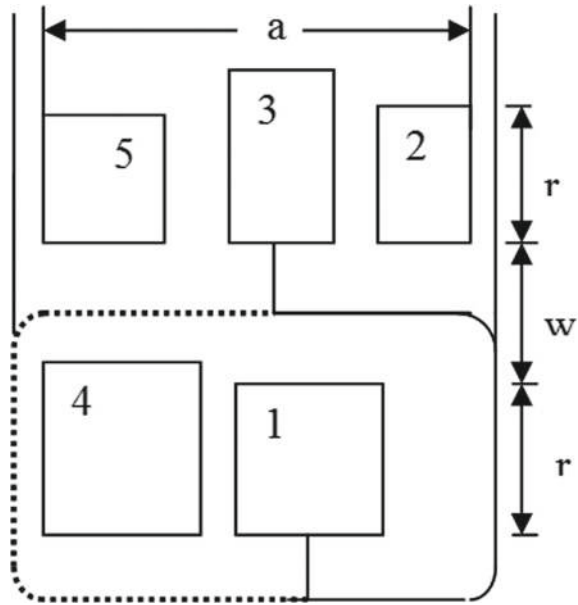
25.5.2 Determination of Length of Paths

First, the coordinates of the points of operating the machines are determined. When calculating coordinates the dimensions of the machines (Table 25.1), the allowable distances between the adjacent machines (d_{ij}) and the widths of the transport paths (w) are considered. Also, the row width (r) equal to the width of the widest machine in that row is determined. The arrangement of multirow machines is shown in Fig. 25.2. Coordinates of the operating points are determined as shown in Fig. 25.3. According to Fig. 25.3, the matrix of rows and coordinates of operating points are obtained as $[x_i, y_i]$. x_i is the x coordinate of the machine i , and y_i is the y coordinate of the machine i . Based on the values of the coordinates, the matrix of lengths of

Table 25.1 Two dimensions of the machines

Machine	Machine dimensions	
	Length (m)	Breadth (m)
Cutting (<i>C</i>)	4	1.5
Lathe1 (<i>L1</i>)	9	3.5
Lathe2 (<i>L2</i>)	7	2.5
Lathe3 (<i>L3</i>)	7	2.5
Hobbing1 (<i>H1</i>)	5.5	3
Hobbing2 (<i>H2</i>)	6.5	3.5
Milling (<i>M</i>)	3	3
Drilling (<i>D</i>)	2	2
Slotting (<i>S</i>)	1.5	3
Planning (<i>P</i>)	6	5

Fig. 25.3 Determination of length of paths



transport paths between the individual machines L_{ij} can be determined. If several paths between machines i and j are possible, the shortest one is selected (Fig. 25.3). When the machines i and j are located in the same row, the path length is determined according to the formula given below [10]:

$$L_{ij} = |x_j - x_i| \quad (25.2)$$

When the machines i and j are located in different rows, the path length is determined by using the two formulas given below:

$$1L_{ij} = x_i + x_j + w + |y_j - y_i| \quad (25.3)$$

$$\text{and } 2L_{ij} = (a - x_i) + (a - x_j) + w + |y_j - y_i| \quad (25.4)$$

From among the two lengths of paths, the minimum path length L_{ij} is selected:

$$L_{ij} = \min(1L_{ij}, 2L_{ij}) \quad (25.5)$$

After calculating the shortest path between all pairs of machines, the matrix L_{ij} is obtained [8]. The value of the fitness function for all chromosomes in the population can be calculated according to Eq. (25.1).

25.6 Case Study

Lakshmi Engineering Workshop is a batch production-based workshop. The data collected from Lakshmi Engineering Workshop, Guntur, A.P is given in Table 25.1.

25.6.1 Adjacency Matrix

Adjacency matrix which gives the distance between machines if they are placed adjacent to one another is given below:

	C	L1	L2	L3	H1	H2	M	D	S	P
C	0	4	4	4	6	6	6	4	5	6
L1	4	0	4	4	5	5	5	4	4	6
L2	4	4	0	4	5	5	6	4	4	6
L3	4	4	4	0	5	5	6	4	4	6
H1	6	5	5	5	0	6	6	5	5	6
H2	6	5	5	5	6	0	7	5	5	7
M	6	5	6	6	6	7	0	4	5	6
D	4	4	4	4	5	5	4	0	4	5
S	5	4	4	4	5	5	5	4	0	5
P	6	6	6	6	6	7	6	5	5	0

25.6.2 Network Model

Based on the data given in the Table 25.2, a network model as shown in Fig. 25.4 is developed to understand and find the flow matrix between machines. There are 200 parts (170+10+10+10) enter the system at machines C, L1, L2 and L3. 170 parts enter the system at Cutting, ten parts enter the system at Lathe1, ten parts enter the system at Lathe2 and ten parts enter the system at Lathe3. Parts after following the sequence which are shown in Table 25.2 leave the system at machines L1, L2, L3, D, S and P in 20, 20, 20, 30, 60 and 50 numbers, respectively.

Table 25.2 Details of parts produced

Part name	Sequence	Number of units
Gear-1	C-L-H-D-S	45
Gear-2	L-M	15
Eccentric shaft	C-L	60
Flanges	L-D	15
Back knife	C-P	50
Rest pins	C-L-D	15

Fig. 25.4 Network model showing machines and the flow of parts

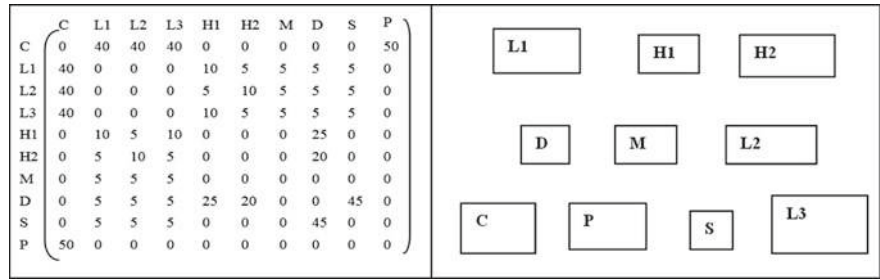
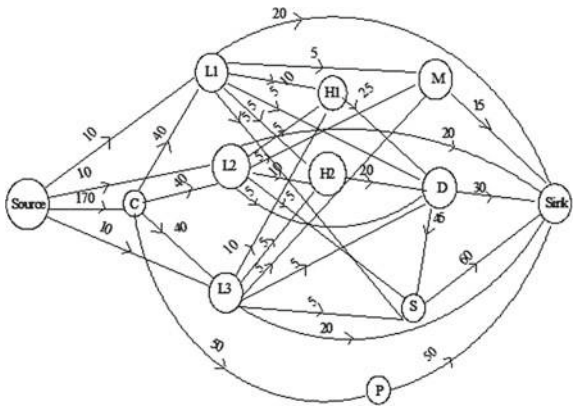


Fig. 25.5 Layout of the existing workshop

From the network model, the flow matrix is obtained and is given below. The cost matrix is assumed as unit matrix. Figure 25.5 depicts the layout of the existing workshop.

A programme is written in C-language for both the GA and ACO techniques to find out the optimum machine layout. The cost for the existing layout obtained by GA and ACO with these inputs is Rs. 10,652.

25.7 Results and Discussion

25.7.1 The Application of GA to the Multi-row 10 Machines Workshop Problem

For the 10 Machines workshop, genetic algorithm (GA) is applied to find out the best arrangement of machines in multiple rows. The values of the evolutionary parameters taken as Probability of crossover $P_c = 0.7$; Probability of mutation $P_m = 0.3$; Population size $P = 180$; Number of generations $G = 250$.

The parameters for multi-row are: Distance between rows = 4 m; Maximum width of transport is taken as $a = 45$ m; Row width = 4 m;

Figure 25.6 shows the results of all evolutions for the multi-row machine layout. From the results obtained from GA, it has been observed that at 48th generation, optimum transport cost of Rs. 6575 is obtained. Figure 25.7 shows the optimum arrangement of 10 machine multi-row machine layout using GA.

Fig. 25.6 Evolutionary process in GA for multi-row machine layout

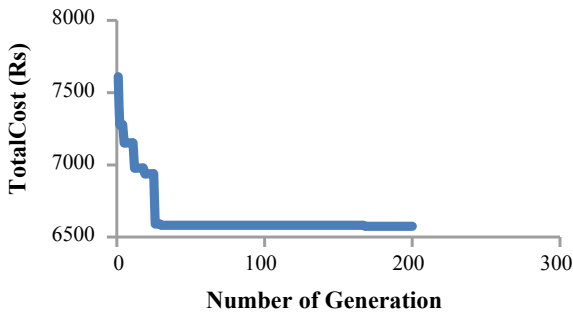


Fig. 25.7 Optimum arrangement for multi-row machine layout using GA and ACO

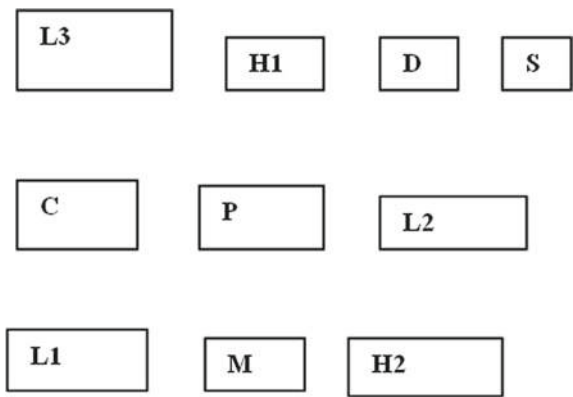
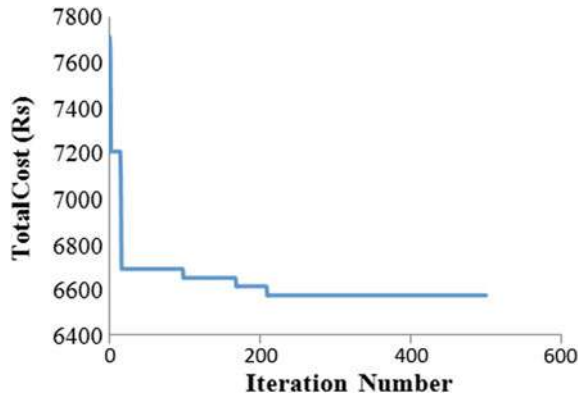


Fig. 25.8 Evolutionary process for multi-row machine layout using ACO



25.7.2 Application of ACO to the Multi-row 10 Machines Workshop Problem

For the same example, ant colony optimization (ACO) is applied to find out the best arrangement of machines in multiple rows. The parameters for applying ACO are Number of ants $n_a = 50$; Number of iterations $n_i = 500$, $\alpha = 1.1$, $\beta = -1$, $\rho = 0.1$, $\xi = 0.02$, $\tau_{ij}(1) = 0.02$ and $Q = 3000$.

Figure 25.8 shows the results of all iterations for the multi-row layout. From the results obtained from ACO, it has been observed that at 210th iteration, optimum transport cost of Rs. 6575 is obtained. Figure 25.7 shows the optimum arrangement of 10 machine multi-row machine layout using ACO.

Thus, for 10 machine workshop, both GA and ACO have given the same optimal arrangement of machines L3–H1–D–S–C–P–L2–L1–M–H2 with total cost of Rs. 6575.

25.8 Conclusions

By means of the presented model and by taking a case study, it is demonstrated that the optimum layout of the machines in the machine layout can be found. The model searches for the optimum layout in rows and finds itself the optimum number of rows.

A case study of Lakshmi Engineering Workshop is taken and optimum arrangement of machines in multi-row is found out by applying genetic algorithm and ant colony algorithm techniques. Optimal arrangement of machines is obtained as L3–H1–D–S–C–P–L2–L1–M–H2 with total cost of Rs. 6575. The total cost for the optimum layout decreased by 38% when compared with the existing layout cost. Hence it is recommended that the Lakshmi Engineering Workshop to use the optimum layout to reduce the total cost.

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Chapter 26

Optimization of Tool and Process Parameter for Injection Molded Component



Pratyush Kar , G. Rajesh Babu and P. Vamsi Krishna

Abstract Injection molding is a standout among the most imperative techniques utilized for forming thermoplastic parts in industry. In molded case circuit breaker (MCCB), Trip-bar is one of the most critical components as safety is concerned which is manufactured by injection molding process. To get it manufactured within the specified warpage and deformities free, several mold flow simulations are carried out using Creo-MoldFlow. The outcomes of the simulation are used to design the mold tool and optimizing the process parameters. The objective of this work is to optimize the process parameters such as filling time, melt temperature, mold temperature for high glass fiber reinforced polyarylamide composites. This consolidates the gray relational analysis (GRA) and CAE flow simulation software, to simulate the process as well as to anticipate the fiber orientation.

Keywords MCCB · Injection molding · MoldFlow simulation

26.1 Introduction

Electrical circuit breaker is a switching device which has manual and automatic control and protection of electrical power system. Trip-bar is one of the most critical components of molded case circuit breakers. It trips the breaker when any of the poles experiences a high instantaneous current. Hence, it is very crucial for the circuit breaker as safety is concerned. The quality of trip-bar produced by injection molding depends on the material, geometry, mold design, and the process parameters.

Many researchers are working on the injection molding process to optimize the process conditions for high-quality product and increase in productivity. The available patents on low-voltage circuit breakers provide a good understanding regarding the working of tripping mechanism and trip-bar functions. C.G. Corporation [1]

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_26

focused on a low-voltage circuit breaker with a residual current trip device, whose poles were arranged in parallel and the residual current trip device was arranged in a molded insulation shell. Christensen [2] worked on multipole circuit breakers tripping devices and manual operating members. The invention was concerned with a multipole circuit breaker which had got manual opening and closing along with automatic opening upon the occurrence of an overload in the circuit at any one of the poles. Ozvelik and Erzurumlu [3] worked on the minimization of the warpage and sink mark in terms of process parameters of the plastic parts having different rib cross-section and layout using Taguchi optimization method for polycarbonate materials. Tang et al. [4] fabricated a mold to produce a thin plate having dimension $120 \text{ mm} \times 50 \text{ mm} \times 1 \text{ mm}$. The thin plate was used for warpage testing. After analysis, it showed the effect of melt temperature was more on warpage. Galantucci and Spina [5] proposed an integrated approach to evaluate gating system configurations to optimize the filling conditions of thermoplastic injection molded parts. The filling pattern of complex part geometry was studied with the help of finite element analysis to improve the product quality which evaluates the component manufacturability at the early stage of the product development cycle, without fabricating prototypes and reducing the experimental tests.

The present work focuses on optimization of injection molding process considering cooling time, clamping force, residual shear stress, and volumetric shrinkage. The optimized process conditions will be used for designing the mold tool. The impact of different gate location on warpage with the optimized process input is studied.

26.2 Methodology

The numerical analyses are based on the 3-D Navier–Stokes flow solver with governing equations for mass, momentum, and energy [6]. Viscosity of polymer melts varies with shear rate, pressure, and temperature. Therefore, viscosity models should account for the variation with shear rate, pressure, and temperature. Such viscosity models include Cross-Exp and Cross-WLF models. The shear-dependent viscosity data are fitted to the Cross model as in Eq. 26.1 [7].

$$\eta = \frac{\eta_0}{1 + \left(\frac{\dot{\gamma}}{\dot{\gamma}_0}\right)^{1-n}} \quad (26.1)$$

Here, η_0 is the viscosity at zero shear and $\dot{\gamma}_0$ and n are data-fitted coefficients. The effect of temperature on viscosity is accounted by means of the Williams–Landel–Ferry (WLF) model as in Eq. 26.2 [7].

$$\eta_0 = D_1 \exp \frac{-A_1(T - T^*)}{A_2 + (T - T^*)} \quad (26.2)$$

$$T^* = D_2 + D_3 \quad (26.3)$$

$$A_2 = A_2^* + D_3 * P \quad (26.4)$$

where D_1 , D_2 , D_3 , A_1 , and A_2^* are constants to be determined while T^* is a reference temperature. Pressure dependence of viscosity is measured by using a throttle apparatus arranged in series with the die of the capillary rheometer. Data obtained from experiments in a pressure range of 40 MPa are fitted according to the Cross model and the D_3 coefficient is estimated as 2.1×10^{-7} K/Pa [7].

26.3 Results and Discussions

Several simulations are conducted using Creo-MoldFlow which is based on hybrid finite element/finite difference method for solving pressure, flow, and temperature fields. The process parameters ranges are provided in Table 26.1. Table 26.2 shows the list of input condition and output responses for each simulation run.

Table 26.1 Process parameters with different levels

Input parameters	Level 1	Level 2	Level 3
Mold temperature (°C)	120	130	140
Melt temperature (°C)	250	270	290
Injection time (s)	2	2.5	3

Table 26.2 Input conditions and output responses for individual simulation

S. No.	Input variables			Output responses			
	Injection time (s)	Mold temperature (°C)	Melt temperature (°C)	Cooling time (s)	Clamping force (Ton)	Volumetric shrinkage (%)	Shear stress (MPa)
1	2	120	250	86.7	10.72	7.22	2.87
2	2	130	270	104.8	4.17	7.9	1.37
3	2	140	290	123.3	1.8	8.64	0.45
4	2.5	120	270	99.2	3.37	7.89	1.18
5	2.5	130	290	115.6	1.61	8.62	0.47
6	2.5	140	250	97.4	9.61	7.18	2.6
7	3	120	290	109.5	1.53	8.61	0.49
8	3	130	250	91.4	8.87	7.15	2.27
9	3	140	270	111.6	3.36	7.07	1.05

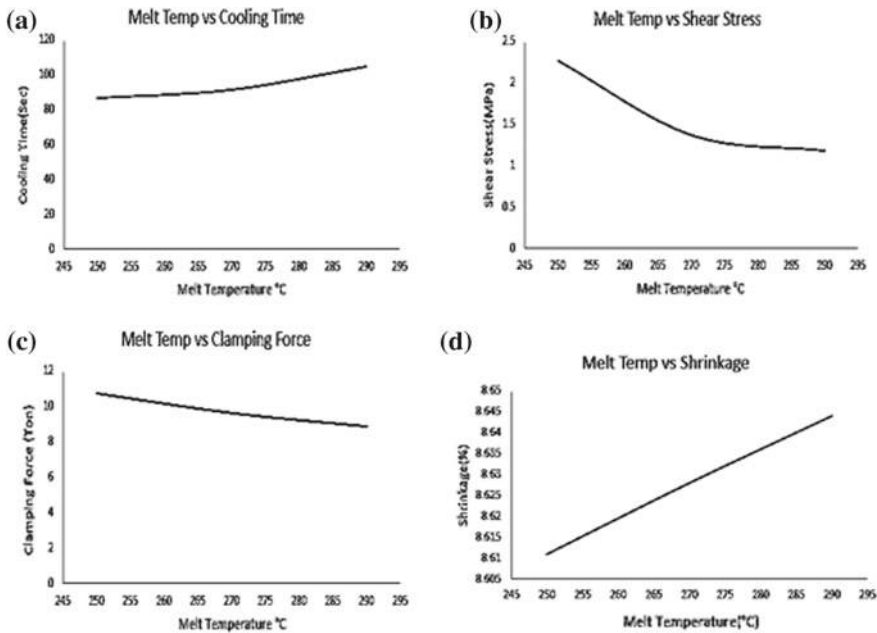


Fig. 26.1 Effect of melt temperature **a** Cooling time, **b** Shear stress, **c** Clamping force, **d** Shrinkage

26.3.1 Effect of Melt Temperature

Figure 26.1 indicates the effect of melt temperature on cooling time, shear stress, clamping force, and shrinkage. The cooling time shows an increment behavior with increase in melt temperature as the heat transfer occurs from molten plastic to the surrounding through the mold. Residual shear stress and clamping force reduce as the flow of plastic is more uniform and obstruction less with increase in melt temperature. Shrinkage increases with increase in melt temperature as thick areas will take more time to solidify.

26.3.2 Effect of Mold Temperature

Fig. 26.2 shows the effect of mold temperature on responses as shear stress, clamping force, cooling time, and shrinkage. It can be observed from the graph that with increase in mold temperature, the shear stress and clamping force values are getting reduced. Induced shear stress can be reduced if the plastic flows inside the cavity with less obstruction and friction. Further, it is observed the cooling time to reach the ejection and shrinkage temperature increases with increase in mold temperature.

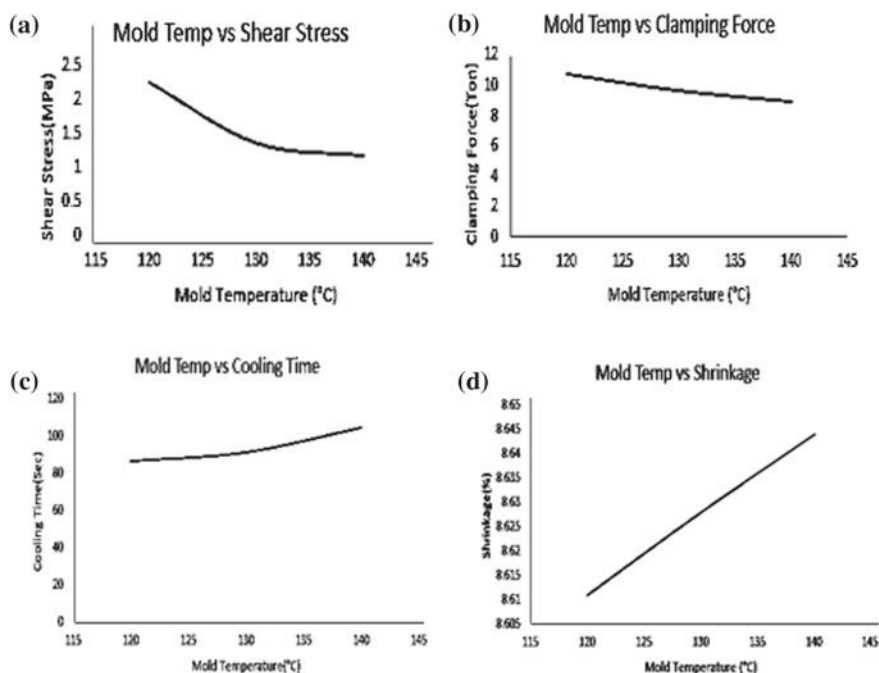


Fig. 26.2 Effect of mold temperature **a** Shear stress, **b** Clamping force, **c** Cooling time, **d** Shrinkage

Since for cooling the heat transfer from melt to the water channels will be reduced by maintaining high mold temperature.

26.3.3 Effect of Injection Time

Figure 26.3 shows the effect of injection time on responses as cooling time, clamping Force, volumetric shrinkage, and residual shear stress. It can be observed from the graph that with increase in injection time, the cooling time is getting reduced. The increase in injection time makes the injection pressure lower causing less friction and reducing the duration of cooling time. The clamping force experiences an increment behavior with increasing injection time as high injection time indirectly lowers the melt temperature. The volumetric shrinkage decreases with increase in injection time due to the thick sections which take more time to solidify. Residual shear stress increases as injection period is more because more injection period lowers the melt temperature making the flow less uniform and with increased resistance.

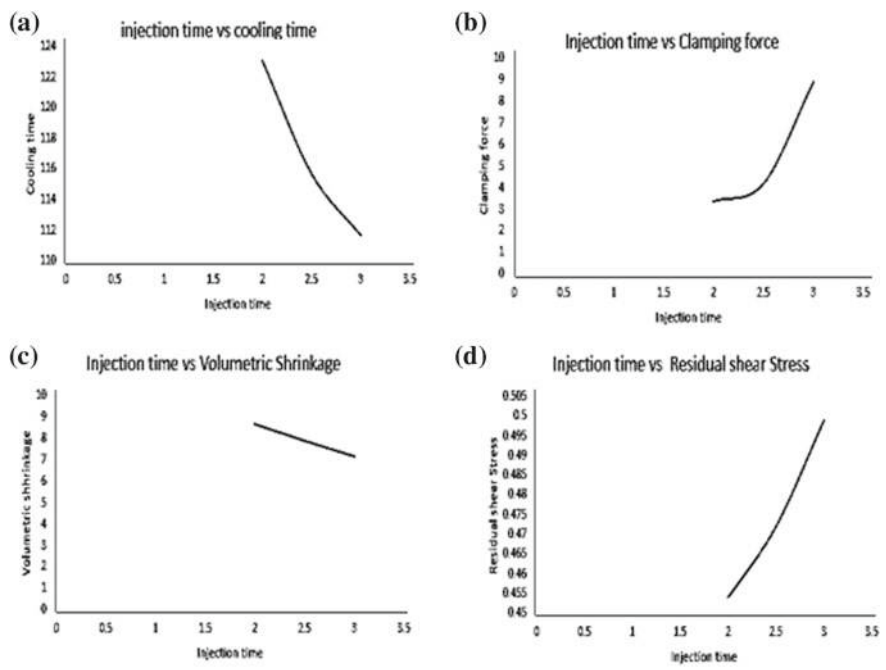


Fig. 26.3 Effect of injection time on **a** Cooling time, **b** Clamping force, **c** Volumetric shrinkage, **d** Residual shear stress

26.3.4 Output Response Analysis

The effect of input parameters on considered output responses, i.e., cooling time, clamping force, residual shear stress, and volumetric shrinkage is analyzed using Taguchi orthogonal experiment design. Signal to Noise ratio (*S/N* ratio) is calculated considering smaller is the better since the objective is to minimize the responses.

Cycle time of a part is very important as the rate of production and the quality of the parts depend on it, which can be reduced by reducing the cooling time. From Table 26.3, it is found melt temperature is most influential in cooling time because of more time in heat transfer time from mold cavity to the surrounding. For clamping

Table 26.3 Influencing parameter rank and *P*-value

	Injection time		Mold temperature		Melt temperature	
	Rank	<i>P</i> -value	Rank	<i>P</i> -value	Rank	<i>P</i> -value
Cooling time	3	0.145	2	0.004	1	0.001
Clamping force	3	0.247	2	0.728	1	0.005
Volumetric shrinkage	3	0.422	2	0.484	1	0.039
Residual shear stress	2	0.108	3	0.252	1	0.057

force, the main influential parameter is melt temperature. It is evident that with increase in melt temperature, the clamping force reduces as the plastic flow becomes more uniform and experiences less obstruction. Shrinkage causes either sink marks or voids in the interior of the molded component and it is observed that the most influential factor is melt temperature. From Table 26.3, it is seen that melt temperature is most influential in residual stresses as its lower value reduces the viscosity of the molten plastic. As viscosity reduces, the molten plastic finds it difficult to flow which induces stresses in the part.

26.3.5 Optimization Using Gray Relational Analysis

In present study, optimization of injection molding process parameters is performed using gray relational analysis. Since cooling time, clamping force, volumetric shrinkage, and shear stress should be minimum, smaller the better criterion is used. The sequential steps are adopted to determine the optimal combinations of the injection molding process parameters [8]. The gray relation grade and coefficient obtained are presented in Table 26.4 where $\mathfrak{Z}_i(k)$ = gray relational coefficient Ψ = weightage varies from 0 to 1. Here, it is considered as $\Psi_{cf} = 0.25$, $\Psi_{vs} = 0.3$, $\Psi_{ct} = 0.5$, $\Psi_{ss} = 0.5$ and n is the number of process responses.

The optimal process conditions as per gray relational analysis are injection time 3 s, mold temperature 140 °C, and melt temperature 270 °C.

Table 26.4 Gray relational coefficients values and grade relational coefficients

S. No.	Cooling time (s)	Clamping force (Ton)	Volumetric shrinkage (%)	Shear stress (Mpa)	Gray relational grade	Rank
	Smaller the better	Smaller the better	Smaller the better	Smaller the better	Smaller the better	
1	1	0.2	0.76	0.34	0.58	5
2	0.5	0.46	0.36	0.56	0.47	9
3	0.34	0.89	0.23	1	0.62	4
4	0.59	0.56	0.36	0.62	0.54	7
5	0.38	0.96	0.23	0.98	0.64	3
6	0.62	0.22	0.81	0.36	0.50	8
7	0.45	1	0.23	0.96	0.67	2
8	0.79	0.23	0.86	0.39	0.57	6
9	0.42	0.55	1	0.66	0.67	1

26.3.6 Feeding System Design

Runner Design

The efficiency of runner is based upon the ratio (A/P). The diameter of circular or semicircular runner is calculated as per the Eq. (26.5) [9]. Here, it is considered the length and weight to be 10 mm and 183 g, respectively.

$$D = \frac{\sqrt{W} * \sqrt[4]{L}}{3.7} \quad (26.5)$$

From the relation, the minimum diameter of runner should be 6.5 mm.

Gate Design

Too small gate affects the resin and significantly influences the available injection pressure and too large gate often results in an unattractive remnant. The width of the gate is calculated using the Eq. (26.6), Where n is material constant with value of 0.8 and A is surface area.

$$W = \frac{n * \sqrt{A}}{30} \quad (26.6)$$

The surface area of part = 13,444.81 mm². So, $W = 3.09$ or 3 mm. $h = n * t = 0.8 * 3.5 = 2.8$ mm. Since after gate removal some marks will exist, keeping this much height for the gate is not feasible, so gate height of 1.2 mm is used.

26.3.7 Warpage Analysis for Various Gate Locations

Various gate location analyses have been carried out to get the best gate location for the warpage values within the prescribed limits, which are shown in Table 26.5. The warpages along X , Y , and Z directions are obtained for the considered gate locations.

The first gate location gives the minimum value of warpage. So, the design of tool will be carried out based on this gate location with the process condition as specified earlier. It turned out to be certain that the gate location influences the melt flow evolution and consequently, the fiber orientation. Simulation results demonstrate that bidirectional flow and asymmetrical fiber distribution are obtained with the double-gate design and placing single gate at any side section of the part.

Table 26.5 Warpage value at different gate location

Coordinates of gate	Type of gate	Warpage in X direction	Warpage in Y direction	Warpage in Z direction
117.05, 5, 0.82	Single gate	0.39	0.25	0.14
1: 97.72, 2.45, 4.72 2: 167.24, 2.12, 4.83	Double edge gate	0.26	2.17	0.48
97.67, 1.47, 4.96	Single edge gate	1.14	0.45	0.5
1: 237.8, -12.4, -9.4 2: 167.35, -11.8, -9.4	Double edge gate	0.9	0.23	1.2
0, -0.37, 0	Single edge gate	0.84	0.22	0.58

26.4 Conclusions

Mold flow analysis is carried out for studying the flow behavior and defects. Optimization of process parameters is performed using Taguchi-based gray relational analysis. The following conclusions can be made:

The optimal sets of process parameters as found out by gray relational analysis are mold temperature = 140 °C, melt temperature = 270 °C, and injection time of 3 s. The effect of melt temperature on response is maximum followed by mold temperature and injection time. The rectangular edge gate having dimensions 3 mm*1.2 mm is recommended for mold tool design. The gate location effects on warpage are investigated and found a single gate at the middle of section is best for flow pattern and to prevent fiber orientation. With this location, the warpage is found to be within prescribed limits.

Acknowledgements This work was supported by Hyderabad Technology Center, GE India Industrial Private Limited, who provided insight and expertise that greatly assisted the work.

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Chapter 27

Flow Path Optimization of Pneumatic Valves Through CFD Analysis



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Abstract Relay emergency valves are typical pneumatic flow control valves which are primarily used in air brake vehicles to speed up the application and release of rear axle(s) brakes. This valve will also have additional provision to apply the trailer brake automatically in the event of accidental decoupling of trailer. The relay emergency valve will graduate, hold, and release of air pressure from the brake chambers to which it is connected. The relay emergency valve is used to reduce the response time of brake applications on heavy-duty vehicles. In order to achieve higher valve response, the valve should yield high flow rate for a wide range of operating pressures. Higher flow output can be achieved with the minimum flow restriction in the valves. To achieve different product functions, the assembly will have subparts like piston, springs, seals, etc. which will restrict the flow passage. To achieve high flow rate without sacrificing the product function, the valve flow area should be maximum. Hence, computational fluid dynamics (CFD) can be utilized as a useful design tool to optimize the flow area of relay emergency valves and also to study the effect of flow restrictions. This paper covers the optimization of the flow path by finding the nominal flow diameters as per ISO 6358. A thorough CFD analysis with several design iterations of the valve has been made to improve and finalize the nominal flow diameter with the required flow rate at the outlet to meet the design requirements. The theoretical results are in good agreement with the experiments.

Keywords Nominal flow diameter · Pneumatic valves · Relay emergency valves

27.1 Introduction

The basic function of a brake system is to slow down a vehicle speed on demand and to maintain its speed during downhill operation and also to hold a vehicle stationary after it has come to a complete stop. Consequently, brakes can be grouped into service brake, which is used for normal braking, secondary or emergency brake, which is used

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_27

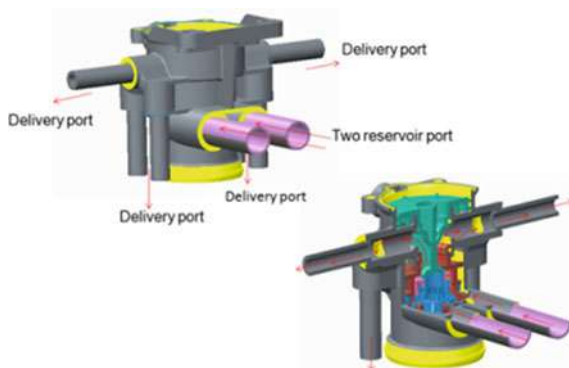
during partial brake system failure and parking brake. This paper completely covers with the relay emergency valve which is used in trailers brake system. Generally, tractor and trailers brake system is controlled by separate pneumatic valves. Relay emergency valve is one which is employed for trailer braking. A separate reservoir is installed for trailer braking application. Air from compressor flows through the supply line when the driver starts the engine. Then it flows through the palm coupling to fill the trailer reservoir through relay emergency valve. The fluid domain was created by considering the cavity inside the geometry by utilizing the CAD modeling package Pro-Engineer.

27.2 CFD Model Description

This section illustrates the procedure followed for building a mathematical model, discretization of a computational domain, and CFD solver setup. Autodesk CFD simulation software is a design oriented CFD modeling code has been used to investigate and optimize the flow path in the relay emergency valve.

Relay emergency valve with complete cross section is shown in the Fig. 27.1 and the valve body with different parts assembled together and the flow path in the reservoir to delivery flow path is focused for the optimization of the flow path. The other parts were designed as per the operating requirements of the valves. The airflow through the 3D valve passages was considered to be isothermal, compressible, and turbulent. The pressure boundary conditions are used to define the fluid pressure at the flow inlet and outlet. The computational model was built such that external structural parts were removed and simplified to reduce the complexity of the problem and the solution time. The inlet and outlet of the valve were extended with the additional caps to obtain fully developed parabolic flow and to reduce entry and exit losses.

Fig. 27.1 Relay emergency valve (two reservoir and six delivery ports)



27.3 Optimization Methodology

Flow rate characteristics of components using compressible fluids can be determined with the help of sonic conductance of the pneumatic component. The conductance of a pneumatic component can be determined from the amount of flow at conditions of standard reference atmosphere, from stagnation pressure and stagnation temperature ratio generating the flow. The flow is assumed to be steady state and valve-opening transient behavior is not considered in the analysis. DIN 1343 is referred to calculate the flow rate at normal atmospheric conditions and ISO 6358 is referred to find the nominal flow diameter through sonic conductance from flow rate at normal atmospheric conditions. Two different methods can be followed for finding the conductance parameter for the given valve. First method is a charge method in which component or product under study where the input pressure is varying and output boundary condition is open to the atmosphere. Ratio of absolute pressure at the outlet to the absolute pressure at the inlet is taken as a pressure ratio. This pressure ratio is varied between 0 and 1. For the different pressure ratio, flow rate at the standard atmospheric condition is calculated from the CFD analysis then this flow rate is converted into normal atmospheric conditions. Then this flow rate is taken for conductance calculation to understand the ability of the product to conduct the gas flow for the given conditions. Then by plotting pressure ratio versus conductance to identify the subsonic flow region and choked flow region. With the help of isosurface plot from CFD analysis, identify the critical regions for increasing the conductance. With the modification, the whole process will be repeated to achieve the required conductance in the value before choking. In the second method instead of varying inlet pressure, the outlet pressure will be varied by keeping constant inlet pressure to identify the conductance of the valve.

27.3.1 Design Iterations

Different concept model has been created to achieve the maximum nominal flow diameter. Figure 27.2 shows the entire three-concept model which has been discussed here.

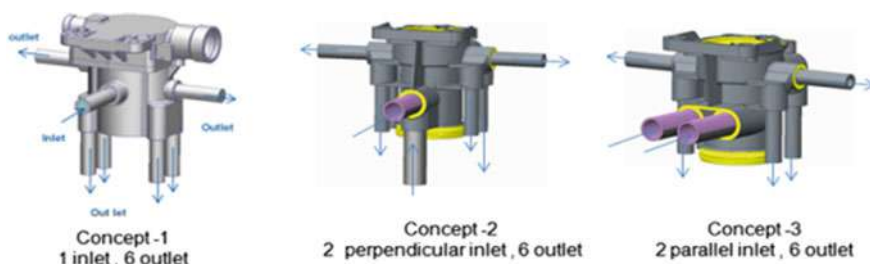


Fig. 27.2 Concept schemes

27.3.2 Concept Details

Concept 1 is having one inlet (reservoir port) and six-outlet (delivery port) internal flow path in concept 1 is having annular flow connected with the four delivery ports and other two delivery ports are connected in different height. Concept 2 is having two perpendicular reservoir ports and three branches of delivery port at both the side. Concept 3 is having two parallel reservoir ports and three branches of delivery port at both the side. Internal construction of valve concept 2 and concept 3 is same. But annular area and branches feeding diameter of concept 2 are 20% lesser than that of concept 2. As all the three concepts are created for same functions, internal flow path is not having much variations as shown for different concepts in Fig. 27.2 and Table 27.1.

CFD analysis conducted for all the three concepts for different pressure ratios to find the flow rate characteristics of the valve. Reservoir port is the one which is supplying air to the relay emergency valve when the brake pedal is actuated. Two reservoir ports are considered for concept 2 and concept 3. Two delivery ports in concept 1 are considered as horizontal and with the different height due to internal construction of the valves. But in concept 2 and 3, horizontal delivery ports are in same height. Then the four delivery ports are considered as vertical. All the port diameters size is same for all concepts.

27.3.3 Velocity Plot

CFD results are presented to understand the flow behavior during inlet is mentioned as reservoir section, inside the valve is represented as horizontal delivery pot, and the delivery is represented as delivery vertical port.

Table 27.1 Input pressure details for the three concepts

Scenarios	Boundary condition		Pressure ratio
	Inlet pressure in bar (Abs)	Outlet pressure in bar (Gauge)	
1	5.066	0	0.2
2	3.3775	0	0.3
3	2.0265	0	0.5
4	1.4474	0	0.7
5	1.128	0	0.9

Figures 27.6, 27.7 and 27.8 represent strong pressure differential exists in the inner valve section, i.e., at end of reservoir port to joining of delivery port. As there is a change in the valve flow area due to internal parts, pressure energy is converted as kinetic energy which resulted in flow acceleration at the downstream portion and hence high velocity which shown in Figs. 27.3, 27.4, and 27.5.

27.3.4 Pressure Plot

Concept 1 reservoir section shows more flow restriction at the port and valve joining portion where the velocity is in higher side compared to concept 2 and concept 3.

Concept 2 and concept 3 are modified to reduce the flow restriction at the port and valve joining portion. Figure 27.4 reservoir section shows horizontal reservoir port is dominating the flow compared to perpendicular reservoir port and this perpendicular input flow is not supporting for this configuration.

Figure 27.5 reservoir section shows the parallel reservoir port is favorable and the flow is gliding through the inner parts of the valve before connecting to the delivery port. Figure 27.4 flow restriction at joining of valve portion and horizontal delivery port is more compared to Fig. 27.5. More flow separation observed in concept 1 and concept 2 compared to concept 3 at the delivery vertical port side.

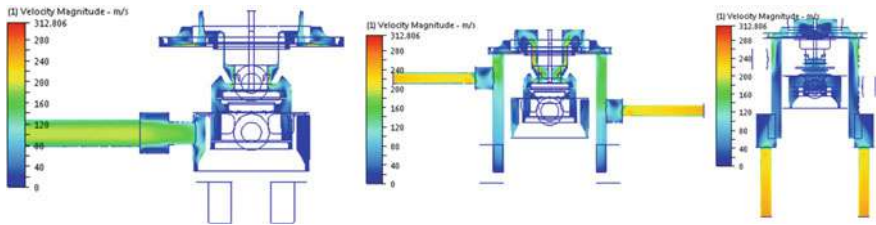


Fig. 27.3 Concept 1—Reservoir section, delivery horizontal port, and delivery vertical port

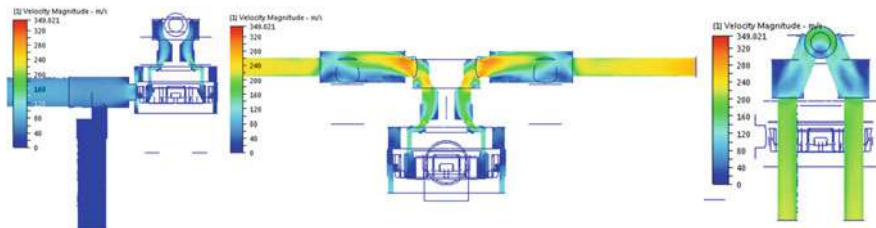


Fig. 27.4 Concept 2—Reservoir section, delivery horizontal port, and delivery vertical port

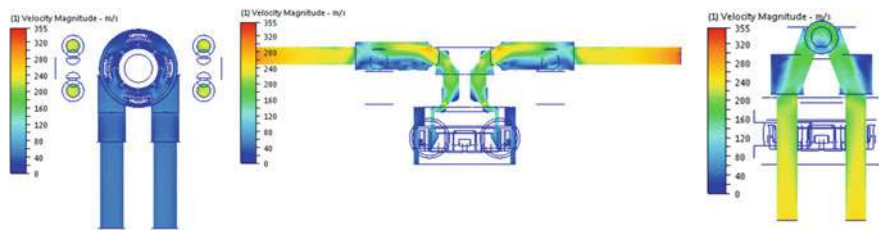


Fig. 27.5 Concept 3—Reservoir section, delivery horizontal port, and delivery vertical port

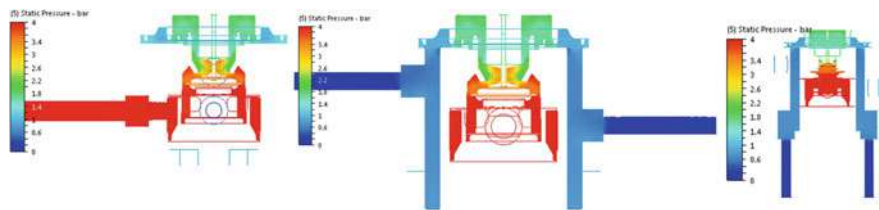


Fig. 27.6 Concept 1—Reservoir section, delivery horizontal port, and delivery vertical port

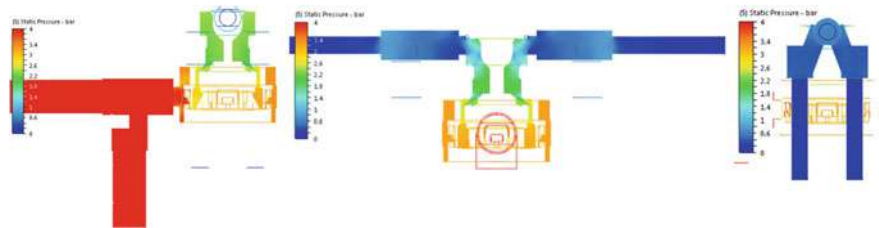


Fig. 27.7 Concept 2—Reservoir section, delivery horizontal port, and delivery vertical port

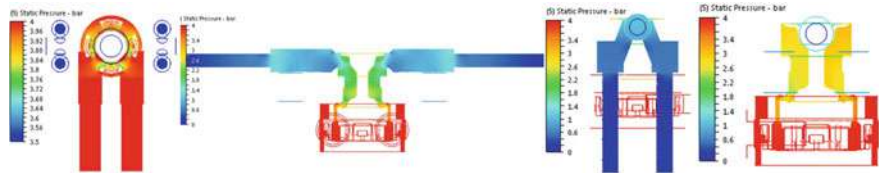


Fig. 27.8 Concept 3—Reservoir section, delivery horizontal port, and delivery vertical port

27.3.5 Isosurfaces Plot

Isosurface plot shown in Fig. 27.9, represents the flow critical region at the velocity of 200 m/s. More flow restriction is observed in concept 1 and concept 2 at the valve inner portion where the functional parts are assembled and it cannot be modified. Concept 1 represents for the same single inlet conditions, and the flow path is having more flow restrictions compared to other concepts. As in concept 2, the perpendicular port is not supporting and the flow output is less, i.e., more flow output is possible to accommodate in the flow path. Concept 3 represents less flow restriction at the inner portion of the valve and flow path is maximum utilized to yield higher flow output.

27.4 Nominal Flow Calculation

Flow rate at the outlet is calculated for three concepts through CFD analysis under standard atmospheric conditions. Then the calculated flow rate is converted to normal atmospheric conditions as per DIN 1343 [4]. Reason for calculating normal flow rate is due to the changes in delivery conditions like pressure, temperature, etc. during CFD calculation. Therefore, it is necessary to normalize all flow rates under normal atmospheric condition.

27.4.1 Flow Rate at Normal Atmospheric Condition

Figure 27.10 shows normal flow rate calculated from CFD analysis. Concept 1 is <0.6% for higher pressure ratio compared to concept 2 and for lower pressure ratio concept 2 is 5% greater than that of concept 1. Concept 3 normal flow rate is 23–26% greater than that of concept 1 and 23–28% greater than that of concept 2. According to ISO 6358, the most important flow-rate characteristic parameter of a pneumatic component is the sonic conductance, C . Equation 27.1 should be taken to calculate conductance parameter in the choked flow region. Where ‘ i ’ represents each data point.

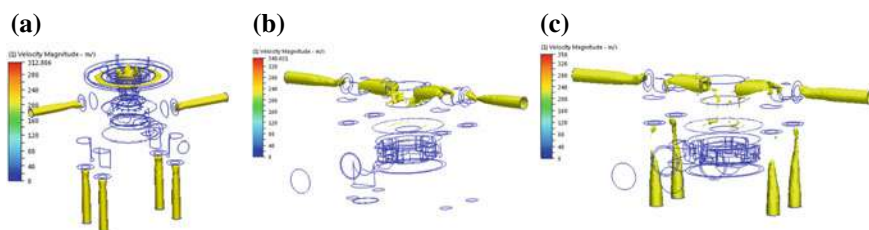


Fig. 27.9 a Concept 1, b concept 2, c concept 3

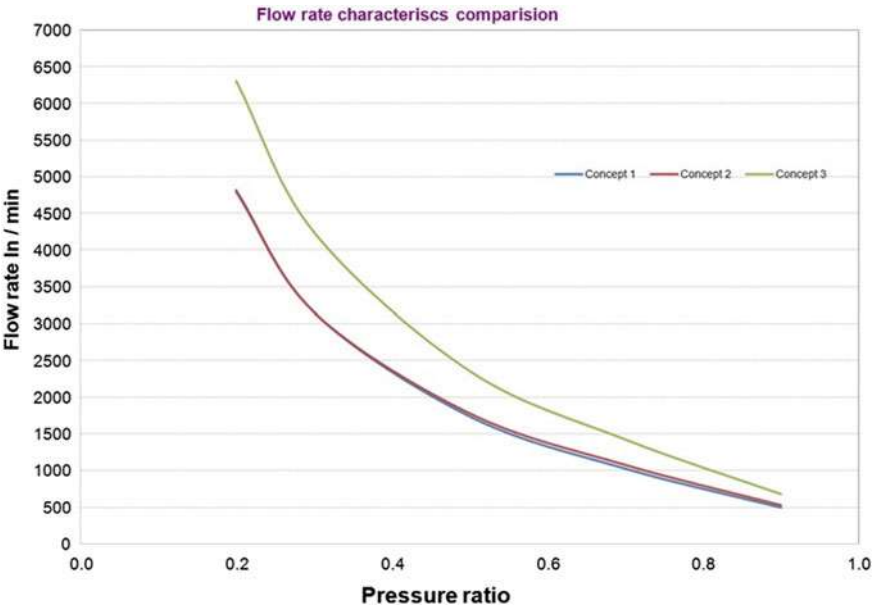


Fig. 27.10 Normal flow rate versus pressure

$$C_i = \frac{q_{mi}^*}{\rho_0 P_{1i}^*} \sqrt{\frac{T_{1i}^*}{T_0}} \tag{27.1}$$

$$\frac{C_e}{C} = \left\{ 1 - \left(\frac{\frac{p_2}{p_1} - b}{1 - \frac{\Delta p_c}{p_1} - b} \right)^2 \right\}^m \tag{27.2}$$

Equation 27.2 should be taken to calculate conductance parameter in the subsonic flow region. Sonic conductance is calculated with the Eqs. 27.1 and 27.2. In all the three concepts, pressure ratio <0.3 indicates the choked flow region where the changes in conductance parameter changes are less for lower pressure ratio and >0.3 indicates the subsonic flow region where the conductance parameter variation is high for higher pressure ratio.

From Fig. 27.11, concept 3 is having high conductance parameter for a wide range of pressure ratio. For high valve flow response, high conductance parameter is important to meet the pneumatic brake system regulations. However, both concept 1 and concept 3 are taken for further study. To reduce development and test lead time, conductance parameter has been validated for concept 1 followed by system-level validation for concept 1 and concept 3 (Table 27.2).

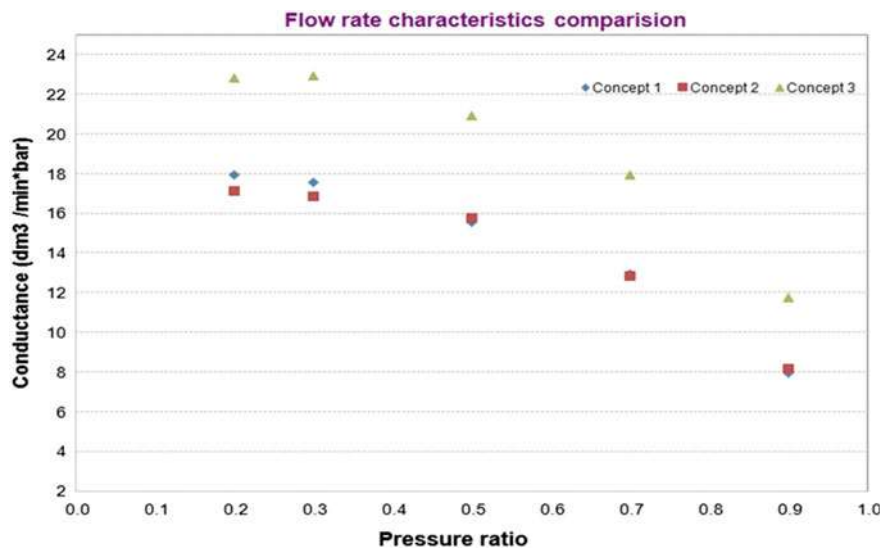


Fig. 27.11 Conductance versus pressure (comparison)

Table 27.2 Nominal flow diameter for three concepts

S. No.	Concepts	Nominal flow diameter in mm
1	Concept 1: One reservoir port	10.6
2	Concept 2: Two perpendicular reservoir port	10.53
3	Concept 3: Two parallel reservoir port	12.14

27.5 Test Bench Validation

Concept 1 design is built to predict conductance parameter of the design through test bench circuit as per ISO 6358-1 clause 5.2. Inlet valve is connected to the controlled pressure source and the outlet valve is connected to the atmosphere where flow meter is employed as per ISO 6358-1 clause 5.2. Variable upstream pressure is maintained to predict the flow rate for the inlet and outlet pressure ratio at different intervals. Test bench results are captured for different pressure ratio and the variation between test bench and theoretical results is 3–7.8%.

27.5.1 Nominal Flow Diameter Calculation

Nominal flow diameter is calculated for different concepts with Eq. 27.3. This effective diameter represents the flow efficiency of the valves for wide range of operating pressure ratios.

$$d_{\text{eff}}[\text{mm}] = \sqrt{\frac{V_N \left[\frac{\text{dm}^3}{\text{min}} \right] * \sqrt{T_1[\text{K}]} }{147.34 * p_1[\text{barA}]}} \tag{27.3}$$

Concept 3 is having higher flow diameter which is 12.6% greater than that of concept 1. But the concept 1 is slightly greater than that of concept 2. Further, concept 3 is proposed for pneumatic brake applications.

27.6 Conclusions

This paper illustrates the calculation procedure for finding the conductance parameter in pneumatic valves. Figure 27.12 shows a good comparison of theoretical results with experimental results. This leads to the conclusion that the methodology can be used to predict flow efficiency through conductance of valves. Further, effective diameter is calculated for all these concepts to understand the minimum flow diameter requirement for pneumatic brake system application. In a pneumatic brake system, the number of pneumatic flow control and pressure control valves is used form reservoir to end brake actuation system. This methodology can be adopted to find the effective flow diameter of all the valves. Further, system-level response can be

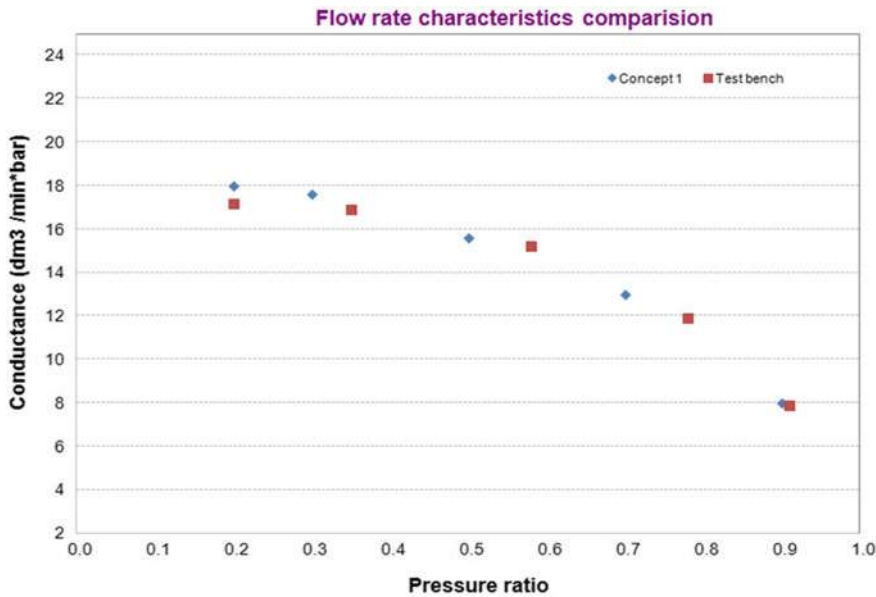


Fig. 27.12 Conductance versus pressure (test comparison)

predicted with the simplified effective flow diameter for every valve during brake actuation. This analysis reduces the development cycle time and provides a solution in a timely, cost-effective manner.

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Chapter 28

Virtual Simulation with Statistical Approach on Performance Optimization



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Abstract Most of the industries are moving toward the “Industry 4.0,” which is the current trend of automation, data exchange, and manufacturing technologies. Industry 4.0 operates on four main objectives which are interoperability, information transparency, technical assistance, and decentralized decisions. This paper focuses on technical assistance that enables integrating virtual simulation and statistical tool benefits and advantages in achieving first-time right design in a foot brake valve. Foot brake valve provides the driver with a graduated control for applying and releasing the vehicle brakes.

Keywords Foot brake valve · Virtual simulation · Statistical approach

28.1 Introduction

A typical air brake system layout is shown in Fig. 28.1, which mainly consists of a pneumatic compressor as a power source, foot and hand brake valves which act as actuating units, and the relay valves as a control unit and actuators.

The compressor provides the compressed air, which is the energy source of air brake system. The air dryer in the layout absorbs the moisture in the pressurized air and also maintains the system pressure. The moisture-free compressed air will be stored in the reservoirs through the system protection valve. During normal brake application, when the driver presses the pedal, the dual brake having two deliveries get actuated [1]. The primary delivery gives signal to the relay valve which actuates the spring break actuator where the pressure is converted into braking force. The braking force is transmitted to the foundation brakes through the slack adjuster where it is converted into braking torque and applies brake on the rear wheels. The secondary delivery of the valve actuates the brake chamber thereby applies the brake on the front wheels. In the case of emergency situations, the driver actuates the hand valve which applies the brake through the spring brake actuator.

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_28

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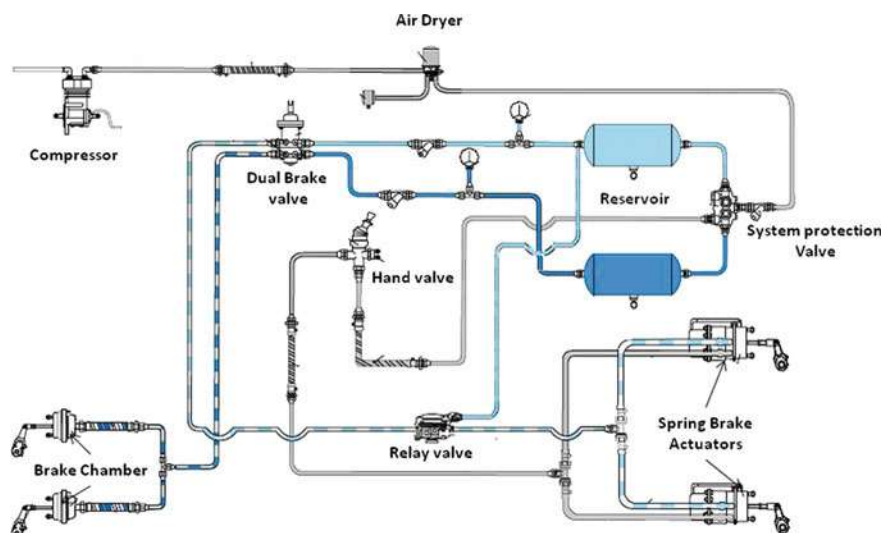


Fig. 28.1 Typical air brake system layout

28.2 Relevance of Investigation

In a typical air brake circuit, the dual brake valve or foot brake valve is used. Dual brake valve is either a floor-mounted, treadle-operated type or firewall-mounted brake valve with two separate supply and delivery circuits for service and secondary braking which are directly operated by the driver. The sectional view of a treadle operated foot brake valve is shown in Fig. 28.2.

Dual brave valves carry sensitive graded increasing and decreasing of pressure in the brake cylinders of a motor vehicle. The brake valve consists of two circuits. Both circuits lie among each other. In the case of failure of one circuit, the other circuit remains unaffected. The primary circuit is connected to the rear axle, whereas the secondary circuit is connected to front axle.

Performance characteristic curve of conventional foot brake valve is shown in Fig. 28.3. The important functional parameters in a performance characteristics curve are crack off travel, jump in pressure, knee point travel, knee point pressure, total travel, and limiting pressure which is marked mentioned in the figure. In a conventional system, the limiting pressure is nothing but the inlet pressure of the system.

In a conventional foot brake valve after a particular travel, the inlet and the delivery pressure will be the same, i.e., inlet pressure and delivery pressure 8.5 bar.

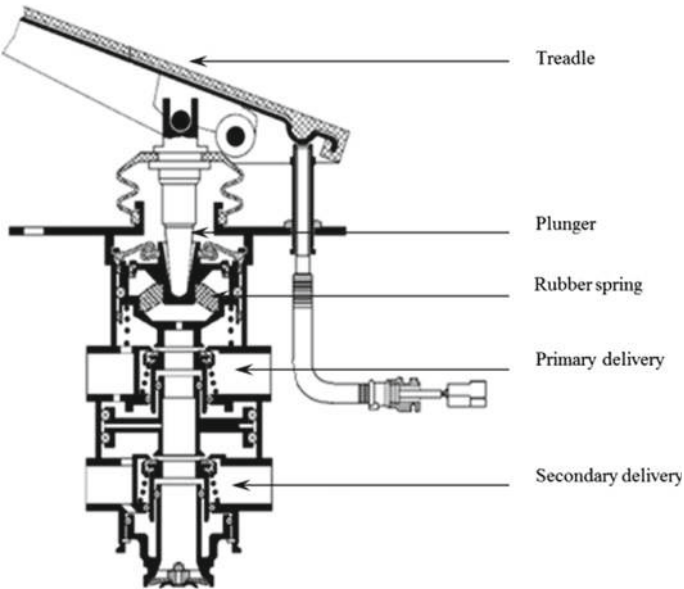


Fig. 28.2 Conventional FBV sectional view

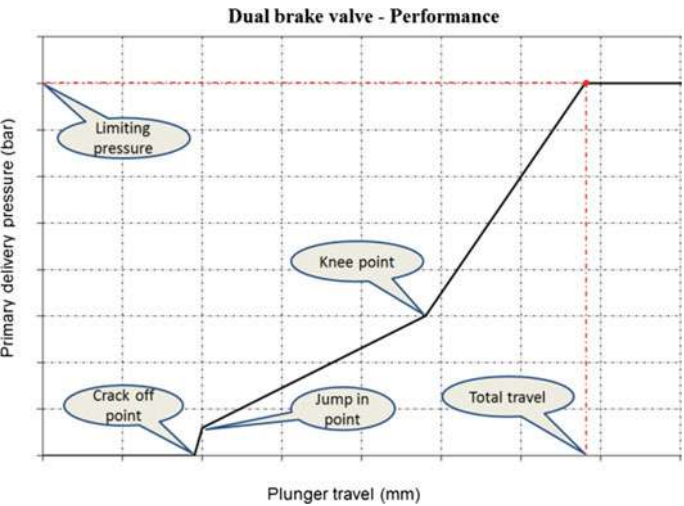


Fig. 28.3 Conventional performance characteristics

28.3 Modeling and Simulation

A new requirement from the customer is to do a pressure limiting function, i.e., for an inlet pressure of 12 bar, the delivery pressure should be 8.5 ± 0.5 bar.

Based on its geometry and construction, the product will be modeled using appropriate mechanical and pneumatic components like piston, spring, and mass available in the library. With reference from [2, 3], modeling of a foot brake is done which is shown in Fig. 28.4. The simulation model results are compared with experimental results for validating the model to carry out DOE. The simulation model is refined to have a correlation percentage of around 96% with test results.

Using the signal library, the major output parameters, crack off travel, jump in pressure, knee point travel, knee point pressure, total travel, and limiting pressure, are derived from the performance graph.

28.3.1 Modeling of Flow Restrictions

The flow passages and annular restrictions in the valve are assumed as orifices in the modeling and are introduced in the corresponding location. The mass flow rate of the orifice is governed by the following equation.

$$m = A \cdot C_q \cdot C_m \cdot \frac{P_{up}}{\sqrt{T_{up}}}$$

where

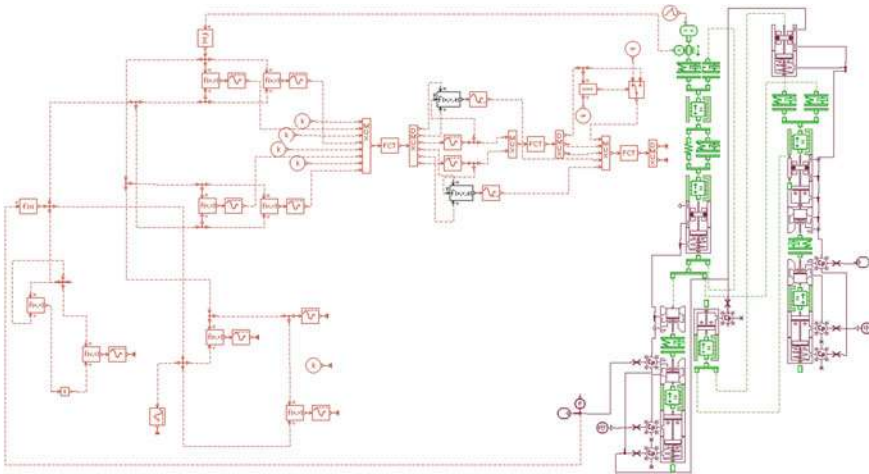


Fig. 28.4 Virtual foot brake valve model—AMESim circuit

- P_{up} Upstream pressure,
 T_{up} Upstream temperature,
 C_m Mass flow parameter,
 C_q Discharge coefficient,
 A Area of a cross-section of orifice.

28.3.2 Modeling of Valve Arrangements

The variation of flow area with respect to spool displacement is calculated by the following equation (Fig. 28.5).

$$\text{area} = (\theta - \sin \theta) \frac{d^2}{8}$$

where

- d Port diameter
 x Spool displacement from closed position.

$$\theta = 2 \cos^{-1} \left(1 - \frac{2x}{d} \right)$$

where

- d Port diameter
 x Spool displacement from closed position.

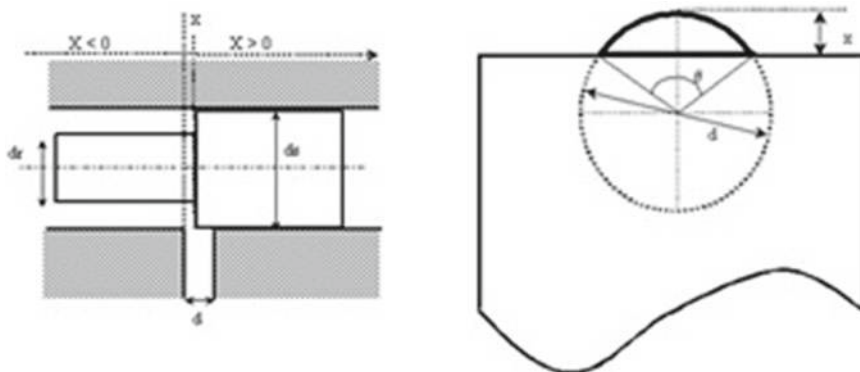


Fig. 28.5 Flow area in pneumatic spool valve

$$\text{area} = \pi * ds * x$$

where

- x Valve lift
- ds Diameter of flapper.

28.4 Methodology

The key functional parameters are arrived directly from performance characteristics for DOE study.

Initial QFD is carried out to address the voice to customers to design requirements. Most significant parameters contributing to performance characteristics are narrowed down to seven factors through QFD and DOE study is carried out. The performance prediction for 128 runs was done and plotted to understand the maximum range of working at each functional location. Figure 28.6 shows the performance characteristics curve for all 128(2⁷) iterations.

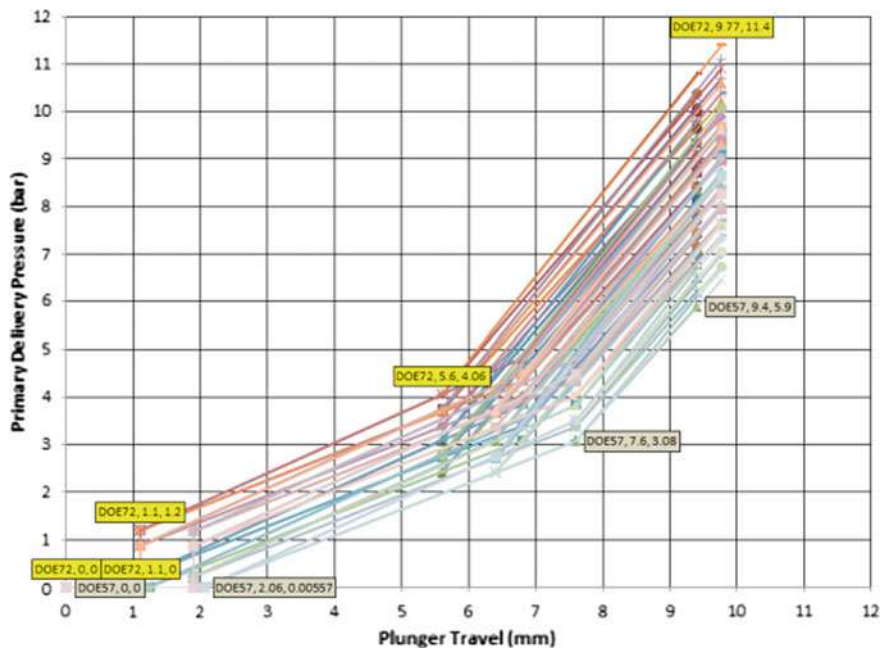


Fig. 28.6 Performance predictions for 128 iterations

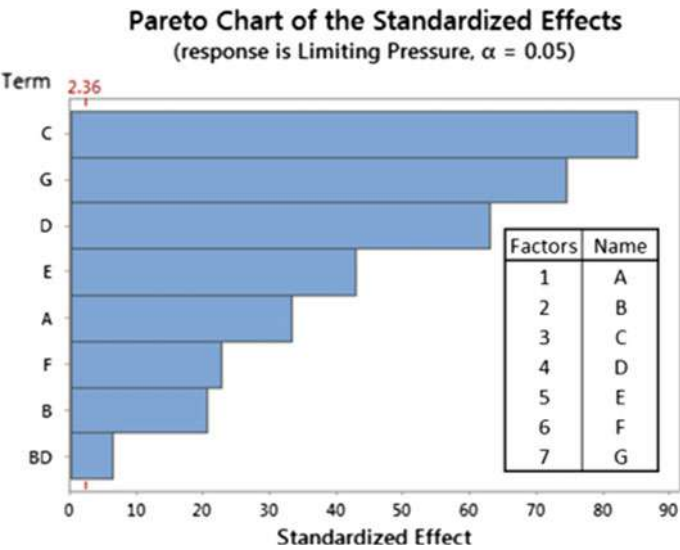


Fig. 28.7 Pareto chart

Considering the extreme tolerances, the limiting pressure maximum and minimum are 11.4 bar and 5.9 bar, respectively. From Pareto chart mentioned in Fig. 28.7, the top contributing factors were identified.

Out of top three contributing parameters for further control of tolerances are not possible in one parameter because of manufacturing limits. Tolerances on spring, gap, and travel are controlled by addition of shim to the existing system for controlling the valve travel. With proposed addition of shim, the performance predictions were done for the extreme cases that are arrived from DOE study which are mentioned in Table 28.1.

The performance characteristics curve for the maximum and minimum conditions compared with initial DOE runs are shown in Figs. 28.8 and 28.9, respectively. The maximum limiting pressure got reduced from 11.4 to 8.45 bar and the minimum limiting pressure got increased from 5.9 to 8.35 bar. Theoretical tolerance band for the proposed band is 0.1 bar. The tolerance of the proposed system is within the specified range of 8.5 ± 0.5 bar.

Table 28.1 Maximum and minimum conditions

Iteration no.	1	2	3	4	5	6	7
57	−1	−1	−1	+1	+1	+1	−1
72	+1	+1	+1	−1	−1	−1	+1

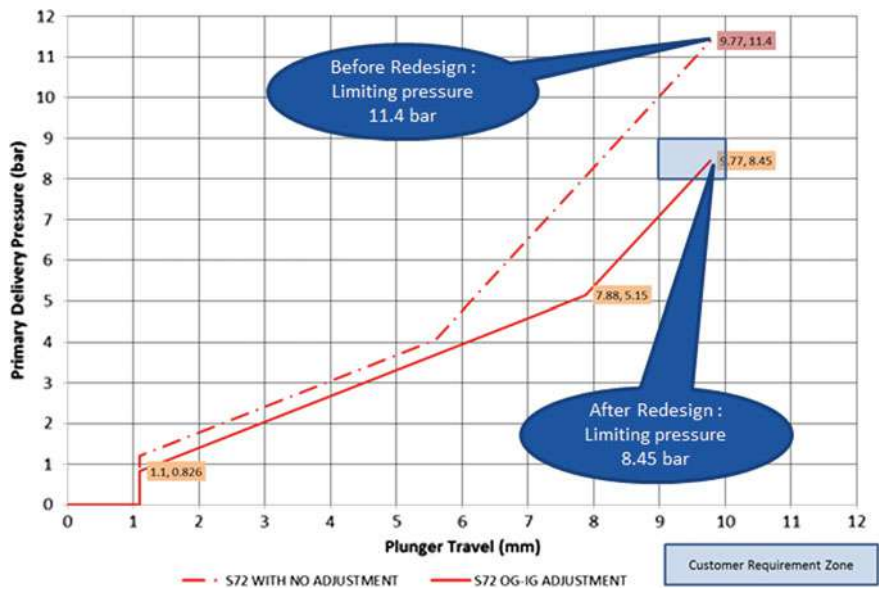


Fig. 28.8 Maximum condition: limiting pressure comparison

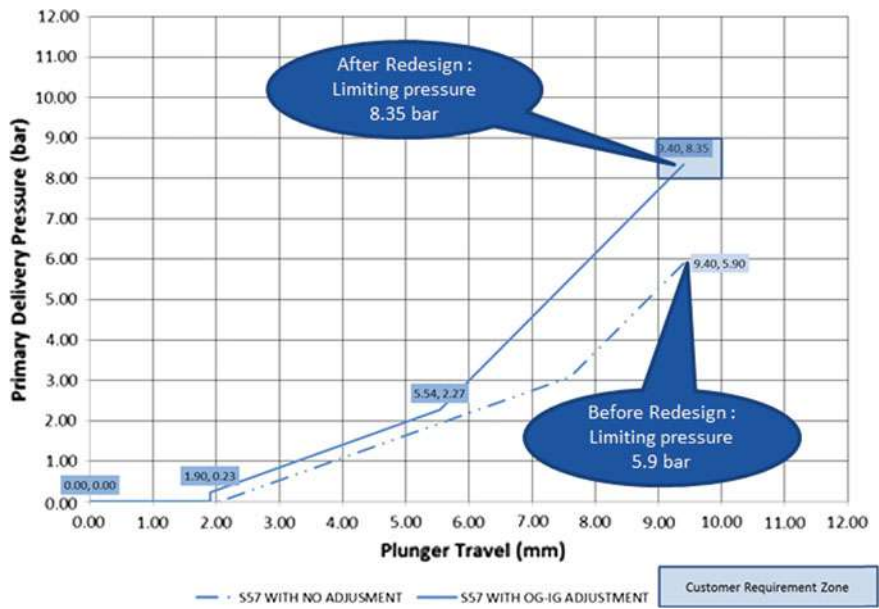


Fig. 28.9 Minimum condition: limiting pressure comparison

28.5 Conclusions

Virtual simulation methodology along with statistical approach has enabled reduction of existing bandwidth from 5.5 to 0.1 bar. To insure that this methodology of optimization has helped in achieving first-time right design in dual brake valve, physical prototype samples were developed and verified against theoretical calculations. The results were found to be well within the tolerance limits.

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Chapter 29

Design, Development, and Modeling of EMLA-Based Wheel Brake Actuation System for an UAV



D. Satish Babu , P. N. Vijay Vittal , Pollov Sarmah
and Veena G. Dikshit

Abstract Wheel brake systems are designed for aircraft became common ever since it participated in transportation and specific applications, as the intricacy and speed of operation amplified and the usage of different airstrips and land surfaces for different conditions. Wheel brake system operation solely depends upon aircraft safe operation on the different surface grounds. The brake system slows the speed of aircraft and stops it in a specific amount of reasonable time and distance. In general, most of the aircrafts having the main landing gear wheels are outfitted with a power-assisted brake assembly and the front or rear landing gear does not having the wheel brake system. This paper presented a development of electromechanical linear actuator (EMLA) based wheel braking system that contained linear actuator, crank arm, master cylinder, and a brake caliper control system. Dynamic model of EMLA-based system simulated with command from flight control computer and actuation system was established. The command control of actuation input and distribution of brake force approach were also discussed. Experiments conducted to estimate the performance of braking system and checked with real system. The simulation studies, and experimental performances show the practicability, effectiveness and implementation of this braking system on a typical unmanned aerial vehicle (UAV). The system was also implemented on a UAV platform, and the actuation requirements for the various braking conditions were quantified.

Keywords UAV wheel brake actuation · EMLA · Brake torque · Actuation controller

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_29

29.1 Introduction

Wheel brake systems are designed for aircraft became common ever since it participated in transportation and specific applications, as the intricacy and speed of operation amplified and the usage of different airstrips and land surfaces for different conditions. Wheel brake system operation solely depends upon aircraft safe operation on the different surface grounds. The brake system slows the speed of aircraft and stops it in a specific amount of reasonable time and distance. In general, most of the aircrafts having the main landing gear wheels are outfitted with a power-assisted brake assembly and the front or rear landing gear does not having the wheel brake system.

A typical aircraft wheel brake system having mechanical or hydraulic linkages allows the pilot to control the brakes by the actuator or cables. The majority of the aircrafts characteristically use disk brakes due to the main operational principle involves converting the kinetic energy of rotary motion into thermal energy through the formation of friction. The disk rotates along with the wheel assembly while a single or multiple piston calipers is stationary resist the rotation by creating friction against the single or multiple disks when the brakes are applied.

Off the shelf, the various brake assemblies are available; all used hydraulic power to operate. Generally, the hydraulic system for the brake assembly is independently realized since it is operational intermittently only when the aircraft is on the ground. For small, light aircraft which do not have any hydraulic systems, the brake actuation is through the pilot directly who will push the rudder pedals which in turn will operate a set of brake master cylinders. To develop the necessary hydraulic pressure to operate the wheel brakes, these master cylinders are used. In an unmanned aerial vehicle (UAV), the pilot is external to the system that will remotely operate the master cylinders through an electromechanical actuator (EMA). The paper discusses a brake actuation system in a typical UAV where an electromechanical linear actuator (EMLA) actuates the brake master cylinders.

29.2 Design of EMLA-Based Wheel Braking System

The schematic of the brake actuation system, which was deployed on the main landing gear of the UAV, is shown in Fig. 29.1. Main landing gear wheels are equipped with a disk brake system having calipers with hydraulic fluid was controlled by a hydraulic control unit as master cylinder. The master cylinders are operated by an electromechanical linear actuator with integrated controller. The controller gets the command through the flight control computer (FCC) either from external pilot system (EPS)/ATOL system/ground control station (GCS). The UAV can identify the pilot intention by acquiring and analyzing the signals from FCC via the communication links and thereafter sending appropriate commands to the actuator controller unit.

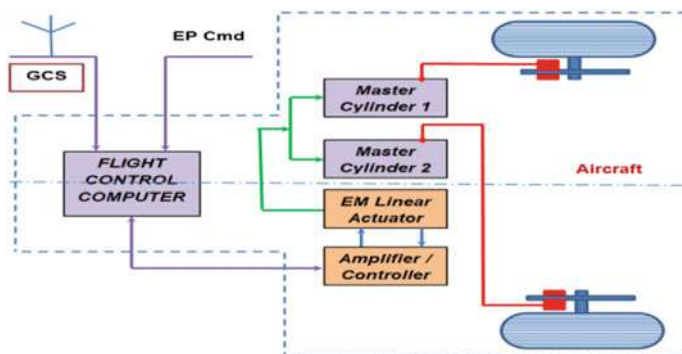


Fig. 29.1 Layout of the EMLA-based wheel braking system

29.2.1 Wheel Brake Actuation System

The actuation system consists of an EMLA with driving amplifier electronics, a master cylinder (both single master cylinder and double master cylinder have been worked upon) and a crank which connect the actuator output rod and the master cylinder piston rod with required link ratio. The unit is mounted on a plate having the *Actuator Fixed End Bracket*, *Crank Hinge Bracket*, *Master Cylinder Mounting Bracket*, and *Amplifier Mounting Bracket* which is assembled on aircraft bulkhead with a set of M6 bolts (Fig. 29.2).

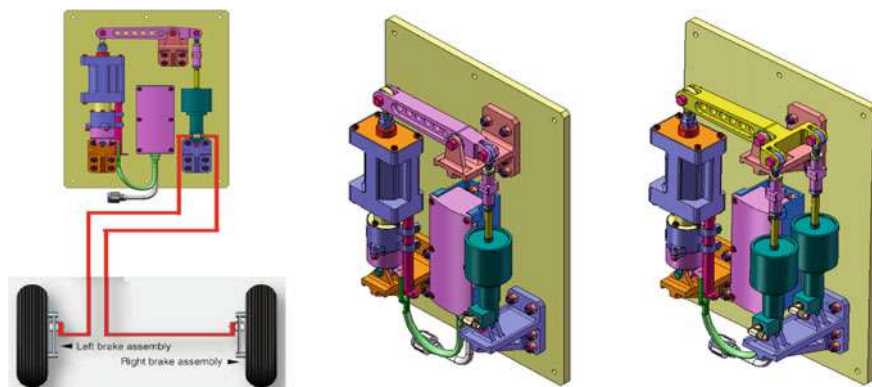


Fig. 29.2 Wheel brake actuation system with single master cylinder configuration and twin master cylinder configuration

The various standard components like brake, master cylinder, and tire details are listed below.

• Master cylinder specifications: – Model: 10–35 (Parker–Cleveland wheels and brakes, USA) – Built-in Reservoir, Cast Aluminum Construction. Bore: 5/8". Stroke: 1.2" Outlet: 1/8" NPT	• Wheel brake: – Disk brake: 5" – Wheel No. 50-1A – Disk thickness: 0.19" – Caliper No. 30-1 – Piston dia: 1 × 1.5"
• Tire: 5.00 × 5-4 TYPE III Tire	

29.2.2 Distribution of Brake Force

To meet the pilot’s demand for deceleration of aircraft, the total braking force is dispersed between the EMLA, master cylinder and the brake caliper. In the existing brake actuation scheme, the braking force dispersion approach is in parallel with a fixed ratio between the EMLA and braking master cylinders. This approach is able to implement in conventional braking arrangement, whereas in the new braking force division methodology with a single master cylinder and twin master cylinder schemes operated by a single EMLA with a specific crank ratio was proposed and the actuator output force was preferentially used. The braking forces variation with rate is shown in Fig. 29.3.

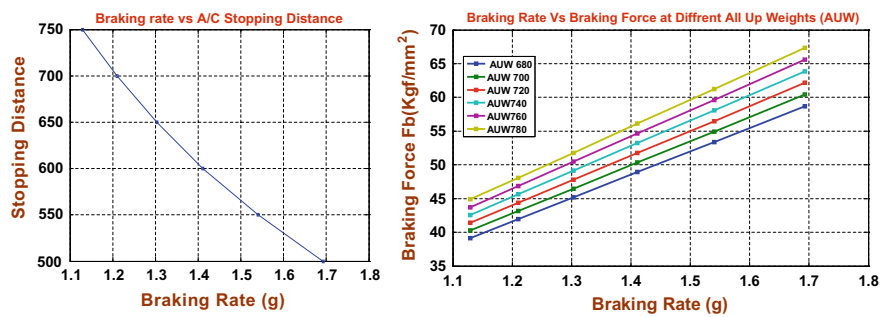


Fig. 29.3 Rate of change of braking force with respect to braking rate at different all-up weights (AUW)

Table 29.1 Estimation of actuator forces for different AUW configurations

S. no.	Parameters	All-up weight (AUW) of aircraft, kg						
		650	680	700	720	740	760	780
1	Kinetic energy of brake, kg f m	56,111	58,704	60,431	62,157	63,884	65,610	67,337
2	Brake torque, kg f m	8.08	8.45	8.70	8.95	9.20	9.45	9.70
3	Pressure on brake piston, kg/mm ²	0.172	0.180	0.185	0.191	0.196	0.201	0.207
4	Force exerted on each master cylinder, (F_m) kg f	34.06	35.64	36.68	37.73	38.78	39.83	40.88
5	Actuator force, (F_a) kg f	11.35	11.88	12.23	12.58	12.93	13.28	13.63

29.2.2.1 Estimation of Actuation Parameters

The requirements of braking forces are estimated in detail for one particular set of input conditions. The various input parameters considered are listed in Table 29.1. The other variables are derived considering different plausible combinations of these parameters.

$$\text{Weight of Aircraft (AUW)} = 650 - 780 \text{ kg}$$

$$\text{Vehicle Speed} = 80 \text{ knots}$$

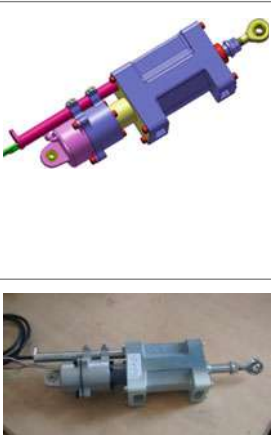
$$\text{Stopping Distance} = 500 \text{ m}$$

29.2.2.2 Details of the EMLA Used for the Brake Actuation System

Based on the considerations of AUW and other design variables, it was estimated that an actuation force of 8.08–9.70 kg f was required to operate the brake actuation system with a fixed crank ratio. To this end, an EMLA driven by a brushless direct current (BLDC) motor and a linear variable differential transducer (LVDT) as a position feedback device was designed, developed, and realized which had a continuous load capacity of 400 N at a no-load rate of 50 mm/s. Table 29.2 lists the specifications of the actuator.

Table 29.2 Specifications of the EMLA for brake actuation

EML actuator specifications			
S. no.	Parameter	Units	Value
1	Travel (elec./mech.)	mm	$\pm 16.5/\pm 18$
2	Force (cont.)	N	400
	Force (peak.)	N	1200
3	Rate (no-load)	mm/s	>50
4	Bandwidth—(no-load) (-3 dB/ 90° Ph lag)	Hz @ mm	≥ 8 @ 2
5	Bandwidth—(load 400 N)	Hz mm	≥ 5 @ 2
6	Accuracy (under 400 N load)	mm	± 0.25
7	Free play	mm	0.02
8	Threshold	%FS	<0.15
9	Stiffness	N/mm	2000
10	Supply	V	28 ± 4
11	Command	V	± 10



29.2.2.3 Servo Loop Modeling

The actuator servo loop has classical three-loop controller structure. Outer is the position loop, next inner loop is the velocity loop, which is incorporated to provide sufficient damping and innermost loop is the current loop to provide assured performance with varying operating parameters such as the drop across the MOSFETs due to ‘ON’ resistance in the H-bridge, variations in the motor and lead wire resistance from motor to motor and variation of resistance w.r.t. temperature, all of which will lead to the loss of stiffness. Block diagram of the servo loop is shown in Fig. 29.4.

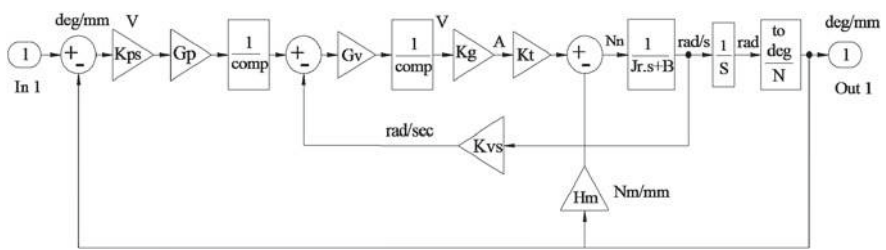


Fig. 29.4 Servo loop modeling of linear actuator

29.2.2.4 Estimation of the Position Gain

The stiffness value for the system is estimated at 2000 N/mm. Under static conditions, minimum gain required will be given by,

$$\text{Stiffness} = K_{ps} G_p K_g K_t \left(\frac{2\pi\eta}{p} \right)$$

where G_p is the position gain (V/V) which need to find from the stiffness as the requirement, K_{ps} is the scale factor (V/mm) for the given command input range (± 10 V), lead of the roller screw/ball screw p and K_g is the current gain and K_t as the motor constant (N m/A).

29.2.2.5 Position and Velocity Gain Estimation

The velocity loop will take the voltage command corresponding to the velocity required in terms of V (rad/s). The overall transfer function for the position loop can be obtained as,

$$\text{TF (Position)} = \frac{0.16 \left(\frac{K_{ps} G_p G_v K_g K_t}{J} \right)}{s^2 + s \left(\frac{B + K_{vs} G_v K_g K_t}{J} \right) + 0.16 \left(\frac{K_{ps} G_p G_v K_g K_t}{J} \right)}$$

where position scale factor (K_{ps}) as 0.5, BLDC motor Inertia ($J = 1.485\text{E}-5$ kg-m²) as per catalogue, current gain K_g A/V, position gain and velocity gain G_p and G_v and viscous friction B (1.41e-6 N m/rad/s).

29.2.3 Actuator Response and Load Test

See Fig. 29.5 and Tables 29.3, 29.4.

29.3 Study of Elements and Simulation

29.3.1 Simulation Studies Layout

To conduct modal studies and interpret the characteristics of the system, a simulation platform was built in Simulink as shown in Fig. 29.6. This consisted of four

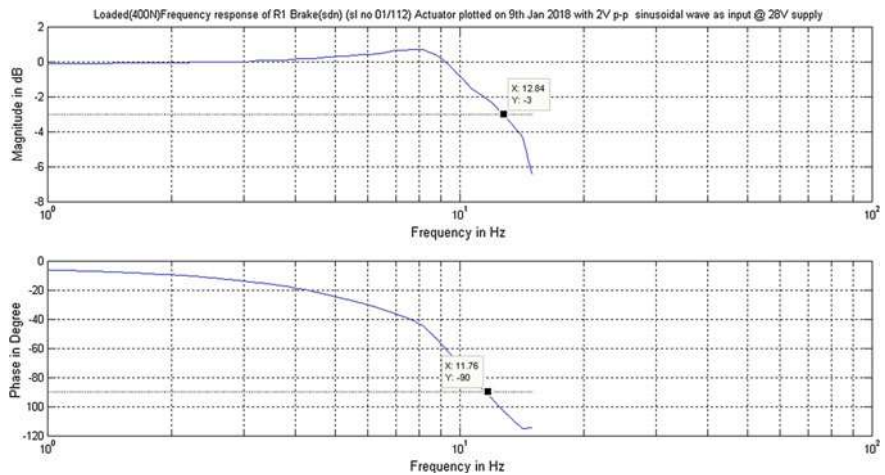


Fig. 29.5 Bandwidth: 2 V_{pp} @ 28 V with 400 N continuous load

Table 29.3 Step response of EMLA

S. no.	Input command	Test result			
		Rate extent	Rate retract	Overshoot	Undershoot
1	2 Vpp/0.5 Hz (10% of full command)	42 V/s	42 V/s	0.12 V	0.07 V
		69.3 mm/s	69.3 mm/s	0.2 mm	0.12 mm
2	5 Vpp/0.5 Hz (25% of full command)	55 V/s	55 V/s	0.15 V	0.15 V
		90.75 mm/s	90.75 mm/s	0.25 mm	0.25 mm

Table 29.4 Load response 2V_{pp} @ 28 V with 400 N continuous load of EMLA

S. no.	Input command	Test result
1	2 Vpp sine @ 28 V supply	−3 dB @ 13.96 Hz/90° @ 13.35 Hz
2	2 Vpp sine @ 24 V supply	−3 dB @ 12.02 Hz/90° @ 12.48 Hz
3	2 Vpp sine @ 32 V supply	−3 dB @ 15.98 Hz/90° @ 13.96 Hz
4	2 Vpp sine @ 28 V supply with 400 N load	−3 dB @ 14.39 Hz/90° @ 13.25 Hz

subsystems: the pilot’s control system, flight control computer (FCC) system, EM linear actuator braking system having controller get inputs from FCC system and the aircraft dynamic model that accepts the pressure input to the brake caliper.

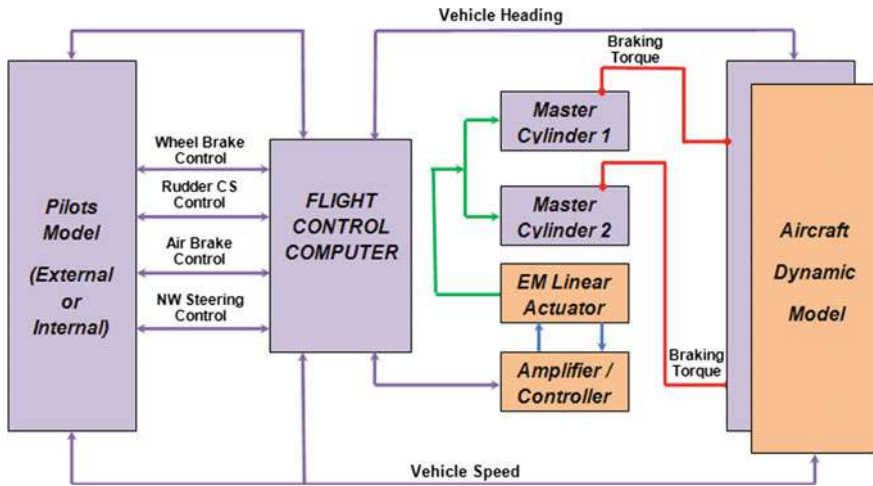


Fig. 29.6 Layout studies of braking scheme with pilot's model, aircraft model, and flight control computer

29.3.2 Modeling of Aircraft Force for Wheel Brake

An aircraft model with movement in longitudinal and axial load conversions was arrived along with drag force due to driving resistances in different runways, surfaces and environmental conditions. The force experienced by the aircraft mass m moving with velocity v along the longitudinal axis experiencing the drag of F_a due to aerodynamic force drag and the main wheels tire force F_l can be represented as follows:

$$mv = F_l - F_a$$

The energy efficiency of the aircraft majorly depends upon the overcoming the drag due to aerodynamics and calculated by mass density of air ρ , reference area A and coefficient of drag C_d as follows:

$$F_a = \frac{1}{2} C_d A \rho v^2$$

29.3.3 Modeling of Tire Component

All the external loads subjected to the aircraft other than aerodynamic loads are the contact surface between the ground and the tires. The tire model represented in the

study by using magic formula having tire normal load as F_{zi} , longitudinal load as F_{xi} , slip rate of tire s_i and the constants B, D derived from the derivations and another constant f as rolling resistance coefficient which not dependent upon the speed are represented below:

$$F_i = F_{xi} - f F_{zi}$$

$$F_{xi} = D \sin(C \tan^{-1} \{ B s_i - E [B s_i - \tan^{-1}(B s_i)] \}) F_{zi}$$

The aircraft normal load affects the tire force and transferred to the landing gear and changes as acceleration changes with cg height of h_g , main landing gear to cg length L_i wheel base of L and gravitational constant g can be shown by the equation:

$$F_Z = \frac{mg}{L} \left(L_i - \frac{\dot{v} h_g}{g} \right);$$

29.3.4 Wheel Modeling

In the wheel brake system of aircraft, the main wheel experiences the force generated between ground and the tire and torque generated by brake caliper. The force equilibrium on the main landing gear wheels can be formed with moment of inertia I , frictional coefficient μ , braking force F_B , and angular acceleration $\dot{\omega}$ and the braking torque applied on each wheel can be represented as:

$$I \dot{\omega} = 2 \mu F_B R \quad T_{Bi} = P_i A_i C_{BF} R_{Ei}$$

The brake torque applied on each wheel T_{Bi} with brake disk effective radius R_{Ei} produced by wheel brake caliper cylinder fluid pressure P_i having cross-sectional area A_i and disk brake factor C_{BF} .

29.3.5 Modeling of Crank and Master Cylinder

In this study, a single-cylinder operation having inbuilt reservoir connected to the actuator output through a crank with fixed crank ratio. This assembly was modeled to estimate the force applied by the EM linear actuator, and the pressure in the master cylinder can represent with crank ratio R_p converted into force applied on cylinder F_p ; pressure generated on brake fluid into the cylinder with stroke S_{MC} , cross-sectional area A_{MC} operated with the preload L_{MC} and the stiffness of K_{MC} with the below equation:

$$F_P = \frac{1}{R_p} (P_{MC} A_{MC} + S_{MC} K_{MC} + L_{MC})$$

29.3.6 Simulation of Wheel Brake Actuation Parameters

EM linear actuator controller unit regulates the pressure in each wheel cylinder by receiving the signals from the pilot to retain stable brake rate. Aircraft pilot's input model and the aircraft model are the key factor of this EM linear actuator force output regulation process by observing the deceleration with continuous and ramp response of the EM linear actuator-based braking system.

Sensitivity study is done for various brake parameters like kinetic energy, braking rate, braking force, time requirement for aircraft to stop, braking torque, and with change in aircraft weight and stopping distances to predict the aircraft performance. The smooth changes in braking rate or deceleration (g) during the wheel braking operation gives the confidence that the pilot will not have significant discomfort. It was observed that the EMLA-based braking system gives sufficient braking force with a rate of 1.7 g. Figure 29.7 shows the simulated forces on each wheel and the calculated total braking force on the vehicle, and Fig. 29.8 shows the variation master cylinder pressure in turn the braking torque to stop the aircraft with specified time limit at different all-up weights, gives the different braking rates at constant braking force distribution, and was the same as designed in graphs.

The master cylinder piston travel is a parameter that can simulate the characteristics of pressure–volume of the brake calipers depends on actuator output shaft connected by crank with a fixed ratio to the master cylinder piston. By maintaining such crank ratio, the master cylinder maintains a smooth travel is regulating pressure in each brake. Figure 29.9 shows the variation of actuator force at different crank

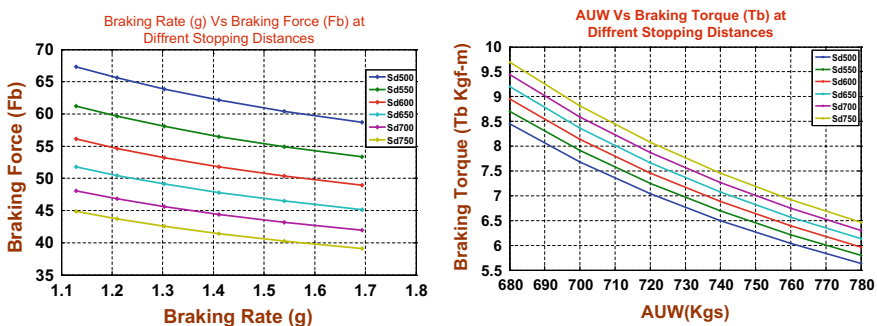


Fig. 29.7 Change in braking rate and braking force at different AUW and stopping distances

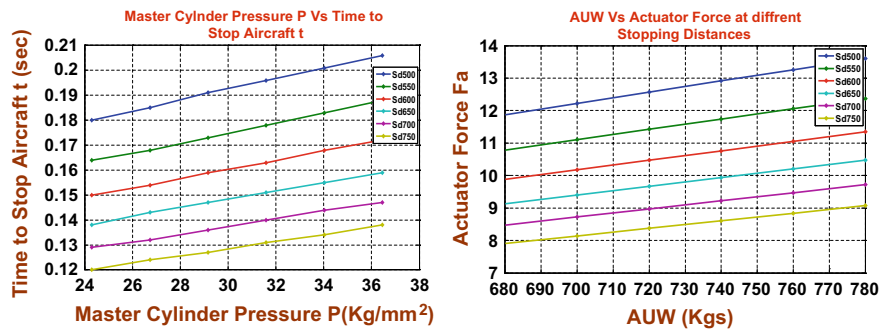
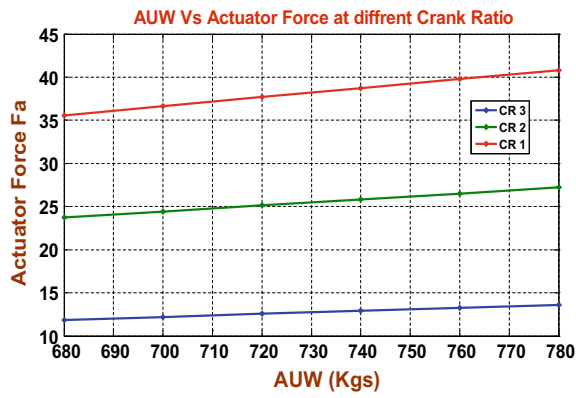


Fig. 29.8 Variation of master cylinder pressure, actuator force at different stopping distances and AUV

Fig. 29.9 Changes in actuator force at different crank ratio and AUV



ratios and AUV. The aircraft weight is expected to change due to consumption of fuel during flight, for this weight range requirement variation in braking torque is about 18%.

29.4 Results and Discussions

The wheel brake actuation system and brake assemblies are integrated in the aircraft (UAV) as shown in Fig. 29.10. The actuation mechanism is actuated with spatial-type test equipment (STTE), the master cylinder which in turn pressurized the wheel brake. The various parameters like actuator stroke, master cylinder stroke, and power supply current and measured brake torque are provided in Table 29.5. The measured brake force meets the maximum brake torque requirement and can be finalized that the brake force generation through EMLA was matching with expectations.



Fig. 29.10 Wheel brake actuation system assembly on aircraft and brake torque measurement test

Table 29.5 Brake torque performance with input of actuator and master cylinder displacement

S. no.	Command (V)	Actuator stroke (mm)	Current feedback (A)	Master cylinder piston displacement (mm)		Average braking torque (kg f m)
				Port side	Starboard side	
1	−8	0	0.1	−0.01	−0.01	0.6
2	−7	1.62	0.1	−0.47	−0.50	0.6
3	−6	3.25	0.2	−0.98	−1.0	0.9
4	−5	4.88	0.3	−1.43	−1.49	1.2 (Brake initiation)
5	−4	6.49	0.4	−1.95	−1.95	2.4
6	−3	8.1	0.4	−2.33	−2.40	3.6
7	−2	9.71	0.5	−2.74	−2.83	4.8
8	−1	11.27	0.5	−3.16	−3.24	6.0
9	0	12.82	0.6	−3.56	−3.64	7.2
10	+1	14.4	0.7	−3.54	−4.03	8.4
11	+2	15.99	0.7	−4.33	−4.41	9.0
12	+3	17.56	0.8	−4.73	−4.73	10.2 (Full brake)

In the experimental study, the system was integrated into the aircraft and main wheels were externally loaded to simulate the actual braking action. The applied load was resisted by the brake actuation system consisting of the EMLA, master cylinders, and the controller. It was observed that a translational movement of 17.56 mm of the EMLA was required to achieve the required brake torque at a constant rate. The wheel brake torque per wheel is measured using a static test rig. The maximum brake torque was measured as 10.20 kg f m per wheel, and based on these test results, wheel brake actuation sequence of operation is implemented in the aircraft as shown in Table 29.5. The actuator command voltage limits are set to −8 to +3 V command input and +3 V command indicated full brake achieved.

29.5 Conclusion

In this paper, an EMLA-based wheel brake actuation system and actuator force control method as well as the brake force distribution configuration was proposed. Then, the system was modeled, and several basic tests were simulated to evaluate the performance of the EMLA feel simulation mechanism, and the brake force distribution strategy.

In this exercise, the brake torque requirements at the wheels were able to quantified in regards to AUW, stopping distance and touchdown speed which otherwise was solely the judgment of the pilot. The brake actuation system was simplified and made compact using twin master cylinder and single master cylinder configurations. It was observed through simulation as well as through test results that the new configurations meet all the requirements of brake actuation system satisfactorily and also no differential braking effect was observed at the wheels of the main landing gear.

Finally, simulations under the pilot's input model and aircraft model with different brake force distribution strategies were carried out and brake torque performance of the proposed system was evaluated by EMLA with fixed crank to operate master cylinder, brake system on UAV under different strategies of AUW, braking rates and stopping distances. The simulations and experimental results showed the feasibility, effectiveness, and implementation of the system proposed.

Acknowledgements The authors sincerely thanks to the Director, Group Director (GD), Divisional Head (DH) and Head, HRD of ADE for giving encouragement and permission to publish this technical paper.

The author gratefully acknowledge the assembly, testing, integration of actuation system, conducting experiments in laboratory and on aircraft by Shafeer Ahmed, Vijay Kumar, Kunjappa, Jerome Marshal, Anoj Jacob, Rajasekhar, Shiva raj, and R1 Project team.

Conflicts of Interest: The authors declare no conflict of interest.

Chapter 30

Design, Fabrication and Simulation of Micro-EDM Machined AISI 316 SS Micro-channel Heat Sink



H. S. Mali , Vivek Baghela and Siddhartha Kr. Singh

Abstract Micro-sized equipment used by many industries like automobile, robotics, micro-electronic mechanical systems, etc., is always prone to generation of heat. The elimination of generated heat from these systems is a difficult and costly affair. Though various approaches may be taken towards the elimination of heat from these systems, a micro-channel heat sink is one of the simplest and cheapest devices employed for the purpose. In the present work, a micro-channel heat sink, using a suitable low-cost material, i.e. AISI 316 SS, is fabricated through the micro-EDM process. Micro-EDM is a non-traditional machining process which allows high precision machining of many difficult to machine materials. Post fabrication, pressure drop and heat transfer in the heat sink are simulated using ANSYS Fluent®. Water, at varying flow rates, is taken as the working fluid, and the effect of surface roughness of fabricated channels is also incorporated in the simulation. Increase in flow rate is found to have a negative influence on heat sink performance as pressure drop increases in addition to decrease in temperature change between inlet and outlet.

Keywords Micro-EDM · Micro-channel · Surface roughness · Heat sink · Pressure drop

30.1 Introduction

In the present day advanced manufacturing context, the components produced must strictly focus on the miniaturized dimensional specifications, in order to obtain compact functional products. A micro-channel heat sink is a type of heat exchanger which is used in numerous applications like chemical processing plants, fuel cells, MEMS and electronic devices. Micro-channel heat sinks are popularly applied for the transfer of heat from heat-producing equipment like IC chips, fuel cell, micro-engines [1]. The advantages of micro-channel heat sink include high heat transfer

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_30

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efficiency, compactness, easy to clean, etc. However, fabrication of micro-channels is a considerable task due to material properties and its compactness. There are many fabrication techniques for fabrication of micro-channels like LIGA, selective laser sintering, chemical etching, diffusion bonding and stereolithography. Apart from these processes, micro-electric discharge machining (micro-EDM) is also used for fabrication of micro-channels. Micro-EDM is non-traditional-type material removal process in which material is removed by the application of electric discharge between two electrodes. One of the electrodes acts as a cutting tool, and the other electrode is the workpiece. A suitable dielectric medium is applied between the two electrodes. And a suitable pulse generation circuit is used to charge the electrodes. As the electrons and ions begin to flow, a plasma channel is created between the electrodes. The flow of electrons and ions in the plasma channel between the tool and the workpiece causes intense localized heat generation leading to the removal of material from both the electrodes due to melting and vapourization. This process is extensively used for machining various hard to cut material.

Design of heat sink plays an essential role on the performance of heat sink and also on pressure drop and heat transfer. Many researchers worked on design of micro-channel heat exchanger. Kee et al. [2] designed and fabricated a counter flow heat exchanger and found that pressure drop is approximately inversely proportional to the cube of channel height. Heat exchanger was fabricated using alumina and CFD models were implemented in Fluent. Cao et al. [3] designed and tested multi plates micro-channel heat exchanger and investigated the effect of surface roughness on pressure drop. Deng et al. [4] designed and fabricated micro-channel heat exchanger for application of natural gas cooling. Sahar et al. [5] investigated on the design aspect and found that the aspect ratio should be less than 2 in order to achieve a low-pressure drop; also the hydraulic diameter has the most significant effect on heat transfer. Vinoth and Senthil Kumar [6] compared three different cross sections for heat transfer and found that the trapezoidal section shows higher heat transfer characteristics. Ashok Kumar et al. [7] studied about surface morphology of micro-machined surface via die sink EDM and showed that obtained surface roughness is in the range of 0.45–2.09 μm . Ramirez et al. [8] studied about surface finish during machining via die sink EDM and also proposed that surface finish is the function of current intensity.

Even though much work has been done in the field of micro-channel heat sinks and exchangers, as evident by the literature presented above, the authors could not find the literature investigating the design and fabrication of micro-channel heat sink on AISI 316 stainless steel and subsequent numerical simulation based on obtained surface texture to estimate the temperature and pressure characteristics along channel length. To fill this gap, current work is undertaken.

30.2 Heat Sink Fabrication

In the present work, for fabrication of micro-channels, the material chosen is AISI 316 stainless steel owing to its corrosion resistance properties along with its high heat conductivity. All these properties make AISI 316 a suitable material for a heat sink. AISI 316 SS is also used in many chemical industries, heat exchange applications and food processing plants. Composition and thermal properties of AISI 316 SS are presented in Tables 30.1 and 30.2, respectively.

To model the micro-channel heat sink, Autodesk Inventor software is used. The complete design of heat sink consists of three parts box, micro-channel plate and top plate. These three distinct parts are then assembled to obtain the final heat sink. For assembling the components, a steel reinforced epoxy is used so as to provide better thermal conductivity. The design and details of heat sink are shown in Fig. 30.1.

The dimensions of the outer box are 30 mm × 30 mm. In this outer box, a pocket is made using a CNC milling machine. The dimension of the milled pocket is 24 mm × 20 mm × 8 mm. The plate on which micro-channels are fabricated has dimensions 24 mm × 14 mm × 8 mm. There are ten micro-channels on the plate having cross section of 0.5 mm × 1 mm, and the channel length is 14 mm.

The machining is performed via micro-EDM process on MIKROTOOLS DT 110i Hybrid micro-EDM machine. For the purpose, Tungsten Carbide (WC) electrodes of diameter 500 μ are used. The experiments are performed in the presence of SUN EDM Oil 100 as the dielectric medium. The actual machining set-up is given in Fig. 30.2. Post machining, FILMETRICS profilometer is used to estimate the surface roughness of micro-channel walls.

30.3 Heat Sink Simulation

ANSYS FLUENT® 15.0 is used to estimate the pressure drop and temperature distribution in micro-channels. Following assumptions are made to perform the simulations:

Table 30.1 Composition of AISI 316 stainless steel

Composition	S	C	Si	Mo	Ni	Cr	Mn	P	Fe
wt%	≤0.03	≤0.03	≤1	2–3	10–14	16–18.5	≤2	≤0.045	Bal.

Table 30.2 Thermal properties of AISI 316 stainless steel

Grade	Density (g m/cm ³)	Thermal conductivity (w/m k)	Specific heat (K J/kg K)	Melting range (°C)	Electrical resistivity (Ωm)
316	7.87–8.07	13–17	0.4521 at 0 °C	1371–1400	0.74×10^{-6}

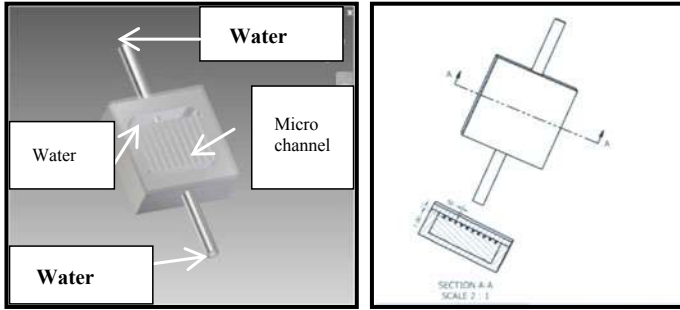


Fig. 30.1 Model of micro-channel heat sink assembly

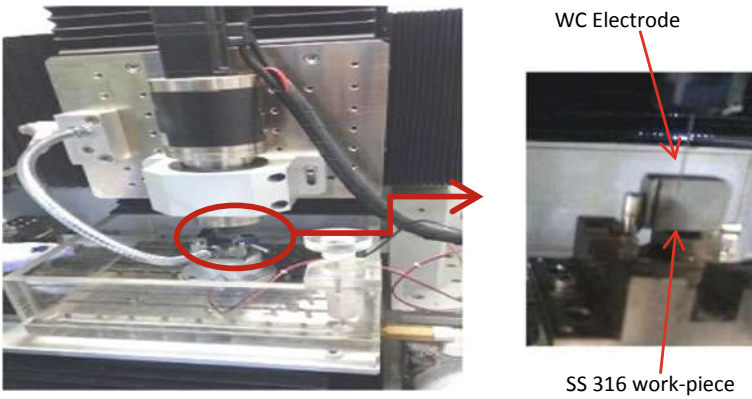


Fig. 30.2 Hybrid mikrotool DT-110i set-up

- The fluid is assumed to be in steady state and incompressible.
- Gravity and radiation heat transfer are neglected.
- Thermo-physical properties of fluid and solid elements are assumed to be invariant with temperature.

Water is taken as the fluid medium. The water flows through the inlet pipe and fills the small reservoir provided at the inlet. This reservoir is provided so that the velocity and pressure at all micro-channel's inlet are uniformly distributed. Further, all the walls other than bottom of micro-channel plate and box are insulated. This ensures that heat can only flow in one direction. Lower surface of micro-channel plate is maintained at 473 K corresponding to the operating conditions.

Figure 30.3 shows the measured value of channel surface roughness. The average surface roughness (R_a) of all the three walls is found to be $34.798 \mu\text{m}$. This value is incorporated in the model and standard $k-\epsilon$ turbulence model is used to perform simulation and estimate the flow characteristics. The rate of fluid flow, i.e. water, is varied from 1 LPH to 1.8 LPH to ascertain the effect of variation in flow rate on

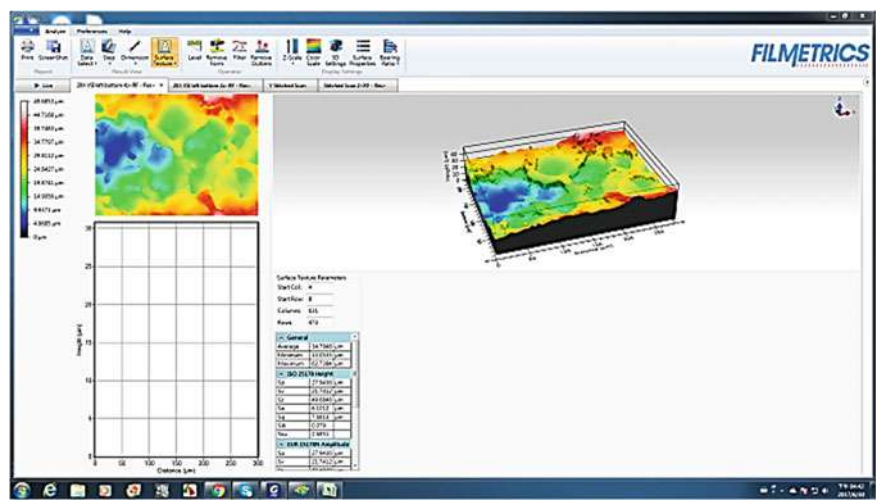


Fig. 30.3 Micro-channel surface roughness

Table 30.3 Boundary condition of water at the inlet of micro-channels

Boundary condition	Value
Inlet pressure (gauge)	10,000 Pa
Water flow rate (LPH)	1–1.8
Inlet temperature of water	300 K

heat transfer and pressure drop. The boundary conditions at the inlet are shown in Table 30.3.

30.4 Results and Discussion

Based on the results obtained, the pressure drop and temperature change in micro-channel heat sink at different flow rates are shown in Table 30.4.

Table 30.4 Temperature and pressure change obtained in micro-channels

Water flow rate (LPH)	ΔP (pascal)	ΔT (K)
1	112	4.5
1.2	137	3.8
1.4	163	3.1
1.6	192	2.9
1.8	227	2.2

Fig. 30.4 Pressure drop variation with flow rate

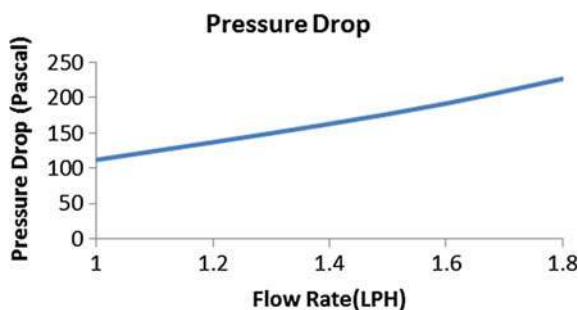
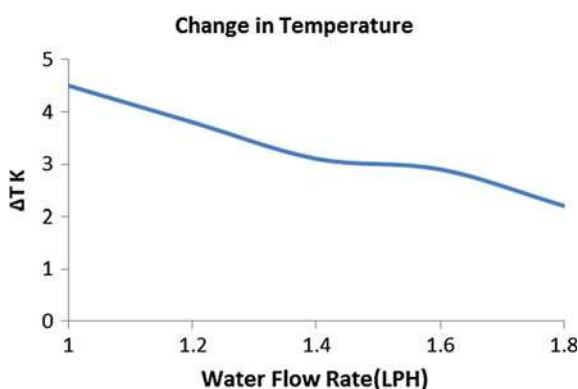


Fig. 30.5 Temperature change with flow rate variation



Figures 30.4 and 30.5 represent the changes in pressure drop and temperature in micro-channels as the water flow rate varies from 1 to 1.8 LPH. As the flow rate increases, the velocity of water also increases. The increase in velocity of fluid leads to a rise in the Reynolds number. As a consequence of the increase in Reynolds number, an overall increase in pressure drop is observed. Also due to the high velocity of water in channels, less time is available for removal of heat from channel surface. As less time is available for exchange of energy, the change in temperature is also observed to be small. Figure 30.6 shows the distribution of temperature in micro-channel plate in micro-channel heat exchanger in both solid domain and fluid domain at water flow rate of 1.4 LPH. As shown in Fig. 30.6, at the bottom surface of plate, the maintained temperature is 473 K. Figure 30.6 also shows the conduction of heat to the top surface and the micro-channel fluid domain.

Figure 30.7 shows the distribution of pressure in micro-channels for the water flow rate 1.2 LPH. The reduction in pressure as water flows from the inlet to the outlet of the micro-channels can be clearly observed from Fig. 30.7. One can also observe the variation of pressure drop at different position of micro-channels. The surface roughness of micro-channels creates an obstruction in the flow of fluid. This obstruction causes the fluid to become more turbulent. This turbulence in the fluid flow leads to a considerable amount of pressure drop [9]. Figure 30.8 shows the change in pressure along the length of micro-channels, from inlet to outlet, at 1 LPH

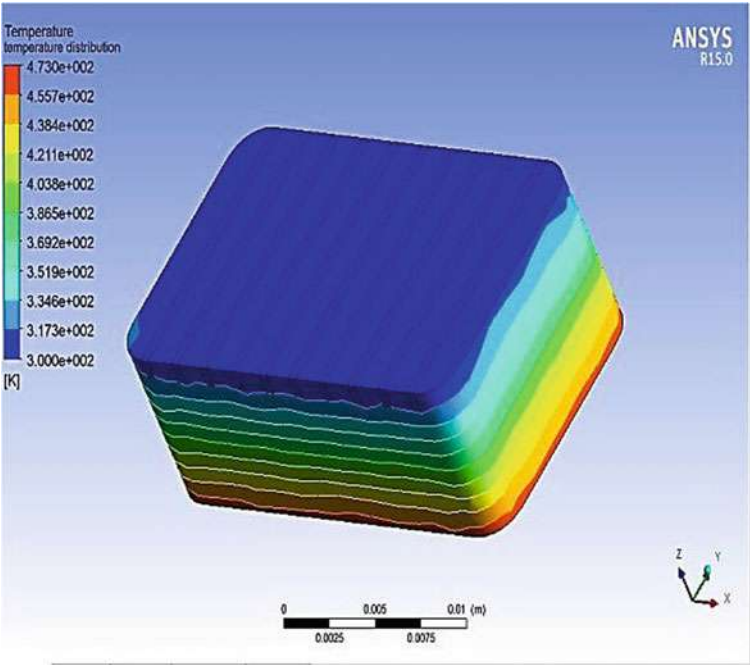


Fig. 30.6 Temperature distribution in micro-channel plate

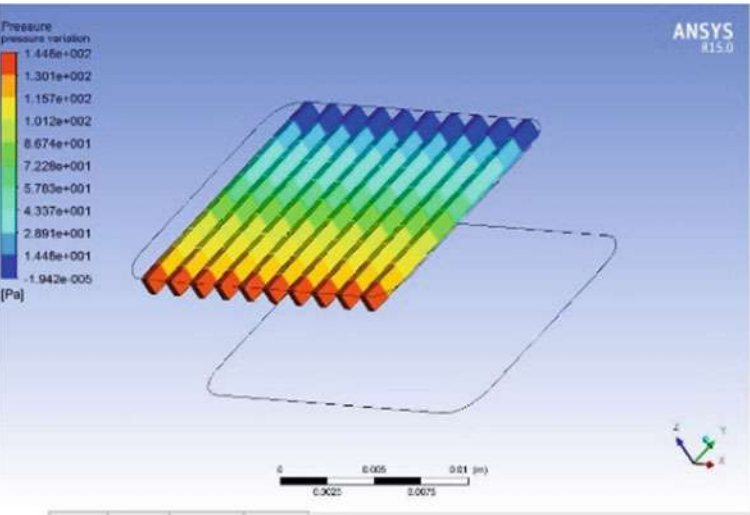


Fig. 30.7 Pressure variation in micro-channels

flow rate. By the plot of points, the range of pressure in different channels at each position can be clearly obtained. It is evident from the figure that as water passes through the channel, its pressure gradually decreases. Less variation in pressure across different channels along the length can also be observed from Fig. 30.8. Figure 30.9 shows the temperature variation of water along the length of micro-channel for 1 LPH flow rate. One can observe from Fig. 30.9 that the temperature of water increases as it passes through the channel. For flow rate 1 LPH, a total gain of 4 K temperature is obtained. For higher values of the flow rate, the change in temperature of the fluid decreases. Figure 30.10 is a vector representation of velocity along the channel length at water flow rate of 1.2 LPH. It can be observed from Fig. 30.10 that at entry, the velocity of fluid is high; thereafter the velocity of fluid slightly decreases in mid-channel. But towards the end of the micro-channels, the velocity of fluid can again be observed to increase. This increase in velocity of fluid can be attributed to pressure drop along the length of the micro-channels. Also, at the mid-section parallel to the length of micro-channels, velocity of water is high than at the boundary of channels. This higher velocity of fluid at the mid-section can be attributed to the formation of boundary layer.

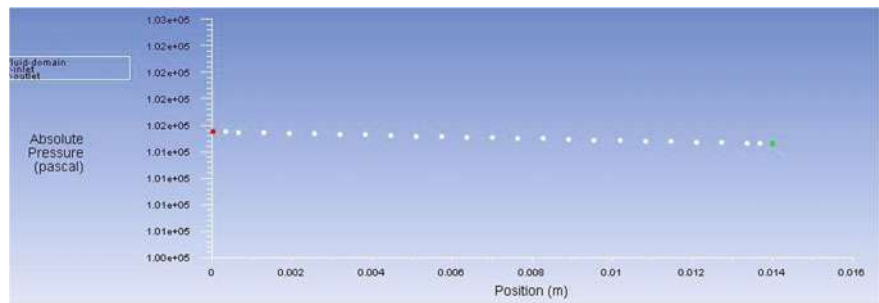


Fig. 30.8 Pressure variation of water in micro-channel

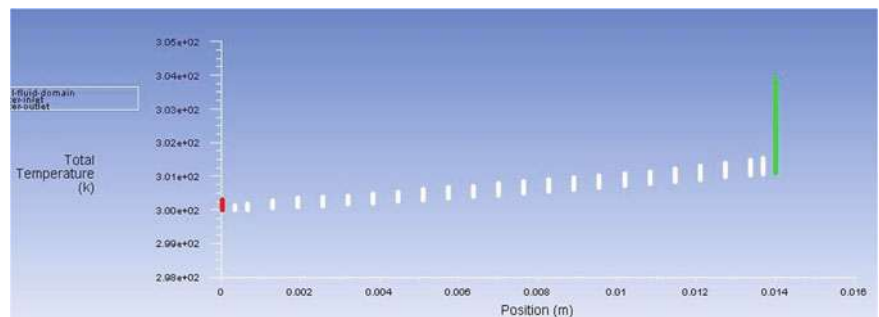


Fig. 30.9 Temperature variation of water in micro-channel

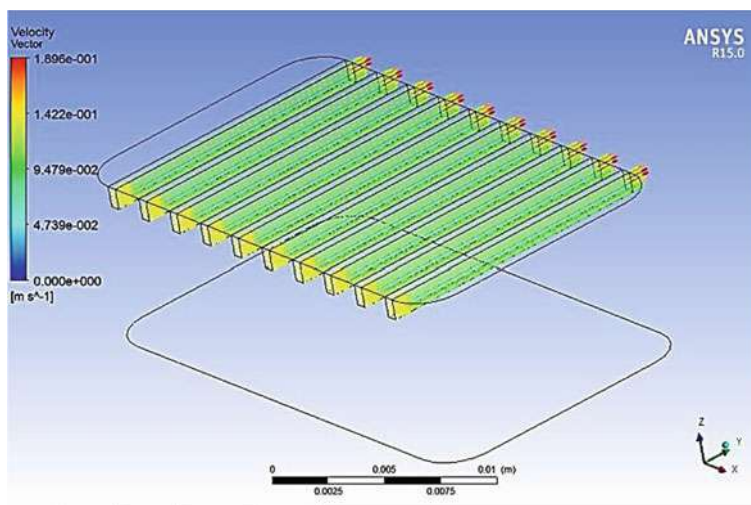


Fig. 30.10 Velocity vector representation in micro-channels

30.5 Conclusions

In this work, a micro-channel heat sink using a low-cost, non-corrosive material, AISI 316 SS, was modelled and fabricated. The fabrication of the micro-channels was performed using micro-EDM process. Surface roughness data of the fabricated heat sink was taken, and this data was used to simulate pressure drop and temperature change within the channels. Based on the performed work, following conclusions may be drawn:

- Pressure drop increases as the flow rate of fluid increases in micro-channels.
- Lesser temperature change is obtained as flow rate of fluid increases.
- Maximum change in water temperature is obtained at the flow rate of 1 LPH and is 4 K.
- Velocity of fluid is found to increase as it passes through the micro-channel allowing less time for heat exchange.
- Increase in velocity of the fluid can be observed towards the exit of the micro-channels due to pressure loss in the channels.

Based on the simulation results, it can be advised to keep flow rate of the fluid on the lower side when using a micro-channel heat sink on a heat-dissipating device for better heat elimination. Further, a reduction in surface roughness of the walls of the micro-channels can have a positive impact as it will reduce turbulence in the flow of fluid which can further lead to a decrease in the pressure loss across the micro-channel heat sink. As a consequence, we may need lower power to operate the heat sink system. In this regard, finishing the walls of the micro-channels is recommended.

Acknowledgements Authors would like to acknowledge the financial support provided by DST SERB through the grant EMR/2016/003372.

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Chapter 31

Geometrical Modeling and Performance Analysis of Textile Composites Using Python Scripted Software Platforms



Pragati Priyanka , H. S. Mali and Anurag Dixit

Abstract Textile-reinforced composites offer enhanced strength to weight ratio, bidirectional strength and impact resistance available with multiple weaving patterns and orientations as three-dimensional (3D) orthogonal, angle interlock, two-dimensional (2D) fabrics, etc. To ensure the applicability of these composites, mechanical characterization is performed prior to use. This study aims at exploring all the geometrical modeling utilities including yarn geometry, weaving patterns, binder yarn details and interpolations between yarn nodes, available with the modeling tool and their characterization using finite element (FE) post-processor tools by assigning boundary, loading and constraint conditions, both Python encrypted tools. Textile fabrics are modeled as representative element volume (REV)/unit cell (using Graphical User Interface, Python script or C++ API functions), exported to Python encrypted FE tool for characterization. Further, REV is repeated in desired orientations to replicate the overall fabric characteristics.

Keywords Fabric geometrical modeling · Unit cell · Finite element analysis

31.1 Introduction

Textile-reinforced composites offer enhanced applicability in multiple domains with high strength and lower weight. Characterization of these materials is based on experimental testing and virtual simulations of practical conditions by producing realistic models of the textile composites [1]. In virtual modeling by using various

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_31

computational tools as TexGen [2], TexComp and WiseTex, the major attention is to be focused on the realistic topology by using various imaging techniques, for example, micro X-ray computed tomography of the textile structure, orientation and the interaction between yarns in the fabrics.

The performance of textile composite at macro-level is highly governed by the modeling accuracy and simulation mechanisms applied at the lower levels in the hierarchy system, i.e., meso-level and micro-level [3]. Accurate modeling approach using virtual tools for textiles needs to address the following stages [4]:

1. Structural and mechanical parameters detailing of the fabric for both woven and non-woven. Further description of the yarn geometrical parameters and structural details in mathematical equation forms.
2. Conversion of these equations into numerical algorithms using core part of the virtual tool.
3. Generation of the textile geometrical model by using the output from algorithms.
4. Display of the generated textile model using Graphical User Interphase (GUI) and visualization using multiple rendering options.

TexGen software, developed at the University of Nottingham is a geometrical modeling tool, acts as a simulation pre-processor for characterization of textiles. TexGen development found its traces back from 1998 [5], redesigned to version 3 with more modules, third-party libraries addition and platform-independent feature by Sherburn in 2006 and released TexGen as an open-source software available at the Web site of SourceForge [2]. TexGen tool processing can be taken as a cluster of several modules (Fig. 31.1) databases named as.

31.1.1 Core Module

This module contains three libraries named Triangle, TinyXML and HXA7241 Octree for textile geometrical modeling. TexGen core contains the separate class to model a textile (CTextile) and to display the output messages (CLogger) and classified as shown in Table 31.1.

31.1.2 Render Module

The module facilitates multiple ways for model visualization in the three-dimensional space of computer graphics and image processing using visualization toolkit as showing yarn nodes, volume, fiber orientation, domain box, X-ray, etc., as in Fig. 31.1a.

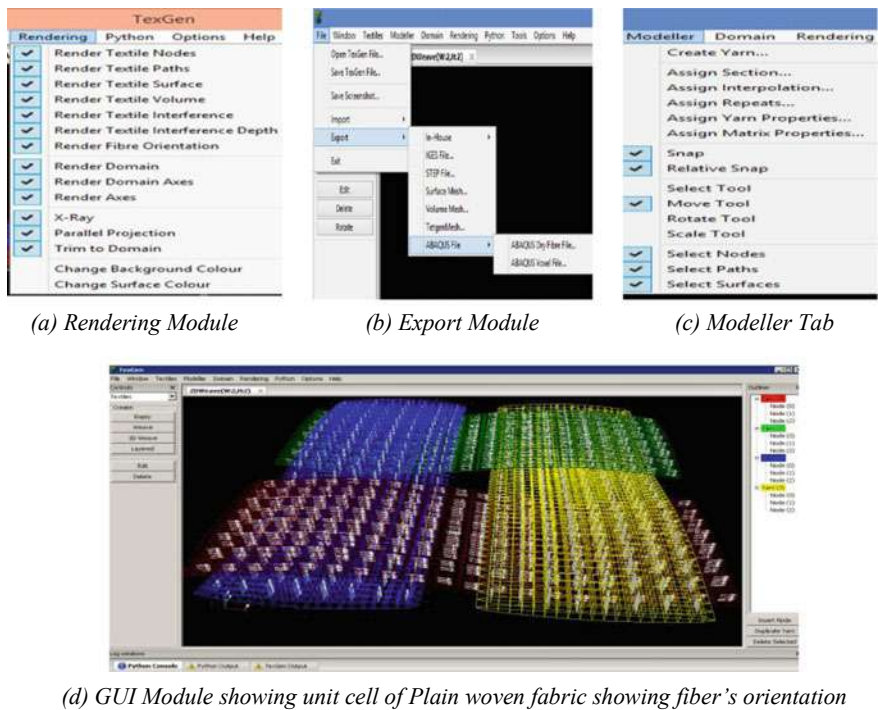


Fig. 31.1 TexGen modules

31.1.3 Graphical User Interface (GUI) Module

GUI module (Fig. 31.1d) in TexGen facilitates the user to model the textile without being worried about the modeling theory and numerical codes involve behind modeling, this works on wxWidgets toolkit. GUI of TexGen offers textile modeling in form of yarn, 2D and 3D woven fabric using empty textile or weave wizard tab.

31.1.4 Export Module

This module (Fig. 31.1b) exports (.tg3) file in different file formats to enhance the software utility, as in Initial Graphics Exchange Format (IGES), Standard for Exchange of Product (STEP), surface mesh with .vtu, .stl file format, volume mesh with tetrahedral mesh (.vtu and .inp format) and Abaqus Input file with dry fiber volume mesh and voxel mesh in .inp file format by using OpenCascade [6] module.

Table 31.1.1 Core module class contents

CTexGen		CLogger						Logger GUI
CTextile		CTextileWeave						
CDomain		CYarn		Repeat []	CInterpolation		2D	3D
CSlave Nodes	CDomain Planes	CYarnSection			CInterpolation			
>Nodes edit	>Edit domain >Delete domain	Constant Sections	Section interpolate >Nodes >Position	>Bezier >Linear >Spline >Cubic	Append [XYZ]	>Plain >Twill >Satin	>Orthogonal >Angle Interlock >Layer to layer	

31.1.5 Python Interface (PI) Module

PI module facilitates interaction of TexGen tool with other Python encrypted tools either pre-processor or the post-processor enabled by Simplified Wrapper and Interface Generator (SWIG) open-source library [7] which provides interface between the Core, renderer and export modules by wrapping of the C++ function and classes. TexGen GUI facilitates to run or record python script through the wizard.

31.2 Textile Geometrical Modeling Using TexGen

TexGen modeling tool utilizes meso-scale modeling approach for unidirectional (non-woven), bidirectional (woven) and multi-directional (knitted, braided and orthogonal) fabrics by single-yarn modeling or unit cell modeling [8, 9]. Accurate description of the modeling parameters as yarn spacing, cross-section, structure, orientation, crimp behavior and mutual interactions is provided through GUI module or by running the Python script.

Unit cell of woven textile fabrics is modeled in certain weaving pattern as plain (Fig. 31.1d), twill, satin, orthogonal and angle interlock pattern by using Peirce model within the specified domain box. For modeling of non-woven unidirectional fabrics, single yarn is modeled with specified cross-section and path, further, it is repeated in desired dimensions. Following section briefs about textile modeling credentials.

31.2.1 Geometrical Modeling Parameters of Textile Yarn

Initially, the textile unit cell can be modeled by creating single yarn and then duplicating it at required position for unidirectional or by creating the weave by defining no of horizontal yarns (warp), no of vertical yarns (weft), spacing, width, fabric thickness and interlacement pattern.

Yarn Sections The cross section of the yarn at normal plane to fiber orientation direction which can be constant throughout the yarn or it can be varied at different nodes by assigning section at each node (Fig. 31.2) as ellipse, lenticular, power ellipse, hybrid and polygon-shaped sections.

$$S(t)x = a \cos(2\pi t), S(t)y = \begin{cases} b(\sin(2\pi t))^n & \text{for } 0 \leq t \leq 0.5 \\ -b(-\sin(2\pi t))^n & \text{for } 0.5 \leq t \leq 1 \end{cases} \quad (31.1)$$

Equation 31.1 with 2a and 2b major and minor axes, gives elliptical section with $n = 1$, for $n < 1$, power ellipse and for $n > 1$, lenticular section.

Yarn Path Yarn path is the centerline of the yarn in fiber orientation direction, defined as a mathematical interpolation function in terms of the polynomial spline

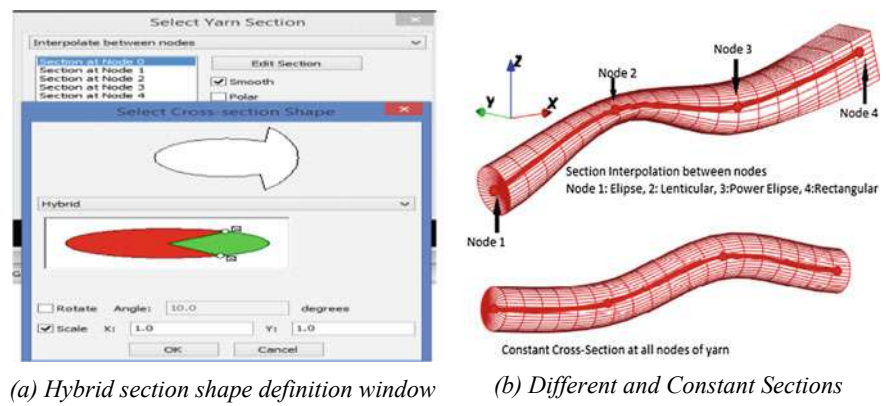


Fig. 31.2 Yarn section assignment at yarn nodes

with minimum C1 continuity order in between the yarn nodes. Figure 31.3 shows bezier, linear and natural cubic spline interpolated yarn path in between the yarn nodes. In polynomials, cubic spline plots the optimal interpolation which is represented mathematically by Eq. 31.2, where $C_i(x)$ is the cubic function.

$$P(x)=\begin{bmatrix} C_1(x), \text{ for } x_0 \leq x \leq x_1 \\ C_i(x), \text{ for } x_{i-1} \leq x \leq x_i \\ C_n(x), \text{ for } x_{n-1} \leq x \leq x_n \end{bmatrix} \tag{31.2}$$

Section Interpolation Between Yarn Nodes An interpolation function, either smooth or polar, is selected to specify how the cross-sectional shape changes in between the sections assigned at two nodes as shown in between the elliptical section at node 1 and lenticular section at node 2 in Fig. 31.2b.

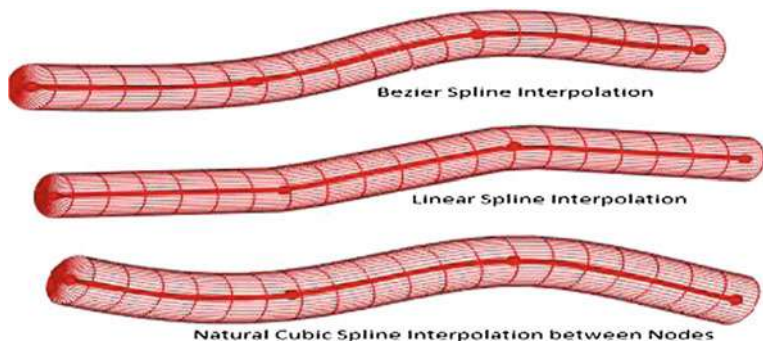


Fig. 31.3 Yarn path interpolation between nodes

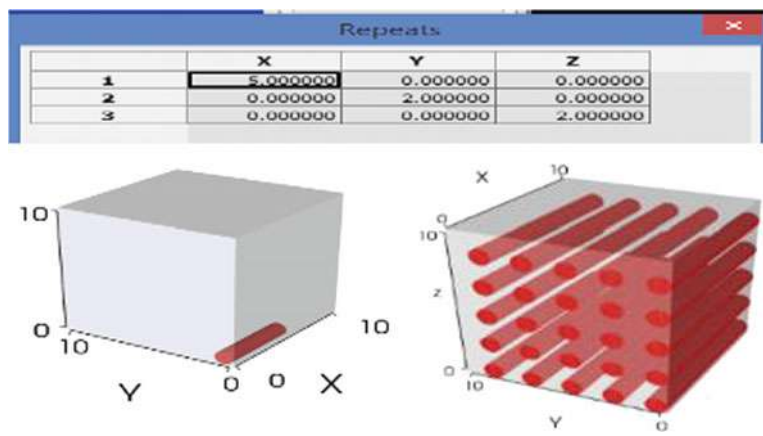


Fig. 31.4 Assign repeats

Domain Definition and Yarn/Unit Cell Repeats Domain box defines the space under consideration for textile modeling, defined by giving input of domain planes (Fig. 31.4) direction cosines and ratios. Domain box can contain only the unit cell or the three-dimensional layered fabric.

Yarn or unit cell generated can be repeated within the domain box created, in all three dimensions by using assign repeats through modeller tab by filling the coordinates of repeat points as shown for single yarn in Fig. 31.4.

31.2.2 Geometrical Modeling of Bi, Multi-directional Woven Textile

Bidirectional weaved textiles as plain (Python script for plain weave unit cell is written below), twill and satin weaves are modeled as unit cell by assigning the geometrical modeling parameters of yarns (defined in Sect. 31.2.1) and interlacing pattern to define the weave type. For multi-dimensional fabrics, binder or knitter yarn’s traveling pattern is defined, enhancing possibility to model the critical aspects of textile composites Fig. 31.5.

31.3 Material Properties Assignment in TexGen

In TexGen tool for textiles geometrical modeling, modeller tab (Fig. 31.1d) facilitates assignment of yarn material properties (volumetric density), geometrical properties (fiber diameter) and matrix material properties. Figure 31.6a shows properties assigned to a yarn with elliptical Section ($2a = 1$, $2b = 0.5$; Eq. 31.1), 10 mm

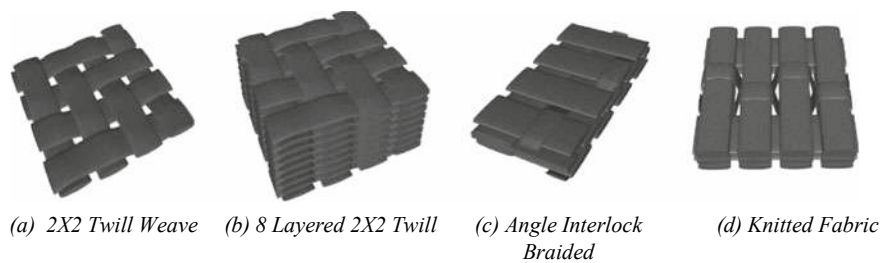


Fig. 31.5 Bidirectional and multi-dimensional fabric

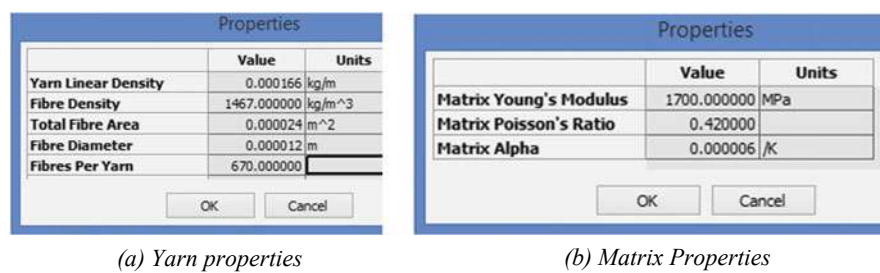


Fig. 31.6 Yarn and matrix material properties

length and Kevlar 49 material. Figure 31.6b shows polypropylene material properties assigned as the matrix. While exporting the textile geometrical model to any finite element tool, domain box represents matrix entity. After assigning yarn and matrix properties in modeller tab, yarn fiber volume fraction and domain volume fraction can be calculated using the tools tab.

31.4 Model Exported to FE Simulation Tool

The textile model developed in TexGen can be exported in multiple formats to further visualize the model performance under various working conditions as discussed in Sect. 31.1.4. Export from textile .tg3 file with matrix (Fig. 31.7c) or without matrix (Fig. 31.7a) to FE package Abaqus® is assisted by Python scripted interface and generating three files with .eld, .ori, .inp extension. The .ori and .eld files contain information regarding element orientation, fiber volume fraction and yarn details. The exported .inp file contains all information about created node coordinates, nodes involved in element type (C3D8R-three-dimensional eight-noded Hex element with reduced integration or C3D8-full integration, Fig. 31.7b), edges created, material properties, boundary and loading condition.

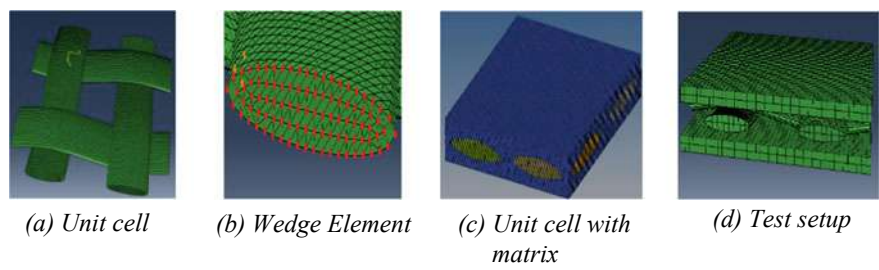
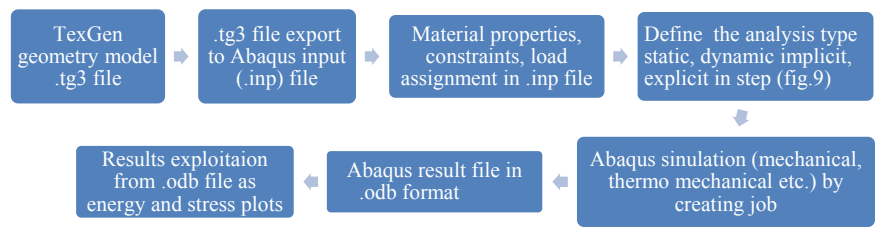


Fig. 31.7 Exported plain weave unit cell with different element types and application purposes

The input file with .inp extension imported in Abaqus can be further processed to simulate dry textile and pre-impregnated textiles behavior under different loadings as under forming, compaction or shear by providing accurate details of model constraints and material properties as input data.

In the following section, compaction (setup shown in Fig. 31.7d) simulation process of a plain weave unit cell is discussed in the flowchart below.



31.4.1 Fabric and Matrix Material Properties

In Abaqus, detailed material properties of matrix and fabric are assigned through the section formulation, and assignment of the material type (isotropic, orthotropic, and transversely isotropic). Further, the nine planar properties such as three each Poisson’s ratios, young’s and shear modulus are defined under the material definition tab.

31.4.2 Boundary and Loading Conditions

To replicate real conditions of the textile during FE simulations, the constraints and boundary condition (BC) applied must fall closer to actual application scenario. Tie

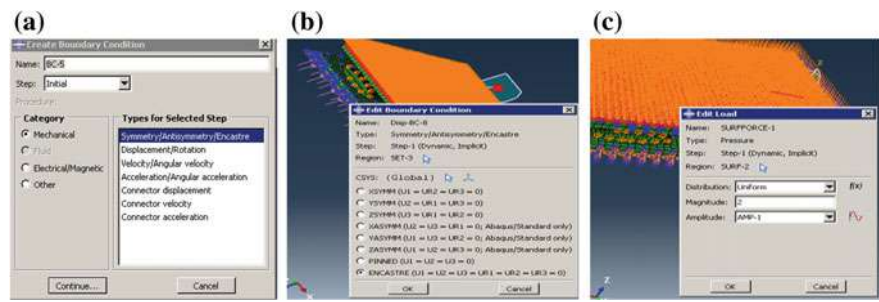


Fig. 31.8 a Boundary condition, b Encastre BC's and c Edit load

constraints and symmetry boundary conditions (Fig. 31.8a, b) cater need of repetition of unit cell in the fabric.

Loading can be applied in concentrated load, moments, pressure, etc., at any node, set of nodes. Figure 31.8c shows pressure applied to the plain weave unit cell constrained in between the two plates assigned with steel material properties.

31.4.3 Job Simulations and Results

After defining boundary condition, constraints and loading criteria in step applied, the jobs are assigned in analysis section of FE simulation. In Job tab, material nonlinearity can be assigned by providing the material-specific behavior in form of user-defined material file written in Fortran codes [10]. Job submission and completion gives stress distribution plots and energy plots as shown in Fig. 31.9a, b.

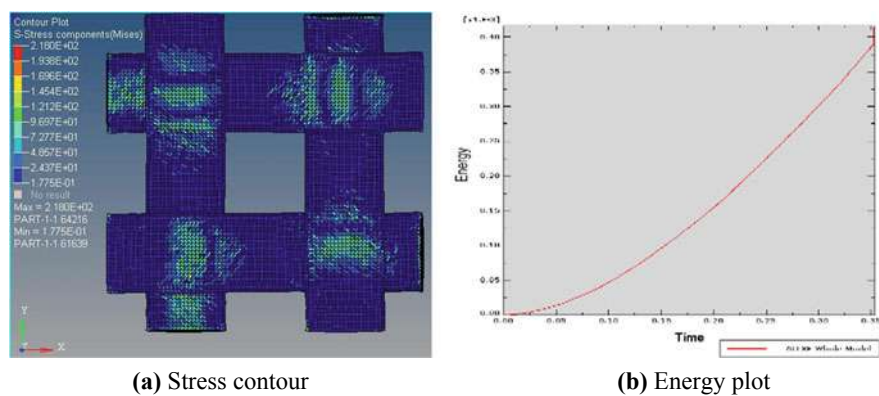


Fig. 31.9 Result visualization of plain weave unit cell compression

The results provide strain obtained in the fabric in terms of the pressure applied and time units, giving the compaction failure strength of the fabric. The section shows one of the capabilities of finite element utilization for TexGen generated textile fabric and composite mechanics characterization, able to predict mechanical, thermal behavior and fluid interaction.

31.5 Conclusions

The paper presents multiple modeling aspects of Python scripted textile geometrical modeling tool TexGen and further, the utility of these model. TexGen facilitates the fiber volume fraction calculation, weave pattern change and yarn structural properties accurate measurement.

The paper presents plain-woven unit cell generated in TexGen and exported to Abaqus input .inp file which is assigned with appropriate boundary conditions, constraints and loads to analyze the fabric behavior under compaction mode by pressure application with certain amplitude. Results obtained are useful in predicting the fabric strength under forming processes which applies compression forces.

Acknowledgements This part of work is financially supported by TEQIP-III.

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Chapter 32

Electromagnetic Transient-Thermal Modeling of High-Frequency Induction Welding of Mild Steel Plates



Ankan Mishra , Sukhomay Pal and Swarup Bag

Abstract High-frequency induction welding (HFIW) is a fast, energy-efficient process that is currently being used to weld pipes, primarily used in oil and gas lines. This work focusses on apprehending the process parameters for the feasibility of welding of flat mild steel plate with a fine refinement of weld structure using HFIW. This multi-physics problem is analyzed by three-dimensionally coupled electromagnetic transient-thermal finite element analysis to understand the electromagnetic heat transfer phenomena and melting. The simulations were done through EMS 2018 add-on package after developing an assembly model in SOLIDWORKS. The magnetic field intensity, magnetic flux density, temperature distribution, and time-temperature plot were obtained and the results are found to be at a good agreement with literature. The skin and proximity effect along with hysteresis losses are considered for the development of the model. Suggestions are made for a better working window with proper welding conditions.

Keywords High-frequency induction welding · Eddy current · Skin effect · Finite element modeling

32.1 Introduction

High-frequency induction welding (HFIW) is a deformation welding process belonging to the electrical resistance welding segment [1]. The induction welding process uses a high-frequency alternating current to generate eddy currents in the work parts leading to Joule's heating, aided by holding force to join them. Also, if the part is magnetic then additional heat through hysteresis loss is obtained until the Curie temperature of the material. Skin effect helps in the flow of current on the surface, and proximity effect attracts the work parts toward each other by an opposite magnetic field [2]. The schematic of the induction welding system is shown in Fig. 32.1. A

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_32

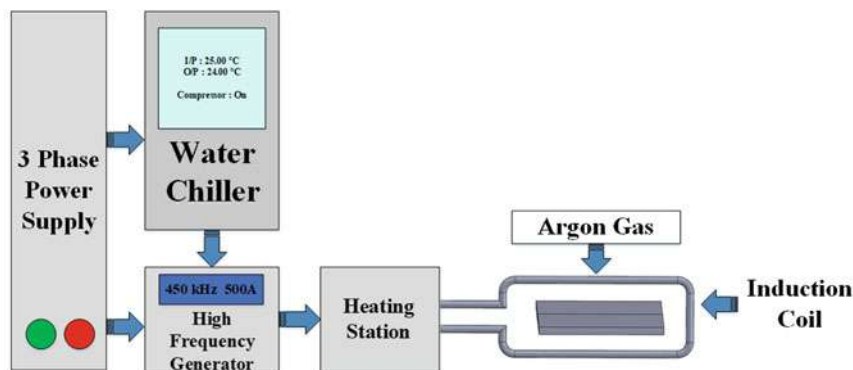


Fig. 32.1 Schematic of high-frequency induction welding setup

water chiller is provided with to cool the work coil and the high-frequency generator to generate high current at high frequency and argon gas shielding to protect from the oxidation of work-parts during the welding are the general components of the setup. The coil geometry is an important factor affecting the generation of heat at the required location. In this analysis, a single turn custom designed induction coil is designed for the HFIW of plates. Along with the coil geometry, a proper work holding fixture is also a requirement for proper holding of the flat plates and the application of the required pressure is also an important aspect. Holding force is not considered in this analysis.

There have been earlier attempts of simulating induction welding process for welding of pipes using coupled electromagnetic and Fourier heat conduction equation through EMS and commercial FEM package. The effect of impeder was found to be a good advancement in the induction welding area. The HAZ zone reduced with increase in frequency, and a better weld is produced [2]. Sequentially coupled electromagnetic transient-thermal analysis along with the motion of steel pipe with respect to coils was utilized to compute the heat treatment of pipe billet. It was found that the steady-state temperature profile generated around the weld has a double ellipsoidal shape. Unsymmetrical distribution of magnetic field in the steel pipe in the axial direction is the main source that the temperature profile is double ellipsoidal [3]. Two strategies were used based on a coupled electromagnetic and thermal problem for simulating induction welding of tubes. In the first strategy, final temperature distribution is directly calculated except temperature beyond the welding point, but with some issues like the hypothesis of constant resistivity results as the material property can be varied from point to point [4]. Time-harmonic electromagnetic and thermal analysis were done to investigate the influence of process parameters on the quality of the weld and to estimate system efficiency and required energy. The value of flux density that was found to be maximum at the V zone is 0.19T, and temperature at the same location was 1400 °C [5]. A thermoplastic matrix composite, polyphenylene sulfide reinforced with carbon fiber was used to fabricate a composite

panel using an induction welding process. The influence of process parameters on the weld like the power of the generator, the coupling distance between the coil and laminate, coil shape, laminate lay-up on the heating rate, and the heat flux distribution were taken into consideration while formulating finite element simulations. Effect of optimal process parameters on weld joints was found to be with good mechanical properties [6]. Micro-alloyed HSLA steels were welded and cyclic heat treated. The analysis of microstructures in the different weld zones after welding and post-heat treatment was analyzed. HSLA steels due to homogeneous microstructure have consistent and similar mechanical properties in all weld zones. The sand or hourglass shape of the weld zone was observed due to the paramagnetic property of steel, and at high temperature, the electrical conductivity reduces significantly leading to heating of the corners more than the central zone. Ferritic grain growth is restricted and due to precipitate hardening, it strengthens ferrite during post-heat treatment of the weld zone [7]. Similar sand or hourglass shape of weld zone was also identified during welding of the pipeline X60, X70, X85 grade steel pipes due to the variation of electrical conductivity across the thickness leading to the hourglass of heating pattern [8, 9]. With the use of a new magneto-dielectric impeder for HFIW pipe welding, a temperature of 1250 °C was achieved at a lower frequency of 350 kHz with better mechanical properties satisfying flattening and hydrostatic test [10]. This is evident that researchers have given less focus on the numerical analysis for the HFIW of plates. The process is proposed to be fast and efficient from the welding for plates rather than being kept limited to pipes. In the past literature, FE modeling of welding of steel pipes, rods, and shapes have been done. The present research work attempts for the FE modeling of high-frequency welding of plates using EMS 2018.

32.2 Theoretical Formulation

Numerical modeling of induction heating is a complex process. This is due to electromagnetic nature of the heat source. In the 1830s, Michael Faraday proposed the concept of electromagnetic induction. Further, Emil Lenz established Lenz's law and Joseph Henry with self-inductance, contributed to this until electromagnetic induction was summarized by Maxwell's equations for electromagnetism in four differential equations (Eqs. 32.1–32.4). Solving Maxwell's equations with isotropic electrical property and magnetic property (Eqs. 32.5–32.7) of the steel plate we obtain eddy current. This eddy current gives the heat through Joule's heating and the heat generated (Eq. 32.8) is taken as input in the modified Fourier equation (Eq. 32.11) for generating the temperature distribution of the weld plates in HFIW process [3–5]. Conduction, convection, and radiation heat loss throughout were also accounted for. Transient heat thermal analysis is considered to calculate the temperature profile. This analysis solves Maxwell's equations to obtain eddy current density or eddy current distribution, which is used again to obtain the heat flux (Eq. 32.8). The heat generated due to hysteresis loss (Eq. 32.9) is also added in the final thermal problem formulation. The Maxwell's equation in addition with isotropic electrical and

magnetic properties are given as follows:

$$\vec{\nabla} \cdot \vec{B} = 0 \quad (32.1)$$

$$\vec{\nabla} \cdot \vec{D} = \rho \quad (32.2)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (32.3)$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad (32.4)$$

$$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E} \quad (32.5)$$

$$\vec{B} = \mu_0 \mu_r \vec{H} \quad (32.6)$$

$$\vec{J} = \alpha \vec{E} \quad (32.7)$$

where $\vec{H}$, $\vec{J}$, $\vec{D}$, $\vec{E}$, $\vec{B}$, ε_0 , ε_r , μ_0 , μ_r , and α are magnetic field intensity (A/m), current density (A/m²), electric displacement field (C/m²), electric field intensity (V/m), magnetic flux density (T), dielectric constant of the material (F/m), relative dielectric constant, magnetic permeability (H/m), relative magnetic permeability, electrical conductivity (1/Ω m), respectively.

The electromagnetic portion will be simulated by solving Maxwell's equations (Eqs. 32.1–32.4), in addition with isotropic electrical and magnetic properties (Eqs. 32.5–32.7) by the EMS 2018 FE software.

The heat generated by induced eddy currents [1] and heat generated due to hysteresis loss are given below in Eqs. 32.8 and 32.9:

$$Q_e = \frac{1}{\sigma} |J_e|^2 \quad (32.8)$$

$$Q_h = k_h f t B_m^n \quad (32.9)$$

where k_h , B_m , n , f , t are material hysteresis constant, maximum flux density (Wb/m²), Steinmetz exponent (varies from 1.5–2.5; for iron 1.6), frequency (Hz), and time (s), respectively.

Due to the skin effect, the magnetic field is unable to penetrate deep and rather decreases exponentially from surface to deep [1]. Approximately 86% of the total power is available on the surface, till the skin depth given by Eq. 32.10. The Q_e is implemented as the heat input in the transient-thermal analysis in the following portion. The thermal problem then is defined by the following heat transfer equation (Eq. 32.11) [3]:

Table 32.1 Input for the numerical modeling

Input parameter	Value	
Coil current	400 A	
Frequency	333 kHz	
Air heat transfer coefficient	10 W/m ² K	
Water heat transfer coefficient	3000 W/m ² K	
<i>Electrical, magnetic, and thermal properties of steel</i>		
Conductivity	103 × 10 ⁵ mho/m	
Core loss function constants (for hysteresis loss)	<i>k</i>	7.3
	<i>α</i>	1.34
	<i>β</i>	2.11
Relative permeability	5000	
Latent heat of fusion	260 J/g	
Melting completion (liquidus)	1460 °C	
Melting onset (solidus)	1410 °C	
Specific heat capacity	470 J/kg K	
Thermal expansion	13 μm/m K	

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \quad (32.10)$$

$$\rho C \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left\{ k_x \left(\frac{\partial T}{\partial x} \right) \right\} + \frac{\partial}{\partial y} \left\{ k_y \left(\frac{\partial T}{\partial y} \right) \right\} + \frac{\partial}{\partial z} \left\{ k_z \left(\frac{\partial T}{\partial z} \right) \right\} + Q_e + Q_h \quad (32.11)$$

The finite element analysis of the process was done using a finite element simulated model on EMS 2018 add-on package by developed assembly model in commercial CAD software. Finite element analysis involves solving of differential equations with boundary conditions on a discretized domain. The geometries which are in contact with air are given with convective heat transfer coefficient. The inputs to the proposed numerical model are given in Table 32.1. The simulation was carried out in a sequential manner which is explained briefly in a flowchart in Fig. 32.2.

32.3 Results and Discussion

The solid model was built and assembled in Solidworks. Material properties of coil and plates, i.e., steel and copper were applied on the corresponding geometries. Fine meshing (0.25 mm as per skin depth) has been done for work plates so as to get skin effect and normal meshing for other geometries. The input parameters mentioned in Table 32.1 were given as input to the EMS 2018 in transient analysis mode and were

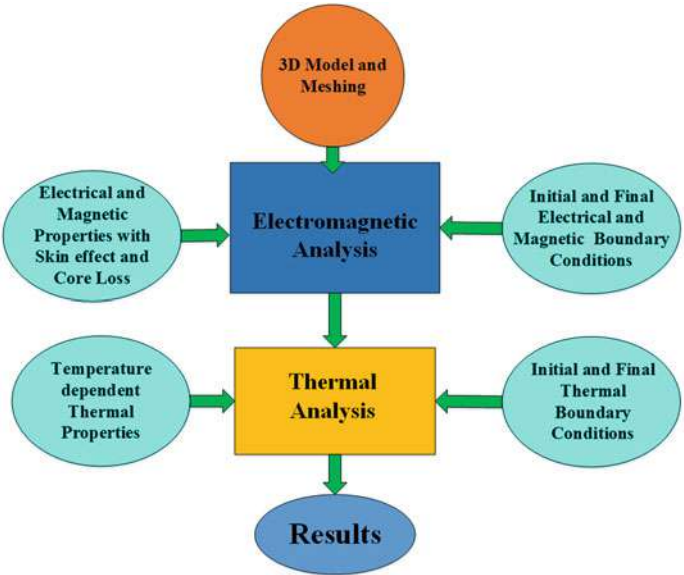


Fig. 32.2 Flow diagram of electromagnetic transient thermal

run for 65s. Magnetic field intensity (H), magnetic flux density (B), temperature distribution profile (T), and time to temperature plot were obtained as shown in Figs. 32.3 and 32.4.

The complete effect of magnetic force, i.e., magnetic field intensity (H) that is being generated due to the coil at the weld joint is shown in Fig. 32.3a. This shows the amount of magnetism that is being present at the weld zone and shape is similar to magnetic lines of force from a regular magnetic bar, which can be clearly seen.

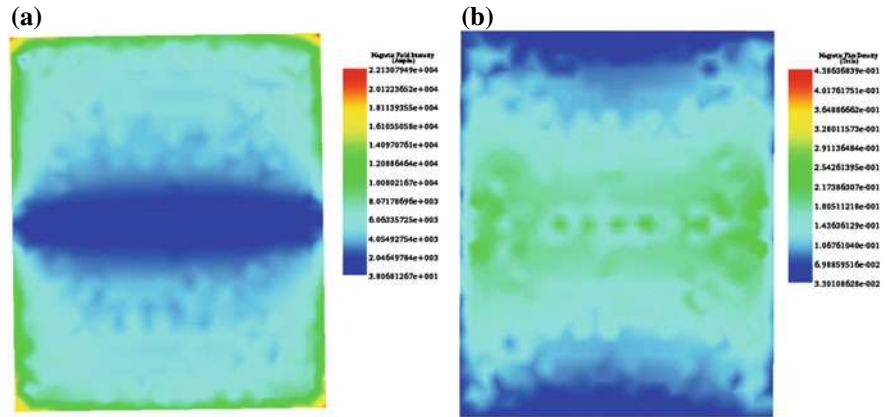


Fig. 32.3 **a** Magnetic field intensity (H). **b** Magnetic flux density (B)

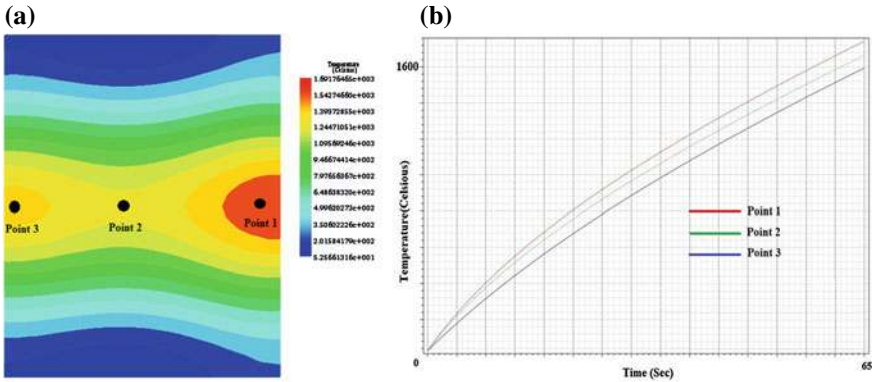


Fig. 32.4 **a** Temperature distribution (T). **b** Time versus temperature plot

The number of lines of force of magnetic field that is being penetrated, i.e., magnetic flux density (B) is dependent on material's magnetic permeability and the source (H) which causes it, related by Eq. 32.6. The magnetic flux density was found maximum to be at the weld joint near the close coil section in accordance with the literature. The maximum value obtained was 0.43 T for 450 kHz input frequency which is in agreement to 0.19 T for 333 kHz. As it can be observed from Fig. 32.3b more amount of flux is present near the closed end of the induction coil. This is due to the change in electrical conductivity of steel at corners at higher temperatures due to more heating at corners [5]. Also, materials ferromagnetic property helps in obtaining higher magnetic flux than paramagnetic or non-magnetic material.

The sand/hourglass shape of the temperature profile is obtained from the numerical result of transient-thermal analysis as shown in Fig. 32.4a. The maximum temperature which was obtained after the simulation was found sufficient, i.e., 1691 °C at point 3 and the average temperature in the required weld zone was around 1500 °C which is for welding of steel plates. The temperature distribution obtained needs further improvement as the temperature at the other ends of the plate also reaches a higher temperature which is not desirable, which can be obtained by use of flux concentrators on the induction coil. The time-temperature plot was plotted as shown in Fig. 32.4b at three points where point 1 is near the open coil/entrance area, point 2 is mid-plate area and point 3 is closed coil area. It can be seen that the temperature and heating rate at point 3 and subsequently less temperature and the heating rate at point 2 and point 1 being least. The reason for both results being due to the paramagnetic property of steel, change of electrical conductivity of steel at higher temperatures at corners due to the higher heating rate at corners than central zone [5].

The simulations were carried out by considering the input conditions from both experimental and numerical literature. From numerical aspect, since the input conditions are given in Table 32.2 were taken from literature [2–5, 10], so the corresponding minimum and maximum temperature, temperature profile, shape of the weld zone, time to temperature plots obtained, were found to be similar to that with

Table 32.2 Literature validation of numerical results

Parameter	Simulation results	Literature results	Conclusion	References
Temperature range, °C	525–1691	779–1626	Matching with numerical literature	[2–5, 10]
Weld zone temperature profile	Sand/hour glass shape	Sand/hour glass shape	Matching with numerical and experimental literature	[3, 7–9]

the literature as shown in Figs. 32.3 and 32.4. Also, from the experimental aspect, the hour/sandglass shape that has been observed [3, 7–9] in experiments, similar hour/sandglass shape of weld zone has been obtained from the numerical result in Fig. 32.4a. The simulations were carried out considering the similar grade of steel and coil. The novel of the work is the attempt of joining plates rather than regular steel pipes which were usually joined using induction welding process. So, there should be less difference in the results of the simulation of pipes to that of plates as the basic source of heat is the same, i.e., induction heating. So, the inputs are taken from literature and the simulation results that were obtained were in accordance and justifies that attempt.

32.4 Conclusion

In the present work, the 3D simulation of HFIW of steel plate by sequentially coupled electromagnetic transient-thermal analysis was carried out. The major heat sources that are considered here are eddy current heating (Joule’s heating) and magnetic hysteresis heating. Maxwell’s equations are solved along with material’s isotropic electrical and magnetic properties to obtain eddy current density and subsequently heat generated due to eddy current. Fourier’s equation of heat transfer was used to obtain the temperature profile of the weld zone. Magnetic field intensity, magnetic flux density, the temperature profile of the weld zone, and time-temperature plot are obtained from the numerical simulation. The magnetic flux density was found to be 0.43 T and maximum temperature of 1691 °C at 450 kHz frequency. The sand/hourglass shape of the temperature profile was found in accordance with the literature reported results. However, for more refined results, optimization of the inputs and incorporation of physical phenomena in the simulation process is going on. The temperature distribution can be improved further by the use of flux concentrator in the required weld area which can further reduce the heat affected zone. Experimental validation with similar process parameters will provide a clear picture of the actual process which is the future scope of this research work.

Acknowledgements The authors acknowledge EM WORKS team for EMS 2018 software and support for this FEM simulation work.

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Chapter 33

Prediction of Machining Responses in Wire EDM on Stainless Steel-316



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Abstract In wire electrical discharge machining (WEDM), material is removed by means of the rapid and cyclic spark that discharges across the gap between the tool and workpiece. In the present work, process parameters of WEDM are tried to be optimize the response variable on Stainless Steel-316 alloy material. SS-316 combinations have been broadly utilized for their predominant properties. For example, high quality, high electrical and thermal conductivities, and low cost. The input parameters considered are pulse-on time, pulse-off time, and current to optimize the responses, viz. surface roughness (SR), volumetric material removal rate (VMRR), dimensional error (DE), and electrode wear (EW). Taguchi's L_{27} orthogonal array was chosen to conduct the experiments according to design of experiments (DOE). SR is measured using surftronic surface tester and VMRR is calculated based on machining time. DE and EW are measured by micrometer. By using artificial neural network, results were predicted and compared with the experimental results.

Keywords Wire EDM · Stainless steel-316 · Artificial neural network (ANN)

33.1 Introduction

Electrical discharge machining (EDM) is broadly accepted technology all over the world with which 3D and complex shapes are machined effectively. EDM is generally utilized as the part beyond words form making ventures, machining heat treated

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materials, and new-age materials, for example, super composites, MMC, earthenware production and so onwards employment with high accuracy, perplexing and complicated shapes and super surface complete can be accomplished by this technique.

Electrically conductive materials are machined with the help of WEDM with no other mechanical powers or strain on the workpiece since there is no contact in between the electrode and workpiece. WEDM is most appropriate for machining the solid materials, for example, zirconium, titanium, molybdenum, and different alloys likewise machine complex shape and profiles that can be complex to be machined by using traditional methods of machining. Workpiece with some hardness and quality can be machined utilizing WEDM and it doesn't decide the choice of hardware, which is in the events of the conventional methods or techniques.

Some of the researchers have studied on an optimization of machining parameters in WEDM in light of Taguchi's quality plan model and analysis of variance (ANOVA). The most impacting factors that are affecting the machining execution are metal evacuation rate, kerf width and surface roughness, pulse-on, voltage and typical (proportion of ordinary flashes to add up to sparkles) were settled [1]. The experiments were conducted to know the influences the machining parameters on the WEDM of the high strength steel for surface harshness and MRR pulse-on, pulse-off, wire feed rate, the dielectric flushing pressure, gap voltage, and the wire tension be varied for experiment using Taguchi's technique [2]. By utilizing Taguchi quality plan, ANOVA and F-test, machining voltage, the current restricting obstruction, beat producing circuit write, and capacitance was perceived as significant parameters that are influencing surface unpleasantness in completing procedure [3]. Parameters, for example, pulse-on, pulse-off, current, and the bed speediness were adjusted and the reaction factors which considered for the examination were VMRR, SR, and accuracy. ANOVA has been done to know the greatness of factor influences [4]. The tests were conducted utilizing Taguchi's L_{16} orthogonal array on stavax material to enhance these procedure factors of WEDM machine. Each arrangement of analysis was led by adjusting factors, for example, pulse-on, pulse-off, current, and bed speediness. Cathode utilized for this experimentation was molybdenum material of 0.18 mm width. Precision, VMRR, and surface unpleasantness were the thought about reactions [5].

The experiments were conducted in the improvisation and utilization of the hybrid artificial neural network and genetic algorithm methodology to optimize EDM parameters. Experimental results are proved satisfactory [6]. An optimization of WEDM by utilizing these response surface methods in SiCp/6061 Al metal matrix composite (MMC) carried out. Input parameters are chosen, viz. pulse-on, pulse-off, wire feed rate, and the servo voltage to think about the procedure factors in regards to kerf width. ANOVA is directed to know the outcome of process variable on machine execution [7]. By utilized response surface methodology (RSM) to estimate the surface roughness in WEDM of pure titanium material. Trial design was planned on Box-Behnken outline. They archived that parameter, for example, pulse-off, wire strain, pulse-on, spark gap, voltage, wire feed rate, and current were the input factors. ANOVA was connected. The test outcomes affirm the created RSM display by 95%

certainty level [8]. The investigation on WEDM in view of gray rational analysis and the statistical analysis was carried out. Taguchi’s quality outline with L_{18} blended orthogonal array (OA) was utilized as. It was discovered that bed speed had extensive impact on MRR kerf width and SR was altogether affected by pulse-on [9]. The examinations on material removal rate and kerf in WEDM in light of Taguchi’s strategy were carried out. The investigational studies were completed by changing open-circuit voltage, wire speed, and flush weight. ANOVA was done to locate the level of criticalness on machining factors [10].

33.2 Experimental Details

Experiments were carried out on Concord DK7732 Wire EDM on Stainless Steel-316 alloy material by varying machine parameters which are pulse-on, the pulse-off, and current. This machine is four axes and controlled by CNC. The machining parameters and their levels are shown in Table 33.1.

Dielectric supply structures supply constant dielectric medium used as the piece of machining. The supply must be controlled route since it is critical to influence a start between gadget to wire and workpiece. Moreover, the dielectric fluid flushes away the particles from this machined surface. Channel is a major bit of this system to disconnect machined particles from the dielectric fluid. The structure is a closed circle system where the dielectric fluid streams from tank by usage of weight of pump and interfacing hoses. The 0.2 mm gap is maintained between workpiece and electrode throughout the experimentation. Wire anode utilized is of 0.18 mm diameter and is comprise of molybdenum material. Analyses are led utilizing new wire for every trial. This investigations are done are as indicate by the DOE design and the parameters for the trials are set in like manner.

Figure 33.1 shows the machining of workpiece by WEDM. Table 33.2 shows the experimental results on design-based and Taguchi’s L_{27} orthogonal array.

Table 33.1 Machining settings used in the experiments

S. no.	Factors	Level 1	Level 2	level 3
A	Pulse-on (μ s)	30	35	40
B	Pulse-off (μ s)	9	10	11
C	Current (A)	3	4	5

Fig. 33.1 Workpiece being cut through WEDM



33.3 Results and Discussions

33.3.1 *Prediction of Machining Responses Using Artificial Neural Network*

ANN is one which delicate registering approach that suit well for the process displaying when the physical wonder of the procedure isn't surely knew, scientific frame characterizing the procedure isn't accessible, and the sensible test information is accessible. The test comes about demonstrated that the procedure parameters like discharge current, idle voltage, pulse duration, wire speed, and dielectric weight are fundamentally affecting on MRR and SR. ANN is utilized to display Wire EDM procedure to foresee the MRR and SR as far as process parameters like discharge. Current, idle voltage, pulse, duration, wire speed, and dielectric weight. Artificial neural systems (ANN) are motivated organically that is, they are made out of components that perform in a closely resembling way to the rudimentary elements of the organic neurons.

$$n = w_{1,1}p_1 + w_{1,2}p_2 + \cdots + w_{1,R}p_R + b \quad (33.1)$$

The information parameters to ANN are standardized as

$$X_n = \frac{X}{X_{\max}} \quad (33.2)$$

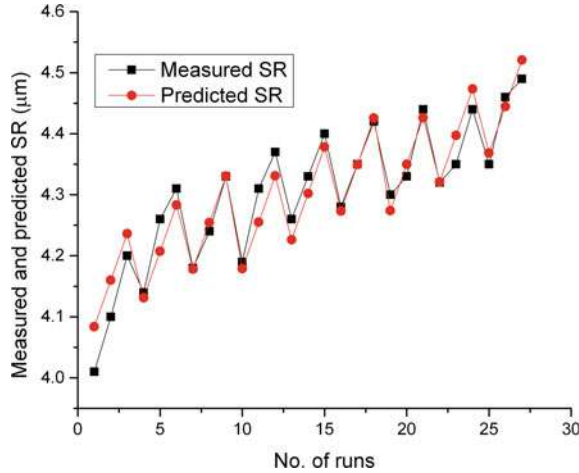
Table 33.2 Experimental result record

Trail No.	Pulse-on	Pulse-off	Current	SR (μm)	VMRR (mm^3/min)	DE (μm)	EW (μm)
1	30	9	3	4.01	9.585	4	2
2	30	9	4	4.10	11.316	5	2
3	30	9	5	4.20	12.788	5	4
4	30	10	3	4.14	8.929	5	2
5	30	10	4	4.26	10.808	6	3
6	30	10	5	4.31	12.167	6	4
7	30	11	3	4.18	8.346	5	2
8	30	11	4	4.24	10.210	7	3
9	30	11	5	4.33	10.978	7	4
10	35	9	3	4.19	10.118	8	3
11	35	9	4	4.31	12.203	9	4
12	35	9	5	4.37	13.896	8	5
13	35	10	3	4.26	9.660	8	3
14	35	10	4	4.33	11.657	9	4
15	35	10	5	4.40	13.343	9	5
16	35	11	3	4.28	9.167	9	3
17	35	11	4	4.35	11.013	10	4
18	35	11	5	4.42	12.888	10	5
19	40	9	3	4.30	11.055	8	3
20	40	9	4	4.33	13.471	9	4
21	40	9	5	4.44	15.407	9	5
22	40	10	3	4.32	10.502	9	3
23	40	10	4	4.35	12.530	10	4
24	40	10	5	4.44	14.421	10	5
25	40	11	3	4.35	9.680	10	3
26	40	11	4	4.46	11.683	11	4
27	40	11	5	4.49	13.366	11	5

where X -Process parameter are to be standardized, i.e., discharge current, idle voltage, pulse duration, wire speed, and dielectric pressure. X_n -Normalized process parameters. $X_{\max}$ -Maximum estimation of the procedure parameter utilized as a part of experimentation.

The present issue incorporates the mapping of the five known causes (discharge current, pulse duration, idle voltage, wire speed, and dielectric weight) to machining execution of Wire EDM. This three process parameters convey among themselves in non-straight manner and the joint effort of parameters isn't known clearly. In this way, while picking a framework write to propose the machining execution of wire

Fig. 33.2 Comparison of experimental and predicted values for SR



electrical discharge machine, capability for compare these difficult non-coordinate relationships must be taken into account for application. From the keeping in touch with, it is found that feed-forward back spread neural frameworks can guide such non-straight relations. Hence, the feed-forward back propagation neural network was chosen for the headway of model for anticipating the DE, SR, VMRR, and EW of Wire EDM machining of Stainless steel-316.

33.3.2 Prediction of Surface Roughness

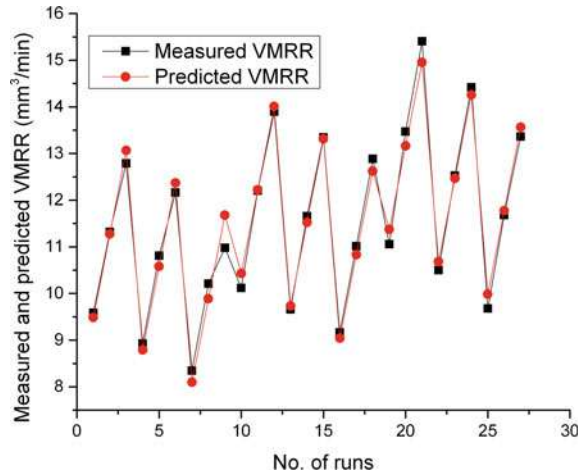
Comparison graph for experimental values versus modeled values was shown in Fig. 33.2. The observation shows that the both plots show a close match and can be concluded that the modeled ANN can be utilized to predict the idea of surface harshness. It is obviously seen that measured surface roughness is associating great with the predicted surface harshness value.

The minimum and maximum deviations were found to be 0 and 2.31%, respectively. In this way, the percentage error values set up the validity of the artificial neural system calculation.

33.3.3 Prediction of VMRR

Comparison graph for experimental values versus estimated values is shown in Fig. 33.3. More the VMRR values the more is the material removal, and hence, the machining time consumed to machine any material or job is less. Further, ANN prediction demonstrates a comparable pattern in every one of the runs contemplated.

Fig. 33.3 Comparison of experimental and predicted values of VMRR



The ANN predicted estimations of volumetric material expulsion rate nearly coordinates with the experimental value. The minimum and maximum errors were seen to be in the range 1.07–5.28%.

33.3.4 Prediction of Dimensional Error

Comparison graph for experimental values versus estimated values is shown in Fig. 33.4. It is observed that the both plots shows a close match and can be concluded that the modeled regression equation can be used to predict the nature of DE.

Fig. 33.4 Comparison of experimental and predicted values of DE

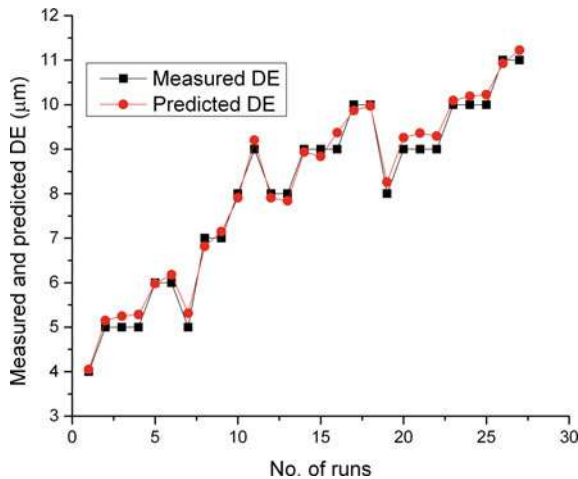
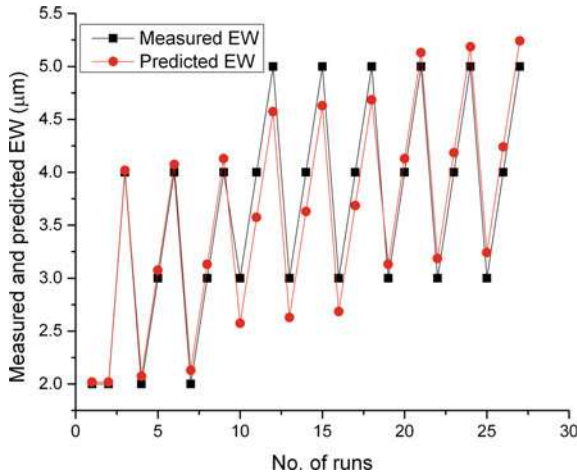


Fig. 33.5 Comparison of experimental and predicted values of EW



Some of the error values are more and the error percentage is within the acceptable limits; hence, the modeled equation can be use to predict the nature of dimensional error. In every one of the cases analyzed, predicted and experimental values are close to each other. The minimum and the maximum errors were given by 0.33% and 8.96%, respectively.

33.3.5 Prediction of Electrode Wear

Comparison graph for experimental values versus modeled values is revealed in Fig. 33.5. It is observed that the both plots show a close match and can be concluded that the modeled ANN can be used to predict the nature of EW. In every one of the cases analyzed, the predicted and experimental values are close to each other. The minimum and maximum errors were given by 0.04% and 8.17%, respectively.

33.4 Conclusions

Experiments were carried out on Concord DK7732 Wire EDM on Stainless Steel-316 alloy material by varying machine parameters which are pulse-on, the pulse-off, and current. Machining characteristics of Stainless Steel-316 alloy material was effectively predicted by utilizing artificial neural network (ANN). Predicted estimations of 70% of data in training set are correlates well with measured one.

Acknowledgements The work reported in this paper is supported by B.M.S. College of Engineering, through the Technical Education Quality Improvement Programme [TEQIP-III] of the MHRD, Government of India.

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Chapter 34

Knowledge Discovery by Decision Tree Using Experimental Data in High-Speed Turning of Steel with Ceramic Tool Insert



A. R. Dhar , N. Mandal and S. S. Roy

Abstract The manufacturing industry is of immense importance. Turning is one of the most basic operations performed across all manufacturing industries till date. Process parameter optimization and modeling in this field, which is very complex, have been investigated by many past researchers. Various methods like statistical techniques, and finite element-based and soft computing-based approaches were used to predict the machinability parameters like flank wear based on the given input cutting conditions like cutting speed, feed rate, depth of cut, etc. Nevertheless, a very few work was done in the area of knowledge discovery with the experimental data. In this work, efforts have been made to extract knowledge automatically using decision tree from the raw experimental data while turning EN24 steel with Cr₂O₃-doped zirconia toughened alumina (Cr-ZTA) ceramic tool insert. After that, the extracted knowledge in the forms of set of fuzzy rules was fed into a custom-made fuzzy logic control (FLC) system developed for predicting flank wear. The results of predictions are validated with experimental test data, and the capability of the system is stated with scope for improvements.

Keywords Tool flank wear · Fuzzy c-means classification · Decision tree · Fuzzy logic control (FLC) system

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_34

34.1 Introduction

The manufacturing industry is of immense importance in the whole world, especially in the fast developing countries, because of its capabilities of generating GDP and employments. Turning is one of the most common and fundamental operations traditionally used in most of the metal cutting industries. The hard turning and high-speed turnings are even preferred over grinding and milling for the reduced tooling cost and higher dimensional accuracy. Development of new tool materials like ceramics, carbides, nitrides, etc., created opportunities for conducting researches in this most common operation by optimizing and modeling the cutting process. By using the optimized models, machinability parameters like cutting forces, surface roughness, tool flank wear, etc., can be estimated based on the given input cutting conditions like cutting speed, feed rate, depth of cut, etc. The most important machinability parameter is tool flank wear because of its enormous contribution on the tooling cost and surface integrity. Building model for the apparent simple operation is one of the most complex jobs because of the involvement of multiple physics like thermal, structural, and material sciences. That is why various methods like statistical techniques [1], finite element-based [2], and soft computing-based approaches [3] exist in the literature till date. The statistical and soft computing-based methods are found to be very accurate but too much sensitive to the data on which they are built. The finite element-based methods, though moderately accurate and built with theories of physics, also suffer from many fundamental assumptions which may affect the virtue of the system. The soft computing-based approaches mostly used computational intelligent algorithms like neural nets, fuzzy logic, genetic algorithm, etc., and their combinations, and they require a lot of training data. Nevertheless, a very few work was invested in the area of knowledge discovery [4] with the raw experimental data, which can be used for training purposes to the machine operators and future model buildings. In this paper, efforts have been made to extract knowledge automatically using decision tree from the raw experimental data while turning EN24 steel with Cr₂O₃-doped zirconia toughened alumina (Cr-ZTA) ceramic tool insert. Experiments were conducted based on response surface methodology (RSM) with three levels for each of the three input cutting conditions, namely cutting speed, feed rate, and depth of cut and response in the form of tool flank wear is measured [5]. In order to build the decision tree, the developed framework classifies the flank wear data using fuzzy c-means (FCM) algorithm. The decision tree is built employing iterative dichotomizer 3 (ID3) algorithm, which is one of the most common machine learning algorithms.

In order to validate the extracted knowledge from the decision tree in the form of a set of fuzzy rules, a fuzzy logic control (FLC) system is developed. Fuzzy logic works on a set of fuzzy rules involving fuzzy inputs which are known as antecedents and fuzzy outputs better known as consequents. The “Mamdani” fuzzy reasoning technique is employed in the FLC system to predict the tool flank wear from the given set of three inputs. The membership functions (MFs) for FLC system are carefully designed to match the crisp inputs and outputs obtained from the experimental data.

The results of prediction are compared with the experimental test data, and the model is found to be considerably accurate.

34.2 Working Methodology

Turning experiments are conducted on HMT lathe using EN24 steel as job and Cr-ZTA as tool insert based on RSM [5]. There are three designated levels for each of the three input parameters, and the response output is flank wear. Levels were decided based on allowable range of operations involving workpiece and tools and the lathe machine available. Increasing the number of levels for better accuracy of a system will in turn increase the number of experiments, which is costly. Toolmakers microscope is used to measure the flank wear. Sixteen such runs are conducted for generating data for the training purpose, and five additional confirmation runs (both including and excluding the training conditions) are carried out to finally test the results. After obtaining the needed data, the steps defined in different modules of this work are followed in order to extract the machining knowledge and validate it. The modules constituting the entire system are discussed in the following subsections.

34.2.1 Fuzzy C-Means (FCM) Classification

Fuzzy classification is done using FCM algorithm, where c is number of centroids (clusters), which are arbitrarily chosen among the data points, and based on the calculated Euclidean distances of each data point from the centroid, the degree of belongingness to the each centroid or fuzzy membership value is updated. FCM is one of the most effective algorithms for data clustering. FCM was proposed by Dunn [6] and later on, it was modified by Bezdek [7]. The standard FCM objective function for partitioning the data into several clusters is given as

$$J_{\text{FCM}} = (U, V) = \sum_{i=1}^c \sum_{k=1}^N \mu_{ik}^p \|x_k - v_i\|^2 \quad (34.1)$$

where $V = \{v_i\}_{i=1}^c$ are the cluster prototype and array $U = \{\mu_k\}$ represents the partition matrix, c is the number of cluster centroids, N is the number of data points, x_k is the k th data point, and v_i is the centroid of i th cluster. $\|x_k - v_i\|^2 = d_{ij} = d(x_k, v_i)$ is the distance measure between the data point and cluster center. μ_{ik} is the fuzzy membership of k th data point to i th cluster. This membership value needs to satisfy the conditions $\mu_{ik} \in [0, 1]$, $0 \leq i \leq c$, $1 \leq k \leq N$, $0 < \sum_{k=1}^N \mu_{ik} < N$, $0 \leq i \leq c$ and $\sum_{i=1}^c \mu_{ik} = 1$, $i \leq k \leq N$.

Parameter $p \in (1, \infty)$ is a weighing exponent on each membership, which determines the amount of fuzziness of the resulting classification and the value is usually set as 2. The value of FCM objective function is minimized firstly, by assigning high membership values to data points which are close to the centroid of its particular class, and secondly, by assigning low membership values to the data points that are away from the centroid. The partition matrix and cluster centroid are updated as the followings.

$$\mu_{ik} = \frac{1}{\sum_{j=1}^c \left(\frac{d_{ik}^2}{d_{jk}^2} \right)^{\frac{1}{p}}} \quad (34.2)$$

$$v_i = \frac{\sum_{k=1}^N \mu_{ik}^p x_k}{\sum_{k=1}^N \mu_{ik}^p} \quad (34.3)$$

In this case, flank wear (in mm) data are classified using the FCM taking $c = 3$, $N = 16$, and $p = 2$. The fuzzy classes for the data points are taken as “low,” “medium,” and “high,” which are actually levels of flank wear. The value of $p = 2$ is taken as common practice though optimal value of p can also be computed in future work. Figure 34.1 shows the classification results of FCM run on the data.

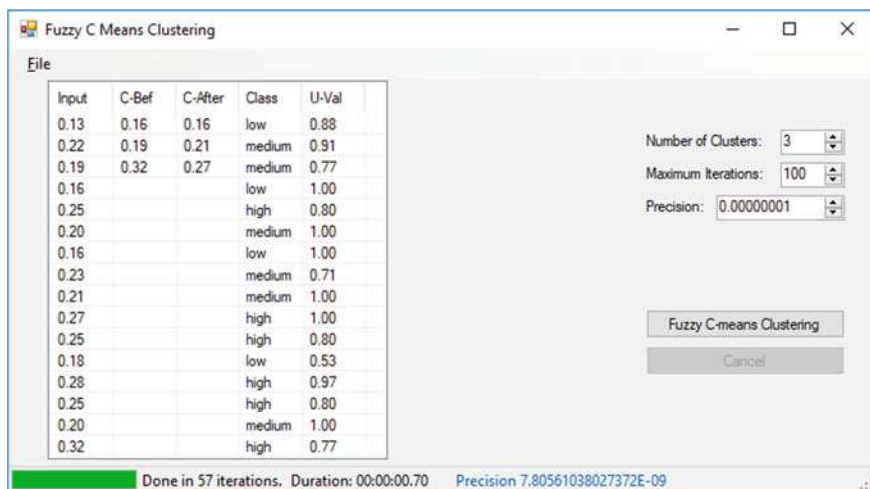


Fig. 34.1 Screenshot showing FCM classification

34.2.2 Decision Tree

Once the classification is done, the same data are used to build a decision tree using ID3 algorithm [8]. Decision tree is a hierarchical data structure that represents data through a “divide-and-conquer” strategy. This was first proposed by Quinlan [8]. The entropy for the entire dataset is calculated by the following formula (Eq. 34.4).

$$E_{\text{system}} = \sum_{i=1}^k -(p_i \times \log_k p_i) \quad (34.4)$$

where k is the number of classes or categories of data, which can be found from the responses, and p_i is probability of the i th class in the dataset, which can be found by dividing the number of i th class instances by the total number of instances. Then the information gain for each parameter or attribute Gain_a is computed by subtracting the sum of the weighted entropy for each value of the attribute ($v_a = u$) from the initial system entropy E_{system} by the following equation.

$$\text{Gain}_a = E_{\text{system}} - \sum_{v_a=u}^{U_a=\{u\}} \left[p_{v_a=u} \times \sum_{i=1}^k -(p_{i|v_a=u} \times \log_k p_{i|v_a=u}) \right] \quad (34.5)$$

where $p_{v_a=u}$ is the weight of the attribute when assigned a value u , which can be found by dividing the number of instances having that attribute value by the total number of instances. $U_a = \{u\}$ is entire the set of values for attribute a . The probability $p_{i|v_a=u}$ is calculated in the same way only considering instances having the attribute value $v_a = u$. The attributes with the highest information gain are selected as root, and iteratively, the tree is built creating branches with each attribute value. The entropy at each branch is calculated now as the system entropy, and the information gain for the rest of the attributes are computed in a similar fashion. On the occurrences of no instances or all same class instances for a particular branch of tree, the output class value is created as leaf node of the tree, denoting a stop of further branching of the node. The decision tree built with the raw data is illustrated in Fig. 34.1. The input values are all categorical in this case, meaning the numerical levels in the input parameters like cutting speed, feed rate, etc., are used as different categories. Now using a straightforward way, rules are extracted from the decision tree. For example, IF CUTTING SPEED is LOW (“140”) and FEED RATE is LOW (“0.12”) FLANK WEAR is LOW. In the current problem, the input parameter depth of cut is eliminated in the process of building the decision tree, and hence, not found in the extracted rules formation (Fig. 34.2).

One obvious issue faced with the current decision tree is some missing evidences or instances marked by “NIL”. There are two ways to go about it. One way, the decision tree can be pruned by removing the branches or leaf nodes with no instances. Alternatively, the leaf nodes can be filled with virtual instances by comparing the values of the sibling nodes. In this case, the second method is adopted because of

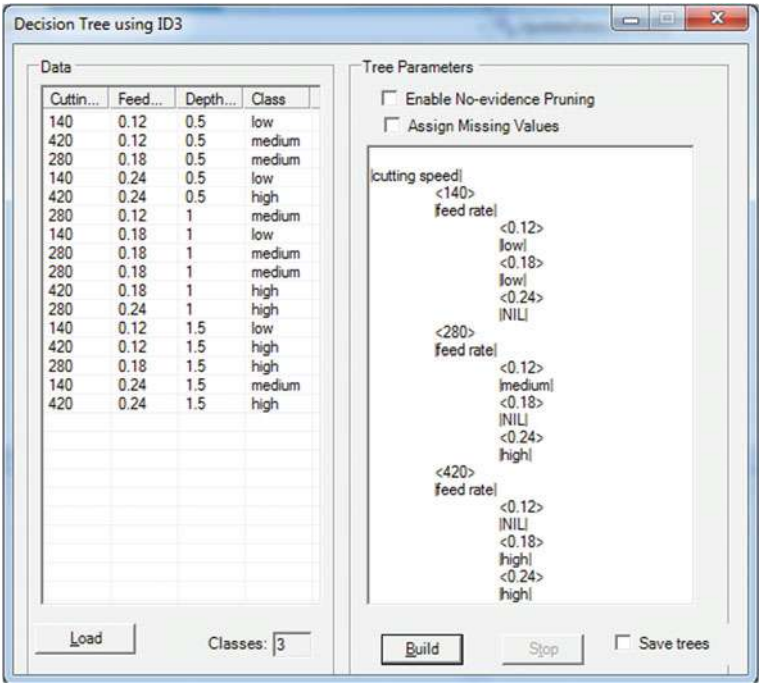


Fig. 34.2 Screenshot showing decision tree built from the raw data

the superior results found in the validation phase. Utmost care should be taken while assigning output class values considering the wear characteristics in turning in order to avoid biasness. This aspect in the decision tree, seemingly human intervened, can also be automated in the context of the current problem.

34.2.3 FLC System

The original aim of the work was to extract a set of rules from the raw data which are not so often expressed in human understandable forms. The decision tree allows building a usable knowledge for the specific turning process by forming antecedent–consequent relationship in linguistic forms, and removing the insignificant parameters from the list of antecedents. Nevertheless, this discovery of rules or knowledge needs to be tested in a system which uses this kind of linguistic rules as inputs. There comes the need for a FLC system, which can work on the current dataset and the extracted rules from the decision tree and predict outcomes which can be validated with experimentation. A tailor-made FLC system is developed employing the “Mamdani” fuzzy reasoning technique [9, 10], where centroid method is used for defuzzification. The developed FLC system uses two half-trapezoidal membership

functions (MFs) and one intermediate full triangular MF for parameter cutting speed. Similarly, two extreme half-trapezoidal MFs and one intermediate triangular MF for other inputs and also for the output are used, namely “low,” “high,” and “medium,” respectively. The MFs are constructed in such a fashion that each one has a mean equal to the value of each level of each parameter. For example, cutting speed which has three distinct values, namely 140, 180, and 420 as levels, leads to the construction of half-trapezoidal MFs with parametric points as (80, 80, 160, 240) and (320, 400, 480, 480), and the intermediate triangular MF is having points as (160, 280, 400). For the flank wear which is the output parameter, three MFs are similarly constructed considering the minimum and maximum values besides the means.

The rules extracted from the decision tree are improvised for missing instances (as discussed in Sect. 34.2.2) and then fed into the FLC system.

34.3 Results and Discussions

In the results section (tab) of the developed FLC system, a sample query with three input values separated with commas is formulated and entered in the input box to ask the output flank wear value. One such sample query is shown in the following figure (Fig. 34.3), where finally crisp flank wear values are obtained by defuzzification. The MFs involved in the input variables of the query along with the triggered rules and the concerned MFs and fuzzy membership value of the output for each rule are illustrated in the system.

The predicted flank wear values are obtained for five such cases in this manner, and the results are validated with the actual flank wear data found in the confirmation runs. The results are shown in Table 34.1. The percentage deviation of each case is shown.

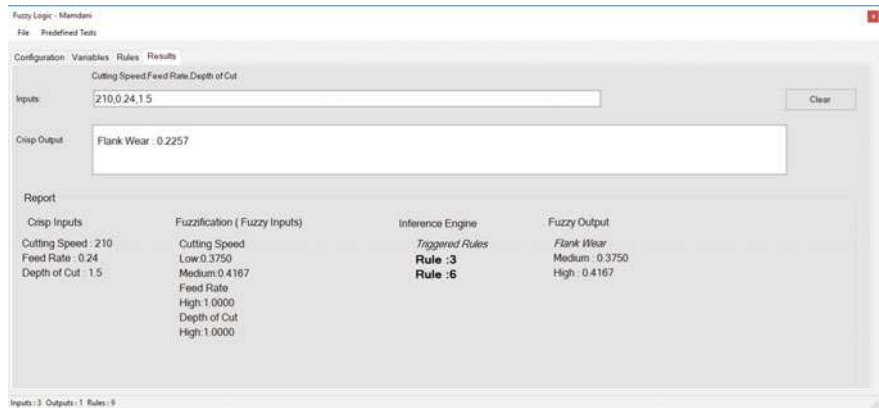


Fig. 34.3 Screenshot showing a sample query with result in FLC system

Table 34.1 Results of prediction

Sl. No.	Cutting speed (m/min)	Feed rate (mm/rev)	Depth of cut (mm)	Flank wear (mm) (actual)	Flank wear (mm) (predicted)	Percentage deviation (%)
1	140	0.18	1.0	0.16	0.156	2.50
2	210	0.24	1.5	0.23	0.226	1.74
3	280	0.24	1.5	0.25	0.261	−4.40
4	300	0.12	0.5	0.18	0.188	−4.44
5	420	0.24	1.0	0.27	0.261	3.33

The results show a maximum deviation of $-4.44(\%)$ which is quite satisfactory in comparison with the results obtained in [5] by statistical regression.

In this context, it should be noted that the extracted rules are not at all generalized. It solely depends upon the current operation including job and tool material and their geometries, the process environment and machine used, and the allowable range of input cutting parameters. Even experimental data with other cutting inserts like coated or uncoated carbides, with other conditions remaining same, may yield a different set of rules, and unlike the present scenario, the depth of cut may be discovered to be a significant factor in determining flank wear in that case.

34.4 Conclusions

Process modeling and optimization of turning operation are indeed complex. Predictive models either use statistical and soft computing approaches which are mostly treated as black boxes too much dependent on the data, or finite element-based methods which are not so accurate because of involvement of multiple physics in operation and prerequisites of critical assumptions on materials properties. Hence, computational methods which are accurate and also explainable to the user at any point are gaining more research attention. Naturally, there is an increasing need to extract useful knowledge from the raw experimental data itself. This work demonstrates a platform to address this issue. The following overall conclusions can be drawn.

1. An attempt has been successfully made to discover the knowledge in turning operation from the raw experimental often not expressed in easily understandable forms to the machine operators. This is accomplished by building a decision tree.
2. The knowledge in the forms of set of rules extracted from the decision tree is validated in a custom-made FLC system. The results indicate the FLC system is quite accurate in prediction indirectly validating the extracted rules.

There are enormous scopes of future work in this regard. They are enlisted below.

- The knowledge in the form of set of rules that was extracted from the decision tree can be later used for other model development like a rule-based expert system.

- Improved methods of fuzzy classification can be tried along with exploring alternative algorithms of building decision tree. Different value for number of clusters (levels of flank wear) can also be tried for further improving the entire system.
- The design and methods of experimentations with different combinations of level can be revisited with more inputs and other machinability parameters like surface roughness, cutting forces, etc.
- Nonlinear MFs like Gaussian or bell-shaped functions can also be tried in FLC system.

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Chapter 35

Decision-Making System for Accepting/Rejecting an Order in MTO Environment



C. H. Sreekar, K. Hari Krishna and P. Vamsi Krishna

Abstract When multiple orders are to be processed in a make-to-order environment, scheduling them properly is of paramount importance. Further, it is also important to foresee whether or not the product can be completed in the stipulated time period. In this present work, FlexSim is used to simulate and determine job processing time, waiting time, machine working time, ideal time, etc. Job and machine status reports are then made from the obtained results, and it gives the shopkeeper ample results regarding the job. The simulation results further help in identifying the optimal sequence and in determining the capacity required in all the machining centers for the jobs to meet their respective due dates. If the time required for the job exceeds due date, then capacity is increased and the job is rescheduled again. Even then if it fails to complete in due date, then it is rejected, else accepted.

Keywords FlexSim · MTO · MTS

35.1 Introduction

Several incoming orders must be considered and evaluated in terms of many different conflicting criteria for acceptance/rejection in a MTO manufacturing system. Hence, an effective evaluation approach is required for improving the decision quality. A sound scheduling system can help a manufacturing organization to achieve its strategic goals. The results of scheduling are useful at the broadest level in capacity planning.

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_35

Based on the relationship between production release and order arrival, production systems can be classified into:

1. Make-to-stock system (MTS)
2. Make-to-order system (MTO)

In MTS manufacturing system, products are produced to stock according to the forecasts of the demand, whereas in MTO manufacturing system, work releases are authorized in accordance with the external demand arrival.

35.2 Production Planning

Production planning may be defined as the technique of foreseeing every step in a long series of separate operations, each step to be taken at the right time and in the right place and each operation to be performed in maximum efficiency [1]. It helps an entrepreneur to work out the quantity of material, manpower, machine, and money required for producing the predetermined level of output in a given period of time.

Production planning compiles records and reports on various aspects of production, such as materials and parts used, products produced, machine and instrument readings, and frequency of defects. These workers prepare work tickets or other production guides and distribute them to other workers. Production planning coordinates, schedules, monitors, and charts production and its progress, either manually or with an electronic equipment. They gather information from customer's orders or other specifications and use the information to prepare a detailed production sheet that serves as a guide in assembling or manufacturing the product. They also locate and distribute materials to the specified production areas. They may inspect products for quality and quantity to ensure their adherence to specifications.

Production planning consists of:

- (a) preplanning (b) routing (c) scheduling (d) dispatching (e) expediting.

35.2.1 Scheduling

Scheduling is a decision-making process to determine when a job is to be started in a machine and when it is to be completed. A job may have to be processed in various different machines or process centers. In sequencing, we consider only the order of priority in the sense which job is to be taken up first and which should be taken up after that and so on [2].

In general terms, scheduling may be defined as the allocation of resources over time to perform a collection of tasks. It is a decision-making function to determine the schedule of events to occur while utilizing the resources allocated to perform the collection of tasks. It determines when each of the tasks is to be performed utilizing

what resources. In scheduling, the time events of starting and completion of the tasks or jobs are essential.

Effective scheduling is essential for successful operations. Normally, scheduling is done after many other managerial decisions have been made. For example, planning for emergency services such as fire protection first requires an analysis of the best locations for fire stations, decisions about the type and quantity of firefighting equipment at each location, and a staffing plan for each station. Sound scheduling can help an organization achieve its strategic goals. For example, a fire department is better able to meet its objective of protecting the community if an adequate number of firefighters are scheduled at all times.

35.2.2 Scheduling Problem

A scheduling problem can be considered as one which refers to answering two kinds of questions.

1. Which resources are to be allocated to perform each task? Allocation decision.
2. When and in what order would each task is to be performed? Sequence decision.

35.2.3 Objectives of the Scheduling Problem

1. Efficient utilization of the resources (or facilities).
2. Rapid response to demands.
3. Conformance to the prescribed deadlines.
4. Completion of the job at the minimum protracted total time.
5. Minimization of the total cost.

35.2.4 Constraints

35.2.4.1 Technological Constraints

This refers to the technological order of tasks to be performed, and one cannot violate the precedence restrictions of the tasks.

35.2.4.2 Resource Constraints

This refers to the quantitative and qualitative capabilities of the resources (or facilities, manpower, machines) and the processing or performance time required for the tasks.

35.2.5 Scheduling Rules

Jobs can be scheduled by any of the following rules:

1. Shortest processing time (SPT) rule: It refers to the sequencing of jobs in the ascending order of their processing time. Here, the highest priority is given to the waiting job with the shortest processing time in the machine under consideration.
2. Earliest due date (EDD) rule: In this case, the priority is given to the job having the least value of due date. The sequencing of jobs is in the ascending order of their due dates, that is, from the lowest value of due dates to the highest value of due dates.
3. First-come-first-served (FCFS) rule: The job that arrived at the workstation first has the priority under a first-come-first-served rule.
4. Critical ratio (CR) rule: The critical ratio is calculated by dividing the time remaining to a job's due date by the total shop time remaining for the job including setup, processing, move, expected waiting time of all remaining operations, including the operation being scheduled. The formula is
$$CR = (\text{Due date} - \text{today's date}) / (\text{total shop time remaining}).$$

A ratio less than 1.0 implies that the job is behind schedule, and a ratio greater than 1.0 implies that the job is ahead of schedule. The job with the lowest CR is scheduled next.
5. Longest processing time (LPT) rule: In this case, the priority is given to the job having the highest value of due date. The sequencing of jobs is in the descending order of their due dates, that is, from the highest value of due dates to the lowest value of due dates.

35.3 Simulation

35.3.1 Introduction

A simulation is the imitation of the operation of a real-world process or system. Whether done by hand or using a computer, simulation involves the generation of an artificial history of a system and the observation of that artificial history to draw inferences concerning the operating characteristics of the real system. The behavior of a system as it evolves over time is studied by developing a simulation model. This

model usually takes the form of a set of assumptions concerning the operation of the system. These assumptions are expressed in mathematical, logical, and symbolic relationships between the entities, or objects of interest, of the system. Once developed and validated, a model can be used to investigate a wide variety of “what if” questions about the real-world system. Potential changes to the system can first be simulated, in order to predict their impact on the system performance. Simulation can also be used to study systems in the design stage, before such systems are built [3]. Thus, simulation modeling can be used both as an analysis tool for predicting the effect of changes to existing systems and as a design tool to predict the performance of new systems under varying sets of circumstances.

35.3.2 *Simulation in Scheduling*

The availability of special-purpose simulation languages, massive computing capabilities at a decreased cost per operation, and advances in simulation methodologies has made simulation one of the most widely used and accepted tools in operational research and systems analysis.

Simulation can be used for the following purposes:

1. Simulation enables the study of, and experimentation with, the internal interactions of a complex system or of a subsystem within a complex system.
2. Informational, organizational, and environmental changes can be simulated, and the effect of these alterations on the model’s behavior can be observed.
3. The knowledge gained during the designing of a simulation model could be of great value toward suggesting improvement in the system under investigation.
4. Changing simulation inputs and observing the resulting outputs can produce valuable insights into which variables are the most important and into how variables interact.
5. Simulation can be used as a pedagogical device to reinforce analytic solution methodologies.
6. Simulation can be used to experiment with new designs or policies before implementation, so as to prepare for what might happen.
7. Simulation can be used to verify analytic solutions.
8. Simulating different capabilities of a machine can help determine the requirements of it.
9. Simulation models designed for training make learning possible without the cost and disruption of on-the-job instruction.
10. Animation shows a system in simulated operation so that the plan can be visualized.
11. The modern system (factory, wafer fabrication plant, service organization, etc.) is so complex that its internal interactions can be treated only through simulation.

35.3.3 *FlexSim description*

FlexSim is a 3D simulation software designed for modeling processes. Processes include manufacturing, packaging, warehousing, material handling, supply chain, and many others. FlexSim is equipped with a powerful array of tools that run the gamut from a “true-to scale” 3D display to a comprehensive collection of statistical reports that can immediately shed light on any aspect of performance in the process. FlexSim makes it easy for decision-makers to visualize—risk-free—results of proposed changes to optimize the flow of products, staffing, resource utilization, floor plan design, and almost any other aspect of the system. FlexSim is a crystal ball to optimize the system before implementing changes in the real life, saving money and time [4].

35.3.3.1 **Properties of FlexSim**

(a) **Powerful Risk-Free Decision Support:**

FlexSim 3D simulation software gives evidence-based support to make informed decisions confidently. It provides the evidence in two ways: through accurate 3D animation and through statistical reporting. FlexSim features a dashboard display to view the system, staff, and operating metrics during the simulation run. The experimenter makes it possible to run “what if” scenarios to compare different options side by side.

(b) **Simple, Four-Step Modeling:**

FlexSim uses a four-step method to model any given process system. First, the process’s CAD-based physical layout is imported (AutoCad DFX or DWG format), and relevant processing objects are added to represent the process. Second, the flow of the items that are to be processed is defined using click-and-drag connections—easy as defining the flow in a flowchart. Third, the user will detail the objects with processing parameters such as process time, routing logic, conveyor speeds, staff requirements, material handling options, and visualization options. Fourth, the relevant evaluative metrics that need to be viewed in the dashboard are defined using easy-to-use pick list options and wizards. Then the model is run. The process coming to life in virtual 3D can be watched.

(c) **Comprehensive and Accurate Statistics for Better Decision Support:**

Because FlexSim is specifically designed to model complex processes, performance and output statistics are selected to accurately reflect those metrics most useful in planning of the production systems. Users can capture data regarding any number

of performance measures, such as machine utilization, transport time, machine state statistics, staff utilization, waiting time, WIP levels, machine breakdown and repair time metrics, and space allocation. Even the impact of unplanned system interruptions can be evaluated. Every aspect of the system can be monitored through accurate and timely performance statistics that the software automatically provides.

35.4 Model Construction

In the current project, a workshop floor layout has been constructed using FlexSim.

- (a) A workshop layout is constructed in FlexSim for simulation. The workshop layout consists of 7 stations like (1) marking, (2) deburring, (3) grinding, (4) welding, (5) gouging, (6) DP test, and (7) radiography. Each station consists of a different number of processors distributed sequentially in “U” pattern, and each station consists of a queue which is connected with the processor.
- (b) Totally 6 jobs have been considered, which has to be scheduled. Each job arrives at the source after a specific interval of time according to the arrival schedule provided at the source. All the jobs are color coded based on their type and a label is provided with a unique value, which is later used for distributing the jobs among processors.
- (c) After arrival of the jobs, they are sent into the system queue. System queue distributes the jobs into various stations based on the type of operations to be performed.
- (d) Each processor is updated with processing times according to the job type; hence, whenever a certain job arrives, it checks the job type and processes it according to its processing time and sends it to the next queue station.
- (e) When a job arrives at the queue, it sends it to the next processor if it is empty, otherwise the job is made to wait. However, when multiple jobs are waiting at the queue, the processor has to decide which job it should take. This decision is made by comparing the label values present on the jobs. The processor is designed to accept the job with the highest label value.
- (f) The processor is also designed to write the in time and out time of the job to the global table using the trigger option available. This global table is later used for developing the job status report and machine status report.
- (g) The flow logic of the flow items is carried out by means of FlexScript which follows the steps mentioned in scheduling algorithms mentioned in the upcoming sections.
- (h) All the processing times are given in minutes.

35.5 Assumptions of the System

1. No machine may process more than one operation at a time.
2. Each operation once started must be performed to completion.
3. Each job once started must be performed to completion.
4. Each job is an entity, that is, even though the job represents a lot of individual parts, no lot may be processed by more than one machine at a time. This condition rules out assembly operations.
5. A known, finite time is required to perform each operation, and each operation must be completed before the next operation begins. The given operation time includes setup time.
6. Due dates are fixed.
7. Whenever a machine becomes idle, a job waiting in the queue is assigned immediately.
8. Machine, crew, and other equipment required to do one operation are treated as one batch.

35.6 Scheduling Algorithm

The algorithm is divided into the following modules: Main scheduling module, Machine selection module.

In addition to these modules, “time reduction module” can also be used. This module reduces the total time required to complete the job, in case of delay in the procurement of raw material.

35.6.1 *Main Scheduling Module*

This is the main module and controls the entire process of scheduling. The steps involved in this module are:

1. Select the order to be scheduled.
2. Select whether the process has to be carried out with or without the increase in capacity.
3. Specify the number of parts available.
4. Specify the due dates and all other details of each part.
5. Print the processing time, cost, and details of each part.
6. If sequence has to be entered, then go to step 7, else go to branch 1.
7. Read the sequence entered.
8. Display the sequence.
9. Read the starting time of first operation.

10. Select the job of the highest priority in the work.
11. Set the job in queue.
12. If all jobs are finished, then go to step 13, else step 10.
13. Check the operation code.
14. Print the working time and idle time.
15. Compute the lateness, completion time, and tardiness values.
16. If all operations are completed, then go to step 17, else 13.
17. Select the highest job in the order.
18. Print the completed values.
19. If all jobs are finished, then go to step 20, else 17.
20. Print the tardiness values of all jobs.
21. If tardiness of all jobs is greater than zero, then go to step 22, else reject the order.
22. Accept the order.
23. If all orders are complete, then stop, else repeat from step 3 (Fig. 35.1).

35.6.2 Time Reduction Module

This module reduces the total time required to complete the job, in case of delay

In the procurement of raw material, the steps involved are:

1. If tardiness values are less than zero, then go to step 2.
2. Select the work centers in which the setup cost is less.
3. Increase the capacity of the work centers.
4. Obtain the new tardiness values by main scheduling module.
5. If CT is less than DD, go to step 6, else repeat step 3.
6. Accept the order (Fig. 35.2).

35.6.3 Machine Selection Module

1. Check the number of machines available for the required operation.
2. Set a job on idle machines available.
3. Complete the processing of operation.
4. If any machine becomes idle, the jobs waiting in the queue will be assigned to that machine.
5. Check whether all jobs are completed or not. If all jobs are completed, go to step 6, else go to step 1.

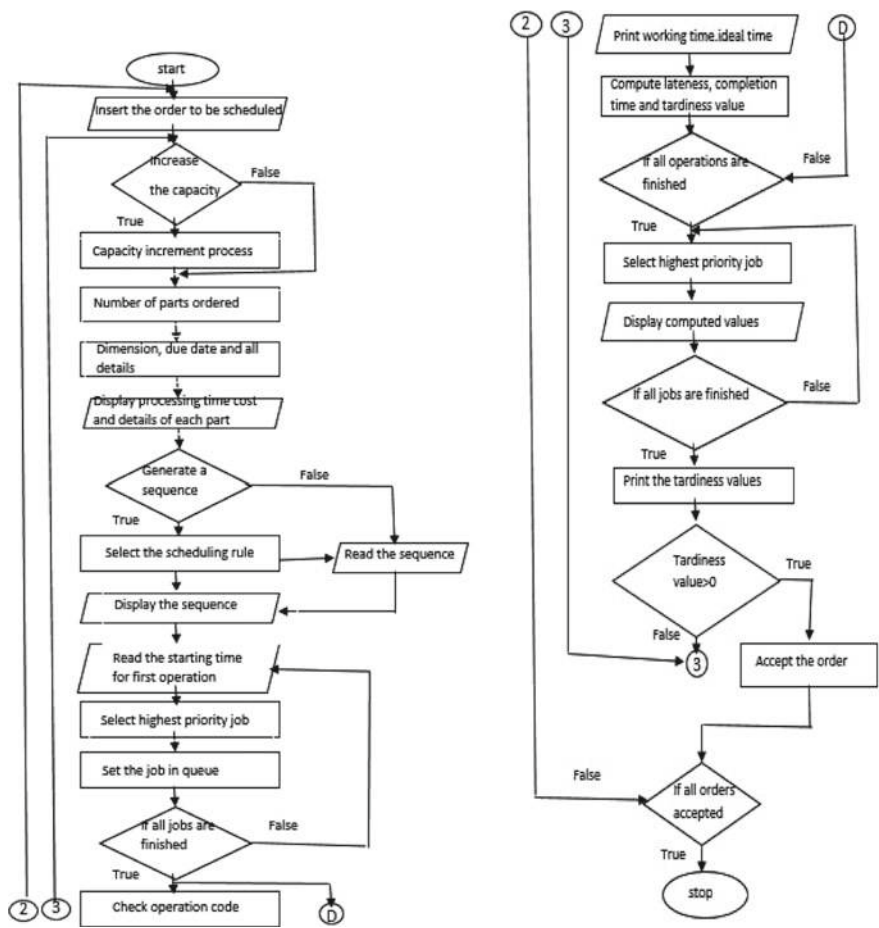
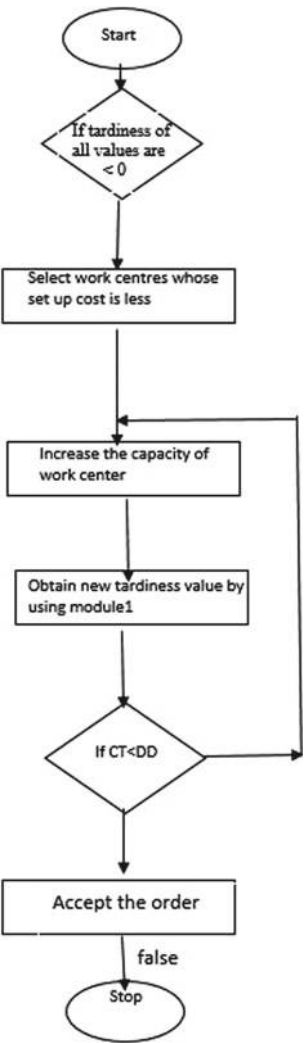


Fig. 35.1 Flowchart of main scheduling algorithm

6. Check whether all operations are completed or not. If all jobs are completed, stop, else go to step 2 (Fig. 35.3).

- Inputs to the system.
- There are three jobs and related to these three jobs are 7 operations. The table depicts the processing time of the operation.
- (a) Processing times of orders (Table 35.1)
 - (b) Assumptions on sequence of operations

Fig. 35.2 Flowchart of time reduction module



The sequence of operations that each job follows is given below:

- order1: 2 → 4 → 3 → 1 → 6 → 7
- order2: 2 → 4 → 1 → 5 → 6 → 7
- order3: 2 → 1 → 3 → 4 → 5 → 6 → 7

Assumptions on due dates
For a better understanding of the process, assumptions are made that all the machines will work 8 h a day. The order date, due date, and today’s date are assumed as follows (Tables 35.2, 35.3 and 35.4; Fig. 35.4).

Fig. 35.3 Flowchart of machine selection module

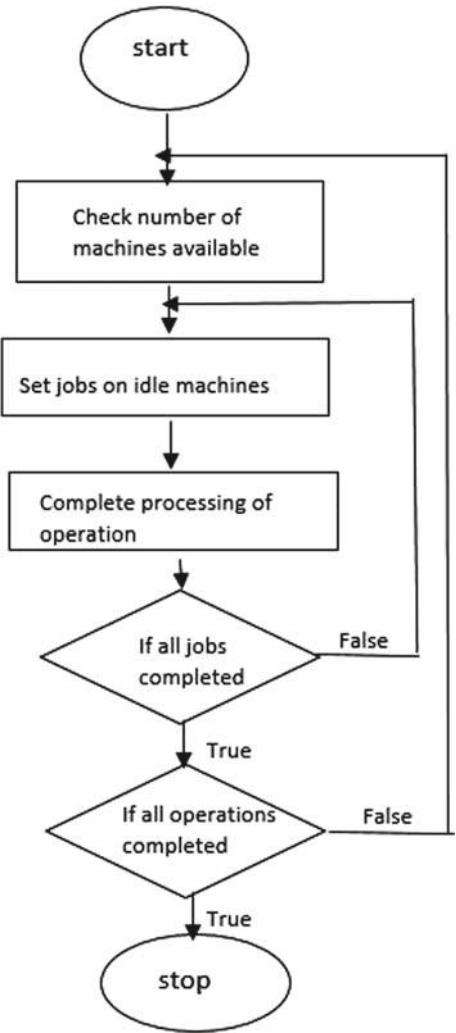


Table 35.1 Processing time of orders

Operational/jobs	J1	J2	J3
Marking	742	742	724
Deburring	664	668	669
Grinding	960	–	870
Welding	1320	1320	1513
Gouging	–	848	828
DP test	675	675	675
Radiography	160	160	160
Total time (in minutes)	4521	4412	5441

Table 35.2 Ordering date and due date of all the orders

Order no.	Ordering date	Due date	Today's date
Order1	April 30	May 26	April 30
Order2	May 3	June 6	April 30
Order3	May 5	June 4	April 30

Table 35.3 Job status

Job	Total processing time	Tardiness	Waiting time (h)	Total cost
J11	27 days 5 h	1 day 5 h	45	5711
J12	29 days 1 h	3 day 1 h	47	5711
J21	17 days 4 h	0	0	5646
J22	20 days 2 h	0	0	5646
J23	23 days 2 h	0	22	5646
J31	30 days 4 h	0	32	7069
J32	33 days 2 h	0	42	7069
J33	36 days 3 h	1 day 3 h	56	7069
J34	39 days 4 h	4 day 4 h	70	7069

Table 35.4 Machine status report

Operation	Working time (h)	Ideal time
Marking	110	9 h 38 min
Deburring	100	0
Grinding	59	6 h
Welding	211	0 h
Gouging	97	92 h
DP test	102	47 h
Radiography	24	192 h

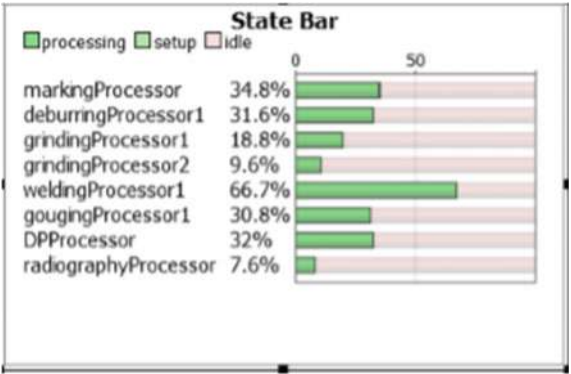


Fig. 35.4 State bar

35.7 Conclusion

In the present work, using FlexSim software simulation is done for different sequencing techniques. Further, job status, machine status, and state bar have been tabulated. The simulation lets the shopkeeper know whether the product can be completed within the due date or not. Thereby, he can decide upon whether or not to accept the job. It also lets us know which sequencing technique is better among others. Thereby, it helps the management to take efficient decisions.

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Chapter 36

Numerical Simulation of Channel Angles and Their Combination Influence on Plastic Deformation Behaviour of Pure Al Processed by Equal Channel Angular Pressing



Ramulu Malothu and Krishnaiah Arkanti

Abstract Equal channel angular pressing (ECAP) is one of the most efficient methods of severe plastic deformation (SPD) for obtaining bulk nanostructured materials. The ECAP die consists of two equal channels that meet at an angle, usually between 90° and 135° . In the present study, the effect of ECAP die channel angles and their combination on the plastic deformation behaviour of pure Al during ECAP under friction and frictionless conditions were investigated. A 2-D finite element modelling was used in order to analyse the plastic deformation behaviour as the material passes through the die. The properties of commercially pure aluminium (Al) have been selected in order to perform FEM simulations. A sound knowledge obtained for the plastic deformation (material flow) and understanding the relationships between plastic deformation and mechanical properties of pure Al.

Keywords Severe plastic deformation · Equal channel angular pressing · Al · Simulation · Finite element modelling · Plastic deformation

36.1 Introduction

Since the last two decades of constructive research in material science and engineering, a number of secondary processing techniques have emerged, with the primary objective of refining the microstructure to characteristic length scales near or below $1\ \mu\text{m}$. An important breakthrough in modern materials science was the application of severe plastic deformation (SPD) techniques for producing ultrafine-grained

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_36

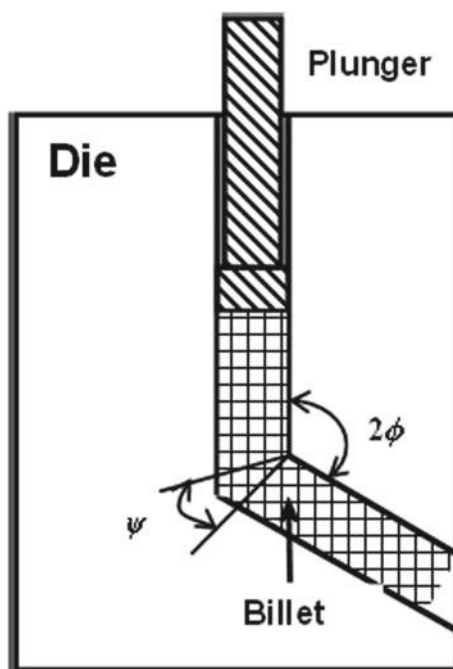
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(UFG) structures with an average grain size in the submicron range and which turn in improvement of physical and mechanical properties of materials with high strength and high ductility. Equal channel angular pressing (ECAP) is at present one of the most promising techniques that can produce bulk UFG materials.

The immediate and future requirements of materials in aerospace and automotive industries can be light metals with high strength and high ductility. The use of aluminium and its alloys are typical in many industrial applications. The strength and ductility of materials, which determine virtually all facets of its mechanical response, are primarily controlled by microstructure, of which the grain size and second phase distribution are important parameters. These in turn are dependent on the processing methods used for the fabrication of these metals and alloys. A typical schematic diagram of the ECAP process is shown in Fig. 36.1.

Due to the geometric constraints of the die, the workpiece deforms in pure shear within a small area at the intersection between the two channels. Indeed, ECAP is one of the well-designed processes for rods and bars to refine the microstructure of various metals and alloys in the modern era. In which very large amount of strains can be imparted onto metal billets by simple shear. The simple shear occurs as the billet passes through the section of an angular die where the entry and the exit channels meet. Since the overall dimension of the billet remains unchanged in this process, the billet can be pressed repeatedly a number of times, so that exceptionally high cumulative strains can be achieved. The microstructural development of ECAP is affected by geometrical as well as operational parameters like channel angle, channel

Fig. 36.1 Schematic diagram of the ECAP process



outer curvature angle, strain imposed, the friction between billet and channel walls, number of passes and processing routes.

Equal channel angular pressing (ECAP) is one of the material processing methods, which is subjected to severe plastic deformation (SPD) through simple shear without changing the overall dimension of the sample. This method was invented in 1972 and first described by Segal in 1974 [1, 2]. A typical schematic diagram of the ECAP process is shown in Fig. 36.1. The ECAP die consists of two equal channels that meet at an angle, usually between 90° and 135° . In the recent past, analytical studies regarding microstructural effects [3] on deformation mode and loading history [4] for estimating equivalent strain [5] in ECAP and numerical studies regarding the effect of material properties [6], die geometry [7, 8] including channel angles [9], processing routes, friction, multi-passes, backpressure, flow stress and grain refinement during ECAP for producing ultrafine-grained materials and also studied on working load, material flow, billet deformation and plastic deformation.

The magnitude of equivalent effective plastic strain (ε_{eq}) after N passes is given by the following relationship for ideal conditions of pressing [10]:

$$\varepsilon_{eq} = \frac{N}{\sqrt{3}} \left[2 \cot \left(\frac{\phi}{2} + \frac{\psi}{2} \right) + \psi \operatorname{cosec} \left(\frac{\phi}{2} + \frac{\psi}{2} \right) \right]$$

where ϕ is the channel angle of intersection and ψ is the channel angle of outer curvature.

Most of the analytical, numerical and experimental work reported to date on ECAP had been used channel angle less than 90° or 90° or a little more than 90° and there has been a little or no attempt to carry analytical or numerical or experimental work on combination of channel angles 120° and 90° and also compare the results obtained when using dies with different channel angles. In the present study, the numerical studies carried out the investigation on the plastic deformation behaviour of the pure Al during ECAP for the individual and also combination of channel angles 120° and 90° .

36.2 Finite Element Modelling

In the current study, the plastic deformation behaviour of Al specimen during ECAP was simulated for one pass using the commercial finite element analysis code ABAQUS/CAE 6.12-1. The deformation behaviour of a square cross-sectioned pure Al samples with dimension $20 \times 20 \times 100 \text{ mm}^3$ in an ECAP die with the channel angles $2\phi = 120^\circ$ and $\psi = 0^\circ$, $2\phi = 90^\circ$ and $\psi = 0^\circ$ and combination of channel angles $2\phi = 120^\circ$ and $\psi = 0^\circ$ and $2\phi = 90^\circ$ and $\psi = 0^\circ$ were simulated under plane strain conditions at room temperature. The sample meshed with four node plane strain element (CPE4R) of total 3125 elements and the analytical rigid element was considered for punch and die. To avoid distortion of the mesh due to a sharp inner corner, a small fillet radius of 5 mm was created on the die. Adaptive meshing was

used in all the simulations. Die was simulated in analytical rigid which has been fixed on reference point, and reference point was demonstrated by rotated cross on top wall of the die. A 2-D problem was considered as the process can be supposed to satisfy the plane strain condition. In the present study, the simulations were carried out at frictionless and frictional condition ($\mu = 0.1$). Heat generated due to the plastic deformation was not considered. The sample material was considered isotropic and homogeneous and the initial relative density uniform throughout the sample. Both the inner and outer channel surfaces were assumed to be rigid and stationary by imposing zero displacement boundary condition along the X- and Y-directions ($U_x = U_y = 0$). A displacement of 80 mm was given to the punch in the extrusion direction. In order to develop the simulations, a deformable sample was used, in which the properties for commercial purity Al (stress–strain relationship, $\sigma = 175 \text{ e } 0.329 \text{ MPa}$) have been taken into account. The material properties like density 2700 kg/m^3 , Young’s modulus 69 GPa and Poisson’s ratio 0.33 were incorporated in the FEM simulation.

36.3 Results and Discussion

Figure 36.2 illustrated von Mises stress distribution of the pure Al during ECAP of die channel angle 120° for (a) frictionless and (b) friction ($\mu = 0.1$) conditions, respectively. It was observed that the stress level was higher at the shear plane where two channels meet as well as along the top of the channel. It was found from the numerical analysis that in case of frictionless condition, the channel outer curvature and top of the channel gap exists but in case of friction condition, the channel outer curvature gap is very less and filled completely at the top of the channel as compared to frictionless condition. The stress levels were very high in contact with the channels as compared to the middle portion of the workpiece particularly; it was more stress levels in case of frictionless condition as compare to friction condition and also continues to follow in contact with the channels. It was also observed that the bottom

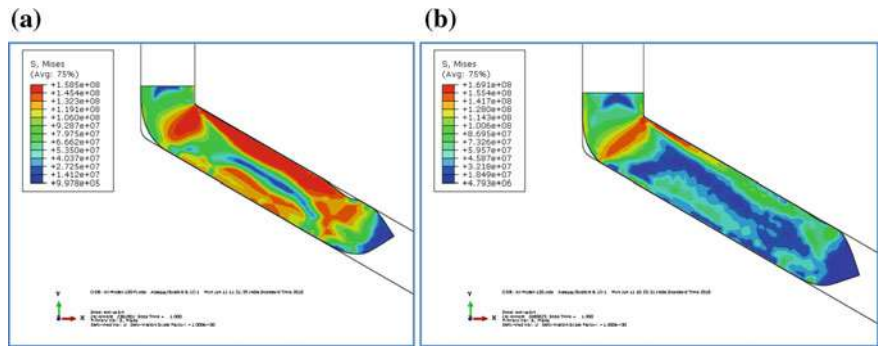


Fig. 36.2 von Mises stress distribution of the pure Al during ECAP of die channel angle 120° along the specimen for the **a** frictionless and **b** friction ($\mu = 0.1$) conditions

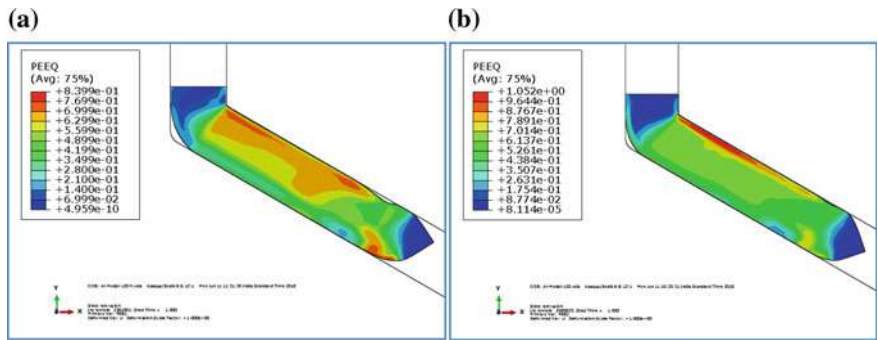


Fig. 36.3 Plastic equivalent strain distribution of the pure Al during ECAP of die channel angle 120° along the specimen for the **a** frictionless and **b** friction ($\mu = 0.1$) conditions

gap remains unfilled in both the cases even though after friction influence during the deformation of the material.

Figure 36.3 shows equivalent plastic strain (PEEQ) distribution of the pure Al during ECAP of die channel angle 120° for (a) frictionless and (b) friction ($\mu = 0.1$) conditions, respectively. The plastic deformation occurs at the plastic deformation zone and further no changes took place in the specimen. Apparently, the strain was not observed in the head and tail portions of the specimen. The bottom gap remains unfilled in both the cases but more gap in the case of frictionless condition.

Figure 36.4 shows von Mises stress distribution of pure Al during ECAP of die channel angle 90° along the specimen for (a) frictionless and (b) friction ($\mu = 0.1$) conditions, respectively. It was observed that the stress level was higher at the shear plane where two channels meet as well as along the top of the channel. It was found from the numerical analysis that in case of frictionless condition, the channel outer curvature gap exists but in case of friction condition, the channel outer curvature gap is almost filled as compare to frictionless condition. The stress levels were very high in contact with the channels as compared to the middle portion of the workpiece particularly; it was more stress levels in case of frictionless condition as compare to friction condition and also continues to follow in contact with the channels. It was

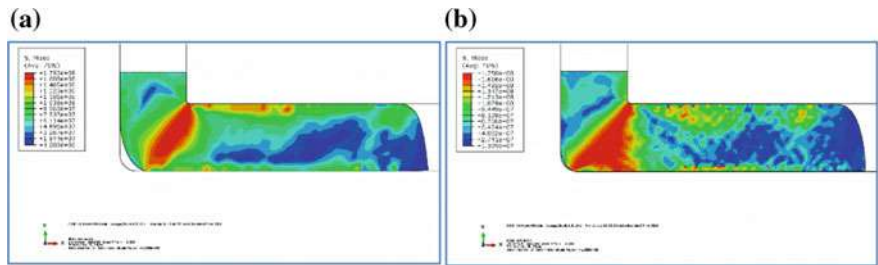


Fig. 36.4 von Mises stress distribution of the pure Al during ECAP of die channel angle 90° along the specimen for the **a** frictionless and **b** friction ($\mu = 0.1$) conditions

observed that the bottom gap remains unfilled in both the cases even though after friction influence during the deformation of the material. It was also observed that very high-stress values found in case of channel angle 90° as compare to channel angle 120° and the load required to press the samples from channel angle 90° was more than the channel angle 120° .

Figure 36.5 shows equivalent plastic strain (PEEQ) distribution of pure Al during ECAP of die channel angle 90° along the specimen for (a) frictionless and (b) friction ($\mu = 0.1$) conditions, respectively. The plastic deformation occurs at the plastic deformation zone and further no change took place in the specimen. Apparently, the strain was not observed in the head and tail portion of the specimen. The bottom gap remains unfilled in both the cases. The channel outer curvature gap is more in the case of frictionless condition as compare to friction condition. The strain distribution in channel angle 90° was very high as compared to channel angle 120° due to the strain produced was lesser for larger channel angles. There was a large amount of variations in strain within the steady-state zone in both the cases and this variation in strain was termed as strain inhomogeneity. Clearly, the strain inhomogeneity was low in case of channel angle 120° but it was very high in case of 90° .

Figure 36.6 shows equivalent plastic strain (PEEQ) distribution of pure Al during ECAP of combination of die channel angles 120° and 90° along the specimen for (a) frictionless and (b) friction conditions ($\mu = 0.1$), respectively. The plastic deformation occurs at the plastic deformation zone and further no change took place in the specimen. Apparently, the strain was not observed in the head and tail portions of the specimen. The strain distribution was more uniform as well as high in the case of friction condition as compare to frictionless condition. It was also observed that very high strain values found in the case of combination of channel angles (120° and 90°) as compare to individual channel angles (120° and 90°).

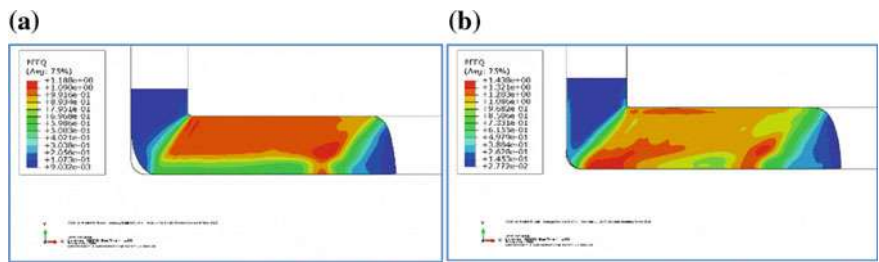


Fig. 36.5 Plastic equivalent strain distribution of the pure Al during ECAP of die channel angle 90° along the specimen for the **a** frictionless and **b** friction ($\mu = 0.1$) conditions

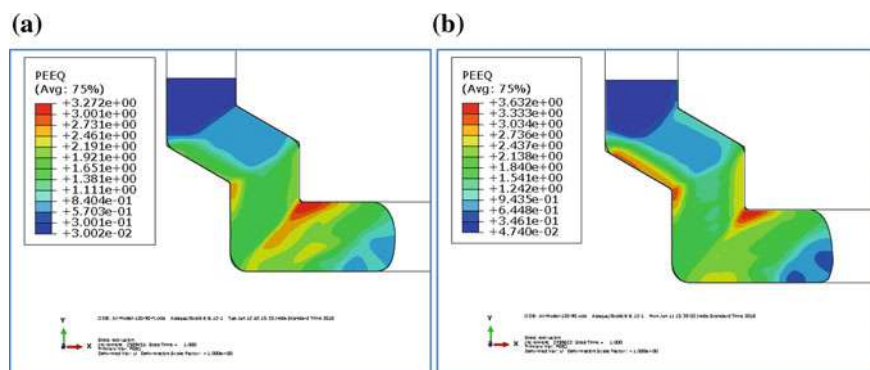


Fig. 36.6 Plastic equivalent strain (PEEQ) distribution of the pure Al during ECAP of combination of die channel angles 120° and 90° along the specimen for **a** frictionless and **b** friction ($\mu = 0.1$) conditions

36.4 Conclusions

Plastic deformation behaviour of the pure Al during ECAP of different channel angles and their combination was analysed under plane strain condition by assuming the generalized Coulomb law in all the cases through finite element modelling using ABAQUS/CAE 6.12-1. In the steady-state region, significant strain inhomogeneity exists across the specimen and die corners were not completely filled due to frictionless condition but die corners were completely filled due to friction condition. Strain inhomogeneity was lesser in channel angle 120° than in channel angle 90° and strain and load decrease with increasing die channel angle. The friction influence in case of combination of channel angles was negligible. The strain generation and distribution were more uniform in the case of combination of channel angles as compare to individual angles.

Acknowledgements This work was carried out as part of research promotion scheme (RPS) sponsored by All India Council for Technical Education (AICTE), Government of India and this assistance is gratefully acknowledged.

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Chapter 37

Teeth Wear Enhancement Along the Tooth Profile of Spur Gear Drive by Balancing the Fillet Stress Through Positive Correction Factor



R. Ravivarman , K. Palaniradja and R. Prabhu Sekar

Abstract This article suggests a modification in the correction factor which will improve the wear resistance for the equalized fillet stress in spur gear drive. As ruled, higher transmission proportional drives have an altering stresses in the fillet of pinion and the wheel. This fillet stress can be equalized by utilizing addendum alteration system. This changed addendum can bring down the fillet stress in the pinion subsequently enhancing the load conveying limit of the drives. Here, the fillet stress is assessed utilizing finite element analysis (FEA) technique for various blends of S+ drives. This correction factor is given in the pinion; wheel with the end goal that the bending strength of the drive is improved. At last, the fillet stress is equalized by varying the correction factors for every one of these drives, and the upgraded estimation of balanced strength with enhanced wear resistance is recommended for the spur gear drive.

Keywords Correction factor · Finite element analysis · Fillet stress · Spur gear drive · Wear

Nomenclature

AD	Line of action of the contact in mm
E	Young's modulus in GPa
F	Load in N
J_w	Wear coefficient in m^2/N
N	Speed at input of the drive in rpm
a_o	Center distance

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_37

b	Face width in mm
h	Teeth wear in mm
i	Transmission ratio
m	Module in mm
p_b	Base pitch in mm
r_{ap}	Radius of addendum for pinion in mm
s	Sliding distance in mm
v	Sliding velocity in m/sec
x'_i and y'_i	Cartesian coordinates of worn tooth
X	Correction factor
Z	Teeth number

Symbols

A	Pressure angle in degree
σ_H	Contact pressure in MPa
σ_t	Fillet stresses in MPa
θ'_i	Tooth thickness at contact in degree
N	Poisson's ratios

Subscripts

N	Direction along normal of tooth surface
a	Addendum
b	Base circle
g	Gear
i	Point of contact at any instant
o	Pitch point
p	Pinion
max	Maximum
w	Working circle

37.1 Introduction

Gear life period which is running unlubricated is reliant both on the fatigue strength and on the wear obstruction of the gear solid. Most of the design standards focus on the fatigue strength failure mode by discounting tooth wear. But it is clear that gears

fail at lower loads mostly because of wear that occur in tooth. To improve the performance properties, the involute gearing with the application of correction factors gets noticed when compared with uncorrected gears. Chernets et al. [1] derived a system for the assessment of the wear in the drives on its toughness and hardness with the influence of correction. Gunay et al. [2] discussed the impacts of addendum adjustment coefficient on spur gear tooth strength through finite element methods (FEMs). However, the displayed aftereffects of investigations on the impact of correction of the teeth on the contact and fillet stress are not inclusive [3] for wear prediction. Archard [4] was the first one to discuss the rubbing of surfaces which will cause wear in the teeth profile while they mesh one among another. Li [5] discussed the error in the machining and assembly of gear drive on load-carrying capacity and rate of loading with modifications in the gear tooth of a spur gear pair. In general, the gear drive will have a non-conformal contact with both the sliding and rolling components in a single gear tooth itself. Hence, it turns out to be necessary to study how the loading actually varies on the gear tooth which depends on deflection, tooth wear, sliding among the adjacent tooth pairs. The current article reports effects from a method which has been established to take into account the bending strength of the gear tooth profile and to deliver a more precise understanding of teeth wear in the tooth profile.

37.2 Corrected and Uncorrected Spur Gear System of Transmission

In a spur gear system of transmission, the two sets of teeth originate its contact during the start of reach, one at highest point of tooth contact-A (HPTC) and other in lowest point of single-tooth contact-C (LPSTC) of the drive as appeared in Fig. 37.1. Over an activity of span, the main set leaves the contact at lowest point of tooth contact-D (LPTC), and consequently, only trailing pair will be transmitting the movement from the highest point of single-tooth contact-B (HPSTC) to C (LPSTC), that is BC set will be engaged until the point of another approaching pair sets up the contact at A to start two sets in contact yet again. Prabhu Sekar [6, 7] made an endeavor to consider the distinctive points of contact for standard spur gear drive-based design model of balanced fillet stress. Thus, as suggested at the point when the transmission ratio (i) is in excess of one, the fillet stresses of the wheel and pinion will be dissimilar. The present work intends to adjust the fillet stress of the wheel and pinion through profile correction. In standard spur gears, the correction factor is maintained at zero. Here the correction factor (Fig. 37.2) is given in such a manner that the maximum fillet stress between the wheel drives is balanced.

Radial distances [6] conforming the contact points for corrected gears are specified by Eqs. (37.1)–(37.4).

$$r_{\text{HPTC}} = r_{\text{awp}} \quad (37.1)$$

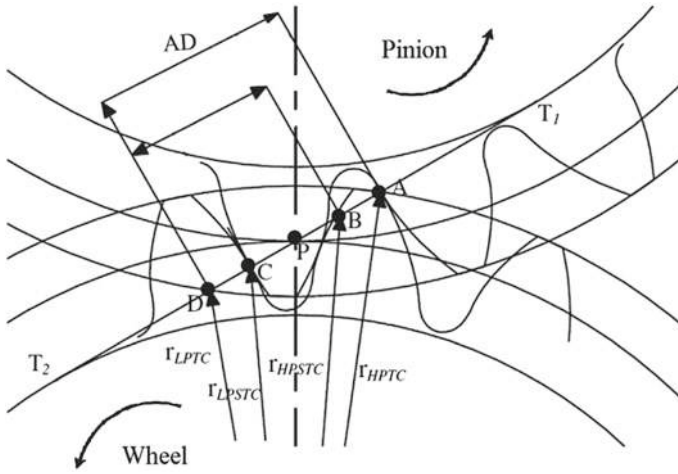
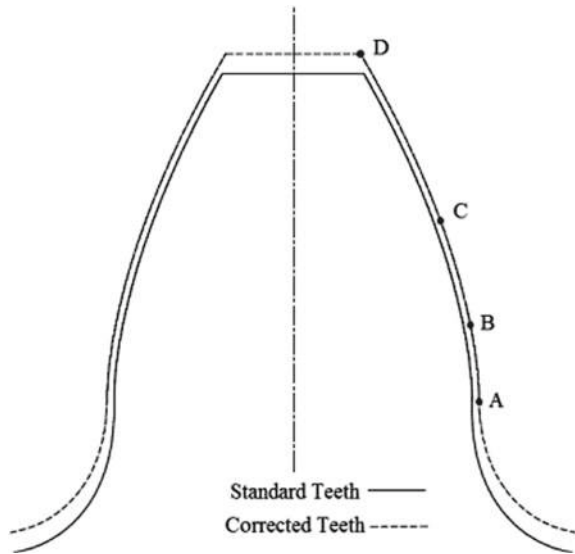


Fig. 37.1 Contacting points of the spur gear drive

Fig. 37.2 Layout of the teeth profile



$$r_{HPSTC} = \sqrt{\left(\sqrt{r_{awp}^2 - r_{bp}^2} + p_{bw} - AD\right)^2 + r_{bp}^2} \quad (37.2)$$

$$r_{LPSTC} = \sqrt{\left(\sqrt{r_{awp}^2 - r_{bp}^2} - p_{bw}\right)^2 + r_{bp}^2} \quad (37.3)$$

$$r_{LPTC} = \sqrt{\left(\sqrt{r_{awp}^2 - r_{bp}^2} - AD\right)^2 + r_{bp}^2} \quad (37.4)$$

$$AD = \sqrt{r_{awp}^2 - r_{bp}^2} + \sqrt{r_{awg}^2 - r_{bg}^2} - a_w \sin \alpha_w \tag{37.5}$$

37.3 FEM of the Spur Gear System of Transmission

A bidimensional component model of the spur gear drive is set up for the examination. Figure 37.3 goes on the defensive contact model of the spur gear design drive taken for investigation. The coding was generated in ANSYS parametric design language (APDL) which has been utilized to create trochoidal and involute profile of the teeth. For cross section of the drive display, 2D-PLANE42 with four nodes is employed. CONTA172 and TARGE169 are utilized for pinion and wheel independently for contact analysis. The wheel and pinion drive parameters taken for FE analysis are shown in Table 37.1. A full-round tip rack shaper is taken for the teeth fillet, and the load dissemination is thought to be uniform with expecting a plain strain condition. Material is supposed to be linear elastic, isotropic, and homogenous. To the extent of the periphery condition, the pinion is controlled in radial direction and the wheel

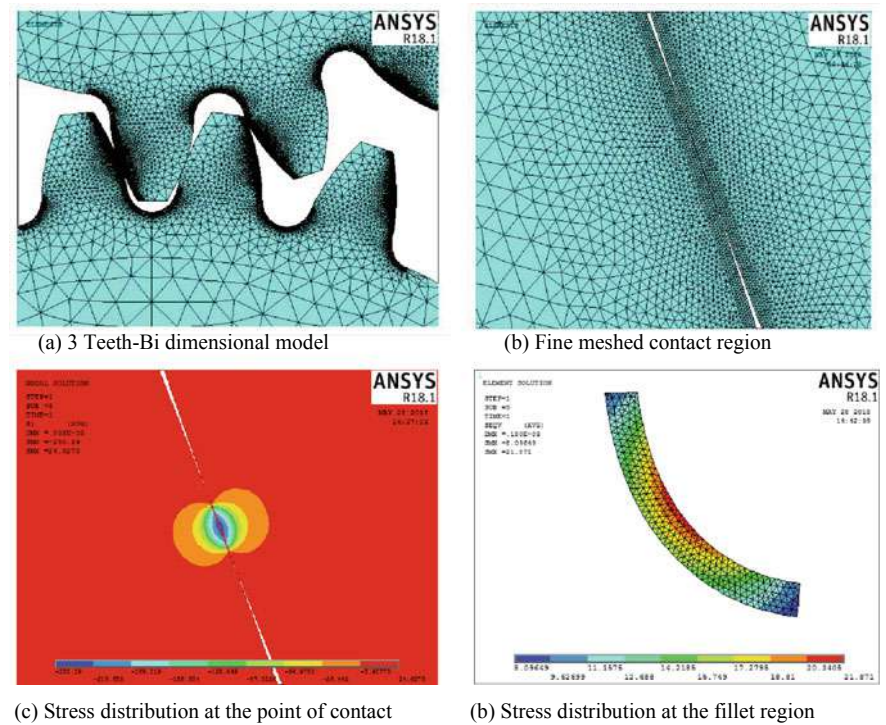


Fig. 37.3 FEA simulated outcomes for the corrected gear drive

Table 37.1 Gear constraints

S. No.	Factors	Values
1	Transmission ratio (i)	1.5
2	Teeth number in pinion (z_p)	20
3	Normal load (F_N) (N)	10
4	Young's modulus (E) (GPa)	210
5	Pressure angle (α_o) ($^\circ$)	20 $^\circ$
6	Rim thickness (mm)	5 m
7	Module (m) (mm)	1
8	Speed at input (N_p) (rpm)	500
9	Addendum height (h_a) (mm)	1 m
10	Poisson's ratio (ν)	0.3

is controlled in both radial and tangential directions. As indicated by convergence investigation, the fillet regions and the contact regions are characterized as critical zones which have been fine-meshed, and the rest of the regions are coarse fit. Figures 37.3c, d show the contact and fillet stress variation along the contact and fillet regions in finite element analysis using ANSYS software. From Fig. 37.3d, it is obvious that the stress is high at the midpoint region of the fillet part which is considered to be the critical zone.

Anderson and Erikson [8] proposed a single-point perception-based contact technique for ascertaining the collected teeth wear is given by

$$h_{i,n} = h_{i,(n-1)} + J_w(\sigma_H)_i (s_p)_i \quad (37.6)$$

Sliding distance (s_p) of the drive at any prompt contact focuses is given by [9]

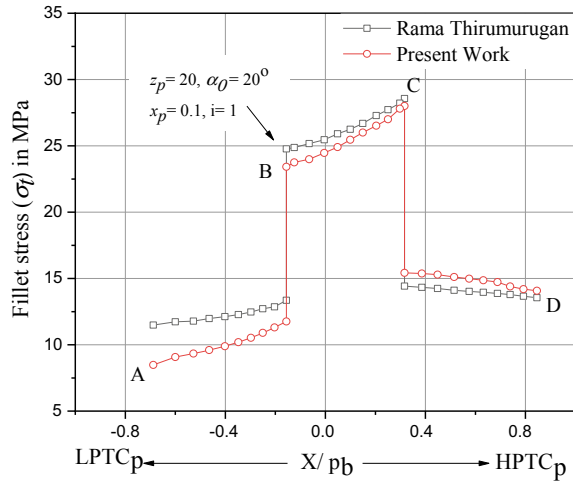
$$(s_p)_i = 2a_i \left(\frac{(v_p)_i - (v_g)_i}{(v_p)_i} \right) \quad (37.7)$$

Here the semi contact width (a_i) [9] created at any prompt contact focuses is specified as

$$a_i = \sqrt{\frac{4F_i}{\pi b} \frac{\frac{(1-\nu_p^2)}{E_p} + \frac{(1-\nu_g^2)}{E_g}}{\frac{1}{(R_p)_i} + \frac{1}{(R_g)_i}}} \quad (37.8)$$

Using FEA, the contact pressure (σ_H) is resolved, and the estimated depth of wear over the tooth surface for every 100 mesh cycle is figured. After that the tooth profile is refreshed. The collected maximum amount of teeth wear is resolved for around 5000 mesh cycles. For discovering the new coordinates of the worn out tooth profile, the equations are given as

Fig. 37.4 Results' comparison with literature [10]



$$x'_i = x_i - h_i \cos \theta'_i \quad (37.9)$$

$$y'_i = y_i - h_i \sin \theta'_i \quad (37.10)$$

From Fig. 37.4, it is clear that the results obtained from the FEA analysis have similar trend to those obtained by Thirumurugan and Muthuverrappan [10]. The results obtained from their study are compared and shown in the plot for $z_p = 20$, $x_p = 0.1$, $i = 1$, $\alpha_o = 20^\circ$. The variation in stresses is noticed in the literature because the study was carried out in loaded model, whereas in the present work, it is done with contact model. Hence, the contact load and their corresponding effects will come into play during analysis.

37.4 Results and Discussion

For the geometry of gears confirmed, the simulated performance of drive was found to be completely reliant on the load distribution along the path of contact. Figure 37.5a predicts the load share distribution between the standard gear drive and addendum modified gear drive with balanced fillet stresses. It is noted that the contact ratio decreases in the corrected gear due to positive correction given to the pinion tooth. Because of this reduced contact ratio, the AB and CD contact regions with double pair decrease, while at the same time, the single-pair contact region (BC) increases proportionally in the addendum modified pinion. In case of the stresses concern, the extreme high maximum fillet stress is balanced in the corrected drive ($\sigma_{t \max p} = \sigma_{t \max g} = 24.625$ MPa). and it is lesser (Fig. 37.5b) than the gear drive of standard case ($\sigma_{t \max p} = 25.734$ MPa, $\sigma_{t \max g} = 24.178$ MPa) which is uncorrected. This is

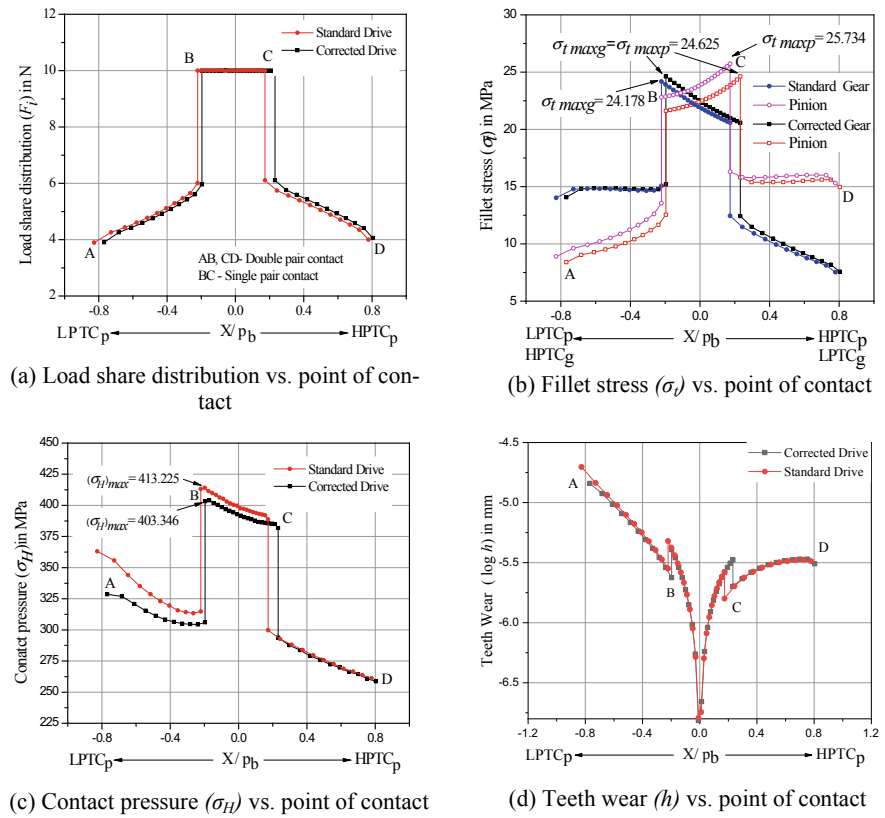


Fig. 37.5 Effect of correction factor along the tooth profile

mainly due to the effect of enlarged critical tooth thickness in the pinion tooth and reduced tooth thickness at the wheel tooth.

The change in pressure variation at the contact points will be accounted due to the blended teeth interaction of standard gears, and corrected one has been compared in Fig. 37.5c and Table 37.2. Their maximal values emerge at the passage when the engagement of single tooth initiates. As correction is given in the drives, the contact

Table 37.2 Data comparison

Correction factor (x_p and x_g)	Balanced maximum fillet stress (σ_t) _{max} (MPa)		Contact pressure (σ_H) _{max} (MPa)
	Wheel	Pinion	
0	24.178	25.734	413.225
0.1 and 0	24.625	24.625	403.346

pressure decreases in the flank region of the double- and single-tooth contacts (AB and BC) because of increment in equivalent radius of curvature at the contact points from a maximum value of $(\sigma_H)_{\max}$ of 413.225–403.346 MPa. The teeth wear curve against loading points is shown in Fig. 37.5d, for standard and corrected gear drives. The trend of teeth wear is similar to those obtained from the standard drives. As shown in Figure 37.5d, the maximum teeth wear occurs at engaging of the teeth loading point (A) in the flank portion. Here in the corrected drives, there is a decrease in velocity ratio and contact pressure along the flank portion which ultimately increases the equivalent radius of curvature. Hence, the teeth wear decreases significantly in the flank portion of the drive. Due to pure rolling at the pitch point, there tends to be no wear and zero sliding. But in the case of the face portion, the wear is slightly increased due to increase in sliding velocity at this region. Thus, it is evident that the corrected drives have improved wear resistance compared to standard gear drives.

37.5 Conclusions

In this article, a detailed examination on the fillet stress of the gear drive is carried out to balance it through correction factor using FEM technique. Balancing of maximum fillet stress is done by applying positive correction over the drive which will in turn reduce the stress in the pinion. This method of reducing the stress will convert the drive into a balanced one among the wheels. It is also proposed that the application of correction in the gearings will lead to improvement in the bending strength and contact strength of the teeth, thereby increasing the load-carrying ability of the drive. As an effect of the above said, the contact pressure will be lowered which in turn decreases the teeth wear and eradicates interferences among the drives. Correction will also increase the equivalent radius of curvature which is a major influencing parameter over the reduction in the amount of contact pressure between the operational surfaces of the teeth profile and thereby enhancing the wear resistance.

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Chapter 38

A Coupled Thermal-Structural Model for Welding of Aluminium Alloy Sheets



Tapas Bajpai, H. Chelladurai and M. Zahid Ansari

Abstract Computational simulations using the finite element method are advantageous in predicting the response of the weldments during the welding processes. These welding simulations help in ensuring the correction in the process design to compensate for the effects of the welding before the commencement of the actual fabrication process. In this work, a coupled thermo-mechanical finite element model is presented for simulating the gas metal arc welding process on thin aluminium alloy plates. For modelling the thermal and mechanical behaviour of the weldments, finite element ANSYS software is used. Temperature-dependent properties of plates are used in the simulation. Effects of conduction and convection due to air and argon gas are considered. For modelling the welding heat source, Goldak's double ellipsoidal heat flux distribution is implemented. With the help of finite element solutions, transient temperature and transient stress distribution in aluminium alloy weldments are estimated.

Keywords Aluminium alloy (AA) · Finite element modelling (FEM) · Heat-affected zone (HAZ)

38.1 Introduction

Aluminium–magnesium alloys are used in the automotive industry, aerospace applications, transport and marine applications. Gas metal arc welding (GMAW) is a widely used process for welding of aluminium alloys as it offers high welding speed, ability to join both thick and thin plates and higher heat input rate [1, 2]. The GMA

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_38

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welding of thin aluminium alloy sheets may lead to severe residual stresses and distortions. These residual stresses and distortions are detrimental as they affect fatigue performance, influence the buckling strength and causes deterioration of dimensional accuracy which eventually leads to costly rectifications. Hence, understanding the origination of residual stresses and distortions in weld structures is very important [3]. Calculating residual stresses and distortions in a three-dimensional body is a difficult task due to the complex thermo-elastic-plastic state developed during the gas metal arc welding process. Therefore, numerical techniques are adopted for solving the thermal-structural problem. The finite element (FE) numerical technique is a popular method to predict residual stresses and deformations in the welding process. With the inception of modern computing capabilities, the FE technique has established itself as an effective tool for determining residual stresses and distortions. Nowadays, this technique is widely used in manufacturing industries as it improves productivity and quality of products and helps in the more appropriate understanding of the influence of various process parameters. FE welding simulations are also beneficial in establishing methods and models for control and design of welding processes.

Zhu and Chao [4] utilised temperature-dependent material properties for simulating residual stresses and distortions in aluminium alloys. Bajpei et al. [2] used the ANSYS software to analyse the thermal-structural behaviour of thin dissimilar aluminium alloys AA5052 and AA6061 plates during GMA welding process. For simulating the welding process, Goldak's double ellipsoidal model was utilised. A 3D finite element model considering temperature-dependent material properties was employed for analysing the thermo-mechanical response of dissimilar aluminium alloy plates. Bhatti et al. [5] investigated the effect of various thermo-mechanical properties of different stainless steel grades (S355–S960) on residual stresses and distortions on T-fillet joints during GMA welding process. A finite element model considering the temperature-dependent thermo-mechanical material properties is employed for performing the simulations. For the elastic-plastic mechanical analysis, bilinear hardening material model with von Mises yielding criteria is used. Kumaresan et al. [6] developed an FE model to predict the transient stress distribution in aluminium alloy plates. Finite element "element death and birth" codes were written in ANSYS software for simulating material addition in the welding process. Mi et al. [7] presented a coupled thermal and metallurgical model for predicting the mechanical behaviour of Ti-6Al-4V plates during tungsten inert gas welding process.

From the literature survey, it is found that there are few investigations were done to predict the transient structural response of the thin weldments using the FE method. Hence, an attempt is made to develop a sequentially coupled thermal-structural finite element model to determine transient residual stresses in AA sheets during GMA welding. To simulate the welding process, ANSYS Workbench v.15 software is used. Goldak's volumetric heat source model is applied during the analysis. In the past work carried out by Bajpei et al. [1, 2], the number of elements used for the FE analysis is more which increased the simulation time. Hence, in this work, an attempt is made

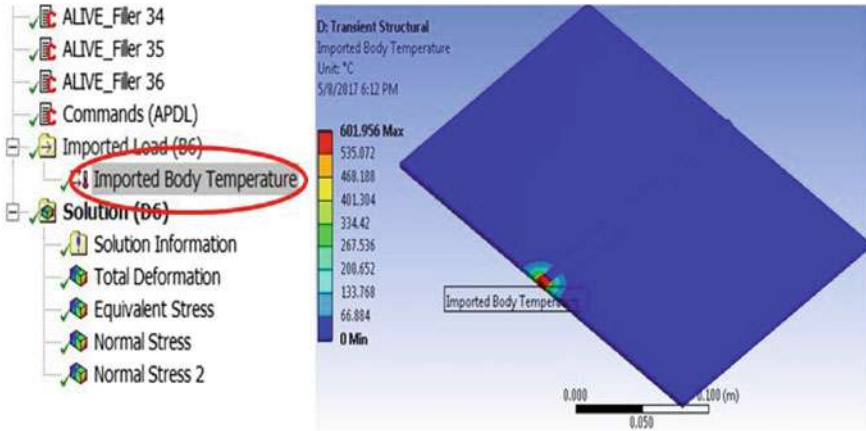


Fig. 38.1 Imported load from the transient thermal component system for structural analysis

to develop an FE model for the welding process in order to decrease the simulation time.

38.2 Numerical Approach for Thermal-Structural Analysis

For simulating the welding process, different component systems are linked together in ANSYS software to perform the thermal-structural computations. The simulation methodology is branched in two different sequentially coupled simulations. For obtaining the temperature distribution on weld specimens, a transient thermal nonlinear analysis is carried out, and subsequently, iterative structural analysis is performed. The obtained thermal results computed at all node points during transient thermal analysis were stored. Subsequent, imported thermal loads as shown in Fig. 38.1 were utilised for executing structural computations.

38.3 Heat Transfer Analysis During Welding

In the welding simulation, the partial differential equation for transient heat conduction in a 3D body is expressed as [8]

$$\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q}_g = \rho c_p \frac{\partial T}{\partial \tau} \quad (38.1)$$

where k_x , k_y , k_z are the thermal conductivities, $T(x, y, z, \tau)$ is the temperature and τ is time.

In the present work, a 3D double ellipsoidal heat source model was adopted to calculate the volumetric heat flux. The rate of volumetric heat flux distributions of the heat source is represented as [9, 10]

$$q_f(x, y, z, t) = \frac{6\sqrt{3}f_f Q}{a_f b_f c_f \pi \sqrt{\pi}} \exp \frac{-3x^2}{a^2} \exp \frac{-3[y + s \times (\tau' - t)]^2}{b_f^2} \exp \frac{-3z^2}{c^2} \left[\frac{\text{J}}{\text{m}^3 \text{s}} \right] \quad (38.2)$$

$$q_r(x, y, z, t) = \frac{6\sqrt{3}f_r Q}{a_r b_r c_r \pi \sqrt{\pi}} \exp \frac{-3x^2}{a^2} \exp \frac{-3[y + s \times (\tau' - t)]^2}{b_r^2} \exp \frac{-3z^2}{c^2} \left[\frac{\text{J}}{\text{m}^3 \text{s}} \right] \quad (38.3)$$

where x, y, z are the Cartesian coordinates, $\tau = \frac{t}{s}$ (seconds), s is the travelling speed in mm/s, r is a characteristic radius of flux, $Q = \eta VI$ is the arc power and η is assumed as 75% [11]. The values of the constants were determined from the experimentation. The details of the experimental set-up and constants values were provided in the research literature provided by Bajpei et al. [1]. During the finite element modelling of heat source, the origin of the coordinate system is placed at the centre of the moving heat source for simulating the movement of the weld heat source.

The heat dissipation between the weld plate and its surroundings takes place by convection and radiation. However, in this investigation heat transfer due to radiation is not considered. The convective heat transfer coefficients for air and shielding gas used in the analysis [1]. For the mechanical analysis, the plate is constrained along the z -axis and negative y -axis. The clamp size in the analysis is 100 mm $\times$ 40 mm. For obtaining the temperature distribution, TEMPERATURE PROBES in ANSYS software were placed at different distances as illustrated in Fig. 38.2.

38.4 Finite Element Analysis

To perform the finite element thermal-structural analysis, bilinear kinematic hardening model is employed for the material of weldments. Figure 38.3 shows the 3D finite element mesh model of butt welded sheets used in the welding analysis. The dimensions of the welded sheet were 200 mm $\times$ 300 mm $\times$ 3 mm. A non-uniform mesh is used to minimise the computational time. Fine meshing was provided near to the fusion zone and coarse meshing for distant zones. For thermal analysis, the meshing of the model is done with three-dimensional SOLID90 elements, while 3D SOLID186 elements were utilised for structural analysis. Finite element ANSYS Workbench software was used to execute the thermal-structural analysis. Total 14,963 elements were used in the analysis. Total elapsed time for thermal computation is about 5 min. The previous computation time is about 12 min with 19,620 elements.

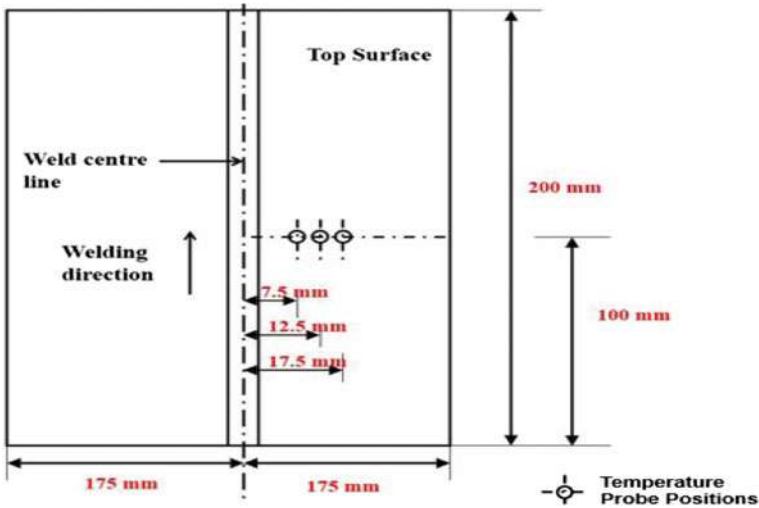


Fig. 38.2 Data points for temperature measurement

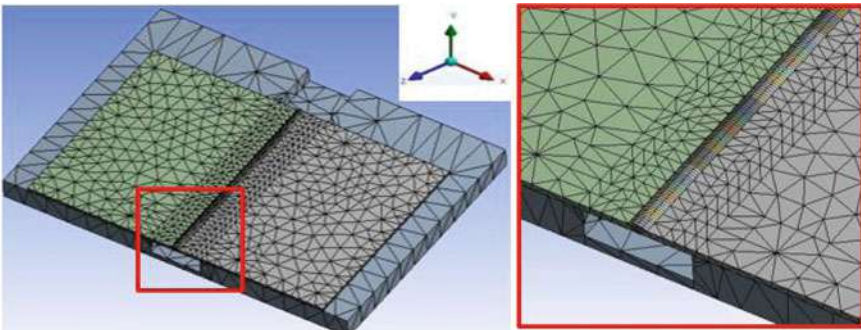


Fig. 38.3 Finite element mesh model of butt welded sheet

38.4.1 Material Modelling

During the simulation study, the material properties used in the FE analysis are taken from [1–3]. The chemical composition (wt%) of material used in the analysis is Al (97.25), Mg (2.7), Fe (0.30), Mn (0.08), Zn (0.07), Si (0.20) and Cr (0.30). Properties of backing plate used in the simulation are $\rho = 7700 \text{ kg/m}^3$, $\alpha = 11.5 \times 10^{-6}/^\circ\text{C}$, $k = 60 \text{ W/m } ^\circ\text{C}$ and $c_p = 0.46 \text{ kJ/kg } ^\circ\text{C}$. Poisson's ratio (ν) value in the analysis was taken as $\nu = 0.33$. The tangent modulus was kept as 0.5% of Young's modulus. Effect of latent heat is also incorporated in the simulation [1].

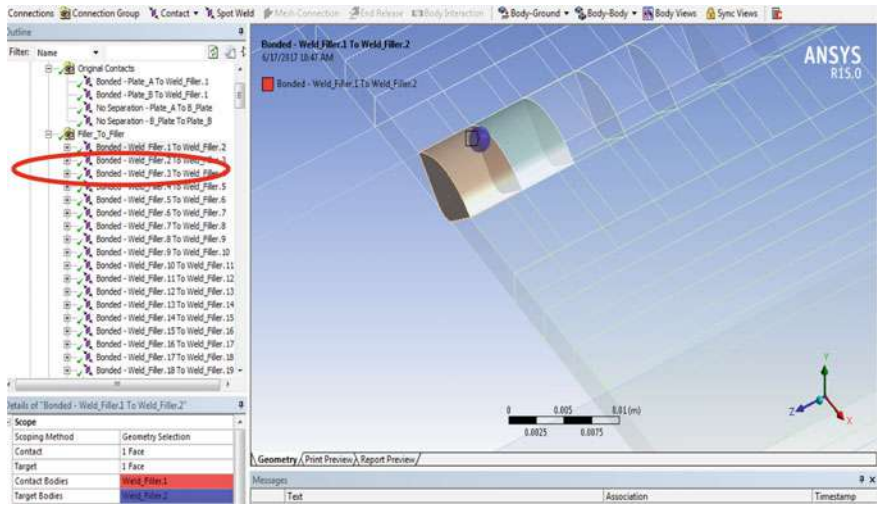


Fig. 38.4 Contact pairs in FEM model between filler–filler material

38.4.2 Contact Modelling

Due to intermittent loading and boundary conditions, surfaces come into and go out of contact with each other unexpectedly. Thus, it is necessary to define the contact regions while simulating welding problems between the filler material and plates in order to enable heat transfer between them. During the simulation, the surfaces (filler and weld plates) to be joined come into contact with each other. Thus, a 3D surface-to-surface contact pair CONTACT174 and TARGET 170 element were used to create contact between the filler material and weld plates [12]. Figure 38.4 shows the contact pair between the filler material and the weld material.

38.4.3 Element Death and Birth Technique

Modelling of filler material addition during welding is significant not only in obtaining transient thermal fields but also to incorporate the structural response of material occur due to the contraction of filler material during the solidification process [12]. In this work, subroutines were written to simulate the filler material addition during the welding process using element “death and birth” codes. Initially, elements in the fusion zone are deactivated, and further elements were reactivated as the weld torch advances.

38.5 Results and Discussion

The transient temperature distributions in the sheets were determined at a distance of 7.5, 12.5 and 17.5 mm away from the weld line in the transverse direction. Figure 38.5 shows the comparison of numerically obtained transient temperatures history recorded at different distances. The peak temperature obtained in the HAZ is about 361 °C. As the heat source moves in the welding direction, the temperature starts reducing due to heat dissipation. The heat is also dissipated due to the presence of clamps and mild steel backing plate.

Figure 38.6 illustrates the transient temperature contour in the aluminium alloy sheets when GMAW heat source moves in the welding direction. From the figure, it is seen that fusion zone experience temperature of about 696 °C indicating melted material. As the weld torch advances, heat is conducted, and thus, the temperature has decreased to the range of 253–165 °C. The results obtained from transient thermal analysis were utilised to perform the transient structural analysis.

Figure 38.7 shows the transient longitudinal and transverse stresses in the sheets during the finite element analysis. From figures, it is seen that 3D residual stress is developed around the weld zone. Tensile stresses were produced in the HAZ due

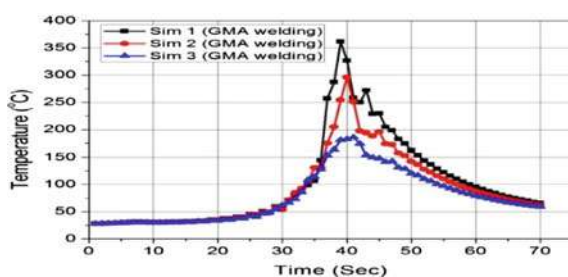


Fig. 38.5 Numerical transient temperatures in AA5052 sheets

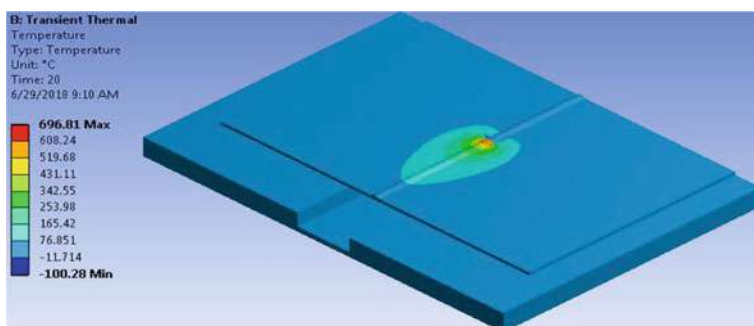


Fig. 38.6 Transient temperature distribution at $t = 20$ s

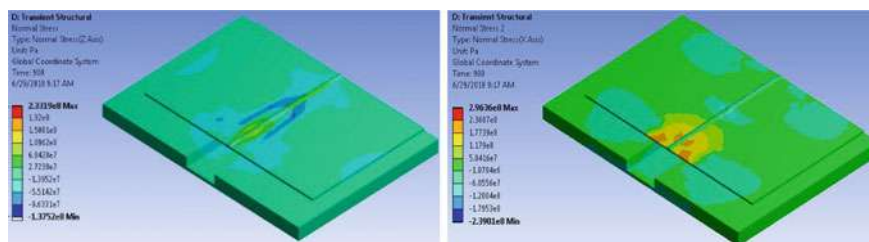


Fig. 38.7 Contour plots of transient longitudinal and transverse stresses in AA sheets

to the contraction of aluminium alloy sheets during cooling. It is also observed that compressive stresses were produced in the sheets away from the weld line.

38.6 Conclusions

Based on the results obtained from the FE numerical simulations, the following conclusions are made:

- FE analysis shows that heat dissipation during welding is majorly due to conduction phenomenon.
- Thermal-elastic-plastic finite element methods coupled with Goldak's double ellipsoidal heat source model can be effectively used to determine the residual stresses of thin sheets during GMA welding.
- Element death and birth technique is an effective technique for modelling filler material addition in GMA welding.

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Chapter 39

Numerical Modelling and Simulation of Single and Multi-spark Impacts in Electrical Discharge Machining



Jibin T. Philip , Basil Kuriachen and Jose Mathew

Abstract In this paper, the 2D surface model of single and multi-spark impacts on electrical discharge machining (EDM), with precise consideration of spark propagation, has been developed and simulated. Theoretical correlation between the input parameters, viz. discharge voltage (V): 30–110 V, discharge current (I): 5–75 A and spark on time (T_{on}): 10–200 μs , were preliminarily established, using the governing equations. The scope of the paper was to model the spark impact phenomenon, so as to determine the most influential factors which can be controlled to produce the required surface finish, for specific applications. Fine/finish machining is achievable at low discharge current, moderate discharge voltage and medium pulse on time, whereas coarse machining requires reverse conditions, preferably. Multi-spark analysis imparts insight into the possibilities in prediction and evaluation of material removal rate (MRR) and surface roughness (R_a) through further design considerations.

Keywords EDM · Modelling and simulation · Single and Multi-spark analyses

39.1 Introduction

Researchers around the globe have attempted to model the EDM process from the early 1970s to fathom the electrical plasma formation, spark propagation and the resultant electrode erosion. Two different mechanisms have been reported so far, to interpret the corresponding material removal mechanism, viz. electromechanical analysis [1] and electrothermal analysis [2–6]. The latter gained popularity and enormous attention, since the material removal in EDM takes place mainly due to

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_39

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the extreme heat pulses generated between the cathode and anode materials. The thermal affections of the EDM process on the workpiece material pose a dire need to perform the temperature distribution analysis to get detailed insight about the thermally affected layers and the corresponding microstructural changes forced on to the material surface. Also, melting temperature curves can delimitate a specific area which is further used for measuring the geometry of the crater [7]. The need and significance in interpretation of single spark EDM process is that, to produce a desired surface integrity, the mechanism should be essentially controlled from the initial stage of spark erosion and formation of cavities. Although, many pioneering works have been carried out in development of single discharge models, conforming to the reality, the efforts by DiBitonto et al. [2] and Patel et al. [3] have and will always remain as the cornerstone for all the models already represented and yet to come. Although, a lot of experimental work has been carried out as the efforts to model the EDM process, no distinguished work was developed so far which could interpret the actual conditions. There are many underlying reasons for this imperative situation. It includes: complexity in proper understanding of plasma channel formation, the stochastic nature of EDM process, the discharge duration being miniscule, etc. [7]. In view of the above, a spark impact model of EDM process has been developed in ANSYS R14.5, with Ti6Al4V as the workpiece material. The effect of the input parameters has been evaluated for determination of the individual effects on crater geometry, for finish and coarse machining considerations. Single spark model was initiated preliminarily which was then efficaciously modified to generate and study the multi-spark thermal effects on the work material. The multi-spark modelling and simulation is a distinctive effort, which has the potential for being the milestone for further advancements.

39.2 Modelling Considerations

39.2.1 Workpiece Domain

Initially, the single spark model was planned and developed, and the workpiece domain was taken as a square section ($250\text{ }\mu\text{m} \times 250\text{ }\mu\text{m}$) in 2D. The schematic representation of the 2D modelling consideration of the domain (black-shaded region with length, x and breadth, y) is as shown in Fig. 39.1. The $q(x)$ represents the heat flux distribution, which follows a Gaussian curve (depicted by the red shaded region), as per the design considerations. Moreover, r_s represents the spark impact radius on the workpiece domain. An extended domain size (larger length and shorter breadth) was selected for multi-spark impact model development and simulation, based on specific parametric values chosen from the single spark analysis model.

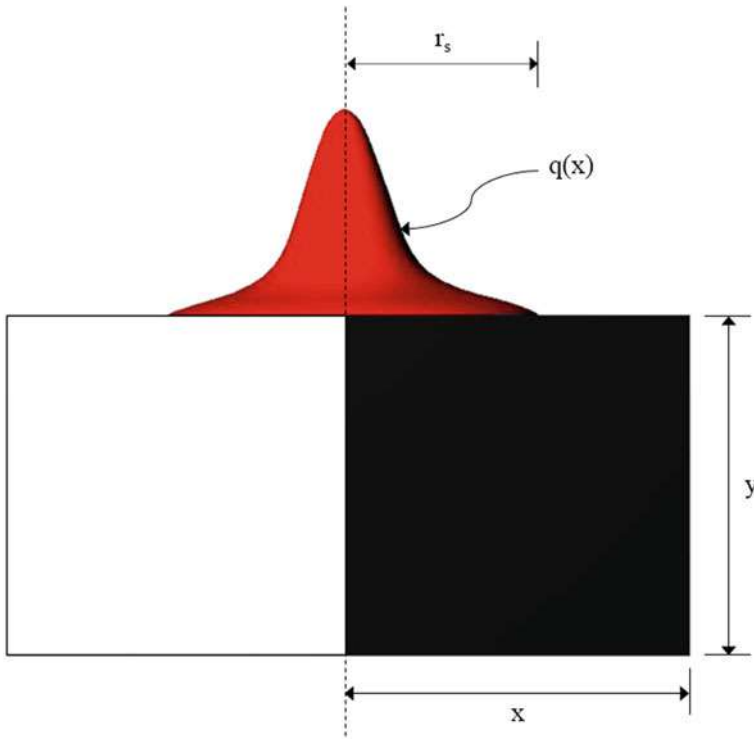


Fig. 39.1 Schematic representation of the workpiece domain and heat flux distribution

39.2.2 Governing Equations

The theoretical parametric correlation was formulated between the input and the output factors based on the existing EDM process governing equations which are discussed below. Foremost, the Fourier heat transfer equation,

$$\partial^2 T / \partial r^2 + 1/r \partial T / \partial r + \partial^2 T / \partial z^2 = 1/\alpha \partial T / \partial t \quad (39.1)$$

where T is the temperature, r (m) is the radial axis, Z (m) is the vertical axis, t (s) is time and α (m^2/s) is the thermal diffusivity of the material.

$$\alpha = K_t / \rho C_p \quad (39.2)$$

where K_t is the thermal conductivity of the material (J/mKs), ρ is the material density (kg/m^3) and C_p is the specific heat (J/kg K).

The spark impact radius [8] is given by,

$$r_s = (2.04e - 3) I^{0.43} T_{\text{on}}^{0.44} \quad (39.3)$$

For Gaussian distribution, heat flux is,

$$q(x) = q_0 \exp\{-4.5 (r/r_s)^2\} \quad (39.4)$$

where q_0 is the maximum heat flux, given by,

$$q_0 = (4.57 F_c V I) / (\pi r_s^2) \quad (39.5)$$

where F_c is the fraction of total EDM spark power going to the electrode, V (V) is the discharge voltage, I (A) is the discharge current and r_s (μm) is the plasma channel radius.

39.2.3 Assumptions

The following assumptions were considered for the simulation of the single and multi-spark analyses of the EDM process.

- A transient thermal model was selected to be developed.
- The workpiece material is homogeneous, isotropic and relieved of residual stresses prior to machining.
- The material properties of the workpiece are temperature-dependent.
- The predominantly accepted Gaussian shape was chosen to be the geometry of the heat source (Fig. 39.1).
- The fraction of spark energy (F_c) transferred to both the electrodes remains constant for each pulse duration.
- Maximum flushing efficiency is taken, i.e. 100%.
- In the case of multi-spark simulation, the secondary spark initiation on the material surface takes place along the periphery of the plasma radius of the previously developed craters.

39.3 Methodology

The modelling and simulation analysis of the EDM process using ANSYS R14.5 software was carried in three distinct steps, namely pre-processing, solution and post-processing. The pre-processing part encapsulates the model development, definition of material properties and domain mesh generation. The solution part is the core, which carries out all the vigorous numerical analysis associated with the defined process. Thereafter, the post-processing is performed to plot, investigate and interpret the results obtained. The entire simulation procedure is governed by the finite element-based approach.

39.4 Results and Discussion

39.4.1 Single Spark Analysis

The temperature distribution plot for a chosen input parametric setting of $V = 30\text{ V}$, $I = 75\text{ A}$ and $T_{\text{on}} = 200\text{ }\mu\text{s}$ is as shown in Fig. 39.2a. The final crater geometry, after elimination of the elements which possess temperatures above the melting point of the workpiece domain, is as in Fig. 39.2b. The crater formed was found to have a paraboloid geometry in contrast to bowl shaped, shallow bowl shaped, hemispherical, crescent-like shaped craters predicted and experimentally obtained by various researchers (Fig. 39.3), for high thermal conductivity materials. It is interesting to note that when AISI W1 tool steel (thermal conductivity, $k = 32\text{ W/mk}$, specific heat capacity, $\text{SHC} = 461\text{ J/kg K}$) and steel with 0.1% carbon ($k = 45\text{ W/mk}$, $\text{SHC} = 420\text{ J/kg K}$) were used as the workpiece material, the crater radius-to-depth ratios of the obtained craters were 2.3278 and 4.6428, respectively. But, in our model where the workpiece domain is Ti6Al4V ($k = 6.7\text{ W/mK}$, $\text{SHC} = 553\text{ J/kg K}$), the crater radius-to-depth ratio was found to be 2.8554, for the aforementioned parametric setting. Although the thermal conductivity of Ti6Al4V is comparatively very low, the crater radius-to-depth ratio of the EDM process developed crater falls in between that of AISI W1 tool steel and steel with 0.1% carbon, having better thermal properties. This peculiar characteristic can be attributed to the fact that, beyond the thermal characteristics of the material, the influence of the process parameters has the capacity to influence the entire erosion phenomenon as a whole.

Now, it is of sublime importance to evaluate the effect of the process input parameters on crater morphology and its dimensions, to get a much clearer perspective of their respective influences. The change in geometry of the crater with various parameter setting has been modelled and simulated. The variation in ratio of crater radius to depth with discharge voltage, discharge current and spark on time has been plotted

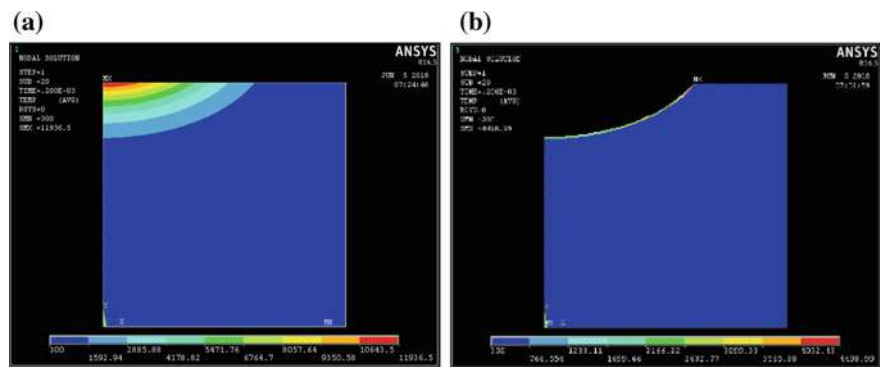


Fig. 39.2 Temperature distribution ($V = 30\text{ V}$, $I = 75\text{ A}$, $T_{\text{on}} = 200\text{ }\mu\text{s}$) **a** plot, **b** crater profile (obtained through elimination of melted elements)

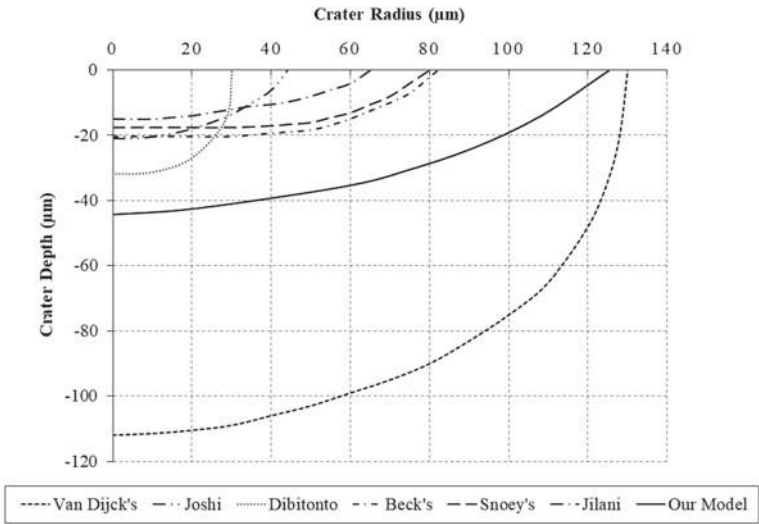


Fig. 39.3 Comparison of crater shapes formed on different materials by EDM spark impacts [2, 4, 5, 8–12]

and is as shown in Fig. 39.4. The impact of current portrays an incremental trend, as the number of surface impinging electrons is increased enormously with current. This results in development of craters with larger radius, due to expansion of plasma channel. But as there is not much enhancement in the acceleration of the migrating electrons, their energy levels remain within definite limits. So, the influence of the current on the depth remains low, albeit the electron numbers are upgraded. The effect of voltage has an inverse trend to that of the current. The increase in voltage

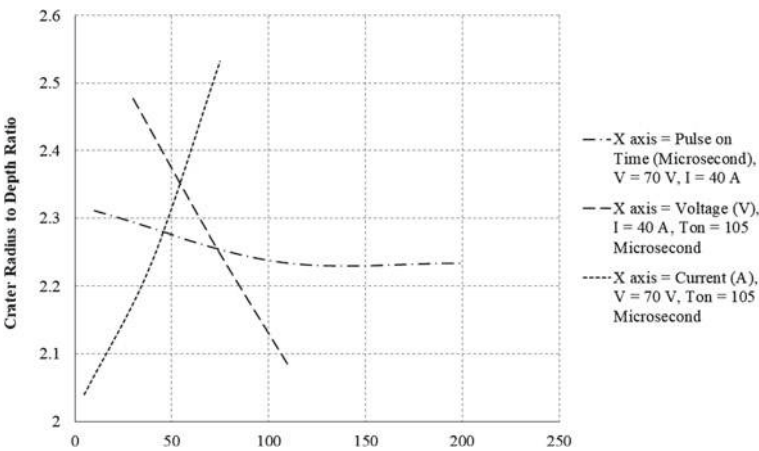


Fig. 39.4 Variation of crater radius-to-depth ratio with the input process parameters

results in enhanced electric field intensity, thereby accelerating the electron movement in the plasma channel. The resultant improved energy of the electrons has a better penetrating effect, upon impact with the workpiece surface. The contribution of voltage results in more material removal along the axial direction than in the radial direction. Hence, the increment in voltage has a greater impact on the crater depth in comparison with crater radius. Further, the effect of pulse on time was found to have less significant influence on the crater radius-to-depth ratio, when other factors are kept constant. The reason being that, when voltage and current are constant, the rate at which the series of electrons hitting the surface and their corresponding velocities remain constant. The resultant effect is that with increment in pulse on time, the melting and vaporization will happen at a confined rate along the radial and axial length of the work material.

39.4.2 Multi-spark Analysis

The single spark analysis has been extended to evaluate the multi-spark phenomenon for the chosen parametric setting. The initiation of secondary sparks on the workpiece domain is considered to be originating at any point along the periphery of spark radius for the previously developed craters. This consideration is taken into account, in view of the fact that the surface exposed to any kind of thermal impacts will undergo changes in its surface material properties. Further, the same craters are used to generate the entire machined surface. The temperature distribution plot for multi-spark impacts is as shown in Fig. 39.5a (for $V = 30\text{ V}$, $I = 75\text{ A}$, $T_{\text{on}} = 200\text{ }\mu\text{s}$). The elimination of the unessential elements will produce the final machined surface as given in Fig. 39.5b. It is clear from the machined surface profile plot, the vitality of crater shapes in producing definite machining surface responses. Shallow craters will contribute better for producing rough surfaces in contrast to deep craters for fine/finish

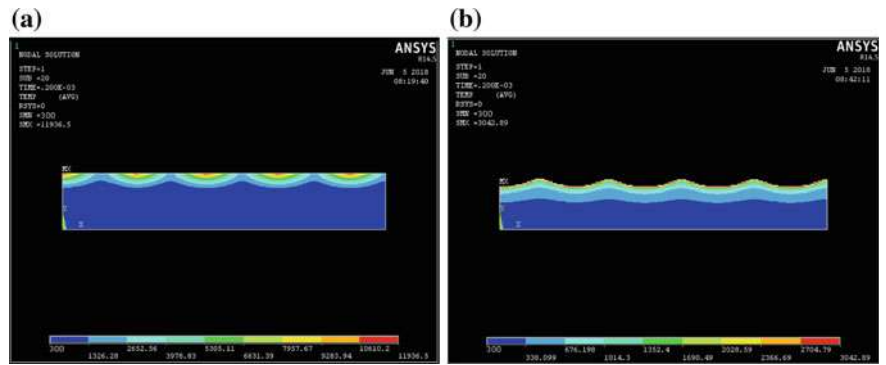


Fig. 39.5 Temperature distribution for multi-spark impacts **a** plot, **b** simulated/developed EDM machined surface

surfaces. Also, it is worthwhile to know that the type of crater overlapping taking place during the EDM process will greatly influence the surface roughness of the machined surfaces. This simulation routine for a generation of multi-spark craters is one of its kinds, to the extent of authors' knowledge and the referred available information. This exquisiteness in modelling and generation of the machined surface profile can be utilized for the determination of material removal rate (MRR) and surface roughness (R_a) though theoretical, numerical or analytical methods. Such an extensive effort has been partly excluded in this paper, for it is beyond the scope of this initiative.

39.5 Model Validation

The developed model was validated by comparison with the modelling and experimental investigation of EDM on AISI W1 tool steel carried out, previously [8]. Figure 39.6 shows the variational plot of crater radius-to-depth ratio with the input process parameters.

It can be observed that the plot of crater radius-to-depth ratio with current, voltage and pulse on time follows a similar trend which was obtained in the present work (as shown in Fig. 39.4). Hence, it is proved that the current and voltage factors have greater influence along with radial and axial directions of the generated craters, respectively. The effect of pulse on time follows a nearly uniform pattern, as the others factors remain constant. The real-time experimentation was excluded in the present work, due to brevity and is strongly considered for future progressive investigations.

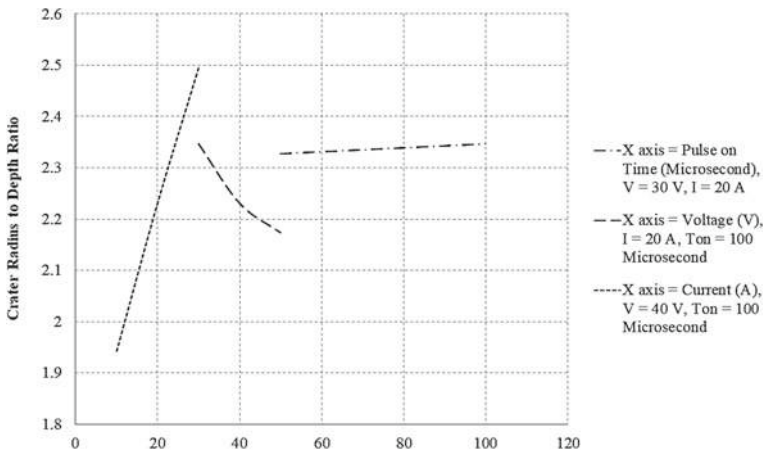


Fig. 39.6 Variation of crater radius-to-depth ratio with the input process parameters [8]

39.6 Conclusions

The numerical modelling and simulation of single and multi-spark craters have been developed under various considerations. The influence of the machining parameters was found to have a greater impact on the generated crater geometry, apart from thermal material properties. The shape of the craters can be varied by different parameter setting, so as to produce machined surfaces of required applicability. The discharge current was found to have greater control over the crater radius, with discharge voltage and pulse on time compensating for it on crater depth. The shape of the craters has an upper hand in the generation of surfaces with specific roughness properties. Moreover, the generation of EDM process machined surfaces through multi-spark modelling and simulation was found to be positively achievable using ANSYS R14.5 subject to a few constraints.

Acknowledgements This work was carried out by the aid of research grant sanctioned from the Science and Engineering Research Board (SERB), DST, Govt. of India (Project Ref. No. ECR/2016/001929). Also, the authors are grateful to Mr. Anjan Karmakar for his significant contribution towards this initiative.

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Chapter 40

Finite Element Simulation and Experimental Investigations to Predict Tool Flank Wear Rate During Microturning of Ti–6Al–4V Alloy



Jiju V. Elias , S. Asams and Jose Mathew

Abstract Mechanical micromachining has gained wide acceptance in the manufacture of miniaturized components for a wide range of applications including aerospace, biomedical, electronics, etc. in recent decades. Microturning is one of the important machining techniques used for manufacturing these components. In micromachining, as the undeformed chip thickness becomes comparable with the cutting edge radius, size effect highly influences the material deformation mechanism. Therefore, the tool experiences a nonlinear variation in cutting forces and specific cutting energy, which accelerates the tool wear. The tool wear mechanism becomes even more complex in the case of micromachining of difficult to machine materials like Ti–6Al–4V alloy. Tool wear is influenced by the combined effect of mechanisms like material adhesion, abrasion, erosion, diffusive wear, fracture, etc. In the present work, the adhesive tool wear model, proposed by Usui et al. is used for the tool wear estimation in micro regime. The tool wear model is calibrated using a hybrid approach based on both finite element simulations and cutting experiments. Validation experiments are done to compare experimental and predicted flank wear rates. Results show that the predicted flank wear rates using Usui model, using calibrated constants, showed better agreement with experimental results.

Keywords Micro turning · Flank wear · Usui tool wear model · Ti–6Al–4V

40.1 Introduction

The demand for miniaturized components with high precision is increasing at faster rate for various industries such as communication, electronic, environmental, biomedical, aerospace and automotive industries. Titanium alloys are extensively used for these applications because of its corrosion resistance, high strength to weight ratio and bio-compatibility. Mechanical micromachining is widely regarded

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_40

as an alternative for the traditional lithographic techniques because of the capability to produce complex 3D geometries and shapes [1]. However, machining in the micro-scale regime poses severe challenges when compared to the conventional macro-scale machining. The factors that affect the micromachining process include size effect, negative rake angle effect, ploughing mechanism, minimum uncut chip thickness effect, etc. These underlying mechanisms in micromachining significantly affect the surface finish, specific cutting energy, tool wear and chip formation. Tool wear during machining operations has great impact on dimensional accuracy and surface roughness of machined parts. It can directly affect the overall operation cost, as it dictates either a more frequent tool change to meet the requirements on surface finish properties or longer machining times by using conservative cutting data to prevent rapid tool edge degradation. The overall production cost can be reduced by performing machining under optimum process parameters and by predicting tool life accurately. One of the earliest attempts in this area is done by Taylor [2], in which he described relation between cutting parameters such as speed, feed, depth of cut and tool life. A number of researches are carried out in the past to formulate an empirical model for the accurate estimation of tool wear. Several researches developed physical and phenomenological wear rate models using intrinsic variables like sliding velocity, contact pressure and interface temperature. Tool wear rate models describe the volume of tool material worn out per unit area per unit time. Takeyama and Murata model [3] which considered both abrasive and diffusive wear mechanisms and Usui model [4] based on adhesive tool wear mechanism are widely used. These models were developed in order to predict the tool wear using less number of calibration experiments. In the current study, Usui's tool wear model is used for the prediction of tool wear rate, during microturning of Ti-6Al-4V alloy because of its highly adhesive nature of Titanium alloys at elevated temperatures. Usui's tool wear equation, given as Eq. (37.1), correlates the tool wear rate with normal stress, interface temperature and sliding velocity at the tool-workpiece interface.

$$\frac{dw}{dt} = A \times \sigma_n \times V_s \times \exp\left(\frac{-B}{T_{\text{int}}}\right) \quad (37.1)$$

where (dw/dt) is the wear rate, T_{int} is interface temperature, V_s is sliding velocity, σ_n is contact pressure and A, B are model constants which need to be calibrated for the corresponding cutting conditions. Literature suggests that Usui's tool wear rate model can predict both crater wear and flank wear with good accuracy. Even though a few researchers have used the same tool wear model constants for macromachining, in the micromachining regime as well, the adequacy of the model in the micromachining regime is not validated by many researchers in the past.

Preliminary experiments show that, in the case of microturning, flank wear is predominant, compared to the crater wear. Hence the scope of the current study is limited to the flank wear. Since the contact pressure on the flank wear land remains nearly constant with tool wear evolution, the effects of contact pressure (σ_n) on the overall tool wear rate can be neglected. In the case of flank wear, the sliding velocity (V_s) can be replaced by cutting velocity (V_c). In addition, the modified Usui's tool

wear rate model proposed by Yen et al. [5], given as Eq. (37.2), which considers the tool flank wear (VB) and clearance angle (γ), is incorporated in the current study.

$$\frac{dVB}{dt} = \frac{\dot{w}}{\tan \gamma}$$

(37.2)

From Eqs. (37.1) and (37.2), the modified Usui’s tool wear rate model can be derived as shown in Eq. (37.3).

$$\frac{dVB}{dt} = A \times \frac{V_c}{\tan \gamma} \times \exp\left(\frac{-B}{T_{\text{int}}}\right)$$

(37.3)

where A and B are model constants and $\frac{dVB}{dt}$, V_c , γ and T_{int} represents the flank wear rate, cutting velocity, clearance angle and interface temperature, respectively. The current study focuses on the prediction of the tool flank wear during microturning of Ti–6Al–4V alloy using uncoated carbide tools, by employing the modified tool wear rate model given in Eq. (37.3). The calibration of modified tool wear model is based on a hybrid approach by using the machining experiment data and FE machining simulations. The tool wear rates are determined by conducting cutting experiments at different machining conditions while the values of interface temperature (T_{int}) are obtained by finite element simulations. The model constants A and B are then found out using regression analysis.

40.2 Micro-turning Experiments

40.2.1 Experimental Plan

In order to determine the tool wear rate under different cutting conditions, micro-turning experiments by varying the cutting speed, feed and depth of cut are planned as given in Table 40.1. The cutting speed, feed and depth of cut are varied in three levels and a total of 27 full factorial experiments are planned for the current study. The levels of different parameters are determined by conducting trial experiments. Feed rate is kept well above the cutting edge radius of the tool to reduce the influence of size effect. The tool wear progression is noted by interrupting the machining operation at specific time intervals and by measuring the tool flank wear.

Table 40.1 Process parameters used for experimentation

Process parameters	No. of levels	Values
Depth of cut (mm)	3	0.1, 0.2, 0.3
Feed rate (μm/rev)	3	10, 15, 20
Cutting speed (m/min)	3	40, 50, 60

Fig. 40.1 Experimental set-up for microturning experiments



40.2.2 *Experimental Set-Up*

Micro-turning experiments are carried out on Hybrid Micromachining Centre (DT-110 MIKROTOOLS Pvt. Ltd., Singapore), at Advanced Manufacturing Centre, NIT Calicut. The machine tool has a maximum travel range of 200, 100 and 100 mm in X, Y and Z directions, respectively. Each axis has an optical linear scale with resolution of 0.1 mm, and close loop feedback control ensures accuracy to submicron dimensions. AC servo motor (100 W) powered spindle head provides a rotational speed ranging from with 1 to 5000 rpm. The experimental set-up used for microturning is shown in Fig. 40.1.

40.2.3 *Workpiece and Cutting Tool*

Ti-6Al-4V alloy (Titanium grade 5) shafts with 6.4 mm diameter and 65 mm length are used as workpiece material, the composition of which is given in Table 40.2.

Uncoated tungsten carbide inserts (SUMITOMO—TCGT090201), with clearance angle of 70 and rake angle of 140 is used as the cutting tool. The average cutting edge radius is measured using 3D optical profilometer (Alicona—InfiniteFocus G5) and is found to be 3.4 μm (approx.).

Table 40.2 Percentage composition of Ti–6Al–4V alloy

Material	Ti	V	Al	Iron	Oxygen	Carbon	Nitrogen
Percentage composition	Bal.	3.5–4.5	5.5–6.8	0.25 (max)	0.2 (max)	0.08 (max)	0.05 (max)

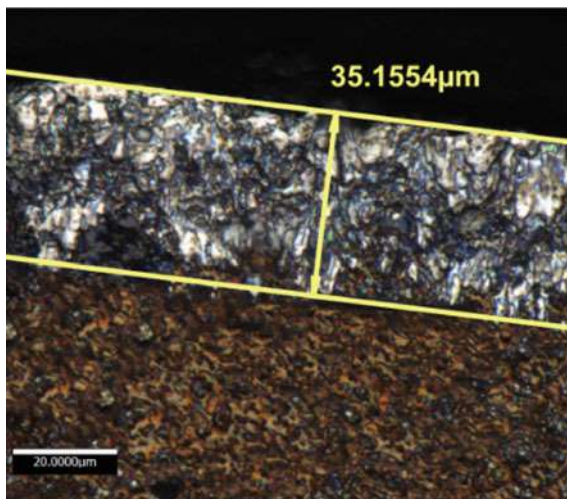
40.2.4 Experiments and Tool Wear Measurement

Micro-turning experiments are carried out on Ti–6Al–4V shafts using uncoated tungsten carbide inserts. Cutting experiments are conducted by varying the cutting parameters as per the experimental plan given in Table 40.1. Machining is interrupted at specific time intervals of 2, 5 and 7 min to measure the tool flank wear. The tool flank wear is measured using 3D optical profilometer as shown in Fig. 40.2.

40.3 Finite Element Simulation

The finite element simulation for microturning of Ti–6Al–4V alloy using uncoated carbide tools is done using commercially available DEFORM 3D software. The 3D model of the cutting tool is made using SOLIDWORKS software and is imported to the DEFORM 3D software to determine the interface temperature for different cutting conditions.

Fig. 40.2 Flank wear observed after 7 min of machining



40.3.1 *Material Model*

The flow stress under different machining conditions is described by material model. During machining high strain, strain rates and temperature affect the flow stress of the workpiece material. Johnson–Cook material model [6] given in Eq. (37.4), which considers thermal softening, strain hardening and strain rate hardening, is used for determining the flow stress.

$$\sigma_{\text{flow}} = [A + B\varepsilon^n] \left[1 + C \ln \frac{\dot{\varepsilon}}{\dot{\varepsilon}_0} \right] \left[1 - \left(\frac{T - T_0}{T_m - T_0} \right)^m \right] \tag{37.4}$$

where ε = plastic strain, $\dot{\varepsilon}_0$ = reference plastic strain rate, $\dot{\varepsilon}$ = strain rate, T_m = melting temperature, T = work piece temperature, T_0 = room temperature, n = hardening coefficient, m = thermal softening coefficient, A = yield strength of the material, B = strain hardening modulus, C = strain rate sensitivity coefficient. In the present study, the Lee and Lin parameters [7] as given in Table 40.3 are used for the material model. The temperature-dependent thermo-mechanical properties are added to the material according to the literatures referred [8] and are listed in the Table 40.4.

Finite element simulations are performed by keeping the workpiece stationary and by moving the tool along the workpiece. The cutting tool is considered as a rigid body while the workpiece is considered as plastic. Relative mesh size is given to the tool with a maximum of 25,000 elements as determined by the mesh convergence study. The workpiece is meshed with a minimum element size of 25% of the feed so that nearly four elements are available in the uncut chip thickness region.

Table 40.3 Johnson–Cook material model parameters for Ti–6Al–4V [7]

Parameters	<i>A</i>	<i>B</i>	<i>C</i>	<i>n</i>	<i>m</i>	<i>T</i> _{melt} (°C)
Values	782.7	498.4	0.028	0.28	1	1660

Table 40.4 Properties of workpiece and tool material in terms of temperature (*T*) [8]

Property	Ti–6Al–4V	WC
Young’s modulus (MPa)	$(0.7412 \times T) + 113375$	5.0×10^5
Thermal conductivity (W/m °C)	$7.039 \times 10^{(0.0011 \times T)}$	$(0.042 \times T) + 36$
Thermal expansion (mm/mm °C)	$(3.1 \times 10^{-9} \times T) + (7.1 \times 10^{-6})$	4.7×10^{-6}
Heat capacity (N/mm ² °C)	$2.24 \times 10^{(0.0007 \times T)}$	$(0.0005 \times T) + 2.07$

40.3.2 Temperature and Heat Transfer Parameters

An ambient temperature of 25 °C is considered for the tool wear simulation. The convective heat transfer coefficient is set as 0.02 N/s/mm/°C, and the tool-workpiece interface heat transfer coefficient is taken as 107 N/s/mm/°C as suggested by several researchers [8].

40.3.3 Friction Model

A hybrid friction model by considering the shear friction law ($\tau = mk$) and Coulomb friction law ($\tau = \mu p$) is used to represent the friction at the tool-workpiece interface. The Coulomb friction law is considered when ($\mu p < mk$) and the shear friction law is considered for ($\mu p \geq mk$). The coefficient of friction between the interface of tool and workpiece depends on the relative velocity between them. The region around the cutting edge is considered as two different contacts. Around the cutting edge, the contact is considered as sticking contact. Along the rake face, a sliding friction contact ($m = 0.7$) and shear friction contact ($\mu = 0.95$) are considered in the current study [8].

The machining simulations are performed for the different cutting conditions by varying the speed, feed and depth of cut and the corresponding interface temperature T_{int} is found out as shown in Fig. 40.3. The cutting simulations are run until steady state is achieved. The interface temperature values for all the 27 experimental runs are obtained from the simulation data. This data is used in conjunction with the experimental tool wear rate to obtain the model constants.

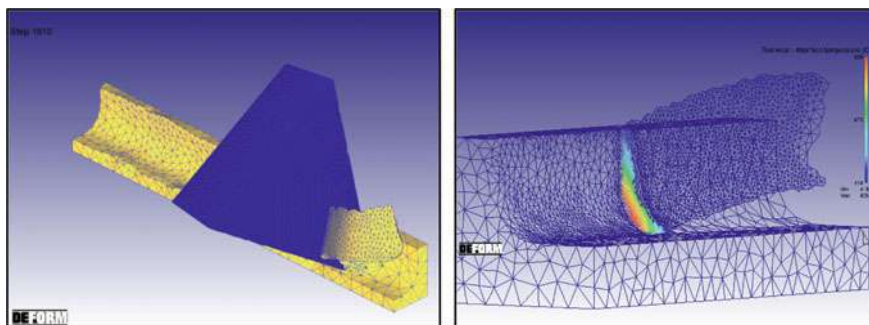
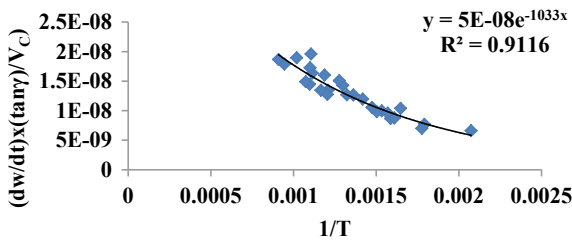


Fig. 40.3 Cutting simulation using DEFORM 3D software

Fig. 40.4 Determination of modified Usui tool wear model constants



40.4 Calibration of Modified Usui Model for Microturning

Usui’s tool wear rate model given in Eq. (37.3) can be rewritten for the regression analysis as follows:

$$\left(\frac{\frac{dVB}{dt} \times \tan \gamma}{V_c}\right) = A \times \exp\left(\frac{-B}{T}\right) \tag{37.5}$$

The constants *A* and *B* are determined by combining results from both experimentation and simulation, using regression analysis. Exponential curve is fit with the function on the LHS in the Eq. (37.5) as Y-axis and 1/*T* as X-axis, with an R-squared value of 91.16%, as shown in Fig. 40.4. Usui’s tool wear model constants *A* and *B* are obtained as, *A* = 5 × 10^{−8} and *B* = 1033.

40.5 Results

Experiments to validate the modified Usui’s tool wear model for microturning of Ti–6Al–4V alloy using uncoated carbide tools are conducted using random machining parameters as given in Table 40.5. The calibrated Usui’s tool wear constants (*A* = 5 × 10^{−8} and *B* = 1033) are used for the tool wear calculation during finite element simulation of microturning process using DEFORM 3D software.

Table 40.5 Process parameters used for experimental validation and results

No.	Speed (m/min)	Feed (μm/rev)	Depth of cut (mm)	Experimental wear rate (mm/s)	Simulated wear rate (mm/s)	Percentage error (%)
1	45	13	0.1	6.01E − 05	5.16E − 05	14.11
2	50	16	0.13	9.82E − 05	8.58E − 05	12.68
3	40	12	0.15	4.48E − 05	3.92E − 05	12.52
4	58	18	0.18	1.51E − 04	1.32E − 04	12.52
5	42	15	0.2	7.48E − 05	6.57E − 05	12.15

On an average, the simulated tool wear data is slightly lower than the experimental tool wear rates. Validations experiments show a maximum absolute percentage error of 12.79%. This variation is mainly due to the underestimation of break-in period and unpredictable factors which causes additional wear during machining. The assumption of constant normal stress may have affected to the accuracy of the modified model. Hence by incorporating these factors, the accuracy of the tool wear model can be improved. It is also observed that the tool wear rate during microturning is high when compared to that of macroturning due to size effect.

40.6 Conclusions

The current study focuses on the tool flank wear rate prediction during microturning of Ti-6Al-4V alloy using uncoated tungsten carbide inserts. Usui's adhesive tool wear model proposed is used for the prediction of tool wear rate. The crater wear, which is found to be negligible in microturning, is ignored. The modified Usui's tool wear model, by considering the tool clearance angle, for the prediction of tool flank wear proposed by Yen et al. is used in the current study. A hybrid method, by combining finite element simulations and cutting experiments, is used to calibrate the proposed tool wear model. Calibrated tool wear model shows fairly good agreement with the experimental wear rates, with a maximum deviation of 12.79%.

Acknowledgements Authors would like to sincerely thank the Department of Science and Technology (DST), Govt. of India and Centre for Precision Measurements and Nanomechanical Testing, Department of Mechanical Engineering, National Institute of Technology Calicut, for providing support to carry out this project under the scheme "Fund for Improvement of Science and Technology" (No. SR/FST/ETI-388/2015).

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Chapter 41

Analysis of a Few Heuristics Proposed Based on Slope Indices to Solve Simple Type—I Assembly-Line Balancing Problems



A. Baskar , M. Anthony Xavier , N. Nithyanandan
and B. Dhanasakkaravarthi

Abstract In an assembly line, any product is subdivided into many tasks which may include subassemblies and processing. These tasks are carried out in several work stations which are responsible for a single or a set of operations. Assembly lines need to be balanced to have even distribution of work for both men and machines. Type—I simple assembly-line balancing problems (SALBP-I) refer to minimization of number of work stations by keeping the cycle time constant. This paper proposes a new set of heuristics that can be used to solve simple type—I assembly-line balancing problems and analyzes them using a few benchmark problems available in the literature. They use slope indices to order the jobs and allot them to different work stations.

Keywords Assembly-line balancing · Type—I problem · Heuristics · Line efficiency · Smoothing index

41.1 Introduction

Let, there are ‘ n ’ tasks that comprise a job that needs to be completed in an assembly line and the corresponding time of completion be t_j ($j = 0$ to n). The order of processing is partially controlled by certain constraints called as the ‘precedence’ that are not to be violated. This defines the tasks that need to be completed before starting a particular task. The tasks are carried out in different work stations that are to be balanced to the possible extent. The maximum time to be spent in a work station defines the ‘cycle time, c .’ The cycle time being constant, the objective is to minimize the number of work stations, m^* .

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_41

The solution procedures include exact methods and heuristic methods. The exact methods are not suitable for larger problems as the computation time grows exponentially with the problem size. SALBP-1 are NP hard [1] and hence, efficient heuristics are required to solve within a reasonable time period. As a result, researchers have developed many efficient heuristic methods over the years.

Positional weight method [2], procedure based on number of predecessors [3], heuristic using trade and transfer [4], heuristic of Hackman et al. [5], precedence matrix method proposed by Hoffmann [6] are a few efficient heuristic methods proposed during earlier periods of research. Many heuristics are available in the literature based on simple as well as combined priority rules. It is generally accepted that Hoffmann's algorithm is still one of the best simple algorithms in this domain, at the cost of execution time.

In the next level, branch and bound method, dynamic programming and evolutionary algorithms are used by many to refine the accuracy of heuristics. Most of the evolutionary algorithms take the results from the simple heuristics as their seed solutions and proceed.

Sivasankaran and Shahabudeen [7] classified assembly-line balancing problems into eight types and conducted a comprehensive review on the available methods.

41.2 Data Set and Heuristics Considered

A precedence diagram is a graphical representation of a project that shows the number of tasks, their respective task times, and the sequence of tasks that need to be completed before a particular task. Figure 41.1 shows the precedence diagram of a SALBP-1 analyzed by Rosenberg and Ziegler [8].

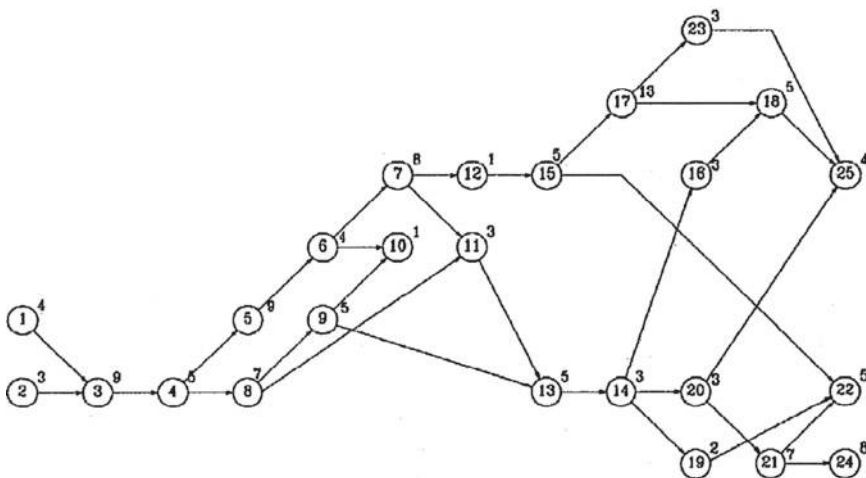


Fig. 41.1 A precedence diagram—Rosenberg and Ziegler

Table 41.1 Cycle time and optimum number of work stations

Rosenberg	<i>c</i>	<i>m</i> *	<i>c</i>	<i>m</i> *	<i>c</i>	<i>m</i> *	<i>c</i>	<i>m</i> *	<i>c</i>	<i>m</i> *	<i>c</i>	<i>m</i> *
25 Tasks	14	10	16	8	18	8	21	6	25	6	32	4

This particular project has 25 tasks with a total time of 125 units. For analyzing the heuristics considered, this particular problem is considered as it is a reasonably large sized tested data set. To have more number of problems, the cycle time is considered for different values, from 14 units to 32 units as listed in Table 41.1.

$$\begin{aligned}
 \text{Strength of the precedence, } D &= 2d/(N(N-1)) \\
 &= (2 \times 32)/(25(25-1)) \\
 &= 0.11
 \end{aligned}$$

Since the same data set is tested for six cycle times, we get a total of six SALBP-1. 'c' represents the cycle time and 'm*' represents the optimum number of work stations for a particular cycle time. The optimum number of work stations for different cycle times is available in the data sets provided by Scholl [9] and is reproduced in Table 41.1. The parameters considered are listed in Table 41.2.

The precedence diagram is transformed into a precedence matrix as shown in the left half of Table 41.3 which can be directly used in a computer program. The matrix is appended with other required parameters (obtained from the precedence diagram) as described and presented in the right half of the same table.

Based on a presumption that simultaneously considering the same parameter before and after a task being considered can result in better algorithms; slope indices (ratios) are being computed for a particular task. They are the ratios of a particular parameter before and after a task.

Table 41.2 Parameters considered

S. No.	Parameter	Notation
1.	Total number of tasks having 'j' as its head (including self)	<i>a</i>
2.	Total number of tasks having 'j' as its tail (including self)	<i>b</i>
3.	Maximum number of tasks having 'j' as its head (including self)	<i>c</i>
4.	Maximum number of tasks having 'j' as its tail (including self)	<i>d</i>
5.	Number of immediate predecessors of 'j' (including self)	<i>e</i>
6.	Number of immediate successors of 'j' (including self)	<i>f</i>
7.	Number of levels prior to 'j' (including self)	<i>g</i>
8.	Number of levels after 'j' (including self)	<i>h</i>
9.	Position weight of 'j' from head (reverse position weight) ... including self	<i>i</i>
10.	Position weight of 'j' from tail ... including self	<i>k</i>

Table 41.3 Precedence matrix and other parameters

Task No. (j)	Task time (t _j)	Predecessor	Successor	Parameter considered									
				a	b	c	d	e	f	g	h	i	k
1	4	–	3	25	1	25	1	1	2	1	12	4	122
2	3	–	3	25	1	25	1	1	2	1	12	3	121
3	9	1, 2	4	23	3	23	3	3	2	2	11	16	118
4	5	3	5, 8	22	4	22	4	2	3	3	10	21	109
5	9	4	6	19	5	9	4	2	2	4	9	30	92
6	4	5	7, 10	18	6	8	5	2	3	5	8	34	83
7	8	6	11, 12	16	7	7	6	2	3	6	7	42	78
8	7	4	9, 11	14	5	7	4	2	3	4	7	28	61
9	5	8	10, 13	12	6	6	5	2	3	5	6	33	51
10	1	6, 9	–	1	9	1	6	3	1	6	1	47	1
11	3	7, 8	13	11	9	6	7	3	2	7	6	58	48
12	1	7	15	7	8	5	7	2	2	7	6	43	36
13	5	9, 11	14	10	10	5	8	3	2	8	5	63	45
14	3	13	16, 19, 20	9	12	4	9	2	4	9	4	66	40
15	5	12	17, 22	6	9	4	8	2	3	8	5	48	35
16	3	14	18	3	12	3	10	2	2	10	3	67	12
17	13	15	18, 23	4	10	3	9	2	3	9	4	61	25
18	5	16, 17	25	2	16	2	11	3	2	11	2	95	9
19	2	14	22	2	13	2	10	2	2	10	2	67	7
20	3	14	21, 25	4	13	3	10	2	3	10	3	69	27
21	7	20	22, 24	3	14	2	11	2	3	11	2	76	20
22	5	15, 19, 21	–	1	17	1	12	4	2	12	1	88	5
23	3	17	25	2	11	2	10	2	2	10	2	64	7
24	8	21	–	1	15	1	12	2	1	12	1	84	8
25	4	18, 20, 23	–	2	21	2	12	4	1	12	1	125	4
	N = 125	d = 32	d = 32										

The weights w1 to w5 are the slope indices as computed below. Weight ‘w’ is the rule proposed by Dar-El; weight, $w = a$ = total number of tasks having ‘j’ as its head (including self). Weight w6 is computed in a different way but, using t_j , k and i that are used for w5 in a slope format. For any task ‘j,’ the slope is the ratio between the right and left parameter values.

Weight, $w = a$; Weight, $w1 = t_j \frac{a}{b}$; Weight, $w2 = t_j \frac{c}{d}$

Weight, $w3 = tj \frac{f}{e}$; Weight, $w4 = tj \frac{h}{g}$; Weight, $w5 = tj \frac{k}{i}$
Weight, $w6 = tj(k - i)$.

Table 41.4 shows the parameters converted as weights for different heuristics. For solving the problems, the weights are arranged in descending order of their weights. The two popular time-tested algorithms proposed by Dar-El [10] and Hoffmann [6] are taken as the benchmarks.

Table 41.4 Weights considered for the analysis

Task No. (j)	Task time (tj)	Predecessor	Successor	Weights					
				t(a/b)	t(c/d)	t(f/e)	t(h/g)	t(k/i)	t(k - i)
1	4	–	3	100	100	8	48	122	472
2	3	–	3	75	75	6	36	121	354
3	9	1, 2	4	69	69	6	49.5	66.375	918
4	5	3	5, 8	27.5	27.5	7.5	16.667	25.952	440
5	9	4	6	34.2	20.25	9	20.25	27.6	558
6	4	5	7, 10	12	6.4	6	6.4	9.765	196
7	8	6	11, 12	18.286	9.333	12	9.333	14.857	288
8	7	4	9, 11	19.6	12.25	10.5	12.25	15.25	231
9	5	8	10, 13	10	6	7.5	6	7.727	90
10	1	6, 9	–	0.111	0.167	0.333	0.167	0.021	–46
11	3	7, 8	13	3.667	2.571	2	2.571	2.483	–30
12	1	7	15	0.875	0.714	1	0.857	0.837	–7
13	5	9, 11	14	5	3.125	3.333	3.125	3.571	–90
14	3	13	16, 19, 20	2.25	1.333	6	1.333	1.818	–78
15	5	12	17, 22	3.333	2.5	7.5	3.125	3.646	–65
16	3	14	18	0.75	0.9	3	0.9	0.537	–165
17	13	15	18, 23	5.2	4.333	19.5	5.778	5.328	–468
18	5	16, 17	25	0.625	0.909	3.333	0.909	0.474	–430
19	2	14	22	0.308	0.4	2	0.4	0.209	–120
20	3	14	21, 25	0.923	0.9	4.5	0.9	1.174	–126
21	7	20	22, 24	1.5	1.273	10.5	1.273	1.842	–392
22	5	15, 19, 21	–	0.294	0.417	2.5	0.417	0.284	–415
23	3	17	25	0.545	0.6	3	0.6	0.328	–171
24	8	21	–	0.533	0.667	4	0.667	0.762	–608
25	4	18, 20, 23	–	0.381	0.667	1	0.333	0.128	–484

41.3 Performance Measures

Perfect balancing of work stations is important to reduce the idle time of individual work stations. If they are not balanced properly, bottlenecks will be a problem in any assembly line. There are three basic measures for the effectiveness of the heuristics viz. (i) Line efficiency (ii) Smoothness index, and (iii) Computation time.

Only the former two measures are considered in this analysis as all the heuristics except the Hoffmann's have the same time complexity. The performance measures are defined as:

$$\text{Line efficiency (LE)} = \frac{\sum_{i=1}^k ST_i}{c \cdot k} \times 100$$

$$\text{Smoothness index (SI)} = \sqrt{\sum_{i=1}^k [(ST_{\max} - ST_i)]^2}$$

ST_i —Total station time of i th station

c —Cycle time

k —Number of work stations.

41.4 Computational Results

The line efficiency and smoothness indices are computed separately for each heuristic algorithm using the benchmark problems in addition to the reference algorithms. Higher values of efficiency and lower value of smoothing index are the indications of perfect line balancing. In all the cases, the ties are broken according to the task number, smaller first. The summary of results is presented in Table 41.5.

It is observed that the heuristic numbers three and seven perform better than others, including the benchmark algorithms in terms of the tested performance measures. When one-way AVOVA was carried out, it was observed that:

Line efficiency : Experimental $F = 0.86 < \text{Critical } F = 2.372$.

Smoothness index : Experimental $F = 0.85 < \text{Critical } F = 2.372$.

Hence, it is concluded that there is no significant difference among the heuristics. However, to confirm their relative performance, pairwise comparisons will be carried out in the extended work. The box plots for line efficiency and smoothing index obtained show that the variation in line efficiencies and smooth indices are relatively less for the weights a , $t(f/e)$ and $t(k - i)$.

Table 41.5 Summary of performance measures for different heuristics

S. No.	Heuristic	Mean efficiency	Mean smoothing index
1	Dar-El heuristic	83.5454	13.8729
2	Hoffmann's heuristic	84.8982	15.7373
3	(ta/b)	85.1931	11.2967
4	(tc/d)	80.8105	15.4748
5	(tf/e)	80.9713	16.2697
6	(th/g)	80.8105	15.6066
7	(tk/i)	85.1931	11.1094
8	$(t(k - i))$	82.0986	13.9643

After the computation, three more cases were analyzed as described below:

- (i) Taking the weight as $t(a/b) + t(k/i)$ (combination of better performing heuristics):

The results show that the mean line efficiency and smoothing index are exactly the same as that of $t(k/i)$ for each cycle time and slightly differ from that of $t(a/b)$.

New, mean efficiency = 85.1931% and smoothing index = 11.1094.

- (ii) The order of one better performing heuristic is reversed to ascending and the measures are again computed.

In such a case, the performance decreases to 82.0986% and 15.8208 from 85.1931% and 11.1094 earlier.

- (iii) The order of one average performing heuristic is reversed to ascending and the measures are again computed.

In this case, the performance increases to 81.9379% and 15.8160 from 80.9713 and 16.2697 earlier.

In the latter two cases, another observation is that the maximum number of jobs available for allotment is four as against five earlier.

41.5 Conclusion and Future Work

This paper discusses a newly proposed set of six simple heuristics based on the slope indices for the simple type—I assembly-line problems. They are tested against the benchmark data set for different cycle times. The results are compared with the popular time-tested algorithms proposed by Dar-El and Hoffmann. Two of the six

heuristics perform better than these two for the tested performance measures, line efficiency, and smoothing index.

To validate the results further, more number of data sets are to be used. Also, further improvements in simple heuristics including tie-breaking rules, different precedence strengths are to be applied for these heuristics also and the effects are to be analyzed. Only forward enumeration is considered here. Backward and bidirectional enumerations are to be implemented for further improvement in the performance.

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Chapter 42

A Thermo-Mechanical Finite-Element Analysis of Resistance Spot Welding of Dual-Phase Steel and Austenitic Stainless Steel



Sagar Rathod, Sunil Ghunage and B. B. Ahuja

Abstract In this paper, a three-dimensional axisymmetric finite-element model for resistance spot welding of automotive steel materials is prepared to analyze the transient thermal and mechanical behaviors of weld pool. Thermal analysis is carried out to analyze the transient thermal properties of the process of resistance spot welding. Based on the results of the thermal analysis, a mechanical (structural) analysis is conducted to evaluate mechanical characteristics of resistance spot welding process. The thermal characteristics and temperature distribution within the body of weld metal have been validated by comparing it with the experimental work. The mechanical characteristics such as distribution of stresses and contact pressure at faying surfaces and electrode–sheet interfaces, the stress and strain distributions in weldment, and the changes during the welding process have been evaluated. The effect of welding parameters on weld strength was investigated. Results obtained through numerical modeling showed good agreement with experimental results.

Keywords Resistance spot welding (RSW) · Finite-element analysis (FEA) · Thermal–mechanical coupling · Weld strength

42.1 Introduction

In automobile industries, resistance spot welding (RSW) is the commonly practiced welding process because of its higher efficiency and lower cost as compared to its peers. Generally, there are approximately 2000–5000 spot welds used for the

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_42

construction of a modern car [1]. While designing new vehicles, the design engineers must consider weight reduction without affecting the safety of occupants. The crashworthiness is an important property of the material used for vehicle body construction, which is also responsible for the safety of occupants. The crashworthiness can be increased by using the materials having high strength. Therefore, the automobile manufacturers and OEMs are trying to find a suitable alternate combination of materials that are used for modern vehicle manufacturing. Many advanced types of steel materials are used for the construction of automobile/vehicles like advanced high-strength steels (AHSS). These modern materials offer higher strength, but its use also results in increased cost of vehicles. Therefore, structural properties need to be optimized by the use of two or more materials for the construction of a vehicular body. This emphasizes the need to find efficient methods of joining these modern materials.

In this study, resistance spot welding has been chosen as it has wider acceptance due to low cost and higher efficiency. The quality of weldments is assessed by the weld strength of the joint and weld nugget size. Additional parameters are maximum temperature generated during the welding operation, the hardness of weldment at various zones like heat affected zone, base metal and fusion zone and changes in microstructure during the welding. From the literature survey, it is observed that the governing parameters for the weld strength are weld current, welding time, and electrode pressure.

In this research work, the simulation model of resistance spot welding is compiled for similar as well as dissimilar welding of dual-phase steel and austenitic steel. The model has been validated with the experimental data. The model is used for the thermal–mechanical coupled analysis using ANSYS18 software. The optimized values of process parameters for the welding of the similar and dissimilar metals were determined using the simulation model.

42.2 Literature Survey

The finite-element analysis of resistance spot welding has been an interesting topic of research since 1990. Since the process is complex due to differences in the mechanical and electrical properties of the materials to be joined, there is a great scope for the numerical analysis of the welding process. Here, a brief review has been taken from the previous research conducted by few of the researchers, and their research details are given to understand the governing parameters and its effect on weld characteristics.

Nied et al. [2] had developed the finite-element model of resistance spot welding of steel materials. The effect of RSW electrode geometry on the mechanical and thermal properties of the weldment was analyzed and the deformation and stresses with reference to temperature change were predicted. The simulation model developed by them was restricted to elastic deformation that does not take into account the contact areas at the electrode–sheet interfaces and between faying surfaces also.

Han et al. [3] have conducted research on heat transfer by the finite-element analysis of RSW of high-strength low-alloy steels. The resistance spot welding was carried out on steel sheets of thickness 1 mm. The model predicted the temperature as a function of time and location for any position in the workpiece. Acceptable correlations between the theoretical and experimental temperatures were made. The temperatures in the spot weld were measured using fine thermocouples embedded in the sheet-steel coupons near the faying surfaces. This model also predicts the weld nugget diameter and the heat-affected zone width.

Saleem et al. [4] have performed the finite-element analysis to analyze nugget formation during the resistance spot welding of steel materials. The RSW model was compiled to compare the results of the published and available data. It was also observed by them that changes in weldment areas are evident by the change in the electrode contact surface area without changing the parameters. The nugget diameter has been changed by varying the tip contact area. When the tip contact area was reduced to minimum permissible value, the nugget formation took place very quickly.

Hou et al. [5] studied a multi-coupled finite-element analysis of resistance spot welding using ANSYS software. Through the thermal histories and temperature distributions obtained from the transient thermal analysis of RSW, the geometry and dimensions of nugget and heat-affected zone were drawn and calculated. During the welding cycles, there was compressive stress between the faying surfaces, which was very helpful for good metallurgical structure, and it forms a condensed weld nugget.

In the past, researchers have studied resistance spot welding of various steel materials and other materials analytically. Few researchers have worked on numerical analysis of RSW of some automobile steel materials such as austenitic stainless steel and aluminum alloys. However, there is ample scope to study the resistance spot welding of some advanced high-strength steels like dual-phase steels, TRIP, and TWIP steels. As there is scope to optimize and simulate the welding process using software programs like ANSYS and NASTRAN, the simulation and optimization of the RSW processes have been carried out using ANSYS environment. The simulation model for RSW is validated by comparing the results with experimental results. The purpose of this research work is to validate a numerical model for dual-phase steels and AISI 304 steels.

42.3 Simulation Methodology

a. Model Description

In this work, ANSYS 18 simulation software is utilized for the finite-element modeling of the process. Resistance spot welding is a very difficult process to simulate and analyze, because it needs to analyze areas like mechanical, structural, and thermal parameters. There are many interactions between different areas such as between

heat transfer and stress distribution, between heat flux supply and temperature distribution, and between current temperature distribution and weld metal deformation. The simulation model contains the electrode geometry and workpiece geometry with some applicable constraints. Two workbenches are used to prepare the simulation model for RSW: mechanical and thermal [6]. Coupled analysis is carried out to obtain the effect of thermal behavior achieved on the workpiece with same mechanical inputs. The model geometry and its boundary conditions are as described in succeeding sections.

b. Geometry and Meshing

The geometry contains four body parts such as upper electrode, upper steel sheet, lower steel sheet, and lower electrode. DP600 steel material sheet is to be weld using copper chrome electrodes. The mechanical, electrical, and thermal properties are assumed to be nonlinear with temperature. The electrode and sheet setup for this model is axisymmetric. The geometry of welding setup used in the simulation model is given in Fig. 42.1. The body sizing type mesh is used resulting into the fine meshing at the area of weld spot. On the remaining portion of the work, course mesh has been used to save the calculation and high-iteration solving time. The mesh size used for course mesh is 0.2 mm and for fine mesh is 0.02 mm.

c. Boundary Conditions and Governing Equations

In this work, an axisymmetric model of resistance spot welding is compiled. Uniform pressure has been applied from the top of the electrode. An electrical current

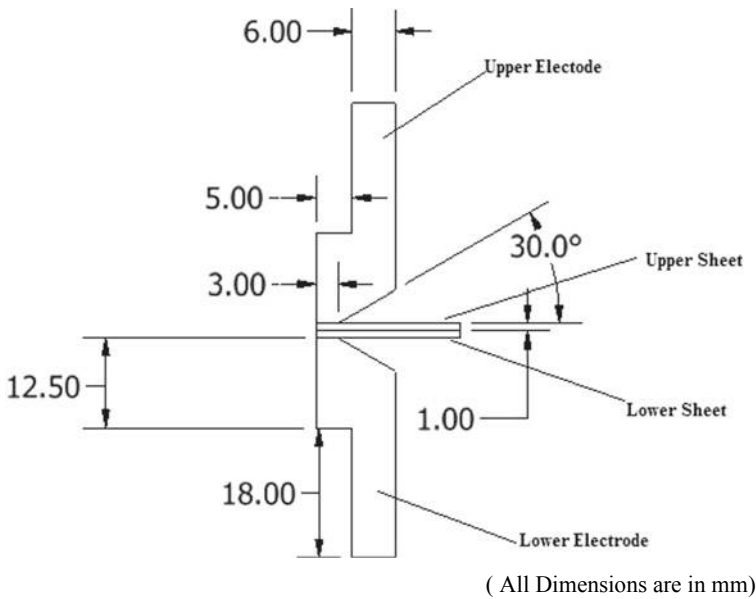


Fig. 42.1 Geometry of FEA model

(50 Hz) flow has been assumed to be uniformly distributed from the top face of the electrode. The current flows across the contact areas through both electrode–sheet and sheet–sheet interfaces. The temperature of the electrode–water cooling interface is maintained at a constant value during the whole welding process. The sliding effect at the electrode–sheet interface is neglected.

In this research work, the transient thermal analysis of RSW process is modeled as an axisymmetric model. However, the model is also a mirror symmetric about the faying surface. Therefore, only half of the resistance weld geometry is considered for modeling, which significantly reduced the computation time of the solver.

The elements used in contact and connections between each body are given in Table 42.1. Four kinds of elements have been used during analysis, two for solid and two for contact purposes. SOLID87 and SOLID90 have been used as thermal and mechanical solid elements, respectively, because it is well suited for modeling irregular meshes. For this part, differential meshing has been used. Fine mesh is used on spot weld area, and course mesh is used for the rest of the part. CONTA174 is a surface-to-surface contact element used because it allows sliding contact in 3D target surfaces. TARGE170 is used with CONTA174 where the pair-based contact is preferred. Here, the pair of two sheets is used in experiments, so TARGE174 can be used for the meshing of the geometric model.

Governing Equations

General mathematical equations used for electrical, thermal, and mechanical aspects of finite-element analysis of RSW process can be provided as below.

For the electrical analysis, the governing equation for calculation of the electrical potential $\varnothing$ in the whole model is as follows [7]:

$$\frac{d}{dr}\left(\sigma \frac{d\varnothing}{dr}\right) + \frac{\sigma}{r}\left(\frac{d\varnothing}{dr}\right) + \frac{d}{dz}\left(\sigma \frac{d\varnothing}{dz}\right) = 0,$$

(42.1)

where

- r —radial distance,
- z —distance in the axis direction of the coordinate system,
- σ —electrical conductivity.

By solving the above-mentioned equation, we can obtain the amount of electrical potential.

Table 42.1 Element types and degrees of freedom

Analysis type	Element type		Degree of freedom
	Solid	Contact	
Electrical–thermal	SOLID87	CONTA174/TARGE170	TEMP, VOLT
Mechanical–structural	SOLID90	CONTA174/TARGE170	UX, UY

During the RSW process, if there is no heat input, then there is joint failure, and if the heat input is excessive, then it leads to deterioration in welding quality due to spatters. Therefore, optimum heat control is a key factor controlling the weld strength.

Heat generated during the resistance spot welding can be expressed as follows [7]:

$$Q = I^2 R t, \quad (42.2)$$

where

q —heat generated,

I —electric current,

R —electrical resistance of the material,

t —time for which the current has been supplied.

The governing equation for axisymmetric transient thermal analysis is as follows [7]:

$$\rho c \frac{dT}{dt} = \frac{d}{dr} \left(k \frac{dT}{dr} \right) + \frac{k}{r} \frac{dT}{dr} + \frac{dT}{dz} \left(k \frac{dT}{dz} \right) + q, \quad (42.3)$$

where

ρ —density of the steel material,

c —specific heat,

T —temperature generated,

t —time of weld cycle,

k —thermal conductivity,

q —rate of internal heat generation, respectively.

In this analysis, material properties like specific heat, thermal conductivity, and material density are assumed to be temperature dependent. Since the thermo-elastic–plastic behavior is nonlinear, therefore, the stress–strain behavior can be expressed as [7]:

$$\{\Delta\sigma\} = [D]\{\Delta\varepsilon\} + \{C\}\Delta T, \quad (4)$$

where vectors $\Delta\sigma$ and $\Delta\varepsilon$ are stress and strain increment and ΔT is temperature increment. Matrixes D and C are material-related constants.

42.4 Model Validation

a. Material Selection

Advanced high-strength steels are the promising materials for the modern vehicles. The material selected for this experimental study is advanced high-strength steel. An attempt has been made to join dissimilar materials like dual-phase stainless steel (DP600) and austenitic stainless steel AISI 304L. AISI 304L is an important commercial alloy as it has excellent corrosion resistance, high strength, good ductility, and toughness. Joining these two materials is necessary as it has wide automotive applications.

The dimensions of the sample have been chosen as 140×40 mm and 1 mm thickness for both of these materials. Dual-phase steel possesses a unique microstructure consisting of ferrite and martensite that offer high strength to these steels coupled with high formability. Its chemical compositions are given in Table 42.2.

Normally, AISI304L steel contains 18% chromium and 8% nickel. Its chemical compositions are given in Table 42.3.

b. Process Parameter Selection and Experimental Work

From the literature survey, it is observed that major factors that researchers have considered for the analysis of weld strength are weld time, weld current, and electrode pressure. These parameters affect the mechanical properties like weld strength, load at failure, and physical properties like diameter of nugget, indentation of electrode, and metallurgical parameters like changes in microstructure and hardness variation. The prime factors affecting heat generation in RSW are electric current and weld time of the welding process cycle. The higher welding current generates large diameter nugget size of spot weld with the sufficient electrode pressure.

For the resistance spot welding of DP 600 steel to AISI304L steel, pneumatically operated spot welding machine (Keje make, 25 KVA) was used. Water-cooled copper chromium electrodes with face diameter 6 mm were used. Welding process parameters used for the resistance spot welding of DP 600 steel to AISI304L are as follows: weld current 7, 8, and 9 kA; weld time—5 cycles (0.10 s), 10 cycles (0.20 s), and 15 cycles (0.30 s); and electrode pressure—1.75, 2.00, and 2.25 bar. Hold time

Table 42.2 Chemical compositions of DP600 steels (in % volume)

C	Si	Mn	P	S	Al	Cr + Mo	Nb + Ti	V
0.17	0.80	2.20	0.080	0.015	<2.0	1.00	0.15	0.20

Table 42.3 Chemical composition of AISI 304L (in % volume)

C	Mn	Si	S	P	Cr	Ni
0.021	1.00	0.70	0.010	0.040	18.00	8.00

Table 42.4 Comparison between shear stress values of the actual experiment results and the simulation model results (DP600-DP600)

Run	Electrode pressure (Kg/cm ²)	Weld time (cycle)	Weld current (Ampere)	Shear stress (MPa)		Difference	Percentage deviation (%)
				Actual	Simul.		
1	1.5	8	7000	242	263	+21	8.6
2	1.5	15	9000	298	279	−19	6.3
3	1.5	22	11,000	328	319	−9	2.7
4	2.25	8	9000	269	277	+8	2.97
5	2.25	15	11,000	305	303	−2	0.65
6	2.25	22	7000	275	256	−19	6.9
7	3	8	11,000	280	297	+17	6.07
8	3	15	7000	289	284	−5	1.7
9	3	22	9000	289	279	−10	3.4

and squeeze time are 50% of the weld time, and off time is selected as 25% of weld time. Sheet material and electrode have well-defined values of electrical resistivity.

c. Model Validation for Similar Metal Welding of Dual-Phase Steel Sheets

To validate the proposed model, shear stress results are obtained by ANSYS software and compared with actual shear stress obtained for the shear tensile strength of the welded specimens. The model prepared in ANSYS is a coupled analysis model with transient thermal and transient structural analyses.

The shear strength comparison of values of welded specimen and numerical simulation is presented in Table 42.4. As the variation in the values of shear stresses in the actual experiments and the simulation model is below 5%, this model created using ANSYS environment is found to be valid for the analysis of RSW of DP600. This validated model was then used for simulation of welding of dissimilar material, DP600 with AISI 304L (Figs. 42.2, 42.3).

The upper sheet is of dual-phase steel (DP600), and the lower sheet is of austenitic stainless steel (AISI304L). Heat flux is applied as 120, 150, and 180 W/mm² for 7, 8, and 9 kA current intensities, respectively. The temperature distribution has been obtained from the transient thermal analysis, and thereafter, the electrode pressures have been applied to the sheets as per input values. The pressure values selected were 1.75, 2 and 2.25 kg/cm² for dissimilar material. These force values are applied to z-axis.

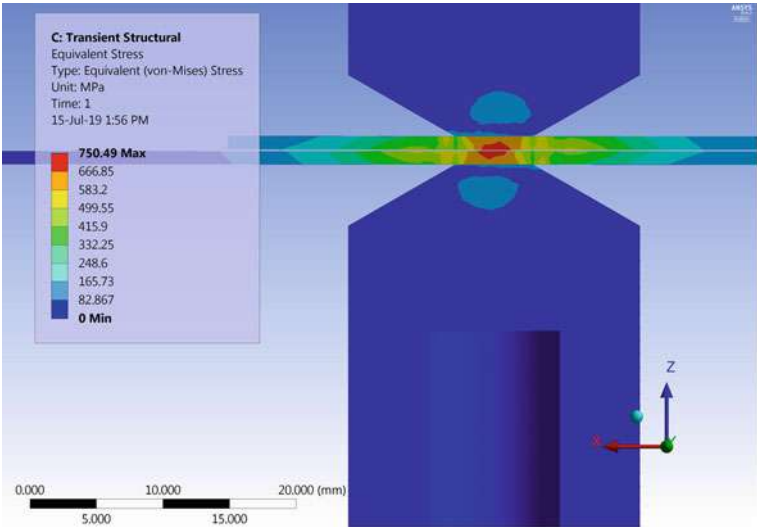


Fig. 42.2 Temperature distribution during welding

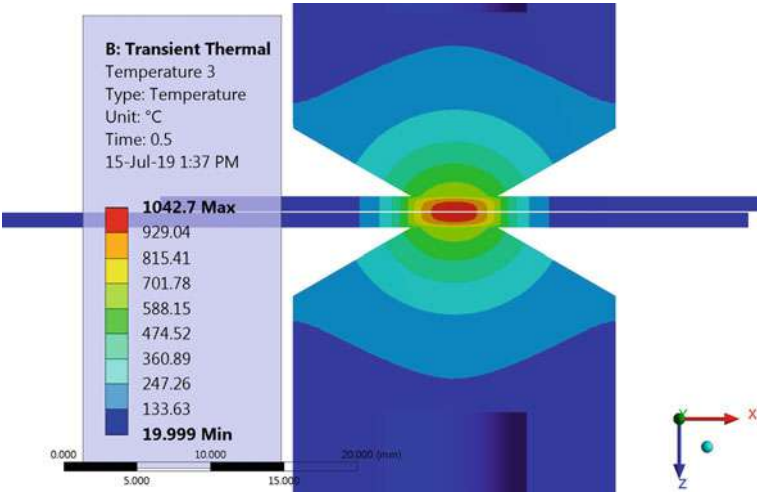


Fig. 42.3 Maximum shear stress distribution during welding

Table 42.5 Parameters and their levels for DOE for RSW of DP600–AISI 304L

Process parameters	Unit	Min (level—1)	Mean (level 0)	Max (level 1)
Weld current	kA	7	8	9
Weld time	Cycles	5	10	15
Electrode pressure	kg/cm ²	1.75	2	2.25

42.5 Design of Experiments for Welding in the Simulation Model

The simulation model has been compiled in ANSYS and is found valid for the analysis of similar and dissimilar metal welding. This compiled model is used for optimization of input parameters for resistance spot welding of DP600 and AISI304L. After validating the model with the experimental data, a set of simulations was designed to investigate the effect of various RSW process parameters on weld strength of the joints [5]. The process parameters and their levels used in this study are listed in Table 42.5. Full factorial design with 27 runs has been used for this study as indicated in Table 42.6.

42.6 Results and Discussion

In this paper, an attempt has been made to investigate the performance weld strength of dissimilar metals using experiment and also using the simulation tool in ANSYS. The main aim of this experimental work was to obtain maximum tensile shear strength in the welded specimen by using the input parameters like weld current, weld time, and electrode pressure. Materials selected for the study are DP600 and AISI304L. It is observed that the shear strength increases with increase in the current intensity, with weld time and with increase in the electrode pressure as shown in Fig. 42.4.

As electrode pressure increases, tensile shear stress values of the welded specimen increase till 2 bar. However, at 2.25 bar, it is observed that the shear strength values decreased as shown in Fig. 42.4a. For higher values of electrode pressure, the contact surface between electrode tip and sheet to be welded is increased and it reduces the resistance between the faying surfaces. As weld current increased from 7 to 9 kA, the shear strength increases and maximum value is possible with current 9 kA as shown in Fig. 42.4b.

For weld time of 5 cycles, as current increases from 7 to 9 KA, shear stress increased from 237 to 278 MPa. As the duration of weld current passing through the circuit increases to 10 and 15 cycles, the trend of increase in shear stress of welded joints is marginal. The maximum shear strength value was obtained during weld time of 15 cycles. The maximum shear strength value obtained is 298 MPa at 9 KA current supply as shown in Fig. 42.4c.

Table 42.6 DOE for simulation experiments of RSW of DP600-AISI 304L

Run	Electrode pressure (Kg/Cm ²)	Weld time	Current (A)	Tensile shear stress (MPa)
1	1.75	5	7000	238
2	1.75	5	8000	261
3	1.75	5	9000	277
4	1.75	10	7000	275
5	1.75	10	8000	285
6	1.75	10	9000	293
7	1.75	15	7000	272
8	1.75	15	8000	284
9	1.75	15	9000	296
10	2	5	7000	241
11	2	5	8000	265
12	2	5	9000	285
13	2	10	7000	278
14	2	10	8000	289
15	2	10	9000	295
16	2	15	7000	275
17	2	15	8000	288
18	2	15	9000	298
19	2.25	5	7000	242
20	2.25	5	8000	256
21	2.25	5	9000	272
22	2.25	10	7000	269
23	2.25	10	8000	274
24	2.25	10	9000	283
25	2.25	15	7000	261
26	2.25	15	8000	272
27	2.25	15	9000	281

Therefore, from the set of experiments conducted, it is concluded that the optimum value for obtaining higher strength is weld current 9 kA, weld pressure 2 bar, and weld time 15 cycles.

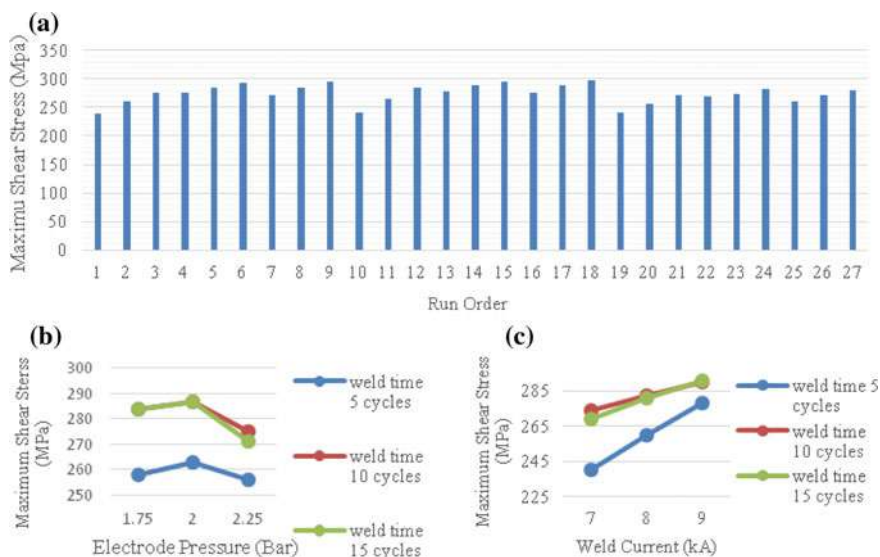


Fig. 42.4 **a** Maximum shear stress values as per run order, **b** maximum shear stress (Mpa) versus electrode pressure (bar), **c** maximum shear stress (Mpa) versus weld current (kA)

42.7 Conclusion

In this paper, a finite-element model using ANSYS 18 environment has been compiled to simulate the behavior of weld specimens in resistance spot welding. The model is used for the thermal–mechanical coupled analysis of similar (DP600-DP600) and dissimilar (DP600-AISI 304L) steels. The main purpose of the simulation is to determine weld strength by optimizing the weld parameters and also comparing simulation results to experimental results. The output values obtained in the simulation model are significant and up to 95% of values agreed with output values of actual experimentation obtained by resistance spot welding. From this analytical simulation experiments, the following conclusions were drawn for RSW of dual-phase steel, DP600 to AISI 304L steel.

1. There is reasonably good agreement between the results of the simulation model and actual experimental data.
2. At low electrode forces and higher values of electric current, the weld strength is observed to be higher.
3. At 2 bar electrode pressure, the maximum value of tensile shear strength is achieved. With further pressure increase in electrode pressure, the strength values are found to be decreasing.
4. The maximum weld strength was obtained at the weld time of 10 cycles. However, the decrease in weld strength was observed at the weld time of 15 cycles. This indicates that excess of heat is generated at the interface causing softening of the heat-affected zone (HAZ) at weld zone and resulting in joint of lesser strength.

5. The weld current is the most contributing parameter among all the parameters for weld strength. The maximum values of weld strength are obtained at 9 kA for dissimilar welding of DP600 to AISI 304L steel.
6. When weld current increases above 9 kA, it generates excessive heat, resulting in expulsion. The excess of heat is given to base metal and part of it is released into the atmosphere by radiation and convection modes. Excess of heat given to base metal, in turn, is responsible for the lower values of the tensile shear strength of the welded specimens.

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Chapter 43

The Effect of Process Parameters on Pulsed Through Transmission Laser Welding of Acrylic and Polycarbonate Sheets



Nitesh Kumar, Nikhil Kumar and Asish Bandyopadhyay

Abstract In the present research, the Taguchi method is applied to obtain the best combination of weld parameters to optimize weld characteristics in through transmission laser welding of two transparent materials namely acrylic and polycarbonate. Taguchi's L9 orthogonal array has been chosen for the experimental design. Laser power, welding speed and frequency have been selected as the process parameters. The individual effects of welding parameters on width (WW) and ultimate load (UL) of weld have been carried out using analysis of variance (ANOVA) and signal-to-noise (S/N) ratio. Next, grey-based Taguchi method has been applied for multi-objective optimization for maximization of ultimate load and minimization of weld width simultaneously. Finally, confirmatory tests have been performed to validate the applied optimization technique.

Keywords Through transmission laser welding · Taguchi method · S/N ratio

43.1 Introduction

Inferable from the broad utilization of plastics in the nourishment business and pharmaceutical industry, welding of plastics has turned out to be one of the quickest developing fields of study. It is now realized that the utilization of dissimilar materials in a joint helps consolidate the properties of the two materials to create an unrivalled quality weld while diminishing the expense. Through transmission laser welding (TTLW) is one of only a few welding techniques which can be utilized for the welding of unique materials. TTLW is currently broadly utilized for joining

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_43

thermoplastics in industry, utilizing laser sources, for example, diode, Nd:YAG and fibre lasers. The parts are situated together before welding and the laser beam goes through the upper part to warm the joint at the outside of the lower part. Bracing of these parts is important to achieve close contact in order that the laser beam melts the material precisely where the parts should be joined. Most thermoplastics can be welded utilizing the appropriate laser components. Special additives/pigments permit laser welding of dark to dark materials as well as transparent to transparent materials. In spite of the fact that the measure of research on TTLW is in the emerging stages, it guarantees high potential. A review of various strategies for enhancement methods for different welding forms has been done before arriving at the decision of the target of the present examination. Acharjee et al. [1] have conducted an experimental investigation of laser transmission welding of acrylics. The Taguchi design has been used as a statistical design for the experiment to set the optimal process parameters. Kumar et al. [2], have applied the Taguchi method with the L9 orthogonal array to find best settings of welding current, welding voltage, welding travel speed and some welding passes. The study of the welding parameters' effect on residual stresses and hardness of weld specimen has been carried out by a statistical technique, i.e. analysis of variance (ANOVA) and signal-to-noise (S/N) ratio. Kumar et al. [3] have attempted to optimize ultimate load in through transmission laser welding of two acrylic sheets with the help of response surface methodology. Hsiao et al. [4] have used the Taguchi method with grey relational analysis for the optimization of plasma arc welding parameters. Anawa et al. [5] have optimized CO₂ laser welding of stainless steel to low carbon steel using the Taguchi method. They have considered power, welding speed and defocusing distance as input parameters and weld pool dimensions as the responses. Kumar et al. [6] have performed multi-objective optimization of through transmission laser welding of dissimilar plastics using RSM. They have considered power, scanning speed and frequency as input parameters with a breaking load and weld width as the responses. Tarng et al. [7] have applied fuzzy logic in the Taguchi method to optimize the submerged arc welding process for multiple performance characteristics with arc current, arc voltage, welding speed, electrode protrusion and preheat temperature as process parameters results are provided to confirm the effectiveness of this approach. Lakshminarayanan et al. [8] have used the Taguchi approach to determine the most influential control factors which will yield better tensile strength of the joints of friction stir welded RDE-40 aluminium alloy. Kumar et al. [9] have applied the Taguchi method with L9 orthogonal array to find best settings of welding current, welding voltage, welding travel speed and a number of welding passes. The study of the welding parameters effect on residual stresses and hardness of weld specimen has been carried out by statistical technique, i.e. analysis of variance (ANOVA) and signal-to-noise (S/N) ratio. Aghakhani et al. [10] have developed a model using parameters such as wire feed rate, welding voltage, nozzle-to-plate distance, welding speed and gas flow rate on weld dilution.

Through transmission laser welding of two transparent plastic is not an area where enough investigation has been done. Hence, the urgent need for research work is required in this field. Taking this into consideration, the present research is planned

to study the effects of welding parameters on the ultimate load and weld width of welded transparent acrylic and transparent polycarbonate sheet. Optimization has been conducted to achieve the quality weld.

43.2 Experimental Set-up and Methodology

In the present study, Electrox EMS 100 laser marking set-up has been used for the welding of plastics. The 12 W Nd: YVO₄ (neodymium-doped yttrium orthovanadate) laser with 1064 nm wavelength has been used for the laser transmission welding purpose. The dimensions of acrylic and polycarbonate plastic sheets of size 100 mm × 35 mm × 0.5 mm have been chosen. Overlapping length is 70 mm. No additional absorbent has been used. The laser is used with a spot diameter of 70 μm. The experimental set-up used for the welding process is shown in Fig. 43.1. The selected process parameters and their limits are given in Table 43.1. For optimization of the responses in through transmission laser welding, MINITAB 16 software has been used. The schematic diagram of through transmission laser welding process and the joining pieces is shown in Fig. 43.2. The pictorial view of a welded sample before and after the break test is shown in Fig. 43.3



Fig. 43.1 A pictorial view of experimental set-up: **a** workstation; **b** laser beam delivery unit; **c** computer interface and **d** hydraulic pump with pressure gauge

Table 43.1 Process control parameters and their limits

Parameters with unit	Notation	Levels		
		1	2	3
Power, W	P	9.5	10.5	11.5
Scanning speed, mm/s	S	0.7	1.35	2
Frequency, kHz	f	270	335	400

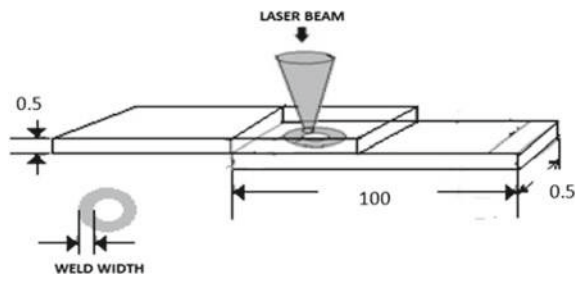


Fig. 43.2 Schematic diagram of through transmission laser welding process

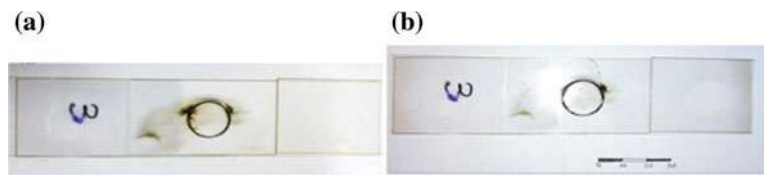


Fig. 43.3 A pictorial view of a welded sample **a** before break test and **b** after break test

43.3 Results and Discussions

The results of the ultimate load (UL) after the pull test and the measured value of average weld width (WW) using optical transmitting microscope are shown in Table 43.2.

The maximum and minimum values of ultimate load have been obtained for sample no. 7 and sample no. 5, respectively, while the maximum and minimum values of weld width have been found for sample no. 4 and sample no. 9, respectively.

Table 43.2 Measured values of weld width and ultimate load of welded samples

Exp No.	P (W)	S (mm/s)	f (kHz)	WW (μm)	UL (N)
1	9.5	0.7	270	222	371.72
2	9.5	1.35	335	150	297.05
3	9.5	2	400	202	236.65
4	10.5	0.7	335	247	363.70
5	10.5	1.35	400	160	186.61
6	10.5	2	270	156	314.70
7	11.5	0.7	400	258	378.90
8	11.5	1.35	270	175	309.50
9	11.5	2	335	103	305.36

43.3.1 Effect of Process Parameters on the Responses

To study the effects of process parameters on weld width and ultimate load, main effect plots have been drawn by using data in Table 43.2. Figure 43.4a and b show the main effect plots for weld width and ultimate load, respectively. It is found from Fig. 43.4 that with an increase in scanning speed, weld width decreases as the heat input per unit length decreases. As the power increases, the weld width also reduces due to the increase in the depth of penetration. However, with increase of frequency, weld width first decreases and then increases. From the main effect plots shown in Fig. 43.4a, the optimum parametric setting for minimization of weld width is power 9.5 W, scanning speed 0.7 mm/s and frequency 400 kHz.

Similarly, Fig. 43.4b can be concluded. The optimum parametric setting for maximization of ultimate load is power 11.5 W, scanning speed 0.7 mm/s and frequency 270 kHz.

43.3.2 Multi-objective Optimization

In grey relational analysis, experimental data, i.e. measured features of quality parameters are normalized to values ranging from zero to one. Grey relational coefficients are calculated to represent the correlation between the desired and actual experimental data. Then, the overall performance of the multiple response depends on the calculated grey relational values. Thus, the multiple response process optimization problem is converted to single objective optimization problem with the response being the overall grey relational grade. The Taguchi method is used for optimal factor setting for maximizing overall grey relational grade. The raw experimental data are given in Table 43.3. The grey relational generation is done based on the following calculations. For larger-the-better characteristics, [4]

x_ij = (y_ij - miay_ij) / (max_i y_ij - min_j y_ij) (43.1)

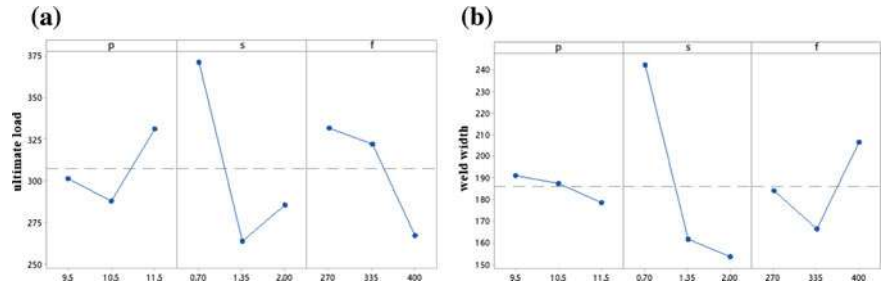


Fig. 43.4 a Main effect plots for weld width and b main effect plots for ultimate load

Table 43.3 Normalized values, grey relational coefficient values and grey relational grades of the responses

Experiment No.	Normalization	Normalization	Grey relation coefficient	Grey relation coefficient	Grey relation grade
	Weld width	Ultimate load	Weld width	Ultimate load	
1	0.232	0.923	0.3943	0.8665	0.6304
2	0.697	0.574	0.6227	0.5399	0.5813
3	0.361	0.26	0.4389	0.4032	0.42105
4	0.0709	0.921	0.3519	0.8635	0.6077
5	0.632	0	0.576	0.3333	0.45465
6	0.658	0.666	0.5938	0.5995	0.59665
7	0	1	0.3333	1	0.66665
8	0.537	0.639	0.5181	0.5807	0.5494
9	1	0.618	1	0.5668	0.7834

For smaller-the-better characteristics, [4]

$$x_{ij} = \frac{\min_i y_{ij} - y_{ij}}{\max_i y_{ij} - \min_j y_{ij}} \quad (43.2)$$

where y_{ij} is the i th quality characteristic in the j th experiment.

Normalized values for ultimate load are calculated by using Eq. (43.1) and those of the weld width are calculated by Eq. (43.2) and presented in Table 43.3.

Grey relational coefficients are expresses as [4].

$$\xi_{ij} = \frac{\min_i \min_j |x_i^0 - x_{ij}| + \xi \max_i \max_j |x_i^0 - x_{ij}|}{|x_i^0 - x_{ij}| + \xi \max_i \max_j |x_i^0 - x_{ij}|} \quad (43.3)$$

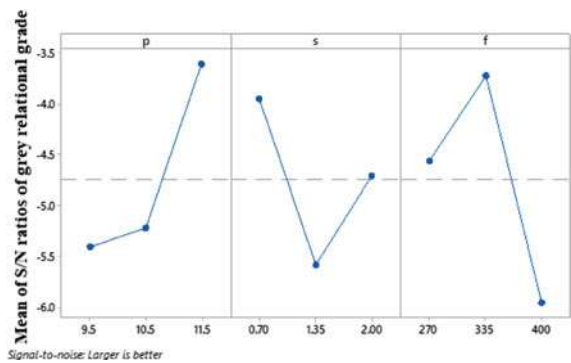
where x_i^0 is the ideal normalized result (i.e. best normalized result = 1) for the i th response and $\xi [0,1]$ is a distinguishing coefficient, which is used to weaken the effect of $\max_i \max_j |x_i^0 - x_{ij}|$ if its magnitude get too large.

The grey relational grade is calculated by using the following equation [4].

$$y_i = \frac{1}{m} \sum_{i=1}^m w_i \xi_{ij} \quad (43.4)$$

The values of grey relational coefficient and grey relational grade are given in Table 43.3 which has been calculated using Eqs. 43.3 and 43.4, respectively. Equal weightage has been given to the two response parameters in calculating the grey relational grades. The main effect plots for the S/N ratios of grey relational grades are shown in Fig. 43.5. The optimum combination of process parameters obtained as

Fig. 43.5 Main effect plots for S/N ratios of grey relational grade



per the plot is power = 11.5 W, scanning speed = 0.70 mm/s and frequency = 335 kHz for maximization of ultimate load and minimization of weld width simultaneously.

43.3.3 Analysis of Variance

The purpose of the analysis of variance is to investigate which welding parameter/s significantly affect/s the responses. This process is carried out by comparing the F-test value of the parameter with the standard F table value at the 5% significance level. The associated *P*-values of less than 0.05 for the models (i.e., $\alpha = 0.05$, or 95% confidence level) indicate that the model terms are statistically significant. Percentage contribution of process parameters on the responses are determined from Seq SS values of ANOVA table. Percentage contribution is calculated using the following expression [5].

$$\text{Percentage contribution of a term} = \frac{\text{Seq SS for the term}}{\text{total Seq SS}} \times 100 \tag{43.5}$$

where SS = Sum of squared deviation
Seq SS = Sequential sum of square deviation

43.3.4 Output of Analysis of Variance

The statistical technique of analysis of variance is applied on measured value (Table 43.2) to identify the significant welding parameter/s influencing the weld width and ultimate load. The results of ANOVA for weld width and ultimate load are given in Tables 43.4 and 43.5 respectively. From Table 43.4, it can be concluded

Table 43.4 Analysis of variance for weld width

Source	DF	Seq SS	Adj SS	Adj MS	<i>F</i>	<i>P</i>	Percentage contribution
<i>P</i>	2	254.9	254.9	127.4	0.08	0.927	1.25
<i>S</i>	2	14432.9	14432.9	7216.4	4.45	0.183	70.96
<i>f</i>	2	2410.9	2410.9	1205.4	0.74	0.573	11.85
Residual error	2	3240.2	3240.2	1620.1			
Total	8	20338.9					

Table 43.5 Analysis of variance for ultimate load

Source	DF	Seq SS	Adj SS	Adj MS	<i>F</i>	<i>P</i>	Percentage contribution
<i>p</i>	2	2890	2890	1445	1.05	0.488	8.98
<i>s</i>	2	19283	19283	9641	7.00	0.125	59.91
<i>f</i>	2	7257	7257	3628	2.64	0.275	22.55
Residual error	2	2753	2753	1377			
Total	8	32183					

that the scanning speed is the most significant factor on the weld width followed by frequency then power. Similar effects have been found for ultimate load Table 43.5.

43.3.5 Confirmatory Tests

After the identification of the optimum levels of the welding parameters, these results are applied to verify the improvement of the combined outcome. The confirmatory experiment is conducted at the optimum combination of values, i.e. power = 11.5 W, scanning speed = 0.70 mm/s and frequency = 335 kHz. After testing, it is found that the optimized combination does indeed produce improved results, weld width = 102 μm and ultimate load = 395.8 N.

43.4 Conclusions

In the present study, lap welding of two transparent plastic sheets each of 0.5 mm thickness one being acrylic and the other polycarbonate has been done under varying welding process parameters, by through transmission laser welding. Based on the observations and analyses made in the study, the following conclusions are drawn:

- The width of the weld seam decreases with an increase in power and also with an increase in scanning speed but does not respond consistently with the increase in frequency.
- ANOVA indicates that scanning speed is the most significant factor for both weld width and ultimate load.
- The combination of process parameters obtained for simultaneous optimization of weld width and ultimate load is power = 11.5 W, scanning speed = 0.70 mm/s and frequency = 335 kHz.
- The confirmatory test validated the improvement in the weld width and ultimate load in the case of welding of acrylic and polycarbonate

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Part II

Product Design and Development

Chapter 44

Design and Development of Combination Tool for Drilling and Tapping Operation on PVC



Yogesh G. Kamble , P. D. Pantawane, B. Rajiv and B. B. Ahuja

Abstract Form or friction drilling is becoming popular due to its ability to quickly form the hole with bushing on the sheet material. The formed hole is to be tapped with form tap, and thus, tapped holes on the sheet materials are used in sheet metal working processes. This process requires two separate operations of form drilling and form tapping and consumes extra time causing more cost of production. This paper proposes the design and development of combination tool, made of high-speed steel (M2) tool which combines the friction drilling and form tapping operation in a single step. The main objective of designing the combination tool is to minimize the operation time and cost of production as both operations can be performed on the same machine using a single tool. The combination tool have been designed keeping in mind polymeric sheet material and the geometry of form section and ta section decided according to the requirement. The two sections of combination tool, viz. form section and tap section used to carry out the operation in that sequence. Form section of combination tool has the geometry vis-a-vis friction drilling tool. The rotating conical tool when contacts the work piece generates the frictional heat resulting in softening of work piece material. Further advancement of tool into the work piece leads to plastic flow of material along the direction of feed which reproduces the cylindrical hole resembling cylindrical portion of form section. Tap section which is located just above the form section carries out the second sequence of operation, i. e., form tapping. Process involves the reproduction of threaded profile on the drilled holes which completes the cycle of forming and tapping the holes on the sheet materials.

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Keywords Combination tool · Form drilling · Tapping · PVC polymer material

44.1 Introduction

The mechanics of hole formation in form drilling is due to thermal softening followed by penetration of tool inside work. Drilling is a hole-making process as it removes material due to a combination of extrusion and negative rake cutting. However, all conventional drilling methods suffer from various limitations that they inherit from the machining processes. Friction drilling is performed by a sharp cone (90° angle) of circular shanked stainless steel tool. The tool is rotated at high speed to develop sufficient temperature, and thrust is applied to form a hole without chips, and hence, some investigators prefer to call thermal drilling. The friction and plastic deformation produces heat and increases the work piece temperature [1]. The formation of bush is an addition in this process which can be used as a natural bush in sheet metal or it can be used for tightening screws after tapping. Normally, the bush thickness is two times the thickness of the sheet metal of a polymeric material and three times thickness of a ductile material. As combination tool is the summation of form drilling tool and tapping tool, initially lower part of tool will create a bushing on sheet metal, tubing, or thin-walled profiles [2]. The bushing thickness can be threaded by the further travel of combination tool, providing a more solid connection for attachment than attempting to thread the original sheet of a material. It produces bushes for a tapped and untapped hole. All work materials from the hole contribute to form bushing. In additionally, no cutting fluid or lubricant is necessary, which makes friction drilling a totally clean, environmentally friendly process.

A radial drilling machine is used to do friction drilling holes and tapping trials on a polymeric material. A radial drilling machine will help to minimize adverse effect on the sheet metal work piece to improve the bushing quality of a hole. Considering the actual experimental conditions in study of drilling, the spindle speeds used for sheet of different metals as variation in rpm. These were determined experimentally to enable the penetration and forming of the hole. For material like titanium, with low thermal conductivity, most of the frictional heat is retained in the tool–work piece interface. The effect of frictional heating is relatively important [3]. This causes increased temperatures in the work piece and results in undesired material damage and improper bushing formation. For the aluminium type of alloy, the thermal conductivity is high. A large portion of the frictional heat is transferred into the work piece, and the effect of friction heating is relatively small. Low temperature causes insufficient increase in ductility and softening, resulting in high thrust force, denting of the work piece, and improper bushing formation. These effects show the selection of low and high spindle speed for titanium and aluminium, respectively.

44.1.1 Selection of Tool and Work Piece Material

To conduct several experiments on a polymeric tube, high-speed steel (M2) as a tool material is selected. High-speed steels are designated as Group M steels according to the AISI classification system. Tungsten is present in all types from M1 to M10, except M6, and cobalt is not present in any these steels [4]. As the temperature generation in the process is estimated to be in the range 400–500 °C so that High Speed Steel is chosen for experimentation. HSS M2 grade is the “standard” and most widely used industry. The chemical composition of tool material is as shown in Table 44.1. The mechanical properties of tool are mentioned in Table 44.2.

High-speed steels have similar performance as compared to tungsten high-speed steels. However, the initial cost of high-speed steels is at the lower side. Commercial available PVC polymeric materials have definite characteristics which drive the machining behaviour. Simultaneously, the mechanisms involved while drilling PVC polymeric materials are specifically different from those observed when drilling homogeneous materials such as metals. The application of PVC polymeric material in the field of manufacturing has witnessed different parameters during the drilling operation that can influence the drilling factors and the material elongation and damage. To know the damage caused by elongation, it is essential to know the properties of a material. The physical properties of the work piece material are as shown in Table 44.3.

By knowing above, a highly adaptable PVC polymer is compatible with different additives; also, it can be plasticized to make it flexible for use in building applications and various window frames.

Table 44.1 Chemical composition on high-speed steel (M2)

Carbon (C)	0.78 (%)
Manganese (Mn)	0.39 (%)
Sulphur (S)	0.002 (%)
Phosphorous (P)	0.003 (%)
Chromium (Cr)	4.17 (%)
Molybdenum (Mo)	4.91 (%)
Vanadium	1.95 (%)
Tungsten (W)	5.86 (%)

Table 44.2 Mechanical properties of high-speed steel (M2)

Hardness	239 HBN
Ultimate tensile stress	739.76 N/mm ²
Elongation	17.26 (%)

Table 44.3 Properties of PVC polymeric material

Physical properties	Value
Tensile strength	2.60 N/mm ²
Notched impact strength	2.0–4.5 kJ/m ²
Thermal coefficient of expansion	80×10^{-6}
Melting point	212 °C (413 °F)
Density	1.38/cm ³

44.1.2 Friction Drilling Tool

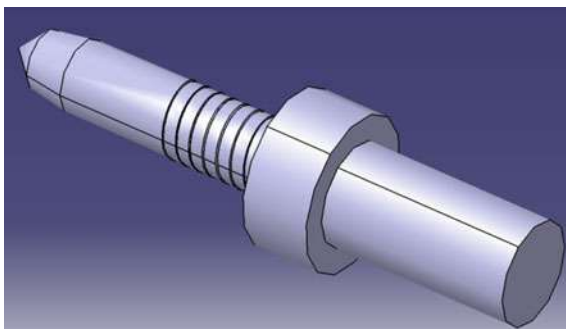
Friction drilling tools mainly used for hole and bush formation on the sheet material basically defined by three regions, viz. 1. centre or tip, conical and cylindrical. Centre or tip has the 90 degree angle with short length, which prevents the early penetration of the tools and helps to generate the frictional heat. The conical region mainly used for fractional heat generation has the angle 30–40 degree and sufficient length of the order of 5 mm, 6 mm depending on material properties and thickness. The cylindrical region which gives the final shape and size to the formed hole has the standard length and diameter. Friction drilling tool has the advantage of forming hole as well as bush on the sheet material which can be threaded in further threading operation. These threaded holes have many applications in sheet material joining processes like furniture, flanges, and vehicle bodies. The complete operation cycle consists of drilling and threading which is time consuming and needs more investment on set-up and tools. Conventional tool used for friction drilling is shown in Fig. 44.1

44.1.3 Development of Combination Tool

In this work, attempt has been made to develop the combination tool which can perform drilling as well as threading operation in order to reduce the operation time and hence cost of production. Tool geometry of a combination tool is an important

**Fig. 44.1** Tool used for friction drilling

Fig. 44.2 Pictorial 3D view of a combination tool



parameter in order to generate sufficient frictional heat and proper hole shape, size and thread quality. Also, the length and quality of bushings generated are affected by tool geometry. In view of this and also considering the workpiece material properties, several trials have been undertaken with different geometry of the tool. Results in terms of bush quality, dimensional accuracy, etc., tested geometry for the given material proposed. The combination tool of high-speed steel (M2) is manufactured on a CNC machine (MTAB make). The pitch of threading section of tool is kept as 1.25 mm. The dimensions of the tool used in this study are as the length of the cylindrical region of tool, $h_n = 10$ mm and diameter 7.9 mm (M8), The length of the conical region = 7.3 mm, the length of the tool tip = 1.22 mm. The tool angles used for this study are tool friction angle, i.e. conical region angle 36 degree, and tool tip angle 90 degree. The 3D model of a combination tool is shown below Fig. 44.2

A number of trials designed carried out to verify the influence of the two main parameters affecting the form drilling and form tapping operation: the spindle speed and the penetration (feed) speed. These parameters are swept from 800 to 1200 RPM and from 0.5 to 1.00 mm/rev, and for tapping, parameters are swept from 50 to 100 RPM and from 0.75 to 1.25 mm/rev, respectively. In the whole process, one can measure cutting force, torque, and temperature. Once the operation completes hole diameter, gap, cup size, thread size, and profile can be measured to formulate the best combination.

The steps and its description of combination tool are as follows:

1. The central area (tip area) of the tool tip: It rotated over the surface of the work piece while advancing further; therefore, thrust force increases.
2. Tool and work piece contact area is vital as the process progresses, the contact area between the conical tool tip and the work piece increases, and thus, the friction and the associated torque start growing rapidly because the radius is increased.
3. Cylindrical portion of the tool contacts the burr portion of higher thickness, resulting in a slight increase in the thrust force and rapid increase in the torque values.
4. On a cylindrical portion, threading profile is present with required pitch of the thread.

Fig. 44.3 Manufactured combination tool



5. For ductile materials as a work piece, it has been observed that the torque decreases when the cylindrical portion of the tool starts to deform the burr region of lower thickness. Also, thrust force experiences a steady decrease in its value.
6. The top burr is crushed by the tool resulting in higher torque and axial force. This area corresponds to the tool chip breaker area. This chip breaker area is not required in case combination tool. Retraction of tool appears in the last final step. At the time of tool retraction, the thrust force decreases rapidly to zero, although there is a minimal friction between work piece and tool as shown by the measured axial moment (Fig. 44.3).

As soon as the constant speed and feed provided, it appears to be a peak thrust force which occurs with travel of combination tool from the initial contact with the work piece. Form tapping process involves the reproduction of threaded profile on the drilled holes, which completes the cycle of forming and tapping the holes on the sheet materials.

44.1.4 Selection of Process Parameters

The input process parameters like cutting parameters, tool geometries (including conical angles), type of tools and tool materials specifications show the effects like thrust force, torque, tool wear, quality of thread profile, delaminating, surface roughness and thread strength which are called output process parameters. So, it is necessary to select the proper process parameters for obtaining the best performance on the drilling operations, best hole quality, which represents minimum damage of the machined components and satisfactory machined surface [5]. For form drilling operations, the following process parameters are used (Table 44.4).

Table 44.4 Form drilling operating parameters

Circular PVC tube (Thickness)	mm	1	2	3
Speed	RPM	800	1200	1600
Feed	mm/rev	0.50	0.75	1.00

Table 44.5 Tapping process parameters

Circular polymer tube (Thickness)	mm	1.5	2.25	3.00
Speed	RPM	50	75	100
Feed	mm/rev	0.75	1.00	1.25

The maximum depth of thread is significantly higher than in the case of conventional threading (cutting) tools, being limited by the quality of the coolant and the tool length. The maximum pitch of the thread depends on the material properties of the work piece. Undeniably, not all the materials are suitable for form tapping, that is, due to the intrinsic deformation involved, the material should have a minimum ductility, minimum failure resistance at 5% and should not exceed a maximum mechanical resistance of 1400 N/mm². Table 44.5 shows the process parameters required to do tapping process on a circular polymeric tube.

To do threading trials, different feed rates (0.75–1.25 mm/rev) were conducted at constant spindle speed (100 r/min) as a conservative criterion, and the criterion with the lowest feed rate was selected since it generates low amplitude and damped peak.

44.2 Experimental Set-Up

A radial drilling machine with tapping attachment, is used for experimental trials over PVC polymer material tube. The work piece is clamped over the specially designed clamping device. The circular tube of $\varnothing 38$ mm, length 100 mm with thickness of 1.5, 2.25 and 3 mm selected for experimentation. Overview of the set-up is shown in Fig. 44.4.

At the time of an experimentation, dynamometer can also be used to measure the thrust force involved mainly in drilling and also in tapping process. A maximum spindle power of 10 kW required which is more than enough for the application. To ensure good clamping, a fixture is designed and manufactured for radial drilling machine itself. This required for the alignment of the work piece with respect to the axes of the machine and avoids the deflection of the sheets during the hole and thread making. Figure 44.5 shows the main pictorial view for the actual threads which are in the damaged condition. Also, the experimental set-up includes work piece materials, fixture, and tools. Three thickness of PVC polymeric material were investigated, combining two types of PVC polymer of common circular polymer tubes.

Fig. 44.4 Threaded portion on a PVC polymeric material



Fig. 44.5 Formation of burr on PVC polymer material tube



The formation of shoulder in form drilling method can be used as natural bush in sheet metal or it can be used to tighten the screws after tapping operation. According to basic fundamentals of drilling and tapping operation, the cutting parameters such as cutting velocity and feed rate, tool geometry, tool types and tool materials have shown greater effect on the thrust force, torque while drilling conventional as well as PVC polymeric tube laminate.

As mentioned earlier, the conventional friction drilling tool has two different sections, viz. a conical and cylindrical surface. The conical portion penetrates the hole and softens the sheet material; meanwhile, a cylindrical segment is responsible for the final hole diameter. On the subject to several form drilling tools in cases, carbide tools (90% WC and grain size ($1\text{ }\mu\text{m}$)) were employed for achieving $M8 \times 1.25$. However, in case of proposed study, high-speed steel (M2) shows different challenges for getting required good dimensional hole and tapping operation [2]. It has been noted that, the hole quality in conventional drilling with twist drills results diminishing of hole quality with tool wear. On the contrary, in friction drilling, just the vice versa case is experienced because of the progressive adhesion of the work material onto the tool bit surface. Afterwards, the hole needs to be threaded by friction tap area on the same tool with pitch 1.25 mm. In respect of the tool, taps are often

polygonal geometries with at least five lobes. Taps are made of same material high-speed steel (M2) material. According to few cases, anti-friction coatings or internal lubrication are advisable as of temperature generation is at higher side. But in the mentioned work in the paper as speed and feed is at lower side and also workpiece material is PVC polymer tube, the temperature generator is at lower side. To carry out threading operation, it has been observed that it requires tapered cutting edges with smaller diameter to initiate the material removal [1]. According to the hole machined with the 7.3 form tool, a form tap for M8 designed and manufactured. In tapping, operation, the maximum depth of the thread and the maximum threaded pitch are the most important parameters while conducting the experiment.

44.3 Conclusions

The following conclusions may be drawn from several trials and literatures on use of combination on PVC polymeric material. Under the optimal friction drilling condition, the bushing length of the drilled hole found nearly two times longer than the sheet thickness. While in case of metallic material, it comes around three times of sheet thickness. Combination tools have several findings considering its effect on a circular polymeric tube material.

- Excessive flash over PVC tube is clearly observed once tapping cycle completes. It needs to be removed manually.
- The bush formation after trial has been found to be two times its original thickness in PVC polymeric material tube.
- High-speed steel (M2) has not shown any change in its original geometry after trials.
- Burr formation on the work piece material has been observed on several threaded holes after tapping operations.

Acknowledgements The authors would like to thank Prof. H. Bagchi, Professor Emeritus, COEP, and Department of Production and Industrial Engineering, College of Engineering, Pune, for providing the necessary support to carrying the ongoing research work.

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Article ID: IJMET 08 04 028 (April 2017)

Chapter 45

Processing and Characterization of a High Entropy Alloy in Application to Golf Club Head



N. A. Srinidhi and M. Ramachandra

Abstract High entropy alloys are known for their excellent mechanical and wear-resistant properties. The main objective is to incorporate the concept of high entropy alloys in the manufacturing of golf club head to enhance the compressive strength, impact strength, hardness and wear-resistant properties. An equiatomic, multi-principal elements were processed to form AlCuFeNiCr high entropy alloy using permanent metal mould casting technique. Mechanical tests were conducted and studied. Pin-on-disc wear test was conducted to determine wear rate using L9 orthogonal array. Results proved that AlCuFeNiCr alloy exhibits superior mechanical and wear-resistant properties compared to other golf club heads which are processed using conventional alloys. Influences of input parameters on output parameters using Minitab software were studied. Scanning electron microscope, energy-dispersive spectroscopy and optical microscope were utilized to study microstructure, elemental composition and worn surfaces, respectively. BCC phase was noticed which conveys better hardness properties.

Keywords High entropy alloys · Pin-on-disc wear test · Design of experiments · Mechanical properties

45.1 Introduction

45.1.1 High Entropy Alloys

High entropy alloys are gaining extraordinary research interest in the field of engineering and material science in particular because they exhibit excellent and potentially desirable properties [1]. Dissimilar to conventional/traditional combinations,

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_45

which are having one or two base materials, these high entropy alloys involve numerous principal elements with the conceivable number of compositions, widening astonishingly more than traditional alloys.

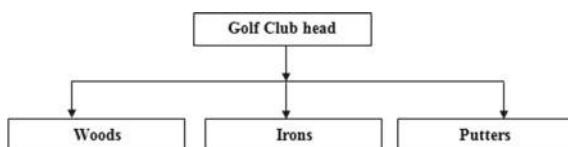
High entropy alloys are those combinations which are having at least five principal elements with equiatomic proportions or nearly equiatomic ratios or it can also be defined as those alloys which contain at least five elements with the atomic percentage between 5 and 35% (major elements) and <5% (minor elements) [2, 3]. The basic principle is that at high temperatures, the stability will be enhanced when there is a significant mixing of solid solution phases in these high entropy combinations compared with the inter-metallic compounds. High entropy alloy exhibits properties which are considered as the major aspects for the wide variety of applications in wear-resistant and high-temperature areas, etc. It deals with few structural properties such as wear, mechanical, oxidation and electrochemical. Mechanical properties include hardness, yield strength, elastic modulus fatigue and creep. Since it has got applications in high-temperature area, resistance to creep, solidification and oxidation play a major role in the selection of material. These results are very important to prove high entropy alloys are better than conventional alloys. There are some applications with the moving counterparts, and wear-resistant materials are used which are hard, thermally stable, tough and chemically non-reactive in nature. In different corrosive conditions, the corrosion resistance is affected by the composition of an alloy and its microstructure. High entropy alloy show a very good performance in sodium chloride and H_2SO_4 solutions as well [4, 5]. Chromium, cobalt, nickel, etc., in high entropy alloys enhance the corrosion resistance properties in acidic medium than conventional alloys. Molybdenum exhibits pitting corrosion resistance properties. Aluminium and manganese do not project a positive effect on pitting corrosion resistance properties. High entropy alloys exhibit lower density and greater strength and have wide application in energy industries and transportation. Since this has capacity to withstand high temperatures, high entropy alloys can be used to manufacture the engine blades over conventional materials such as titanium and steel alloys [6]. High entropy alloys are generally used in cryogenic applications in the storage of liquefied gas and can retain the properties in the 77 K as well [6]. The uniqueness in this research activity is the material combination to manufacture a golf club head which exhibits better compressive strength, impact strength and higher wear-resistant properties, whereas the materials such as titanium and other metal matrix composites used in conventional golf club heads show high density, lower hardness and less compressive strength.

45.1.2 Golf Club Head

A head is a part of golf club which is mainly used to strike the golf ball into a distance. A club comprises of a long shaft with a grip and club head. There are 14 golf clubs in a standard set. The golf club heads are categorized into three types.



Fig. 45.1 Golf club head



In this article, the processing of iron heads was discussed using AlCuFeNiCr high entropy alloy. The pictorial representation of a golf club head is shown in the Fig. 45.1.

45.2 Experimental Procedures

The corresponding principal elements were mixed in proper proportions according the principle of high entropy alloys, and a composition AlCuFeNiCr was melted and processed using casting technique in a high-temperature gas furnace and poured into a metal mould in a controlled temperature or an optimum temperature ranges from 700 to 750 °C; otherwise, the casting exhibits less strength and creates blowholes. Pre-heating of metal moulds was carried out to avoid the thermal damage, and it makes the molten metal to move quickly and easily. The ingots of $\phi 25$ and 190 L were prepared. ASTM standards were incorporated in machining of ingots to conduct compression test, impact test, hardness test and pin-on-disc wear test.

45.2.1 Brinell Hardness Test

The Brinell hardness test was led on as-cast AlCuFeNiCr high entropy alloy to determine the hardness value. The specimen of geometry $\phi 20$ mm and L30 mm was prepared as indicated by ASTM E10 standards [7]. A load of 500 kgf was applied with a ball indenter of dia 10 mm. Subsequent to the loading of specimen for 30 s, the diameter of the indentation on specimen was observed. The general equation which was used in arriving the BHN is as follows.

Brinell hardness number (BHN) can be calculated by using the formula:

$$\frac{2P}{\pi D \left[D - \sqrt{D^2 - d^2} \right]} \quad (1)$$

where

D = Diameter of the indenter in mm

d = Indentation diameter

P = Load applied in kgf

45.2.2 Compression Test

Compression test was carried out to determine the compressive strength, break load, percentage decrease in length and percentage increase in area. The specimen was prepared using conventional lathe of geometry $\phi 20$ mm and L60 mm according to ASTM E9 [8]. Universal testing machine of capacity 200KN was utilized to obtain the results. Initial length and diameter of 60 mm and 20 mm were taken respectively. Stress versus strain graph was plotted.

45.2.3 Izod Impact Test

Izod impact test was carried out to determine the impact strength of as-cast AlCuFeNiCr high entropy alloy of dimensions $\phi 12$ mm and L75 mm. Impact tests were carried out in studying toughness of the material. Brittle materials have low toughness. Impact value directly depends on the temperature. Usually, impact energy is low at lower temperatures. The dimensions of specimen have a role in the results because it allows a different kind of imperfections. Stress risers in the material reduce impact energy. The sample was tightly held in the post with a cylindrical fixture. The hammer was fixed at 90° before releasing.

45.2.4 Pin-on-Disc Wear Test

Golf club head processed using high entropy alloy should have high wear resistance properties. The specimen for pin-on-disc wear test of $\phi 6$ mm and L40 mm was prepared as per the G99 standards [9]. L9 orthogonal array was designed to conduct the experiments to calculate frictional force, wear rate, coefficient of friction. Results were analysed using Minitab software to arrive the percentage influence of input

parameters such as sliding speed, load and time on output parameters. wear rate, frictional force and coefficient of friction.

45.2.5 Microstructure Analysis

AlCuFeNiCr samples were machined according to the standards using various machining techniques. Surface of the sample was made smooth using different grades of emery paper P400, P600, P800 and P1000 to remove the scratches. Semi-finished surface was polished using diamond paste on a velvet cloth for 30 min to get mirrored finish.

Scanning electron microscope and energy-dispersive spectroscopy were utilized to check the morphology and elemental composition of the specimens.

45.2.6 Golf Club Head (AlCuFeNiCr)

The golf club head was fabricated using the high entropy alloy. Since the head was fabricated with the novel alloy, the properties which it exhibits are expected to be better than the heads which are fabricated using conventional alloys. It gives the significant properties like low density, promising compressive strength compared to other alloys. The fabricated golf club head (AlCuFeNiCr) is shown in the Fig. [45.2](#).

45.3 Results and Discussion

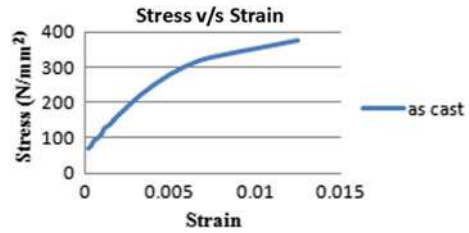
45.3.1 Brinell Hardness Test

The diameter of the indentation on specimen was observed to be 2.71 mm using Brinell microscope which is an instrument which makes proper measurements of

Fig. 45.2 Fabricated model



Fig. 45.3 Stress versus strain curve



indentations on the specimens after the test. Using the condition which involves D , d and P , the hardness values were determined. The hardness was found out to be 85HB. It can be represented as 85HB 10/500/30.

45.3.2 Compression Test

Compressive test was successfully carried out and discussed below. It was observed that the length has gradually decreased and the percentage reduction was 3.6% and area increased with a percentage of 6.1%. Load was noted down for every deflection ΔL . The AlCuFeNiCr alloy fractured at 118.4 KN and the ultimate compressive strength was calculated and found to be 377 MPa.

In Fig. 45.3, it is observed that the stress value is ranging from 68.75 to 377 MPa and fractured at 118.4 KN.

45.3.3 Izod Impact Test

Izod impact test was carried out to determine the impact strength of the as-cast AlCuFeNiCr alloy. The impact energy was noted to be 3.9 N-m, and impact strength was calculated to be 34.69 J/m². Observations were made on the impact properties of AlCuFeNiCr alloy and other conventional alloys. It was observed that the AlCuFeNiCr alloyed golf club head exhibits better impact strength. Impact strength can be calculated using the expression:

$$\text{Impact Strength} = \frac{\text{Impact energy in J}}{\text{Cross sectional area in m}^2} \quad (2)$$

45.3.4 Pin-on-Disc Wear Test

Wear test was carried out to find out the wear rate of as-cast AlCuFeNiCr high entropy alloy. L9 orthogonal array was implemented with three levels and three factors.

Input parameters Load (5, 10, 15 N)
Sliding speed (300, 500, 700 rpm)
Time (30, 35, 40 min)

Output parameters Wear rate, frictional force, coefficient of friction

Influence of input parameters on output parameters was obtained using Minitab software and discussed below.

Figure 45.4 shows that the wear rate gradually increases with increase in the load. The wear rate is minimum at 5 N and gradually increases with increase in load from 5 to 15 N. At the maximum load, the wear rate reaches to 56.6694 microns/min, and as time increases, the wear rate decreases. In experiment 7, the average wear was obtained to be 2055.183 microns which was conducted for 40 min and the wear rate corresponding to the average wear was calculated to be 51.3795 microns/min, whereas in experiment 8, the average wear and wear rate were noted to be 1700.08 microns and 56.6694 microns/min, respectively. By comparing these two experiments, it can be concluded that the time is inversely proportional to wear rate. In the experiment 9, the wear rate reduced because of the larger sliding over the disc which causes hardening of surface layer composition of debris.

Figure 45.5 shows the frictional forces of various experiments conducted using pin-on-disc wear test which were compared with one another. It proves that the frictional force increases with load. In the experiments 1, 2, 3 with constant load of 5 N, the average frictional force was 2.24 N, and in the experiments 4, 5, 6 with a constant load of 10 N, the average frictional force was seen to be 3.592 N, and for 15 N of load in the experiments 7, 8, 9, the frictional force was 5.972 N. This analysis shows that the load is directly proportional to the frictional force.

Figure 45.6 represents coefficient of frictions of various experiments plotted on a bar graph. In experiments 1, 2, 3 with the load of 5 N, the average coefficient of friction was 0.449. The coefficient of friction with 10 N in the experiments 4, 5, 6 was noted to be 0.3592, whereas in the trials 7, 8, 9 the average coefficient of

Fig. 45.4 Wear rate in microns/min

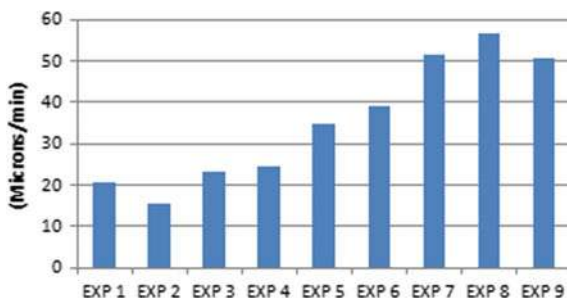


Fig. 45.5 Frictional force in Newtons

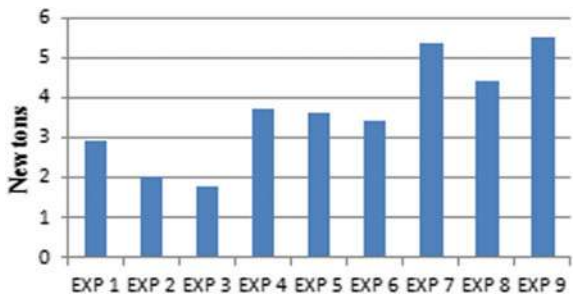
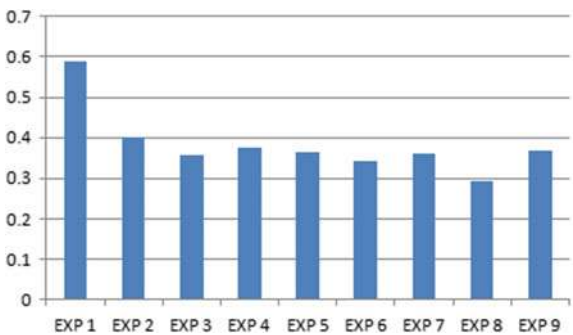


Fig. 45.6 Coefficient of friction



friction was found out to be 0.339. These experimental results prove that the load is inversely proportional to the coefficient of friction [10, 11]. It is observed that there is an increase in coefficient of friction after the experiment no. 8; this is due to the less surface roughing when there is an entrapment of waste debris.

Minitab 18 software was utilized to determine which factor/experiment is playing a significant role in wear rate. It was noticed that load of 15 N, sliding speed of 500 rpm and time 30 min were the major significant factors in causing the maximum wear rate. The percentage influence of load, sliding speed and time on wear rate, frictional force and coefficient of friction was developed and tabulated as shown in Table 45.1.

Load is considered to be the significant factor on wear rate and frictional force as the increase in load increases the wear rate and frictional force which is already been discussed in the previous sections. The load is inversely proportional to coefficient

Table 45.1 % contributions of input parameters on output parameters

Input parameters	Wear rate	Frictional force	Coefficient of friction
Load	90.75	88.77	39.17
Sliding speed	2.57	5.12	24.35
Time	6.63	0.34	5.71

of friction, and the % contribution of load is less compared with the other output factors [10, 11].

Optical microscope was used to study the worn surface. The images below give the detailed information.

In Fig. 45.7a large and dark spots were observed. The presence of these spots is the evidence of adhesive wear which occurred between the pin and the disc [12]. In Fig. 45.7b, small amount of chromium element is left without melting. This is

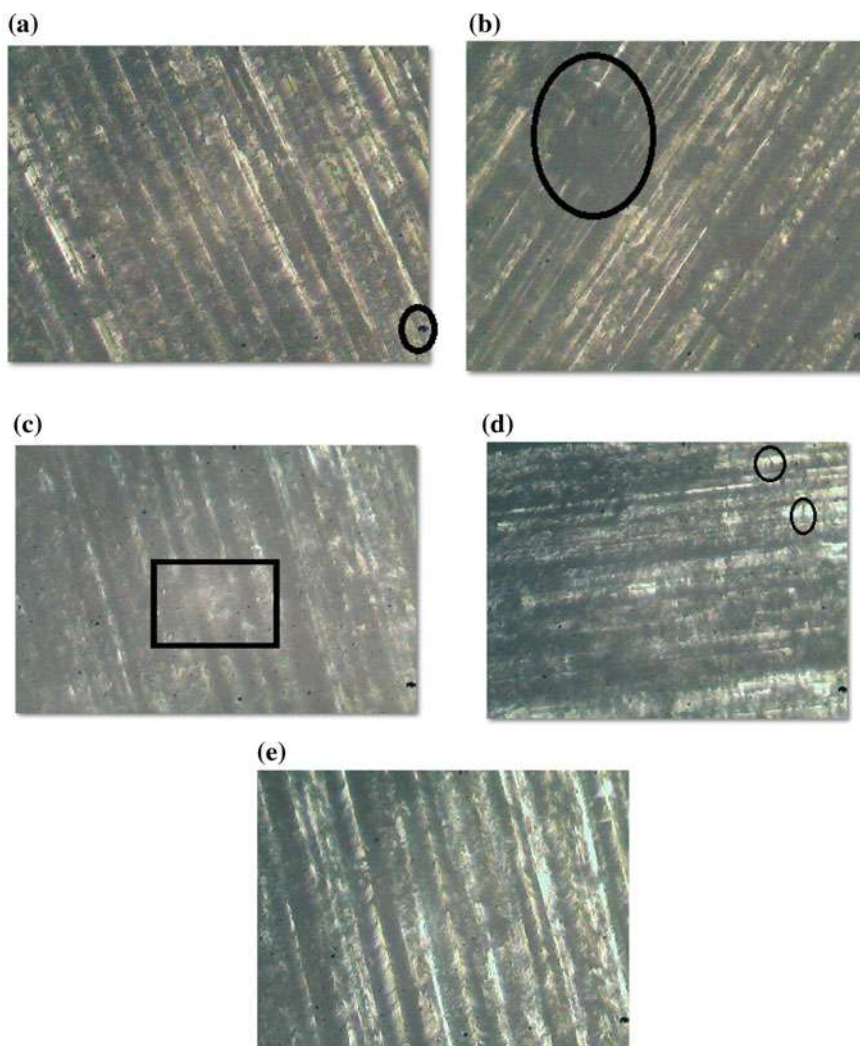


Fig. 45.7 a Dark spots. b Chromium element. c Debris on the wear surface. d Micro-cracks on wear surface. e Groves on the surface

because of the formation of oxides which increases the melting point and causes some material to remain same without changing its phase.

In Fig. 45.7c, some debris particles were observed and these particles are responsible for the formation of the cracks due to the abrading action. In Fig. 45.7d, the micro-crack in turn shows the evidence of the plastic deformation [12].

In Fig. 45.7e, the grooves were observed. Maximum number of grooves and depth indicates the more wear rate. The grooves which are formed are sharp, and it depends on the load applied during the experiment. Hence, higher the load more will be the grooves and larger the depth [13].

45.3.5 Microstructure Analysis

Scanning electron microscope and energy-dispersive spectroscopy were utilized to study the morphology of the surface and to check the elemental composition, respectively. The sample preparation has been explained in detail in the previous sections.

In Fig. 45.8a, b, morphology and peaks are seen which gives out the percentage composition of each elements. Here, it clearly shows that there are five elements of which three (Al, Cu, Ni) are considered to be major elements and two (Fe and Cr) are minor elements.

The detailed study on the microstructure of as-cast AlCuFeNiCr alloy was made and discussed below.

In Fig. 45.9a, the dark portion represents martensite region and there is a high nickel and cast iron deposition region at the grey region mentioned above. The black and grey regions imply ordered BCC phase [14].

In Fig. 45.9b, casting defects were identified which caused mainly due to the higher temperatures interactions and lack of pre-heating of metallic mould. Aluminium-rich region is projected in the image. There are white spots which has the domination of either aluminium or copper. The observed intermetallic shapes which looks like a Chinese script comprises of several micro-cracks due to the internal tensions occurred during cooling phase [14, 15].

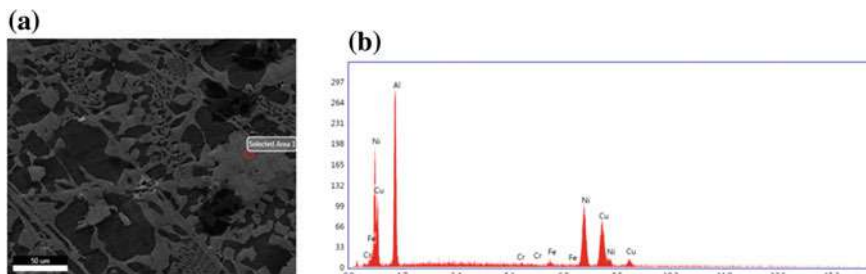


Fig. 45.8 a SEM image. b Elemental composition

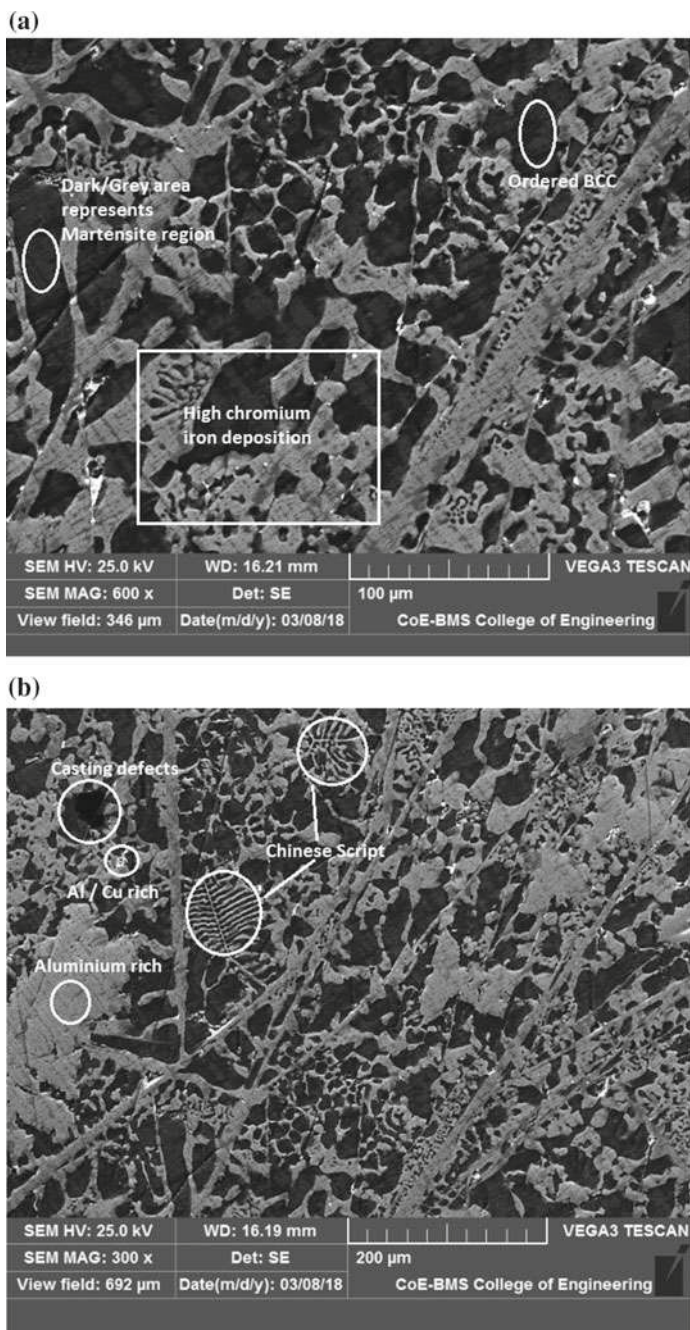


Fig. 45.9 **a** Martensite region, high chromium iron deposition. **b** Aluminium-rich region, casting defects, Chinese script, Al/Cu-rich region. **c** Stress ruptures. **d** Light, dark and grey regions

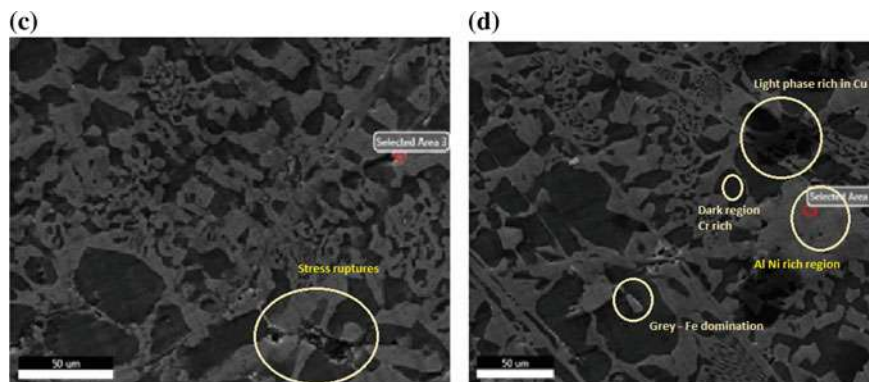


Fig. 45.9 (continued)

In Fig. 45.9c, stress rupture at the grain boundaries which damages the high entropy alloys. It starts from the grain boundaries, and it continues along the phase and links to the granules on the adjacent grains and causes the intergranular fracture [16]. In Fig. 45.9d, there are three different regions; copper dominates the light region, chromium is rich in dark region, and grey region is rich in iron.

45.4 Conclusions

A high entropy alloy (AlCuFeNiCr) was prepared utilizing permanent metal mould casting technique. The samples were machined according to ASTM standards utilizing both conventional lathe and wire-cut edm. Mechanical tests were conducted on AlCuFeNiCr alloy to decide compressive strength, hardness and impact strength. It was observed that these alloys display better properties compared with traditional combinations. The compressive strength of 377 Mpa, Brinell hardness of 85 HB/10/500/30 and an impact strength of 34.69 J/m² demonstrate that the high entropy alloy is better to manufacture a golf club head. It additionally displays low density properties. Pin-on-disc wear test was conducted to decide the wear properties of AlCuFeNiCr high entropy alloy, which demonstrates that as the load increases, the wear rate also increases. Influence of input parameters on output parameters was resolved utilizing Minitab software. In wear rate and frictional force, load has the most noteworthy influence as load is specifically relative (directly proportional) to the wear rate and frictional force. The influence of load on coefficient of friction is less, and sliding speed has a promising contribution in coefficient of friction. Worn surface analysis and microstructure examination have been carried out utilizing the pictures taken from optical microscope instrument and scanning electron microscope separately. BCC phase was noticed which conveys the better hardness properties. Stress ruptures along the grain boundaries, and defects were also seen. Overall, the golf club

head which was manufactured using high entropy alloy exhibited better mechanical properties, wear-resistant properties. In the present market, the heads are being fabricated using conventional alloys, for example titanium-reinforced metal matrix composites which usually show higher density, low hardness, etc. Implementing the concept of high entropy alloy can overcome the limitations.

Permanent metal mould casting technique was used to prepare the test samples. Though it is economical, there were few voids, and cracks and blowholes observed. Investment casting technique can be selected in the later stages to avoid the defects. Material combinations can be varied to improve the mechanical properties. Case hardening is suggested to improve the hardness of the surface. Force analysis can be carried out to determine the equivalent von Mises stress, strains and total deformation by applying different forces which usually a golf club head experiences.

Acknowledgements The authors wish to express their deep sense of gratitude to Centre of excellence in advanced material research and TEQIP, BMS College of Engineering, Bangalore, for providing necessary facilities.

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Chapter 46

Design and Development of Improved Ball End Magnetorheological Finishing Tool with Efficacious Cooling System



D. A. Khan , Z. Alam , F. Iqbal and S. Jha

Abstract Ball end magnetorheological finishing is a magnetic field-assisted nano-finishing process in which the electromagnet is used to generate the required magnetic field. The electromagnet is energized by DC current which also generates heat in the electromagnet. Nano-finishing is a time-taking process which requires the machine to run for a long time such as hours or days. Sometimes the material which is to be finished requires higher magnetic field for finishing. Due to these two reasons, it requires continuous cooling of the electromagnet so that the finishing operation can be prolonged without stopping the finishing operation. To see the above requirements, a new cooling system has been developed in which the electromagnet is cooled continuously by the electrically insulating oil circulated at low temperature. In this new cooling system, the electromagnet is cooled from the inside too for effective cooling of the inner layers of the copper wire. Results on this newly designed tool show that it continuously works without getting heated.

Keywords Electrically insulating oil · Cooling system · Ball end · Magnetorheological · Finishing

46.1 Introduction

Ball end magnetorheological finishing tool or simply called BEMRF tool is an assembly that consists of rotating spindle, electromagnet, cooling coil and rotary valve as shown in Fig. 46.1. Spindle is a centrally rotating body made of magnetically soft ferromagnetic materials. It has an axially drilled hole through which magnetorheological (MR) polishing fluid is supplied to the spindle tip. Spindle is passed through a cylindrical electromagnet coil. Electromagnet coil is the key element of BEMRF tool assembly which is placed over a hollow non-magnetic bobbin in which the spindle

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_46

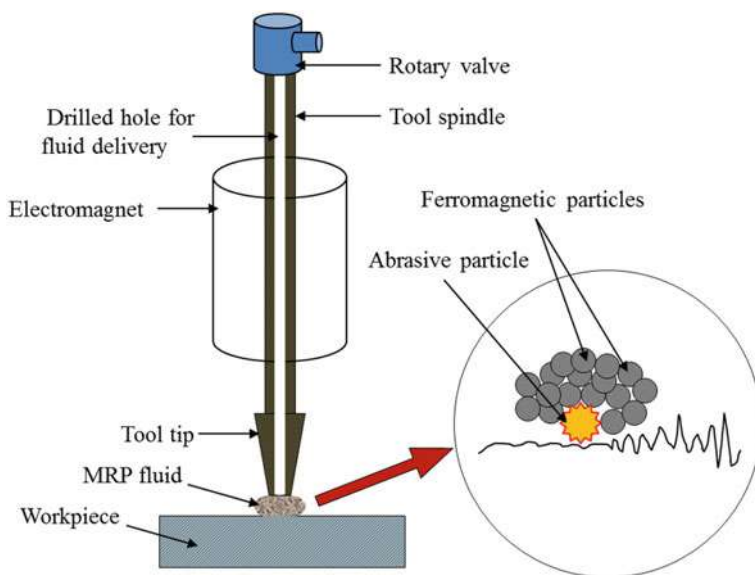


Fig. 46.1 Schematic of ball end MR finishing tool

can rotate [1]. The electromagnet is fabricated by placing the turns of copper wire on the entire length of the bobbin in multiple layers in the form of a copper coil. When the DC power is supplied to the electromagnet, the magnetic field generates around the copper coil and the magnetic lines of forces forming the loops pass through the center of the electromagnet. These magnetic lines of forces, passing through the center of the electromagnet, magnetize the ferromagnetic spindle axially placed in it [1–3].

The strength of the magnetic field depends on many factors such as magnetizing current, number of turns of copper wire, length of the copper coil, the inner and outer diameter of the coil. Due to the resistance offered by the copper wire, some part of the energy supplied to the electromagnet in the form of DC power converts into heat that increases the temperature of the electromagnet. Continuous current supply keeps on increasing electromagnet temperature. The electromagnet copper wire could withstand a certain temperature so it becomes necessary to control the temperature rise in the electromagnet.

The inventor of BEMRF tool has wrapped copper tube around the electromagnet coil through which the coolant was flown to carry out the heat generated in the electromagnet. The pressurized MR polishing fluid is supplied using the rotary valve at the top of the spindle, which facilitates fluid delivery system along with connecting tubes in their stationary position while the spindle is in motion [2].

46.2 Problems Associated in Finishing with Existing BEMRF Tool Design

For the experimental purpose, same type of BEMRF tool, as mentioned in the literature and explained in the previous section, was developed. As mentioned earlier, in the developed BEMRF tool also, for the electromagnet cooling, a copper tube was wound around the electromagnet in the form of a coil. A coolant at low temperature was circulated in the copper tube to take out the heat generated in the electromagnet. During the functionality test of the developed tool, two major problems have been faced with this design of BEMRF tool. First and major problem was that in this design when copper cooling coil was wrapped around the electromagnet, the cooling was improper, and it was limited only to the outer layers of the electromagnet. The heat developed in the inner layers accumulated and resulted in a continuous temperature rise of the inner layers. Due to which, it was impossible to operate the tool at higher magnetizing current and/or continuously for longer finishing time. For the safe operation, it was required to switch off the magnetizing current and to stop the finishing operation.

At medium current values, such as 4 or 5 A, it was very difficult to run the tool as it required switching off the current after every 20 min or so to cool the electromagnet which restricted the continuous operation of the tool.

With the existing cooling system, the second major problem was that at the normal atmospheric condition, the moisture presented in the air condensed on the cooling tube. As the cooling tube was in close contact with the electromagnet so there was always a risk of short circuit in the electromagnet copper wire due to huge amount of condensed moisture. The problem becomes severe in the month of July–August in New Delhi (India) when the atmospheric air is full of moisture.

Due to low magnetic flux density in the working gap, non-magnetic materials require high magnetic field and longer finishing time in fluid-based magnetic field-assisted finishing processes. For BEMRF tool, it raises a need for robust tool design with an efficacious cooling system for finishing of materials such as copper and aluminum which can make it capable to operate at higher magnetizing current and for longer finishing time without interrupting the finishing operation [4–6].

46.3 Improved Oil Immersed Electromagnet Cooling (OIEC)-Based BEMRF Tool

To meet the above requirements, the BEMRF tool has been improved by incorporating an effective cooling system in which the electromagnet was immersed in cold electrically insulating oil. The electrically insulating oil was cooled separately in a constant temperature bath and was continuously circulated at low temperature in a

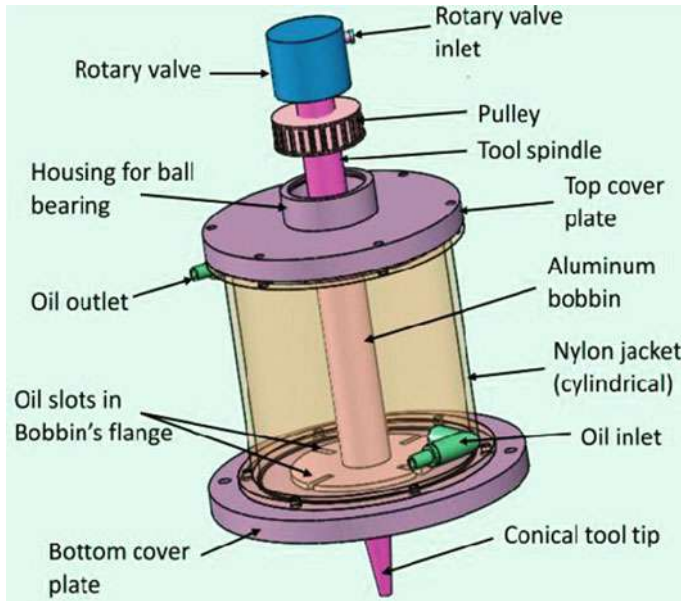


Fig. 46.2 Conceptualized 3-D CAD model of BEMRF tool

nylon cylindrical jacket enclosing the electromagnet. The oil is circulated at a constant rate by the hydraulic pump. The conceptual three-dimensional CAD model of the OIEC-based BEMRF tool is shown in Fig. 46.2.

For better and efficient cooling, a constant gap is kept between two sets of copper layers through which the cold cooling oil can flow to cool the electromagnet from the inside. The gap between two sets of electromagnet layers is maintained using a cage-shaped structure of copper wire between these sets of electromagnet layers. Oil slots are provided in the bobbin's flanges through which the oil enters in the gaps maintained between two sets of electromagnet layers. Cooling oil is electrically insulating in nature and possesses good cooling properties, and also at the same time, it does not affect the generated magnetic field. The construction details of OIEC-based BEMRF tool are discussed ahead:

Following are the main components of OIEC-based BEMRF tool:

(1) Spindle, (2) bobbin, (3) stationary electromagnet with constant oil gap, (4) nylon cylindrical jacket, (5) top and bottom cover plates with bearing housings, (6) resistance temperature detectors (RTDs).

(1) **Spindle:** Spindle is the rotating part of the BEMRF tool assembly. It is cylindrical in shape and has an axially drilled hole through which the MR polishing fluid can be pumped. At the upper end of the spindle, a rotary valve has been attached to facilitate the rotational movement of the spindle keeping the fluid delivery system intact in its position. Spindle is made up of material which possesses good magnetic properties.

Spindle is supported on two ball bearings to facilitate its rotation. One end of the spindle is made conical in shape for enhancing the accessibility to finish the deep pockets and narrow spaces as shown in Fig. 46.3. When the electromagnet is energized, the generated magnetic field magnetizes the tool spindle, and the tool spindle works like a permanent magnet. As soon as the current is switched off to the electromagnet, the spindle demagnetizes.

- (2) **Bobbin:** A bobbin is a hollow cylindrical part having two flanges at the opposite ends, Fig. 46.4. Electromagnet coil was wound over the cylindrical part of the bobbin and supported over the flanges at the ends. Four radial slots have been cut in each flange to provide access to the cooling oil to the annular space of the electromagnet as shown in Fig. 46.4. The bobbin is made of aluminum so that it is light in weight, and due to high thermal conductivity, it also transmits some part of the heat generated in the electromagnet to the cooling oil. Because of

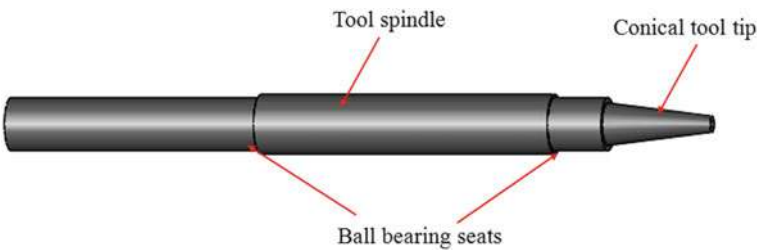


Fig. 46.3 BEMRF tool spindle

Fig. 46.4 Aluminum bobbin

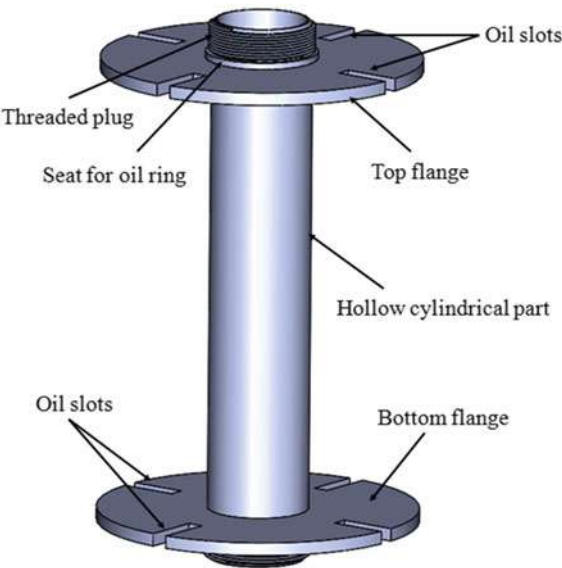
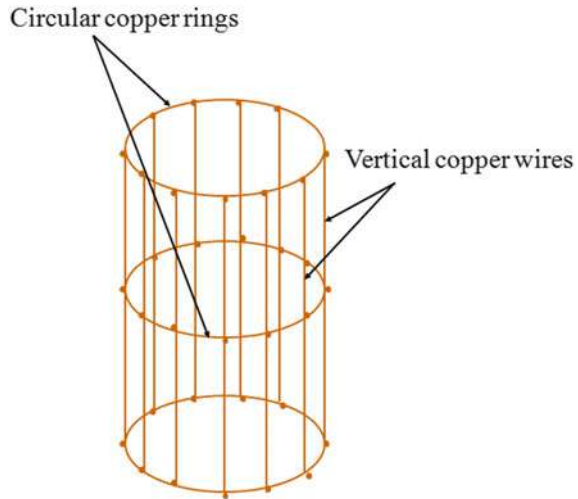


Fig. 46.5 Cage-shaped structure of copper wire



non-magnetic material, the bobbin does not magnetize when the electromagnet is energized. At the extreme ends of the bobbin, the externally threaded plugs have been used to fasten the top and bottom cover plates with the bobbin. Tool spindle is passed through the bobbin and rotates axially in it.

- (3) **Stationary electromagnet with constant oil gap:** The electromagnet was wound over the bobbin in the form of multiple layers of copper wire one over another. When an electric current is passed through the two terminals of the electromagnet coil, some energy losses occur that result in heat generation in the copper wire and increase the electromagnet temperature.

At high current values, the heat generated inside the electromagnet is proportionally high, so there is a need to cool the electromagnet from inside also. To cool the electromagnet from the inside, a cage-shaped structure of copper wire, as shown in Fig. 46.5, is put after every few layers of the electromagnet. In such a way, it maintains a gap equal to the wire diameter (almost 2 mm) between two sets of electromagnet layers, Fig. 46.6. This arrangement of cooling also makes the electromagnet safe from short circuit because the electromagnet is placed in a nylon cylinder, and in this way, it is out of reach of the condensed atmospheric moisture.

This design of the electromagnet with the inside oil passage helps in reducing the electromagnet inside temperature when cold cooling oil flows through these passages. The electromagnet fastened with the bottom cover plate is shown in Fig. 46.7.

- (4) **Nylon cylindrical jacket:** Cylindrical jacket is made of nylon due to its lightweight and less moisture condensation on it as compared to metals. Nylon jacket encloses the electromagnet and provides sufficient space for cooling oil to flow. Low-temperature cooling oil enters from one side of the cylinder near the bottom through the nozzle and escapes from the top part of the cylinder at

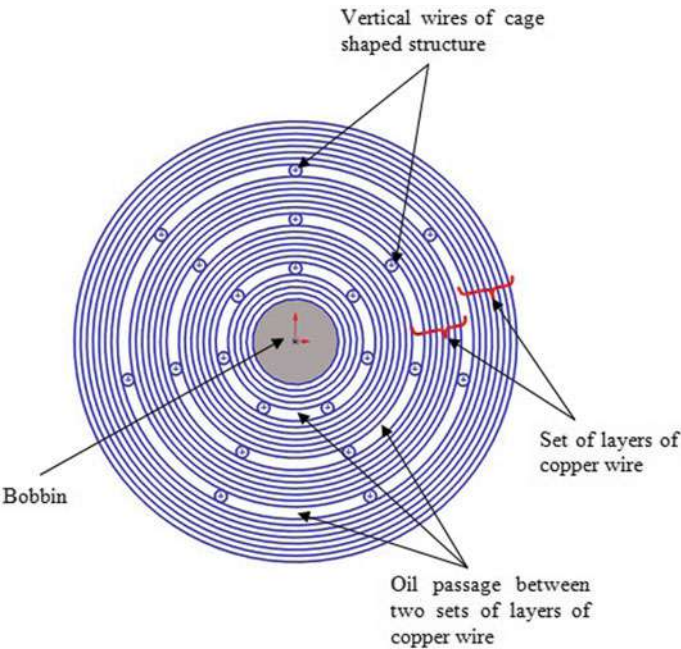


Fig. 46.6 Constant gap for cooling oil flow between electromagnet copper wire layers

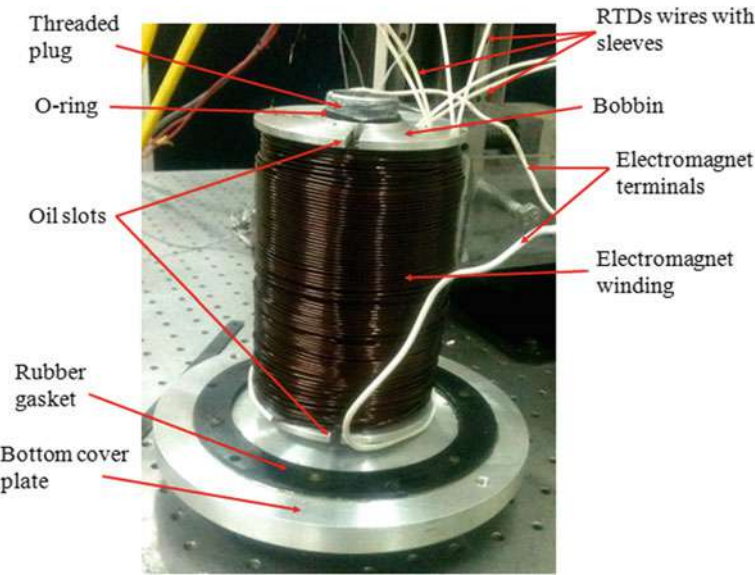
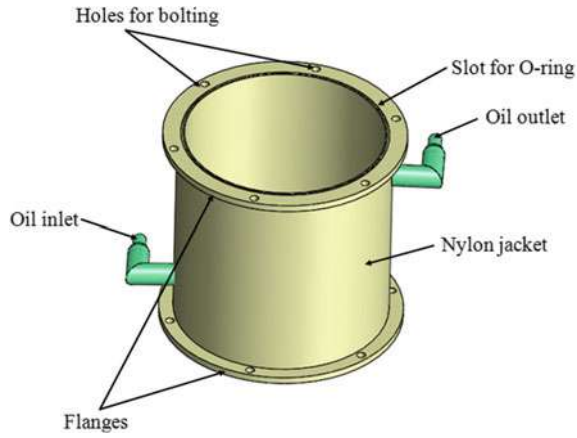


Fig. 46.7 Electromagnet fasten with bottom cover plate

Fig. 46.8 Nylon cylindrical jacket with inlet–outlet oil connectors



the opposite side, Fig. 46.8. The oil is cooled in a low-temperature oil bath and supplied to the nylon cylinder by the flexible PVC tubes.

The low-temperature cooling oil enters from bottom side of the nylon cylinder through connector, i.e., oil inlet and cools the electromagnet from outside as well as inside through the gap created between two set of layers of copper wire, and finally comes out through the other connector, i.e., oil outlet at the other end of the nylon cylinder. Figure 46.9 shows improved BEMRF tool in its developing stage, containing electromagnet enclosed in nylon cylindrical jacket.

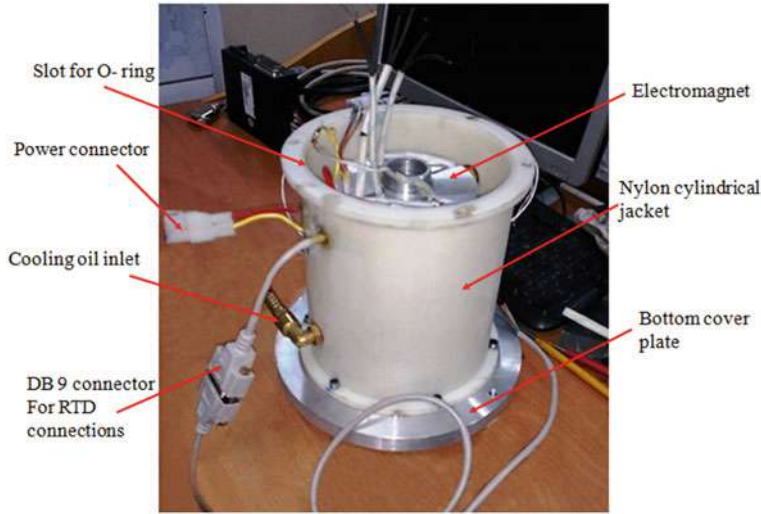


Fig. 46.9 Improved BEMRF tool in its developing stage

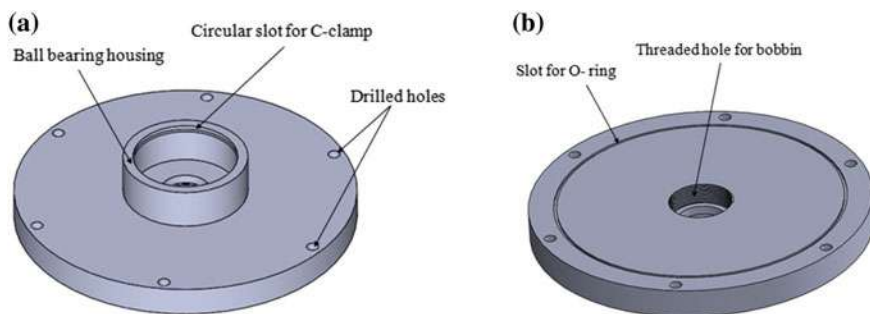


Fig. 46.10 Top cover plate **a** front view **b** back side view

- (5) **Top and bottom cover plates with bearing housings:** Top and bottom plates cover the two ends of the nylon cylinder. These plates have centrally drilled hole through which the bobbin is fastened with the plates and keep the bobbin fix in its position. Both the plates have housing for the ball bearings in which the tool spindle moves. Intensive care has been taken to keep the axis of both the bearings coincident with the axis of the spindle. The top and bottom cover plates were the same in all respect except the size of the plates. The bottom plate has bigger diameter to fix it with the frame structure.

The CAD model of the top cover plate is shown in Fig. 46.10. O-rings were rested between plate and nylon cylinder to make the system leak-proof. Both the plates were checked for their parallelism.

- (6) **Resistance temperature detectors (RTDs):** Resistance temperature detectors (RTDs) are used to sense the temperature accurately and precisely inside the electromagnet. These are resistance-based sensors whose resistance is a function of temperature. A small amount of excitation current is passed through it, and the voltage which is proportional to the resistance is converted and calibrated for the temperature. Pt100 RTDs has been introduced in electromagnet coil at the time of winding to measure the electromagnet temperature when it is energized. In these Pt100 RTDs, the material of the main element is platinum, initially that offer 100 Ω resistance value at 0 °C. In the electromagnet coil, three Pt100 sensors were introduced to measure the temperature generation of three different layers of the electromagnet, as shown in Fig. 46.9. The temperature sensors are introduced in innermost, middle and outermost layers of copper wire to measure the temperature of electromagnet at these locations. The output wires of Pt100 RTDs are connected with digital temperature indicator which shows the value of the temperature generated at any of the above three locations in the electromagnet.

CAD model of complete BEMRF tool assembly is shown in Fig. 46.11. The tool spindle is rotated by a DC motor using pulley and belt drive. BEMRF tool assembly rests on a metallic frame structure. The frame structure is made in such a way that at

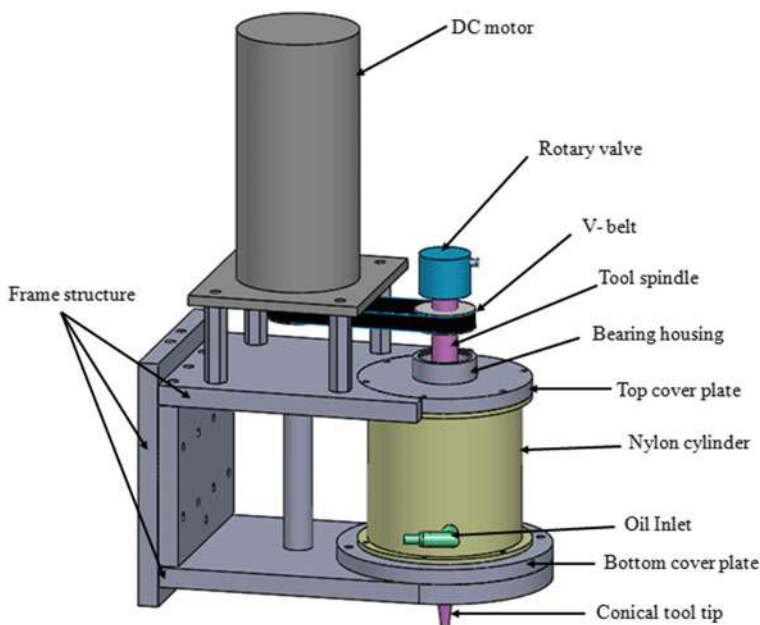


Fig. 46.11 Complete BEMRF tool assembly

any point of time if any repair work is required then the BEMRF tool assembly could be detached from the frame structure without unmounting the frame structure from machine body. Picture of actual tool assembly in the machine is shown in Fig. 46.12. All the parts of the BEMRF tool assembly except tool spindle and cooling cylinder are made of aluminum due to non-magnetic nature and light in weight. In all respects, the tool spindle should be exactly perpendicular to the machine base keeping in mind the parallelism and perpendicularity of the mating surfaces.

46.4 Performance Evaluation of Developed OIEC-Based BEMRF Tool

Developed BEMRF tool has been tested for its magnetic field generation and for the cooling effect. For this purpose, the tool is tested for the current ranging from 1 to 8 A and at varying working gap ranging from 1.2 to 2.1 mm. The magnetic flux density is measured using digital Gauss meter. The results of magnetic flux density in the working gap for a mild steel sample are shown in Fig. 46.13.

From the results as shown in Fig. 46.13, it is observed that as the current increases the magnetic flux density increases. Increasing the working gap decreases the magnetic flux density in the working gap. The results of magnetic flux density have been



Fig. 46.12 Actual picture of BEMRF tool assembly

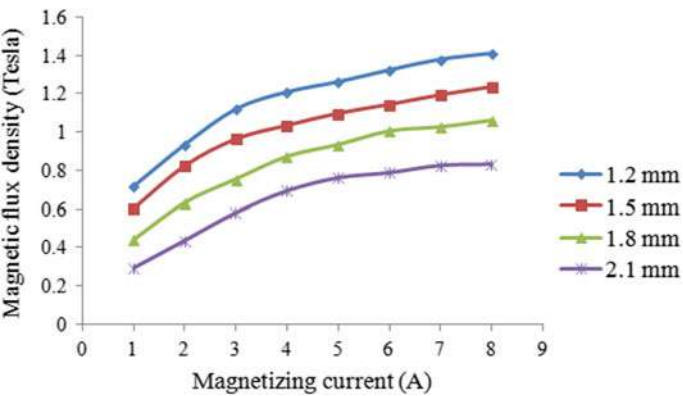


Fig. 46.13 Magnetic flux density in working gap for mild steel sample at varying current and gaps

repeated on constant current and gap values to observe the repeatability of the magnetic field generation. From the repeatability test, it is observed that every time the tool is showing almost the same values of the magnetic flux density. This proves that the developed tool is generating consistent magnetic flux density and could be used for the scientific investigations.

The electromagnet was designed to generate 1.2 T magnetic flux density in a working gap of 1.2 mm. The electromagnet design parameters were selected in

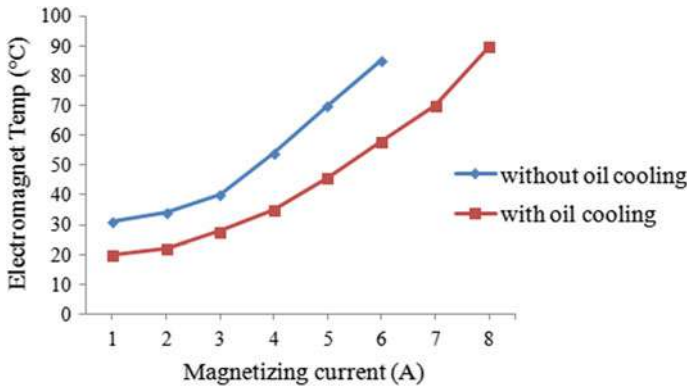


Fig. 46.14 Temperature variation of the electromagnet inner layer with and without oil cooling

such a way that the electromagnet generates the desired magnetic field (1.2 T) at 3 A magnetizing current (The design of electromagnet has not been presented in this manuscript). But from the actual measured values of magnetic flux density (Fig. 46.13), it is observed that 1.2 T magnetic flux density is achieved at 4 A current. This is because of energy losses in the electromagnet copper wire in the form of heat and eddies which have not been considered in the design calculations. Although the electromagnet is designed for 1.2 T magnetic flux density but providing efficient cooling, the magnetic flux density has been achieved up to 1.41 T at 8 A current.

Effect of oil cooling of the electromagnet is shown in Fig. 46.14. The temperature of innermost layer of copper wire is measured by Pt100 resistance temperature detector. For the comparison purpose, the temperature values of the electromagnet have been measured with and without providing the oil cooling. The current is supplied for 30 min time, and then the maximum temperature attained by the innermost layer is recorded as shown in Fig. 46.14.

From Fig. 46.14, it is observed that for the same current values the temperature developed in BEMRF tool with oil cooling is much lower than the temperature developed in BEMRF tool without cooling. Since the electromagnet temperature is associated with the magnetizing current, the current value increases as the temperature of the electromagnet increases. At higher values of current (beyond 3 A), the temperature rises with a higher rate. In the condition when cooling is not provided, the temperature of the electromagnet keeps on increasing with time for a particular current value and the temperature attained in 30 min is shown in Fig. 46.14. But at the same time when oil cooling is provided, temperature stabilizes within 30 min time and remains constant for rest of the time.

In the case when oil cooling is not provided, at 6 A current, the electromagnet attained 85 °C temperature in 30 min of current on time and keeps on increasing with the time. But on the other hand with the oil cooling, it is possible to supply 8 A current to the electromagnet with a temperature rise of 90 °C which stabilizes within

30 min. This shows that with the oil cooling, the electromagnet runs safely even at higher magnetizing currents.

Developed oil immersed electromagnet cooling (OIEC)-based BEMRF tool exhibited the ability to withstand higher magnetizing current without getting heated beyond the permissible limit of the copper wire. Since the maximum temperature attained in the electromagnet remains constant for rest of the time, it makes the tool capable of performing continuous finishing operation without switching off the current. Also, in this design, the electromagnet is safe from condensed atmospheric moisture as it was a big problem with the previous BEMRF tool design presented in the literature.

46.5 Conclusions

- A new system of oil immersed electromagnet cooling (OIEC)-based BEMRF tool is developed in which the electromagnet is cooled by low-temperature electrically insulating oil.
- The electromagnet was designed in such a way that a radial gap was maintained between two sets of electromagnet layers through which the cooling was provided to the inner layers of the copper wire to cool the electromagnet from the inside.
- Newly designed and developed BEMRF tool worked well in terms of prolonged finishing operation without coil heating and could be operated at higher magnetizing current.
- The new design of BEMRF tool is safer from seepage of condensed moisture in the electromagnet coil since the electromagnet coil is immersed in transformer oil.

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Chapter 47

Analyzing Enablers of Emission Reduction Strategies of Cement-manufacturing Industry of India under Fuzzy Environment



Sachin Balsara , P. K. Jain and Anbanandam Ramesh

Abstract Cement production is the most emission-intensive industry. The Indian cement industry is the second-largest greenhouse gas emitter. Continuous emissions are responsible for global warming and extreme climate change. The cement industry is currently under pressure to reduce emissions. Therefore, we seek to evaluate the enablers of emission reduction strategies of Indian cement industry. In the present study, fuzzy AHP is used to assess the enablers of emission reduction strategies of cement industry. The FAHP helps in establishing the priorities of the enablers of emission reduction strategies. According to the study outcomes, there are 17 enablers related to emission reduction practices. FAHP analysis showed that enabling factors: Litigation risk, health issue, local public or societal pressure for emission reduction, cut in subsidies and increased taxes on fossil fuels and demand for low-carbon product are on top priority as per global rank.

Keywords Climate change · Greenhouse gas · Emission · Cement · AHP

47.1 Introduction

Natural and anthropogenic factors are responsible for Earth's climate change. Human activities have a drastic effect on climate change, which is evident from the increasing greenhouse gas (GHG) in the environment leading to global warming. Continued

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_47

greenhouse gas emissions (GHGEs) will further increase warming and change the climate. Sustained and substantial reductions in GHGEs will limit climate change and its associated risks [1].

In the context of climate change policies, enablers are understood as processes patterns and activities that produce a definite impetus for the climate. Enablers are considered as a variable that motivates the attainment of climate change mitigation strategies in any industry.

For industry, reduction of GHGEs is particularly critical in the pulp and paper, chemicals and petrochemicals, steel and iron, aluminum and cement the five sectors which are most energy and emission-intensive [2]. Cement is a fundamental requirement of modern society. It is the primary building material and become synonymous with construction activity. Cement used extensively in infrastructure development, industrial sector, urban housing, and employment generation [3]. Infrastructure is a backbone of socio-economic development of any country and its institutions. Cement is always considered as a barometer of progress in a developing country. The per capita consumption of cement is accepted as an important indicator of the country's economic growth. In addition to these, the primacy of cement industry would continue, as all over the world cement remains paramount for the infrastructure development and near the future, no other material would possibly substitute it [4]. Thus, for the economic growth and expansion of any country, cement-manufacturing industry plays a vital role.

In the world, manufacturing of cement is one of the remarkable emission and energy-intensive industry. It is one of the major sources of anthropogenic GHGEs among the industrial activities [1] Production of cement accounts for about 5–8% of total global anthropogenic CO₂Es [5].

Thus, this study has consecutive goals:

- To identify the common enablers of climate change mitigation strategies of cement-manufacturing industry.
- To evaluate the relative importance weight of each common enablers of climate change mitigation strategies of cement-manufacturing industry.

However, it will be impossible to implement all the enablers of mitigation strategies simultaneously to manage, control, and reduce the GHGEs from the cement industry. Hence, industry should identify some enablers of emission reduction, which have essentially to manage, control and reduces the GHGEs from the cement industry by employing fuzzy analytical hierarchy process (FAHP). The result might improve environmental performance assuring accessible implementation of essential emission reduction strategies.

47.2 Literature Review

Various main factors and enablers of emission reduction strategies in the cement industry are described as:

47.2.1 Business Factor

The government removes subsidies and enhanced taxes on carbon fuels turning a carbon subsidy into one of carbon taxation [6]. There are several issues being faced by the cement industry that needs to be resolved, including the continuous increase in energy supply cost and the availability of high-quality raw materials and their prices [7]. Some lawsuits charging on organization with negligence, public nuisance, or trespass because organization generates significant carbon emissions [8], but there is an opportunity also for existing organizations to modify their processes and products/services to be more sustainable [9].

47.2.2 Government Regulations and Policies

Regulation compliance playing a crucial role in driving the necessary environmental improvements [10]. Perform achieve and trade (PAT) is a mechanism based on market to strengthen cost-effectiveness through certification of excess energy savings in energy-intensive industry that can be traded [6] beyond these government may impose a high penalty for environmental pollution on industry [11].

47.2.3 Internal Factors

Top management involvement and commitment to emission reduction is an important internal factor to the organizations [12]. Cost reduction through an extensive collection of functional tasks including fossil fuel substitution with alternate fuel or substitutions with other sources of renewable energy, clinkers substitution by fly ash, blast furnace slag, etc., is another important internal factor [9]. Environmental awareness of employee is also a significant driver of emission reduction [12]. As a part of an overall commitment to sustainable development, Corporate Social Responsibility (CSR) and Ethical responsibility are internal responsibilities of organization to reduce GHGs [13].

47.2.4 Market Pressure

The market is shift to low-carbon products and there is a challenge to differentiating product based on green activities relative to a competitor is important mitigation measure [8] while Kolk and Pinkse [12] studies that change in the pattern of demand for low-carbon goods as a result of increasing consumer awareness and market pressure. Some companies enhance their brand image and corporate reputation by lowering

emissions [8]. Okereke [10] stated that one of the significant sources of external pressure is from media and environmental NGOs apart from government.

47.2.5 Stakeholder Pressure

Investors are now demand more carbon disclosure from organizations [9] while societal pressure for emission reduction is another an important enabler to mitigate emission and Health issue generated from uncontrolled emission from alternative fuels, dust generated from cement plant is further considered as a significant pressure to reduce emissions [13].

47.3 Solution Methodology

In this study, FAHP has been used to prioritize and explore the enablers to implement emission reduction strategies in Indian cement industry. Although multi-criteria decision making (MCDM) methods can accomplish the goal of the study but to handle uncertain and imprecise factors and data, the fuzzy approach has been used. The fuzzy approach also has the ability to resemble human-decision nature.

47.3.1 Fuzzy AHP

AHP, initially developed by Thomas L. Saaty in 1980, is a very accepted technique to handle MCDM complications including multiple quantitative and qualitative factors. Using scale experts is allowed to specify their liking. This scale is very helpful for an individual or for experts to generate a decision. Giving importance weights for each factor and sub-factors is the purpose of AHP. In this study, fuzzy AHP is used for analyzing enablers of emission reduction strategies cement-manufacturing industry of India.

To manage efficiently the fuzziness of the data involved in the decision, fuzzy AHP is the fuzzy extension of AHP. Fuzzy AHP can adequately manage quantitative and qualitative data in MCDM problems. For the preferences of one criterion over other triangular fuzzy numbers (Table 47.1) are used since triangular membership function are fewer complexes when splitting values (low, med and high MF) comparing other membership functions. Then, the synthetic extent value of the pairwise comparison is calculated by using the extent analysis method. Based on this, the weight vectors of alternatives, sub-factors, and the main factors are decided. As a result, the final priority weights of the alternatives are decided based on the different weights of the main factors and sub-factors. The highest priority would be given to the factor with the highest weight.

Table 47.1 Triangular fuzzy conversion scale

Intensity of importance	Fuzzy number	Linguistic variable	Membership function	Reciprocal membership function
1	$\tilde{1}$	Equally important/preferred	(1,1,3)	(1/3,1,1)
3	$\tilde{3}$	Weakly important/preferred	(1,3,5)	(1/5,1/3,1)
5	$\tilde{5}$	Strongly more important/preferred	(3,5,7)	(1/7,1/5,1/3)
7	$\tilde{7}$	Very strongly important/preferred	(5,7,9)	(1/9,1/7,1/5)
9	$\tilde{9}$	Extremely more important/preferred	(7,9,11)	(1/11,1/9,1/7)

The FAHP approach used by the authors is elaborated as follows:

Let $P = \{p_1, p_2, \dots, p_n\}$ and $Q = \{q_1, q_2, \dots, q_m\}$ represent the object and the goal set, respectively, then using the extent analysis method given by Chang [14]. The m extent analysis values for each object are denoted as:

$$N_{oi}^1, N_{oi}^2, \dots, N_{oi}^m, \quad \text{where } i = 1, 2, \dots, n.$$

where all the N_{oi}^j ($j = 1, 2, \dots, m$) are triangular fuzzy numbers.

The value of the extent analysis of the i th object for m th goal is represented as N_{oi}^m .

Step 1: The fuzzy synthetic extent value (F_i) with respect to the i th criterion is defined as,

$$F_i = \sum_{j=1}^m N_{oi}^j \otimes \left[\sum_{i=1}^n \sum_{j=1}^m N_{oi}^j \right]^{-1} \quad (49.1)$$

$$\sum_{j=1}^m N_{oi}^j = \left(\sum_{j=1}^m n_{1j}, \sum_{j=1}^m n_{2j}, \sum_{j=1}^m n_{3j} \right) \quad (49.2)$$

$$\left[\sum_{i=1}^n \sum_{j=1}^m N_{oi}^j \right]^{-1} = \left(\frac{1}{\sum_{i=1}^n n_{3i}}, \frac{1}{\sum_{i=1}^n n_{2i}}, \frac{1}{\sum_{i=1}^n n_{1i}} \right) \quad (49.3)$$

Step 2: The degree of possibility of $N_1 = (n_{11}, n_{12}, n_{13}) \geq N_2 = (n_{21}, n_{22}, n_{23})$ is defined as

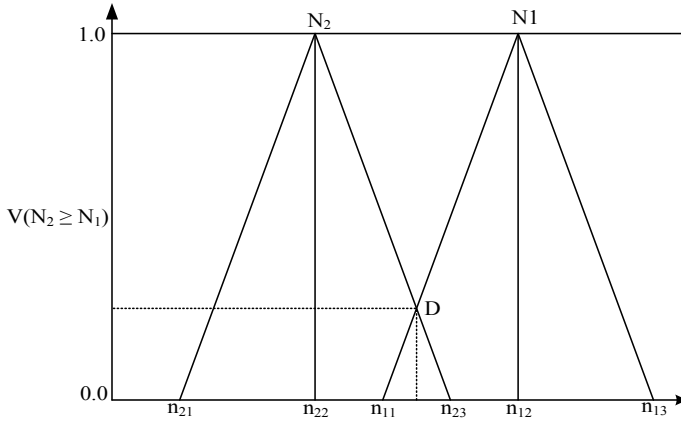


Fig. 47.1 Intersection between N_1 and N_2

$$V(N_1 \geq N) = \sup_{x \geq y} [\min(\mu_{N_1}(x), (\mu_{N_2}(y)))] \quad (49.4)$$

when a pair (x, y) exists such that $x \geq y$ and $\mu_{N_1}(x) = \mu_{N_2}(y) = 1$, then we have

$$V(N_1 \geq N_2) = 1 \quad (49.5)$$

since N_1 and N_2 are convex fuzzy numbers so,

$$V(N_1 \geq N_2) = 1 \quad \text{if } n_{11} \geq n_{21} \quad (49.6)$$

$$\text{and } V(N_2 \geq N_1) = \text{hgt}(N_1 \cap N_2) = \mu_{N_1}(d) \quad (49.7)$$

where d is the ordinate of the highest intersection point D between μ_{N_1} and μ_{N_2} (shown in Fig. 47.1).

When $N_1 = (n_{11}, n_{12}, n_{13})$ and $N_2 = (n_{21}, n_{22}, n_{23})$ then ordinate of D is computed by

$$V(N_2 \geq N_1) = \text{hgt}(N_1 \cap N_2) = \frac{n_{11} - n_{23}}{(n_{22} - n_{23}) - (n_{12} - n_{11})} \quad (49.8)$$

For the comparison of N_1 and N_2 , both the values of $V(N_1 \geq N_2)$ and $V(N_2 \geq N_1)$ are required.

Step 3: The degree possibility for a convex fuzzy number to be greater than k convex fuzzy numbers $N_i (i = 1, 2, \dots, k)$ can be defined by

$$V(N \geq N_1, N_2, \dots, N_k) = V[(N \geq N_1) \text{ and } (N \geq N_2) \text{ and } \dots \text{ and } (N \geq N_k)] \quad (49.9)$$

$$= \min V(N \geq N_i), \quad i = 1, 2, \dots \quad (49.10)$$

$$\text{If } m(P_i) = \min V(F_i \geq F_k), \quad (49.11)$$

for $k = 1, 2, \dots, n; k \neq i$, then the weight vector is given by

$$W_p = (m(P_1), m(P_2), \dots, m(P_n))^T, \quad (49.12)$$

where $P_i (i = 1, 2, \dots, n)$ are n elements.

Step 4: After normalizing W_p , we get the normalized weight vectors

$$W = (w(P_1), w(P_2), \dots, w(P_n)) \quad (49.13)$$

where W is a non-fuzzy number and this gives the priority weights of one alternative over others.

47.4 Application of the Proposed Method

The FAHP framework can prioritize enablers of implementing emission reduction strategies in Indian cement industry. The framework consists of two phases.

Phase I Identifying the enablers of implementing emission reduction strategies of Indian cement industry from literature resources and with the industrial expert consultation.

In this study, 17 enablers under five main factors were identified through intensive discussion with experts who were actively involved in processes of cement manufacturing and through literature review as shown in Table 47.2.

The selected industrial experts were remarkably proficient in their discipline, having industrial experience of above ten years. The questionnaire was created to obtain experts' viewpoint on the cement industry. The hierarchy of decision consists of three levels; the ultimate objective of the decision process which is at the first level in decision hierarchy determined as "Enablers of implementing emission reduction strategies of Indian cement industry". The second level consists of the main factors and the third level is the sub-factors (Fig. 47.2).

Phase II Calculate the weight of the barriers using FAHP. In second-phase pairwise comparison matrix, the main factors and sub-factors were constructed from the scale given in Table 47.1. The pairwise comparison matrixes of the main factors are given in Table 47.3 and the pairwise comparison matrices of the sub-factors are given in Tables 47.4 and 47.5.

Table 47.2 Identified main factor and their enablers

Main factor with code	Description of enablers	References
Business factor (BF)	Cut in subsidies and increased taxes on fossil fuels (BF1)	[6]
	Litigation risk (BF2)	[8]
	Opportunity to modify the product and processes (BF3)	[9]
Government regulations and policies (GR)	Environment regulation compliance (GR1)	[8–10, 12]
	PAT Scheme by BEE (GR2)	[6]
	High penalty for environmental pollution (GR3)	[11]
Internal factor (IF)	Top management involvement and commitment to emission reduction (IF1)	[8, 9, 12]
	Cost reduction through material substitution and operational improvement (IF2)	[1, 9, 12]
	Environmental awareness of employee (IF3)	[9, 12]
	Corporate social responsibility (CSR) and ethical responsibility (IF4)	[1, 8, 10–12]
Market pressure (MP)	Greenmarket competitive pressure (MP1)	[8, 12]
	Demand for low-carbon products (MP2)	[12]
	Enhanced brand image and corporate reputation/improved public image (MP3)	[1, 8, 10, 12]
	Media and NGOs attention to climate change issue (MP4)	[11, 10, 12]
Stakeholder pressure (SP)	Investor demand (SP1)	[8–10]
	Local public or societal pressure for emission reduction (SP2)	[11]
	Health issue (SP3)	[11]

47.5 Result and Discussion

With respect to the FAHP analysis, a three-level hierarchy structure of the driving factors illustrated in Fig. 47.2 was developed. A pairwise comparison of the factors was conducted using the standard scale given in Table 47.1. From the second column of Table 47.5, it is evident that business factor (BF) ranks first in the main factor level, with an evaluation of 0.3652. Litigation risk (BF2), cut in subsidies and increased

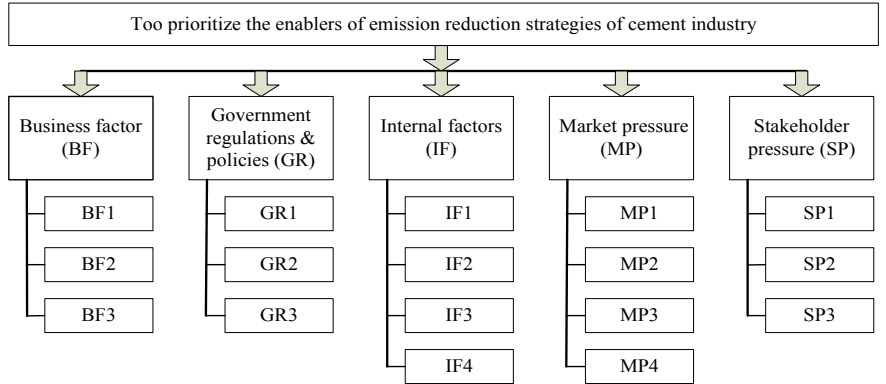


Fig. 47.2 Hierarchical structure of the enablers of emission reduction strategies of Indian cement industry

Table 47.3 Pairwise evaluation matrix for the main group factors

	MP	GR	SP	BF	IF	Rel. wt.	Rel. rank
MP	(1,1,3)	(1,3,5)	(0.2,0.33,1)	(0.142,0.2,0.33)	(5,7,9)	0.1996	3
GR	(0.2,0.33,1)	(1,1,3)	(0.2,0.33,1)	(0.11,0.142,0.2)	(5,7,9)	0.1391	4
SP	(1,3,5)	(1,3,5)	(1,1,3)	(0.2,0.33,1)	(7,9,11)	0.2738	2
BF	(3,5,7)	(5,7,9)	(1,3,5)	(1,1,3)	(7,9,11)	0.3652	1
IF	(0.11,0.142,0.2)	(0.11,0.142,0.2)	(0.091,0.11,0.142)	(0.091,0.11,0.142)	(1,1,3)	0.0224	5

Table 47.4 Pairwise evaluation matrix for the business factors

	BF1	BF2	BF3	Rel. wt.	Rel. rank
BF1	(1,1,3)	(0.142,0.2,0.33)	(1,3,5)	0.231	2
BF2	(3,5,7)	(1,1,3)	(7,9,11)	0.718	1
BF3	(0.2,0.333,1)	(0.091,0.11,0.14)	(1,1,3)	0.05	3

Table 47.5 Relative ranking of main factors and sub-factors of emission reduction strategies of Indian cement industry

Main factor	Relative weights	Sub-factors	Relative weights	Relative rank	Global weights	Global rank
BF	0.3652	BF1	0.231	2	0.084361	4
		BF2	0.718	1	0.262214	1
		BF3	0.05	3	0.01826	10
GR	0.1391	GR1	0.332	2	0.046181	8
		GR2	0.124	3	0.017248	11
		GR3	0.543	1	0.075531	6

(continued)

Table 47.5 (continued)

Main factor	Relative weights	Sub-factors	Relative weights	Relative rank	Global weights	Global rank
IF	0.0224	IF1	0.336	1	0.007526	13
		IF2	0.189	3	0.004234	16
		IF3	0.307	2	0.006877	14
		IF4	0.168	4	0.003763	17
MP	0.1996	MP1	0.189	3	0.037724	9
		MP2	0.407	1	0.081237	5
		MP3	0.033	4	0.006587	15
		MP4	0.371	2	0.074052	7
SP	0.2738	SP1	0.05	3	0.01369	12
		SP2	0.312	2	0.085426	3
		SP3	0.638	1	0.174684	2

taxes on fossil fuels (BF1), opportunity to modify the product and processes (BF3) obtain global rank first, fourth and tenth.

As per Table 47.5, with an evaluation of 0.2738, stakeholder pressure (SP) ranks second in the main factors while sub-factors, health issue (SP3) and local public or societal pressure for emission reduction (SP2) have global rank second and third.

Next market pressure (MP) accompanied with an evaluation of 0.1996 ranks third among the main enabling factors, related sub-factors, involving a demand for low-carbon products (MP2) and media and NGOs attention to climate change issue (MP4) obtains the fifth and seventh global rank.

Role of Government regulations and policies (GR) among main factors acquired the fourth priority as shown in Table 47.5. High penalty for environmental pollution (GR3) and environment regulation compliance (GR1) are related sub-factors, which also obtain the higher weight in the sub-factor level.

Finally, as per Table 47.5, with an evaluation of 0.0224, Internal factor (IF) obtain the last position among the main factors.

47.6 Conclusions

The outcome of the study helps us in focusing on and identifying significant enablers of emission reduction measure in the Indian cement industry. Hence, by working on these identifying key enabling factors for reducing the GHGEs, Management of cement industry can build the green corporate image and producing low-carbon cement at an economical rate. However, according to the industrial point of view, it is very challenging to give identical weight to all enabling factors. Therefore, cement-manufacturing industry must know which enabler should be given more

weight regarding the adoption of emission reduction strategies through employing the FAHP technique.

As per the fuzzy AHP, the global rank of the key enabling factors; BF2, SP3, SP2, BF1, MP2, GR3 and MP4 are 1, 2, 3, 4, 5, 6 and 7 respectively. Thus, this key enabler should be considered for implementing the emission reduction strategies among Indian cement industry. For future research work, the analysis can be done by using other MCDM techniques like PROMTHEE, Preference Vector Method, Best Worst Method, TOPSIS, and ANP.

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Chapter 48

Role of Product Development Process for NPD Success in Indian Manufacturing Industries: Quality, Cost and Technological Aspects



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Abstract ‘MAKE IN INDIA’ necessitates the boost in Indian manufacturing industries to be involved in new product development (NPD) for achieving industrial sustainability. Product development process (PDP) is one of the most vital factors of NPD for developing new products as per customer demand. The successful adoption of PDP requires the top management support (TMS), external collaboration (EC) and market analysis (MA) to develop the suitable environment for successful NPD by producing the high-quality products with technological developments in reduced cost. This study develops a framework comprising of the aforesaid factors and measures by structural equation modelling (SEM) approach with the primary data collected from 263 experts of Indian manufacturing companies. The analysis infers that PDP is escalated by TMS, EC and MA to develop new products trading off among product cost, quality and technological developments for NPD success. The positive influence of TMS on EC has also been explored.

Keywords New product development · Product development process · Top management support · Quality

48.1 Introduction

New product development (NPD) has become one of the key components in order to sustain in the competitive market environment. The rapid growth of the global market and the ever changing market environment endures firms to be involved in NPD activities [1]. There are several factors critically affecting the firm’s performance of NPD is renowned as the critical success factors (CSFs). Product development process (PDP) is identified as one of the most essential factors of NPD to develop quality

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_48

products at minimum cost as per customer demand [2]. Segmentation of PDP into several stages enhances the methods of NPD by utilizing the resources optimally for developing the products as per increasing demand. Top management support (TMS) is an inseparable part of different phases of NPD that helps to overcome technologically turbulent environment [3]. The culture of external collaboration (EC) identifies the requirements and priorities of customers essential for PDP [4]. Similarly, market analysis (MA) is an effort of accumulating the customers' needs through structured observations of their requirements. Like success factors, the measures for quantifying the success of NPD are equally essential. The customers are always keen to have high-quality products in minimum cost having technologically updated features [5]. Though there are several articles recognizing the essentiality of these mentioned factors and measures, there is a research gap in developing the comprehensive framework linking these factors and measures in a single model. This study evinces the requirement of empirical researches for bridging this gap. The main objective of this study is to develop a comprehensive framework considering the TMS, EC and MA for PDP to develop high-quality products in minimum cost with technological developments in turn escalating the NPD success. The empirical study utilizes the primary data collected from 263 design and development experts of Indian manufacturing companies.

48.2 Theoretical Background for Hypotheses Development

PDP comprises of series of activities from idea generation to commercialization of the newly developed products in the market [6]. Managerial support and motivation, EC and MA make PDP effective for continuing development activities. The role of PDP for NPD success is undoubtedly essential and been discussed in previous literature. This section deals with the theoretical background for developing hypotheses linking the factors and measures for framing the structural model to achieve NPD success.

48.2.1 Top Management Support (TMS)

TMS is an essential phenomenon of all phases of the development process to handle the difficulties that arise during PDP [7]. The need of TMS for successful NPD is well established by systematic review to sustain in the increasing competitive market environment. Yeh et al. [8] ranked the factors of NPD and identified the managerial support as important one and correlated this factor with other variants. TMS influences the knowledge-sharing process to improve the firm's innovative capacity. Top managers' relationship with external networks including suppliers, customers and competitors helps in setting the PDP for NPD. It motivates the integration of external bodies offering joint resource planning and coordination essential for PDP [9]. These

interpretations clearly depict the fact that H1a: TMS positively motivates PDP for NPD success. H1b: TMS improves the EC of the firm for better PDP.

48.2.2 External Collaboration (EC)

EC includes the involvement of various departments of NPD teams, suppliers and customers for effective use of knowledge and information through committed coordination [10]. It produces a high degree of innovativeness in PDP escalating the NPD success. Supplier involvement helps in recognizing and solving the potential problems upfront. Again, customer involvement utilizes the innovative capabilities of the potential customer in joint decision-making with the NPD teams [11]. This practice involves the suppliers as well as customer's integration in PDP making the NPD tasks effective for fulfilling the customer requirements essential for NPD success. This infers that H2: EC encourages PDP of the firm for NPD success.

48.2.3 Market Analysis (MA)

MA is a process of accumulating the first-hand information about the potential characteristics of the newly developed products for customer satisfaction [12]. The technical similarities and dissimilarities also have been recognized through market analysis which helps to set the changes required in PDP for developing high-tech products with unique characteristics in lower cost without compromising the quality as well. From this, it can be stated that H3: MA positively influences the PDP for NPD success.

48.2.4 Product Development Process (PDP)

Development process of new products has been identified as one of the success factors critical for operational success [2]. PDP helps the firm in utilizing the resources optimally for developing the products as per customer demand. It consists of five major stages including concept testing, prototype testing, pretest market, test market and launch of the product [6]. Sufficient investment in PDP makes the firm technologically updated. Effective PDP offers high-tech products in reduced cost with better quality as well. The above discussions infer the statement that H4a: PDP effectively reduces the product cost (C). H4b: PDP positively influences product quality (Q). H4c: Effective PDP offers the technological developments (TD) to the new products. There is insufficiency of the empirical researches in the previous literature to recognize and develop a comprehensive framework considering managerial support, EC and MA for PDP in Indian manufacturing sectors. Realization of the necessity of

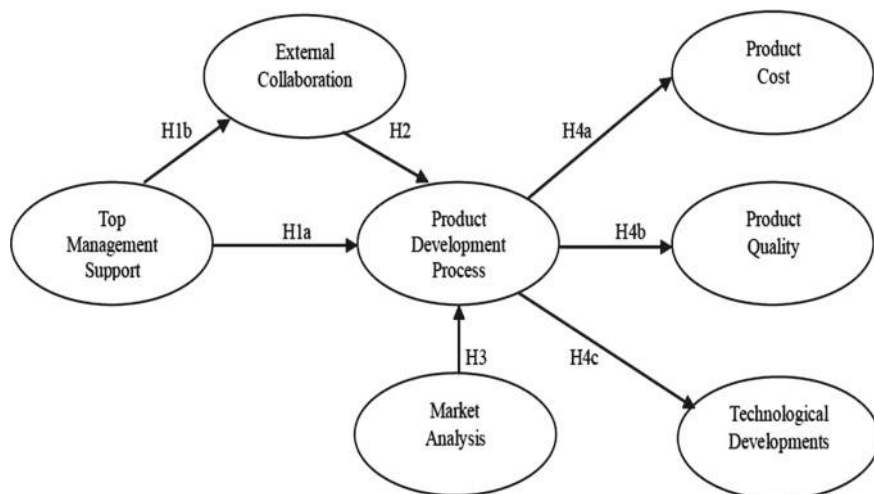


Fig. 48.1 Path model of constructs representing hypotheses

high-end PDP in small and medium enterprises (SMEs) in India is still inadequate. The socio-economic structure of India makes the development process of the new products a conventional one which has become one of the obstacles for industrial sustainability. This scenario leads towards gradual decay in the future growth resulting incompetence in the global scenario. This study is an effort to realize the impact of TMS, EC and MA for adopting high-end PDP to develop high-quality products in minimum cost with technological developments for structuring the concept of ‘MAKE IN INDIA’ a feasible one. The above theoretical background helps in framing a comprehensive path model illustrated in Fig. 48.1, exploring the developed hypotheses to relate the mentioned constructs.

48.3 Methodology

48.3.1 Methods

Analytic hierarchy process (AHP) is one of the multi-criteria decision-making method developed by Thomas L. Saaty (1987) used to rank the factors as per their importance and priority based on experts’ opinion [13]. In this study, the ranking of the factors such as TMS, EC and MA is performed as per their contribution for successful adoption of PDP. Similarly, the priority of success measures has also been calculated to recognize the most important criteria for quantifying the success. Structural equation modelling (SEM) is used for analysing hypotheses about

relations among latent constructs which have been developed from previous literature and collected data from company experts [14]. This study utilizes maximum likelihood (ML) estimation method for hypotheses testing for constructing the relationship model using PDP and its associated variables for analysing their effects on NPD success of the firm by using IBM SPSS AMOS 21.0.

48.3.2 *Questionnaire Development*

This exploratory research needs to develop a semi-structured questionnaire comprising of the constructs and their indicators as observed in the available literature for gathering detailed information to develop the framework. A questionnaire comprising of both, open-end and closed-end sections, is been framed for accumulating experts’ opinion and suggestions. The closed-end portion gathers the information regarding the degree of importance and implementation using seven-point Likert scale. In open-end section, the provision for providing suggestions of experts about the manifests is also mentioned in the list.

48.3.3 *Sample and Data Collection*

A semi-structured questionnaire has been developed for accumulating data from 263 experts of Indian manufacturing companies. Though various types of organizations are selected as targeted samples, all of them are manufacturing companies developing engineering products. A pilot study has been conducted in the eastern part of India, mainly in Kolkata and Howrah districts. Mostly, the Directors, Vice Presidents and Managers of design and development departments are considered as the right persons to be questioned as their responses are dependable for future analysis. Demographic profiles of the reliable respondents compatible for this study have been mentioned in Table 48.1.

Table 48.1 Demographic profile of respondents

Classifications	Total	%
Geographical location: Indian manufacturing industries	263	100
<i>Respondent’s profile</i>		
Executive	69	26.23
Manager	83	31.56
Senior manager	48	18.25
Vice president	36	13.69
President	27	10.27

(continued)

Table 48.1 (continued)

Classifications	Total	%
<i>Organization type</i>		
Fabrication	46	17.49
Electrical equipment	33	12.55
Industrial valves	32	12.17
Textile machineries	27	10.27
Fire fighting equipment	26	9.89
Hydraulics and pneumatics	25	9.50
Burner and heater	22	8.37
Material-handling equipment	21	7.98
Cell and battery	14	5.32
R&D sectors	9	3.42
Air ventilators	8	3.04

48.4 Results and Discussions

48.4.1 Weight Calculation Using AHP

This study incorporates the identification of the most important success factor among TMS, EC and MA for controlling PDP to achieve NPD success by calculating their weights using AHP based on Saaty's scale representing 1 = equally important, 3 = somewhat more important, 5 = much more important, 7 = very much important and 9 for strongly important factors. The prioritization of the success measures such as cost, quality and technological developments on the basis of PDP for developing new products is obtained by calculating the priority vectors by AHP as well. In case of success factors, the judgemental matrix is formed by comparing the importance of the three decision criteria such as TMS, EC and MA as represented in Table 48.2. The values of priority vectors depict the weightages of the respective factors for performing PDP.

Finally, comparing the importance of the three success measures of NPD which are cost, quality and technological developments, prioritization is performed by calculating the priority vectors as shown in Table 48.3.

Table 48.2 Judgmental matrix of NPD success factors

Success factors	TMS	EC	MA	Priority vectors
TMS	1	3.11	3.11	0.607
EC	0.32	1	0.69	0.172
MA	0.32	1.44	1	0.220

$$\lambda_{\max} = 3.01, CI = 0.005, CR = 0.008$$

Table 48.3 Priority vectors of cost, quality and technological developments for PDP

Success measures	Cost	Quality	Technological developments	Priority vectors
Cost	1	0.28	1.82	0.212
Quality	3.56	1	4.22	0.654
Technological developments	0.55	0.24	1	0.135

$$\lambda_{\max} = 3.026, CI = 0.013, CR = 0.022$$

According to the calculation, the success factors influencing PDP for developing new products, the priority vector is obtained as (0.607, 0.172, 0.220) for TMS, EC and MA as enlisted in Table 48.2. These values infer that managerial support is the most important factor critical to the firm's success for controlling PDP of NPD followed by market analysis and entrepreneurial culture. In case of success measures, i.e. cost, quality and technological developments on the basis of the PDP for enhancing NPD success, the priority vector is calculated as (0.212, 0.654, 0.135) as listed in Table 48.3. This interprets that the quality of the product is the most vital measure for developing the new products to fulfil the customer needs. Cost of the product is the second most essential measure followed by technological developments for measuring the NPD success considering the PDP of the Indian manufacturing firms.

48.4.2 Analysis of Measurement Validity

The seven-point Likert scale is used as a survey instrument to quantify the degree of importance in the implementation of the indicators as per the experts' opinion. In case of importance, 1 denotes strongly disagree and 7 denotes strongly agree about the importance in the implementation of the indicators; whereas, in case of success measure, 1 denotes very low and 7 denotes the degree of implementation rate. The reliability testing of the survey data is performed by composite reliability (CR), average variance extracted (AVE) and Cronbach's Alpha reliability testing using IBM SPSS 21.0. The reliability index portrays the reliability as well as the internal consistency of the collected data sample. The CR value greater than 0.5 is considered as highly reliable. For AVE, values greater than 0.5 indicate the reliability. In case of α , values should be either greater or equal to 0.8 is treated as reliable. The principal component-based EFA is performed for identifying the indicators having higher loadings for the respective constructs and the standardized regression weights (SRWs) of them are calculated as listed in Table 48.4 which also incorporates the reliability indices as well.

Table 48.4 List of latent constructs along with their manifest variables including path estimates and reliability indices

Latent constructs along with their manifests	FL	SRWs
Top management support (TMS) [CR = 0.69, AVE = 0.57, α = 0.853]	–	–
1. Support and motivation (m1)	0.935	0.34
2. Willingness of taking risk (m2)	0.921	0.75
3. Support entrepreneurship culture (m3)	0.842	0.64
4. Commitment (m4)	0.806	0.53
5. Frequency of annual meeting (m5)	0.767	0.51
6. Delegation of top management	0.522	
7. Leadership by example	0.488	
External collaboration (EC) [CR = 0.79, AVE = 0.61, α = 0.936]	–	–
1. Involvement of customers in NPD (m6)	0.856	0.33
2. Involvement of suppliers in NPD (m7)	0.823	0.79
3. Collaboration for know-how (m8)	0.818	0.74
Market analysis (MA) [CR = 0.63, AVE = 0.50, α = 0.829]	–	–
1. Market plan generation (m9)	0.901	0.82
2. Customer satisfaction (m10)	0.886	0.86
3. Consumption pattern (m11)	0.852	0.71
4. Target market, growth pattern (m12)	0.830	0.67
5. Market testing (m13)	0.814	0.84
6. Market research (m14)	0.798	0.94
7. Advertisement and promotion (m15)	0.775	0.41
8. Competitor monitoring (m16)	0.753	0.58
Product development process (PDP) [CR = 0.76, AVE = 0.64, α = 0.921]	–	–
1. Consistency of NPD rate (m17)	0.953	0.59
2. Investment in PDP (m18)	0.944	0.86
3. Updated technological innovations (m19)	0.926	0.79
4. Adequate number of new products (m20)	0.906	0.70
5. Availability of new products that are first to market (m21)	0.899	0.80
6. Involvement of customers in NPD (m22)	0.861	0.63
7. Suppliers involvement in design (m23)	0.842	0.66
8. Training provided on NPD (m24)	0.795	0.74
Product development process (PDP) [CR = 0.76, AVE = 0.64, α = 0.921]	–	–
9. Concurrent workflow (m25)	0.750	0.76
10. Hands-on working experience (m26)	0.712	0.88
11. Active use of in-house knowledge database (m27)	0.681	0.88
12. Early involvement of manufacturing department in NPD (m28)	0.614	0.70

(continued)

Table 48.4 (continued)

Latent constructs along with their manifests	FL	SRWs
13. Failure mode effect analysis	0.579	
14. Design for manufacturability (DFM)	0.540	
15. Practice of lean manufacturing	0.521	
16. Adoption of just-in-time (JIT) culture	0.508	
17. Refinement of product after launch	0.489	
Product cost (C) [CR = 0.78, AVE = 0.69, α = 0.935]	–	–
18. Profit margin	0.892	0.36
19. Domestic market share	0.867	0.54
20. International market share	0.819	0.39
21. Percentage of sales by NPD	0.791	0.42
22. Development cost	0.768	0.73
23. Attain return on investment	0.744	0.65
24. Attain profitability goal	0.725	0.81
25. Attain margin goal	0.704	0.46
26. Meet unit share goal	0.659	0.55
27. Revenue growth	0.624	0.49
28. Meet revenue growth	0.612	0.72
29. Profitability relative to competitors	0.570	
30. Profitability relative to spending	0.514	
31. Net sales growth	0.507	
Product quality (Q) [CR = 0.83, AVE = 0.72, α = 0.947]	–	–
m39: Meet quality guidelines	0.948	0.68
m40: Achieved product performance goal	0.914	0.70
m41: Achievement of design goals	0.849	0.83
Technological developments (TD) [CR = 0.80, AVE = 0.71, α = 0.939]	–	–
1. Technological breakthrough (m30)	0.890	0.49
2. Beating competition technologically (m31)	0.867	0.51
3. Expanding product family (m32)	0.821	0.86
4. Rate of failure (m33)	0.784	0.69
5. Frequency of product launching(m34)	0.692	0.53
6. Reduction of risks (m35)	0.627	0.88
7. Beating competition to market	0.602	0.54
8. Cannibalization effect	0.528	

Note 1. Principal component-based EFA was performed for factor extraction based on loading values. 2. CFA is performed to calculate the individual regression weights of manifests. 3. Maximum likelihood methods are applied for measurement model estimation. 4. All *t*-values are significant to $p < 0.001$

48.4.3 *Measurement Model*

Confirmatory factor analysis has been performed for estimating the model fit. The crucial step of SEM is the estimation of path values for analysing the model fitness. Besides, there are multiple methods of assessing the model fitness such as chi-square including degree of freedom, goodness-of-fit (GFI), adjusted goodness-of-fit (AGFI) and root mean square of error approximation (RMSEA). As per the fitness test, the model fitting values are within the desired range as $\chi^2 = 590.096$, degrees of freedom = 298, $\chi^2/\text{degrees of freedom} = 1.98$, RMSEA = 0.037, GFI = 0.932 and AGFI = 0.909 [15]. Values of reliability indices such as CR and AVE values range from 0.63 to 0.83 and 0.50 to 0.72. Again another vital reliability index α values range from 0.829 to 0.939, greater than 0.8, depict the satisfactory data reliability for future analysis. The values of reliability indices and SRWs with factor loading >0.6 have been represented in Table 48.4. It has been observed that for TMS, the SRWs range from 0.34 to 0.75, for EC 0.33 to 0.79, for MA 0.41 to 0.94 and for PDP 0.59 to 0.88. The SRWs for success measures range from 0.36 to 0.81 for cost, 0.68 to 0.83 for quality and 0.49 to 0.88 for technological developments. It shows that the values are positive, and it can be interpreted that these indicators positively related to the respective success factors for NPD success of the firm.

48.4.4 *Structural Model*

Analysis of measurement model is followed by the structural model development. In case of structural model, also the fitness tests are performed showing the appropriate model-to-data fit as $\chi^2 = 605.236$, degrees of freedom = 310, $\chi^2/\text{degrees of freedom} = 1.95$, RMSEA = 0.031, GFI = 0.962, AGFI = 0.928 [15]. The structural model representing the hypothesized relationships among the constructs and their antecedents has been developed by using IBM SPSS AMOS 21.0 software as shown in Fig. 48.2. The positive path value in between TMS and PDP supports the hypothesis (H1a) depicting the positive role of TMS to control PDP (path estimate = 0.80, $p < 0.01$). The influential impact of TMS on EC (H1b) is depicted as significant by the path estimate 0.78 ($p < 0.01$). The effect of EC on PDP (H2) has been supported by the positive path value (path estimate = 0.61, $p < 0.01$). The MA also has a significant impact on PDP (H3) as recognized by the obtained path estimate = 0.59 ($p < 0.01$). Similarly, the path estimates between PDP and product cost, quality and technological developments are 0.58, 0.82 and 0.51, respectively, significant for $p < 0.01$. The values of path estimates between the constructs for both input and output ranges, as represented in Table 48.5 are positive depicting that proposed hypotheses are supported. The developed structural framework is shown in Fig. 48.2.

Figure 48.2 symbolizes the detailed framework of the interrelationship model containing all the latent constructs along with their indicators. Here in this study, the

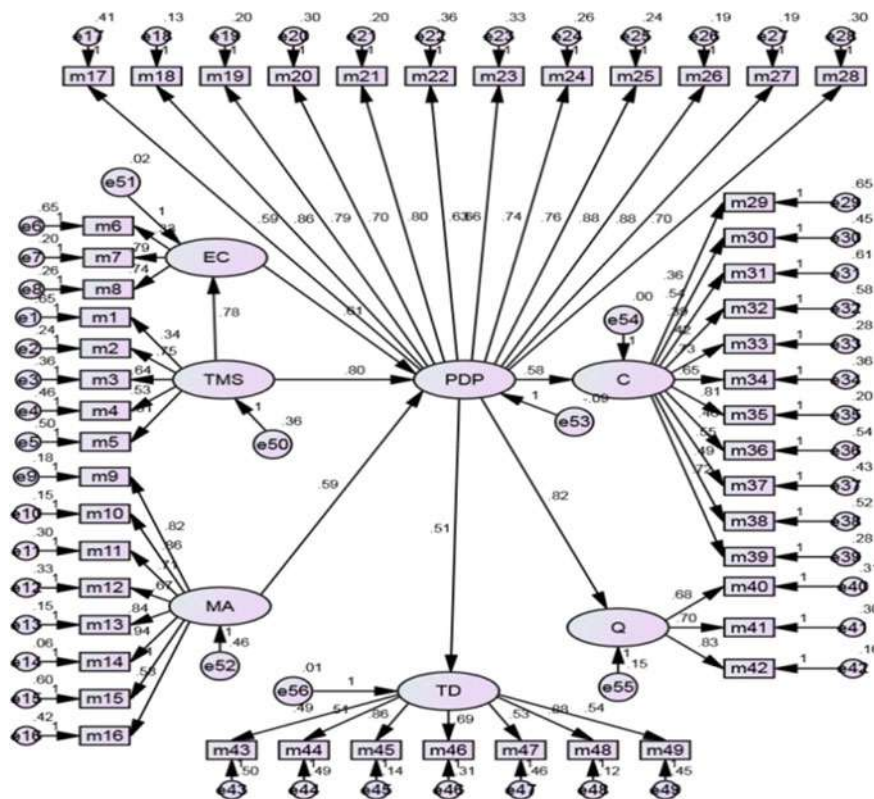


Fig. 48.2 SEM model after execution depicting the interrelationships of latent constructs and their manifests including path estimates

Table 48.5 Statistics of path estimates

Path descriptions	Hypotheses	Standardized estimates	t-values
TMS → PDP	H1a	0.80 (***)	10.335
TMS → EC	H1b	0.78 (***)	13.202
EC → PDP	H2	0.61 (***)	8.360
MA → PDP	H3	0.59 (***)	10.809
PDP → C	H4a	0.58 (***)	16.518
PDP → Q	H4b	0.82 (***)	16.007
PDP → TD	H4c	0.51 (***)	6.858

Notes ***indicate the significance at p-value <0.01

latent constructs are TMS, EC, MA, PDP, product cost, product quality and technological developments. Among these, TMS, EC, MA and PDP are input constructs; whereas product cost, quality and technological developments are output constructs. This model clearly represents the regression weights of the indicators measuring the latent constructs as mentioned in Table 48.4. Similarly, the path estimates among the constructs are listed in Table 48.5 which is also clearly represented in this structural model. These estimates are proving the impact of one construct on another as assumed from the theoretical background. Each indicator of a construct is associated with standardized error depicting the difference between the actual and estimated path values for individual path coefficients. The direction of arrows also has important implications in this model as it shows the direction of impact indicating the effect of one factor on another. This structural model is an interrelationship model clearly portraying the impact of managerial support, EC and MA for escalating PDP to produce new products with high quality in minimum cost with technological developments.

48.5 Conclusions

The aim of this study is to realize the contribution of PDP for NPD success to boost Indian manufacturing sector for shaping 'MAKE IN INDIA' feasible. For implementing PDP in the practical field, support and motivation from top management, EC and MA are highly essential as identified in this research. This study prioritizes these factors and develops an interrelationship model among them to realize their importance in implementation in the practical field. First, the ranking of these constituents is performed to prioritise them for PDP of the firm. The TMS is found as the vital most factor of PDP followed by MA and EC. Like, success factors, success measures are also ranked which depicts that quality is the most essential measure of NPD success followed by cost and technological developments.

Second, the indicators of PDP are recognized which helps to realize the PDP in Indian manufacturing industries. It introduces the importance of these indicators as hands-on working experience and active use of in-house knowledge database succeeded by investment in PDP, availability of new products that are first to market, updated technological innovations, concurrent workflow, training provided on NPD, adequate number of new products and early involvement of manufacturing department in NPD, suppliers involvement in design, involvement of customers in NPD and consistency of NPD rate. These are the essential antecedents need to be implemented for enriching PDP of the firm. The adoption of these practices can be beneficial for PDP of Indian manufacturing industries.

Third, it recognizes TMS, EC and MA for escalating the PDP of the firm. In case of TMS, the willingness of taking risk is identified as the most essential indicator to be implemented in the firm followed by support entrepreneurship culture, commitment, frequency of annual meeting and support as well as motivation from management bodies. TMS encourages the collaboration with external bodies for enriching the PDP as well which is then measured by the involvement of customers in NPD, suppliers'

integration for NPD and collaboration for know-how. Among these, the involvement of suppliers has the highest impact to measure EC of the firm. Similarly, MA is performed efficiently to influence the PDP by adopting the practices of market research followed by customer satisfaction, market testing, market plan generation, consumption pattern, target market, growth pattern, competitor monitoring and advertisement and promotion. Finally, the NPD success of the firm is manifested by product cost, its quality and the technological developments achieved by realizing the PDP in the firm with managerial support, EC and MA. By realizing these effects, the study offers the way to achieving NPD success leading the manufacturing firms of India towards industrial success and survival.

Acknowledgements The research work was substantially supported by a grant from the Department of Science and Technology (DST) of India as a DST INSPIRE Fellowship.

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Chapter 49

Design of Open Battery Pack Interface for Electric Vehicle Personalization



F. Chen , J. Zhang , M. Wu , X. Chu and Uday Shanker Dixit

Abstract Satisfaction of personalized customers' requirements of battery packs determines the competitiveness of electric vehicles. To meet the personalized requirements of battery packs, this paper develops a method for the design of an open interface for the connections and disconnections of battery pack modules to the electric vehicles. First, location and initial design scheme of the open interface are proposed based on relation analysis between customer demands and the electric vehicle module. To better facilitate the functional interactions between battery pack modules and electric vehicles, adaptability of open interface is then enhanced through design evaluation and modification. Influence of uncertainties due to unknown battery pack modules to the performance of electric vehicle is minimized through structure optimization of the open battery pack interface. The design can support electric vehicle personalization, better lifecycle flexibility, open technology innovation, and open business model.

Keywords Personalization · Open interface · Electric vehicle · Battery packs · Uncertainties

49.1 Introduction

Battery packs are critical to the design of an electric vehicle. Many specifications of electric vehicles including safety, cost, driving distance, charging and discharging time, lifecycle maintenance, etc., are influenced by their battery packs [1–3]. For

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_49

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most of the existing electric vehicles, battery packs are connected to electric vehicle through a closed interface. All the specifications of their battery packs are determined in the design process of electric vehicles and not allowed to be changed [4].

The open interface is a new type of product interface. Compared to the traditional interface (i.e., closed interface), in which all the specifications of add-on modules are defined in product design processes and cannot be changed in the product operation, open interface can be used to connect undefined/unknown modules provided by third-party vendors [5]. Product with open interfaces can better support product diversity, adaptability, sustainability, and flexibility [6–9]. Since different personalized modules provided by third-party vendors can be connected to product platform, open interface can also support product personalization and open product innovation. For an electric vehicle, design of an open battery pack interface has the following advantages:

- Electric vehicle personalization. Since different battery packs can be designed and produced according to customer's individual requirements by third-party vendors, personalization of electric vehicles can be achieved.
- Better product lifecycle flexibility. Different battery pack modules can be easily assembled and disassembled by customers and/or third-party vendors to the electric vehicle during the product lifecycle span.
- Open electric vehicle innovation. Advanced battery technologies from third-party vendors can be integrated with the electric vehicle through the connection of battery pack modules.
- Open business model. Instead of original electric vehicle manufacturer and preselected battery pack where supplier provides all the components in electric vehicles, third-party vendors can also participate in the development of electric vehicles.

For the design of open battery pack interface, adaptability and robustness need to be well considered; the interface adaptability measures the capability of interface to support connections of different functional modules, while the robustness is used to measure the functional performance of interface under the influence of uncertainties due to unknown functional module. In this paper, design of open battery pack interface is carried out by considering both functional adaptability and performance robustness.

49.2 The Open Battery Pack Interface Design

The framework of this study is shown in Fig. 49.1. The design method of open battery pack interface of electric vehicle includes: (1) identification of the interface location and development of initial design scheme; (2) interface optimization considering functional adaptability; and (3) interface optimization considering performance robustness.

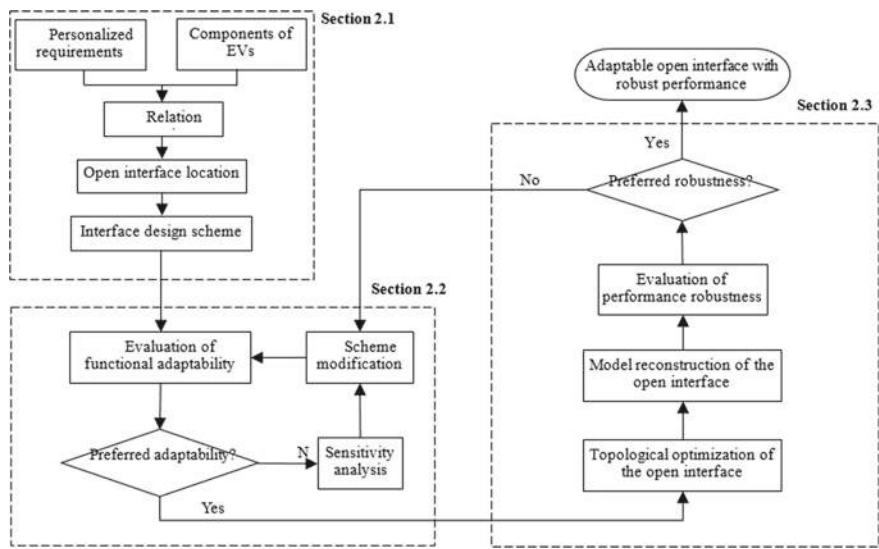


Fig. 49.1 Framework of the open battery pack interface design

49.2.1 Design of the Initial Interface Scheme

For the development of interface design scheme, personalized requirements and physical component of electric vehicle are required to be analyzed first. In this study, the personalized customer requirements include power capacity of the electric vehicle, types of battery used in the electric vehicle, safety degree, charge and discharge time, driving distance of per charge, minimum ground distance of electric vehicle, and the maximal speed of the electric vehicle. Major physical components include battery cells, enclosure, battery management system (BMS), chassis, motor, car body, braking system, power transmission system, etc.

Generally, personalized requirements are achieved through design of physical components. Three types of relations between requirements and components are observed: no relation, medium relations, and high relation. In this study, relations between the personalized requirements and those physical components are modeled by a matrix shown in Table 49.1.

From Table 49.1, we can found that battery cell, enclosure and BMS have the strongest relations with the personalized requirements of customers. Therefore, to better meet the different personalized requirements of customers through adapting physical components of an electric vehicle, the following components are considered:

- Battery pack module: the battery cells, enclosure, and BMS need to be grouped into a personalized battery pack module.
- Common platform: rest components of electric vehicle can be considered as a common platform.

Table 49.1 The matrix for modeling relations between demands and components 0: no correlation; 1: medium correlation; 2: high correlation

	Battery cell	Enclosure	BMS	Chassis	Motor	Car body	Braking	Power trans- mission
Power capacity	2	2	1	0	0	0	0	1
Types of battery	2	1	2	0	0	2	2	1
Safety degree	2	2	2	1	1	0	0	0
Change and dis- charge time	2	1	2	0	0	2	2	0
Driving distance	2	1	1	0	1	0	0	0
Minimum ground distance	1	2	0	2	0	1	0	0
Maximal speed	1	0	2	0	2	0	0	2

- Open battery pack interface: an open battery pack interface is required to connect the personalized battery pack module and the common platform.

Based on the connection analysis among the physical components, the open interface is located between the battery pack enclosure and the chassis. According to [5], electric vehicle parameters are usually considered as constraints for the open interface design. In this study, electric vehicle parameters are provided in Table 49.2. According to Table 49.2 and the personalized requirements of customers, in this study, the preferred parameters of battery pack module including length, width, height, and weight of battery pack module are summarized in Table 49.3.

An initial open interface design scheme was designed as shown in Fig. 49.2. The initial design scheme mainly includes sliding track, movable hangers, sliding blocks, and several hexagon screws with different sizes. By using the interface described in Fig. 49.2, parameters of battery pack module can be adapted within certain range for satisfying customers’ different requirements. Changing ranges of the battery pack module parameters are summarized in Table 49.4.

Table 49.2 Parameters of electric vehicle [5]

Item	Parameter	Value
Automobile space	Long	4700 mm
	Wide	1800 mm
	High	1640 mm
	Minimum ground distance	125 mm
Battery pack	Type	Lithium iron; Fe-battery, etc.
	Capacity	20–85 kWh
	Driving distance	120–400 km
	Weight	≤900 kg
Battery pack reserved space	Long	≤2000 mm
	Wide	≤1650 mm
	High	≤250 mm

Table 49.3 Preferred parameters of battery pack module

Parameter	Symbol	Range	Unit
Battery pack length	L	(0,2000]	mm
Battery pack width	W	(0,1650]	mm
Battery pack height	H	(0,250]	mm
Battery pack weight	M	(0,900]	kg

49.2.2 Enhancing Functional Adaptability of the Open Battery Pack Interface

To better satisfy personalized requirements, the interface needs to be optimized for achieving better ranges of battery pack module parameters. In this section, functional adaptability of the initial design of the open battery pack interface is evaluated first. Then, the design scheme of the open interface is modified for enhancing functional adaptability of the open interface.

Functional adaptability evaluation of open interface. For an open interface, functional interactions between add-on modules and the platform need to be facilitated [10]. The functional adaptability of open interface is used to evaluate the interface capability to support the functional interactions between add-on modules and platform. In this study, functional adaptability is considered for the design of open battery pack interface.

Due to constraints of physical components of the open battery pack interface, feasible parameters of battery pack modules can only partially overlap/meet the preferred parameters. On the functional level, the interface functional adaptability, A_F , can be calculated as

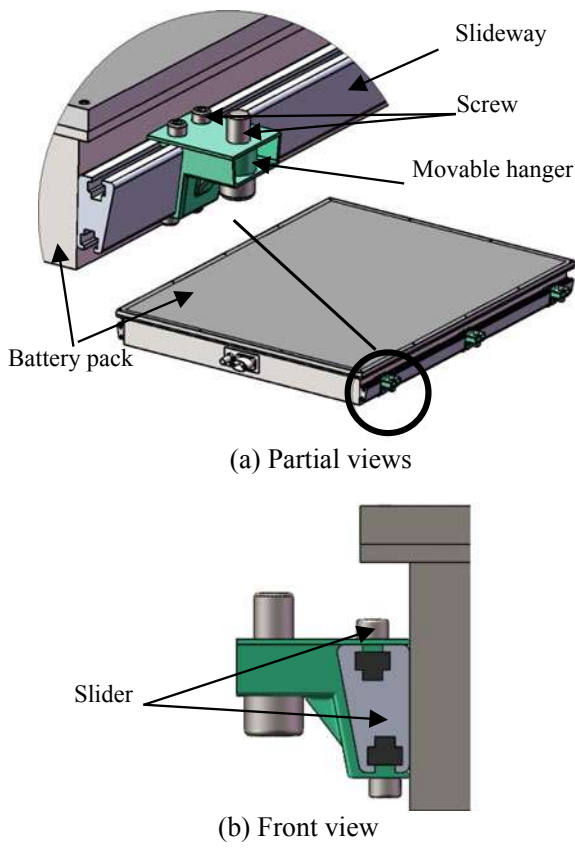


Fig. 49.2 Initial design of open battery pack interface

Table 49.4 Parameters of the initial battery pack module

Parameter	Symbol	Range	Unit
Battery pack length	L1	(0,2000]	mm
Battery pack width	W1	1650	mm
Battery pack height	H1	250	mm
Battery pack weight	M1	(0,400]	kg

$$A_F = R_O / R_R \tag{49.1}$$

where R_R represents the range of required or ideal inputs and outputs of the interface, R_O represents the overlapped/met range of the feasible inputs and outputs with R_R .

Generic methods for the calculation of functional adaptability were provided in our previous work [10].

Improvement of open battery pack interface. In Sect. 49.2.1, the open battery pack interface of the electric vehicle is preliminarily designed. By considering the functional adaptability, it is found that the parameter ranges of battery pack modules need to be optimized for better satisfaction of personalized requirements. Through sensitivity analysis of the design to the functional adaptability, to improve the functional adaptability of the open battery pack interface, the open battery pack interface is designed as shown in Fig. 49.3. The battery pack interface, which uses the open interface design, evaluation and operation method to guide the design, is described in detail in the literature [5]. Parameter ranges of the modified design scheme of the open interface are summarized in Table 49.3. As shown in Fig. 49.3, the improved design of the open battery pack interface is mainly composed of the following parts: a personalized battery pack, an adaptable tray, a T-slot (distributed in the four corners of the adaptable tray), a clamping plate, and positioning plate. The positioning plate and the clamping plate can be moved through the T-slot in the four corners of the adaptable tray so as to position and clamp the individualized battery packs of different sizes. In the modified design, three positioning plates on the corner of the adaptable tray are used for the positioning plate with the T-slot. The personalized battery pack is positioned on the adaptable tray through three fixed positioning plates, clamping plate to secure the personalized battery pack. Parameter ranges of the modified design scheme of the open interface are summarized in Table 49.5.

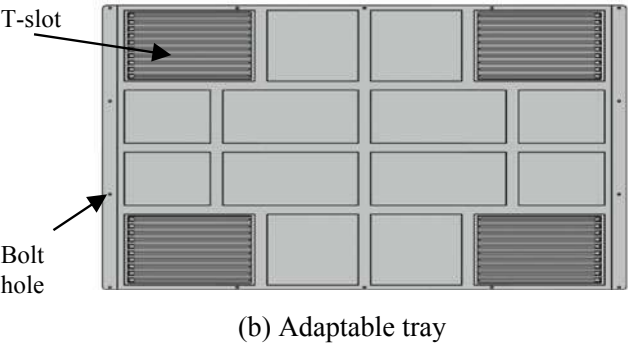
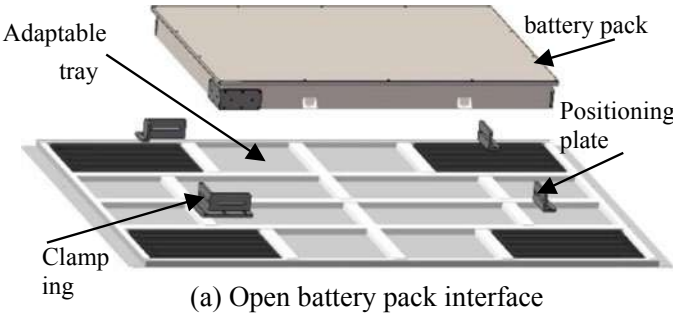


Fig. 49.3 Modified design of open battery pack interface

Table 49.5 Ranges of inputs and outputs of the modified interface design

Input/Output parameter	Symbol	Range	Unit
Battery pack length	L2	[1280,2000]	mm
Battery pack width	W2	[960,1650]	mm
Battery pack height	H2	(0,250]	mm
Battery pack weight	M2	(0,900]	kg

Table 49.6 Functional adaptabilities of the initial and modified design

Design	Functional adaptability	Improvement
Initial design	0.361	+84.2%
Modified design	0.665	

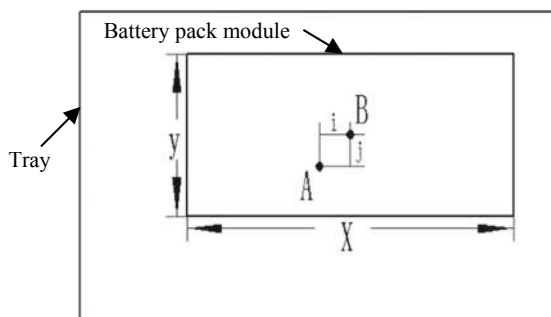
Functional adaptabilities of the initial and modified design scheme. The open battery pack interface of electric vehicle is used to connect the personalized battery pack modules provided by third-party vendors. Open battery pack interface is required to facilitate the connection of battery pack with different specification parameters to the platform. According to the preferred battery pack parameters shown in Table 49.3, the parameters of the initial battery pack module shown in Table 49.4, and the parameters of the modified battery pack module shown in Table 49.5, by using the Eq. (49.1), the functional adaptability of the initial interface design scheme and the modified design scheme is obtained as 0.361 and 0.655, respectively. We can find that the functional adaptability has been improved by 84.2%, which means a much higher capability to satisfy personalized requirements of customers (Table 49.6).

In Sect. 49.2.2, functional adaptabilities of two open battery pack interfaces are evaluated. The results show that functional adaptability of the modified design is much better than the initial design.

49.2.3 Robust Optimization of Open Battery Pack Interface

Open battery pack interface is used to connect personalized unknown battery pack modules provided by third-party vendors. Uncertainties associated with the unknown battery pack modules need to be considered carefully for better product quality. In this section, uncertain parameters due to unknown battery pack module are modeled first. Robust performance is obtained through topology optimization of the open interface.

Fig. 49.4 Uncertainties of battery pack modules



A represents the gravity center of the tray; B represents the gravity center of the battery pack module

Determination of uncertain parameters. The open battery pack interface can be used to connect with battery pack modules with different specification parameters including length, width, height, and weight of the battery pack modules as shown in Table 49.5.

In this study, since the different specification parameters can change in certain ranges, the following uncertainties due to specification parameter variations need to be considered:

1. The weight of different battery pack module leads to the uncertainties of vertical load on the adaptable tray. According to Table 49.5, the vertical load of different battery pack module to adaptable tray, F , can vary from 0 to 9000 N;
2. The length and width of different battery boxes lead to the uncertainties of the loading position of the tray (see X , Y in Fig. 49.4). According to Table 49.5, the changing range of X and Y can be $1280 \leq X \leq 2000$, $960 \leq Y \leq 1500$.
3. The gravity center A of the adaptable tray has an offset with the gravity center B of the battery pack module (see i , j in Fig. 49.4). In this study, the following ranges are considered: $0 \leq i \leq 50$, $0 \leq j \leq 50$.

Evaluation of performance robustness of the open battery pack interface. Uncertainties associated with battery pack modules can influence the performance of the open interfaces. To ensure the required function of the adaptable tray, in this study, the maximum stress of the adaptable tray cannot exceed the yield strength of the selected material.

In order to ensure that the maximum stress under different battery pack modules not exceeding the ultimate yield strength of the selected material, robustness of the performance of the open interface need to be optimized. In this work, the performance robustness index R of open interface can be calculated as:

$$R = \sigma_{\max}(m \leq m_0) \quad (49.2)$$

where mass of the open battery pack interface m should be smaller than the m_0 . $\sigma_{\max}$ is the maximum stress found in the open adaptable tray.

In general, the maximum stress of the interface is not allowed to exceed the yield strength limit of the selected material. Moreover, the maximum stress of the open interface should be “smaller the better.”

Topology optimization of open battery pack interface. To minimize the performance robustness R , which has been defined in Eq. (49.2), the topology optimization method is used for the optimization of the adaptable tray. In this study, the topology optimization tool plug-in in ANSYS software is used to carry out finite element simulation and optimization.

In this work, the modified design scheme of the adaptable tray is simplified under the condition of retaining the structure function. As shown in Fig. 49.5a, it is a simplified adaptable tray. The topology optimization problem for the entire tray can be described as:

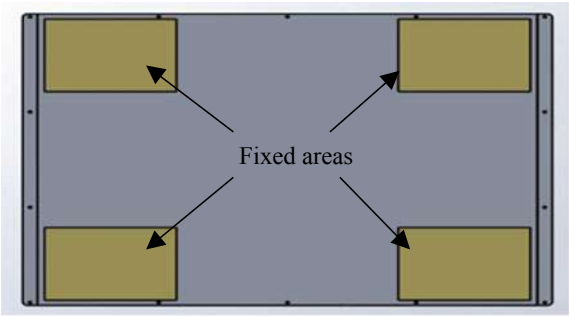
- Objective: minimize $\sigma_{\max}$
- Optimization area: the adaptive tray
- Subject to: maximal stress ≤ 350 Mpa; Mass of the open interface $m \leq 300$ kg
- Exclusion zone: 14 bolt holes and the four fixed areas.

The optimization result is shown in Fig. 49.5b. Reconstruction of optimized result was carried out for better manufacturability. The reconstructed model based on the optimization results is shown in Fig. 49.5c.

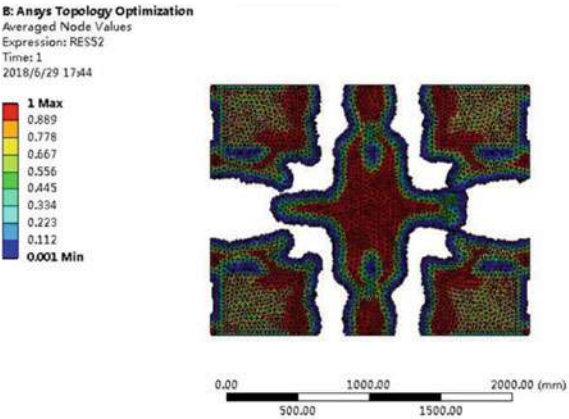
Performance robustness of the modified design and the final (reconstructed) design. In this work, ANSYS is used to find the maximum stress of the modified design shown in Fig. 49.3 and the final design shown in Fig. 49.5c. Uncertainties associated with the battery pack modules are considered for the identification of the maximal stress on the adaptable tray. First, discretization of uncertain parameters in ANSYS software needs to be done; then the maximal stress of the adaptable tray can be obtained by using the central composite experimental method. The results obtained are shown in Table 49.7.

According to Table 49.7, we can identify the maximal stresses of the modified design and the final design, respectively. The performance robustness of the two designs is summarized in Table 49.8. It is found that the maximal stress of the final design has been reduced and the performance robustness has been largely improved through optimization.

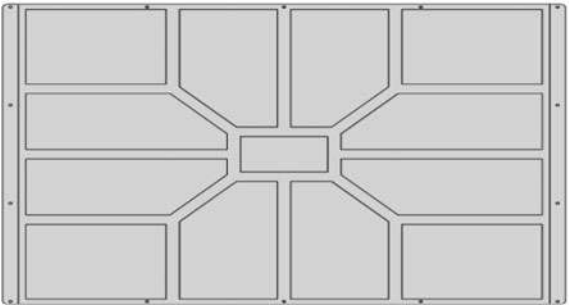
Fig. 49.5 Optimization of the open adaptable tray



(a) Model for topology optimization



(b) Topology optimization result



(c) Reconstructed model

Table 49.7 Maximum stress value

No.	X (mm)	Y (mm)	<i>i</i> (mm)	<i>j</i> (mm)	Maximal stress of modified design (MPa)	Maximal stress of final design (MPa)
1	1640	1305	25	25	122.63	68.89
2	1640	1305	0	25	129.02	69.85
3	1640	1305	50	25	115.09	72.87
4	1640	1305	25	0	124.5	69.62
5	1640	1305	25	50	180.84	67.48
6	1280	1305	25	25	117.37	67.06
7	2000	1305	25	25	153.98	83.82
8	1640	960	25	25	168.44	94.29
9	1640	1650	25	25	90.25	57.72
10	1386.5	1062	7.395	7.395	131.08	78.3
11	1386.5	1062	42.605	7.395	127.97	80.93
12	1386.5	1062	7.395	42.605	120.81	79.86
13	1386.5	1062	42.605	42.605	121.78	77.42
14	1893.5	1062	7.395	7.395	141.87	91.19
15	1893.5	1062	42.605	7.395	146.02	96.64
16	1893.5	1062	7.395	42.605	131.08	90.44
17	1893.5	1062	42.605	42.605	184.83	96.57
18	1386.5	1548	7.395	7.395	87.07	57.92
19	1386.5	1548	42.605	7.395	117.41	56.6
20	1386.5	1548	7.395	42.605	65.99	60.21
21	1386.5	1548	42.605	42.605	112.83	58.83
22	1893.5	1548	7.395	7.395	141.65	67.18
23	1893.5	1548	42.605	7.395	125.17	72.71
24	1893.5	1548	7.395	42.605	112.06	67.13
25	1893.5	1548	42.605	42.605	130.98	72.88

Table 49.8 Comparison of robustness of modified design and the final design

Design	Maximal stress	Improvement
Modified design	184.83	+47.7%
Final design	96.64	

49.3 Conclusions

Open interface can be used to connect personalized modules provided by third-party vendors. Design of open interface can support product personalization, adaptability, extendibility, openness, changeability, etc. To better satisfy customized requirements, in this paper, a framework is proposed to the design of open battery pack interface for facilitating the connection of personalized battery pack module to electric vehicles.

Location and initial design scheme of the open interface were proposed based on relation analysis between customer demands and the electric vehicle module first. To better facilitate the functional interactions between battery pack modules and platform of the electric vehicle, functional adaptability is considered for the design of the open battery pack interface. In addition, uncertainties associated with battery pack modules, including weight, length, and width of battery pack modules, can lead to stress variations of the open battery pack interface. To ensure the required quality of the electric vehicle, maximal stress of the interface due to those uncertainties are minimized by using topology optimization method. Design results have shown that the proposed method is effective to design of open battery pack interface with good adaptability and robustness.

Acknowledgements The authors wish to thank the National Natural Science Foundation of China (No: 51505269), and the Sailing Talent Program, Pearl River Scholar Program, and the Science and Technologies Program (No: 2017B090922008) of Guangdong Province, China for providing financial supports to this research.

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Chapter 50

Enhancement of Static and Dynamic Characteristics on Micro-lathe Bed by the Use of Alternate Form Design and Composite Materials



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V. Prabhu Raja , C. Raja Kumaran , K. N. Manojkumar
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Abstract Machine tools are operated at higher speeds vibrations are generated between workpiece and tool, results dimensional variation and poor surface finish on machined components. Hence, machine tools are developed with high dynamic stiffness by using high stiffness and damping composite material to reduce such effects and to achieve better dimensional accuracy with good surface finish on machined components. The machine tool structure considered in this study is a cast iron (CI) micro-lathe bed. The numerical model of the reference cast iron micro-lathe bed taken up for study was developed and experimental validation of the same was done. Static finite element analysis of the validated numerical model with worst-case cutting forces and moments was carried out for three different materials, namely gray cast iron, epoxy granite, and nettle polyester, and the results were compared, and the need for form design was justified. Static characteristics are improved through the use of cross sections and rib configurations with higher bending and torsional stiffness. Dynamic characteristics are improved through the use of stone- and fiber-based composite materials with higher specific stiffness and damping properties. The improvements in static and dynamic characteristics of the newly developed structures are investigated.

Keywords Composite · Form design · Micro-lathe bed · Finite element analysis

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_50

50.1 Introduction

High structural stiffness and damping are desired properties in machine tool applications, as they lead to higher cutting speeds which improve productivity (by reducing the cycle time). The high operating speed generates more vibration at the joints and interfaces of the machine tool elements [1]. It affects dimensional accuracy and shows poor surface finish on machined parts. It is desirable that the challenges due to vibration in machine tools are brought to a minimum level. In general, cutting machine tool industries develop machines using cast iron and steel as materials to build structures for material removal process through cutting action. Further, traditional machine tool materials exhibit either static stiffness or damping and vice versa [2]. To improve the static and dynamic performance of machine tools, alternate materials and new design concepts with enhanced stiffness and better damping properties are needed. Composite materials show promise as alternate materials for machine tools. Composite materials consist of two materials—one material, responsible for stiffness and another responsible for damping and so the composite exhibits both improved modulus and damping properties owing to which composite materials are increasingly used in the manufacture of machine tools. Ferrocement, as the name implies, consists of steel and cement, and this material offers three to four times higher damping ratio than CI [3]. Polymer concrete materials exhibit improved dynamic stability in machine tools due to higher damping ratio (four to seven times) than that of cast iron. Mineral casting materials consisting of granite as filler, called epoxy granite, this also enhance both damping ratio four to seven times and thermal stability two times than that of CI machine tool structure [4].

The requirement of high specific stiffness combined with high damping can be satisfied by employing fiber-reinforced polymer composite materials [5]. The commonly used fiber-reinforced polymer matrix composites are carbon-epoxy, glass-epoxy, glass-polyester composites and hybrid composites (sandwich structure composed of e-glass fiber faces and CI column core, and carbon-epoxy faces joined with welded steel structures using adhesives and bolts). Sandwich structures contain steel faces, inserts, aluminum rings, and castiron columns for improvement in the structural stiffness of the composite. The realization of new fiber-reinforced composite material (CFR) structures improves static, dynamic, and thermal properties of precision machine tool that can produce precise products [6]. The mass reduction potential for sandwich structures such as CI and steel structure filled with polymer concrete, e-glass fiber faces, and CI column core, is typically about 20% including better damping, whereas CFR can reach 60% mass reduction without sacrificing on stiffness and damping of the structural components.

Mostly in current research, synthetic fibers like glass, carbon, and aramid are used in fiber-reinforced composite. However, synthetic fibers are hazardous to human health and environment. The manufacturing cost of the above fibers is also high. In order to overcome the above problems, natural fibers can be used for making the composite. Natural fibers have a greater edge over synthetic fibers in that they are readily available, environmentally friendly, and cost-effective. Plenty of research

works are being carried out on natural fibers because of their inherent higher specific mechanical properties, lightweight and higher performance to weight ratio [7, 8]. Hence, they find new applications in automotive, aerospace, packaging, building, marine, electronics, and other engineering fields [9]. The principle of light weight material selection and form design can satisfy requirements of machine tool structures such as higher stiffness and damping [5].

In this study, static and dynamic characteristics of machine tool structures are enhanced by the use of alternate form designs and composite materials. Using constant area/volume approach, best cross section from a set of selected cross sections is determined. Finite element method is employed to identify the best rib configuration among the available standard rib configurations. Using constant stiffness approach, taking the cast iron bed as reference, composite lathe beds are developed for equal stiffness. Numerical model of the cast iron lathe bed was developed, and experimental validation was carried out. The cutting forces at the worst operating condition are evaluated and the same are used to perform finite element analysis on the developed numerical model for the evaluation of deformation, maximum stress, and factor of safety. Numerical models of composite lathe beds are also developed, and finite element analysis was carried out for the worst-case cutting forces. The results of cast iron and composite late beds are compared and the improvements in static and dynamic characteristics are recorded.

50.2 Development of Numerical Model of CI Lathe Bed

50.2.1 Static Analysis on CI Bed

To validate the numerical model of CI bed, experimental static deflection and modal analysis was carried out. Further, the results are compared with numerical results. Figure 50.1 shows reference CI bed, solid model, boundary condition, and deformation model.

Static performance of the reference CI micro-lathe bed was carried out using strain gauges, dial indicator, NI data acquisition (DAQ) system with LabVIEW software as shown in Fig. 50.2. Deflection and strain values are noted for incremental loads of 10–300 N.

ANSYS Workbench software was used to carry out numerical static deflection analysis on the lathe bed. Loads in the range 10–300 N were used for the analysis. Figure 50.1 shows the cast iron lathe bed model with boundary conditions and deformation results. Comparison of static test deflection and strain results obtained from experimental and FEA shows that a good correlation was observed with a maximum deviation of 10% between the experimental and numerical analysis.

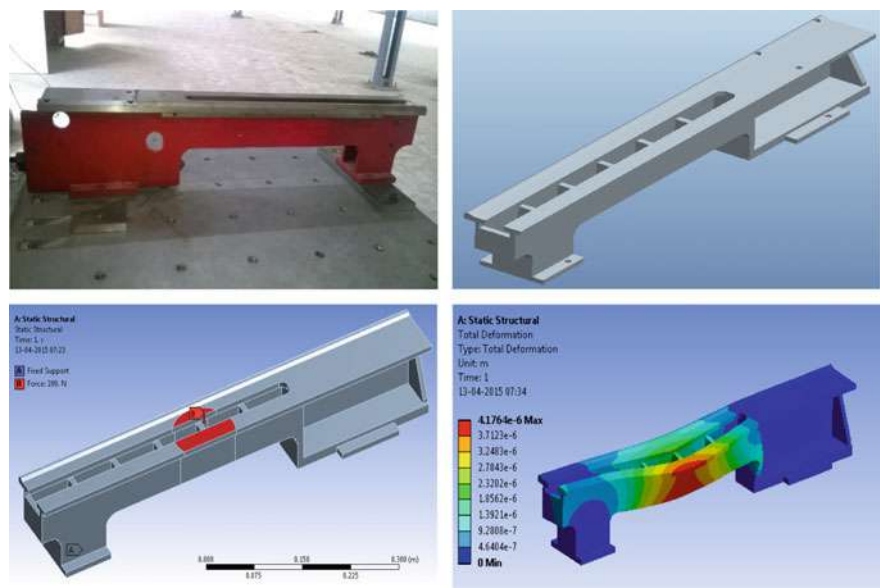


Fig. 50.1 Cast iron micro-lathe bed physical and FEA model

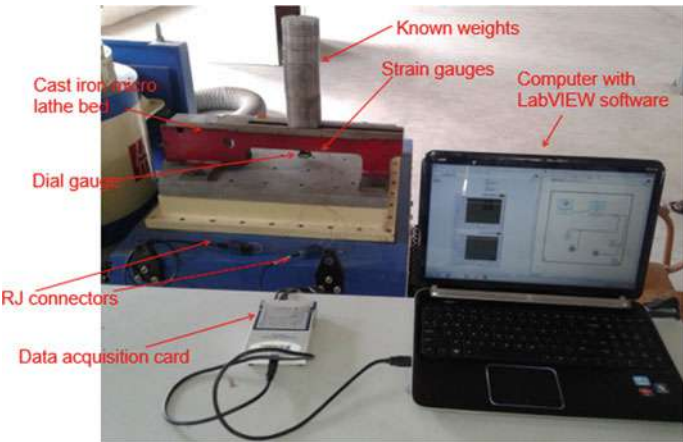


Fig. 50.2 Experimental static deflection test setup

50.2.2 Dynamic Analysis of CI Bed

Dynamic analysis of reference CI micro-lathe bed was carried out using impact hammer, accelerometer, NI DAQ with LabVIEW software. The experimental setup is shown in Fig. 50.3.

The first six natural frequencies and the corresponding mode shapes are evaluated and are shown in Fig. 50.4. The fundamental natural frequency determined by the finite element method was found to have close correlation with experimental value, with a deviation of 3.8%. Hence, the numerical model developed is deemed to be a reliable model and can be used for further analysis.



Fig. 50.3 Experimental modal analysis test setup

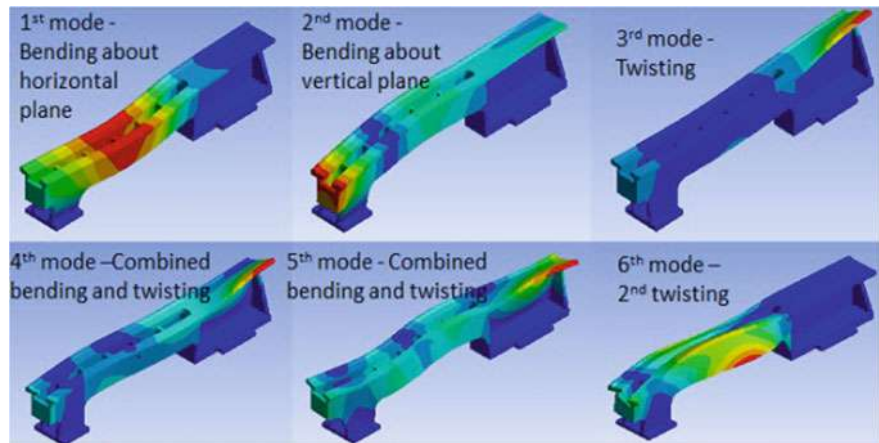


Fig. 50.4 First six mode shapes of cast iron lathe bed

Table 50.1 Properties of materials

Material	Gray cast iron	Epoxy granite (12:88) [4]	Nettle polyester (40:60) [10]
Young’s modulus (GPa)	110	30	3
Poisson’s ratio	0.3	0.25	0.22
Tensile strength (MPa)	130–400	20–45	20–40
Compressive strength (MPa)	500–1200	65–150	98–120
Damping ratio	0.017	0.032	0.036
Density (kg/m ³)	7200	2250	1188

50.2.3 Alternate Materials

The main aim of this work is to improve the dynamic properties (natural frequency and damping) of machine tool structures without affecting stiffness. Composite materials help to achieve improved dynamic properties in structures. In this study, two different composite materials are used, namely epoxy granite and nettle polyester. Table 50.1 shows the properties of alternate materials chosen for this study. Both epoxy granite and nettle polyester have higher damping ratio and lower density than gray cast iron material.

50.3 Design of Composite Lathe Bed

Finite element analysis (FEA) of the micro lathe bed model under the worst-case cutting forces and moments of CI bed and composite materials were also carried out as shown in Fig. 50.5 and the comparison of results are shown in Table 50.2.

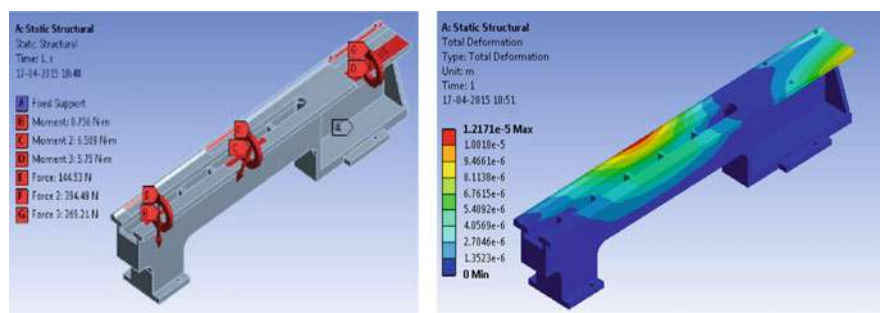


Fig. 50.5 CI micro-lathe bed FEA BCs and deformation plot

Table 50.2 Comparison of FEA results of CI, EG, and NP lathe beds for standard ‘I’ section

Sl. No.	Lathe bed material	Mass (kg)	Relative weight	Total deformation (μ)	Maximum principal stress (MPa)
1	Cast iron	11.39	1	12.2	3.5
2	Epoxy granite (12:88)	3.56	0.31	44.5	3.5
3	Nettle polyester (40:60)	1.88	0.17	443.1	3.4

From the table, it is evident that the maximum deflection in composite lathe beds for the same geometrical dimensions is quite high compared to cast iron micro-lathe bed. This is because of the lower value of Young’s modulus of the identified composite materials and hence there is a need for form design.

50.3.1 Identification of Best Form

To improve static performance of the newly designed micro lathe bed, through form design principle “hollow hexagonal” as shown in Table 50.3, was found better torsional and bending stiffness than other forms and the same form has adopted to design lathe beds.

Table 50.3 Identification of best rib configuration

Form number	Form design	Form number	Form design
1		6	
2		7	
3		8	
4		9	
5		10	

50.3.2 Identification of Best Rib Configuration

In order to study the effect of various rib configurations, the cast iron micro-lathe bed with straight transverse ribs (existing rib configuration) is considered as reference and other values are normalized based on it. Six different possible rib configurations were developed, and finite element analysis was carried out. Based on analysis, lathe bed with the combination of longitudinal and transverse stiffeners (rib configuration 3 in Table 50.4) is identified the best rib configuration with improved bending and torsional stiffness.

50.4 Design and Analysis of Lathe Bed for Constant Stiffness

The proposed CI, EG and NP lathe beds were subjected to worst case cutting forces and moments, based on FEA results the hollow hexagonal cross section with combined longitudinal and transverse vertical ribs designed composite EG and NP lathe beds provides enhanced bending and torsional stiffness. Figure 50.6a, b shows the improvements in static and dynamic characteristics were evaluated.

Table 50.4 Comparison of rib configurations

Sl. No.	Rib configuration	Form design	Relative weight	Relative bending stiffness	Relative torsional stiffness
1	Existing transverse vertical ribs		1	1	1
2	1		1.022	1.00	1.05
3	2		1.116	1.03	1.31
4	3		1.217	1.32	1.49

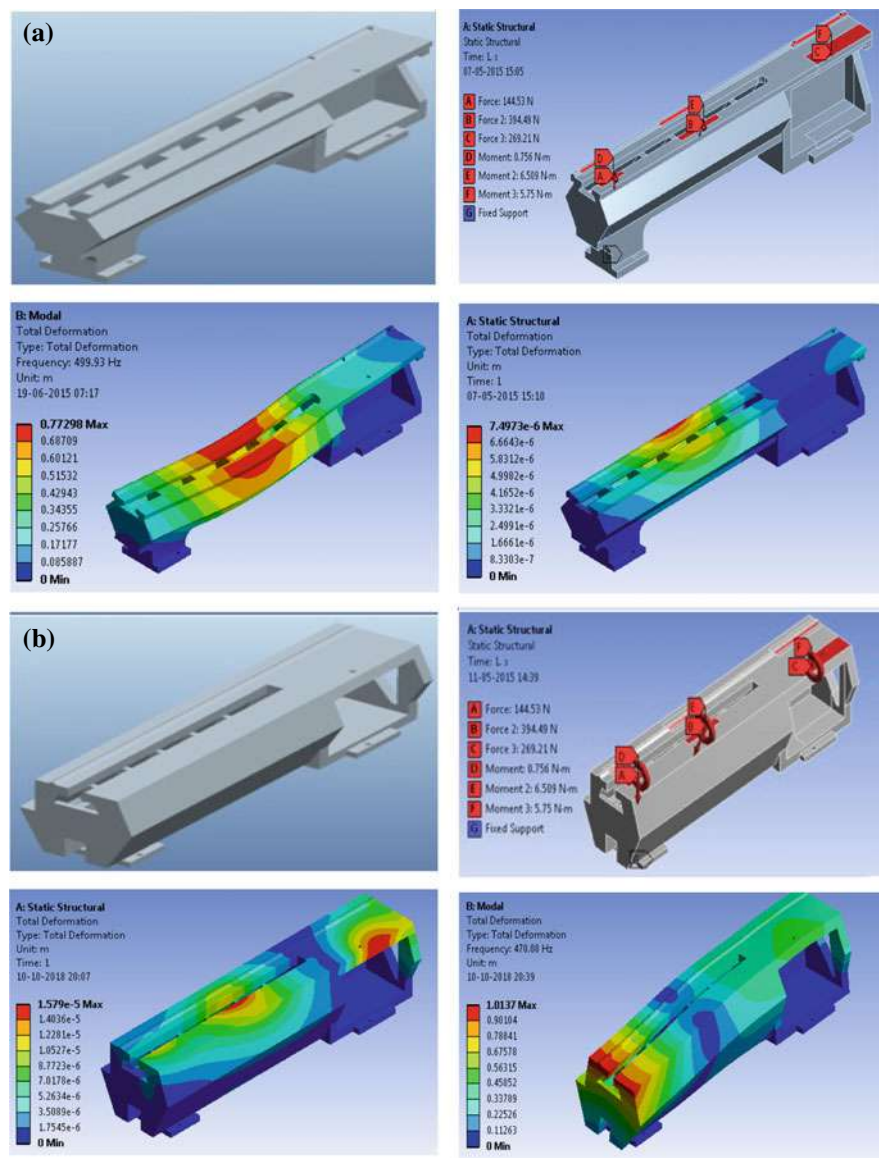


Fig. 50.6 Composite micro-lathe beds static and dynamic response FEA plots

Table 50.5 Comparison of FEA results of CI and composite lathe beds for “Hexagonal” section

Sl. No.	Lathe bed material	Relative weight	Total deformation (mm)	Maximum principal stress (MPa)
1	Cast iron	1.00	12.20	3.5
2	Epoxy granite (12:88)	0.70	7.50	1.0
3	Nettle polyester (40:60)	0.80	15.80	0.5

50.5 Results and Discussion

Table 50.5 shows the comparison of FEA results of CI, EG, and NP lathe beds. The newly designed EG and NP composite micro-lathe beds showed 33 and 6% weight reduction without affecting the stiffness of the micro-lathe bed structure by the use of form design principle. Further, the fundamental natural frequencies were found to be higher for composite lathe beds. The studies indicated that EG and NP composites could be considered as a suitable alternate to cast iron structures in machine tools.

50.6 Conclusions

In this work, epoxy granite (12:88) and nettle polyester (40:60) were identified as alternate materials owing to their high damping characteristics, low density and comparable strength to weight ratio. The static and dynamic characteristics of the micro-lathe beds are improved through the use of modified cross section and rib configuration with higher bending and torsional stiffness. Also, with the use of composite materials, weight reduction was achieved without affecting the stiffness. Hence, composite lathe beds could be used as a suitable replacement for cast iron lathe beds.

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Chapter 51

Development of Indigenous Direct Drive Rotary Guide Bush Device and Establishment of Three-Spindle Synchronization for Sliding Headstock Automat



S. Deepak , Nagesh Nadig and S. R. Chandramouli

Abstract CNC sliding headstock automat (SHA) has two spindles that work simultaneously along with turning tools and rotary tools to manufacture complex components with higher productivity. The concept of Swiss-style machining is adopted, wherein the job is fed as bars and held in a collet system and cutting happens close to guide bush support. This facilitates slender parts machining with ease. In the present context, guide bush device is developed with an exclusive built-in motor design, compact enough to optimize the end piece. At times, jobs that are machined in the main spindle with the support of guide bush need to be further carried by the sub-spindle. In such a situation, three-spindle synchronization is a challenge. An effort is put to understand the limitations of mechanical and electrical sub-systems to bring out a methodology for three-spindle synchronization.

Keywords Guide bush · Sliding head automat · Spindle synchronization

51.1 Introduction

Lathe is actually the first machine tool which came into being as a useful machine for metal cutting. It formed the basis of production of all the other machine tools such as automatic lathes, CNC sliding headstock automats. These machines are those which produce component automatically with less human intervention. The sliding head machine is of twin spindle head design, one for the primary operation and the other

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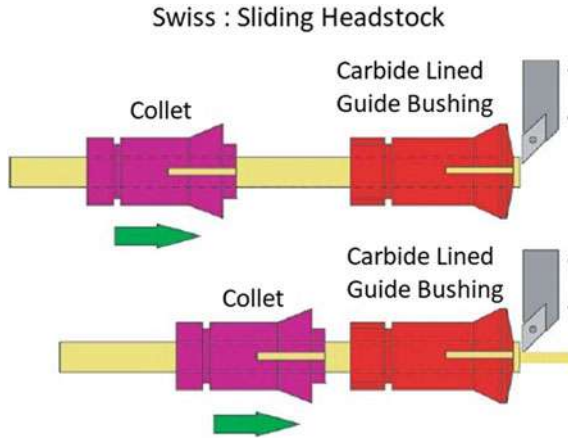


Fig. 51.1 Working of sliding headstock automat

for the secondary operation to bring in productivity of Swiss-style automat. These machines are preferred for slender parts with component sizes varying from $\varnothing 3$ to $\varnothing 32$ mm diameter. Sliding head machine consists of two spindle units, guide bush device, tool post and multiple axes. In conventional lathe that has fixed headstock, the workpiece is held in chuck or collet and extends into the machine as a cantilever or can be supported on the end by the tailstock. Sliding head machines are distinguished from conventional machines by having a sliding headstock. Bar stock which is feed by an automatic bar feeder passes through a chucking collet in the main headstock, which clamps on to it as shown in Fig. 51.1. The bar emerges into the tooling area through a guide bush, which locates the bar radially during machining. The headstock moves precisely back and forth in the z-direction (feed direction), taking the bar with it. Turning tools mounted on gang slides contact bar very close to the guide bush support, usually within 1–3 mm. The forward motion of the bar provides the feed for the cutting action. Gang slides carry multiple turning tools for OD (outer diameter) turning, stationary tool holders, which house tool holders for drilling, tapping, reaming operation and rotary tool (i.e. live tools) for milling, drilling, tapping and reaming application to be done on main spindle side (i.e. primary operation). Main spindle side consists of three axes such as Z1-axis for to-and-fro motion of bar stock, X1-axis for giving depth of cut and Y1-axis for tool change to carry out multiple cutting operations [1]. Back working tool stations consist of tools for boring, drilling, tapping, reaming and broaching operation which are provided for sub-spindle operations (i.e. secondary operation). Sub-spindle consists of two axes such as Z2-axis for component pickoff from main spindle after the primary operation and X2-axis for giving depth of cut and tool change to carry out multiple cutting operations. The guide bush is used for supporting the workpiece to maintain precision geometrical tolerance throughout the machining of the workpiece. With a

sliding head machine, the guide bush supports the workpiece so close to the tools that the deflection due to the cutting forces is essentially zero. As a result, you can take heavy depth of cuts and still maintain a precise dimension on the part. There are three types of rotary guide bushing device, one is driven mechanically by spindle itself [2–4], the second is driven electrically by externally mounted motor [5, 6], and third is also driven electrically by built-in motor independent of main spindle. Motorized rotary guide bushing device has few advantages such as reduction in mechanical losses and optimum end piece length over the other two guide bushing device. This led to the development of indigenous direct drive rotary guide bushing device.

51.2 Design of Direct Drive Rotary Guide Bush Device

Direct drive rotary guide assembly consists of built-in motor, spindle, spindle encoder, etc. Since guide bush assembly needs to accommodate raw material of size Ø 32 mm, accordingly guide bush [7] is selected. Spindle, which is a rotating member, is designed by taking cutting force of the main spindle into consideration.

Bar diameter	32 mm
Speed	2300 rpm
Cutting speed	230 m/min
Depth of cut	4 mm
Feed	230 mm/min

Power of the motor = UK_hK_sQ/E

(51.1)

Tangential Cutting Force = 6120 N/v

(51.2)

Torque at spindle = 975 N/n

(51.3)

51.2.1 Motor Selection

Based on experience and as a thumb rule, it is assumed that only 25% of main spindle cutting force will be acting on the guide bush.

Table 51.1 Power and torque calculations

Power required (kW)	Tangential cutting force (kgf)	Torque required (kgf m)
1.165	29.3	0.49

Suitable built-in motor, i.e. BiI 80S/20000-B, is selected from FANUC catalogue [8] based on power and torque requirement as shown in Table 51.1. Power and torque graph of built-in motor is shown in Figs. 51.2 and 51.3 that are taken from FANUC catalogue [8].

Fig. 51.2 Motor power diagram [8]

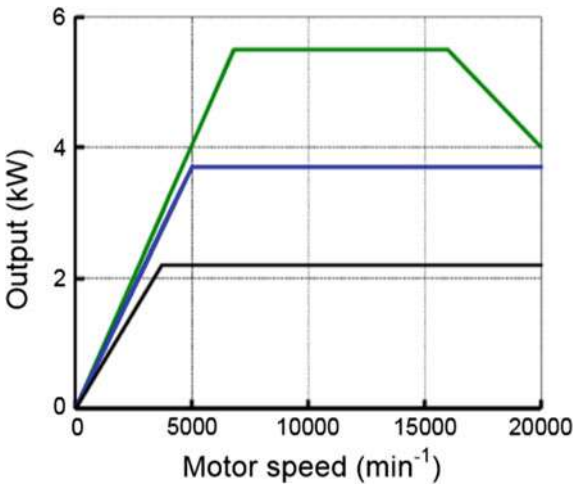


Fig. 51.3 Motor torque diagram [8]

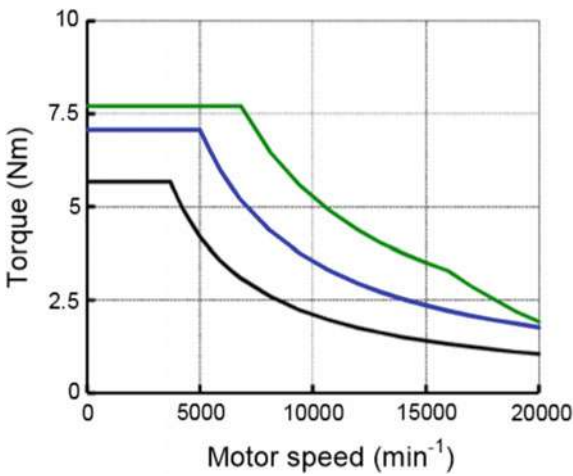
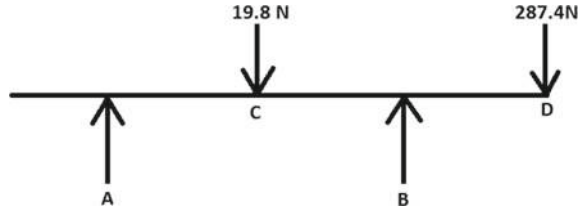


Fig. 51.4 Free body diagram



51.2.2 Spindle Calculation

The spindle is a rotating machine element, which is circular in cross section, and is used to transmit power and motion to guide bush. The spindle is hollow shaft supported on bearings to transmit power [1]. The shaft is generally acted upon by bending moment, torsion and axial force. In machine tools normally, during the design of shafts, requirements from stiffness considerations are more stringent than strength consideration. Rotor weight acts at point C, and the cutting force acts at point D as shown in Fig. 51.4.

Spindle Material = En 353

- d Diameter of shaft
- D_o Outer diameter of shaft
- D_i Inner diameter of shaft
- M Bending moment
- T Twisting moment

$$\text{Shaft Diameter} = d^3 \frac{16}{\pi \tau_{\max}} \sqrt{M^2 + T^2} \quad (51.4)$$

From Eq. 51.4, minimum spindle diameter is 20 mm.

51.2.3 Bearing Selection

A bearing is the critical component of any high-speed spindle design; our requirement states that spindle must provide the high rotational speed of 8000 rpm. Angular contact ball bearing is used in high-speed spindle designs as they provide precision load-carrying capacity and speed required. They are designed to provide both axial and radial load-carrying capacities when properly placed. Angular contact ball bearing is available with a different preloading, and due to high-speed requirement, light

preload bearing is selected. Bearings are mounted in back-to-back arrangement to achieve good accuracy and rigidity at both the front and the rear ends of the assembly. Bearing calculation such as static, dynamic load-carrying capacity and service life was carried out as per the standard procedure using NSK catalogue [9] for spindle diameter starting from 20 mm. The bearing of inner diameter 70 mm for the front end and inner diameter 65 mm for the rear end satisfies the conditions of speed and strength; hence, the said bearings are selected based on NSK selection procedure [9].

Bearing Designation:

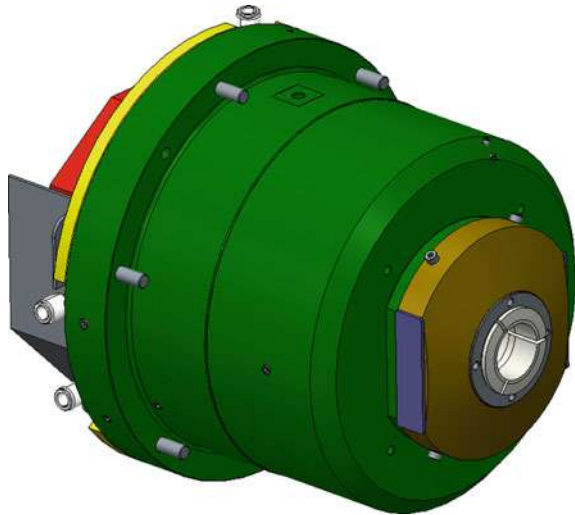
- (1) Type: angular contact ball bearing
- (2) 7914CTYNSULP4 (inner diameter: 70, outer diameter: 100, width: 16)
- (3) 7913CTYNSULP4 (inner diameter: 65, outer diameter: 90, width: 13).

3D Modelling

Creo Parametric 2.0 version was used for 3D modelling of direct drive rotary guide bush assembly.

Figures 51.5 and 51.6 show the complete assembly of direct drive rotary guide bush. Design of every component of the assembly was done considering design standards [10]. All the component detailing/drawing was prepared with appropriate geometrical dimension, tolerances and released for manufacturing [10]. Direct drive rotary guide bush assembly consists of a total of 34 components.

Fig. 51.5 Direct drive rotary guide bush assembly



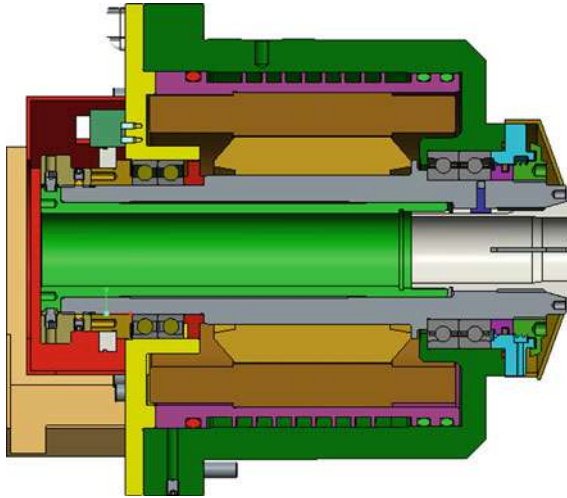


Fig. 51.6 Direct drive rotary guide bush assembly

51.3 Development of Direct Drive Rotary Guide Bush Assembly

Component manufacturing as per bill of materials is made with a quality focus. All components are inspected against drawing before assembly. The first step is to machine rotor Inner Diameter and shrink fit rotor on to the spindle/sleeve as per instructions from Fanuc catalogue [8], then both front and rear bearings are assembled on to spindle. The stator is shrink-fitted to cooling jacket [8]. Accordingly, all the components are assembled into housing. At the rear end of assembly, sensor ring is shrink-fitted and sensor head is placed above sensor ring on rear bearing housing with gap of $150\text{ }\mu\text{m}$ to obtain speed and orientation feedback. Rear bearing is axially arrested by lock nut. The stator is oil cooled by circulating oil in the cooling jacket. Air purge is also provided for the front bearing to restrict entry of coolant and dust into bearing. Figure 51.7 shows the prototype of direct drive rotary guide assembly developed in ACE.

Fig. 51.7 Prototype of direct drive rotary guide bush assembly developed at ACE



51.4 Experimentation

Step 1: Direct drive guide bush assembly is assembled on to fixture and tested/run with the help of test rig to thermally stabilize bearing. The results are tabulated in Table 51.2, and the graphical representation is shown in Fig. 51.8.

The temperature of both front bearing and rear bearing got stabilized at 8000 rpm; hence, the bearing thermal stabilization of direct drive guide bush assembly is completed.

Step 2: Two-spindle synchronization between main spindle and direct drive rotary guide bush is executed.

Direct drive rotary guide bush is assembled on to the machine as shown in Fig. 51.9. Initially, direct drive guide bush assembly is driven independently. Direct drive guide bush is driven in synchronous with the main spindle by using “spindle synchronous control for guide bush” [8]. In this function, the feedback data from master spindle axis is transmitted to slave spindle axis by spindle inter-amplifier communication and it becomes the position command for the slave spindle axis for the synchronization on a continuous basis as shown in Figs. 51.10 [8] and 51.11 [8].

Step 3: Three-spindle synchronization between main spindle, direct drive rotary guide bush and sub-spindle is executed.

In three-spindle synchronization, the main spindle motor operates as the guide bush motor, sub-spindle motor and the main spindle, guide bush and sub-spindle are integrally rotated and driven by the main spindle motor [8]. Pictorial representation is shown in Fig. 51.12 [8].

Table 51.2 Bearing thermal stabilization results

S. No.	Time	Speed (rpm)	Front bearing temperature (°C)	Rear bearing temperature (°C)	Remarks
1	10:00	200	21	21	
2	10:05	500	23	22	
3	10:10	1000	24	22	
4	10:20	1500	24	24	
5	10:37	2000	26	26	
6	10:47	2500	27	25	
7	10:52	3000	27	26	
8	11:02	3500	27	29	
9	11:07	4000	27	34	
10	11:17	4500	28	42	
11	11:18	4500	29	45	Stopped
12	11:35	4500	25	25	
13	11:40	5000	28	42	
14	11:45	5000	38	50	Stopped
15	12:00	5000	25	25	
16	12:06	5000	34	50	Stopped
17	01:08	5000	23	22	
18	01:18	5500	37	32	
19	01:23	5500	50	33	
20	01:47	6000	27	25	
21	01:57	6500	49	26	
22	01:58	6500	50	26	Stopped
23	02:28	6500	25	23	
24	02:33	7000	27	25	
25	02:37	7000	30	50	Stopped
26	03:16	7000	23	22	
27	03:21	7500	28	34	
28	03:26	8000	39	42	
29	03:45	8000	40	43	
30	05:45	8000	41	44	Stabilized

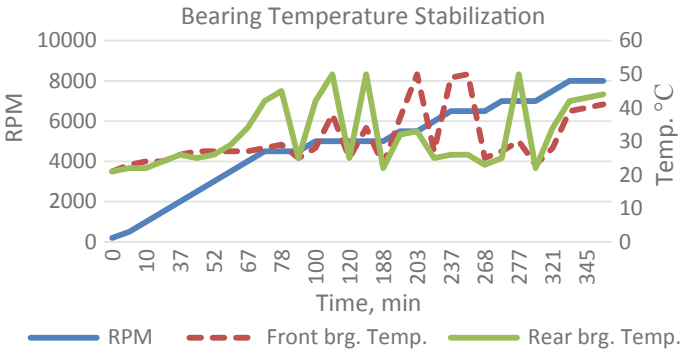


Fig. 51.8 Bearing thermal stabilization



Fig. 51.9 Direct drive rotary guide bush assembled on to the machine

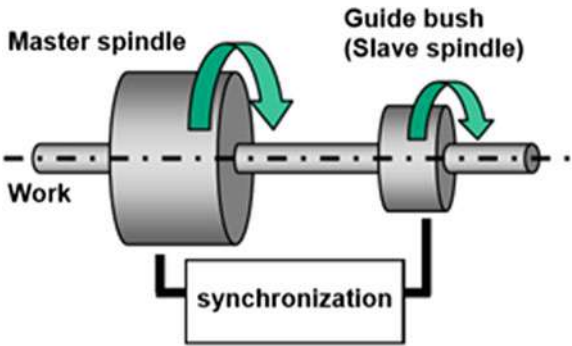


Fig. 51.10 Two-spindle synchronization [8]

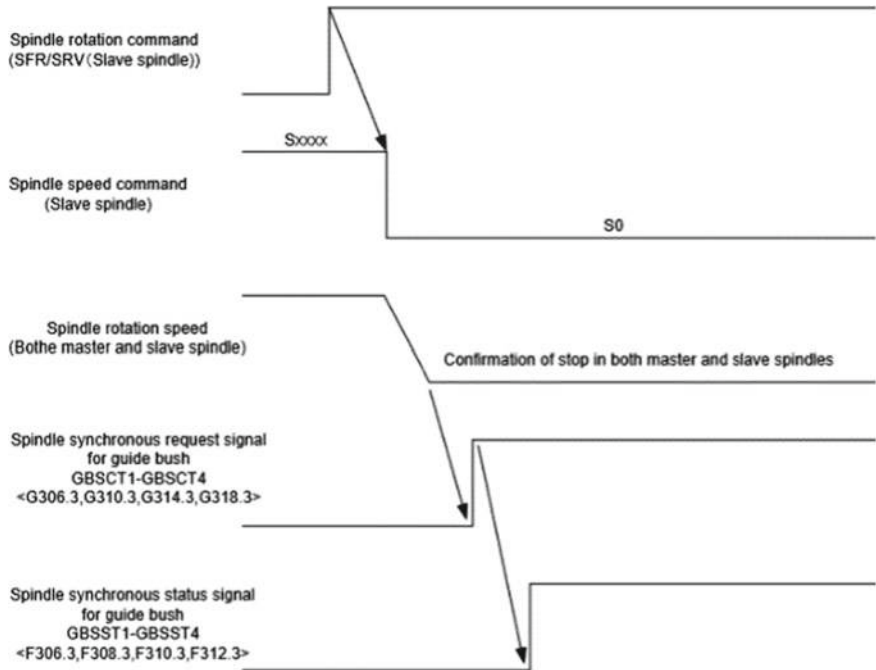


Fig. 51.11 Turning on spindle synchronous control for guide bush [8]

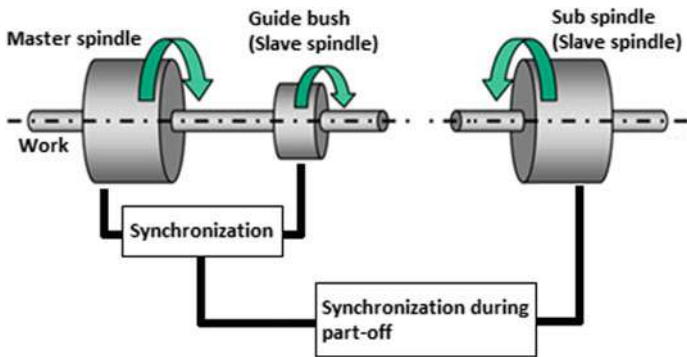


Fig. 51.12 Three-spindle synchronization [8]

Step 4: The practical metal cutting test is conducted on various types of cross-sectional bars to prove three-spindle and two-spindle synchronizations including direct drive rotary guide bush assembly.

Case 1: Round cross-sectional component (automotive component) was continuously produced from 3-m round bar of outer diameter 25 mm as shown in Fig. 51.13 on sliding headstock machine successfully with both two-spindle and three-spindle synchronizations.

Case 2: Square cross-sectional component (door hinge component) was continuously produced from 3-m square bar of 22 mm across flat as shown in Fig. 51.14 on sliding headstock machine successfully with both two-spindle and three-spindle synchronizations.

Case 3: Hexagon cross-sectional component (hydraulic fitting component) was continuously produced from 3-m hexagon bar of 27 mm across flat as shown in Fig. 51.15 on sliding headstock machine successfully with both two-spindle and three-spindle synchronizations.

Fig. 51.13 Round cross-sectional finished component



Fig. 51.14 Square cross-sectional finished component



Fig. 51.15 Hexagon cross-sectional finished component



51.5 Conclusions

Design and development of direct drive rotary guide bush device are to replace synchronous guide bush device in order to reduce power load on the main spindle and to demarcate two independent drive systems. The novel concept of developing direct drive rotary guide bush device is made practical because of parallel development taken place in electrical, electronic field, especially for multi-spindle synchronization. Considering the advantages of direct drive rotary guide bush device, it is designed, developed and practically proved for its commercial application. Challenges that are faced and addressed are the development of compact size cooling jacket for stator housing and to establish an oil cooling system for built-in motor. Multiple materials such as cast iron (SG iron) and medium carbon steel (C45) were tried for the cooling jacket and tested for practical implementation. It is found that C45 made cooling jacket works better and addresses cooling jacket issues made out of cast iron (SG iron) such as blow holes leading to leakages. During cutting operation, support to the raw material and partial sharing of cutting load is observed with that of main spindle; hence, two-spindle and three-spindle synchronizations are achieved.

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Chapter 52

Development of 3-Axis Micro-Step Resolution Desktop CNC Stage for Machining of Meso- and Microscale-Features



Shweta Patil and Sandip S. Anasane

Abstract Requirement of miniaturization in manufacturing equipment for micro-scale components and product is drastically increasing. Recently, desktop CNCs have proved to be more convenient than other micro machines due to its precision and accuracy. This paper presented a low cost indigenously designed and developed micro-step resolution desktop CNC stage which can be used for machining precision components at comparatively reasonable cost. Mechanical resolution of 1 μm has been achieved by designing and developing precision drives and controller for obtaining micro step by controlling stepper motors and drivers. The work is also focused on designing noise- and vibration-free mechanical structure of the CNC stage to achieve micro resolution motion. Universal G-code Sender is used as GUI and sends G-codes to machine. The developed desktop CNC stage which can have the potential use for various precision machining applications, especially, aimed for micro-machining domain. The developed stage can also be employed for the precise movement of workpiece and/or tool for various micro- and mesoscale-machining processes such as laser machining, electrochemical micro-machining, as well as tool based micromachining.

Keywords Micro-step resolution · Stepper motor · G-code · Universal G-code sender

52.1 Introduction

Increasing need of CNC system to manufactured complex parts with precision and accuracy but at considerable expensive. For manufacturing micro-featured products with precision requirement, the general-purpose machine tools are not efficient either in terms of size and energy. Smaller machines will consume less power, space, and time that help to reduce the cost also. Such idea led to the concept of Desktop CNC

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_52

stage. The idea on development of desktop CNC came forward to reduce the cost and complexity in CNC systems as well as to provide user-friendly solution for precision micromachining.

As Increase in industrial requirements of miniaturized products with high precision, good quality and at low costs are increasingly demanded. Machine tools have to fulfill the requirements such as dimensional accuracy and precision. Standard machine tools are used for machining of micro-featured products but these machine tools consume more space, power, and time. More importantly, accuracy requirement in micromachining is the major issue in standard machine tools. By applying concept of Desktop CNC stage with micro-step resolution, resource such as materials, time, and energy are required less.

The energy required to operate the micro-step resolution desktop CNC machine is reduced as well it requires less material and components to make the machine, hence bringing down the cost greatly. As the size reduces, weight of moving component also reduces so that vibration and noise generated during operation also reduced. This Desktop CNC machine becomes lighter and denser than conventional machines, so it becomes easily portable. The productivity and manufacturing speed also increases due to possible faster operation. The resolution of stage is up to $1\text{ }\mu\text{m}$ hence this 3-axis micro-step resolution desktop CNC stage which will use for various precision machining especially for micromachining applications such as laser machining, electrochemical micromachining as well as tool based micromachining.

In this paper, an inter-disciplinary design project toward the development of a micro-step resolution desktop CNC stage is presented. Micro-step resolution desktop CNC machine provides high stability due to designed structure and material selection. The micro-step resolution is achieved by incorporating the features of drive system, stepper motor and motor driver. The system analyzed G-code and the interpreted by microcontroller. Micro-step resolution Desktop CNC stage is capable of 3-axis simultaneous interpolated operation. With the help of micro-step resolution desktop CNC technology, machine tool is able to make micro-precision products down to microscales in a precise manner.

52.2 Literature Review

The growing demand of miniature products promotes the development of desktop CNC which consume less power, space and time. To develop micro-step resolution CNC stage, machine's structure design and analysis are most important, which is the objective of this paper. S. B. Chandgude and Patil discussed about the selection of layout, which is suitable for small-scale machines. This selection process is carried out by structural analysis using Boolean algebra technique (BAT). By using this technique one optimal machine tool structure is selected which fulfills all the geometrical constraints. Author suggested model is XYOZC, which fulfills all requirements of machine tool structure [1].

S. Mekid et al. applied Taguchi technique for designing micro mechanical system so as to make the structure robust and stable in a variety of conditions. Authors used methodology to approach design optimization using Taguchi technique, finite element modeling, and also considered a micro physical problem that helps to improve machine structure. Author's purpose is to improve the structural design, which is less sensitive to external influence [2]. Structural design should be robust and rigid. These features are analyzed by ANSYS software tool. Xiaozhou Li et al. choose gantry structural design by comparing various design parameters like stiffness, static, and dynamic stability. Gantry micro milling machine tool structures were analyzed [3].

In this paper, micro stepping is the main important feature so that micro-scale product can be machined. Gheorghe Baluta et al. presented micro-stepping mode for controlling stepper motor to improve the performance of stepper motor. To improve the positional accuracy, micro-stepping control system is useful and it also reduces resonance effect in stepper motor drive. Micro stepping is more important while machining at low speed, this micro-stepping control system reduces low speed ripple [4]. As CNC controller is the heart of the CNC machine, which controls most of the functions of CNC machine. Rajendra Rajput et al. investigated and analyzed the various types of CNC controllers, i.e., used in the CNC machines especially in milling and turning. Perfect machining in minimum time is the requirement of manufacturing industry. Different controllers like UMAC, FANUC, etc. are analyzed with various parameters like cutting speed, feed, and depth of cut [5].

Xiang Zhang et al. successfully develop the 5-axis machine tool, which is applicable to complex 3D micro milling. Author used UMAC multi-axis motion controller and PID controller making machine to have high dynamic performance [6]. For the purpose of studying machine tool miniaturization technology and micromachining mechanism, a three-axis desktop milling machine tool is presented by Q. Y. Wang et al. The work presented in this paper is focused on meso and micro features such as micro gears, micro thin walls, and microelectrode. Authors also analyzed the profile and roughness of these micro features. To analyze machining performance, different experimentations were carried out on machine tool to make complex microstructures is demonstrated [7]. After setting up machine, it was necessary to check the geometrical accuracy that can be used for precise milling. Author Adam Janasek et al. performed test on the milling machine to check geometrical accuracy of machine [8].

52.3 Methodology

To design and develop micro-precision desktop CNC stage are required in-depth knowledge of mechanical and electronic system. Electronics system provides control signals to mechanical system as a result of which the motors are actuated. The software system provides a command set to the electronics system; it also generates controls for mechanical system. The block diagram of overall process is shown in the Fig. 52.1, and the detail explanation is given below.

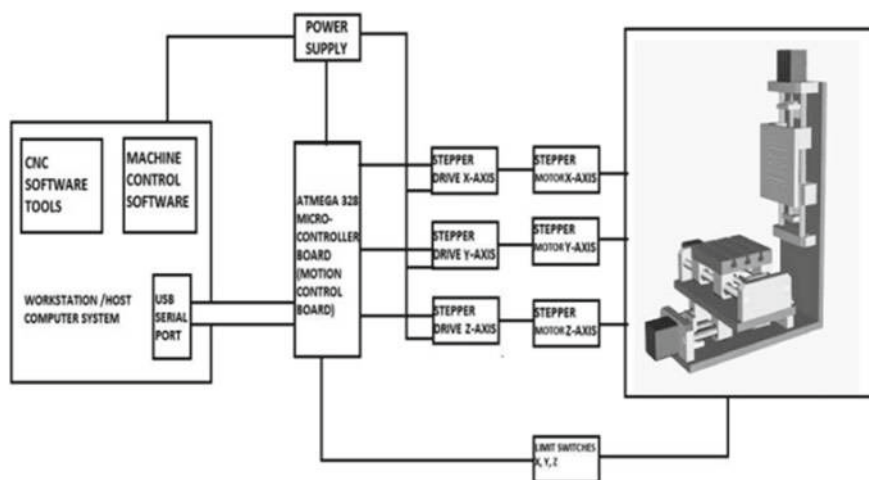


Fig. 52.1 Block diagram of overall process

52.3.1 Machine Structure

Machine structure is key feature of the machine tool. Desktop CNC structure is the integration of all machine components. Desktop CNC structure needed more stiffness than general-purpose machines at higher speed. The machine structure is more rigid so that the static and dynamic stiffness increases. A designed structure gives more stiffness; results in variety of operation that can be done in a precise manner. The desktop CNC machine operated on small parts using very small diameter tools means that they are very sensitive to external and internal vibrations. If vibrations are not reduced or eliminated, the cuts made and the parts created will be dimensionally inaccurate and relatively useless. So that the entire metallic machine structure must be rigid and the assemblies must function kinematically and geometrically flawlessly. Various machine structure designs are compared using different requirements such as rigidity, damping capacity, and stability and chosen one optimal machine structure for machine which fulfill all requirements as shown in Fig. 52.2. The advantages of selected design is more workspace for machining of different micro features by utilizing variety of clamping holding arrangement as per machining process. Also this design is more rigid and more stable which can be used for precise machining. Specification of desktop CNC stage is presented in Table 52.1. And photographic view of developed CNC stage is shown in Fig. 52.3.

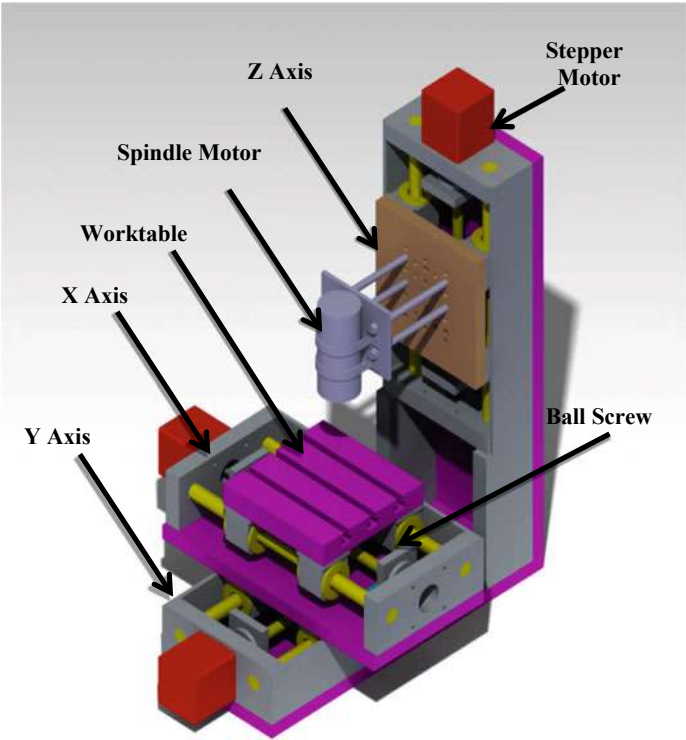


Fig. 52.2 Machine structure of micro-step precision CNC stage

Table 52.1 Specification of micro-step resolution desktop CNC stage

Parameters	Specifications
Number of axes	3 (X, Y, and Z)
Material	Aluminum alloy and steel
Drive system	Ball screw
Accuracy	1 μm
Worktable size	150 $\times$ 150 $\times$ 25 mm
Table flatness	0.005 μm
Work envelope	100 mm $\times$ 100 mm $\times$ 100 mm

52.3.2 Static and Dynamic Analysis

To investigate static and dynamic stability of designed 3 axis desktop CNC stage finite element analysis (FEM) was done. Mesh is generated on structure as shown in Fig. 52.4a. There is negligible deflection due to its own weight of machine structure. By applying 10 N force at the machining position in Z-direction, the result showed that back frame of desktop CNC structure undergo less than 0.01 μm deflection in



Fig. 52.3 Picture of the 3-axis desktop CNC machine

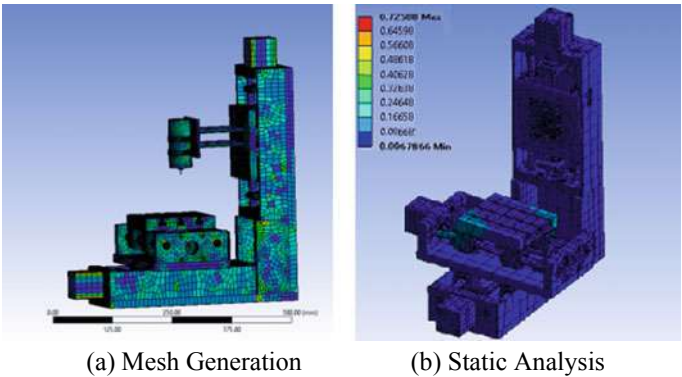


Fig. 52.4 Mesh generation and static analysis with a 10 N concentrated force

z-direction and at work table displacement change about 0.06 μm . The finite element model is shown in Fig. 52.4b. It seems that the 3-axis micro-step resolution desktop CNC stage has good stiffness due to good frame design.

Table 52.2 Parameters of feed system

Feed shaft	Travel (mm)	Resolution (μm)	Positional accuracy (μm)
X-axis	200	1	1.5
Y-axis	200	1	1.5
Z-axis	200	1	1.5

52.3.3 Drive Design

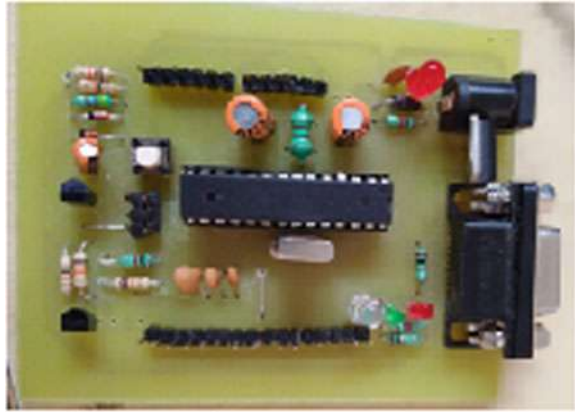
For 3-axis micro-step resolution desktop CNC, drive system is required to move tool and worktable in precise manner. Ball screw is selected as drive system for CNC because it gives precise linear motion than any other drives system like belt drive system and lead screw. Mechanical resolution of 1 μm is achieved by selection ball screw with smallest pitch. Class C3 ball screw is used which is most precise level. Selected ball screw have 12 mm diameter and 2 mm pitch. The parameters of feed system are depicted in Table 52.2. Ball screw also reduces the friction and backlash.

52.3.4 Selection of Motor and Driver

Selection of motors is most important because it is the main component of the driving control mechanism method. Stepper motor is selected because it has high holding torque and micro-stepping mode. A stepper motor and driver are as shown in Fig 52.5. The shaft of motor is controlled by electrical pulses applied in sequence. Motor rotation directly depends on the pulses applied. Speed depends on frequency of pulses and rotation is proportional to the number of pulses. For controlling the motion of axis, NEMA 23 stepper motor is used. NEMA 23 gives torque 10 kg-cm. Steppers are



Fig. 52.5 Stepper motor and stepper motor driver

Fig. 52.6 Controller board

the most affordable actuating solution for machine tools currently. Micro stepping is done in stepper motor so that system can move the smallest distance. For micro stepping, stepper motor driver is used. Microcontroller is send to step and direction signals to stepper motor driver. These signals are converted into rated high-voltage electrical signals, so that stepper motor runs accurately. TB6600 is micro-stepping drive is used for smooth operation and is chosen to drive the NEMA 23 stepper motor. Micro stepping is achieved in TB6600 using a synchronous PWM output drive.

52.3.5 Design and Developing of Controller

There are different types of controller available in markets. But to get high accuracy and precision, the controller will be designed and developed. Atmega 328p microcontroller is selected to control the electromechanical device. It acts as brain of the CNC system which receives controls of all the motion of the system. Software system sends commands to microcontroller. Controller is flashed with interpreter code which was written in the C language and which was converted form G-code. G-code is used to control tool path by sending control signals to stepper motor. Fabrication of controller will be with minimized cost and increased accuracy. Microcontroller board is shown in Fig. 52.6 and electronic circuit diagram in Fig. 52.7.

52.3.6 Software and GUI

There are different software chains used in CNC-based manufacturing are represented in the Fig. 52.8. Computer-aided design (CAD) software is used to design part file

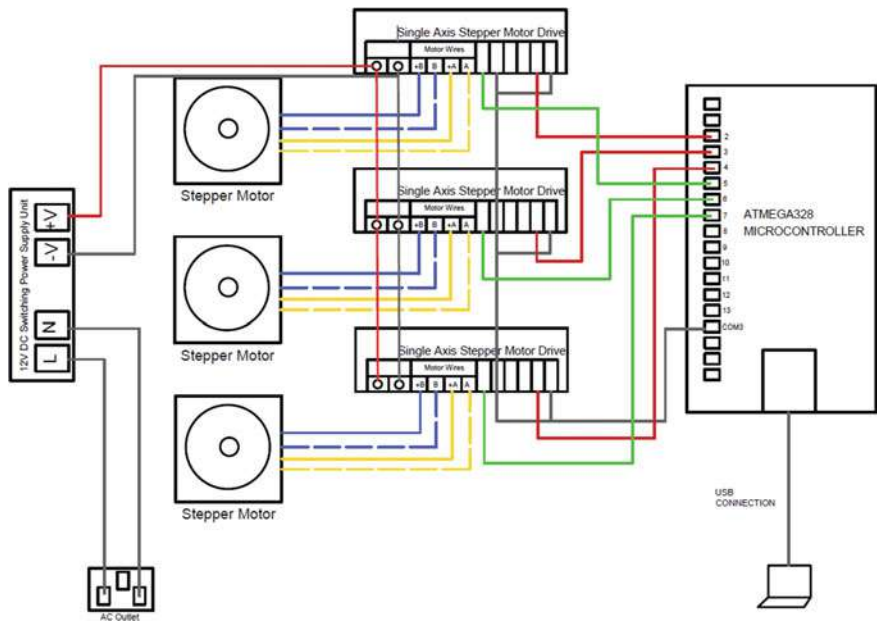


Fig. 52.7 Electronics circuit wiring

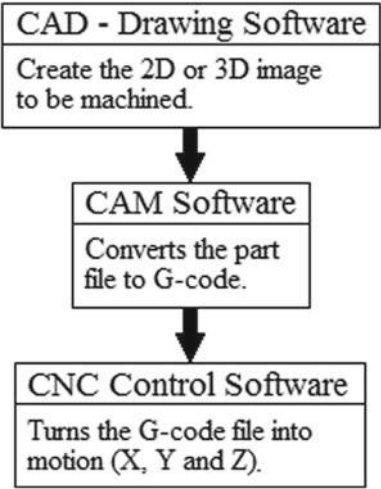


Fig. 52.8 Software tool chain

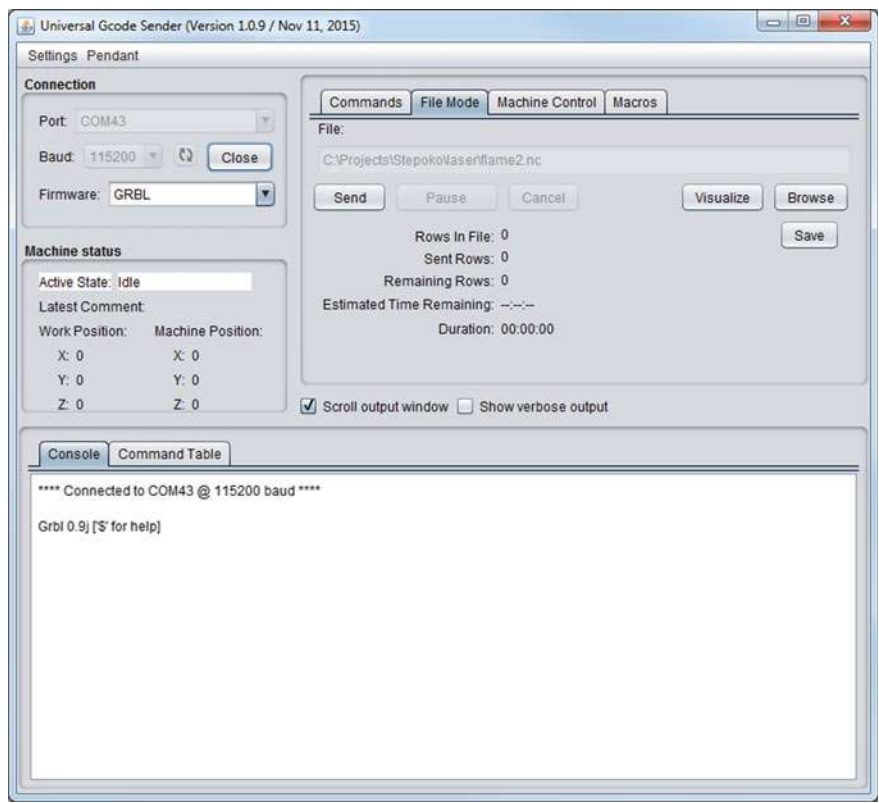
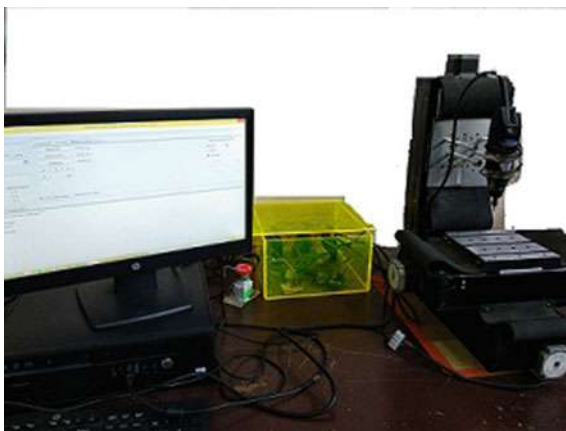


Fig. 52.9 CNC control software window

which is to be machined. Computer-aided manufacturing (CAM) software is used to generate tool path in machine readable code, i.e., G-Code. Generated G-code .nc file is send to microcontroller so that the correct motion of stepper motor is obtained. All used softwares are freely available.

Universal G-code Sender is used to send the .nc files to microcontroller board which contains G-codes to create the tool path. USB is used to communicate between computer and microcontroller and send G-code through it. G-code firmware is used and flashed on controller which is responsible for tool path control. Each line of .nc file is read by G-code firmware and it sends electronic signals as per the instruction to stepper motor through driver and the motion of axes is controlled. Universal G-code sender GUI is shown in Fig. 52.9.

Fig. 52.10 Developed CNC stage along with controller



52.3.7 Machine Integration

The development of key machine components discussed above was emphasised on the structure stability and micro-step motion precision and accuracy. The successful development of these sub-systems is important to meet design requirements. The developed desktop CNC stage along with controller hardware is shown in Fig. 52.10.

52.4 Machining Results

To check machine capability, cutting test was needed in order to test the processing capacity. Three machining tests had done by using the strategy, which was suitable for micromachining. Four design pockets of hexagonal, ellipse and triangular were tested. The features can be machined in the range of 1–2 mm. Aluminum plate of thickness 1 mm has been used.

The hexagonal shape pocket milling has been carried out by using flat end milling cutter of 300- μm diameter. Hexagonal shape pocket has edge length 800 μm and depth 300 μm . The cutter moves down for each depth in helix and cutter cuts work piece layer-by-layer along the Z-axis. The machined hexagonal shape pocket is as shown in Fig. 52.11.

By using same tool specification elliptical pocket was proceed with major axis of 2 mm, minor axis 1 mm and depth of 300 μm is shown in Fig. 52.12.

In Fig. 52.13 triangular pocket size of 1.5 mm base and 2 mm height, depth 300 μm . Square pocket size of 700 μm side and depth 300 μm has also been machined and shown in Fig. 52.14.

Fig. 52.11 Hexagonal pocket size of 800 μ m edge, depth 300 μ m

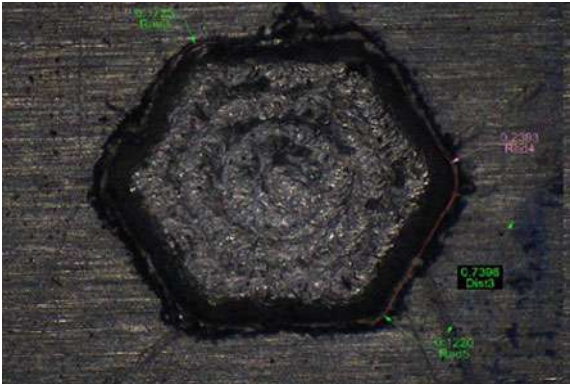


Fig. 52.12 Ellipse pocket size of 2 mm $\times$ 1 mm $\times$ 300 μ m

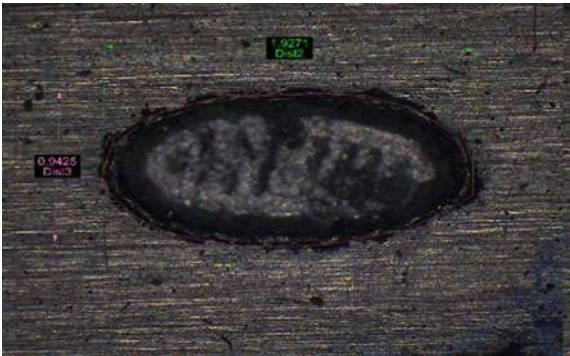


Fig. 52.13 Triangular pocket size of 1.5 mm $\times$ 2 mm $\times$ 300 μ m



Fig. 52.14 Square pocket
size of $700\text{ }\mu\text{m} \times 700\text{ }\mu\text{m} \times$
 $300\text{ }\mu\text{m}$

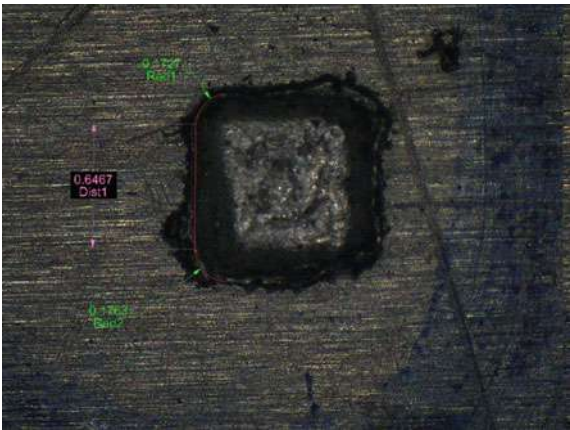


Table 52.3 Percentage
deviations of features

Features	Percentage of deviations (%)
Hexagonal pocket	0.025
Elliptical pocket	0.036
Triangular pocket	0.07
Square pocket	0.11

All the measurements were taken with the help of “Rapid-I” vision measurement system with magnification of 67 X.

The actual dimensions measured are presented in Table 52.3. From the measurements taken, it has been observed that there are some deviations compared with actual CAD model and machined features. The percentage of deviations is within the range of 0.07–0.11, which proves the capability of indigenously developed desktop CNC stage for intended application. Rounded corners are formed in Square and triangular pockets. This will be minimized by tool compensation factor. The percentages of deviation of micro features are presented in Table 52.3.

In Fig. 52.15. Spur gear machined in aluminium is shown with size of 3.60 mm pitch, 4.8 mm overall diameter and $500\text{ }\mu\text{m}$ is machined. Spur gear machined in titanium material is machined with size of 4.765 mm and $150\text{ }\mu\text{m}$ as shown in Fig. 52.16. The feed rate is 1500 mm/min and depth of cut is $50\text{ }\mu\text{m}$ (Fig. 52.17).

Fig. 52.15 Spur gear size of 4.765 mm overall diameter and 500 μm thickness machined in aluminum material



Fig. 52.16 Spur gear size of 4.765 mm overall diameter and 150 μm thickness machined in pure titanium material

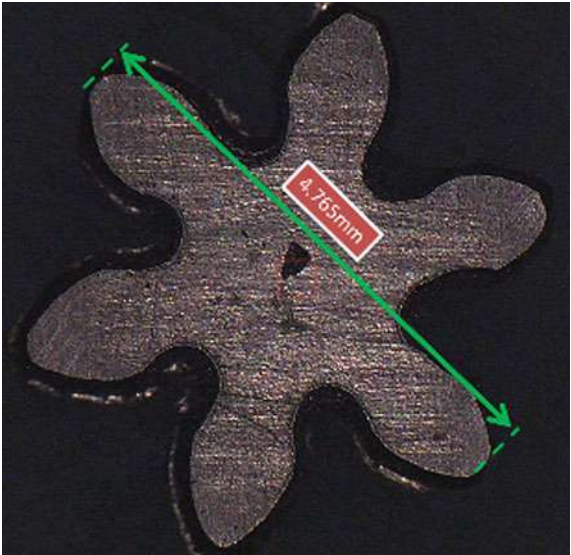
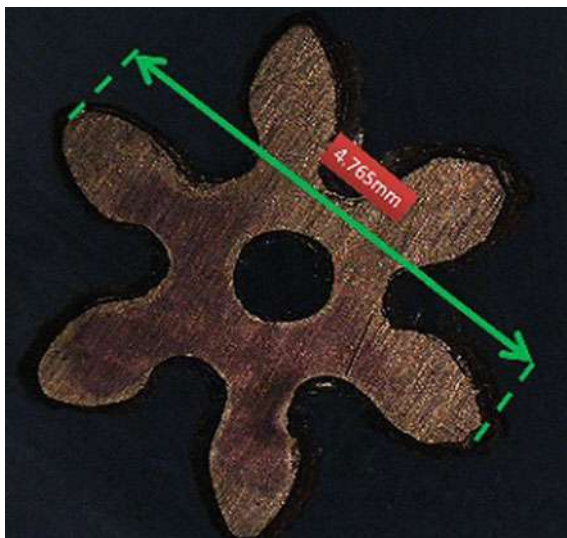


Fig. 52.17 Copper spur gear size of 4.765 mm overall diameter and 400 μm thickness



52.5 Conclusions

Developed desktop CNC stage has more stability and rigidity. So that various micro features can be machined precisely and accurately. Mechanical resolution of 1 μm is achieved by designing and developing precision drives and controller to obtain micro step by controlling stepper motors and drivers. Using Universal G-code sender software G-codes are sent to machine via controller. By using high-speed electrical spindle with speed of 20,000 rpm, various micro-featured are machined. Four design pockets in range of 1–2 mm are tested. Dimensional deviation is about 0.1%, deviation can be minimized by tool compensation. Also precise aluminum, titanium and copper gears are also machined. Micro-step resolution desktop CNC stage proves precision and accurate meso- and micro-featured machining. This indigenously developed low cost easy to use CNC stage has potential in machining of meso- and microscale-features and components.

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Chapter 53

Design and Development of a Pump-Driven Variable Buoyancy Engine (VBE) for Autonomous Underwater Vehicles/Gliders



B. K. Tiwari and R. Sharma

Abstract A change in buoyancy for an ‘Autonomous Underwater Vehicle/Glider(AUV/G)’ in conjunction with wings can be used to convert vertical motion to horizontal motion, improve propulsive efficiency and power consumption, design a compact shape, and increase the AUV/G’s endurance (range and duration of the operations). Thus, an efficient buoyancy control can extend the ocean-sampling missions from hours to weeks or months, and to thousands of kilometers of range. In this regard, we propose to develop vehicles that employ the ‘Pump-Driven Variable Buoyancy Engine (PDVBE)’ to have the capability to self-ballast and buoyancy control. Herein, we present the design and development of a computer-aided design (CAD) model for pump-driven variable buoyancy engine (VBE) for the AUVs/Gs. The proposed CAD model is modular in architecture and its various modules like design, simulation, control, and testing are scalable. Furthermore, the CAD model is integrated into the overall design of the AUVs/Gs. Finally, we present the various applications that are currently under investigation to demonstrate the applicability of our proposed design model.

Keywords CAD model · Pump-driven variable buoyancy engine · Buoyancy controller · AUV · AUG

53.1 Introduction

The ‘Underwater Vehicles (UVs)’ are compact in size and designed in the power critical mode. Because of the constrained space, limited size available on board and also to have the capability to remain operational for long time in the submerged conditions; the energy storage space gets limited. Additionally, an AUV/G can dive to and sit on the bottom of the lake/river/sea/ocean for long data-gathering missions (e.g. to acquire acoustic/video/chemical data) [1]. Although, both the buoyancy and

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_53

depth control are possible with the use of a thruster, the use of thruster has two disadvantages: A high energy consumption and restricted use because of the closeness to the bottom in case AUV/G is in grounding mode of operation to save power with thruster. Because of its enormous economic importance, the CAD has been a major driving force for research in engineering structures design, and it is important to note that the ship design and production community has been one of the earliest users of computer methods and CAD models and software. Herein, our aim is to design a CAD model for the VBE for AUV/G and the detailed analysis of the method of the buoyancy control and selection of the pumps for VBE. In our proposed CAD model, we consider four important design features in the VBE: (1) the capability to launch the vehicle/glider, (2) the capability to control operating depth, (3) the capability to hover, and (4) the capability of dynamic trim control. In practice, VBE can be installed at the forward end or aft end or both including any location in between too. The proposed CAD model is modular and it is roughly the same for any VBE of aft location or forward location. The important modules of our proposed CAD model are (1) reservoir module, (2) pump module, (3) plumbing module, and (4) control module.

The remaining of this paper is organized: Sect. 53.2 presents the geometrical description of the AUVs/s; Sect. 53.3 discusses the pump module; Sect. 53.4 reports the working method of the VBE in-detail; and Sect. 53.5 concludes the paper and identifies the future scope of research.

53.2 Geometrical Description of the AUVs/Gs

Since, normally the underwater operations are performed by completely submerged vehicles (i.e., AUVs/AUGs/ROVs, etc.), their shape and size influence the net buoyancy and other force acting on the vehicle, e.g., drag during the forward/heave motion of the vehicles, and the range and endurance, etc. Therefore, the shape and size of the vehicle can ‘only’ be decided based on their operational requirements. Herein, we focus on the design of VBE for AUV/AUGs and the type of VBE selected to suit the specific design of the AUV/AUG. Following [2], the fundamental design shape of AUV/G is shown in Fig. 53.1, and the relationships are

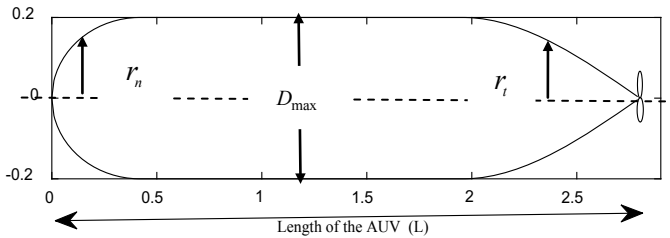


Fig. 53.1 Geometric fundamental design shape of AUV/G

$$r_n = \frac{D_{\max}}{2} \left\{ 1 - \left(\frac{L_n - x}{L_n} \right)^{n_n} \right\}^{1/n_n}; \quad \text{for } 0 \leq x \leq L_n, \quad (53.1)$$

$$r_m = \frac{D_{\max}}{2}; \quad \text{for } L_n \leq x \leq (L_n + L_m), \quad (53.2)$$

$$r_t = \frac{D_{\max}}{2} \left\{ 1 - \left(\frac{x - L_n - L_m}{L_t} \right)^{n_t} \right\}; \quad \text{for } (L_n + L_m) \leq x \leq L \quad (53.3)$$

where r_n is the nose radius, r_m is radius of the middle body, r_t is the tail radius of the AUVs, n_n is the nose-shape coefficient, n_t is the tail-shape coefficient, and L is the total length of the AUVs, which is the sum of the nose length (L_n), middle length (L_m), and tail length (L_t). The net buoyant force (ΔB) acting on the AUV/G is

$$\Delta B = W - B = (m - \rho \times \nabla)g \quad (53.4)$$

where W is the weight of vehicle B is the buoyancy of the vehicle, ∇ is the volume displaced by the vehicle, g is the gravitational acceleration, and ρ is the density of the fluid. If the net buoyant force $\Delta B = 0$, then the vehicle is neutrally buoyant, if $\Delta B < 0$ then it is positively buoyant, and if $\Delta B > 0$ then the vehicle is negatively buoyant. So, in order to control the depth based upon the operational requirements, we need to change ΔB , and this can be done either by changing the mass of the vehicle m or by changing the volume displaced ∇ by the vehicle. Our idea is to use the hydraulic pump to transfer the fluid from and to the ballast tanks/cylinders/spheres to control the ∇ and these transfers of the fluid control the total mass m of the vehicle.

53.3 Pump Module

Herein, we focus on this module because this is the component which plays an important role for the transfer of fluid. The pumps are mainly classified as hydraulic pumps (which can deal with water and oil) and pneumatic pumps (deal with compressed air). Here, we focus on the hydraulic pumps, which are further classified as dynamic pumps [i.e., non-positive displacement pump (N-PDP) and positive displacement pumps (PDPs)] and static pumps. The working principle of the dynamic pumps is based on the acceleration of the fluids and the PDPs are based on the use of mechanical forces to move the fluid through the system [3]. The PDPs include both the rotary as well as reciprocating pumps and almost all the N-PDPs are rotary types. Some of the important PDPs and N-PDPs are listed in Table 53.1A. Selection of the pump for VBE is based on the parameters such as the depth of operation of the UV which is integrated with the VBE through the hydrostatic pressure, types of fluid used for the buoyancy control (e.g. oil to change the displaced volume by the vehicle or water to change the overall weight of the UV), and the maximum flow rate of the fluid (i.e., rate of change of buoyancy) and the maximum buoyancy control capacity. For

Table 53.1 A—Some of the important PDPs and N-PDPs and B—Classification of the PDPs based upon their performance parameters

(A)		
Types of pumps	Positive displacement pump	Non-positive displacement pump
Reciprocating pumps	Piston pump (both axial and radial), single acting, double acting, plunger pump and diaphragm pump, and hand pump, etc.	Details are not available
Rotary pump	Gear pump (external or internal gear pump), lobe pump, vane pump, flexible impeller pump and peristaltic pump, gear pump (external or internal gear pump), flexible impeller pump and peristaltic pump, etc.	Dynamic pump (i.e., centrifugal and axial flow propeller pumps, etc.)
(B)		
Parameters	Reciprocating PDPs	Rotary PDPs
Capacity	Low	Low/Medium
Pressure	High	Low/Medium
Maximum pressure	689 bar	276 bar
Requires relief valve	Yes	Yes
Flow type	Constant	Constant
Characteristic	Pulsating	Smooth
Space needed	Requires larger space	Requires small space
Initial costs	High	Low
Maintenance costs	High	Low
Energy costs	Low	Low
Fluid recommended	Viscous liquids, dirty chemicals, tacky glue and adhesives, oil, and lubricating fluids	Optimum for viscous fluids require clean, clear, and non-abrasive

a VBE designed for the deepwater operations of the UVs, the high-pressure pumps can be used, i.e. the PDPs are the better choice since they are designed for the high pressure. Also, even for the low or medium depths of operation of the VBE, they are suitable because for them, it is easy to control the mass flow rate and hence rate of change of buoyancy. For the VBE which transfers the oil in order to control the buoyancy by changing the overall volume displaced, the pumps which designed for the viscous fluids can only be used. Furthermore, for the VBE designed to change the buoyancy by controlling the overall weight of the vehicle (i.e., by filling/ejecting the water in/out of the ballast tank); the self-priming and low-viscosity pumps are better choice. In our opinion, the PDPs are the best choice for VBE, since they consist of almost all the basic requirements of the VBE such as self-priming, able to produce

constant delivery for a given speed and compact size and light in the weight, etc. And, the efficiency of PDPs depends on the design, fluid viscosity, operating pressure, and speed. The reciprocating pumps are more efficient than the rotary pumps (i.e., efficiency of the precision gear pumps is less than or equal to 90% and for piston pumps normally it is greater than 90%).

The technical details of the PDPs, based upon their performance parameters are listed in Table 53.1B, and we observe that the reciprocating PDPs can be used for 689 bars, i.e. up to 6000 m water-depth in ocean. Herein, we analyze two reciprocating PDPs, firstly, rack–pinion arrangement with a syringe-type piston-operated ballast tank as shown in Fig. 53.2a and other one is single-acting reciprocating piston pump.

53.3.1 Syringe-Type Piston Pump

The rack–pinion arrangement with syringe-type piston-operated ballast tank is shown in Fig. 53.2a. In this, the cylinder itself acts as the ballast tank and the piston can move in or out of the cylinder by the rack and pinion arrangement (i.e., which acts as a linear actuator) or by means of some other electric linear actuator. The thrust required at the piston head (T_{ph}) is computed [4]:

$$T_{ph} = \Delta P \times A_{cs} = ((P_{atm} + \rho_{sw} \times g \times h) - P_{in}) \times \pi/4 \times d^2 \quad (53.5)$$

where P_{atm} is the atmospheric pressure, ρ_{sw} is the seawater density, g is the gravitational acceleration, h is the operating depth, and d is the diameter of the cylindrical ballast tank. From Eq. (53.5), it is clear that the thrust required at piston head is a function of the d inside which piston is moving and also on the pressure difference against which the piston is acting. However, the l is restricted with respect to the length of AUV/G. Hence, it has been observed that the syringe-type positive displacement piston pump can be used only for the low capacity of buoyancy change.

53.3.2 Single-Acting Reciprocating Piston Pump

A single-acting reciprocating piston pump is shown in Fig. 53.2b and in it, the inlet of the cylinder in piston that is actuating is connected with the storage tank and piston is connected to the crankshaft through the connecting rod, when the crankshaft is rotating for the first half, i.e. for the 180° . The piston moves outside of the cylinder and suction valve opens and discharge valve closes so that the fluid fills inside the cylinder. And during the other half rotation of the crankshaft, piston moves inside the cylinder during which discharge valve opens and suction valve remains closed. This pump can be used for the VBE of the large capacity of desired buoyancy changes. Since it has a ballast tank separately and that can store a large amount of the fluid. Later, this can be discharged at the high pressure. However, this type of pump consists of many

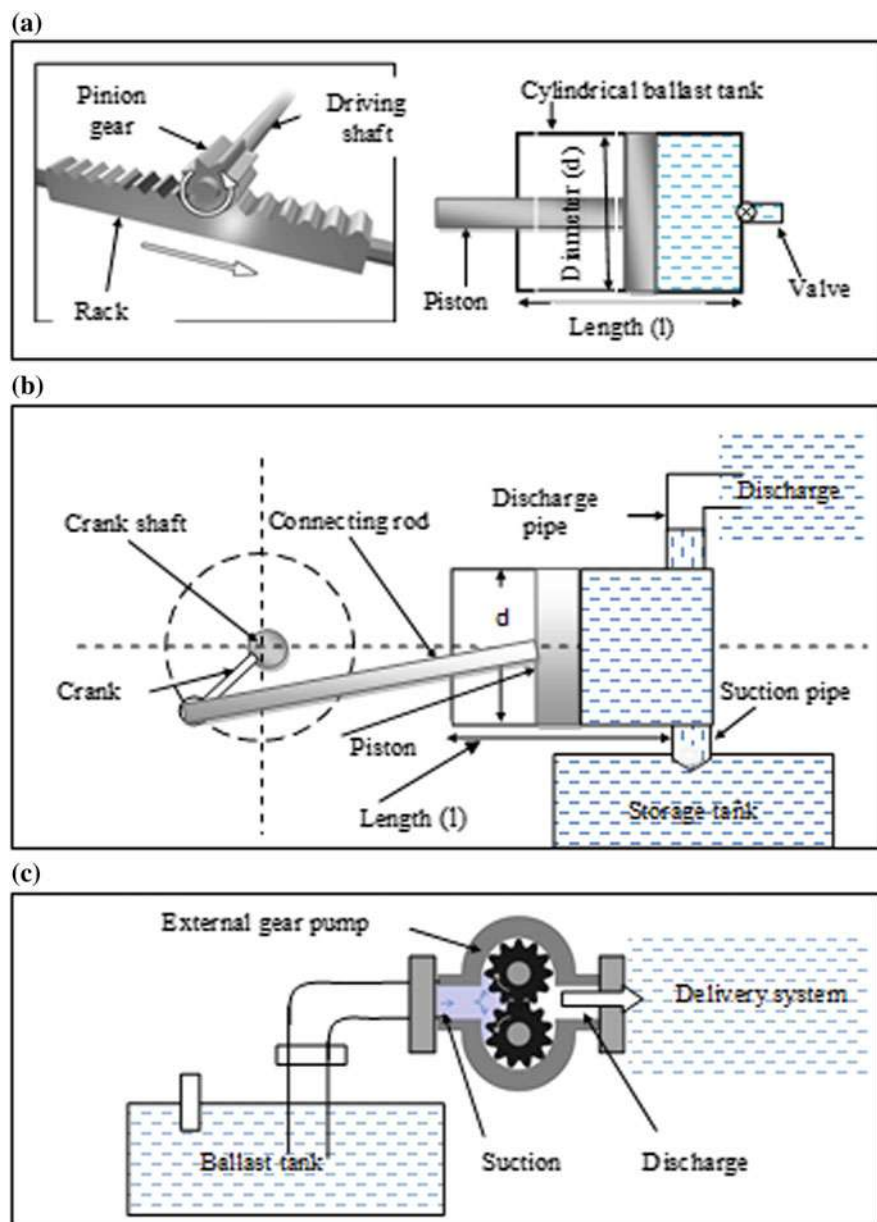


Fig. 53.2 **a** Rack-pinion arrangement with a syringe-type piston-operated ballast tank, **b** single-acting reciprocating piston pump, and **c** external gear rotary pump

component such piston, crank, connecting rod, and crankshaft, etc., and hence, it is a complicated mechanism, large in size, and pulsating transfer of fluid will results into the vibration [5]. Although, the piston pump is more efficient than rotary pump, the rotary pump is better especially for the size constraints of the VBE, because of their compact size and smooth flow conditions.

53.3.3 Rotary Pumps

It's important to reduce the weight-to-buoyancy capacity ratio of the VBE and perform efficiently at the required pressure (i.e., designed depth of operation of the VBE). And, rotary pumps (i.e., gear pump, the screw pump, and the moving vane pumps) are capable of transferring more fluid than the reciprocating pump of the same weight or same amount of the fluid with low weight. An external gear rotary pump is shown in Fig. 53.2c, but these type of pumps are required to be fitted with a relief valve in order to protect the pump and piping system, because of their ability of the fixed amount of fluid transfer irrespective of the external resistance.

For a rotary pump the mass flow rate is

$$\dot{m} = Q \times \rho_{\text{fluid}} \times N \quad (53.6)$$

where Q is the volumetric flow of the fluid per revolution (i.e., cc/rev), ρ_{fluid} is the density of the fluid (kg/m^3), and N is the speed of the motor shaft connected to the pump in rpm. Hence, in order to control the rate of buoyancy change (i.e., for a designed rotary pump Q is fixed and for the same fluid ρ_{fluid} is also constant), we need to control the speed of the motor. The performance parameters of the 'liquiflo' H-series model (i.e., H1F) gear pump which can be used for the fluids such as water (i.e., at 1 cP viscosity) and oil (i.e., at 100 cP viscosity) have been utilized in our design from [6].

53.4 Working Method of the VBE

The schematic conceptual diagram of the proposed CAD model for the VBE is shown in Fig. 53.3. A hydraulic system is connected to the motor, speed of the motor is controlled by the controller, and power required for the motor is supplied from the power storage system. The pressure management system (i.e., relief valves, etc.) is connected with the hydraulic pump and the pressure sensor gives the signal for operating depth. Hence, we can control the flow between the ballast tank and seawater based upon our requirements of the buoyancy, i.e. positive, negative or neutral in order to raise, sink, or to hover. The conceptual diagram of the control module is shown in Fig. 53.4. Our proposed CAD model for the VBE is applicable to the AUG, AUV: single-hull and multi-hull, as shown in Fig. 53.5.

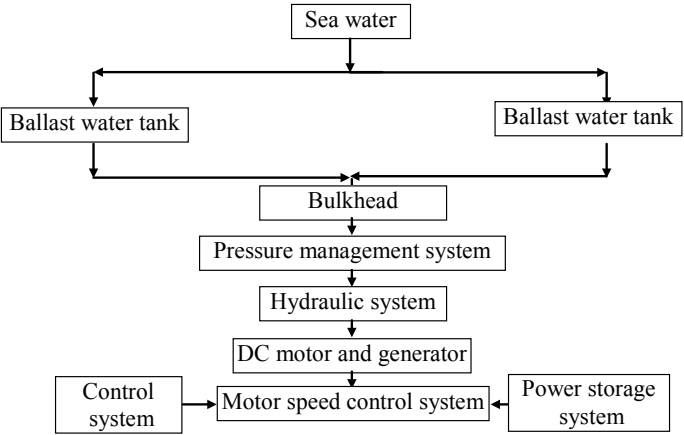


Fig. 53.3 Conceptual diagram of the proposed CAD model for VBE

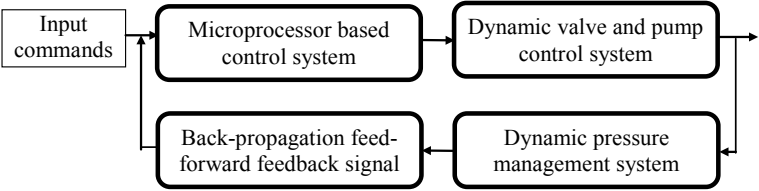


Fig. 53.4 Conceptual diagram of the control module

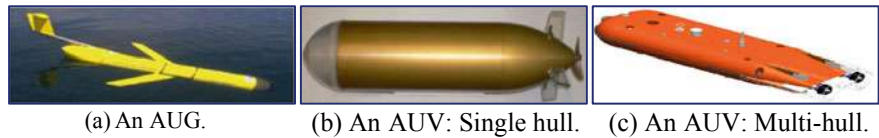


Fig. 53.5 Applicable ranges of our proposed CAD model for VBE

53.5 Conclusions

This paper has presented a computer-aided design (CAD) model for pump-driven variable buoyancy engine (VBE) for autonomous underwater vehicles/gliders. The proposed model has shown satisfactory results with the following features: (1) the capability to launch the vehicle/glider, (2) the capability to control operating depth, (3) capability to hover, and (4) the capability of dynamic trim control. The dynamic trim control maintained the AUV in trim under a variety of operating conditions freeing up control authority for maneuvering and hence increasing vehicle’s speed and enhancing depth control.

Acknowledgements This research was supported by the internal research grants of IIT Madras through research scheme: OE14D212 and from Marine Systems Panel, NRB, India, via a sponsored project: NRB-263/MAR/12-13.

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Chapter 54

Application of Value Analysis and Value Engineering for Cost Reduction of Global Pumping Unit



Aniket Bhosle , Avinash Sah and D. K. Shinde

Abstract At present, the competition in Indian retail market for fuel dispensers is leading to a constant battle for an adequate market share. The Indian retail market for fuel dispensers is favouring the lowest bidder. Recent tenders have seen aggressive bidding from all the bidders, which leads to price war between different organizations. Value analysis and value engineering are the concepts, which help in understanding the correct path to proceed with improvements, which will ultimately result in higher customer satisfaction or we can say increased market shares. A case study of Global Pumping Unit is taken for the research, in which the design of components with removal of excess material and change in material have significantly reduced cost of whole pumping unit using VAVE methodology. In this study, a total of three components namely rotor, pump body and stator housing were taken to apply VAVE which resulted in 30, 4 and 10% of cost reduction.

Keywords Value analysis (VA) · Value engineering (VE) · VAVE · Global pumping unit (GPU) · Cost reduction

54.1 Introduction

In the era of today's competitive world, every organization is investing more on increasing the customer satisfaction of their product or service. Customer mainly wants the use and aesthetic functions. But sometimes to fulfil the customer wants, the possibility of increase in cost for required changes in performance/functionality coupled with rise in commodity prices is more while walking on this path. Hence, it is important from an organization point of view to not only reduce the costs incurred but also constantly introduce new innovations at all levels and across all functions in a company. Miles [1], the originator of Value Analysis and Engineering Technology in

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_54

his book writes about why so much unnecessary cost exists in everything we do. The value methodology commonly applied under the names value analysis (VA), value engineering (VE) and value management (VM) [2], helps us to identify and eliminate these unnecessary costs. VE is an organized effort directed at analysing the function of goods and services for the purposes of achieving basic functions at the lowest overall cost, consistent with achieving essential characteristics [3] while VA as defined by the Society of American Value Engineers International (SAVE International), is “the systematic application of recognized techniques by multi-disciplined team(s) that identifies the function of a product or service; establishes a worth for that function; generates alternatives through the use of creative thinking; and provide the needed functions, reliably at the lowest overall call” [4]. In this study, we have applied VAVE techniques for a few components of Global Pumping Unit to reduce the cost so that the Product can easily penetrate the market needs during its life cycle.

54.2 VAVE Application Over Product Life Cycle

Value Analysis and Value Engineering are sequential steps adopted to increase the value of the product or service. There can be a number of ways to implement a value analysis/value engineering where different changes are made in various key drivers of the value (say performance, cost, customer satisfaction, resources, etc.). VAVE is generally taken as an internal project in an organization, specifically, by cross functional departments which focuses more on continuous improvement or value addition. However, it is very important for an organization to know the difference between value analysis and value engineering. Answers to questions such as ‘when’ and ‘how’ will make implementation of VAVE more efficient. For any certain project, the VA and VE studies are applied by a multi-disciplinary team to improve its value. SAVE International sets 6 sequential phases for performing a successful VE study [5]. VA and VE can give extremely good results if implemented at correct time over the lifespan of product and with systematic Job plan. It is very important to understand various key drivers of value and the intensity of a small change on the value. VAVE projects in an organization can be adopted during any stage of the project development cycle; although, the greatest benefit and resource savings can be typically achieved by applying VA and VE in the appropriate phases of development cycle. To get detailed answer of ‘when to implement VA/VE?’ we need to analyse more on product life cycle and understand various phases.

Initially, the product life cycle concept is centred on the need to produce a coherent framework that could account for the relative success or failure of an individual product introduced into the market, when best to change strategies such as pricing or product manufacture, and determining when product should be discontinued [6]. Product Life Cycle is basically divided into four main phases viz: Introduction phase, growth phase, maturity and saturation phase and decline phase. As time passes product sales increase at first (Introduction phase), then more quickly (growth phase), then once again more slowly (maturity and saturation phase), and finally decrease

(decline phase). Nowadays, the focus is more on increasing the length of maturity and saturation phase. The retention power of a product/service depends on the continuous improvement in its value. Hence, at a point of time in product life cycle, it becomes necessity for an organization to identify and eliminate these unnecessary costs otherwise product will be phased out before its expected life cycle.

Mathematically, value can be defined as ratio of performance or function or customer satisfaction with cost or resources incurred to develop such level of product/service. And hence, they are the key drivers for changes in the value.

$$\text{Value} = \frac{\text{Performance/Function/Customer Satisfaction}}{\text{Cost/Resources}}$$

Value engineering focuses more on various engineering modes under the development of a new product. It may deal with design, functionality, aesthetics, etc. of a product or service. The value engineering helps to make the end result more efficient in comparison to cost without changing quality aspects as well as business need during the development phase (introduction phase) as shown in Fig. 54.1. As in this phase, making changes to design/functionality/aesthetics will be easier and can be done at least cost. Value engineering generally involves various ways such as:

- a. Part design configuration change [7]
- b. Change in engineering process/methodology incorporated.

Value analysis, on the other hand gives best results during the growth, maturity and decline stage as shown in Fig. 54.1. Value analysis focuses more on reducing the cost of given product or service by adopting changes (say in business process or material composition). As the business has reached in the operations, it inculcates higher cost in changing initial designs or functionalities and hence it is advisable to go for improvement through value analysis. This in general, involves ways such as:

- a. Source/Supplier substitute
- b. Material substitute [7]

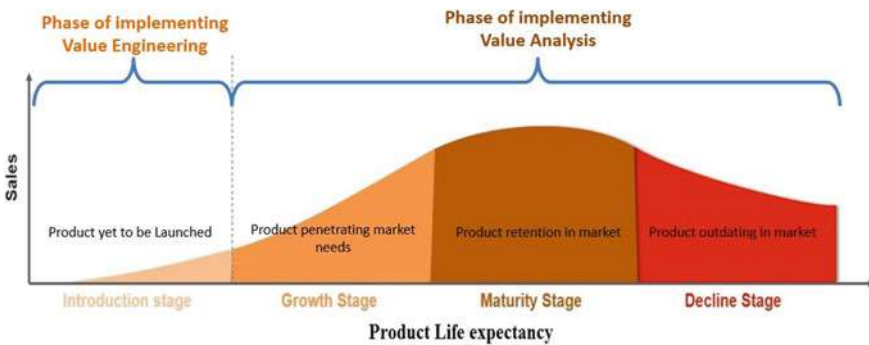


Fig. 54.1 Product/service lifespan [7, 8]

- c. Removal of non-performing assets
- d. Elimination of under-utilized human resources.

It can be easily observed from Fig. 54.1 about the various phases which a product/service undergoes during its life. VAVE must be applied after complete involvement of owner(s) and consultant(s) decision-making, to achieve optimal results. Hence, optimum results can be expected when resources are set aside for VAVE early in the design process, focusing on owner and consultant impact [7]. It is advisable to go for VAVE implementation as early as possible to ensure potential savings before commitment of funds into it, approval of systems, services or designs. Organisations look at two major things while implementing VAVE for their products, first is the investment required to implement any change and second is resistance to change. As shown in Fig. 54.2, the two lines are:

Acceptance Line—represents investment required to implement any change
Resistance line—represents resistance to change.

From Fig. 54.2, it can be observed that as time increases the cost of investment for any change (acceptance line) reduces. It is highest during the Introduction phase, whereas the resistance to change (resistance line) increases as the product moves towards decline phase. The area between these two lines shows the savings from VE and, gradually, this savings decreases as product/service gets older. It is always good to adopt VE at early stage to gain maximum savings, whereas VA can result in savings but is less than as compared to VE at an early stage.

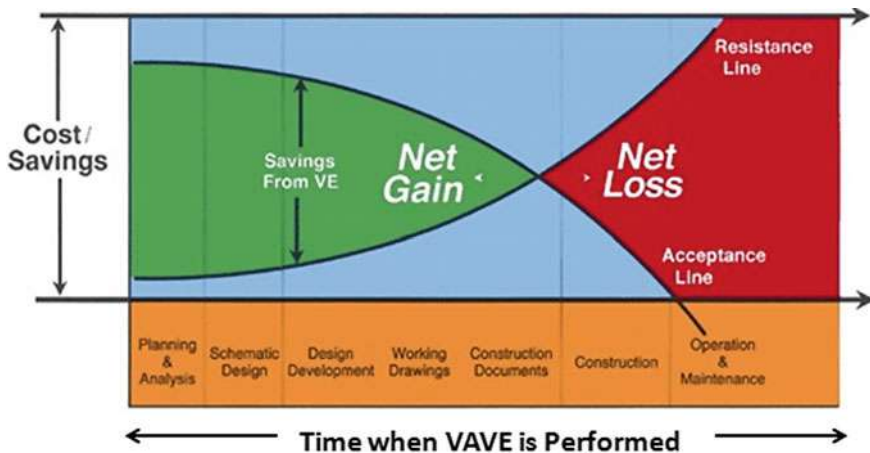


Fig. 54.2 VAVE over product lifespan [8]

54.3 Case Study of GPU

In this paper we have discussed a case study of Global Pumping Unit (GPU), which is manufactured at Gilbarco Veeder Root (GVR) India Pvt. Ltd., Coimbatore, Tamil Nadu (India). They also manufacture variety of fuel dispensers. During fuelling operation, the fuel line is connected to a pipe that runs to the bottom of the fuel storage tank. Fuel is drawn up by suction (vacuum creation), but first passes through a filter that protects the mechanism. The pump then pressurizes the fuel to 3.5 bar (maximum) and pushes it through a gas separator, which gets vented to atmosphere [9]. GPU is selected for case study as it is most popular and relatively fast-moving product (Fig. 54.3).

GPU is GVR pumping unit launched into the market in 2014 which is currently in growth stage of product life cycle and is facing a cost challenge from the competitors. To address a threat posed by competitors in terms of penetrating the market needs, we have decided to apply VAVE techniques for GPU.

A cost breakup for all the parts of GPU is shown in Fig. 54.4. We have applied value analysis technique for cost reduction on following components:

- A. Rotor
- B. Pump body
- C. Stator housing

The motivation for selection of pump body and rotor can be easily understood by observing Fig. 54.4. They have been selected, as both of them holds major cost of GPU and carries good scope of improvements in their value. However, after analyzing 'top cover' for VE, it has been observed that savings per part were only Rs. 5–10 which is very less compared to its original cost. Further, 'blade' and 'main float valve assembly' are very critical in design to make changes and hence were not selected

Fig. 54.3 Global Pumping Unit



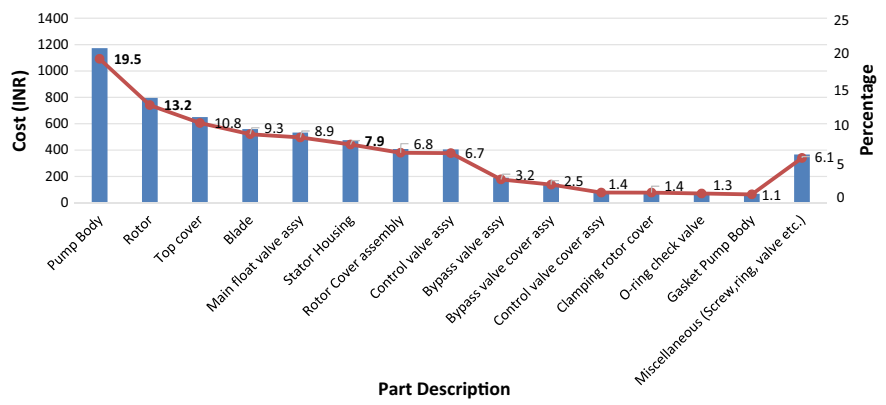


Fig. 54.4 Cost breakup for various parts of GPU

for VAVE. Stator housing has shown good scope of improvement in its design by the engineering department. This made us to select this part for applying VAVE.

54.3.1 Rotor

Data Collection and Analysis:

It is observed that the unnecessary increase in cost is due to extra machining efforts required to achieve final product. Therefore by application of VA technique, design modification for rotor is suggested. As the rotor revolves, the vanes trap fluid between the rotor and the casing draw the fluid through the pump. Currently the rotor casting is made as a solid cylindrical part. There are ten slots for vane blades which need to be milled with two passes for each slot (slot width is 4.70 mm). The proposal is to provide the slots in the casting with width of 4.00 mm to broach these slots to be finished size (4.70 mm) individually. This will completely eliminate the milling operation, and then broaching operation will provide a much better slot finish (Fig. 54.5).

VE Job Plan: Refer Table 54.1.
Achievement: With the same function the rotor full casting as replaced by Rotor—as cast slots.
Results of VE Job Plan are

$$\begin{aligned} \text{Tool development charges/piece} &= \frac{\text{Tool development cost}}{\text{No.of considered years} * \text{No.of pieces per year}} \\ &= \frac{333,774}{1 * 7947} \\ &= \text{Rs. 42} \\ \text{Total cost per piece} &= \text{Tool Development Charges/piece} + \text{Material Cost} \end{aligned}$$

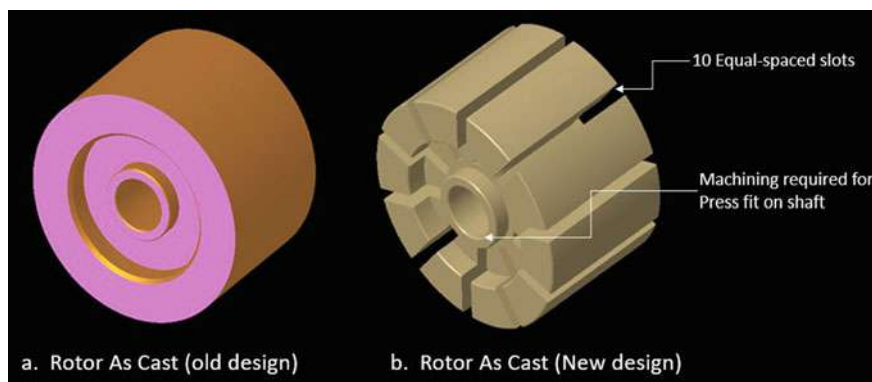


Fig. 54.5 Design of rotor

$$= 42 + 243$$

$$= \text{Rs. } 285$$

$$\text{Net savings} = 403 - 285$$

$$= \text{Rs. } 118$$

$$\text{Percentage savings in cost} = 118/403$$

$$= 30\%$$

54.3.2 Pump Body

Data Collection and Analysis:

In GPU, there are two major parts top cover and pump body. Pump body is the major contributor in the overall cost of GPU. Pump body houses Inlet check valve, Vane rotor cartridge, bypass valve and the control valve. Currently, it is made up of aluminium die cast A413 and machined as necessary. To achieve the desired cost savings, we are changing the pump body material from A413 to A380 (Fig. 54.6).

VE Job Plan: Refer Table 54.1.

Achievement:

With the same function, the material of the pump body is changed.

Results of VE Job Plan are

Current Material : Aluminum die cast A413

Proposed Material : Aluminum die cast A380

Mass of the pump body = 3.8 kg

Material cost per kg for A413 = Rs. 270

Table 54.1 VE Job Plan

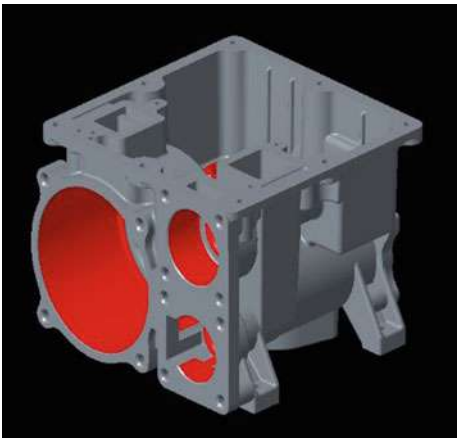
Phase	VE Job Plan	Rotor	Pump Body	Stator Housing
INFORMATION	What is it?	Rotor (Fig. 5)	Pump body (Fig. 6)	Stator housing (Fig. 7)
	Current Material?	FG 260 Cast Iron	Al Die Cast Alloy A413	FG 260 Cast Iron
	Proposed Material?	FG 260 Cast Iron	Al Die Cast Alloy A413	FG 260 Cast Iron
	What does it cost?	Rs. 403/-	Rs. 1042/-	Rs. 327/-
	How many parts?	One	One	One
	What does it do? (List all Functions)	1) Generate Suction 2) Lock the vane blades 3) Hold the pump shaft	1) Contains Rotor Stator Housing, rotor, shaft etc.	1) Hold the rotor and Vane blade assembly 2) Guide the Shaft
	Current usage quantity?	7947 per year	7947 per year	7947 per year
	What is the Forecast?	For 1 year	For 1 year	For 1 year
EVALUATION	What is the primary function?	Generate Suction	Contains Rotor Stator Housing, rotor, shaft etc.	Hold the rotor and Vane blade assembly
	What else will do	Rotor-As cast slots	Pump body-Weight reduction	Stator housing-Weight reduction
	What will that cost	Rs. 285/-	Rs. 1000/-	Rs. 295/-
	Which alternative way show difference between Cost & Use Value?	Greatest Offered by: Rotor-As cast slots	Greatest Offered by: Pump body-Weight reduction	Greatest Offered by: Stator housing-Weight reduction
PLAN	Which ideas are to be developed?	First Choice: Rotor-As cast slots	First Choice: Pump body-Weight reduction	First Choice: Stator housing-Weight reduction

(continued)

Table 54.1 (continued)

Phase	VE Job Plan	Rotor	Pump Body	Stator Housing
	What other functions (work or sell) and specification features must be incorporated?	<i>Function</i> -Same as existing	<i>Function</i> -Same as existing	<i>Function</i> -Same as existing
		<i>No. of parts</i> -No change	<i>No. of parts</i> -No change	<i>No. of parts</i> -No change
		<i>Space required</i> -Same	<i>Space required</i> -Same	<i>Space required</i> -Same
		<i>Durability</i> - Certainly	<i>Durability</i> - Certainly	<i>Durability</i> - Certainly
		<i>Aesthetic</i> - No Change	<i>Aesthetic</i> - No Change	<i>Aesthetic</i> -No Change
SELLING	What do we need to sell our ideas and forestall road-blocks?	Supplier discussion for feasibility study	Detail feasibility on critical wall thickness	Supplier discussion for feasibility study
		Feasibility study of Manufacturing engineering with supplier for stock requirements and required tolerances	Feasibility study of Manufacturing engineering with supplier for stock requirements and required tolerances	Feasibility study of Manufacturing engineering with supplier for stock requirements and required tolerances

Fig. 54.6 Design of pump body



Material cost per kg for A380 = Rs. 250

$$\begin{aligned} \text{Tool development charges/piece} &= \frac{\text{Tool development cost}}{\text{No.of considered years} * \text{No.of pieces per year}} \\ &= \frac{858,276}{1 * 7947} \\ &= \text{Rs. 108} \end{aligned}$$

$$\begin{aligned} \text{Total cost per piece} &= \text{Tool Development Charges/piece} + \text{Material Cost} \\ &= 108 + 892 \\ &= \text{Rs. 1000} \end{aligned}$$

$$\begin{aligned} \text{Net savings} &= 1042 - 1000 \\ &= \text{Rs. 42} \end{aligned}$$

$$\begin{aligned} \text{Percentage savings in cost} &= 42/1042 \\ &= 4\% \end{aligned}$$

54.3.3 Stator Housing

Data Collection and Analysis:

Stator housing is used to hold the rotor and vane blade assembly. The radial wall thickness of backside boss is reduced from 12 to 7.5 mm. Because of this change, the weight of the casting got reduced from 3.44 to 3.119 kg (Fig. 54.7).

VE Job Plan: Refer Table 54.1.

Achievement:

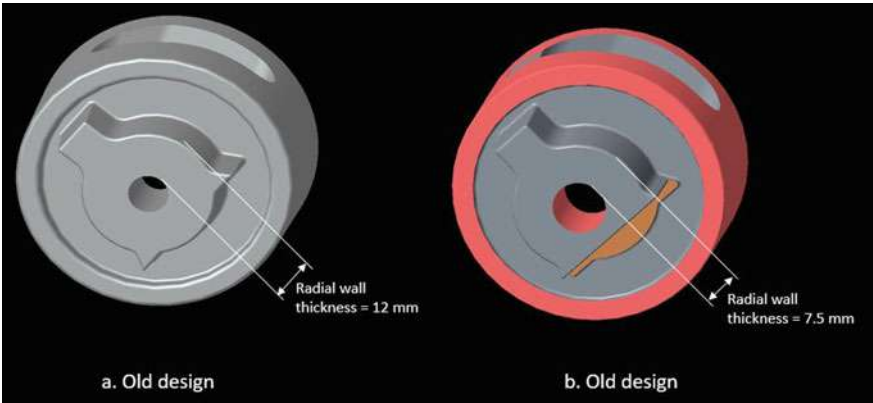


Fig. 54.7 Design of stator housing

With the same function, the weight of the stator housing is reduced.
Results of VE Job Plan are

$$\begin{aligned}\text{Tool development charges/piece} &= \frac{\text{Tool development cost}}{\text{No. of considered years} * \text{No. of pieces per year}} \\ &= \frac{254,304}{1 * 7947} \\ &= \text{Rs. 32}\end{aligned}$$

$$\begin{aligned}\text{Total cost per piece} &= \text{Tool Development Charges/piece} + \text{Material Cost} \\ &= 32 + 263 \\ &= \text{Rs. 295}\end{aligned}$$

$$\begin{aligned}\text{Net savings} &= 327 - 295 \\ &= \text{Rs. 32}\end{aligned}$$

$$\begin{aligned}\text{Percentage savings in cost} &= 32/327 \\ &= 10\%\end{aligned}$$

Combined results:

In general there are six methods by which value of the product can be increased viz. Part design configuration change, change in engineering process, source/supplier substitute, material substitute, removal of non-performing assets and elimination of under-utilized human resources. In this study, we have seen change in engineering process of rotor resulted in cost reductions of 30%. Also material substitute of pump body results in 4% of cost reduction and removal of non-performing assets of stator housing results in 10% of cost reduction (Fig. 54.8).

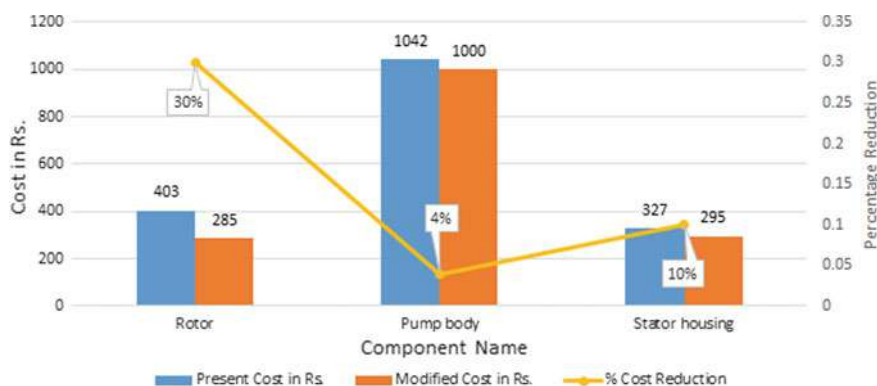


Fig. 54.8 Results of various VAVE techniques applied on selected parts

54.4 Conclusion

After successful implementation of VAVE methodology over the products, they can now survive more on their maturity stage of product life cycle. It increases the life of product and hence the profitability of the organization. The VAVE projects require a team of active members who are desperate to work with innovative ideas which comes by thinking out of the box. Here we can see the cost reduction for pump body is very less, then also organizations look for such kind of opportunities because in the era of competitive world, a little cost reduction can help product to capture larger area of market and hence chances of winning the tenders increases. Organization must not see VAVE project as a destinations to reach and stop. However, organizations should motivate their employees to take efforts and be ready with a number of upcoming projects which can increase the value of their products/services. Small change in cost may not affect much at current scenario but will always help organizations to build the brand value and customer's trust. Power of customer retention increases simultaneously, and thus results in growth of company.

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Chapter 55

Development of Prototype Variable Geometry In-Pipe Robot for Reconfigurable Applications



S. Pon Vignesh Pappu , M. Ajin and Gopal Satheesh Kumar

Abstract From smart cities to cyberville, infrastructure development of the world has been growing in leaps, off late. While all the domains have shown sweeping progress, handling of waste leaves much to be desired even with all the efforts of the researchers in this domain. The methodology adopted to design reconfigurable robot for moving inside the sewage pipeline that has the capacity to remove the blockage is discussed in this paper including the structural and kinematic analysis of the robots that were done for the design. Many configurations were considered and simulated to recognize the improved characteristics of each model and finally a prototype is fabricated for experimentation. Reconfigurability is achieved through implementation of ultrasound sensors since the robot is to be tested in simulated environment that does not include the wet conditions. Five-layered fuzzy-based integrated control architecture developed for achieving the desired motion and reconfiguration is also presented. The membership function and the corresponding output are also presented for the case of varying pipe diameters of underground sewage pipelines.

Keywords Clog removal · Sewage pipe · Static analysis · Structural analysis · Robot · Reconfigurability · In-pipe robot · Variable geometry

55.1 Introduction

Sustainable development is the key for all round growth of any country. Technology has helped not just the top-notched people, but even at the grassroots level. This is an effect of adding a human touch to the engineered solutions like safety equipments. Sometimes, these sanitation workers enter manholes to clear the clogs which might happen in the pit or inside the connecting pipelines. Sometimes, the workers are exposed continuously to human waste and hazardous substances that take a heavy toll

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_55

on the health of the workers resulting in many premature deaths [1]. Making available appropriate technological solutions is the only answer to this problem. Thus, it is clear that the need to enter manholes still holds. It also leads to the identification of the need that the technological support that goes along is to be improved upon and hence this work. A reconfigurable robot is presented as a solution for in-pipe line applications. Variable geometry aspect of the reconfigurable robot is included in this work.

55.2 Literature Survey

Human–robot interface (HRI) is a fertile research area where constant improvements are being sought and implemented. Robots have been time and again used for known/unknown terrain navigation and path planning applications and have met success of varying degrees [2]. This problem is viewed as a known terrain with uncertain operating conditions for which numerous algorithms have been proposed [3].

Inspection by motor-driven camera platform is slow and costly and suffers from the many disadvantages [4]. As a solution, a complete autonomous sewer robot named MAKRO was developed with a vision to make life-long sewer robots, i.e., robots which permanently work in the sewer with docking stations, preferably in manholes. Other robots like MINBOT-I and MINBOT-II [5] were developed for search-and-rescue purposes in underground coal mine applications. They are made to be water and explosion proof, since they encounter problems like corrosive & explosive environment, limited ingress and so on. Both these research works set goals for future researchers. Krishna Prasad and Bala Karthikeyan [6] later developed a prototype system for cleaning and removing blockage of large sewer pipes which is tested in real sewer pipeline during normal sewer operation. Manipulators with various designs like DSAM would be attempted to work for its dexterity. In case there is a need, the backward-motion control [7] could be adopted for the developed robot without trailer.

Among the research happening in the world and Indian scenario [8], an appropriate solution that is ready and easy to implement is unavailable. Hence, the objective of the research work is to design, develop, and analyze a reconfigurable robot for scavenging applications inside the manholes and also for the connecting underground pipelines.

55.3 Design, Structural, and Dynamic Analysis

Three designs were proposed initially and studied to optimize the performance of the robot for better traction control. The design shown in Fig. 55.1b has a hollow passage which allows sewage water to pass through with minimum resistance. This allows for the robot to demand minimum force requirement for traction and holding torque. All the three designs underwent structural analyses.

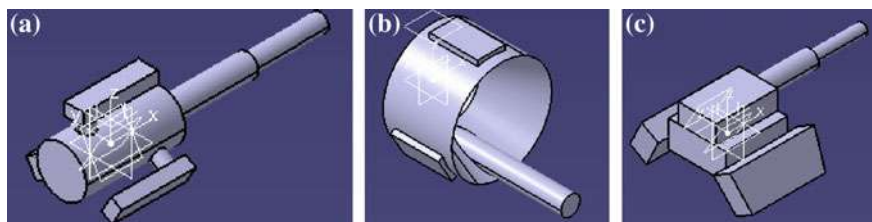


Fig. 55.1 Initially proposed designs for the robot

Based on various preliminary analyses, the final robot design is decided to be based on a cuboidal-shell shape where the wheels are placed along the diagonals of the two opposite faces of the cuboid shell. Finally, the spider configuration was tested numerically for unwanted forces. The leg lengths of the spider were made reconfigurable to accommodate variable geometry approach through the legs [9].

The wheels are distributed at an angle of 90° as shown in Fig. 55.2. Length of the rotating blade is nearly equal to the diameter of the pipeline ($<$ diameter of the pipeline). The sizes of some of the important elements which make up the robot structure are decided based on the design constraints of the sewer pipelines. Sewage robot is tested in a dry sewage pipe operated using RF mode. The robot moves without any problem, and the movement inside the pipe is observed with the camera. Initially, the robot is operated in forward direction and reverse direction. The ultrasonic sensor senses the obstacle distance and produces warning. Once the obstacle is detected, the rotor starts to rotate. Initially, the high torque motor is actuated and transmitted to the wheel via square gear. So, the power at the wheel will be more. The wheels are adjusted for the inner diameter of the sewage pipe. The load must be distributed equally to all the wheels both on the upper set and bottom set to get a steady motion. The rotor in the front crushes the obstacles and the brushes attached at the bottom are designed to sweep out the crushed particles or any other debris to be removed.

The unique design of the robot when checked for rigidity should not deform due to twisting moments while moving inside the sewer pipe. Torsional rigidity analysis

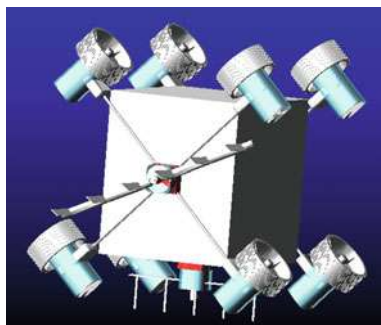
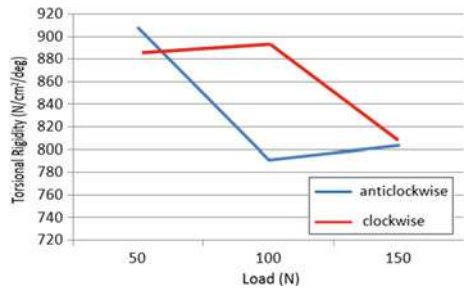


Fig. 55.2 Design with cross-spider configuration

Fig. 55.3 Torsional rigidity versus load



is the product of modulus of rigidity (C) and polar moment of inertia of the shaft (J). It is performed with both clockwise and anticlockwise moments, for three forces of magnitudes 50, 100, and 150 N acting on the structure. And, torsional rigidity is also defined as the torque required to produce a twist of one radian per unit length of the shaft. Simulating the robot shell, it is identified that 100 N is the critical load at which torsional rigidity reaches the maximum value. The values of torsional rigidity obtained from the analyses are in conformance with the design parameters as shown in Fig. 55.3.

Then, the model is fabricated and tested in practical cases for its working and stability analysis. The motion of the wheels and path of travel is observed. The range of RF receiver reduces in a closed environment such as a sewer pipeline, so it requires good wireless control system to make the robot operate in deeper operating conditions. Torque contributed by each of the eight motors is reduced due to traction. This is because of the rough and rugged surface of the sewer pipe. The torque of the front motor used for actuating the rotor blade should be more when the obstacles are tight or closely packed.

Static and kinematic simulations ran successfully after creating a contact between sewage pipe and wheels and the results were discussed in. The sewer robot is expected to be used in pipes of diameter 400 mm to clear clogs which are composed of soil, leaves or small plants, cloths and papers. Hence, the result ensures that the traction analysis to be done in later stages would provide valid results for each wheel. Even with bending stiffness analysis, it is evident for both downward and upward acting forces, and the bending stiffness remains the same for all three loads, i.e., 50, 100, and 150 N.

The simulation result obtained for the downward acting forces over the surface shows that the critical bending occurs at about 1000 N, which is insignificant in the case of the robot. It is assumed that to clear the clog the robot requires a force of atleast 50 N to push and remove it. In order to control and coordinate the activities of the robot, a control algorithm is developed and is discussed in detail in the next section.

55.4 Fuzzy-Based Control of Robot

The layered control architecture based on state-space representation was presented in [10], and as an extension, a Fuzzy-logic-based control algorithm is proposed to unite the layered approach depending on the task allocated to the robot as shown in Fig. 55.4.

The control architecture is multi-layered that includes the functional layer for individual robot and a supervisory layer for the central station for designating the robots for task allocation which is common in swarm and distributed robotics. A membership function (MF) is a curve that defines how each point in the input space is mapped to a membership value. Two ultrasound signal readings (U1 and U2) are taken as inputs and four pipe diameters ranges are taken for validating the results for fixing the membership functions. The following Fig. 55.5 shows the output. It correlates the working of the developed algorithm for the ultrasound sensor reading to the corresponding diameter for which the mechanisms have to open to engage the walls of the pipe. This forms the lowest layer of the control architecture for traction control.

Figures 55.6 and 55.7 show the sample fuzzy graph output obtained in comparison with pipeline diameter and ultrasonic signals. The results show concurrence of the results obtained from the earlier figures.

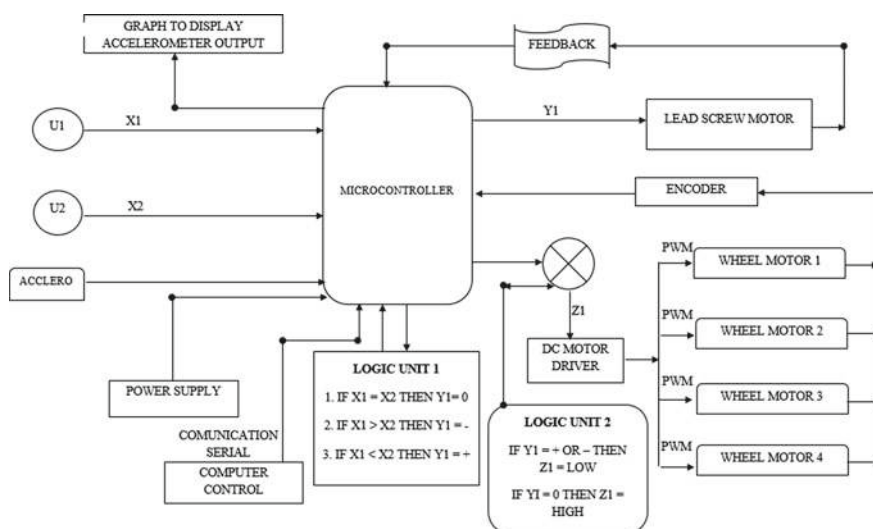


Fig. 55.4 Five-layered fuzzy-based integrated control architecture

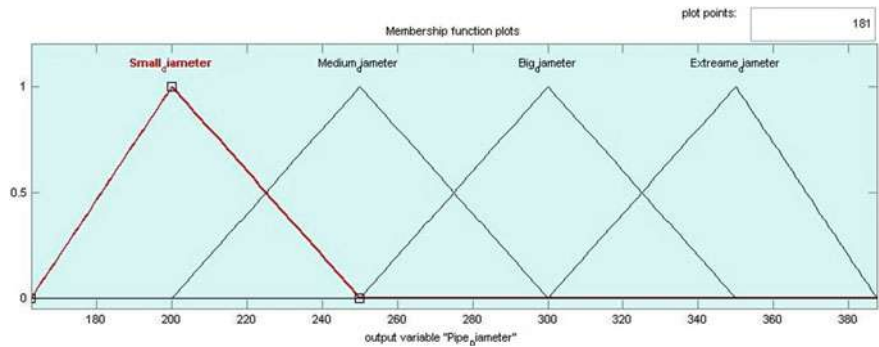


Fig. 55.5 Membership functions for output

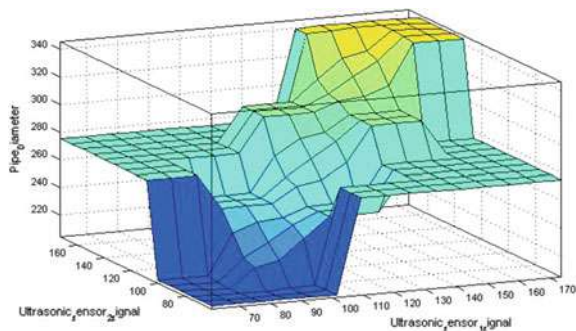


Fig. 55.6 Surface-based output

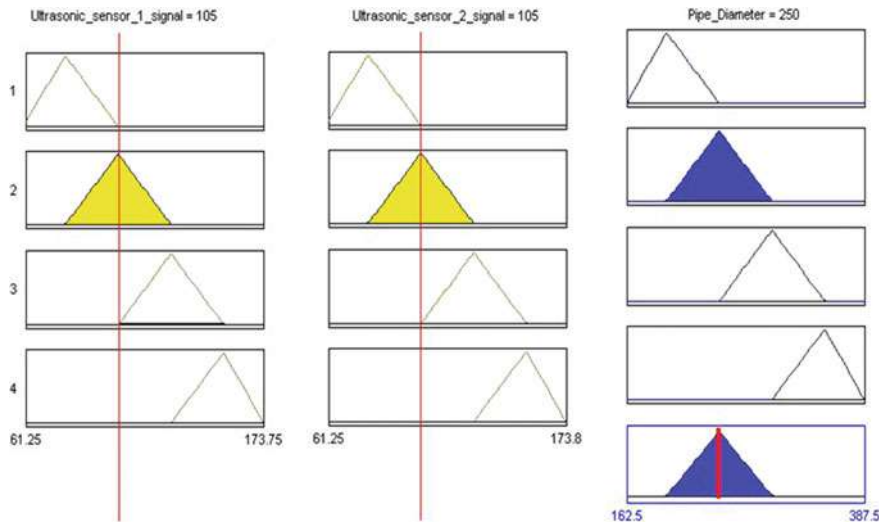


Fig. 55.7 Fuzzy-rule-based output

55.5 Experimental Setup and Result

Many models were progressively designed with improved performance on the desired characteristics [10] two of which are shown in Fig. 55.8. The second model is adopted for the prototype fabricated for experimentation.

An in-pipe cleaning and inspection robot called RsiRci (Reconfigurable Sewer In-pipe Robot for Cleaning and Inspection) was assembled at SSN Robotics laboratory and is shown in Fig. 55.9. RsiRci robot's arms can contract and expand allowing reconfigurability. Each of its two arm sets is controlled by hybrid linear actuator. Its faceplate head is mounted with two ultrasonic sensors to sense the pipe diameter. It has accelerometer sensors fixed on its arm to allow the robot for self-localization and also for the map building of the sewer environment. All the eight arms have wheels fixed at each end, thus allowing the RsiRci robot to operate both in-pipe and outdoor effectively.

The proposed design is unique in comparison with those already implemented. It operates with a lead screw mechanism, making the robot adapt to any pipe diameter ranging from 200–350 mm (commonly used UGD pipe diameter in Tamil Nadu, India). The stability to move inside the varying pipe diameter is provided by lead

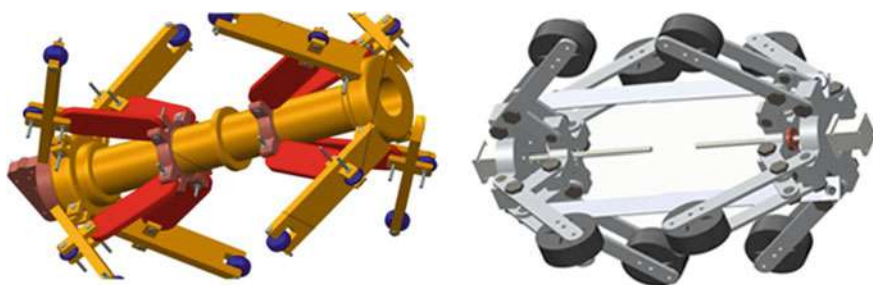


Fig. 55.8 CAD model of proposed robot



Fig. 55.9 Model of fabricated RsiRci robot

Table 55.1 Specification of RsiRci robot

Robot weight, R_W	~6 kg
Length of the robot, L_R	300 mm
Spanning diameter, D_R	350 mm
Tractive effort, T_E	7.789 N
Speed of the robot, S_M	0.3 m/s
Radius of wheel/tire, R_D	32.5 mm
External arm length	145 mm
Internal arm length	70 mm
Arm thickness	3 mm
Faceplate diameter	90 mm
Small faceplate diameter	42 mm
External arm rotation angles	<90°
Flange height at above the faceplate	25.47 mm

screw mechanism which also prevents the arm closing or opening during the working condition. Specification of the fabricated RsiRci robot is listed in Table 55.1.

55.6 Conclusions

The progressive design variations that were evaluated for performance characteristics were presented in this paper. The structural analysis for the legs for each robot design and also for the complete assembled robot was also performed and modifications were done to the design accordingly. As an outcome, a modular reconfigurable sewage cleaning robot is designed, dynamically analyzed, fabricated and is being tested currently. A fuzzy-based control algorithm is proposed to arbitrate between the proposed five-layered control architecture for the purpose of controlling the robot toward achieving the objectives of motion and reconfiguration. As an outcome of the experimental analysis, modifications would be made to the proposed control architecture and the algorithm after observing the results. It is expected that this robot could be added with the capability of doing multifunctional tasks such as inspection, cleaning, and surveillance operations, by making corresponding changes in the control architecture.

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Chapter 56

Six Sigma in Battery Assembly of Skid-Steer Loader



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and Abhishekshivram

Abstract Lead–acid battery is the vital part of any automobile, which is used to start the engine, and powers the electric controller and accessories. Quality of the battery determines the smooth starting of engine and working of electric accessories in the vehicles. A damaged battery due to wrong assembly method leads to reduced battery life and frequent starting problems in the working field. This paper eliminates cost of poor quality involved in the assembly of battery on the machine by eliminating the rejection, rework, battery surface damages, and operator fatigue by using Six Sigma methodology with improved quality by using an effective battery mounting bracket design.

Keywords Six Sigma · DMAIC · Battery design and assembly · Skid-steer loader

56.1 Introduction

Six Sigma is a highly disciplined process that focuses near-perfect process [1]. Continuous process improvement with low defects is the goal of this method. The aim of the Six Sigma is to make a process effect with 99.99996% defect-free. This means Six Sigma process produces 3.4 defects per million opportunities or fewer as a result. It has a business management philosophy works to maximize performance and minimize cost [2]. The key aspect of Six Sigma is its structured approach to reducing the cost of quality through robust identification of root causes. By eliminating the causes of excessive variability, we can reduce all the cost due to the defect.

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design
and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_56

56.1.1 Problem Background

Battery is one of the major parts in skid-steer loader vehicle assembly. Improper battery mounting bracket design leads to battery bulge and plays rework in the assembly of skid-steer loader. The main objective of the project is to eliminate the battery and bracket rejection and rework the assembly line of skid-steer loader vehicle using a modified mounting bracket design. The objective of the paper is to reduce the efforts required by eliminating the rejection, rework, and operator fatigue.

56.2 Literature Review

Antony [1] attempts to explain the usefulness of Six Sigma in various potential areas in which it can be implemented to improve the quality of products and service. From the paper, it is inferred that Six Sigma can be implemented in various fields and service operations. Prashar [3] demonstrated the application of Six Sigma methodology for identification of problems and reducing the costs generated as a result of generating poor quality. Boonsuk and Frank [5] designed an automated fixture for a rapid machining process which includes the design of sacrificial support length, shape, size, number, and location to maximum allowable deflection of the part and also maximizing machinable surface area. Pavlov [6] presented on lead–acid battery technology and its influence on battery performance parameters. He described the technology in the form of integral theory and explained the interrelation between the technological process and the performance characteristics of batteries. Pendokhare and Quazi [7] examined the use of Six Sigma methodologies in order to satisfy customer needs and to maximize the profit. They proved that the integration of Six Sigma methodology, lean, knowledge management, and cost of poor quality was an approach for such a systematic innovation management and improvement of processes. Roy and Liao [10] reported that stability analysis is the major factor which can be used to determine the design of brackets and clamps, and they developed a methodology to analyze the stability of the components in the automated fixture design environment. Breyfogle et al. [4] explained Six Sigma deployment methodology for different business processes that can be used in various potential areas. Six Sigma methodologies are applicable for all domains like software, health care services, etc., even though researchers have applied in educational field [9].

56.3 Methodologies

Six Sigma is a data-driven problem-solving methodology which reduces defect rates in any process and uses one of the two approaches—DMAIC (Define, Measure, Analyze, Improve, and Control) or DMADV (Define, Measure, Analyze, Design,

and Verify). DMADV methodology is an improvement system used to develop new process or products at Six Sigma quality. In this, the process is falling below specification and looking for increment so that DMAIC methodology is chosen [8].

56.4 Experimental Analysis

56.4.1 Define Phase

The reputed company in Chennai runs a single shift per day. The data collection is carried out during this shift. The shift starts at 7.45 AM and ends at 4.15 PM. Based on the cycle time and productivity hours, the number of assembly per month is 120. Out of 120, an average of 15–18 batteries gets rejected due to defects.

There are two reasons for rejection of battery and bracket. Battery is rejected due to bulge and crack formed on the top surface of battery, where the mounting bracket will come in contact with battery. Bulge occurs either when torque occurs on the bracket or after the road test of the vehicle. The bracket gets rejected due to bend formation at bottom side of the bracket where the bolt is seated and secured to the lower part of the chassis. Performing the battery assembly with the existing bracket and process becomes undesirable. In the last one year, the company runs with an average rejection rate of 10/month.

56.4.2 Measure Phase

Figure 56.1 shows monthly rejection for battery and bulge. The calculation for the rejection rate is as follows:

$$\begin{aligned}\text{Total productivity time per shift (min)} &= \text{Total shift time} - \text{Non productivity time} \\ &= 510 - 50 \\ &= 460 \text{ min}\end{aligned}$$

$$\begin{aligned}\text{No. of component targeted per shift} &= \text{Total productivity time per shift} / \text{Cycle time} \\ &= 460 / 60 \text{ min} \\ &= 7\end{aligned}$$

$$\text{Actual no. of batteries produced/month} = 120 \text{ Nos}$$

$$\text{No of rejection per month} = 18 \text{ Nos}$$

$$\text{Total rejection rate} = 18/120$$

$$\text{Defects per unit (DPU)} = 18/120$$

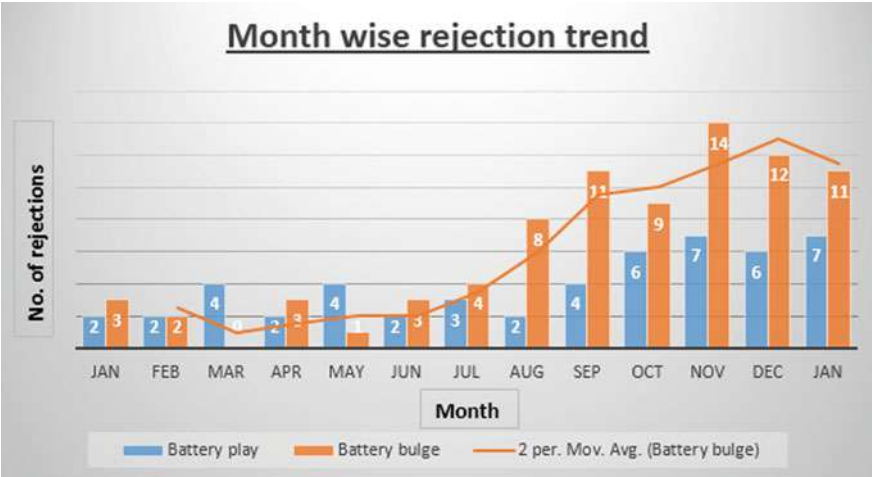


Fig. 56.1 Monthly rejection for battery and bulge

Defects per opportunities (DPO) = $18/3 \times 120$

Defects per million opportunities (DPMO) = $DPO \times 1,000,000$

= 50,000 (As per DPMO calculator)

From the sigma table, the current performance level is 3.1.

56.4.3 Analysis Phase

There are plenty of causes which can result in either battery bulge or bracket bend. The issue of battery bulge occurred either at the time of assembly or after road test. Sometimes the bracket also gets bent during the assembly process. The torque value applied remains same for all assembly. There are many reasons that cause battery bulge. To find out the exact root cause for battery bulge, a fishbone diagram has been drawn. The below fishbone diagram in Fig. 56.2 will give a brief idea about possible root cause analyses of battery bulge.

Prime Cause Analysis

We found that the issue is with the clamping force acting on the battery during torqueing operation. While torqueing the battery mounting bolts, the downward force acting on the battery bracket creates a bending moment at the free end of the bracket. This will bend down the bracket due to less bending strength of bracket which creates

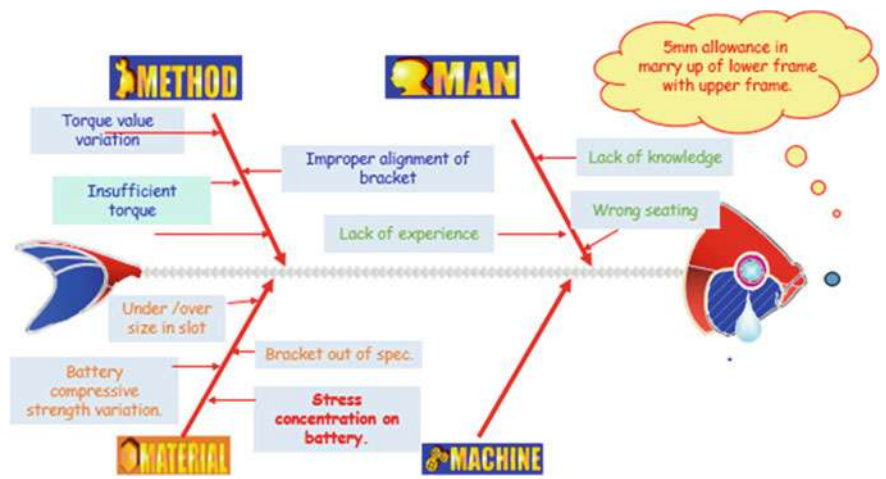
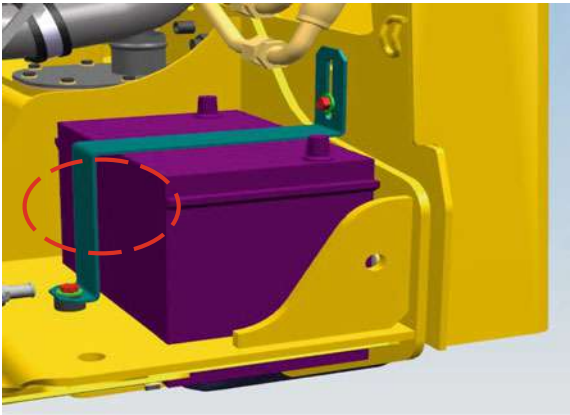


Fig. 56.2 Root cause analysis in battery bulge

Fig. 56.3 Battery bulge area



a limited contact area with battery at free end of bracket. This bending moment on the bracket end acts as a compressive force on battery top. Due to limited contact area, the compressive force acting on the battery surface exceeds the compressive strength of the battery material, and stress concentration happens on the battery top surface. This will result in battery bulge and crack. The current bracket design was not able to withstand the bending moment and distribute the clamp force evenly to the battery which results in stress concentration on that particular corner of battery where usually bulging happens as shown in Fig. 56.3.

Since the current bracket makes less contact area with the battery, stress develops on the battery and battery bulge occurs. As the trial conducted with added plate to increase the contact area, there was no battery bulge during assembly and road test. The comparison between three trial results is listed in Table 56.1.

Table 56.1 Comparison of three trial results

Trail name	Major observations
Torque changing	Not recommended due to battery found loose in lower torque value. And battery bulge at higher value
Weld boss height increasing/decreasing	Not recommended because only 50% occurrence has been reduced and created another problem with battery seating
Contact area—increasing/decreasing	Battery bulging fully eliminated with the same torque value. The only issue is bracket bend at the point of torque application

Based on the comparison, added plate completely eliminates the battery bulge. The best solution to avoid the battery bulge issue is to modify the bracket with increased contact area. To avoid the bending of bracket, gusset plates can be provided at the bottom.

56.4.4 Improve Phase

The design calculation of bracket dimensions is done below with its 3D model and drawings.

Compressive force acting on battery (P) = 5625 N

Yield strength of battery = 10 N/mm²

Considering factor of safety (special case) for LDPE = 4

Design strength = Yield strength/FOS

= 2.5 MPa

Minimum contact area is required with battery by the bracket to withstand the clamping force with factor of safety 4.

Actual length of contact of bracket with battery (L) = 25 mm

Total area of contact required with battery = ($P * FOS$)/Yield strength

= (5625 * 4)/10

= 2250 mm²

Maximum possible width of contact of bracket with

battery measured at 9 Nm = 25 mm

Length of contact required = 2250/25

= 90 mm

So a plate of 25 mm width and 90 mm length with thickness same as battery bracket (3 mm) will be added to the top face of bracket to withstand the clamping force by the battery without bulge in *Y*-direction. The same dimension plate will be added to the bracket inside face to withstand the clamping force in *X*-direction.

So,

Length of the gusset plate $X = 30.48 \text{ mm}$

Force acting on gusset, $F1 = 10542 \text{ N}$

Consider we are going to add two gusset plates $n = 2$

Therefore, $F = 5271 \text{ N}$

Assume the tensile load acting on gusset $F = P/A$

Factor of safety = 4

Assume tensile force of wrought Iron = 234 N/mm^2

Area $A = 5271/234$

= 22.53 mm^2

The bracket material is assumed as a bonded structure for Pro Mechanical Analysis, and the normal load of 500 N is applied as shown in Fig. 56.4. Material is

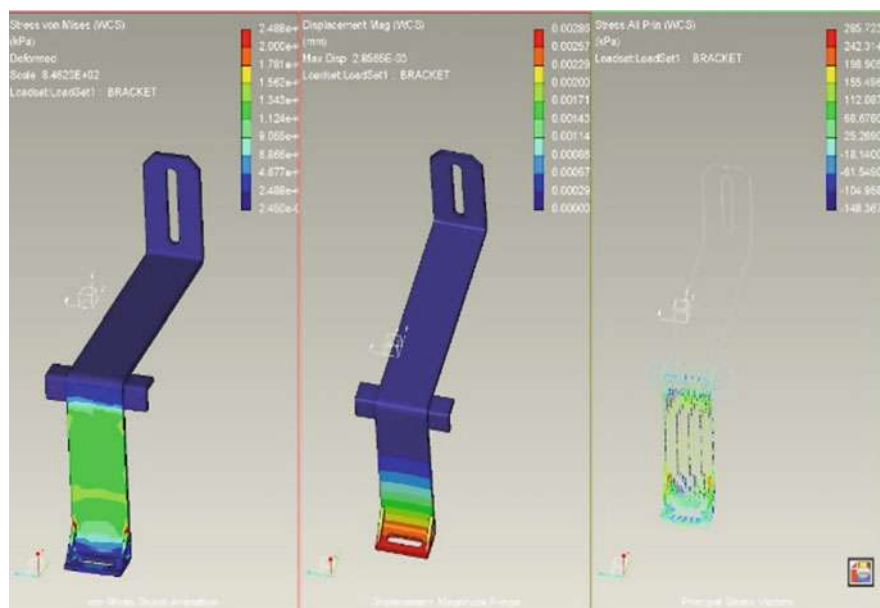


Fig. 56.4 Battery clamp analysis result

Fig. 56.5 Implemented battery and bracket assembly



EN 8 with elastic modulus of 202,158 MPa, Poisson's ratio = 0.3, and density = 7840 kg/m³.

Based on the assumptions (material, acceptance criteria-yield strength), the design is passing as per the analysis performed. Considering factor of safety of 2.5, the structure was tested for 2.5 times the normal load. As per caterpillar specification, factor of safety for steel is 2.5 and load is applied 2.5 times higher the actual load. During analysis, von Mises stress distribution in the result image lies in green and blue colors. That means the bracket is safe with 2.5 times higher load, and hence, the design is safe. From the analysis, all designs are safe. Similarly, von Mises stress distribution for battery, chassis, top locators, and bottom locators was analyzed.

Figure 56.5 shows the factual view of the newly designed battery bracket assembled on the battery to the lower frame.

56.4.5 Control Phase

In this phase, battery and bracket assembly defects were kept under observation. After implementing the new battery bracket, the rejection trend is monitored.

$$\text{No of rejection per month} = 1 \text{ No}$$

$$\text{Total rejection rate} = 1/120$$

$$\text{Defects per unit (DPU)} = 1/120$$

$$\text{Defects per opportunities (DPO)} = 1/3 * 120$$

$$\begin{aligned} \text{Defects per million opportunities (DPMO)} &= \text{DPO} * 1,000,000 \\ &= 2778 \text{ (As per DPMO calculator)} \end{aligned}$$

From the sigma table, the current performance level is 4.3. Therefore, by rejection rate because of the battery bulge and play, the bracket bend becomes zero percentage after implementation of the project.

56.5 Conclusion

Based on manufacturing, procuring, and assembling criteria, bracket design has been selected as the best feasible solution and implemented in the assembly. According to the trial experiments and data observed, the bulge on the battery and bracket bend were eliminated. By the implementation of this bracket design, the required assembly can be achieved without any defects. It also avoids rework time. This bracket reduces the operator fatigue, and a high degree of dimensional accuracy is achieved with high quality and customer satisfaction. The rejection of battery and bracket due to design failure has been successfully eliminated by implementing a newly improved bracket design. The defect was reduced, and the performance of the company is improved from 3.1 sigma level to 4.3 sigma level.

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Chapter 57

Concept Design and Development of Position Sensor in Door Control System



G. Dinesh Kumar , L. Ragunathan and A. N. Rajaraman

Abstract Automotive companies are more concerned about safety and comfort. Indian manufacturers are still dependent on conventional pneumatic systems to control door operations, for normal open and close. In case of obstacle sensing between the doors, the current system is capable of sensing obstacle with some restriction that includes higher operating cost, high lead time for assembly, and finally dissatisfies customer requirements. New position sensor to be designed and developed to control door system effectively, which is easily mountable, easily adaptable to all types of mounting, and cost-effective solution. Therefore, a new linear position sensor compatible with existing ECU, Valves, and system requirement is designed and developed.

Keywords Position sensor · Pneumatic door cylinder · ECU · Automotive

57.1 Introduction

In the field of automotive electronics and vehicle control systems, door control system is associated with an advanced motor vehicle. An advanced motor vehicle consists of a number of electronic control units (ECU) with door control system (DCS). Door control system is responsible for controlling and monitoring various electronics in the vehicle's door. Door control system is either present in individual door separately or in combination. In most of the cases, the driver's door module acts as a master controller and others act as a subsystem in the communication protocol.

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_57

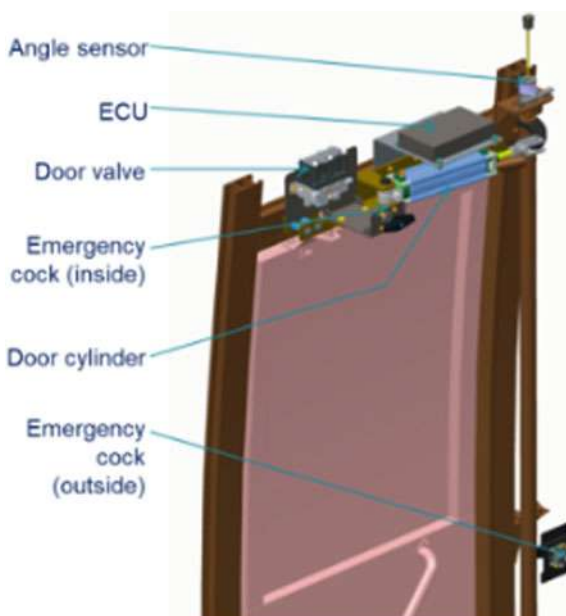
57.1.1 Door Control System Layout and Components

During the opening of doors, the combined action of rotary and linear movements is made to slide the door wings to the side of the doorway, and vice versa for close. This makes clear space (meeting the regulations) for both entrance and exit doors. Pneumatic systems are used to operate the doors for public vehicles and are operated by double acting cylinders along with the combination of pneumatic valves and electronic controls. In this system, opening and closing of the doors are controlled by switches (ON/OFF), and these switches provide signal to the door valves thru' electronic control. When the switch is turned ON, the doors will open and remain open until, the close command is given and vice versa. When the switch is turned OFF, the doors will close.

Figure 57.1 shows the electro-pneumatic door control system layout with double acting cylinders operated simultaneously with the help of door control valves, door position sensor, an electronic control unit, and wiring harness.

Obstacle sensing is one of the key features (safety aspect) of the proposed door control system. This is one of the safety features, which call for Automotive Industry Standard and Regulations for bus body code approval. Obstacle sensing and position of the door are detected by a rotary sensor. A rotary sensor is a multi-terminal resistor with an adjustable (linear or rotary type) contact forming a voltage divider. The measuring instrument called as a potentiometer is essentially a voltage divider used for measuring electric potential (voltage). These are also used to find the steering

Fig. 57.1 Door control test rig



position, crankshaft position, throttle valve position, gear shaft position, etc., in many of the automobile applications.

Basic Functions of Door Control System

- Door open and close operation
- Door opening and closing with damping (independent variable parameter)
- Door turns to 'forceless' in the opening direction during obstacle
- Door turns to 'open condition' in closing direction during obstacle
- Emergency inoperable at vehicle speed >5 kmph (speed integration)
- Halt brake integration, CAN-enabled system, and Configurable parameters
- Emergency deactivation—emergency cock returns back to normal position.

Existing door control system consists of a rotary sensor (positioning) for an obstacle detection, which gives an analog voltage output of required V DC. This signal acts as trigger for ECU to operate the valve to actuate the door valve for open condition and vice versa for close condition also.

Installation is mounting and integration is linking electronics of a rotary sensor with door shaft and ECU are found to be difficult, as it requires a diagnosis module for setting voltage range. For open position, A to A_1 V and for closed position, B to B_1 V is required, which needs a skilled person and takes more time for serial production and stabilization. It will also increase inventory time and cost in importing electronic coils. Hence, there is a need to develop a new linear position sensor for door control system.

Problem Statement In Indian road transportation, the rotary sensor is not suitable due to the road condition. For Indian vehicle application, the rotary sensor mounted on to the shaft, is getting repositioned from an initial pre-locked position, because of the uneven road surface. This results in a change of the corresponding voltage of the rotary sensor and hence misbehaves. Moreover, existing rotary sensor is not suitable for different types of door such as single and double jack knife door, in-swing door, and out-swing door. Door control system is not only mounted on the top, it is also mounted on the middle and in bottom position depending on the vehicle layout and requirement. Hence, this rotary sensor will have difficulty in mounting on to the shaft at middle- and bottom-mounted systems.

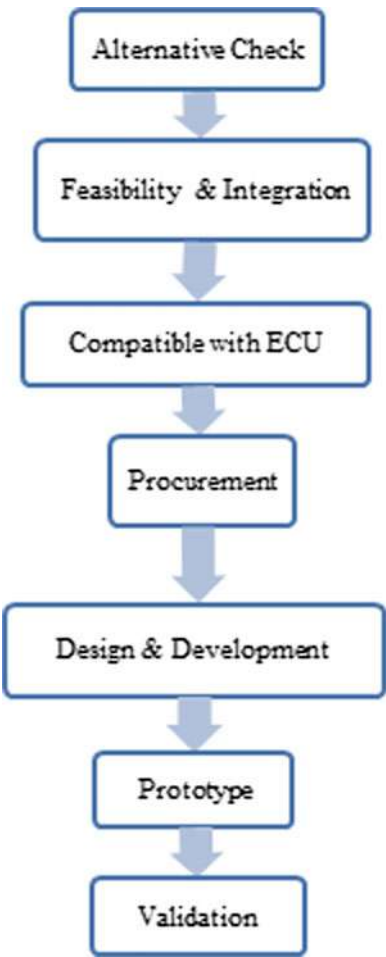
Objective

- To develop a position sensor for obtaining the position of door and door cylinder at a lower cost with an analog output.
- To develop a position sensor, suitable for different sizes of door cylinder of various stroke lengths.
- To design and develop a sensor with a pneumatic cylinder, integrated as a single assembled kit having high rigidity and robustness.
- To validate and test the developed prototype.

57.2 Methodology

Check for the alternative sensors available to replace with the existing with minimal changes in design. Confirm the feasibility and integration of the sensor (compact in size—preferably) to be mounted on the pneumatic cylinder. Pin configuration of the sensor to be compatible with existing ECU. Once above parameters are met, procure the sensor to check the functional requirement. Design an integrated system with the procured sensor to the cylinder and develop the same. Prototype sample can be made to visualize the physical touch of the assembly. Part and assembly level validation and testing can be done on the sample to prove the concept. All the required tests can be done for the approval of the design at various design phases (Fig. 57.2).

Fig. 57.2 Flow chart of the methodology



57.2.1 *Alternative Sensors*

During testing, different position sensor such as

- Linear potentiometer sensors
- Inductive sensors (LVDT)
- Hall-effect sensor
- Magnetostrictive sensors
- String or wire draw potentiometer
- Potentiometer
- Capacitive sensors, and
- Optical sensors have been tried to suit the displacement of various stroke of pneumatic cylinder.

57.2.2 *Sensor Selection—Compatible with ECU*

The sensor that satisfies the ECU requirement with analog output and mating pin configuration has been studied. After testing, it is found that linear potentiometer, string potentiometer, and inductive sensors (LVDT) are found to be suitable for the application with minimal changes from existing design.

Linear Potentiometer Sensor

- Linear potentiometers are economically priced (moderate).
- These devices rely on mechanical contact to provide position feedback, and thus exhibit repeatability and some hysteresis.
- Availability of the parts (i.e., resistive strip) is found to be difficult.
- For customized parts, suppliers for resistive strip (length) and wiper are not available. But body length is higher for smaller stroke variant cylinders.

LVDT

- LVDTs are very versatile, non-contacting, and robust; they offer long-term reliability in harsh or hostile environments.
- Body is lengthier and has internal parts of copper wire and brass. Involve winding process, which will increase the basic cost and hence, affects the serial production.
- It works on AC principle. To convert from AC to DC; modulator and demodulator are required, which need signal conditioning circuit and it adds up extra cost to the sensor [1]. Hence, LVDT is not a cost-effective solution.

String Potentiometer

- Cost effective among these options.
- Child parts can be easily manufactured and assembled.
- Sensing element is readily available in the market according to application requirement at a lower cost.
- Different sensing range and customized dimensional requirements are available.
- Compact in size and lesser space utility as compared with linear potentiometer and LVDT.

57.2.3 Design Concept

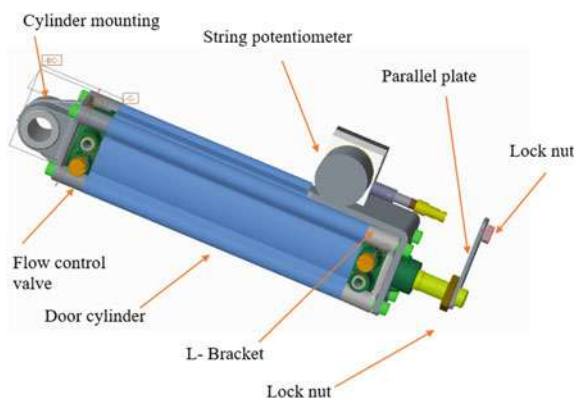
3D model of Door Control System with string potentiometer (Fig. 57.3).

The string potentiometer is mounted on the door cylinder at one end near piston rod with help of *L*-shaped bracket and clamping.

Mounting of string potentiometer is assembled to *L*-bracket and is bolted from the bottom side of bracket and housing. *L*-bracket subassembly is mounted on the cylinder with screws. To keep the string and piston rod parallel, there is a plate connected. The position of parallel plate is maintained by a lock nut at a particular distance to maintain the initial voltage value at door closed position. The string-puller is fitted to a parallel plate with help of nut.

- A Parallel plate is mounted on the piston rod with lock nut and mounted at a particular position to maintain the initial output voltage, and other side is locked by a lock nut.
- At the initial position, the potentiometer shaft is at '*C*' ° and door is in closed position. A parallel plate is mounted on piston rod at a distance of '*D*' mm from a string-puller, which gives the initial output analog voltage of '*A*' V constant.
- As driver operates the open switch, the door valve is actuated and operates door cylinder by compressed air. The piston moves gradually outward, which pulls

Fig. 57.3 3D model of door cylinder with integrated string potentiometer



the string. Potentiometer shaft rotates on pulling, which results in output voltage proportional to the distance moved by the piston, i.e. door opening condition.

- For 'X' mm length cylinder, the effective stroke is minimum than the complete stroke. As parallel plate reaches the maximum stroke, potentiometer shaft is rotated by 'C₁'° then it gives 'A₁' V as an output voltage.
- As driver operates the close switch, the valve in the system actuates and executes return stroke of cylinder, i.e. door closing position.
- At this position, the string will rewind on to the spool because of torque provided by spiral spring due to spring return force.

57.2.4 Design Calculation

Calculation for Electrical Measuring Stroke Length

The basic concept of a potentiometer is used, which follows the voltage divider circuit output voltage formula, which is as follows (Table 57.1),

$$V_o = V_i x R_1 / (R_1 + R_2); V_o = V_i (\rho L_1 / a) / (\rho L_1 / a) + (\rho L_2 / a)$$

$$V_o = V_i x L_1 / (L_1 + L_2) \quad (57.1)$$

$$R\rho = L/A \quad (57.2)$$

V_i	Input applied voltage;
V_o	Output voltage;
$R = (R_1 + R_2)$	total resistance 'S' kΩ;
ρ	Resistivity of the resistive track in potentiometer in ohm meter;
$L = (L_1 + L_2)$	Total length of resistive track;

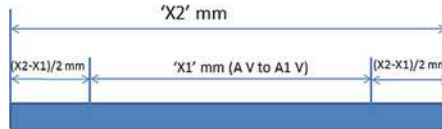
Table 57.1 Electrical measuring stroke length

Cylinder variants	Stroke		Voltage range for 'D' mm linear POT	Voltage range
	Effective stroke	Maximum stroke	Effective voltage	Maximum voltage
X	X ₁	X ₂	A–A ₁ V	A–A ₁ V

L_1 distance moved by the wiper contact in mm.

Calculation of Number of Turns of Multi-turn Potentiometer V_o

Using ' T ' turn wire wound potentiometer, (' S ' $k\Omega$) designed a string potentiometer for required maximum stroke length of ' X_1 ' mm. For maintaining initial and final range of output voltage, overall length of the string is to be taken of ' X_2 ' mm. For ' T ' turns it will give an angular movement of ' C '°.



Calculation of Spool Diameter

For ' T ' turns i.e. $C^\circ = D$ mm single turn i.e. $360^\circ = D/C = D_1$ mm

This distance travel for a single turn is equal to the perimeter of spool or drum on which wire rope is wound.

The perimeter of spool = D_1 mm; $(\pi \times D) = D_1$ mm; $D = 'D_2'$ mm (calculated)

Diameter of the spool $D = 'D_2'$ mm

Calculate the Spiral Spring Design

From the given extraction force and arbor, diameter torque is to be calculated. And by putting all other values in given Eq. (3), length of flat spiral spring is calculated.

$$T = \frac{\pi E b \theta t^3}{6L} \quad (57.3)$$

T = Torque developed (N-mm)

F = Force applied for extraction (N)

b = Width (mm)

θ = Number of turns to revolve

t = Thickness of spring (mm)

L_2 = Effective length of spring (mm)

E = Young's modulus of steel spring (N/mm²)

$T = D \times F$ (Spool diameter x Force applied for extraction)

$T = 'Q'$ mm

Then, this value of torque is put in Eq. (3) and re-arranged

$$L_2 = \frac{\pi E b \theta t^3}{6T} \quad (4)$$

$$L_2 = 'L_3', \text{ mm}$$

From this calculation, it is found that the total length of the flat spring is ' L_3 ' mm.

57.3 Outcome

This includes the various test conducted on the final product of string potentiometer with door cylinder. The tests carried out are

- Functional test,
- Performance test, and
- Limited endurance test.

57.3.1 Functional Test

In this test, to check open and close operation, the sensor is assembled with cylinder power supply, multimeter, regulated pressurized air supply of 'P' bar pressure, working table connecting wires, hose pipes, and control valve.

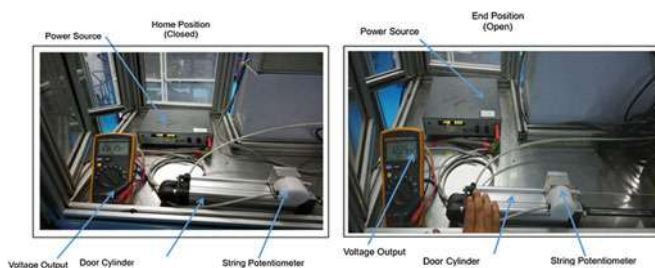
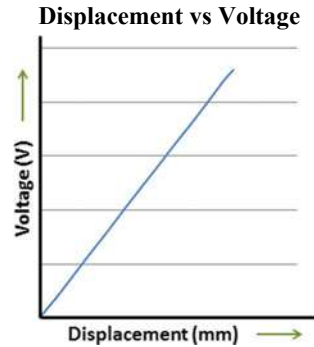


Fig. 57.4 Results of displacement versus voltage



57.3.2 Performance Test

To carry out the performance test, DAQ (Data Acquisition) system, BNC cable, string potentiometer, and LabVIEW installed in a computer system were used.

- With help of DAQ system, string potentiometer is connected and supply voltage of 'A' V is given as an input to terminal 1 (red) and to the ground terminal 3 (white).
- An output from the string potentiometer terminal 2 (green) is taken and provided as an input to DAQ system channel with the help of BNC cable.
- LabVIEW program includes DAQ assistance which displays and records in an excel file.
- This output voltage is multiplied with scale factor to convert into the corresponding displacement.
- With help of the x - y chart, the output voltage versus displacement is plotted (Fig. 57.4).

57.4 Conclusion

The design and development of the door control system were carried out. This solution is the most cost effective and most feasible for serial production. Endurance test for this sensor has to be carried out for further analysis in order to know the modes of failure and reliability. This integrated system can be implemented to any type of mounting of the control system in the vehicle.

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Chapter 58

Remote Monitoring of Axle Loads for Heavy Commercial Vehicles



M. Richard Alexander , V. Hudson and Pozhilan

Abstract A practical approach on remote monitoring of axle loads for heavy commercial vehicles with mechanical suspension was presented. At present, there are motor vehicles act implemented in most of the countries against operating commercial with overload. The major root cause for many truck accidents is overloading. When a truck is loaded more than its rated load, it will increase the risk of affecting vehicle stability while driving. Due to overloading of trucks, the driver will loss the steering control during turning and braking, and this loss of control leads to major accidents on roads. In some cases, even when the cargo load is within the rated load, any one of the individual vehicle axle will get overloaded due to non-uniform distribution of payload. Uniform distribution of loads is important for safe vehicle operation. Overloading of trucks will reduce the life of tires and parts related to suspension from their actual service life, which leads to increase in service cost for the vehicle. The best way to overcome these problem is to monitor each axle loads by avoiding traditional method of weigh bridge measurement. Axle load monitoring system presented in this paper works under the technique of capturing axle load with respect to the deflection of mechanical spring suspension. In addition, the system supports remote data capturing using telematics device. This will avoid the physical presence for measuring.

Keywords Axle load · Fleet · Gross vehicle weight · Overloading · Payload · Pulse-width modulation · Rated load · Telematics

58.1 Introduction

India has a market for 1.2 billion consumers which indicates that it is one of the faster-growing economies in the world and thus it requires efficient transportations and logistics industry. Around 60% of goods are transported by roads in comparison with other modes of transportation. There has been a significant increase in the

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© Springer Nature Singapore Pte Ltd. 2020

M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_58

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commercial vehicles on the road in the recent times. Increase in commercial vehicles is a reflection of increasing demand for the movement of goods.

In order to meet the market demand or to increase their margins, many truck owners are permitting overloading of their trucks risking the consequences of in-transit issues as well as its negative impact on the infrastructure. Overloaded trucks destroy roads and highways; they result in causing road accidents. There is also a significant negative impact in terms of casualties, road blockages, and negative productivity of the transport operations due to incidents involving vehicle overload.

In order to reduce overloading of trucks and accidents, Government has taken some major steps. Rule has been implemented to punish the vehicle operators with fines and canceling permit of vehicle when the vehicle was operated in overload conditions. Regional transport department will monitor the vehicle load status during vehicle operation on public roads. Inspection stations were formed by government near toll plaza's to monitor overloaded trucks.

Implementation of strict rules from government with high penalties will have control on overloading vehicle. Vehicle operators will think on avoiding overload of trucks if the penalty for law-breaking is higher than the monetary benefit that comes from operating vehicle with additional load more than the rated load. Major drawback of current inspection station used by regional transport department is that they are available only at particular areas and also not on major scale, additionally it requires an inspector to catch hold of overloaded trucks. To overcome such problems, on board axle load monitoring through telematics device will play a major role for remote monitoring of vehicles.

58.2 Problem Statement

Every truck is designed to carry a defined load. Each truck manufacturer will have their design calculation to define rated load of the vehicle based on the engine capacity, axle strength, suspension, etc. The rated load is the safe load with which the vehicle can be operated.

In current practice, loading of cargo is done at one location and measurement of vehicle load is done at other locations using standard weigh bridge measurement, where fleet operators and National Highway Traffic Safety Administration does not have any control on overloading.

One use case example is a Truck OEM facing breakage of front axle. As the outcome of root cause analysis, design calculation, and testing results show that their axle will withstand the defined rated load. Now, the suspect by the OEM was that customer would have overloaded (above the rated load) the axle. As discussed earlier, it is possible to overload an individual axle even when the cargo load is below the rated load due to non-uniform loading pattern.

So to evidence the overloading, OEM wants a technology to monitor axle load remotely. As the breakage was happened in front axle, design concept in this paper

covers only for front axle load monitoring but the same pattern can be extended to other axles of the vehicle to monitor complete vehicle load details.

58.3 Design Concept

In order to have clear idea on the requirement, design concept generation process for remote monitoring of axle load can be divided into five sections. Final design concept should fulfill below mentioned activity (Fig. 58.1).

- Measurement of parameters with respect to loads
- Data transfer to telematics device
- Data collection in telematics server (cloud)
- Data processing at backend (relating captured data to applied load)
- Display of axle load value to the customer.

In this concept, height sensor is used for permanent detection of height variation between axle and chassis of a vehicle. It is actuated via a lever equipped with an additional link optionally. The electrical output signal of the sensor is proportional to the angle position of the lever and nearly independent from the ambient temperature.

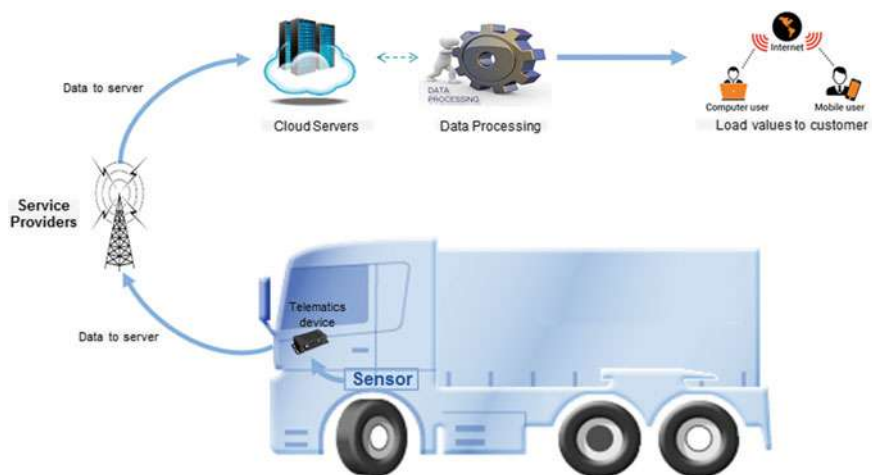


Fig. 58.1 Overview on remote monitoring of axle loads

Evaluation and conversion of the measured field is done by the application-specific integrated circuit (ASIC) providing an electrical signal at the sensor output in the form of duty cycle (PWM output), which is temperature compensated and proportional to the angular position of the lever.

Height sensor will continuously monitor the vehicle's height, i.e. is the height between the axle and chassis, and send its readings to telematics ECU. Telematics ECU will transfer the data to backend server through GSM connection available in the telematics ECU. From the data received in backend, change in height value from reference value (unladen value) will be calculated which is the actual deformation occurred due to loading of cargo (payload), also it is the exact replica of deformation occurred due to loading in the mechanical spring suspension which is fitted between chassis and axle. By relating the deformation value with stiffness of the spring, load on the axle can be calculated. Finally, calculated axle load value can be broadcasted to end-user through web applications.

Connectivity between Telematics ECU and sensors will be through wiring harness. Each sensor requires power, ground, and output connection and for telematics ECU required connections are power, ground, ignition, and input of sensor data. Wiring harness design will have automotive grade connectors. Length and branching for wiring harness design are studied from the vehicle. Branches that are passing through engine area should have high temperature protection sleeves covered over the harness. Wire size should be selected based on the current rating of circuit.

Two types of link rods were used, one is the lever which will get attached to sensor at one end and on the other side it will have the rubber bush which will hold another link rod in the direction perpendicular to the lever, other end of the link rod will be connected to the axle through clamp bush (Fig. 58.2).

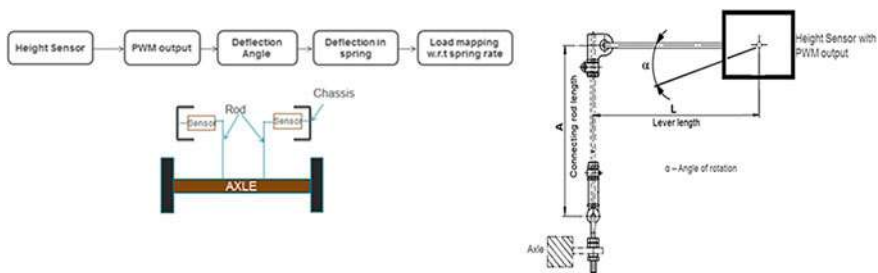


Fig. 58.2 Fitment model of the design concept

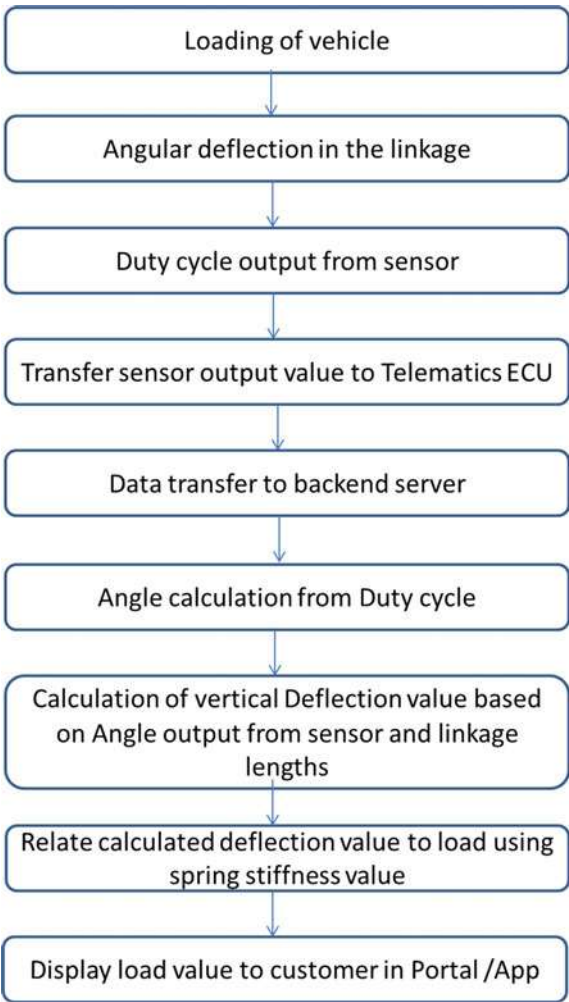
In software architecture, backend server act as a space for receiving and storage of data. Backend server will have acknowledgement mechanism to ensure the data transfer between server and the telematics device fitted in the vehicle. The data to backend server will be in the form of script language. The data in the backend server will be processed and displayed to user through graphic user interface portal. Backend server is facilitated with auto scale up feature to manage data's of a more number of vehicles in the future. Portal is the Web application that processes the data from backend server and displays the necessary data to customer in readable format. Also, a provision can be made in the portal to generate an alert to configured mobile number/email id whenever individual axle load exceeds the rated load.

Certain values like sprung mass, unsprung mass, unladen load distribution on each axles are requested from the customer to perform the design calculation. Data from the customer was verified at vehicle level to confirm on the give data. Initially, gross vehicle weight was measured by placing the vehicle in weigh bridge with no payload, then to measure the weight distribution in each axle at unladen condition each axle was placed separately in weigh bridge and the values are noted for each axle. It is not possible to place the drive axle alone in the weigh bridge, so it can be calculated from weight measured by placing drive and tag axle together in the weigh bridge and subtracting it with weight measured by keeping tag axle alone in the weigh bridge. Measured values are in-line with the data shared by the customer. Stiffness of the spring used in front axle is required for load calculation, data for spring stiffness is provided by the customer. Based on the customer data, calculation of reference value for that particular vehicle was calculated. This reference value is required for calibrating the setup after installation of sensor and link rods in vehicle. Also this reference value is used for backend calculation where the sensor output value will be referred against this reference value to calculate the actual deflection of the spring due to addition of payload in the vehicle.

After installation of complete setup, calibration of sensor needs to be done. Laptop can be connected to serial port interface provided in the wiring harness to fetch the data from the telematics ECU directly, which can be used for calibration. Once the serial interface was connected, sensor fitment can be adjusted using clamp bush screws till the sensor output value gives the deflection value same as the reference value which is a calculated for that particular vehicle.

Complete workflow of this design concept is described in below flow chart (Fig. 58.3).

Fig. 58.3 Workflow of design concept

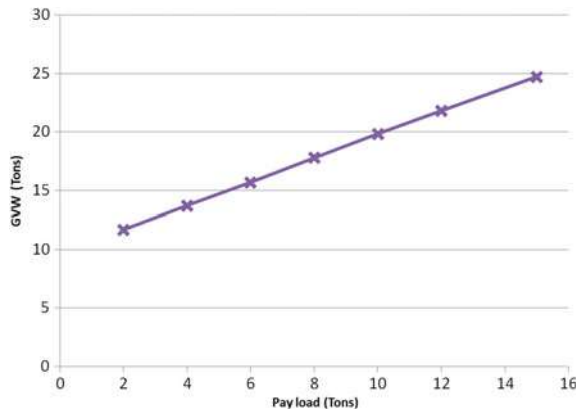


58.4 Test Results

After fitment and calibration, concept was tested at vehicle level. Loading of vehicle was done using known load blocks with incremental addition of payloads in terms of 2 tons till payload of 15 tons as shown in Fig. 58.4. Standard weigh bridge measurement was used to measure the actual load on front axle for the applied load which will be cross-checked against the value calculated using the developed axle load monitoring system.

Reason behind the use of payloads till 15 tons is because the vehicle used for testing is 25 tonner vehicle which has unladen weight (self-weight) of 10 tons so as per OEM suggestion allowable payload is 15 tons.

Fig. 58.4 Loading pattern



Front axle load was sensed by the sensor and the data was transferred to configured IP through telematics where the data processing was done as per the formulation generated to calculate the load. Calculated load was displayed in the portal. Portal load data was verified against the real value.

In order to capture left and right side distribution of load in front axle, two sensors were fitted in front axle with one on left side and one on right side. In this paper, results were shared as single front axle value by combining both right and left sensor output values.

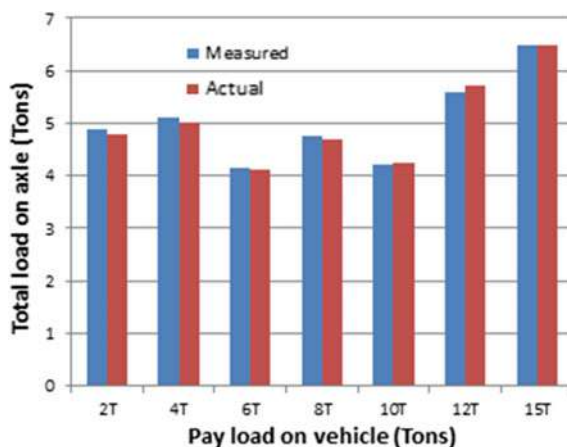
In Fig. 58.5, actual load values measured in weigh bridge were compared to the load value captured using developed axle load monitoring system. The results of 6 ton and 10 ton confirms the systems accuracy is same irrespective of loading condition (load and un-load) because the actual load on front during 4 ton pay load is more than 5 ton when 6 ton pay load was added actual load on front axle reduced to less than 5 ton due to loading pattern. As a result, developed system also captures this reduction in front axle load with same accuracy level maintained in 4 ton payload test.

58.5 Conclusions

Setup as per the design concept was made and tested at vehicle level; the level of accuracy with this method is around 98% with comparison of actual and measured value as shown in Fig. 58.5. This remote monitoring system concept can be extended to other axles of vehicle by adding additional sensors and installation setup with which load data of complete vehicle as well as individual axle can be captured and analyzed.

The main advantage with this developed axle load monitoring system are listed below

Fig. 58.5 Front axle load comparison (actual vs. measured)



- Axle load monitoring system avoids physical presence of humans near vehicle to monitor the weight.
- Actual load on the vehicle can be measured instead of knowing whether the vehicle is over loaded or not.
- With extension of this concept to other axles of vehicle, individual axle loads can be captured in a single shot, which saves time compared to standard weigh bridge measurement.
- This system will support all cargo loaders to place and distribute the weight uniformly so that no individual axle of the vehicle gets overloaded.
- Axle load data can be captured anywhere at any time through remote monitoring
- Government can have control on truck overloading, if data can be sent to dedicated government server to monitor truck overloading.
- Government can issue penalties to the lawbreakers remotely with the data captured from their vehicle.
- This system can alert fleet operators when vehicle axles are overloaded.
- Overload details along with location data of the vehicle where it got overloaded can be given to customer based on the output from telematics device which has in-built global positioning system (GPS) feature.

Chapter 59

Influence of TMTM as the Secondary Accelerator on Blooming Resistance of NBR-PVC Blends



R. Ananthanarayanan and S. Shanmugham

Abstract NBR-PVC copolymer blend provides a combination of better chemical resistance and weathering resistance. In NBR-PVC blends, di-accelerator system is normally used to achieve desired properties with better cross-linking density. Tetramethylthiurammonosulphide (TMTM) is widely used as the secondary accelerator to achieve various properties like heat ageing and high cross-linking density. When TMTM level is exceeding the soluble limit, blooming phenomena observed in the di-accelerator system. This study deals with the effect of accelerator system on blooming phenomena in NBR-PVC material. Various combinations of tetramethylthiurammonosulphide (TMTM), Mercapto-benzothiazole disulphide (MBTS) and N-Cyclohexyl-2-benzothiazole sulfenamide (CBS) were worked out to eliminate the blooming effect with desired functional properties are discussed in this paper.

Keywords NBR-PVC blends · TMTM · Blooming · Ozone resistance · Compound mixing · Secondary accelerator

59.1 Introduction

Dust seals are mainly used to prevent the dust and water entry to the devices used in brake system. Dust seals are mainly made with elastomer and thermo plastic elastomer (TPE). Main material properties required for dust seal are resistance to water, weathering (ozone) resistance and resistance to mineral oil. Current dust seals made with chloroprene based elastomer in the black colour is not identifiable to the customers as it is similar to the mating part colour and hence customers forgot to assemble the dust seal in the device during service, due to which water and dust entry to the devices which affects the durability of the interior components like springs and sheet metal components. To avoid this failure, dust seal colour changed from black to red with the current material. Chloroprene dust seal with red colour does

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shanmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_59

Fig. 59.1 Blooming phenomena of dust seals



not meet the resistance against the water. Pugh matrix is used to sort the optimum material selection, and NBR-PVC blends are the best option in terms of economic and functional application.

NBR-PVC blends have become technologically important to rubber companies for replacement of chloroprene elastomers as a cost-effective and better property, especially oil resistance [1]. The presence of PVC facilitates improved ozone resistance of NBR, which has given the possibility of using as blends in automotive parts for pneumatic brake systems such as boots, bellows and dust seal application. To achieve better physical properties in NBR-PVC, sulphur donor accelerators like MBTS and TMTM are used as cross-linking agents for better cross-linking density. In WABCO India, we came across blooming defect in dust seal made of NBR/PVC blend during the development process. Blooming is characterized by discolouration or fading on the material surface. Dust seals are mainly used to prevent the dust and water entry to the devices used in the brake system. Main material properties required for dust seal are resistance to water, weathering (ozone) resistance and resistance to mineral oil. High sulphur content is essential for NBR-PVC blend to accelerate the curing [2]. But usage of high sulphur content in NBR-PVC blends resulted in blooming Fig. 59.1.

A discoloration or change in the appearance of the surface of the rubber due to the migration of solid or liquid materials/substances is termed as blooms in rubber [3]. The substance of crystalline in nature, which blooms must have limited but appreciable solubility in rubber and be present more than the solubility limit. This excess will exist as discrete particles throughout the mass of the rubber either because it has never dissolved or because, having dissolved at the temperature of vulcanization; it has crystallized out on cooling.

Sulphur blooming is a major issue in the rubber industry and is often avoided by substituting soluble sulphur with 'insoluble' sulphur. In this case, the processing temperature should be kept well below the transition point of the sulphur from its insoluble to soluble form, and it is around 105 °C. One more major addition in rubber that results in blooming is accelerators as they have limited solubility in organic rubbers. EPDM and butyl rubber vulcanisates are highly prone to accelerator bloom. Stearic acid, oils and lubricants are other additives that can cause blooming depending on their concentration and compatibility with the base rubber. To avoid blooming, it is often advised to use accelerators with higher solubility by using a combination of

accelerators instead of a single one and optimize the amount or use polymer-bound accelerators.

To eliminate blooming, various compounding with different accelerator systems and dosages are produced and tested for functional requirements. Blooming in NBR-PVC blend can be eliminated by reducing the sulphur content, but curing behaviour of the compound significantly gets affected by the decrease in sulphur content. An optimum amount of sulphur content with sulphur donor accelerators for dust seal is manufactured through rubber compounding and use of tri-system accelerator groups.

59.2 Methodology

59.2.1 Materials Used

Compound formulation details of NBR-PVC masterbatch blend and amount of chemical additions used shown in Table 59.1. Nitrile butadiene rubber (NBR) and polyvinyl chloride (PVC) were mixed in the ratio of 95:5.

Usually, NBR/PVC blends are mixed in the ratio of 70:30 or 50:50 for obtaining improved mechanical properties and ozone resistance [3]. For dust seal application, PVC with NBR blend was selected for obtaining the ozone characteristics. 95:5 (NBR: PVC) compound used with Mercapto-benzothiazole disulphide (MBTS) as a primary accelerator.

Table 59.1 Formulation of master batch

Ingredients	Value
Low ACN NBR	55.0
Medium ACN NBR	30.0
PVC NBR	15.0
Sulphur Rhenogran S-80	1.20
Zinc oxide active	3.0
Antioxidant	1.5
Precipitated silica	50.0
PEG 4000	4.0
Plasticizer DINP	15.0

Table 59.2 Accelerator dosage level for various trial compounds

Ingredients	T1	T2	T3	T4	T5	T6
Sulphur	1.2	0.5	0.5	0.5	0.5	0.5
MBTS	2.0	1.0	0.0	0.0	1.0	1.5
CBS	0.0	0.0	1.0	2.0	1.5	1.5
TMTM	2.0	1.0	1.0	2.0	1.0	1.0

59.2.2 *Compound Mixing and Cure Characteristics of NBR/PVC*

Compound masterbatch is mixed using ban bury intermix. Final batch mixing was carried out on two roll mills as per the ASTM D3182. Accelerators such as MBTS, CBS and TMTM were used to minimize the sulphur content. Cure characteristics determined by using rheometer as per the ASTM D2084. Various compounds were compression moulded at 170 °C under pressure of 150 kg/cm² in an electrically heated press to prepare the test slab and buttons. Dumbbell test specimens were cut from the sheets for different testings to evaluate material properties.

59.2.3 *Measurements*

With reference to the works of [1], the following range of 1–2 phr of accelerators was taken for experimental trials Table 59.2. Effects of various dosages with respect to mechanical properties and ageing characteristics were studied.

Tensile strength and elongation at break were measured on dumbbell specimens as per the ASTM D412. Tensile strength test was carried out based on FIE, 10 KN tensometer at 500 mm/min crosshead speed. The hardness test is done by using IRHD Wallace tester (International Rubber Hardness Degree) as per the ISO 37. The ageing property was studied after ageing in 100 °C for 70 h in the hot air oven.

59.3 *Outcome*

59.3.1 *Effect on Mechanical Properties and Vulcanization*

From Table 59.3, mechanical properties are better in the initial compound T1 (MBTS 2.0 phr and TMTM 2.0 phr with 1.2 phr of sulphur) compared to the other test samples. High mechanical properties are mainly due to the presence of high sulphur content leading to high compression set. Combination of high mechanical properties and high compression set are not ideal for the current application. High sulphur levels or

Table 59.3 Mechanical properties of the compounds

Properties	T1	T2	T3	T4	T5	T6
Hardness	75.2	74.4	74	75.2	70	70
Tensile strength	16.3	14	15	14.5	13.2	11.5
% Elongation at break	550	500	450	330	335	375
Compression set @ 24 hours 100 °C	35	42	45	38.7	32.8	29.5

higher dosage of TMTM in the NBR-PVC blend gives higher mechanical properties compared to the trials with low sulphur levels and tri-accelerator system.

From Table 59.3, T6 shows the lowest tensile strength and hardness compared to other compounds due to the low sulphur level and low TMTM accelerator level resulting in high cross-linking density but low polysulphide linkage. Low polysulphide links generally provide better compression set.

These results proved that the dosage of TMTM is influenced in increased mechanical properties [3]. We achieved higher mechanical properties in T4 comparable to T1 compound. The individual contribution of CBS is not identified in our results, but MBTS and CBS combination resulted in significant influence on mechanical properties.

Although the high sulphur content in di-accelerator systems (T1) provides better curing comparable to tri-accelerator systems such as T5 and T6 compounds (Figs. 59.2 and 59.3), compression set properties are not sufficient in the current application scenario. Also, higher sulphur content initiates reversion and more prone to blooming in the normal application. Mass content studies were not carried out in the current work, and hence effects of individual composition are not discussed. Their mass content will give more detailed overview over the oxidative nature, thermal stability and degradation steps.

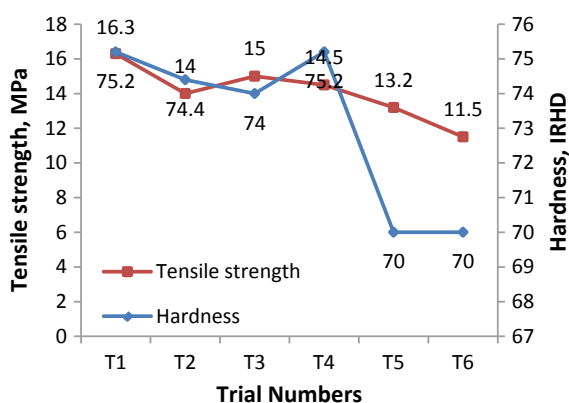
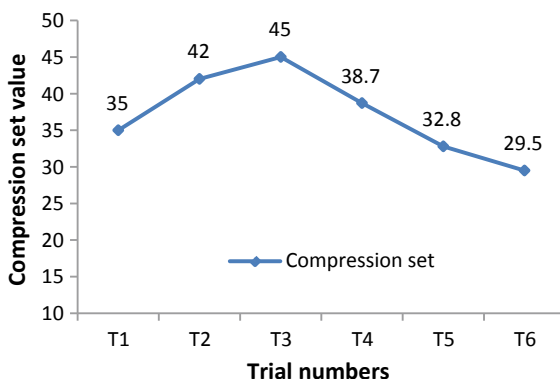
Fig. 59.2 Tensile strength and hardness of various trial compounds

Fig. 59.3 Compression set value taken for all samples for evaluating the curing efficiency



Scorch effects were also studied and verified to understand the curing properties of the various trial compounds. Normally, Mercapto accelerator groups provide low scorch safety as it normally gives accelerator effect in low temperature itself [4].

For delaying the scorching effect, CBS is normally added in the system to delay the scorch time as it forms simple S-N linkages. Along with CBS, MBTS was added in the system of T5 and T6 to improve the cross-linking density and also to get scorch safety. From Fig. 59.3, it is evident that the cross-linking density has improved comparably to other binary accelerator system. Sample T1 shows third-best combination due to the presence of high sulphur content which improves the cross-linking density and is also highly prone to migration of sulphur atom under adverse conditions [3]. Combination of moderate mechanical properties and lower compression set is optimum for the current dust seal application.

59.3.2 Effect of Ageing on the NBR/PVC Compounds

From Table 59.4, compound T4 is showing higher ageing resistance among all the compounds. Thermal stability of the NBR-PVC blend showed better stability when analysed through the TG scan—DSC studies [5]. Moreover from their studies, better thermal stability of the blend is due to the combination of Cl size and electronegativity and due to the absence of double bond in PVC [5]. Addition of TMQ as antioxidant further improves the thermal stability.

Table 59.4 Ageing properties of the compounds

Change in mechanical properties	T1	T2	T3	T4	T5	T6
Change in hardness	7	4.8	7	1.5	5.8	3
Change in Tensile strength, %	−14.8	9.9	−7.7	−2.3	−12.1	−7.9
Change in Elongation break, %	−32.9	−11	−17	−9	−14.3	−12

From the test results (Table 59.4), it is evident that the T4 and T6 sample possesses better stability comparable to the results of Pruneda et al. [5]. Contradictory to the previous research data [1, 5], we observed poor ageing characteristics in the remaining test samples due to the presence of high sulphur (T1), low accelerator dosage (T2) and tri-accelerator dosage (T5, T6). Poor thermal stability might be caused by the usage of volatile matter resulting in mass loss and increased curing time.

T4 sample shows better stability better than the samples with MBTS due to the presence of low sulphur and higher accelerator dosage. Similar effects were observed in the earlier test results published by Oliveira et al. [6] worked in the binary accelerator system and sulphur levels where low activation energy is recorded [6].

Based on the Rheograph studies, T6 samples show better scorch safety and flat curing plateau compared to other compounds. Flat curing plateau is generally preferred for avoiding reversion effects when subjected to higher temperatures.

59.3.3 Colour Test for Identifying Blooming

Blooming effect is studied through visual observation on various trial compounds. Dust seal produced from the various compounds is subjected to 48 h of natural ageing at room temperature. After 48 h of ageing, dust seals were visually examined for surface defects.

Figure 59.4 shows the various dust seals produced from the compounds, and T1 and T4 show signs of blooming defect. Most noticeable is T2, T3, T5 and T6 compounds dust seal did not show any signs of blooming. These characteristics observed in the samples T2, T3, T5 and T6 should be related to the effect of low level of TMTM accelerator which arrests the free sulphur atoms and restricts the migration of sulphur precipitation [3].

Even though TMTM restricts the migration of sulphur content, high dosage of 2.0 phr TMTM results in the provision of extra sulphur donors resulting in sulphur precipitation (T4 compound). In T1 samples, high sulphur content and high accelerator dosage favours the migration of sulphur atoms resulting in blooming.

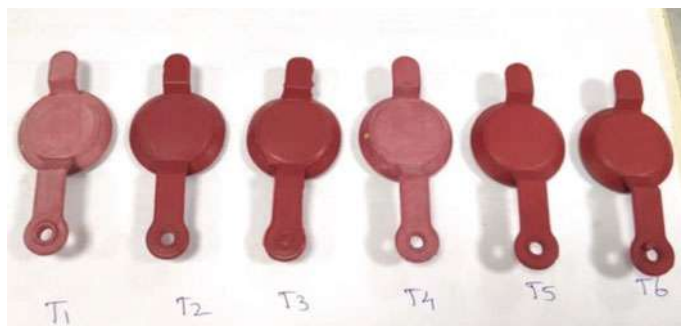


Fig. 59.4 Visual examination of dust seals

59.4 Conclusions

Higher sulphur content resulted in higher blooming phenomena. Reducing the sulphur content resulted in the reduction of blooming phenomena in the di-accelerator system (T2 and T3) but resulted in increasing the compression set. High dosage of accelerator in the di-accelerator system (T4) resulted in reverting of blooming phenomena due to the higher level of TMTM. Selection of tri-accelerator system for red dust plug compound eliminated the blooming phenomena as well as reduction of compression set. The dosage of 1.0 phr of TMTM and 1:1 ratio of MBTS and CBS provided an optimum solution for obtaining the blooming free compound with lowest compression set for external application such as dust seals.

Acknowledgements The authors would like to thank WABCO India Limited to support this project and like to express my gratitude to Mr. V. P. Balaji from WABCO for the valuable technical guidance and support.

The authors also would like to thank Mr. Nermaiselvan and M/S Roop Polymer management for the excellent support provided to carry out experimental trials.

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Chapter 60

Rapid Product Development from an Existing Product Using Reverse Engineering Method



G. Sen  and B. Doloi

Abstract Reverse engineering is a progression of the process which creates a 3D CAD model of an existing part without any engineering drawing. This technology can be utilized when an object having complex shape needs to be copied to produce a prototype. Reverse engineering starts with the collection of point cloud data of object by scanning and digitizing to develop a 3D CAD model. This 3D CAD model is used to produce the object/parts using various rapid prototyping (RP) techniques like fused deposition modelling (FDM). There are several application areas of reverse engineering. It can be utilized to duplicate a product, when unique original drawing or documentation is not accessible. In different cases, it can be used in re-designing an existing part, examination and alterations for enhancement of a product. The aim of this paper is to discuss the reverse engineering process and the different stages involved in reverse engineering and development of a prototype of step cylinder and gear from the existing product.

Keywords Reverse engineering · Rapid prototyping (RP) · Fused deposition modelling (FDM) · Gear · Step cylinder

60.1 Introduction

In the present competitive worldwide market, industries are always looking for new technology to shorten the product development time for new product developments that fulfil all customer expectations. Industries have invested in CAD/CAM, rapid prototyping and a range of new technologies that provide business benefits. Reverse engineering (RE) is the method that provides business benefits in shortening the product development cycle. Reverse engineering is characterized as the way towards generating a geometric CAD model from 3D points obtained by scanning/digitizing existing parts/items. The procedure digitally capturing the physical entities of a component, referred to as reverse engineering (RE), is frequently characterized by

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_60

researchers with respect to their specific task [1]. According to Abella et al. [2], the basic idea of creating a section based on an original or physical model without the utilization of engineering drawing is called RE. Yau et al. [3] describe RE as the way towards re-establishing a new geometry from an existing part by digitizing and modification of a current CAD model. Reverse engineering is presently utilized for manufacturing, industrial design and jewellery design and production.

There are different RP techniques which offer a quick and simple transition from generation of the CAD model to the manufacture of physical item. FDM is one of the RP techniques, but not like other RP systems, which include laser, powders, resins; this FDM process utilizes heated thermoplastic polymer in the form of filaments which are expelled from the tip of the nozzle in a temperature-controlled condition.

Lee et al. [4] described a reverse engineering system for rapid prototyping of products having complex surfaces. The system consists of three main components: a 3D optical digitizing instrument, surface generation software and a rapid prototyping machine. The surface generation software exports 3D image into STL format, which is used to develop a prototype using the sinterstation 2000 machine. Son et al. [5] had proposed a laser scanning system which automatically generates a solid surface 3D image irrespective of surface complexity. The automated system reduces the scanning time, improves the scanning quality and minimizes error. Sood et al. [6] studied the effects of orientation, layer thickness, raster angle, raster width, raster-to-raster gap with the help of Taguchi's method on dimensional accuracy. Significant factors and their influences have been found out using Taguchi's method. Hur and Lee [7] determined the slicing accuracy based on the user-defined cusp height. Cusp height is the maximum deviation from the layered part to the CAD surface measured in the direction normal to the CAD surface. More cusp height means lesser smooth surface and vice versa. In this technique, a small allowable cusp height is for important faces which are required to be smooth and a larger cusp height is for other relatively unimportant faces. Cao and Miyamoto [8] used a direct slicing approach from AutoCAD solid models to fulfil the requirement of quicker and precise slicing in RP.

The current research focuses on the development of the CAD model of step cylinder and gear from the existing product using a non-contact-type laser scanner and also analyses the deviation of geometrical features with the original product. The development of physical product and its comparison with the CAD model have been also discussed in this paper.

60.2 Experimental Setup

Sense 3D scanner is used to scan the object. It is a non-contact-type scanner which works on a laser triangulation principle. Figure 60.1 represents the photographic view of a 3D scanner.

FDM utilizes heated thermoplastic filaments which are expelled from the tip of the nozzle in a temperature-controlled condition. For this, there is a material deposition subsystem termed as the head which comprises of two liquefier tips: one tip for

Fig. 60.1 Photographic view
of a 3D scanner



support material and other is for deposition of material for making of parts. The material is supplied to the head in the form of a flexible strand of solid material from a supply source (reel). One pair of pulleys or rollers is utilized as the material advance mechanism to grip a flexible strand of modelling material and advance it into a heated dispensing or liquefier head. The head is attached to the carriage that moves along the X–Y plane. The build platform moves along the Z direction. The drive motion is provided to selectively move the build platform and dispensing head relative to each other in a predetermined pattern through input of the drive signals to the drive motors from CAD/CAM system. Figure 60.2 represents the photographic view of the FDM machine.

60.3 Development of the CAD Model from the Existing Object

Reverse engineering (RE) is a process of developing the CAD model directly from the existing object. The data capture of the existing product is done by the scanner. There are two types of scanners available: one is contact-type scanner and another is non-contact-type scanner. The example of a contact-type scanner is coordinate measuring machine. A 3D laser scanner is an example of a non-contact-type scanner. In this present research work, a non-contact-type 3D laser scanner has been used. The development of the CAD model involves various stages. The stages are shown in Fig. 60.3.

Fig. 60.2 Photographic view of the FDM setup

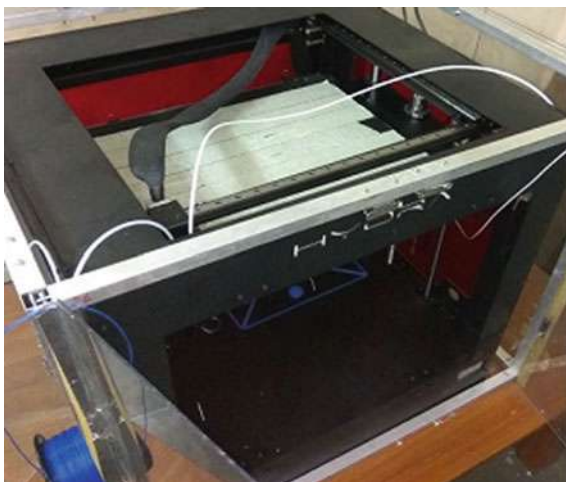
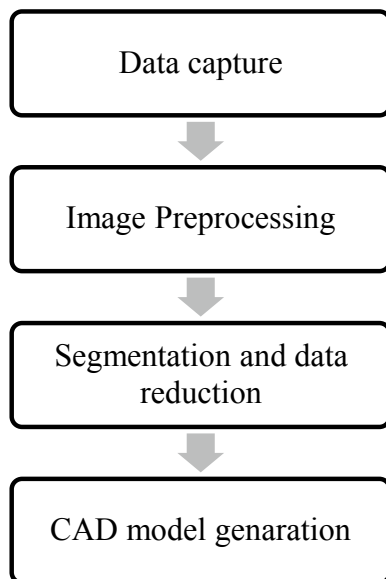


Fig. 60.3 Flow charts of reverse engineering



60.3.1 Data Capture

In the scanning phase, a 3D laser scanner is used to scan the object and to extract the geometrical features of the object with accuracy. This method is called laser triangulation. Laser triangulation is a technique which utilizes location and angle between light sources and photo-sensing device to derive position. A high-energy light source, laser, is focused and projected at a pre-specified angle on the surface

of interest. A photosensitive device, usually a camcorder, detects the reflection of the surface, and then by utilizing geometric triangulation from the known angle and distance, the location of a surface plane relative to a reference plane can be calculated. By triangulation method, data can be captured at quick rates. The exactness is dictated by the resolution of the photosensitive device and the distance between the surface and the scanner.

60.3.2 Image Processing

The point cloud data is then used to generate the surface in a 3D system software.

60.3.3 Segmentation and Data Reduction

Segmentation has been an essential part of the process of surface modelling from scanned point data. It is not possible to control the area of interest in a 3D sense scanner. Some part of the surrounding data is also captured in point cloud data. It is the process of partitioning a point cloud into meaningful regions or extracting important features from the point data. The point data segmentation is done by edge-detection methods. The process segmentation has been done in a 3D system sense software.

60.3.4 Generation of the CAD Model

After reduction of data in the segmentation phase, the CAD model has been generated using a 3D system sense software. In this present work, the diametrical deviation and circularity error of the CAD model of step cylindrical part as well as gear have been determined and variation in these with respect to the actual part has been analysed.

Development of the CAD Model of the Cylindrical Geometrical Feature A step cylindrical object has been taken for the experiment. Figure 60.4 shows the photographic view of the step cylinder. The CAD model is developed based on the image of the object scanned by a 3D scanner around the object.

The CAD model has been exported using Autodesk Fusion 360 software as shown in Fig. 60.5 for measurement of diametrical deviation of the CAD model with respect to the original product. Table 60.1 shows the diametrical deviation of the CAD model at various heights.

From Fig. 60.6, it has been observed that diametrical deviation is more at the bottom portion of cylinder (up to 20 mm). In the upper portion (80–95 mm), the variation in diametrical deviation is less.

Fig. 60.4 Photographic view of original cylindrical object



Development of the CAD Model of Gear A gear has been taken as an existing product for the experiment. Figure 60.7 shows the photographic view of the gear. The CAD model of gear is developed based on the scanned image by a 3D scanner around the object.

The CAD model has been exported using Autodesk Fusion 360 software as shown in Fig. 60.8 for measurement of the outer circle and inner circle diameter of the CAD model with respect to the original product. The internal diameter of the original gear is 25.23 mm and the outer diameter is 59.38 mm. The inner diameter of the CAD model is 24.528 mm and the outer diameter of the CAD model is 56.784 mm as shown in Fig. 60.8.

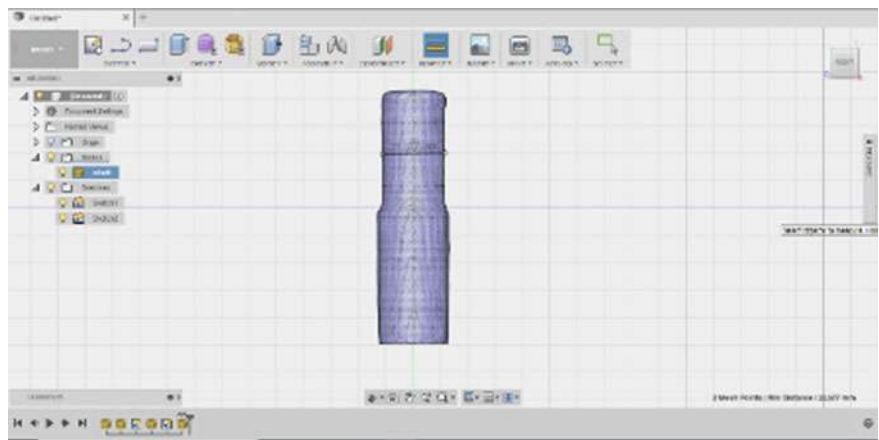


Fig. 60.5 Photographic view of the CAD model of the cylindrical part

Table 60.1 Diametrical deviation of the CAD model of the cylindrical part

Height from the base (mm) (H)	Diameter of the object (mm)	Diameter of the CAD model (mm)	Diametrical Deviation (mm) (DD)
0	25.32	24.18	1.14
5	25.83	25.33	0.5
10	25.21	25.61	0.4
15	25.65	25.26	0.39
20	25.36	25.14	0.22
25	25.26	25.43	0.17
30	25.33	25.36	0.03
35	25.61	25.38	0.23
40	25.12	25.38	0.26
45	22.31	22.29	0.02
50	22.13	22.41	0.28
55	22.07	22.32	0.25
60	22.54	22.49	0.05
65	22.41	22.18	0.23
70	22.17	21.91	0.26
75	22.21	22.18	0.03
80	22.34	22.19	0.15
85	22.38	22.52	0.14
90	21.93	22.57	0.64
95	22.28	22.06	0.22

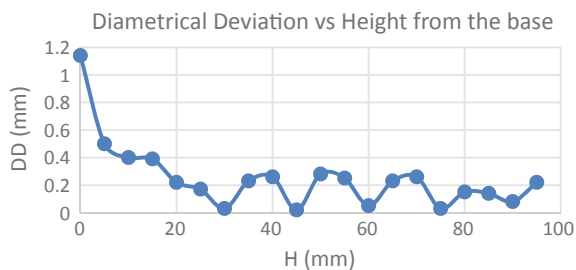


Fig. 60.6 Variation in diametrical deviation versus height from base



Fig. 60.7 Photographic view of a original gear

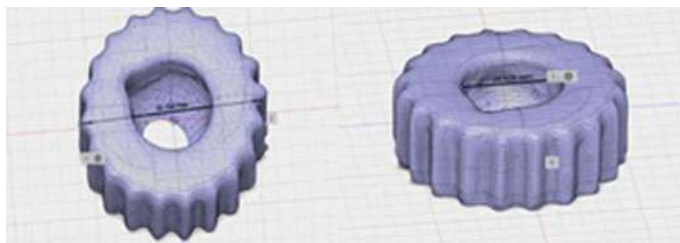


Fig. 60.8 Photographic view of the CAD model of gear

60.4 Development of a Prototype Using the CAD Model

The CAD model has been generated using reverse engineering to develop a new prototype. There are various rapid prototyping techniques to produce the prototype using the CAD model. In this current research, a prototype has been developed using fused deposition modelling (FDM). FDM Hydra 400 machine has been used to develop a prototype. CURA 2.0 software has been used to slice the CAD model. Polylactic acid (PLA) has been used as the thermoplastic polymer in this research.

Ultimaker CURA 2.0 software has been used to select the process parameters. There are five process parameters such as layer thickness (mm), bed temperature (°C), extruded temperature (°C), print speed (mm/sec), and fill density (%).

60.4.1 Development of Cylindrical Prototype

The CAD model of the cylindrical part developed by scanning has been sliced by CURA 2.0 software. CURA 2.0 software generates G code programme. The head and bed of FDM machine follow this G code programme using a stepper motor to develop the cylindrical prototype.

Figure 60.9 shows the photographic view of the CAD model for generating STL of cylindrical prototype file in CURA 2.0. Figure 60.10 shows the outer layer and inner layer structures of step cylindrical part. Table 60.2 presents diametrical deviation and circularity error of fabricated cylindrical part with respect to the CAD model.

Figure 60.11 shows the graphical representation of the variation in diametrical deviation of step cylinder at different heights from the base of the fabricated cylindrical part with respect to the CAD model. The diametrical deviation is less in the middle part compared to the top and bottom portions of step cylinder.

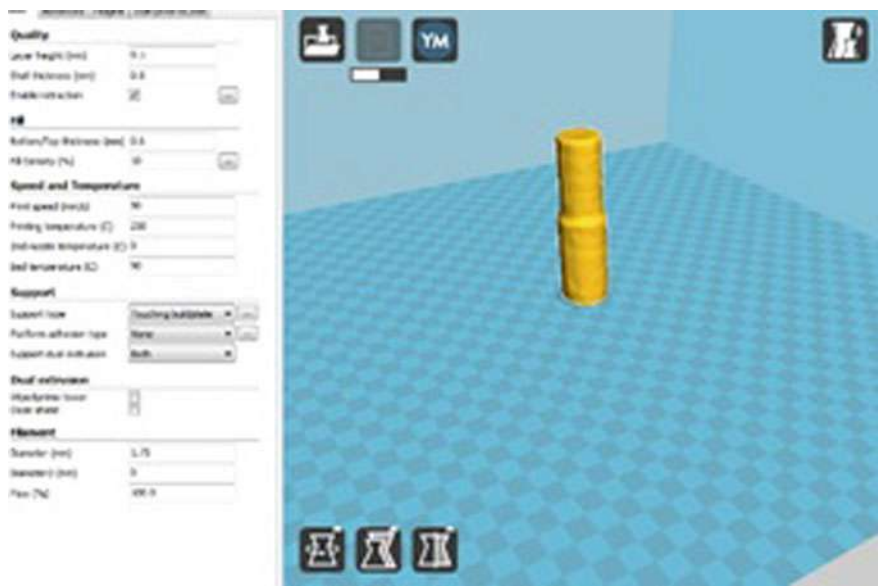


Fig. 60.9 Step cylinder STL file in CURA

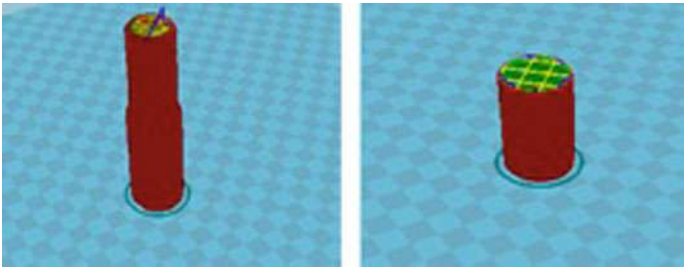


Fig. 60.10 Outer layer and the inner layer

Table 60.2 Diametrical deviation and circularity error of cylindrical prototype

Height from the base (mm) (H)	Diameter of the CAD model (mm)	Diameter of prototype (mm)	Diametrical deviation (mm) (DD)	Circularity error (mm) (CE)
0	24.18	24.90	0.72	0.34
5	25.33	25.52	0.19	0.21
10	25.61	25.84	0.23	0.08
15	25.26	25.57	0.31	0.14
20	25.14	25.28	0.14	0.11
25	25.43	25.37	0.06	0.17
30	25.36	25.62	0.26	0.10
35	25.21	25.73	0.52	0.16
40	25.38	25.77	0.39	0.34
45	22.29	22.68	0.39	0.41
50	22.41	22.28	0.13	0.26
55	22.32	22.24	0.08	0.37
60	22.49	22.13	0.36	0.42
65	22.18	22.07	0.11	0.13
70	22.56	22.17	0.39	0.06
75	22.18	22.22	0.04	0.21
80	22.19	22.34	0.15	0.31
85	21.52	22.17	0.35	0.44
90	22.57	22.14	0.43	0.46
95	21.06	21.21	0.15	0.43

Fig. 60.11 Variation in diametrical deviation versus height from base

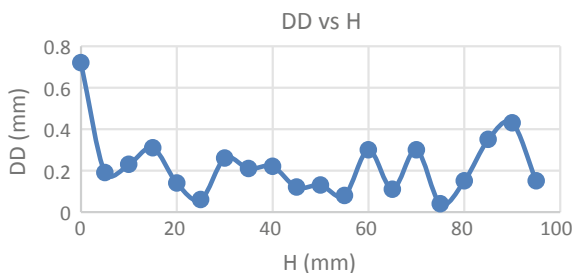


Fig. 60.12 Variation in circularity error versus height from base

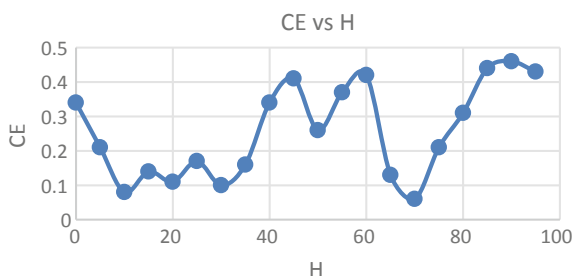


Figure 60.12 shows the graphical representation of circularity error at different heights from the base. Circularity error is less in the lower part of the step cylinder compared to the upper part. Figure 60.13 shows the photographic view of the cylindrical prototype.

60.4.2 Development of Gear Prototype

The CAD model of gear developed by scanning has been sliced by CURA 2.0 software and G code programme. The head and bed of FDM machine that follow this G code are generated to develop the gear prototype using FDM technique of rapid prototyping.

Figure 60.14 shows the photographic view of the CAD model for generating STL file of gear prototype using CURA 2.0. Figure 60.15 shows the outer layer and inner layer structures of gear the CAD model. The inner diameter and the outer diameter of gear prototype are 23.69 mm and 55.78 mm, respectively. Figure 60.16 shows the photographic view of a developed gear prototype.

Fig. 60.13 Photographic view of a cylindrical prototype



60.5 Conclusions

Reverse engineering (RE) concept has been applied to develop the CAD model of existing product of different geometrical feature using a 3D laser scanner. Fused deposition modelling (FDM) method has been utilized to develop a prototype of various geometry for testing a 3D model generated by the RE concept.

From the study of the CAD model of step cylinder, it has been observed that the average diametrical deviation of the CAD model for step cylinder is 0.318 mm. The diametrical deviation is more in the bottom portion of cylinder (up to 20 mm). In the upper portion (80–95 mm), the diametrical deviation is less.

From the analysis of the prototype of a step cylinder, it is observed that the average diametrical deviation of developed step cylinder is 0.27 mm. The diametrical deviation of step cylinder is less in the middle part compared to the top and bottom portions. The average circularity error of developed step cylinder is 0.28 mm. The

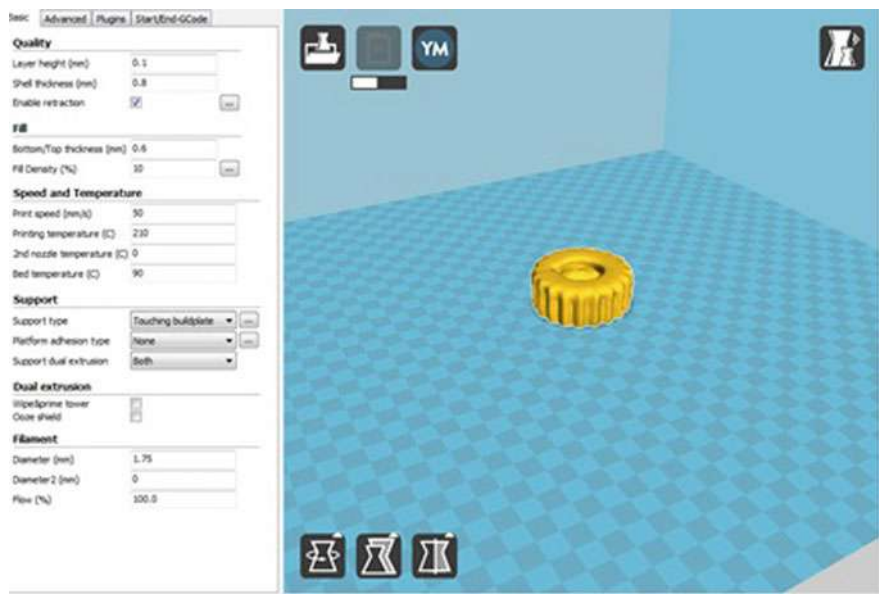


Fig. 60.14 Photographic view of STL of a gear file in CURA

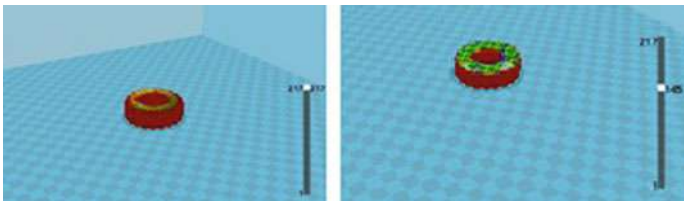


Fig. 60.15 Outer layer and the inner layer



Fig. 60.16 Photographic view of a gear prototype

circularity error is less in the lower position of the step cylinder compared to the upper part.

From the study of the CAD model of gear, it has been found that the deviations in the inner diameter and the outer diameter are 0.702 mm and 2.596 mm, respectively. From the analysis of a prototype of gear, it is observed that the diametrical deviations in the inner and the outer diameters are 0.838 mm and 1.004 mm, respectively.

Reverse engineering can be used to develop a new product from an existing one when the original design is not available. It may also be used to modify the design of an existing product. Due to material and machine restriction, it is not possible to develop the original product in the present work. But rapid prototyping techniques like selective laser sintering can be used to develop the metallic product from the CAD model generated by reverse engineering concept from an existing product. The recent development in this area may open a new horizon in the sector of rapid and green manufacturing.

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Chapter 61

Productivity Improvement by Reduction of Cycle Time Through Implementing Clustering: A Case Study



Satbir Singh and Sandeep Singhal

Abstract Productivity is the performance paradigm implying the transformation of man power and material sources into essential goods and utilities. Instant study is concerned with a small-scale production unit near Ambala, Haryana, manufacturing quality tractor parts and fulfilling the monthly requirement of big customers such as Swaraj tractors, Standard, Preet, and Sonalika tractors. The primary goal of the implemented research was to examine the determinants required for reduction of cycle time and betterment of productivity at the manufactory level. The recommended clustering for manufacturing the intended components is designed by developing a universal setup to target the non-productive elements, i.e., setting time. Needful was achieved by stimulating the monthly production and dropping the component manufacturing cost by way of reducing its cycle time by employing clustering principle. *p*-chart for fraction defectives employed as a statistical tool. Experimentation reveals that validating improved processes and tooling, a total productivity improvement of above 10% was observed.

Keywords Clustering · Cycle time · Monthly production-rejection · Manufacturing cost · Productivity improvement

61.1 Introduction

61.1.1 Productivity Relevance and Scope

The phrase “Productivity” has graced a familiar and house word as virtually everyone remarks about it. Even then “productivity” has a different meaning to different persons. As an outgrowth, its extent varies from efficacy to effectiveness, to degrees of absenteeism, and turnover, to output measures, to the level of consumer satisfaction, to intangibles such as workflow disruption, and intangibles like loyalty, morale, and

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_61

job comfort. Productivity is the efficiency criterion with which material and human resources are dually converted into required service-program, utility, and subsequent goods. The productivity means stability among all the elements of production which end up in maximum output with the least attempt (**Peter Drucker**). The rate of production to an individual category of input is partial productivity, e.g., labor productiveness. According to the European Productivity Council (**EPC**), productivity is non-stop variation of commercial and social life to changing situations. Concept-wise productivity and production totally differ. Production concerns to total output, and at the same time productivity is a relative aspect wherein output evermore expressed in terms of input. Production means the output in money form, while productivity is the efficiency of the production system.

61.1.2 Clustering Principle

The technique of performing a bunch of manufacturing activities in a special setup, i.e., in a typically shared working plane to contribute to a reduction in setting time, viz. non-productive time is apprehended as clustering.

61.1.3 Current Status of Research

Productiveness measurements and evaluation have extensively emphasized in recent years. However, little attention turned into targeted at optimizing productivity. Productivity planning is a complex activity due to the restriction put by the extreme diversity of manufacturing. Extensive work on productivity analysis and measurement has been put forward by many researchers. Even then an unlimited quantity of research work is presently being achieved to optimize the available resources. The contribution of some of the researchers toward productivity improvement is as under.

Productiveness improvement approach and quality profile must alter emulating the operating development, quality measures, and overall economic performance [1]. It has been maintained that the decline in work-in-process (WIP) inventory has contributed toward gain in productiveness [2]. Water productiveness increases with water-saving irrigation, however, yield diminishes. Yield and field-level water productivity can be better via enhancing total factor productivity. Conserving water in one region and irrigating land in some other vicinity increases rice production [3]. Due to new technologies, TFP accelerated in common through 4.4% in step with 12 months for the length 1964–1993. For the duration of the years 1964–1982, trend has been alternatively increasing, while it has been declining by 0.33% annually throughout the years 1983–1993 [4]. Marketing needs to be treated as manufacturing. Productiveness can also be accelerated by way of reducing the cycle time and statistically controlling the production operation [5]. Up to what degree monetary productiveness is contributed through agglomeration economies? From the findings,

it's miles clear that market accession through enrichment in infrastructure is a critical deciding issue for company stage productiveness [6]. Best for units with low export intensities, productivity adds on drastically as a result of outsourcing of materials [7]. In dry rain-fed regions, supplemental irrigation (SI) is the existence sustaining technique with sound potential for boosting agricultural production and enhancing productivity. By directing rainwater to the flora through runoff, water harvesting can improve agriculture [8]. With modernized and optimized irrigation structures, water productivity is projected to be expanded via alleviation in irrigation water and additions in crop yield [9]. As a strategy prospect, it is tricky to interpret whichever aspects carry the productivity comparison among firms and which are included in higher productivity [10]. Guidelines contributing individual manufacturers to handle heterogeneity in the prices can result in a sizeable reduction in measured total factor productivity (TFP) and output [11]. An across the world-related data-based analysis of aggregate labor productivity exhibits that low agricultural-labor productivity and an excessive share of employment are vitally answerable for low gross productiveness as far as the developing countries globally are involved [12]. The results indicated that productiveness increase turned into vitally attributable to technological advancement and annual productivity growth became recorded 3.3% throughout the comprehensive universities [13]. The principal potential benefit of business liberalization has been increasing productivity in domestic enterprises [14]. For method validation and advancement, problem-resolving method, viz. defining, measuring, analyzing, improving, and controlling (DMAIC) turned into applied for evaluation of the problem. For system substantiation and betterment, Taguchi's experimental layout has been used [15]. The software was used to systematically present and analyze the methodology to increase productivity at the company level [16]. Curtailing cyclic time needed to produce the part; productiveness improvement may be stimulated by figuring out various wastes held up in the course of different manufacturing lines and selecting the important one [17]. Cycle time optimization performs a life-maintaining role in productiveness dependence. Productiveness has been increased through the use of the optimized processing parameter systematically. By reducing the cycle time, substantial enhancements are carried out [18]. One of the prime objectives is to recognize and eliminate waiting or idle time for improving productivity and lessening production cost [19]. The experimentation revealed that employing recommended processes together with advanced tools; month-to-month production was improved via above 16% because of decreased cyclic time [20]. Determinants reallocation across enterprises together with shifting of laborers plus resources to plants with higher productiveness contributed significantly toward an increase in labor as well as capital productivity including total factor productivity, particularly throughout the period 2003–2014 [21]. The intended approach stimulated productivity growth by using the advanced manufacturing concepts, which led to decreased monthly rejections emerging out of producing essential parts [22]. Production twist is the principal cause of misallocation. Determinants of changes in allocation performance might be involving increased competition in homey markets, stronger credit supply, and legal concerns [23]. Results imply that administration possesses features regarding both of the technology and input [24]. The stated research results in

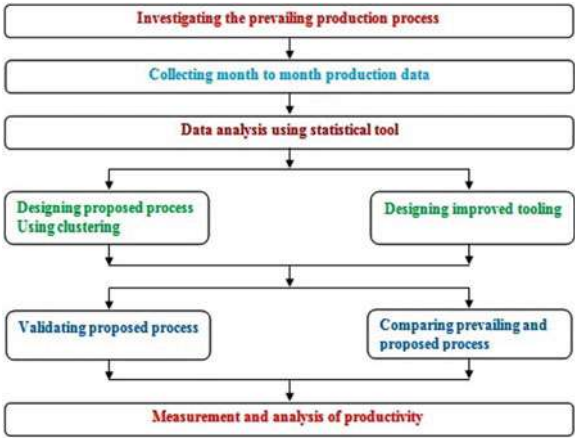
reduced cost in addition to the denial of unfruitful processing and also offers proper working conditions. Implementing correct sequence of the methods mentioned above increases manufacturing productiveness [25].

As a consequence of reviewing the contemporary literature, it's far relevant to affirm that productivity has an outstanding task to fulfill in manufacturing companies. Most of the sound strategies practiced via researchers precisely focused on reducing lead time to decrease the machining price. Contrarily the current scenario covets evaluation for **multi-objective functions** such as lessening flow distance to decrease the material handling expenses, space required, labor cost, cost of tools, and the quantum of machine required. Manhood researchers' focus on nurturing the partial productivities, but the primary requirement of the present period is to amplify the overall productiveness, viz **holistic productivity**, without negotiating and conceding on **purposes, characteristics, and protection**.

61.2 Methodology

The objective of the suggested methodology is to implement a case-based approach to demonstrate how the available resources while used correctly and effectively can help the enterprise get rid of waste and improve product excellence and productivity. To carry out numerous operations on multiple machines, enforcing traditional manufacturing techniques consumes extra time, calls for ample space, labor cost, material handling price, transportation/delivery cost, and greater tooling expenses; therefore to restrict the tremendous rise in total production cost and drastic fall in productivity, some advanced techniques implemented for manufacturing the desired parts. Resources applied efficiently, and workpiece holding devices made are capable of quick loading-unloading to speed up the production process. The management of the corporation expected improvement in productivity by increasing the monthly production of vital components by targeting their cycle time and also reduce monthly rejection rate by clustering different possible operations, using combination tooling.

In producing critical components, the suggested methodology entails data compilation for the month-to-month production of the company with similar rejection, i.e., defectives. The quality status of the manufactured parts comes to be monitored and analyzed by employing the technique of statistical sampling inspection and making use of acceptance sampling through attributes, which include, "GO and NOT GO" gauges. While analytical strategies are of exceptional importance among diverse methods for statistical control, *p*-chart practiced as an interpreter for statistical analysis. The underlying flow diagram displays the revolutionary methodological steps needed to assure the betterment of productivity.



Flow diagram for methodology adopted

61.2.1 Investigating the Prevailing Production Process

Distinctive components manufactured via the enterprise with machinery used and equipment installed were studied to investigate and analyze the present manufacturing method. A principal component produced within the organization, i.e., lift arm is shown in Fig. 61.1.

Description of existing operations, different machines needed, and time consumed to carry out the required operation on the individual machine, the cost of producing the above-cited element is exhibited in Table 61.1 as under given.

Table 61.1 indicates operation time chart for the prevailing system including special operations with their collection necessary for producing the stated issue, multiple machines required, the hourly charge of machines, time elapsed on every



Fig. 61.1 Lift arm top view

Table 61.1 Existing operation time chart for lift arm

Op. No.	Operation description	Machine	Avg. cycle time (min)	M/c rate (Rs. per hr)	Part cost (Rs.)	Helper cost (Rs.)	Cost of material handling (Rs.)
1	Single and double eye ends rough milling, rough boring bar Ø41.8 on single eye end, drilling Ø18.5 on one end of double eye end, drill Ø6.8 tapping for M8 on fork end, 15° chamfering on Ø41.8, Ø18.8 hole milling at double eye end, and finish reaming Ø19 H11. All the above operations carried out on lift arm outer edge	HMC	5.0	480	40.00	1.50	1.00
2	Maintain dimension 49.1 mm between two faces of fork end	V-MILL CONV	2.5	150	6.25	0.75	–

(continued)

Table 61.1 (continued)

Op. No.	Operation description	Machine	Avg. cycle time (min)	M/c rate (Rs. per hr)	Part cost (Rs.)	Helper cost (Rs.)	Cost of material handling (Rs.)
3	Rough milling on single eye end and facing of the fork end (inner face), face milling Ø80, drill hole Ø18.5, hole mill Ø18.8, on fork end, outer face on the single eye and inner face on fork end finish mill, the inner side of fork end reamer Ø19H11	VMC	3.0	375	18.75	0.90	0.50
4	Tapping M8X1.25 on fork end	R-DRILL CONV	5.0	150	12.50	1.50	–
5	Broaching of splines	BROACH CONV	3.0	150	7.50	0.90	0.50
6	Deburring and visual checkup	Manual	1.0	–	1.00	–	–

machine to complete the essential operation, and machining cost per component in producing the so-referred to as vital element lift arm. Following assumptions were looked after for making ready the operation time chart proven shown in the table.

- For different machines, hourly rate shown in Table 61.1 includes machine buying cost, interest, salaries acquired, tooling cost, the power consumed, miscellaneous expenditures/overheads and depreciation.
- Machining cost was figured out by multiplying the hourly rate of a machine with time elapsed on each of the machines for performing various operations needed to produce the stated element.
- The manufacturing cost of a component includes machining cost, assistant/helper cost, material handling, and transportation cost.
- Attendant/apprentice rate granted per hour was Rs. 18.
- Material handling and carrying/shipping cost fixed was a Rs. 1/-for elongated movement while Rs. 0.50 similarly for a smaller distance inside the machines.

61.2.2 Data Collection and Statistical Analysis for Monthly Production

Relevant data for production with resulting rejections for the crucial component arranged for eighteen-month period on a month-to-month basis fulfilling the existing and proposed method to check the anomaly in the monthly production and appraise the defectives, viz. rejection occurring in the vital part manufacture. Taking month-to-month production with N number of components as the volume of sample, and with n number of defectives, is to be investigated statistically practicing *p*-chart in which

fraction defectives on monthly basis, $P = n/N$, higher limit of control, $UCL = \bar{p} + 3\sigma$,
and lower limit of control, $LCL = (\bar{p} - 3\sigma)$ where Standard deviation, $\sigma = \sqrt{\frac{\bar{p}(1-\bar{p})}{N}}$ and Centerline, $CL/\bar{p} = \sum \frac{n}{N}$.

Table 61.2 exhibits demographic attributes for *p*-chart obtained from existing monthly basis production/rejection pattern of the essential part for the last 9 months, viz. from May 2013 to Jan 2014.

The regular partial productivity listed for various additives, viz. materials, labor-capitals-machines, and overall productiveness employing month-to-month data collected for production-rejection of the principal part, i.e., lift arm, based on the information given below by the accounts division of the corporation applying existing manufacturing system has been exhibited in Table 61.3.

Selling Price per component (Rs.) : 390 = 00
Material cost per component (Rs.) : 265 = 00
Production cost per component (Rs.) : 93 = 55

Table 61.2 Statistical attributes for p -chart using exiting process

Sample No.	Month with Year	Monthly production as sample size (N)	Defectives as faulty components (n)	Fractional defectives (P)	UCLP for p -chart ($\bar{p} + 3\sigma$)	LCLP for p -chart ($\bar{p} - 3\sigma$)
1	May 2013	920	45	0.0489	0.0711	0.0281
2	June 2013	1015	55	0.0542	0.0700	0.0291
3	July 2013	870	45	0.0517	0.0716	0.0276
4	Aug 2013	975	50	0.0513	0.0704	0.0286
5	Sep 2013	925	46	0.0497	0.0710	0.0282
6	Oct 2013	890	41	0.0460	0.0714	0.0277
7	Nov 2013	875	42	0.0480	0.0716	0.0276
8	Dec 2013	970	50	0.0515	0.0705	0.0286
9	Jan 2014	880	39	0.0443	0.0715	0.0276

Centerline, $CL/\bar{p} = \sum \frac{n}{N} = 0.0496$

Table 61.3 Monthly existing material, labor-capital-machine, and overall productivity

Month with Year	Production revenue (Rs.) [x]	Material cost (Rs.) [y]	Labor-capital-machining cost (Rs.) [z]	Material productivity [x/y]	Labor-capital-machine productivity [x/z]	Overall productivity [$x/y + z$]
May 2013	341,250	243,800	86,066	1.400	3.965	1.035
June 2013	374,400	268,975	94,953	1.392	3.943	1.029
July 2013	321,750	230,550	81,389	1.395	3.953	1.031
Aug 2013	360,750	258,375	91,211	1.396	3.955	1.032
Sep 2013	342,810	245,125	86,534	1.398	3.962	1.034
Oct 2013	331,110	235,850	83,260	1.404	3.976	1.038
Nov 2013	324,870	231,875	81,856	1.401	3.969	1.036
Dec 2013	358,800	257,050	90,744	1.396	3.954	1.032
Jan 2014	327,990	233,200	82,324	1.406	3.984	1.040

61.2.3 Implementing Designed Process and Tooling

To substantiate the above-explained methodology, underneath referred to fundamental aspects are taken into deliberation to reduce cycle time and monthly defectives which result in a reduction of component manufacturing cost.

- Surface accuracy improved by enforcing superior tooling, i.e., carbide combination tools.

- By using a small total of settings and replacing conventional machines with CNC machines, complete rigidity ensured, resulting in a reduced imprecision.
- Stable supporting and robust holding in addition to firm clamping of the work-in-process part on VMC affirmed by an developed expandable universal setup to hold and clamp particular sizes of the vital part, viz the lift arm, used for numerous variants of the tractor.
- By way of drilling the required holes correctly in a stable casted boss in place of cast holes, concentricity is better assured.

By carrying out particular operations on an especially developed universal setup implementing clustering principle results in the reduced idle period, i.e., setting time, which hence manages decreased parts manufacturing costs. Improved tools, viz. carbide tooling adds tool life as an outgrowth of inordinately maintaining the superior cutting ability and additionally performing a supplemental number of processes in continuation with the same cutting point without re-sharpening. In manufacturing the vital part, viz lift arm, the number of machines needed become reduced to four of five by executing clustering principle by way of using the developed universal setup. The vital component so-called lift arm clamped in the above-stated fixture is shown in Fig. 61.2.

The operations time chart for the improved process is exhibited in Table 61.4. As an outcome of cheaper helper and machining cost due to curtailed cycle time, the unit production cost has gone drastically down in comparison with earlier costs, through engaging lower hourly rate machine. Operation number one, three, and four, of Table 61.1, performed in clustering employing single setup on a vertical machining center enforcing advanced tooling. A unique operation numbered as one in Table 61.4 manifests the clustered operations stated above. Manufacturing cycle time for the essential component reduced drastically due to performing gang milling

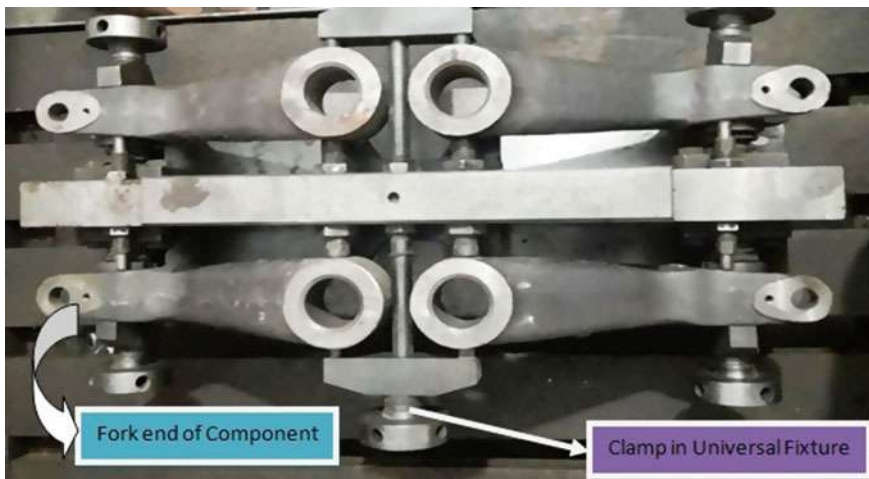


Fig. 61.2 Lift arm (4 numbers) firmly clamped in the developed universal setup

Table 61.4 Improved operation time chart

Op. No.	Operation description	Machine	Avg. cycle time (min)	M/c rate per hr (Rs.)	Part cost (Rs.)	Helper cost (Rs.)	Material handling cost (Rs.)
1	Single and double eye ends rough milling, finish mill of single and double eye end, maintain 54.5 mm dimension, rough boring bar Ø41.8 and finish bore Ø42, on single eye and chamfer 30°xØ50, drill Ø6.8 for M8tapping on fork end, tapping M8X1.25 on fork end, Ø18.5 hole drilling double eye end, and Ø19 H11 finish reamer. All above operations to be performed on the outer edge lift arm in clustering	VMC	6.0	375	37.50	1.80	1.00

(continued)

Table 61.4 (continued)

Op. No.	Operation description	Machine	Avg. cycle time (min)	M/c rate per hr (Rs.)	Part cost (Rs.)	Helper cost (Rs.)	Material handling cost (Rs.)
2	Face milling for Ø80, rough milling of the single eye and maintain dimension 50 mm. Maintain dimension 49.1 mm between two faces of single eye end and obtain 17 mm dimension as the thickness on the outer edge of fork end	V-MILL CONV.	5.0	150	12.50	1.50	0.5
3	Chamfer 15° on Ø42, on opposite face of single eye end	PILLER DRILL M/C	1.5	150	3.75	0.45	–
4	Broaching of splines	BROACH CONV.	3.0	150	7.50	0.90	0.5
5	Deburring and visual checkup	Manual	1.0	–	1.00	–	–

Table 61.5 Statistical attributes for the improved process

Sample No.	Month with year	Monthly production as sample size (N)	Defectives as faulty components (n)	Fractional defectives (P)	UCLP for p -chart ($\bar{p} + 3\sigma$)	LCLP for p -chart ($\bar{p} - 3\sigma$)
1	April 2014	1060	28	0.0264	0.0398	0.0108
2	May 2014	1180	29	0.0246	0.0390	0.0116
3	June 2014	1005	25	0.0249	0.0402	0.0104
4	July 2014	1125	30	0.0266	0.0393	0.0113
5	Aug 2014	1070	26	0.0243	0.0397	0.0109
6	Sep 2014	1030	26	0.0252	0.0400	0.0106
7	Oct 2014	1000	25	0.0250	0.0401	0.0104
8	Nov 2014	1122	29	0.0258	0.0394	0.0112
9	Dec 2014	1015	25	0.0246	0.0400	0.0105

Centerline, $CL/\bar{p} = \sum \frac{n}{N} = 0.0253$

on the V-Mill to complete operation number two of Table 61.4, achieving comparable gains in a month-to-month production and consequent productivity.

Table 61.5 indicates statistical attributes for developing p -chart for the successive nine-month period, i.e., April 2014 to Dec 2014 using improved monthly basis production/rejection pattern.

As a result of implementing the proposed process, partial productivity measure comprehended on a monthly basis in its rare forms, viz. materials, labor-capital-machines, and overall productivity taking under-referred factors for the reduced manufacturing cost into consideration is presented in Table 61.6.

Selling Price per component (Rs.) : 390 = 00

Material cost per component (Rs.) : 265 = 00

Manufacturing cost per component (Rs.) : 68 = 90

61.3 Results and Discussion

Consequent upon the implementation of proposed study and experimentation, the company has started a post-manufacturing inspection for every tenth piece instead of each fifth one by way of excellent surface finish together with better accuracy attained on manufactured parts. Resultantly, costs for inspection and quality control got exceptionally reduced. VMC takes 6.0 min including the air cut time in performing the clustered operations. As a consequence of enforcing the advanced

Table 61.6 Improved monthly material, labor-capital-machine, and overall productivity

Month with year	Production revenue (Rs.) [x]	Material cost (Rs.) [y]	Labor-capital-machining cost (Rs.) [z]	Material productivity [x/y]	Labor-capital-machine productivity [x/z]	Overall productivity [x/y + z]
April 2014	402,480	280,900	73,034	1.433	5.511	1.137
May 2014	448,890	312,700	81,302	1.436	5.521	1.139
June 2014	382,200	266,325	69,245	1.435	5.519	1.138
July 2014	427,050	298,125	77,513	1.432	5.509	1.136
Aug 2014	407,160	283,550	73,723	1.436	5.522	1.140
Sep 2014	391,560	272,950	70,967	1.435	5.517	1.139
Oct 2014	380,250	265,000	68,900	1.434	5.519	1.138
Nov 2014	426,270	297,330	77,306	1.433	5.514	1.137
Dec 2014	386,100	268,975	69,934	1.435	5.520	1.139

manufacturing system and tooling, i.e., engaging expandable universal setup for manufacturing vital part, viz. lift arm, the production cycle time presently computed as 16.5 min is presented in Table 61.4. Initially, employing the existing manufacturing process, the stated cycle time was 19.5 min as indicated in Table 61.1. The month-to-month production tremendously enhanced by above 15%, while part production costs got drastically curtailed by higher than 26%. By carrying out the operation of gang milling on vertical milling through applying advanced tooling on VMC, better surface finish with close dimensional tolerances is effortlessly achievable. Improved tolerance limit of ± 0.02 mm obtained on manufactured parts is closer than the existing limit of ± 0.05 mm, and therefore arising fewer defectives comparatively, i.e., reduced rejections up to 2%, earlier the corresponding became touching 4%. Succeeding high-speed steel tools on conventional as well as CNC machines, by carbide combined tools, has produced over 14% reduction for tooling cost. Average productivity for material increased by 2.58%. Average capital, labor, and machine productivity have multiplied by more than 39%. An improvement of above 10% turned into recorded for the average total productivity.

Advanced tools, viz. carbide tooling, withstand incredibly high temperatures and retain cutting points active for a relatively long interval of time without re-sharpening, due to an excellent red hardness and toughness property, consequently, maintain a longer tool life. Cutting speed and feed considerably improved to perform the similar operation. Diminishing the idle period, i.e., tool changing time (TCT) by applying combined tooling managed a fantastic saving in production cycle time. For statistical comparison of the present and advanced process, the fractional defectives from the samples of altering sizes are credited to the employment of statistical control chart specially *p*-chart. The horizontal line at right angles to the vertical/ordinate axis manifesting the value as 0.0496 in Fig. 61.3, mangled as CL is the chart centerline.

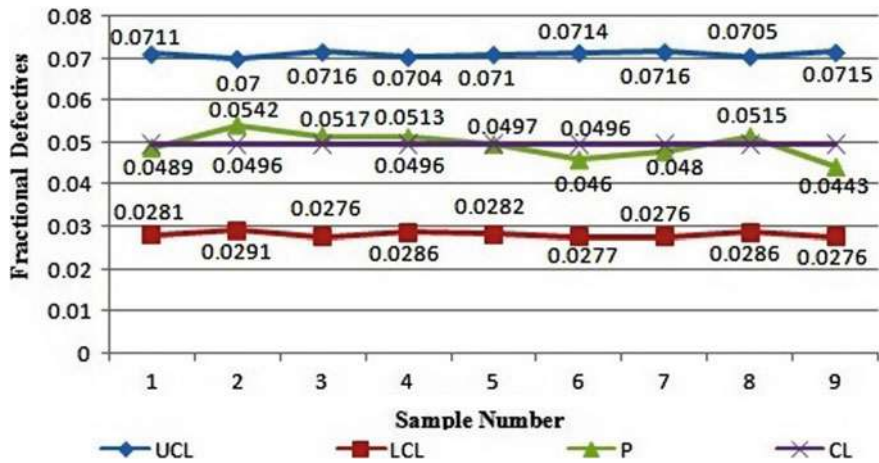


Fig. 61.3 Control chart for existing attributes

The graph signifies that fractional defectives for sample number 2nd, 6th, and 9th exhibited by triangular dots are tumbling away from centerline CL in Fig. 61.3 compared with same triangular points exhibiting fractional defectives for improved process, viz. Figure 61.4 which are plunging in the nearby space of centerline, CL.

After deciphering the graphs, it was inferred that the recommended manufacturing process statistically regulated in a better style. The non-similarity of p -charts for the revised and existing process has rendered and manifested reduced variation of monthly fractional defectives for the improved process.

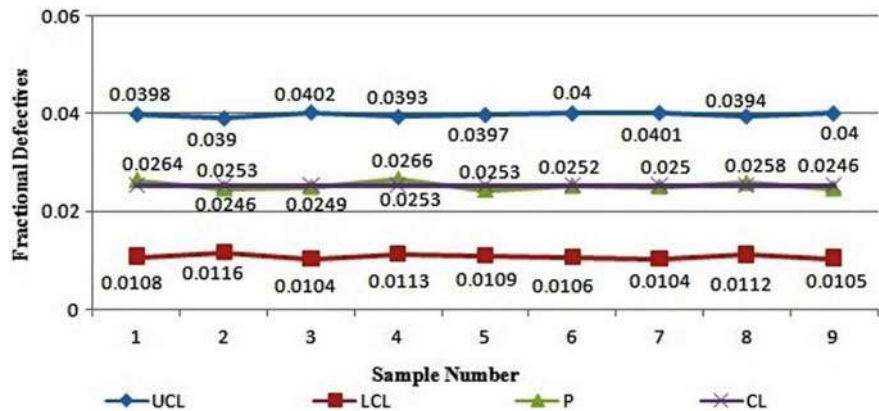


Fig. 61.4 Control chart for improved attributes

61.4 Conclusions

By implementing clustering principle, i.e., performing particular manufacturing operations in a unique setup in a single plane using combined carbide tooling through employing a typically developed expandable fixture, relevant machines needed were reduced to 4 of 5, which resulted in alleviation of the capital invested and space required. The suggested methodology in the instant paper vitally contributed toward the management of the corporation in the aspect of an above 14% decline in tooling costs. Approximately, a saving of 3.0 min changed into achieved, in the component production cycle time by dint of the reduction in non-productive time, through executing clustering. Reduced number of setups managed a close tolerance limit undoubtedly of ± 0.020 , achieving higher accuracy on the component. Organization profit straight away multiplied because of improved production with an equivalent drop in defectives. Consequently, the material and the labor productivity got remarkably improved, emerging as meaningful production incentives for workers, ensuring work complacency, and resultantly growing personnel retention. As the statistical control chart for improved process produces fewer variations of attributes, it infers better statistical control. A comparable rise of more than 10% recorded for average total productivity.

The implemented study was carried out in a multiproduct manufacturing enterprise. The company is practicing global tactics in the form of an ISO quality management system. Though, enhancement of umbrella productivity utterly fulfilled; facility layout is scarcely manageable to install in such a multi-objective industry, which might considerably hamper reduction of cycle time. Research findings are valuable to leaders of manufacturing enterprises world across. The skills generated as an outcome of implementing such comparisons are beneficial and deployable in related firms. Hence, contribution of the proposed implementation might be vital for the sizable structure of information globally.

Acknowledgements The author is grateful to Ambala College of Engineering and Applied Research (ACE) for providing necessary facilities. I would also like to express a broad sense of gratitude to my Supervisor Dr. Sandeep Singhal, Associate Professor, Department of Mechanical Engineering, NIT Kurukshetra for imparting his able guidance without which this work would have been impossible.

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Chapter 62

Experimental Investigation of Core Shear Properties and Facing Sheet Fracture Stress of Spherical Sandwich Structure



V. Pandyaraj and A. Rajadurai

Abstract In this study, two different models interlock and stagger spherical structure were designed and developed by using molding process. The matrix used for fabrication is vinyl ester and the woven glass fiber cloth of grade 300 and chopped strand mat (CSM) of 450 grades are used as reinforcement. The size of the model is 330×44 mm. Two pitch distances of 24 and 32 mm were selected as spacing between the centers of spheres. The shear stress of core and the fracture stress of facing sheet were tested using three-point bend test. From the test result, it is understood that the interlock model (CSM 32 mm) possess a high shear stress of core having a value of 465 kPa, which is less or more equivalent to honeycomb structure and fracture stress of facing sheet of 37.29 MPa. Whereas the stagger model with woven 32 mm pitch shows low core strength of 69 kPa and fracture stress of facing sheet of 5.58 MPa.

Keywords Spherical sandwich structure · GFRP · CSM · Interlock · Stagger

62.1 Introduction

A sandwich structure is developed by joining two thin skins to a light-weight thick core. The material with high thickness and low density is normally preferred as core because it gives good bending stiffness. In sandwich structure, the core carries the shear load and the face sheet carries the bending load. In tension and compression, the face sheets are strong and stiff compared to core because to provide good section modulus. The core material has high specific material properties in particular flexural strength and stiffness properties relative to overall density of the sandwich structure. Therefore, the structures are efficient in carrying out the bending loads. In addition to that they also provide resistance to buckling while undergoing compression and shear in panels. Finite element analysis method is also used to study the properties of the sandwich structure.

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_62

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Bart-Smith et al. [1] carried out the bending performance of sandwich structure by using the thin cellular metal cores. A map was created to study the failure load for face yielding, core shear, and indentation. The stiffness yield load and limit load were found analytically, numerically, and experimentally, and the results were compared and were found to be same. Mudassir [2] carried out the flexural test of two different composite sandwich laminate foam with two different orientation of fiber. From the experimental result, it is found that the orientation of phenolic foam has higher flexural stiffness when compared to that of polyurethane sandwich panel.

Venugopal et al. [3] found the effective of static four-point bending load of nomex flex core and carbon-reinforced polymer face sheet combination. The bending properties of finite element model predictions were compared with the experimental data. 2D and 3D FE model results correlate with experimental results of sandwich specimen and the maximum percentage error was 11.27% only. Foo [4] investigated the failure of sandwich panel using low-velocity impact test and understood that the core density and strain hardening of the material affects the impact strength of the sandwich structure. Jan [5] found out the flexural behavior of the honeycomb structure analytically and experimentally and compare them both.

Pandyaraj and Rajadurai [6, 7] made spherical sandwich structure of four different GFRP configurations namely Regular; inverted; interlock and stagger. They tested the compression behaviour of these specimens and found that the interlock model exhibited the maximum compression strength.

Pandyaraj and Rajadurai [7] also investigated the flexural behaviour of Regular and inverted spherical sandwich structure and found that the regular model exhibited a higher core ultimate shear stress and facing fracture stress.

62.2 Fabrication of the Specimen

In this study spherical structure namely interlock and stagger are taken for investigation. The shear stress of core and fracture stress of facing sheet of each orientation was tested and analyzed by taking 160 mm as the supporting span distance. The matrix used for fabrication is vinyl ester and the woven glass fiber cloth of grade 300 and chopped strand mat (CSM) of 450 grades are used as reinforcement. The size of the model is 330×44 mm. Two pitch distances of 24 and 32 mm were selected as spacing between the centers of spheres. The diameter of the sphere is 16 mm, so in order to have a proper spacing between each sphere, and the distance had been taken as 1.5 and two times of the diameter as spacing distance. The flatness at the top of the sphere is 8 mm in order to have a good surface contact between the core and the skin. The geometrical parameters for the interlock and stagger model are facing sheet thickness = 1 mm, height of the core = 16 mm, diameter of the sphere = 16 mm, flatness at the top of the sphere = 8 mm, thickness of the wall = 1 mm for woven, and 2 mm for CSM. The pitch distance between the center of each sphere is taken as 24 and 32 mm, respectively.

Figure 62.1 shows the fabricated model of interlock and stagger of chopped strand

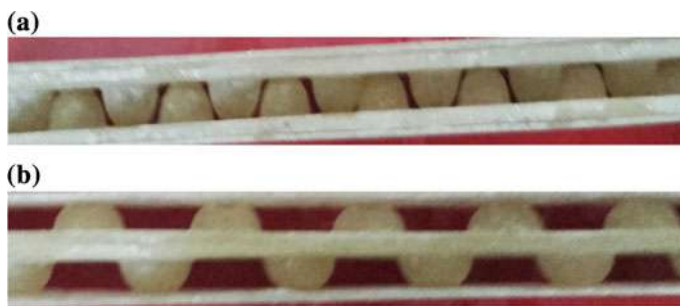


Fig. 62.1 Fabricated model of **a** interlock and **b** stagger of chopped strand mat with 32 mm pitch distance

mat with 32 mm pitch distance.

62.2.1 Three-Point Bend Test

The three-point bend test was carried out by using a universal testing machine. Figure 62.3 shows the bend test experimental setup. The structures were mounted on the table of the machine and they are loaded gradually. The ultimate load and the deflection were noted down. The shear stress of the core and fracture stress of facing sheet were found out according to ASTM C-393. Figure 62.2 shows the specimen mounted and specimen loaded while taking the experiment for interlock and stagger model [7]. The formulae for calculating the shear stress of the core and fracture stress of the facing sheet are listed below.

$$F_s^{\text{ult}} = P_{\text{max}} / (d + c)b \quad (62.1)$$

$$\sigma = P_{\text{max}}S / 2t(d + c)b \quad (62.2)$$

where

- F_s^{ult} shear stress of core, MPa,
- P_{max} maximum force prior to failure, N,
- d thickness of sandwich, mm
- c thickness of core, mm ($c = d - 2t$) and
- b width of sandwich, mm
- σ fracture stress of facing sheet, MPa
- t thickness of facing sheet in mm, and
- S length of the span, mm

Table 62.1 shows the comparison of the shear stress of core and fracture stress of facing sheet for interlock and stagger model.

Fig. 62.2 Three-point bend test, **a** specimen mounted and **b** specimen loaded

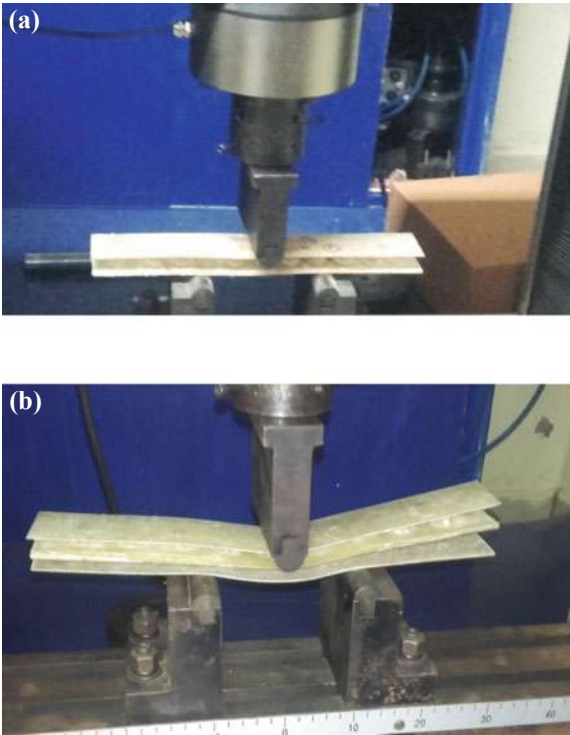


Table 62.1 Comparison of the shear stress of core and fracture stress of facing sheet for interlock and stagger model

Name sample	Reference for sample	Shear stress of core ‘KPa’	Fracture stress of facing sheet ‘MPa’
Interlock	ILW-24	151	12.11
	ILW-32	195	15.69
	ILC-24	298	23.85
	ILC-32	465	37.29
Stagger	STW-24	111	8.92
	STW-32	69	5.58
	STC-24	384	30.73
	STC-32	355	28.42

62.3 Results and Discussion

Figure 62.3 illustrates the comparison of shear stress of core for interlock and stagger models. The shear stress of core is better for interlock CSM 32 mm pitch (ILC32) and it is low for woven interlock 24 mm pitch (ILW24) while comparing within interlock models. For stagger model shear stress of the core for chopped strand mat 24 mm pitch (STC24) is higher and woven 32 mm pitch (STW32) is lower. However, on comprehensive analysis, it is understood that the shear stress of the core is high for interlock (ILC32) spherical model and is low in stagger (STW32) model in one-to-one comparison.

Figure 62.4 illustrates the comparison of fracture stresses of facing sheet for interlock as well as stagger model. The fracture stress of facing sheet is better for interlock CSM 32 mm pitch (ILC32) and it is low for woven interlock 24 mm pitch

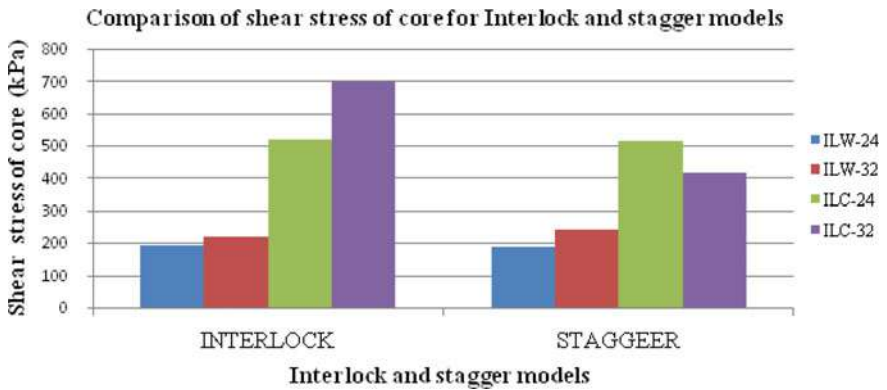


Fig. 62.3 Comparison of shear stress of core for interlock and stagger models

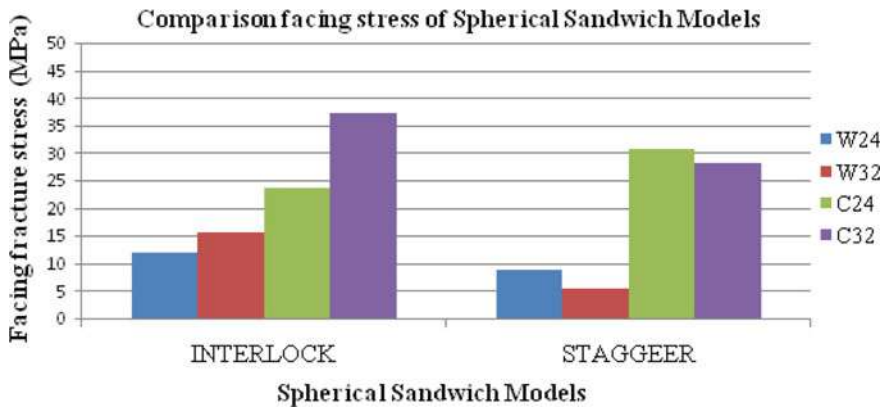


Fig. 62.4 Comparison of fracture stresses of facing sheet for interlock as well as stagger model

(ILW24) while making a comparison within interlock models. For stagger model, the fracture stress of facing sheet of chopped strand mat 24 mm pitch (STC24) is higher and woven 32 mm pitch (STW32) is lower. On overall analysis, it is understood that the fracture stress of facing sheet is maximum for (ILC32) spherical model with 37.29 MPa. Whereas the facing stress of (STW32) is 5.58 MPa, which is less in one-to-one comparison. From reference [5], it is understood that the value for shear stress of core is of 0.9 for honeycomb structure made of aluminum material, but the proposed model is 45% less than it which is made of GFRP. So this structure can be used for applications where low load is required.

62.4 Conclusions

A novel sandwich structure of spherical shape, namely interlock and stagger model fabricated by using GFRP, had been selected for the study purpose. From the experiment results, it is understood that the shear stress of core and the fracture stresses of facing sheet are higher for interlock and lower for stagger model. The increased strength is due to the better bonding between the facing sheet and the core in sandwich structure. For light load application, the proposed structure would be used in the form of partitioning wall and roofing sheet where specific strength requirement is high.

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Chapter 63

Design Analysis of Brass Cartridge Case for Water Disruptor Application



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and Himanshu Shekhar 

Abstract Brass cartridge cases are being popularly used in various types of ammunition since last 100 years, e.g., armaments of small-arm cartridges, artillery shell, and power cartridges for fighter aircraft. Cartridges are filled with propellants and pyrotechnic composition. With suitable means of ignition, it generates hot gases with pressure and temperature. These gases are utilized to perform certain work on the system. Brass cartridge case for water disruptor applications plays a significant role in destruction of the suspicious objects. Further, it has cumulative consequence to make them non-operational. This paper discusses the design aspects of brass cartridge for disruptor application of suspected improvised explosive devices (IEDs). Performance evaluation of brass cartridge was carried out in closed vessel (CV) using the data acquisition system. Parameters such as maximum pressure and time to reach maximum pressure have been evaluated in this CV. The brass material properties such as tensile strength, percentage elongation, and yield strength were determined using universal testing machine (UTM). In CV test, the cartridge experienced 16.08 MPa. The same cartridge when fired in velocity test rig (VTR) it was subjected to 63 MPa internal pressure. Using the data obtained by above methods, an attempt has been made to determine stress, strain, and deformation of the cartridge case theoretically and numerically using ANSYS software. The results obtained by both the methods were compared. It was seen that the results are in good agreement with each other. It is observed that the percentage error for Von-Mises stresses is 13.2% using numerical and theoretical. The percentage error between numerical and theoretical values of

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_63

hoop stresses are 11.5% for stress and for strain it is 9.6%. The main objective of this paper is to carry out design and analysis of cartridge case analytically as well as numerically. The results of FE analysis for stress and strain are in good agreement with theoretical calculated results and numerical analysis as percentage error is less than 13. This draws the inference for validating numerical and theoretical results. The novelty in this research work is that the cartridge case is subjected to propellant pressure generated inside the case with pressure of 63 MPa. This pressure is measured using data acquisition system in a specially designed test rig.

Keywords ANSYS · Axisymmetric model · Brass cartridge case · Deformation · Disruptor · FEM · IEDs power cartridge · Thin cylinder theory

Symbols, Nomenclature, and Units

A	Area, mm ²
E_{case}	Modulus of elasticity of case, GPa
P	Max pressure generated in the case, MPa
r	Radius of the case, mm
t_c	Cartridge wall thickness, mm

Greek Symbols

π	Constant (3.14)
ε	Strain developed in the case
ν	Poisson's ratio
σ	Stress developed in the case MPa

Subscript

h	Hoop
l	longitudinal
i	Internal
o	Outer

63.1 Introduction

63.1.1 Cartridge Cases

Design and analysis of any component play a very important role before its actual use in the system. The main objective of the paper was to carry out design and analysis of cartridge case. This design validation aids to demonstrate how the system will perform its intended function. The analysis gives fair idea whether the component will survive during field trials. Production of ammunition of cartridge case brass material has an outstanding record of service. Power cartridge in disruptor application utilizes brass as material. On initiation of cartridge, it provides the gas pressure due to propellant burning action. This phenomenon of converting solid to gas poses a rapid chemical reaction liberating energy. Power cartridges are known as gas operating devices containing propellant or pyrotechnic. On suitable initiation, gases were utilized to perform destruction task of suspected improvised explosive devices (IEDs) [1]. The water-jet is produced and is effective against suspected objects including IEDs. The various materials are used in the manufacturing of cartridge cases like brass, steel, and aluminum depending upon cost, compatibility of explosives, and its availability. Pichipila et al. has carried out metallographic inspection of rifle brass cartridge using scanning electron microscopy (SEM), optic microscopy, and X-ray energy dispersive spectrometry (EDS) [2]. During firing process, the deformation of steel cartridge case was very intricate. It was hard to reproduce the interaction between chamber and cartridge case. To study this problem, the semi-closed bomb test was significant. Nevertheless due technical restrictions, the method was not spontaneously achieved the deformations and force pertaining to cartridge and different parts of chamber. The semi-closed bomb test method was utilized to launch a projectile of steel cartridge with different chamber pressures was reported by Zhao et al. [3]. During rolling of Cu–30Zn of brass material, the strain change path on grain refinement and cryogenic temperature effect was determined by Konkova et al. [4].

The novelty in this research article is that the internal pressure realized by the propellant was measured using data acquisition system in a specially designed test rig. The pressure measured in this test rig is 63 MPa, which considered a valuable input for design analysis and its numerical analysis. In the earlier work, the pressure was recorded in closed vessel which was less, i.e., 16.08 MPa. An experimental, numerical analysis, and power cartridge characterization using brass material for disruptor application was discussed by Parate et al. [5]. Both numerical and experimental study was conducted by Eugen et al. [6] to assess efficiency for drawing of 60820 temper Al alloy using cartridge tubes. The validation of theoretical evaluation was carried out by superimposing the stress graph triaxiality ratio vs. plastic strain and graph of 60820 temper Al alloy fracture envelope for a zero load parameter.

63.1.2 Objective of This Paper

Objective of this paper is to investigate the design aspects of brass cartridge and to determine stress, strain, and deformation of the cartridge case theoretically and numerically using ANSYS software. Theoretical and numerical results were compared. The finite element analysis (FEA) technique using ANSYS software was employed for the analysis of brass cartridge case for water-jet application. The objective of this work is to carry out FEA, i.e., numerical analysis of brass cartridge case and validate its results with theoretical calculations.

63.2 Cartridge Case Design Assumptions

Important design considerations of brass cartridge case are the volume where propellant mass loaded. The cartridge is designed for single-shot purpose. The best source of energy for achieving disruption of IEDs is the propellant.

63.2.1 Material Selection of Cartridge Case and Its Manufacturing

The cartridge case is made of brass material. The various points to be considered in cartridge case design are the maximum pressure developed. The cartridge length was selected on the basis of the propellant weight, booster weight and means of ignition. The choice of brass cartridge material is due to the following reasons

- (a) Best combination of ductility and strength
- (b) Cold working properties
- (c) Ease for machining (% Pb 2–3.5)
- (d) Meet the obturation requirements
- (e) Good for thermal conductivity
- (f) Compatible with propellant and pyrotechnic compositions
- (g) High coefficient of thermal expansion.

The requirement of case was to accommodate the squib, propellant weight and sustain the high pressure and temperature gas. The cartridge cases were usually made up of “cartridge brass,” which is 57/39 or 70/30 copper-zinc alloy with the proportion of other elements. Case is lacquered to protect it from corrosion and to add aesthetic look. Whenever, the diameter and length are small, the cases are machined from the brass rod using machining.

63.2.2 Design Considerations for Cartridge Case

Thickness of cartridge case is determined based on calculations made using thin cylinder theory where $d > 10t$.

The following assumptions are laid down

- (a) The cylinder is made of homogenous material
- (b) Circumferential or hoop stress is uniformly distributed
- (c) Young modulus in tension or compression is same
- (d) Plane section perpendicular to the longitudinal axis of cylinder before loading remains plane after loading.

63.3 Materials and Methods

63.3.1 Description of Brass Cartridge Case

Brass cartridge case is composed of Zn 40.95%, Cu 56%, with other impurities such as Sn, Ni, Mn, Al, Sb not exceeding 0.7%, and Fe 0.35% [7]. The mechanical properties of brass material are modulus of elasticity 100 GPa, tensile yield strength 280 MPa, Tensile ultimate strength 395 MPa, hardness 90–115 HV, percentage elongation 12, coefficient of thermal expansion 1.8×10^{-5} /°C, density 8400 kg/m³, and Poisson's ratio 0.33.

The tensile test was conducted on brass material using a suitable test specimen. The behavior of material is represented in stress-strain graph. The stress-strain plot was obtained by testing a standard test specimen on universal testing machine. The standard test piece is fabricated and subjected to gradual loading till it gets fractured. The curve is shown in Fig. 63.1. The mechanical properties are obtained using universal testing machine.

63.3.2 Dimensions of Cartridge

The dimensions of cartridge height and flange thickness are selected as per the dimensions of the breech module where it is fitted. The dimensions of the cartridges are given in Table 63.1.

The photos of brass cartridge case with other important engineering dimensions of cartridge case are depicted in Figs. 63.2 and 63.3, respectively.

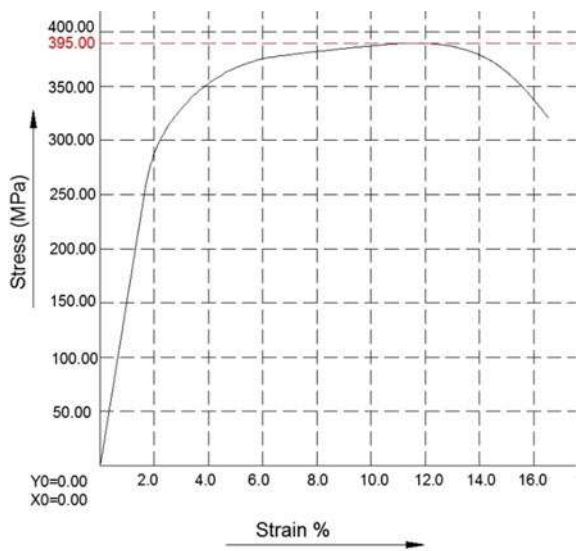


Fig. 63.1 Stress–strain curve for brass material

Table 63.1 Dimensions of the cartridges

Dimensions	Unit (mm)
Total length	54.5
Outer diameter	20
Internal diameter	17
Thickness	1.5

63.4 Finite Element Method (FEM) Analysis

A static structural FE analysis of the cartridge case was carried out using ANSYS software. Actual drawing dimensions of the case were used for modeling. Since the case is symmetric about central axis, an axisymmetric model of the case was created and pressure was applied on inside the edges. The model is rotationally symmetric about Y-axis. The model is constrained at edge marked “A” so to simulate the fitting of the cartridge in real-time scenario. The model with boundary conditions is illustrated below in Fig. 63.4.



Fig. 63.2 Photos of brass cartridge case

63.4.1 *Meshing*

The meshing is done on the geometry after applying load conditions to generate FE model. The meshed geometry is shown in Fig. [63.5](#).

Mesh information such as size, nodes, and elements are mentioned in Table [63.2](#).

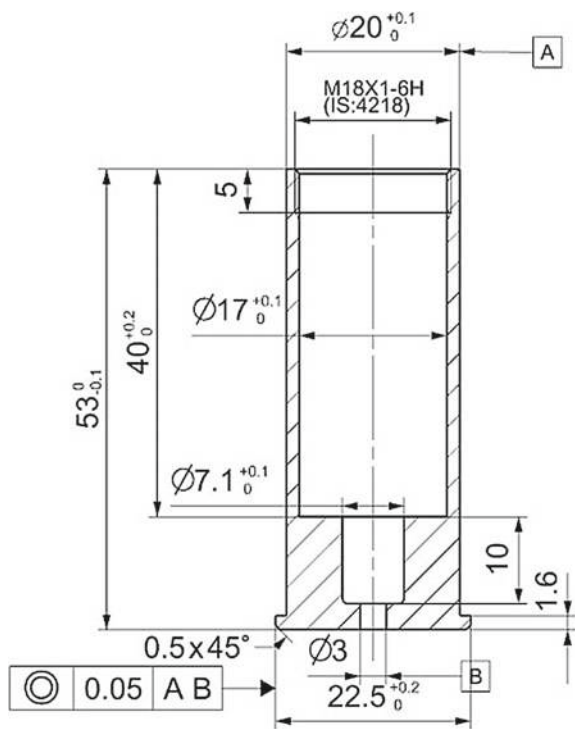


Fig. 63.3 Engineering drawing of cartridge case

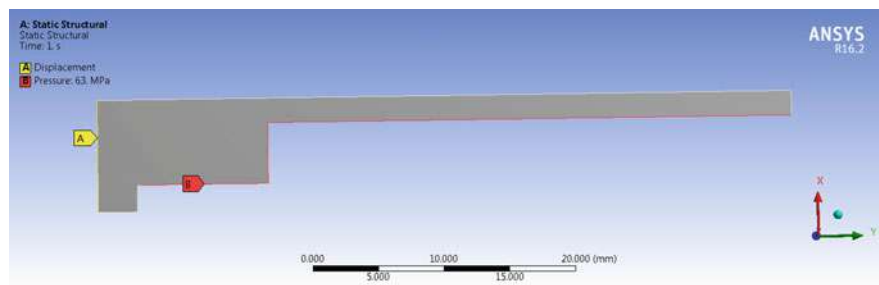


Fig. 63.4 The model of axisymmetric cartridge case

63.5 Theoretical Calculation

63.5.1 Theoretical Estimation of Stresses

The various stresses like hoop stress, longitudinal stress, and equivalence stress with strains are determined by taking mechanical properties mentioned at Sect. 63.3.1

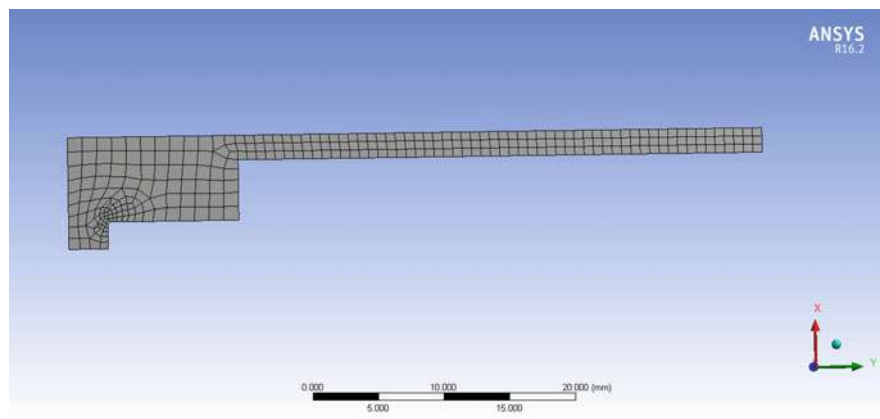


Fig. 63.5 The meshed geometry

Table 63.2 Mesh size, nodes, and elements

Mesh physics	Mechanical
Relevance (global fineness)	49
<i>Sizing</i>	
Advanced size function (sizing of mesh within a model)	Kept on for proximity and curvature-based features in the model
Relevance center (size of mesh based on physics)	Coarse
<i>Statistics of mesh generated</i>	
Nodes	1072
Elements	303

obtained during tensile testing and actual values of dimensions as given at Table 63.1. The calculations as an example are illustrated as below

(a) Hoop stress (σ_h)

The design of the cartridge case based on Hoop stress which is predominant [8, 9]

$$\sigma_h = \frac{Pr_i}{t_c} \tag{63.1}$$

where

- t_c Cartridge wall thickness = 1.25 mm
- P Max pressure generated in the case = 63 MPa [10]
- r_i Internal radius of the case = 8.5 mm

$$\sigma_h = \frac{8.5 \times 63}{1.5} = 357 \text{ MPa}$$

$$\text{Strain in the case} = \frac{\sigma_h}{E_{\text{case}}} = \frac{357}{1 \times 10^5} = 357 \times 10^{-5}$$

(b) Longitudinal Stress (σ_l)

Longitudinal Stress will be half the value of Hoop stress

$$\sigma_l = 178.5 \text{ MPa}$$

(c) Equivalent Von-Mises stress acting on cartridge

$$\begin{aligned} \sigma_{\text{eq}} &= \sqrt{\sigma_h^2 + \sigma_l^2 - \sigma_h \times \sigma_l} \\ &= 309.17 \text{ MPa} \end{aligned} \quad (63.2)$$

(d) The outer diameter (OD) of the case (t_c) can be determined as follows

$$d_o = d_i + 2t_c = 17 + 2 \times 1.5 = 20 \text{ mm} \quad (63.3)$$

63.5.2 Theoretical Estimation of Strains

(a) Hoop strain (ε_h)

Hoop strain is calculated by the following equation

$$\varepsilon_h = \frac{1}{E}(\sigma_h - \mu\sigma_l) \quad (63.4)$$

where

E Young modulus of brass material = 100 GPa

μ Poisson's ratio = 0.33

$$\varepsilon_h = \frac{1}{100 \times 10^3}(357 - 0.33 \times 178.5) = 2.98 \times 10^{-3}$$

(b) Longitudinal strain (ε_l)

Longitudinal strain is calculated by the following equation

$$\varepsilon_l = \frac{1}{E}(\sigma_l - \mu\sigma_h) \tag{63.5}$$

where

$$\varepsilon_l = \frac{1}{100 \times 10^3}(178.5 - 0.33 \times 357) = 0.606 \times 10^{-3}$$

$$\text{Total strain} = \varepsilon_h + \varepsilon_l = 3.58 \times 10^{-3}.$$

63.6 Result and Discussions

63.6.1 Results

The numerical and theoretical values were determined as explained in above paragraphs. The various plot contours for hoop stress, Von-Mises, radial strain, and hoop strain obtained by FE analysis are given at Figs. 63.6, 63.7, 63.8 and 63.9, respectively. Figure 63.6 shows the maximum and minimum hoop stress are 315.9 and – 5.2 MPa. Figure 63.7 shows the maximum and minimum Von-Mises stress are 356 and 11.8 MPa. Figure 63.8 shows the maximum and minimum radial stress as 176 and –109 MPa.

Figure 63.9 shows the maximum and minimum hoop strain are 0.0033 and 0.00115.

These values are given in Table 63.3. From the table, it is observed that the percentage error for Von-Mises stresses is 13.2% using numerical and theoretical. The percentage error between numerical and theoretical values of hoop stresses are 11.5% for stress and for strain it is 9.6%.

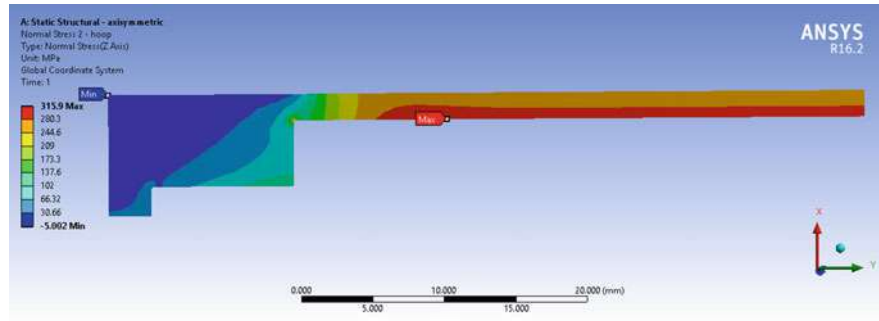


Fig. 63.6 Stress plot (hoop stress)

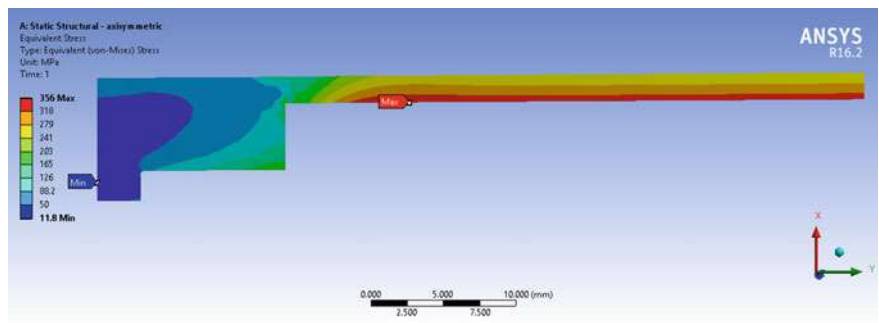


Fig. 63.7 Stress plot (Von-Mises)

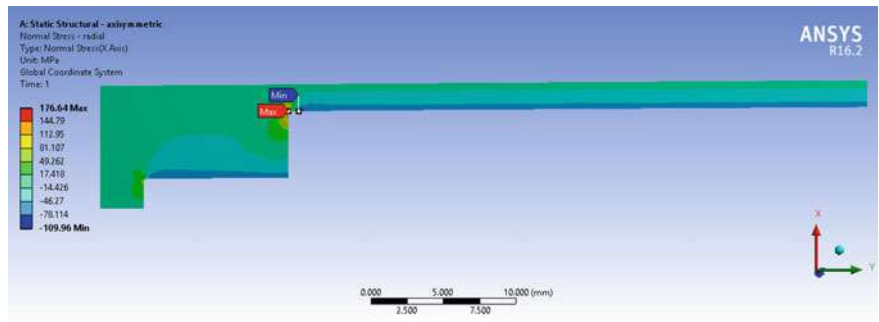


Fig. 63.8 Radial stress plot

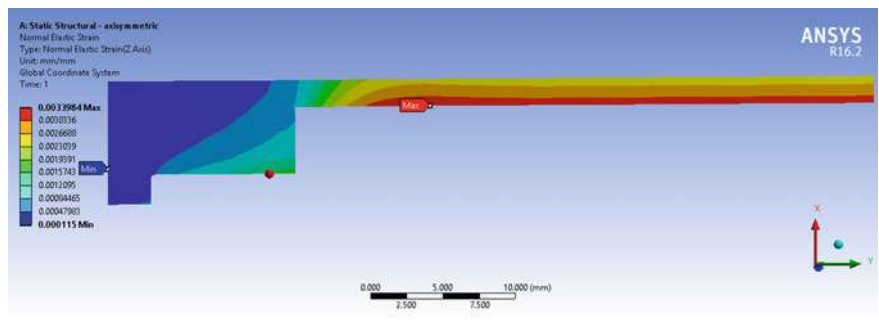


Fig. 63.9 Strain plot (hoop strain)

Table 63.3 Numerical, theoretical values of stresses and strains

Calculated stresses and strains				
Mechanical parameters	Theoretical	FEA	Units	% error
Cartridge hoop stress	357	315.9	MPa	11.5
Cartridge hoop strain	2.98×10^{-3}	3.30×10^{-3}	–	9.6
Von-Mises stresses	309	356	MPa	13.2

63.6.2 Discussions of Results

There is marked difference between the values of stresses, particularly hoop stress. FEA gives lower values of stresses as compared to theoretical calculation using thin cylinder theory. However, the following aspects should be considered before drawing any conclusions:

- (a) Thin cylinder theory assumes uniform wall thickness, whereas the geometry used for FEA is a stepped construction with varying wall thickness. More material is present in steps, thereby reducing the stress. This leads to lower level of stresses in FEA.
- (b) However, Von-Mises stresses are greater in FEA than theoretical analysis. This is due to inclusion of radial stresses in FE analysis.
- (c) Boundary conditions are applied in a different manner in FEA. FEA was used to simulate actual conditions those will be applied to the component while in service. Thus, it would give more realistic results for predicting the performance.
- (d) Effects of stress concentration at the corners, notches will be captured in FEA. The plot of radial stress is shown in Fig. 63.8. Stress is higher at the corner where stress concentration is likely to occur.
- (e) The FE analysis considers the effect of radial stresses which are neglected in thin cylinder theory.

63.7 Conclusions

The FE analysis using ANSYS was carried out on brass cartridge case to determine various stresses, strain, and deformations. Theoretical calculations were also made using thin cylinder theory to have general estimation of stresses in the cartridge case. Based on the above analysis, a manuscript presents results of combination of experiment, theoretical, and computer simulation for design and analysis of power cartridges using a brass material.

The numerical results with the theoretical data were correlated. The following inferences were drawn:

FEA predicts the results more accurately due to the following reasons

- (a) Actual product geometry is subjected to analysis applying actual in-service boundary conditions.
- (b) Effect of radial stresses and stress concentration is also considered.

The results of FE analysis for stress and strain are in good agreement with theoretical calculated results and numerical analysis as percentage error is less than 13. This draws the inference for validating numerical and theoretical results.

In both the approaches of analysis used, equivalent (Von-Mises) stress is crossing the yield limit of the material. So the cartridge case will yield causing permanent

deformation. But it will not break as stresses are below ultimate strength. This is acceptable as this is a one-shot device.

Acknowledgements Authors like to state their thanks to anonymous reviewers and editorial board for acceptance of manuscript. Further we are grateful to Dr. Venkateswara V Rao, Outstanding Scientist and Director ARDE, Pashan, Pune-411 021 for his kind permission to publish this work. The authors also extend their sincere thanks to Shri. V K Dixit, Sc. 'H', Associate Director for his invaluable guidance and support. We also thank Mrs. Kalpana Tipare for carrying out the mechanical testing of brass specimen.

Conflict of Interests The authors wish to confirm that there are no known conflicts of interests regarding the publication of this paper. There was no significant financial support for this work that could have influence its outcome.

Funding The author(s) received no financial support for the research, authorship, and/or publication of this article.

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Chapter 64

Design and Analysis of Hydraulic Fixture for WABCO Body Housing



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Abstract In order to have interchangeable parts in mass production, jigs and fixtures play a vital role in manufacturing process. A fixture is a special tool designed for specific purpose and specific component for operation. The present work deals with the design of machining fixture for milling and drilling operations for a Wabco body housing. The cutting forces involved in the operations are taken into consideration for designing the fixture. The present fixture designed is hydraulic operated and used for operation like face milling, drilling and boring of the body housing. Design standards are taken from Makino for designing this machining fixture. In the design process based on the geometry of the component to be machined, the machine, the table layout and corresponding clamping slot positions are then selected. Since the final component cannot be produced by a single operation, it is necessary to plan for various operations to get the final shape. The fixture is then designed by considering all the clamping forces from various cutting operations. An emphasis was also made to optimize the manufacturing cycle with respect to the original design and it was possible to successfully improve the cycle time involved in the manufacturing process while still meeting all the required standards.

Keywords Fixture design · Body housing · Wabco · Manufacturing cycle

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_64

64.1 Introduction

Fixtures are devices designed to repeatedly and consistently maintain the orientation of a work piece during processes such as machining, assembling, welding, inspection, etc. They play an integral role in manufacturing by functioning as a special tool used for locating and firmly holding a workspace in the proper position during a manufacturing operation. As a general rule, it is provided with devices that support and clamp the work piece. It is fixed to the machine bed by clamping in such a position that the work is in the correct relationship to the machine tool elements. The main purpose of the fixture is to locate the work quickly and accurately support it properly and hold it securely, thereby ensuring that all parts produced in the fixture will come out alike within the specified limits and in this way, accuracy and interchangeability of the parts are provided. It serves to simplify, otherwise complicated operations, so that cheaper and relatively unskilled labour, may be employed to perform operations previously reserved for skilled machines. By maintaining or even improving the interchangeability of the parts, a jig or fixture contributes to a considerable reduction in the cost of assembly, maintenance and the subsequent; the potential of standard machines and the quality of the parts produced. One important goal is to design a fixture in such a way as to make it foolproof, and thereby contributes to added safety for the operator, as well as for the work. The objective of the paper is to design and analyse hydraulic fixture for machining “WABCO BODY HOUSING” to avoid chip accumulation during machining, to provide internal hydraulic pipelines, air seat check arrangement and foolproofing while loading. The present fixture designed to be hydraulic operated and capable to perform operations like face milling, drilling and boring of the body housing (Fig. 64.1).

64.2 Requirement, Material Used and Specifications

The aim is to design and analyse hydraulic fixture for machining Wabco body housing. An attempt is made to avoid the accumulation of chips during machining and to provide foolproofing while loading. The material used for the fixture is aluminium

Fig. 64.1 Wabco body housing



Table 64.1 Component specifications

Component	Wabco body housing
Material	Aluminium alloy (JED-029 M1)
Input condition	Cast part
Hardness	50–100 BHN
Weight	0.27 kg
Operations	Face milling, thread milling, drilling, tapping and boring
Machine	Vertical machining centre slim 3 PC

having hardness of 50–100BHN. The Wabco body housing designed is cast and has a weight of 0.27 kg (Table 64.1).

Two components, namely Component OP 10 and Component OP 20 are used for testing.

64.3 Conceptual Layout

Fixture is to be designed for face milling, end milling, thread milling, oil hole drilling, M6 tapping and boring. The holes are to be drilled on the machined face. Hole sizes and lengths depend on the nature of the operations to be carried out. By taking all above factors into considerations, we have generated a conceptual layout of fixture as follows:

- Component to be loaded on the rest pads with the help of Rough Guide Pins.
- Component to be located and oriented with the help location and orientation assembly.
- Hydraulic push cylinder pushes the component against the rest pads.
- Sequence valve open work support actuated.
- Part seat check to be accommodated on rest pads (Fig. 64.2).

64.4 List of Operations (Cycle Time Calculations)

Cycle time estimation is prepared by considering customer's tact time (time available to complete the task.)

- Cycle time = cutting time + non-cutting time
- Cutting time = approach length + drawing length
- Non-cutting time = loading–unloading time + positioning time + atc time + table indexing time + pallet changing time



Fig. 64.2 Components OP 10 (left) and OP 20 (right)

64.5 Methodology

1. Identifying the machining areas of the component as per the requirements of work and machine elements.
2. Determining the supporting, locating, clamping surfaces and calculating cutting and clamping forces.
3. Force calculations used to select hydraulic elements of the fixture.
4. Input data of the component and machine specifications are considered in the design of the fixture layout and analyse the accessibility, geometric constraints, tolerances, deformation and tool interface analysis.
5. Starting from the part modelling and making sub assembly and main assemblies.
6. Checking the feasibility to manufacture the fixture parts.

64.6 Procedure of Fixture Design

In order to design the fixture, certain input data is required. The details of the component such as material properties are taken in order to perform the FEM analysis. Whereas, the input conditions of the component combined with the pre-machined and machined drawings are performed to decide the locating, clamping and resting surfaces. The cutting force calculations are made by taking into account the processes related details such as the depth of cut, spindle speed, feed and width of cut. The cutting details such as the tool geometry, tool dimensions and tool material in order to make the calculations for calculating the cutting force, interference with any fixture parts, and to check the reach of the tool to the machining area of the components. Machining details such as the maximum movement of the spindle, the dimensions of the bed and the indexing details are made use of to check the interference with any machine parts and check the mounting of the fixture on the machine. The type of

clamping, i.e., manual clamping or power clamping is selected based on the capacity and demand. This selection of the type of clamping also affects the investment cost. Details pertaining to the process capability help determine the accuracy and the rigidity of the design. The design was also compensated for the ability to adapt to different designs.

The different parts that make up the fixture are rest pad, link clamp assembly, anti-fall assembly, push cylinder assembly block assembly and cylindrical and diamond pin.

64.7 Calculations

64.7.1 Input Calculations

- Spindle speed (n) = 12,740 rpm
- Feed (S_m) = 7650 mm/min
- Feed per tooth (S_z) = 0.12 mm/tooth
- Cutting length = 621 mm
- Cutting time = 0.08 min
- Width of cut (b) = 32 mm (32 * 0.8)
- Depth of cut (t) = 2 mm
- Diameter of cutter (D) = 40 mm
- KF_s = Specific cutting force (N/mm²) from table as per tensile strength is 600 MPA
- Hardness of material to be machined also from 0.12 feed/tooth, i.e., mm/tooth
- η = Cutting efficiency or machine coefficient (Take 0.8 or 0.85).

In case of milling operation, tangential force is to be developed while cutting the work.

64.7.2 Cutting Force Calculations of Face Milling (OP-10)

- **Cutting speed:** $V = \pi Dn/1000 = 1600.95$ m/min
- **Metal removal rate:** $Q = b t S_m/1000 = 489.6$ cm³/min
- **Cutting power:** $N = (t \times b \times S_m \times KF_s)/(60 \times 10^6 \times \eta) = 5.76$ kW
- **Tangential cutting force:** $P_z = 6120$ N/v = 220.32 N
- **Torque at spindle:** $T_s = 975$ N/n = 35.1 N-m
- **Clamping force:** F = Cutting force (Max) $\times$ FOS/Coefficient of friction (b/n clamp pad and part surface) = 0.66 kN
- **Work support:** $R = 1.5 \times$ Clamping force = 0.99 kN.

64.7.3 Cutting Force Calculation for Hole Ø7 Drilling for OP-10 and OP-20 Fixtures

- Cutting Speed (v) = 160 m/min
- Drill dia (D) = 7 mm
- Feed per tooth (S_z) = 0.2 mm
- Spindle speed = 7280 rpm
- Feed (S_m) = 1460 mm/min
- Feed per rev (S) = 0.2 mm/rev
- K = Material factor select from table as per tensile strength and hardness of material to be machined
- **Cutting Speed:** $V = \pi Dn/1000 = 160.09$ m/min
- **Cutting Power:** $P = D^2 \times K \times n (0.647 + 17.29 \times S)/10^5 = 18.67$ kW
- **Torque:** $(Ts) = 975 \times (kW)/n = 25.004$ N-m
- **Thrust force (F_z)** = $570 \times k \times D(S)^{0.85} = 1.53$ kN
- Clamping force: F = Cutting force (Max) $\times$ FOS/Coefficient of friction (b/n clamp pad and part surface) = 4.6 kN.

64.8 Selection of Hydraulic Elements

The hydraulic elements are selected from standard catalogues available as per the requirements and applications. In this fixture assembly, following items as shown in Table 64.2 are used for the stated applications. Apart from the above items, standard fittings for oil connections and plugs are used. The straight stud fittings are used for connecting oil ports with the cylinders and G1/8” plugs are used to plug the holes.

64.8.1 Selection of Clamping Cylinder (LKA0400)

This cylinder is used to actuate the link clamp. This is double-acting cylinder, thus two lines are required for suction and discharge of oil. The piston of this cylinder is linked to the clamp lever to transmit the cylinder force to the component for holding the component securely while operation. Thus, the basis for selection of this cylinder is the force required to counter the thrust force during face milling. The thrust force

Table 64.2 Hydraulic elements used

Item	Make	Application
LKA0400	KOSEMK	Clamping (OP-10)
LL0550-CA	KOSEMK	Clamping (OP-20)
DP0241-AS	KOSEMK	Push cylinder (OP-10)

and torque formulae for drilling and milling are taken from the “MAKINO Machining Data Handbook” for the selected feed/rev and drill diameter, Thrust force = 1.5 kN. To nullify this force so that the component will not get displaced during machining, at least 4.6 kN of force is required (Considering FOS = 3). The cylinder force depends upon the stroke and pressure. From the cylinder specification table cylinder force of LKA 0400, two cylinders are suitable for our application as it gives 2.3 kN force at 7 MPa pressure.

64.8.2 Selection of Clamping Cylinder (LL0550-CA)

The cylinder force depends upon the stroke length and hydraulic pressure. From the cylinder specification table in the catalogue, LKA0550-CA cylinder is suitable for our application as it gives 6.72 kN force @ 7 MPa pressure, specification of the cylinders. From the catalogue, we obtain the empirical relation (push force) = $p \times 0.96 = 7 \times 0.96 = 6.72$ kN.

Thus, LL0500-CA has been selected for clamping the body in OP-20 operation.

64.8.3 Selection of Clamping Cylinder (DP0241-aS)

The main function of push cylinder is to produce butting force so that the component cannot deviate during cutting. So butting force is equal to the cutting force. This cylinder gives 1.01 kN force at 7 MPa pressure.

64.8.4 Selection of Sequence Valve BLG2830-0G (5.0 MPa)

The sequence valve is used for proper sequencing of actuation of cylinders. This valve operates multiple actuators in sequence to perform positioning and clamping. When incoming port pressure reaches the sequence setting pressure value, the pressure is supplied to outgoing port. The sequence of actuation is given in the assembly drawing. The general sequence of actuation of cylinders occurs in the following order, firstly Push cylinders (along x and y coordinate axes) should actuate to place the component in precise position and orientation, next the clamp cylinders must actuate after push cylinders at full pressure to ensure that the component is correctly kept against the rest pads, next the work supports should actuate to provide adequate supporting force during the machining operation. In this fixture, the sequence valve is used first to clamp the component then sequentially actuate work supports. Here, model BLG2830-0G (for one revolution/turn of the knob 2.8 MPa of pressure can be changed) sequence valve is selected for easy connection to the required cylinders from sequence valve outlet.

64.9 Analysis of the Fixtures

64.9.1 Selection of Fixture Body

Since a vertical machining centre is used for the component machining for operation 10 and 20, i.e. tool axis is in vertical direction. For the component to be loaded perpendicular to the cutter axis, the base plate should be horizontal to the machine bed. For this purpose, the selected base plate is horizontal. This structure should withstand the maximum cutting forces applied, i.e. if the base plate is loaded with a 4600 N of thrust force, its maximum deflection should be within 50 μm .

64.9.2 Stress Result

By this analysis, the stress included is 11.2 N/mm^2 at maximum force of drilling at the farthest point (hole). This value is within below the permissible limits, i.e. value is less than yield strength of material, i.e. 620.42 N/mm^2 .

64.9.3 Displacement Result

By this analysis, the deflection is coming 10.4 μm at maximum force of drilling (4600 N) at the farthest hole point. The permissible limit is 50 μm . So, this value is within permissible limit.

64.9.4 Strain Result

From this analysis, the strain produced is low (6.37922e^{-088}) as deformation caused due to the stress is less and negligible.

64.10 Conclusions

A fixture is designed for a Wabco body housing. The following results are obtained after designing and trying out the fixture.

- The functional objective of designed fixture is achieved to hold the work piece properly during machining and by incorporating hydraulic elements, fixture has been made operator friendly.

- Previously cycle time was 10 min (customer info.). This fixture's cycle time is 1.5 min/component. So total cycle time per component is reduced by 6 times.
- Previously, 3 set-ups were required to complete one finished component (Customer info.) This fixture has been designed for 2 set-ups (OP10 and OP20). In OP10, vertical machining centre with Fourth-axis rotation helped to reduce one set-up.
- Another important aspect in designing the fixture is to reduce non-production time, i.e. set-up time. Set-up time is also reduced, as it is being automatic pallet change machine.
- During machining, deflection of structure through analysis is 10.4μ , which is less than permissible limit is 50μ .
- The stress value of structure through analysis is 11.2 N/mm^2 which is less than yield strength of plain carbon steel 620.42 NN/mm^2 .

Chapter 65

Development of Alignment Fixture for Precision Assembly of Aerospace Control Surfaces Incorporating Process and Assembly Variations



N. Sankaranarayanan, Ch. Venkateswarlu, G. Ravinder and Shivpal Singh

Abstract Actuator alignment is crucial and of paramount importance in meeting high precision pointing and reliable control performance of Aerospace vehicle emission requirements. In this paper, a holistic approach has been made for precise alignment of the Aerospace control surfaces with the actuators through the development of an Assembly Fixture with the help of CAD software taking into account variations in fabrication and assembly for meeting the stringent specification requirements. The Assembly Fixture resulted in reduction of lead time for 3D inspection and also ensured repeatable functional performance of the actuator assembly.

Keywords Control surface · Actuator · Assembly jig · Alignment · Cant angle · Orthogonality

65.1 Introduction

Accurate and reliable attitude stabilization is one of the most important problems in the current Aerospace vehicle attitude control system. The actuator uncertainties including misalignment during installation and torque magnitude deviation influence the mission objectives significantly [1]. Aerospace Missions typically require achieving the programmed manoeuvres within the short span of mission duration. In the real-time mission scenario, designing an attitude control system is, in particular, a tedious process when simultaneously addressing all these issues for meeting the high precision pointing requirement and desired control performance. There is obviously a strong dynamic interaction and influence between the translation dynamics and the attitude dynamics of an Aerospace vehicle. Considering the high pointing accuracy requirements for imaging infrared seeker, even small misalignments of the actuator axes could be detrimental. The unbalanced torques caused by actuator misalignment may likely to produce undesirable angular rotation to the vehicle.

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© Springer Nature Singapore Pte Ltd. 2020
M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering,
https://doi.org/10.1007/978-981-32-9487-5_65

The actuator misalignment often occurs when actuators are fixed on the body of the vehicle, which introduces the control-dependent disturbance into the coupled dynamics. Actuator axis alignment errors are likely to cause mission performance degradation in proportion to the relative magnitude of the alignment errors [2].

Hence it is of paramount importance that the actuator misalignment should be controlled within the specifications during the assembly stages.

In this paper, a holistic approach has been attempted to minimise the control surface misalignment of an Aerospace vehicle through the deployment of a special in-process assembly jig which takes into account all the manufacturing and assembly variations.

65.2 System Description

An Aerospace vehicle deploys 04 numbers of rear control surfaces for stabilisation and steering. Since the guidance system is based on imaging infrared stabilised seeker head, a tight roll control is imperative. The system comprises of four Electro-Mechanical Actuators with fins and two servo electronics units. The servo electronics unit drives the motors as per the commands from Control and Guidance Computer (CGC). Each unit drives two geared DC servomotors each of which comprises of a DC servomotor with permanent magnets along with an integral planetary reduction gear head.

The right-angle bevel gear transmits the reduction gear output to the control surface whose axis is orthogonal to the corresponding motor axis. The potentiometer assembly measures the control surface deflection. The details of system specification of control surface actuator are given at Table 65.1.

Table 65.1 System specifications for control surface actuator

Type of control	Aerodynamic (4 nos of rear control surfaces)
Fin deflection rate	4.8 rad/s
Fin deflection	$\pm 20^\circ \pm 1.2$
Command	± 10 V (from command guidance control)
Feedback	± 10 V (to servo electronics unit)
Ortho angle	$90 \pm 0.5^\circ$
Cant angle	$0 \pm 0.5^\circ$

65.3 Assembly Fixture Development for Control Surface Actuators of an Aerospace Vehicle—A Case Study

An Aerospace vehicle with four control surface configuration is required to be precisely assembled to the actuator assembly for better manoeuvrability and enhanced controllability of the vehicle during the mission. The control surfaces are required to be within the permissible limit which is carried out with the help of the assembly Jig. The detailed assembly drawing specifying the permissible geometric tolerances of Aerospace control actuator is shown in Fig. 65.1. In order to maintain the stringent assembly requirements of the dimensional and geometric tolerances of control surfaces within the specified limits, a special purpose assembly jig has been fabricated with the help of CAD software taking into account the manufacturing and assembly variations. Tolerances within an assembly are defined during the setting of engineering specifications in the design phase. However, during assembly process execution, certain assembly variations arise from the individual components, inherent manufacturing process variations, joining and assembly sequence being adopted [3]. Variations always exist due to imperfections in the manufacturing processes and materials, and various other random errors. These deviations propagate and accumulate as parts are assembled and can quickly drive assembly dimensions out of specification.

The process of dimensional variation analysis (DVA) comprised of 3D CAD simulation model which simulated the nominal size and shape of the component parts in the assembly. Virtual fixtures are added to the simulation model of an assembly, which is then used to align the component parts of an assembly to the specified tolerance configuration [4].

The assembly model incorporated and overlaid the realistic achievable features of geometric and form tolerances of the individual components manufactured, thereby simulating the real-time scenario. This 3D model simulation was of great utility in finalising the width dimensions of the Fin Alignment Arm Link of the fixture which completely enveloped the fin surfaces.

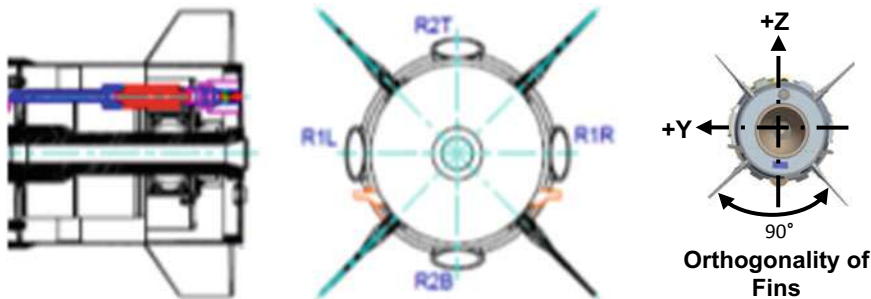


Fig. 65.1 Assembly view of control fins

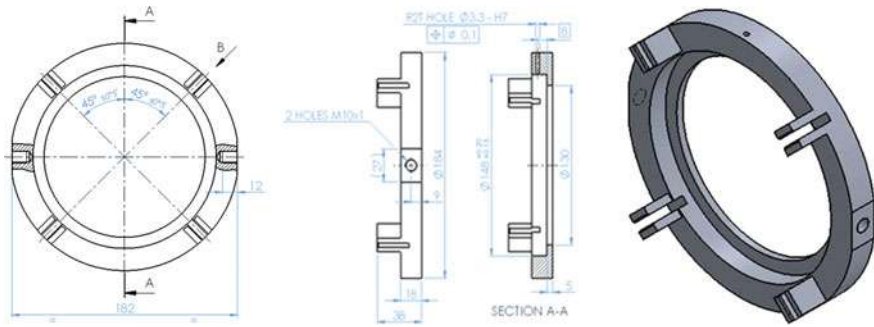


Fig. 65.2 Fin alignment fixture

The validation of the virtual fixtures being largely empirical in nature is dependant upon the 3D simulation model incorporating the dimensional variation of the assembly and its components.

Assembly Fixture planning is to conceptualise a basic configuration through analysing all the available information regarding the material and geometry of the workpiece, operations required and assembly tooling required for the operations.

The following outputs were included in the Assembly Fixture design plan:

- Assembly type and complexity
- Orientation of workpiece within fixture
- Ease of operation
- Locating datum faces
- Design specification limits.

The Assembly Fixture designed as shown in Fig. 65.2 is being employed for precise lateral and tilt alignment of Aerospace fins by incorporating and utilising the precision locating features on the Aerospace component. The dimensional feature of two long fork arm supports of the Assembly Fixture has been designed taking into account the maximum permissible limit variation of Cant Angle and Orthogonality of the individual control surfaces. Further, adequate provisioning features have been incorporated into the fixture with accessibility for ease of assembly of the actuators to the control surfaces.

65.4 Fixture Verification and Functional Performance Evaluation

In general, the assembly tooling primarily controls the position, location and form of the required assembly; however, this tooling has an additional function as a checking mechanism, thereby acting as a quality gate as well for the completed assembly. The CAD model was used to construct the allowable limits of size for dimensional

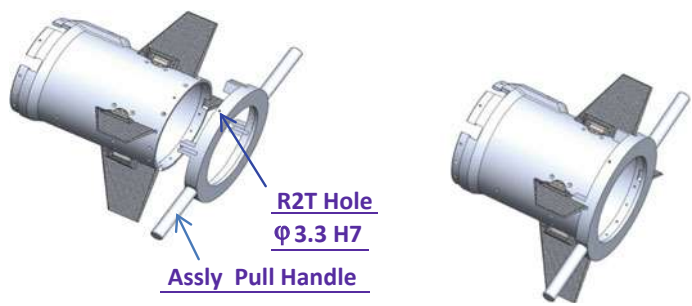


Fig. 65.3 Final assembly of alignment fixture

and form variations for the independent fin assembly. The tooling hole (TH) system is the assembly system that is widespread in aeronautic field and consists of jigs mated with the assembly components by means of pins that are inserted into precise tooling holes (TH) located on the mating assembly components. The tolerances of the TH are very critical since they influence the location and orientation among the different components of the assembly and, therefore, the functional requirements of the assembly [5].

The fabricated Assembly Fixture has been positioned on the actual component assembly as shown in Fig. 65.3 at the correct locator ($\phi 3.3\text{-H7}$) position on the component with the help of high precision locating pin. The Assembly Fixture has been suitably configured to facilitate non-interference during assembly operations and also to envelop the fin assembly at its worst case maximum permissible GD&T Tolerance limit condition. The mechanical assembly of control surfaces was carried out with the help of four M3 mounting screws.

Aerospace fixtures have broadly two metrology requirements namely: (a) fixture setting and recertification and (b) assembly conformance, where the success of (a) is largely dependent on task (b). Measurement systems employed for both of these must have a low uncertainty of measurement to ensure build tolerances are maintained.

Qualification of Assembly Fixture was accomplished through 3D CMM dimensional inspection on the initial prototype hardware assembly by verifying critical key parameters like Cant Angle and Orthogonality which were found to be within the specifications as shown in Table 65.2. The measurement process was subsequently repeated on a number of fabricated assemblies in order to establish consistency of

Table 65.2 Critical dimensional inspection report

S. No.	Specification	Measured value
1	$90 \pm 0.5^\circ$	$90 \pm 0.05^\circ$
2	$\phi 3.3\text{H7}$	$\phi 3.30$
3	8	8.06
4	$\phi 148(+0.15/+0.20)$	148.17

repeatable performance, thereby establishing the confidence in the assembly methodology.

The functional performance of the control actuation system was estimated for (a) null Position (electrical and mechanical) evaluation (b) command and feedback assessment and (c) repeated folding and unfolding operations which quantitatively validated the Assembly Fixture.

65.5 Conclusion

This paper highlights the stages involved in the development of precision in-process CAD-based Assembly Fixture development for Aerospace actuator control surface assembly taking into consideration manufacturing and assembly variations of the components. The development of the Assembly Fixture proved to be advantageous with respect to rapid ease of assembly in meeting the stringent system specifications and further obliterating the lead time involved in 3D inspection. It also reduced the dependency on manual, mechanical in-tool checks for component placement and assembly conformance.

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Chapter 66

Product Design Development and Structural Stress Analysis of Chain Cutting and Riveting Tool for Automotive Vehicle Application



G. Ponsanjoy , M. V. Tamilselvaa , R. Ramanathan
and K. Ganesh Babu

Abstract In the past, the rivets of roller chains, used in two-wheelers, were removed manually with great effort, using a hammer and a chisel or by grinding the head of the pin, resulting in the damages of the outer pin, link plates, and the bushes of the roller chain, which in turn reduces the efficiency of the chain drive. In order to overcome this and make it easier for removing, riveting back, and gauging of master link, we intend to design and develop a tool for removing the riveted pin from the roller chain without damaging the chain drive components. The same tool can also be used for riveting the pins as well as gauging master link in a roller chain drive by slightly changing some components of the tool. The chain drive is held by the tool, and the total operation is done manually by a single person, without the help of any additional fixtures.

Keywords Chain drive · Outer plate · Fixtures · Alignment bolt · C holder

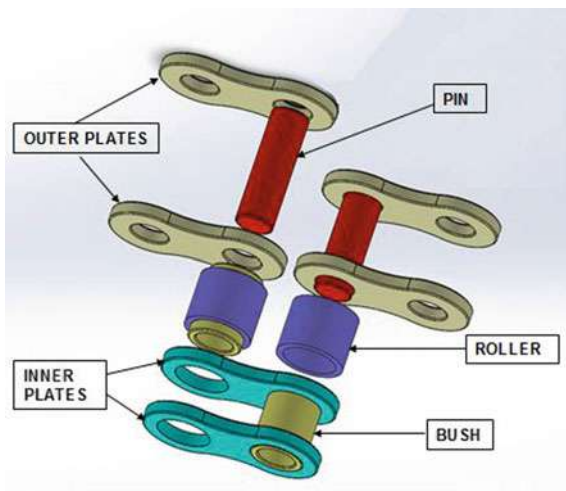
66.1 Introduction

Chains used on a motorcycle, bicycle, or any other machines are reliable machine components that transmit power by means of tensile forces. In general, they are classified according to the material used or manufacturing process involved or their application. Based on their field of application, they can be divided into six types as (1) power transmission chain, (2) small pitch conveyor chain, (3) free flow chain, (4) large pitch conveyor chain, (5) precision conveyor chain, and (6) top chain.

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Fig. 66.1 Basic components of the transmission chain



66.1.1 Typical Structure of Power Transmission Chain

A typical configuration of 420, 428, and 520 type chains, for which we have designed the cutting and riveting tool, is a power transmission chain that is shown in Fig. 66.1. A chain assembly consists of five main components which are inner plate, outer plate, roller, bush, and pin. While manufacturing the chain, the inner plates are press-fitted on both sides of the bush. The pin is inserted with the bush, and its ends are press-fitted on the outer plates. Roller is free to rotate on the bush. This chain is called roller chain. It is driven by a toothed wheel called a sprocket. It is a simple, reliable, and efficient means of power transmission.

66.1.2 Literature Survey

Shoji et al. [1] proposed some methods of weight reduction for the roller chain. The method recommended by them was based on FEM, and the stress and deformation of the roller chain link plate were evaluated. The weight reduction was achieved up to 10%, though the stresses are higher than the existing model.

Velhal [2] observed the gaps in the literature, and hence, the development of a complete model for finite element analysis of the drive chain link was pursued. Work was carried out to evaluate the strength of the chain link, when used in a demanding application of severe loading conditions, throughout the useful life of the chain by using finite element methodology.

Ogunwole [3] presented a manual rivet extractor and designed and fabricated it to facilitate disassembling of blind rivets of sizes 4.0 mm $\times$ 19 mm and 4.8 mm $\times$ 19 mm from joints, machines, and other structured assemblies without causing

damages to the components. It is a robust and handy device; when operated carefully, it is expected to be very useful to the operator.

Kondhalkar and Choudhari [4] presented in their work the determination of safe loads, and the design as well was tested for safe operation in the automobile. As part of their work, evaluation of the design using analytical methodology was carried out and followed by the experimentation to validate it. An existing chain link was used for benchmarking of the research work. Finite element analysis tools like HyperMesh and ANSYS were suitable to reaffirm the performance of the link under tensile loads. Recommendations of the best-suited geometry or material were presented.

Augustine and Girish Kumar [5] established a processing method for analysis of chain drive that is as important as its production. Testing of chains requires machines on which a specimen is mounted and testing was carried out. Testing was performed at hourly, daily, or monthly intervals depending upon the type of test. Testing machines were designed to a particular dimension of the product.

66.1.3 Critical Review of the Literature Survey

Many researchers expressed the importance of chain drive in an automotive system. The extensions of chain components due to tensile load were carefully studied. Few papers described about the reduction of wear and tear in the chain components to enhance the life cycle of the chain drive system. Optimization in chain component design has been analyzed by a few operators, and the modification in design and manufacturing is sought for. All researchers have utilized the better usage of the software at a minimum level, where the abovesaid parameters were addressed thoroughly. Only in one paper, an effort was made to introduce a tool for the removal of rivets from any component with lesser manual effort (less than 1 kN force—calculated from the torque applied on the bar). Even though many researchers alluded about chain drive systems, their stresses, material optimization, its life cycle improvement, etc., none of the papers reported about the design and development of a new tool for simultaneous removal of rivet as well as inserting a rivet, gauging, and riveting it with the outer plate in the chain. In motorcycle manufacturing and servicing industry, the time taken for the assembly of chain and adjustment of chain is to be minimized in such a way that the rate of productivity can be enhanced. For maintenance and servicing of the chain, a simpler tool is in need for manual removal and riveting.

66.1.4 Present Design Development Work

In the past, during the maintenance of the chain, the rivets of the roller chains were removed manually with great effort using a hammer and chisel or by grinding the head of the pin, which is time-consuming coupled with laborious work. Hence, a new tooling system is proposed that will remove the pin from the drive chain without

damaging the outer plate and roller of the chain. This system can also be used for riveting the chain links to the required length.

66.1.5 Objectives of the New Design

Proposed design development work has the following objectives: (i) compact and easy to handle, (ii) lesser time for removal or riveting, (iii) ergonomically designed, (iv) enhanced efficiency, (v) operable by an individual, and (vi) low-cost tool.

66.2 Testing of Chain Rivets

Estimation of how much force is required to remove the rivet pin from the chain was done by carrying out the following processes: (i) design and fabrication of suitable fixtures and (ii) testing for estimation of the force required to remove the pin from the chain link by placing the chain link in the fixture and then subjecting it to compression tests in spring testing machine. Figure 66.2 shows the method of fixing the link and collecting the data from the universal testing machine/spring testing machine. Thus, the compressive force required to remove the rivet pin from the chain was noted. For pushing the rivet pin from chain drive, the force required to be found as $[w] = 2000 \text{ kg} = 19,620 \text{ N}$. After knowing the force required to push the pin and the force that a man can apply through hand to a handle, the size and pitch of the screw



Fig. 66.2 Force estimation using spring testing machine

were calculated by iteration method, assuming a suitable value for the coefficient of friction at the screw.

For designing the tool, on-site measurement of the chain drive and its surroundings in a two-wheeler was considered.

66.3 Schematic Diagram of Fully Assembled Tool

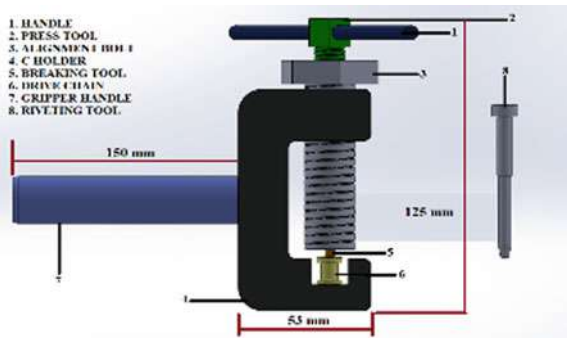
Figure 66.3 shows the assembled view of cutting and riveting tool. The parts of the said tool are as follows: (1) handle, (2) press tool, (3) alignment bolt, (4) C holder, (5) breaking tool, (6) drive chain, (7) gripper handle, and (8) riveting tool. These tools are used for two processes: removal of the pin and riveting of the pin.

66.3.1 Removal of the Pin

The drive chain (6) is placed on the slot of the C holder (4). The top outer link of the drive chain is clamped by the alignment bolt (3) by threaded action on the C holder, thus forcing it to seat in the slot correctly without changing the center line.

There are 3 sizes of the breaking tool used. The breaking tool is selected according to the diameter of the pin of the drive chain. Now the appropriate breaking tool (5) is inserted into the top hole of the alignment bolt. Then the press tool (2) with the handle (1) is inserted into the same hole and screwed down with little force; through this in turn forces the pin of the drive chain out from the chain. Figure 66.4 shows the pin removal from chain assembly.

Fig. 66.3 Assembled view of the tool



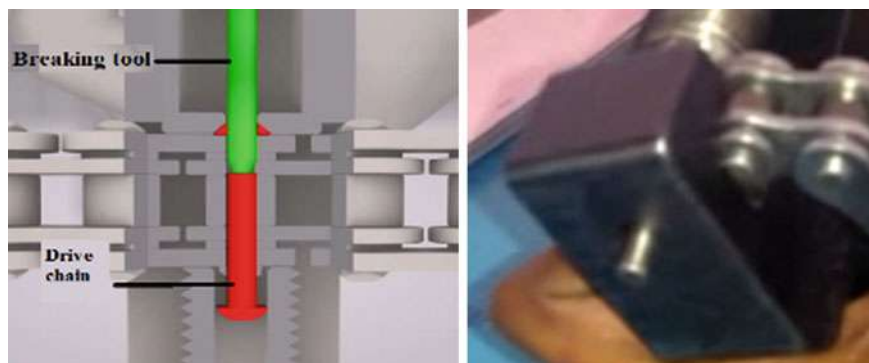


Fig. 66.4 Pin removal process in two-wheeler drive chain using the proposed tool

66.3.2 Riveting of the Pin

Figure 66.5 depicts the riveting of master chain link connecting the two ends of the two-wheeler drive chain.

The separated drive chain (6) with master link is placed on the slot of the C holder (4). The top outer link of the master link is clamped by the alignment bolt (3) by threaded action on the C holder, thus forcing it to seat in the slot correctly without changing the center line.

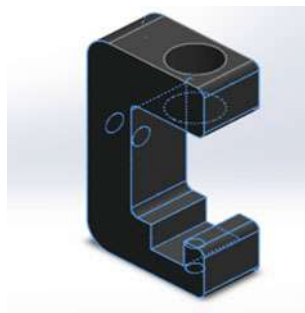
The riveting tool (8) is inserted into the top hole of the alignment bolt. Now the press tool (2) with the handle (1) is inserted into the same hole and screwed down with little force. The tip of the riveting tool flares the rivet in the master link and rivets it. This rivet holds the two ends of the chain together. After design, they are analyzed by using ANSYS software before manufacturing.

66.4 CAD Model Generation

SOLID WORKS and ANSYS were used to create modeling and simulation. First, the CAD model was created as per the designed dimensions, and the files were converted into respective portable formats which the ANSYS software can accept.

Fig. 66.5 Place where riveting is to be done



Fig. 66.6 C holder**Fig. 66.7** Press tool

66.4.1 C Holder and Press Tool

Figure 66.6 shows the CAD model of C holder, and Fig. 66.7 shows the model of press tool.

66.4.2 Alignment Bolt and Gripper Handle

Figures 66.8 and 66.9 show the CAD model of alignment bolt and gripper handle, respectively.

66.4.3 Breaking Tool

Figure 66.10 shows the CAD model of breaking tool, with different end diameters.

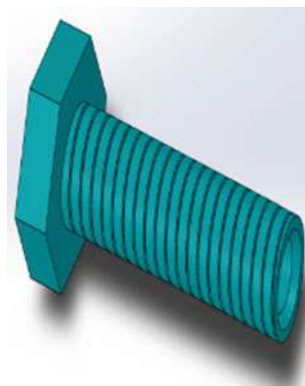
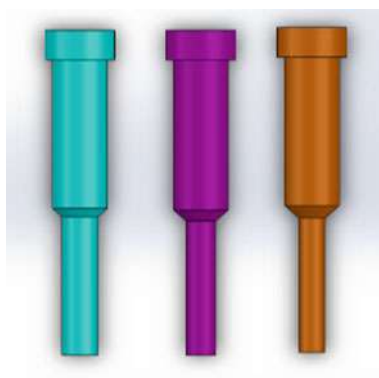
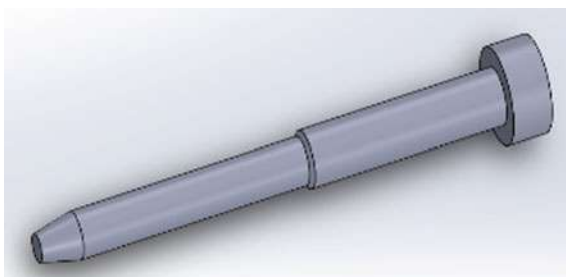
Fig. 66.8 Alignment bolt**Fig. 66.9** Gripper handle**Fig. 66.10** Breaking tool with diameters of 3.30, 4.30, and 5.30 mm

Fig. 66.11 Handle**Fig. 66.12** Riveting tool

66.4.4 Handle and Riveting Tool

Figures 66.11 and 66.12 show the CAD model of handle and riveting tool.

66.5 Modeling and Analysis of Components

Finite element models of the components are made as they have the advantage of including all the components in the model, making it very close to real system. The components were discretized with solid elements, and free meshing is used to mesh the components. Proper boundary conditions and loading are applied at different places to find out the static structural responses and buckling response of the tool components.

66.5.1 Static Structural Analysis of Tool Components

Figure 66.13 shows the maximum von Mises stress of $60,040.7 \text{ N/mm}^2$, which occurs at the tip of the C holder; because of the structural discontinuity in the component (sharp edges), the analytical nominal stress value is 250 N/mm^2 . Number of elements

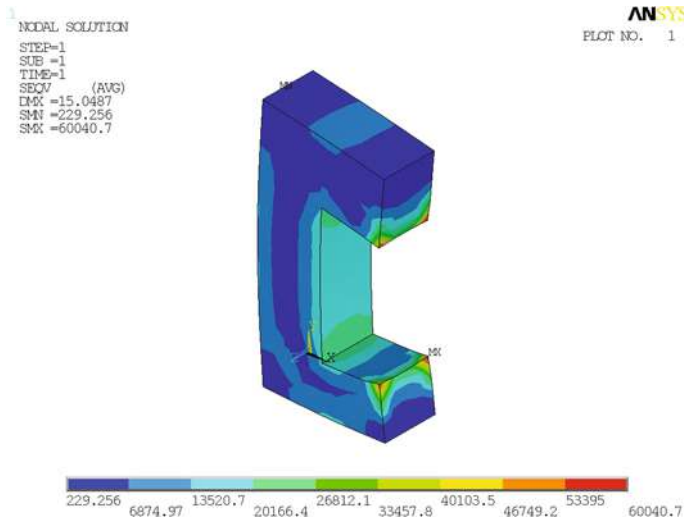


Fig. 66.13 Von Mises stress contour for C holder

is 1711, fixed at the middle of the C holder, and compressive force 20,000 N is applied at the top and bottom ends of C holder. This can be eliminated by filletting the ends of the component.

Figure 66.14 shows the maximum von Mises stress having magnitude of 615,386 N/mm², which occurs at the stepped portion of the gripper handle because of the structural discontinuity in the component (sharp edges). This can be eliminated

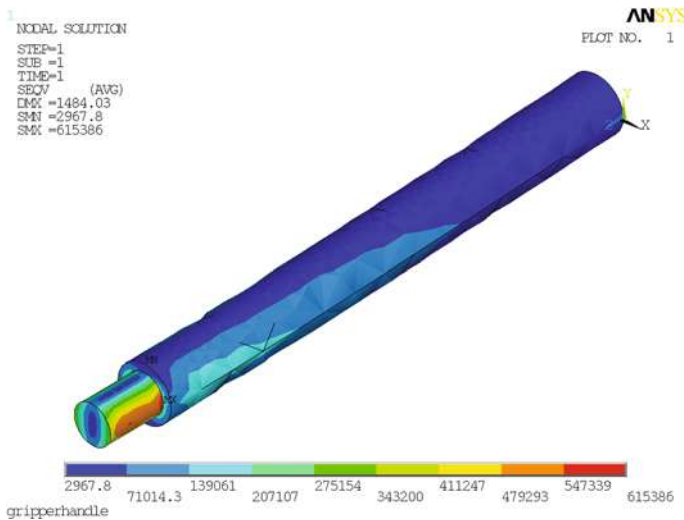


Fig. 66.14 Von Mises stress in gripper handle

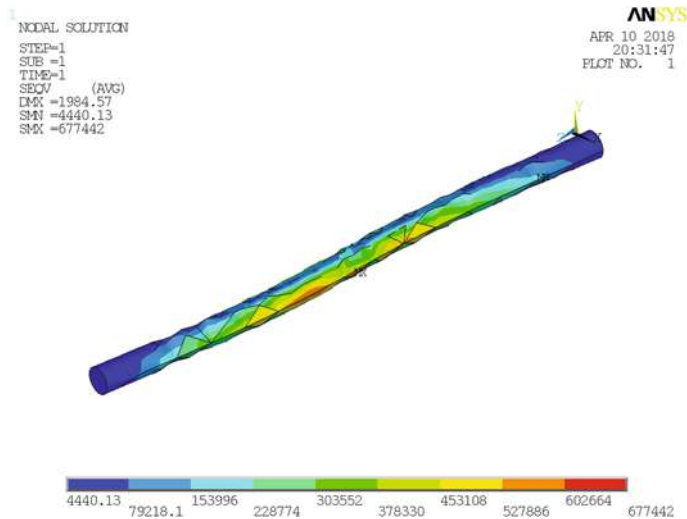


Fig. 66.15 Von Mises stress in handle

by filleting the ends of the component and increase in diameter. For gripper handle, number of elements is discretized into 10,977, one end is fixed, and bending load 130 N is applied at the other end. For handle, number of elements is 3700, middle of the bar is fixed, and both ends are applied with a tensile load of 100 N.

The von Mises counter for the handle is presented in Fig. 66.15. The maximum von Mises stress occurs at the fixed end having the magnitude of 677,442 N/mm². Because load applied at the two ends of the bar is high and the diameter of the bar is small, the stress is high at the fixed point. This can be eliminated by increasing the diameter of the bar at the fixed end or the length of the bar may be reduced. Diameter of the bar has been increased.

66.5.2 Buckling Analysis of Tool Components

Figure 66.16 shows the maximum von Mises stress 6772.29 N/mm² which occurs at the fixed end of the alignment bolt and crippling load of 189,032 N. This can be eliminated by increasing the diameter of the component. Hence, the diameter of the component is increased. Number of elements is 7289 for alignment bolt, and the load applied on it is 4000 N, and number of elements is 14,812 for breaking tool, and the load applied on it is 20,000 N. In both these components, one end is fixed, and at the top of the other end, compressive load is applied.

Figures 66.17, 66.18, and 66.19 show the maximum von Mises stress, which occurs at the tip of the pushing tool because of the structural discontinuity in the

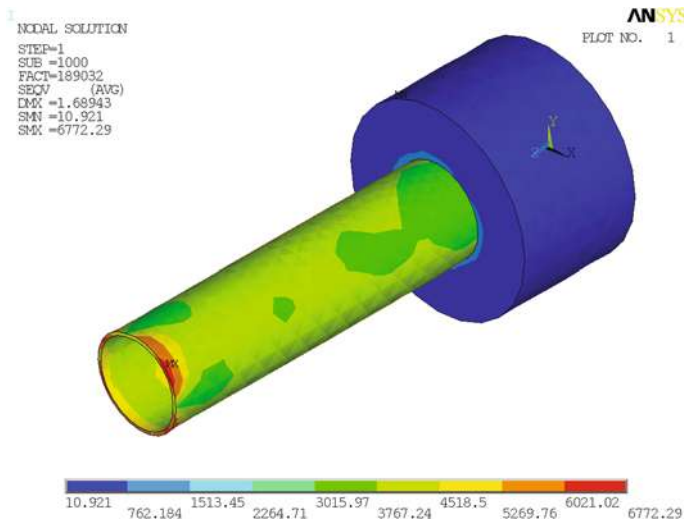


Fig. 66.16 Von Mises stress for alignment bolt

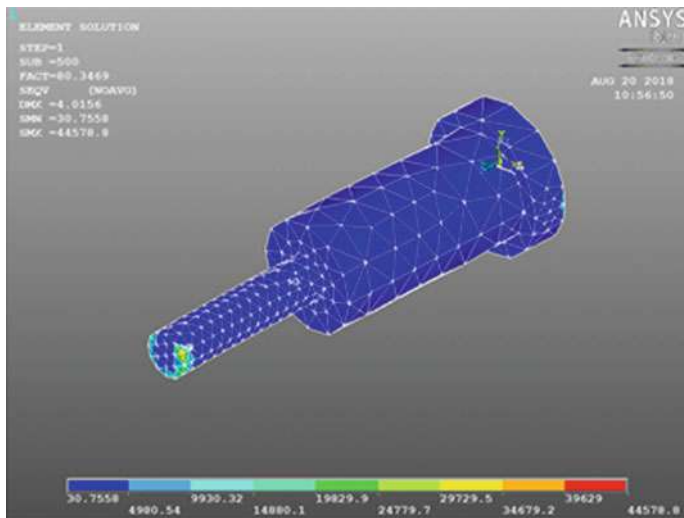


Fig. 66.17 Von Mises stress breaking tool diameter of 3.30 mm

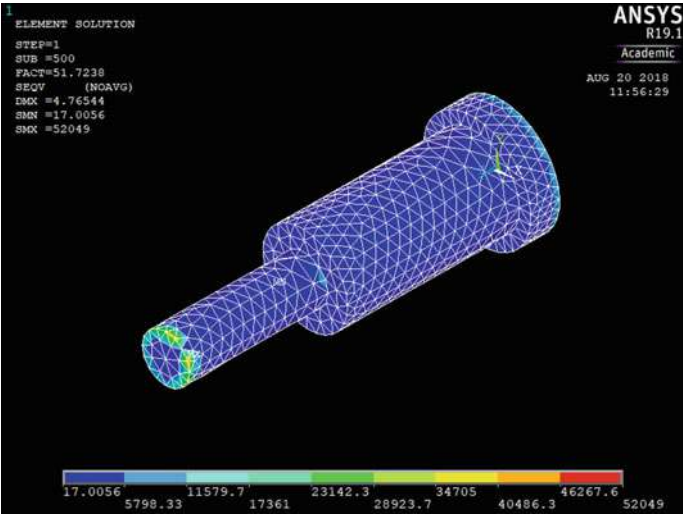


Fig. 66.18 Von Mises stress breaking tool diameter of 4.30 mm

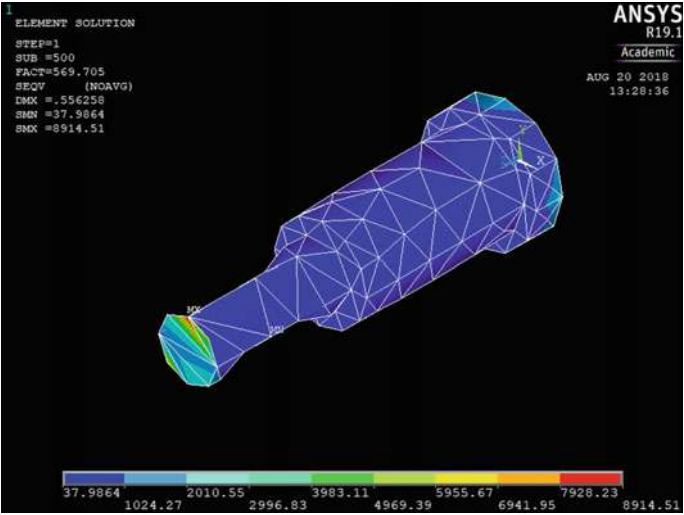


Fig. 66.19 Von Mises stress breaking tool diameter of 5.30 mm

component (sharp edges). This can be eliminated by filleting the end of the component.

66.6 Materials Used

All the components are made of EN-31 material and hardened, except C holder and gripper handle that are made of EN-8 and mild steel, respectively.

66.7 Result, Discussion, and Conclusion

From Fig. 66.13, the stress of 229 N/mm^2 is found to be the minimum value and the calculated analytical value is 250 N/mm^2 . C holder is designed for a standard yield stress value of 725 N/mm^2 for the given material. The breaking strength of the same material after hardening is 1002 N/mm^2 , and hence, the design is very safe. In Figs. 66.14 and 66.15, the minimum stresses are also high. The load applied was very high about 40,000 N for this analysis. The designed load for gripper handle and handle are 130 N and 100 N, respectively, as it is the average force which can be applied by a human being. Based on this, the calculated bending stress values for gripper handle and handle are 25 N/mm^2 and 710 N/mm^2 , respectively, which are safe for the chosen materials. For breaking the tool of sizes 3.30, 4.30, and 5.30 mm diameters, the analysis was repeated. The maximum stress value is 1024 N/mm^2 for the given material which is found at almost all the portions of the tool. Hence, it is safe. The theoretical value of maximum compressive stress is 125 N/mm^2 as calculated. Thus, a low cost cutting and riveting tool has been implemented for two-wheelers to reduce manpower and effort. Compared to the other related tools, this tool is simple in design and easy to manufacture, and hence, the cost of our tool is less. The cost is less than Rs. 2000/-. The concept used is modular, so it can be modified for different sizes of the drive chains. The von Mises stress and crippling load observed from the finite-element analysis predict the strength of the tool design and manufacturing.

66.7.1 Future Enhancement

1. Due to small size of the components, various difficulties were encountered during manufacturing. Such precision jobs can be done in CNC machines. The designed dimensions can be changed for suitable application, standardization, and cost-effectiveness.

2. 3D printing technology can be applied for manufacturing certain components. In 3D printing, the size of the component can be optimized and greater precision can be achieved.
3. The operation can also be automated by using the air gun instead of turning the handle of the tool manually.

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Chapter 67

Design and Development of Cartridge-Based Automated Fluid Delivery System for Ball End Magnetorheological Finishing Process



Z. Alam , D. A. Khan , F. Iqbal , A. Kumar and S. Jha

Abstract Ball end magnetorheological finishing (BEMRF) is a nanofinishing process, and the surface characteristic of the finished product plays a major role. In this process, the volume of polishing fluid affects the quality of the finished surface. Therefore, there arises a need for automated supply of precise and controlled volume of polishing fluid in BEMRF process. In the present work, a new fluid delivery system (FDS) is developed that supplies a precise amount of material-specific polishing fluid to finish different materials under varying finishing parameters by automated delivery of fluid stored in cylindrical-shaped cartridge. Apart from this, since the composition of the polishing fluid is material-specific, it requires continuous change of polishing fluid for finishing of different materials. This becomes extremely expensive and time-consuming with the existing system of polishing fluid being mixed in a large stirrer tank and supplied to the tool tip via a peristaltic pump. The cartridge-based FDS helps in addressing these issues by allowing economical and quick change of material-specific polishing fluid composition. The new FDS is also equipped with radio frequency identification (RFID) technique which verifies the material-specific fluid and only then supplies the fluid for finishing application.

Keywords Fluid delivery system · Automated · Ball end · Magnetorheological · Finishing

67.1 Introduction

Ball end magnetorheological finishing (BEMRF) is the latest variant of MR finishing process. It is based on a smart fluid that undergoes rheological changes on the application of external magnetic field. This change in rheological properties (viscosity and yield stress) is known as the magnetorheological effect [1], and hence, the

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_67

smart fluid is termed as magnetorheological (MR) fluid. MR fluid is a suspension of micron-sized magnetizable particles like carbonyl iron particles (CIPs) dispersed in nonmagnetic oil or water-based carrier medium. MR fluid containing polishing grades abrasive particles is known as magnetorheological polishing (MRP) fluid.

In BEMRF process, a tool spindle made of magnetic material contains a centrally drilled hole, and this tool spindle is placed along the axis of an electromagnet. The polishing fluid passes through the centrally drilled hole in the spindle to reach the tool's tip. When the MRP fluid reaches the tip of the tool, a DC power supply energizes the electromagnet. The energized electromagnet leads to the formation of hemispherical profile of the magnetic field below the tool tip. Therefore, the MRP fluid gets morphed into a hemispherical ball at the tool's tip by orienting itself in the direction of magnetic field as shown in Fig. 67.1a.

The gap between tool's tip and workpiece surface is known as the working gap. A magnetic flux density gradient is formed in the working gap when the electromagnet is energized. In the BEMRF tool–workpiece system, the tool's tip is near to the electromagnet, and the workpiece is comparatively far from the electromagnet. Therefore, the flux density is on the higher side near the tool tip and lowers near the workpiece surface, leading to the formation of flux density gradient. Due to this magnetic gradient, the magnetic CIP in the MRP fluid aggregates toward the tool's tip, and in doing so, it pushes the nonmagnetic abrasive particles toward the top surface of the workpiece [2]. While normal force produced due to this action causes

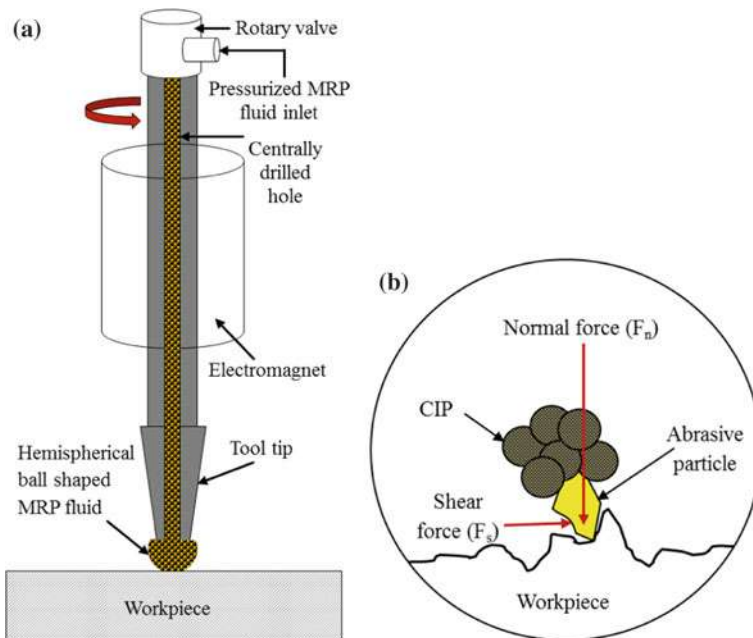


Fig. 67.1 Schematic of **a** BEMRF process and **b** mechanism

the penetration of abrasive into the workpiece, the shear force causes the shearing of the roughness peaks (Fig. 67.1b).

BEMRF process has the capability of nanofinishing a wide variety of materials ranging from ferromagnetic (EN31, mild steel) [3] to nonmagnetic substances like fused silica glass [4], polycarbonates [5], silicon [6], copper [7] and aluminum [8]. Since BEMRF is a nanofinishing process, the surface characteristic of the finished product plays a major role. In this process, the volume of the MRP fluid determines both the quality of the finished surface as well as the size of the finishing spot [9]. In turn, the finishing parameters of the BEMRF process (current and working gap) determine the amount of polishing fluid that is to be used during finishing [9]. Therefore, there arises a need for automated supply of controlled polishing fluid volume in BEMRF process. Apart from this, since the constituents/composition of the MRP fluid is material-specific, i.e., for finishing of different variety of materials different constituents/composition of MRP fluid is required, it requires continuous change of polishing fluid to realize finishing operations on different materials. This makes the BEMRF process both expensive and time-consuming because the existing fluid delivery system (FDS) [3] does not support quick and inexpensive way to change the MRP fluid. Moreover, there should be a system that verifies the material-specific fluid and only then supplies the fluid for finishing application.

In the past, researchers have developed a fluid delivery system for wheel-based magnetorheological finishing process where continuous flow of MR fluid is used [10]. However, the wheel-based MR finishing process has limitations when it comes to finishing of free-form surfaces of complex geometries. The present invention deals with the fluid delivery system designed especially for BEMRF process to improve the capability of finishing of complex geometries of a variety of materials.

67.2 Problems Associated with Existing Fluid Delivery System (FDS)

From the study of effect of polishing fluid volume in BEMRF process conducted by Alam et al. [9], it is evident that controlled and precise volume of MRP fluid is essential to obtain a good sized finishing spot with better quality surface finish. Also, every workpiece material to be finished by the BEMRF machine has a unique composition of the MRP fluid. Some material-specific compositions of polishing fluid for BEMRF process are listed in Table 67.1.

Since the composition of the polishing fluid is material-specific, it requires continuous change of polishing fluid for finishing of different materials. This becomes extremely expensive and time-consuming with the existing system of polishing fluid being mixed in a large fluid reservoir with the help of stirrer and supplied to the tool tip via a peristaltic pump [3]. In the event of change of workpiece, the MRP fluid reservoir is to be cleaned, and the entire fluid is to be changed before the new fluid being pumped to the tool tip. Moreover, the preparation of the polishing fluid

Table 67.1 Material-specific constituents and composition of MRP fluid

Materials	Constituents/compositions of polishing fluid
EN31 steel [3]	CIPs (20%), SiC (20%), paraffin oil + grease (60%)
Copper [7]	Electrolytic iron powder (23%), alumina (14%), base fluid (63%), benzotriazole (additive)
Aluminum [8]	Electrolytic iron powder (25%), alumina (10%), base fluid (65%)
Polycarbonate [5]	CIPs (15%), diamond (15%), deionized water (70%), acetone (additive)
Silicon [6]	CIPs (25%), ceria (5%), glycerol (2.5%), NaOH (0.75%), deionized water (66.75%)
Fused silica glass [4]	CIPs (30%), cerium oxide 1663 (10%), deionized water (60%)

in the stirrer tank for being pumped out by the tool tip makes the process of fluid delivery time-consuming. Also, the existing FDS for BEMRF process is very bulky and lacks any control over the amount of polishing fluid supplied to the tool tip. High volume of polishing fluid often leads to deteriorating surface quality. To prevent the uncontrolled delivery, the polishing fluid may be manually applied to the surface of the workpiece. However, manual application of the polishing fluid can be tiresome, time-consuming and does not ensure that the polishing fluid is evenly applied on the surface of the workpiece.

Another problem associated with the existing FDS is that it does have any mechanism to evaluate or validate the material-specific composition of the polishing fluid before it is supplied to the tool tip for finishing. Polishing fluids having similar constituents but in different volume ratios cannot be differentiated with naked eyes. This often leads to manual or subjective error in selecting the right polishing fluid for the workpiece material. Application of a wrong or undesired polishing fluid on the workpiece may result in wastage of the polishing fluid as well as the workpiece.

67.3 Design and Development of Cartridge-Based Automated FDS

The new cartridge-based FDS addresses the above issues and facilitates automated delivery of precise amount of polishing fluid depending upon the selected workpiece material and finishing parameters. This system is equipped with radio frequency identification (RFID) which aids in verification of material-specific composition of the polishing fluid and only then supplies the polishing fluid to the tool tip. Also, the new FDS is driven by a stepper motor whose drive is controlled by a customized controller having the database of the amount of fluid required for different finishing parameters. This helps in automated delivery of precise volume of polishing fluid to the tool tip once the finishing parameters are set on the machine. Apart from this, the new FDS is cartridge-based which allows for economical and quick change of

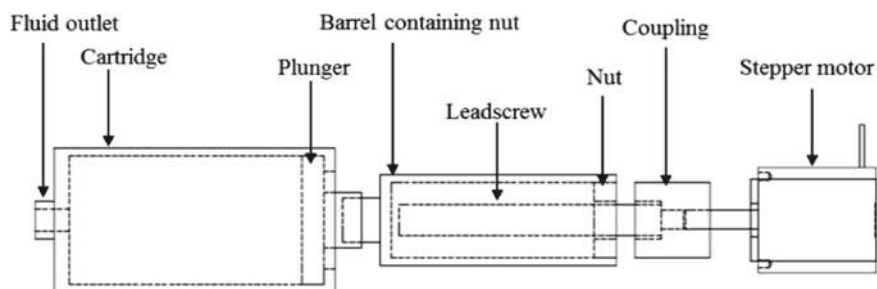


Fig. 67.2 Schematic of the new cartridge-based automated FDS for BEMRF process

polishing fluid composition if the workpiece material changes thus saving both time and money.

Figure 67.2 shows the schematic of various components used in the new cartridge-based automated FDS for BEMRF process. A cylindrical-shaped cartridge is equipped with a small opening at one end and a movable plunger at the other end to push the polishing fluid out of the nozzle. The plunger is connected on the other end with a hollow cylinder which further consists of nut mounted in the inner side. The nut is connected to the leadscrew which is driven by a motor coupled on the other end using a coupling. The rotary motion from the motor is converted to linear motion of plunger via leadscrew and nut arrangement.

The cylindrical-shaped fluid cartridge has a RFID tag placed on the outer wall of the cartridge. The RFID tag contains information about the constituent and exact composition of the material-specific polishing fluid. During application of the cartridge to the automated fluid delivery system, the RFID reader of the BEMRF machine confirms the workpiece material and verifies the related fluid composition present in the cartridge. This system eliminates any error associated with incorrect supply and use of polishing fluid during the finishing operations.

The working of the automated FDS is shown by a flowchart in Fig. 67.3. The stepper motor and its drive are controlled by a customized controller which has access to the database of the amount of fluid required for different finishing parameters [9]. When the workpiece material to be finished is selected by the user, the RFID reader of the BEMRF machine verifies the correct MRP fluid through the RFID tag placed on the MRP fluid-filled cartridge. If the cartridge does not contain the correct constituent/composition of MRP fluid for finishing of the selected workpiece, it asks the user to change the fluid cartridge. Once the right constituent/composition of MRP fluid is verified, it asks the user to provide the finishing parameters/conditions required for finishing the selected workpiece. Depending upon the finishing conditions and the workpiece material, precise amount/volume of fluid is automatically ejected out from the cartridge by the controlled movement of stepper motor [9].

As clear from Fig. 67.2 and the above explanation regarding its working, this new FDS eliminates the error of incorrect fluid usage and supplies precise amount

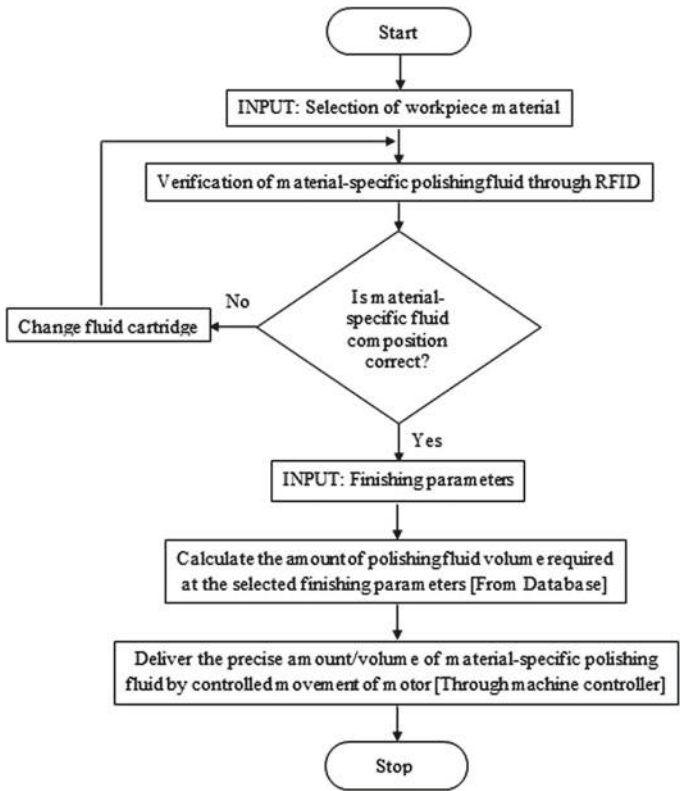


Fig. 67.3 Flowchart showing the working of new FDS for BEMRF process

of material-specific polishing fluid to finish different materials under varying finishing parameters by automated fluid delivery from the cylindrical-shaped cartridge. This system greatly enhances the finishing performance of BEMRF process providing better surface characteristic of the workpiece through controlled and calculated supply of polishing fluid. It is also extremely successful in reducing the wastage and change-over time associated with finishing of different materials by facilitating easy, quick and economical method for changing the material-specific polishing fluid through different cartridges.

67.4 Results and Discussions

Figure 67.4a shows the integration of the automated FDS with the five-axis CNC BEMRF machine. The schematic diagram describes the implementation of the new FDS system for supplying material-specific polishing liquid to the CNC BEMRF machine. This fluid delivery system employs cartridge filled with the polishing fluid

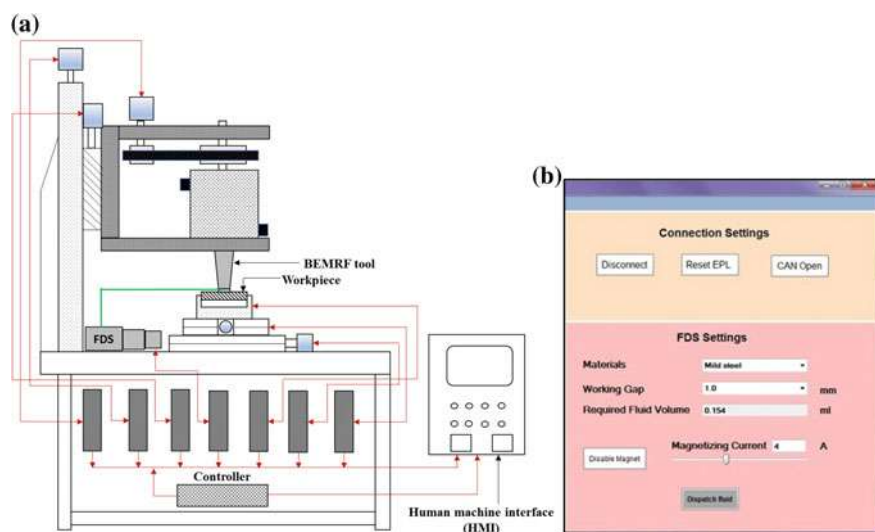


Fig. 67.4 **a** Schematic of integration of cartridge-based FDS with five-axis BEMRF machine and **b** screen shot of HMI for integrated control of FDS settings and other BEMRF parameters

to dispense a pre-defined amount of the polishing fluid to the BEMRF tool tip. Further, the system ensures that correct composition of the polishing fluid is being ejected to the tool tip.

In operation, depending on the material of the workpiece, a cartridge is selected based on the composition and constituents of the polishing fluid filled in the cartridge. This selection is made based on the RFID tag placed on the outer wall of the cartridge. In order to identify the correct MRP fluid for finishing a workpiece material, the cartridge containing the polishing fluid is equipped with an RFID tag which stores information pertaining to the constituents and composition of the polishing fluid. The RFID reader on the five-axis CNC BEMRF machine reads the RFID tag to collect the information pertaining to the polishing fluid in the container. The information is then compared with the polishing fluid data to verify that a correct cartridge is placed in the FDS for finishing the workpiece.

The new FDS also facilitates automated delivery of precise and controlled amount of the polishing fluid, depending upon the finishing parameters. For example, before executing a finishing operation, the user of the CNC BEMRF machine provides certain additional parameters, such as the working gap, pertaining to the finishing of the workpiece. This input is provided by the user through a human-machine interface (HMI) as shown in Fig. 67.4b. The machine controller, based on the additional parameters and the material of the workpiece, determines an amount of the polishing fluid to be supplied to the tool tip. For this, the machine controller accesses the polishing fluid database to determine the amount of the polishing fluid. Based on the determination, the machine controller drives the stepper motor of the FDS so as to dispense a required amount of the polishing fluid to the BEMRF tool tip. For

example, the machine controller may cause the stepper motor to take pre-defined number of steps which in turn causes the plunger to push a specific amount of the polishing fluid.

Further, this FDS prevents any wastage of the polishing fluid in case of change in the workpiece, as the cartridge containing MRP fluid can be replaced with another cartridge, based on the material of the workpiece. Thus, the present fluid delivery system improves the performance of the BEMRF process and reduces any manual error by ensuring the delivery of right amount of material-specific polishing fluid for the BEMRF process.

67.5 Conclusions

From the above work on design and development of cartridge-based automated fluid delivery system for BEMRF process, the following conclusions may be drawn:

- The new FDS supplies precise amount of material-specific polishing fluid to finish different materials under varying finishing parameters by automated delivery of fluid stored in cylindrical-shaped cartridge.
- The RFID tag stores details pertaining to the constituents and composition of the polishing fluid thus facilitating in easy identification of the correct cartridge based on the material of the workpiece.
- It is also extremely successful in reducing the wastage and change-over time associated with finishing of different materials.

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Chapter 68

ARM Controller Based Smart Loom for Generating Basic Weaves



R. Kumaravelu and S. Poornima

Abstract A Smart Loom is a Computing Intelligence based embedded Loom. The benefits of the Smart Loom include the generation of complicated woven fabric without any intricacies in a less span of time. In the present handloom industry, the incorporation of basic weaves on the sample production is very tedious job, since there is no technological intervention in the present scenario. The Basic weaves determines the structure of the fabric. In the proposed system the weaving shedding mechanism is implemented through miniature stepper motors which are operating at low voltage levels. The required basic weaves are stored in the form of Raw Binary Form using Multimedia Memory Card using FAT (File Allocation System) file system. The shedding operation is activated by the stepper motor mechanism using ELJAC principle and it doesn't require any punched cards for the basic weaves generation. The Client/Server mechanism is used for the woven data transfer and weaving operation. In this Development, the Embedded System (ARM Processor LPC2148) Architecture with 32 Stepper motors has been incorporated.

Keywords Jacquard · Memory card · RS485 · Basic weaves

68.1 Introduction

A Smart Loom is a Computing Intelligence based Embedded Loom. The Micro-controller is programmed to activate the heald shaft mechanism in order to produce the woven fabric effect. The Smart Loom facilitates an environment to produce a woven fabric with any basic weaves up to 32 heald shafts. In the handloom based woven fabric for making sample fabric the incorporation no. of heald shaft involves more complication. Hence, facilitating any basic weaves makes less feasibility. In

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_68

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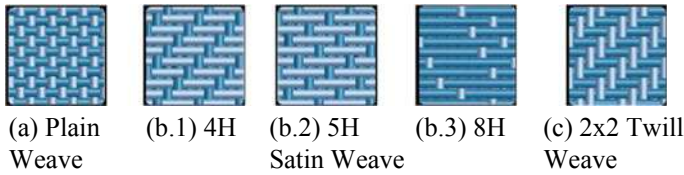


Fig. 68.1 Plain weave models

the existing model, the heald frames are used that too with limitation. In the proposed model the usage of heald frames are avoided rather than all the shafts are individually controlled. A woven fabric is a flat, pliable structure produced by interlacing a series of filling yarns with a series of warp yarns at right angles a specific pattern. The filling yarns are inserted by a shuttle that traverses the yarns from one side of the warp to the other interlacing the two systems of yarns, in a prearranged weave pattern. There are several weave patterns that can be woven. The most common weave pattern there will be variance based on the thread counts and the style of the weaves. The plain fabric comprises the weaves like Fig. 68.1 (a) Plain weave (b) Satin weave (4H Satin, 5H Satin, 8H Satin Weave) (c) Twill weave etc., and the woven structure is furnished.

To produce the plain fabric, the basic weaves are necessary with basic Warp and Weft thread. In general according to the harnessing, the heald shaft count would be varying. Through this product, the heald shafts are eliminated, instead individual warp thread movement is controlled through low voltage stepper motors. Any kind of basic weaves can be produced on the woven fabric with 12 in. width which could help master weavers and academicians/textile students to understand the effect of weaving with different yarns.

68.2 Existing System

The present system uses heald frames in order to generate the woven fabric effect. The heald frame count also limited up to 8. Hence, the inclusion of basic weaves also having limitation.

68.3 Proposed Implementation

68.3.1 Background

The main objective of the smart loom is to generate basic weave effect using embedded system without punched cards and heald frames. Any kind of basic weaves shall

be accommodated to fit the hooks capacity of 32. The end user interface with Master controller facilitate the features like Initialize, Forward, Reverse, Play, Pending Movement, Flag Resetting, Navigation Step Value Adjustment, Pending Movement Step Value Adjustment.

Among the features Play would generate the shedding effect in discontinuous (Multiple thread colour filling effect for the same row) and continuous (Single color filling effect) modes. During the Play option, the memory card gets accessed to fetch weaves information and transferred to the client controller over RS-485 serial bus. Then, client controller parses the information and sends it to the stepper motor through the appropriate I/O board interface. Now, the shedding effect is generated. Upon the acknowledgement from client controller the next sequence of shedding information would be read from memory card and transferred over RS-485. The stepper motors with gear box attachment and pulley arrangement is used for harnessing the warp treads. From the totally available 32 hooks (32 stepper motors) any kind of user-defined harnessing is possible for the 18 in. width fabric and reed size of 40 threads per inch. For the academic and sample preparation purpose frequent pattern changing (using weaves) is mandatory. In the conventional weaving practice, deployment of heald shaft is mandatory and it involves loom setup changes. The conventional process leads to time consuming operation. By our development no setup changing is required. Hence, this smart loom using embedded system would be quite sophisticated for the sample makers and academic people. Every basic weave has binary information in its every cell. The number of cells count in width determines the heald frames count requirement in conventional loom. But, in our development heald frames are unnecessary irrespective of the weave patterns.

68.3.2 *System Interface*

Fig. 68.2—shows the typical system interface of a smart loom using 32 stepper motors.

The proposed system individually controls the warp threads to generate the shedding effect. When plain weave is used the alternate threads are complement of each other and depicted in Fig. 68.3.

68.3.3 *Stepper Motor Signal Strengthening*

The GPIO of LPC2148 microcontroller has to control the movement of the stepper motor. Hence, signal level of the GPIO has to be strengthened. For this purpose the 74HC244 8 bit buffer/line driver with 3-state device is used. Once the signals are strengthened, it is latched with 8-bit latch device 74LS373 in a double level. The first level latch is selected through a decode logic and second level latch is commonly selected through a GPIO signal P0.20.

Fig. 68.2 System interface of smart loom

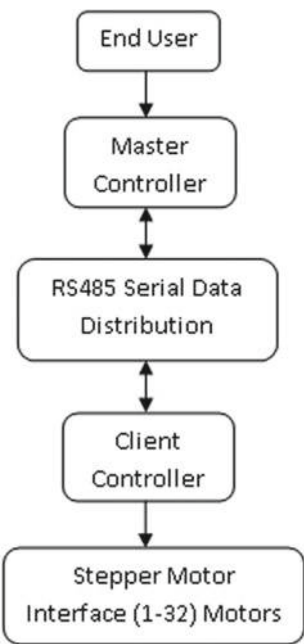


Fig. 68.3 Plain weave pattern for 2 hooks

1	0
0	1

68.3.4 Conversion of Logic Level to Stepper Motor Voltage

After enabling the second level latch the signals are converted from logic value to motor supply voltage using Darlington-Pair photo coupler.

In Fig. 68.4, an illustration of a single GPIO pin logic level is converted to stepper motor voltage and is fed into a single channel of a unipolar stepper motor driver. The usage of Darlington pair model photo coupler provides better efficiency for the stepper motor’s current per phase is high.

68.3.5 Stepper Motor Interfacing

For every stepper motor one ULN2003A driver components is interfaced.

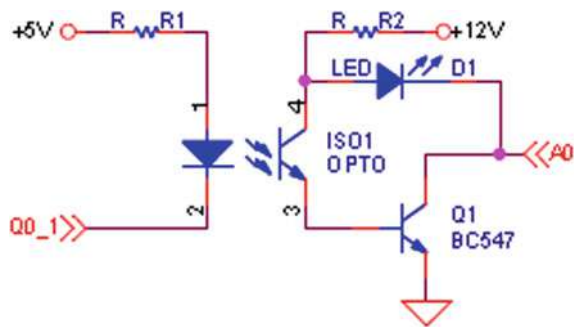


Fig. 68.4 Logic level to stepper motor voltage

The stepper motor logic for controlling the warp thread movement is based on work proposed by kumaravelu et al. [1]. The invention relates to a shedding device for textile technology, comprising an individual thread control system (Jacquard technique) which actuates a weft insertion, with a spring, respectively on the upper and lower ends of the heald in a so-called open-shed operating mode with a 180° lifting course of the lifting element [2]. The total capacity of the electronic card less jacquard is 128 hooks. This is a modular construction with 2 client controller modules [3]. For the circuit protection from back EMF the diodes are connected which is illustrated in Fig. 68.5. During the forward sequence, bit sequence 3, 9, 12, 6 is sent over the data line, accordingly the Unipolar stepper motor wires have been assigned. Similarly, during the reverse sequence, bit sequence 6, 12, 9, 3 is sent over RS485 data line. The jacquard loom requirement is based on the Halaceli [4]. The ARM Processor requirement is based on Vidap and Shahane [5].

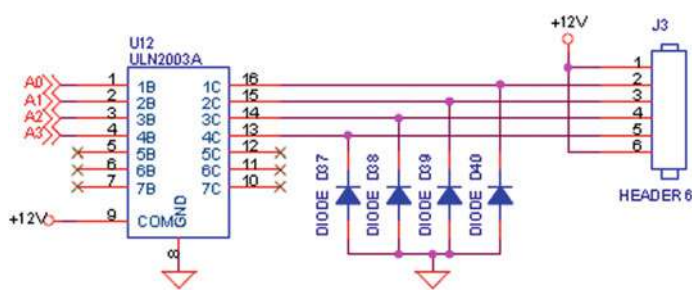


Fig. 68.5 Stepper motor driver interface

68.3.6 Stepper Motor Stepping Algorithm

When the woven design data has digital 1s the stepper motor forward and reverse movement gets activated. For the digital 0s there is no effect in the movement.

Algorithm for Forward Rotation

1. Start the Forward Movement controlling
2. Initialize Stepper pulse sequence data array pdata with the values 3,9,12,6
3. Set PulseCnt to the required count value
4. Read Mdata for the stepper motor from the Master controller
5. Set Pseq index to 0
6. Set Pcnt - pulse iteration count to 1
7. While Pcnt <= PulseCnt
 - 7.1 If Mdata has Binary sequence 1 Then
 - 7.1.1) Read pdata[Pseq] and send it to the Stepper motor driver board to enable the movement
 - 7.2) Set Pseq = (Pseq+1)%4
 - 7.3) Increment Pcnt by 1
8. Stop the Process

As per Fig. 68.3, the basic weaves are constructed with binary values 1 and 0 only. The presence of a value 1 denotes the requirement of a stepper motor movement either in Forward or Reverse direction. The value denotes the necessity of shedding movement. The presence of a value 0 denotes that the stepper motor movement is not required in both of the direction.

Algorithm for Reverse Rotation

1. Start the Reverse Movement controlling
2. Initialize Stepper pulse sequence data array pdata with the values 3, 9 , 12, 6
3. Set PulseCnt to the required count value
4. Read Mdata for the stepper motor from the Master controller
5. Set Pseq index to 3
6. Set Pcnt - pulse iteration count to 1
7. While Pcnt <= PulseCnt
 - 7.1 If Mdata has Binary sequence 1 Then
 - 7.1.1) Read pdata[Pseq] and send it to the Stepper motor driver board to enable the movement
 - 7.2) If Pseq = 0 Then
 - Set Pseq = 3
 - else
 - Increment Pseq by 1
8. Stop the Process

Table 68.1 Mdata binary sequence

Positional weightage	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Value	1	0	1	0	1	0	1	1

The above two algorithms depicts the stepper motor movement direction in either Forward or Reverse direction for the specified number of pulses. For design input any generic image files such as BMP, JPEG, JPG, GIF, WMF are included. The RBF (Raw Binary File) converter utility converts these generic images into raw binary format. The mathematical expression behind the Mdata construction is based on the following expression (Table 68.1):

The mathematical model for the number sequence generation for the bitmap pixel is given below:

$$\text{Mdata} = \sum_{(i=0 \text{ to } n)} 2^i * \text{PixelStatus}$$

where *Pixel status* could be either 0 or 1. The 1 represents dot is present and 0 represents dot is absent. The value n stands for no. of hooks. In this model the no. of hooks represents 32, since 32 stepper motors are used to control the warp thread movement for the shedding effect generation. Every stepper motor I/O board is capable of controlling 8 stepper motors. The **Mdata** identifier is holding the value of every board in order to control the 8 stepper motors movement. In this way, all the 4 stepper motor I/O boards are controlling the 32 stepper motors. The first row denotes the positional weightage in binary number system. The second row denotes the value obtained from the Master controller. The value 1 found in the second row denotes, those stepper motors needs to be turned ON and turned OFF during the Forward rotation and Reverse rotation.

68.3.7 Shedding Device and Harnessing

Figure 68.6, Grouping Grid (21) is used to construct Warp Thread Group and harnessing can be done in Unit repeat, straight repeat, mirror repeat etc., which is found at Harnessing Area (22). In Fig. 68.6, the illustration is given for **32 Hooks** found at Grouping Grid (21). Among 32 hooks of the shedding device (32) the first eight hooks ranging from **1 to 8** is used for the border area of the fabric, the left side is repeated in a straight harnesssed way and right side is harnesssed in a mirrored effect way. For the rest of the hooks ranging from **9 to 32** is straight repeated in the body area of the fabric. In this manner, based on the shedding device capacity (means number of Hooks) the Warp thread groups and size of every Warp thread group can be flexibly arranged. Table 68.2 compares the features of Smart loom with the conventional loom to incorporate the basic weaves sample generation.

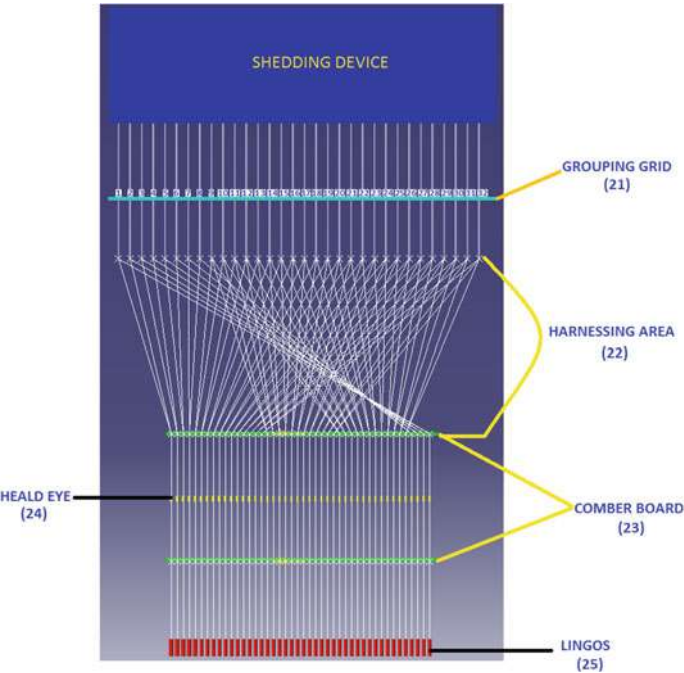


Fig. 68.6 Shedding device with harnessing

Table 68.2 Comparison between smart loom and conventional loom for making the sample

Loom type	Heald shaft requirement	Time to alter the loom
Conventional loom	As per the width of the basic weaves, the heald shaft count is varying	Based on the heald shaft count the warp grouping is done and it is a cumbersome process
Smart loom	No heald shafts are required	Everything is dynamic and any basic weaves are having suitability for the loom without any changes and nothing is difficult

In Fig. 68.6, the Comber Board (23) is used to uniformly maintain the Warp width. It is kept at top and bottom position of the heald eye. In Fig. 68.6, the Lingos (25) are used to control the warp threads vertical movement against gravity force.

Fig. 68.7 Smart loom working model



68.4 Real World Implementation

This 32 Hooks—smart loom is for generating basic weaves using embedded system would help the sample makers/academicians/students to prepare designs/samples very quickly without altering the loom setup. It does not require any punched cards.

Fig. 68.7 depicts the working model of a smart loom, which is capable of producing 12–18 in. width fabric with any kind of basic weaves in weaving. This loom is light weight and easy to port.

68.5 Results and Discussions

In the conventional weaving practice, it is very difficult to incorporate the basic weaves with dynamic changes. Also the no. of heald shafts would vary based on the basic weaves. The developed smart loom doesn't use heald shaft principle, since it uses individual/group controlling of warp threads. For any sampling preparation,

only less time is required, since the design is generated through computer. This smart loom has more usefulness and cost effectiveness in academic/fabric sample making.

68.6 Future Scope

The main objective is to influence the academic people and sample makers, since any basic weaves could be designed and prepared for the weaving process. This system doesn't require any heald shaft for the weaves generation, since every warp thread are individually controlled. This loom is highly compact, and this could be facilitated for the braille weaving also. Also, the warp and weft gives customized support for varieties of thread types with EPI (Ends per Inch) and PPI (Picks per Inch).

Acknowledgements Authors are thankful to the TDT Division of Department of Science and Technology, Govt. of India, for funding the project under S & T scheme (File No: DST/TDD/2K10/91, Dated: 19.05.2011).

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Chapter 69

Manufacturing of Autoclaved Aerated Concrete (AAC): Present Status and Future Trends



Amit Raj , Arun Chandra Borsaikia and Uday Shanker Dixit

Abstract Recent researches have indicated the manufacturing of several useful products by using different industrial wastes all over the world. The materials remain as a waste till their potential to right use is understood. The solid waste in India has emerged as a great threat to the environmental health of the country. Nowadays, building materials are manufactured from various environmental wastes based on needs and availability. An overview has been given on the manufacturing of autoclaved aerated concrete (AAC), one of the potential building materials. The replacement of major raw material (sand) with the solid/industrial wastes in manufacturing of AAC leads toward a sustainable process. In this paper, present status on manufacturing of AAC as well as possible utilization of industrial wastes for its production is presented. A complete industrial manufacturing process of AAC block along with advantages, applications, cost-benefit analysis, challenging issues, and future scopes have been highlighted.

Keywords Sustainable manufacturing · Sustainable material · Pollution · Fly ash · Waste materials · Brick

69.1 Introduction

The conservation of natural resources, eco-friendly manufacturing, energy saving, and comfort living have forced to search a sustainable manufacturing process for a

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_69

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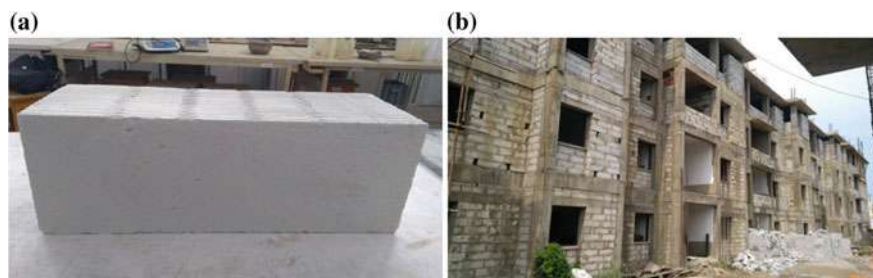


Fig. 69.1 AAC block. **a** a single unit and **b** a building made of AAC block

sustainable building material. The construction industry in India is responsible for the largest share of CO₂ emission into the atmosphere [1]. Clay bricks are manufactured using top fertile soil causing soil infertility, threat to soil erosion, and also a concern of food production. Huge amount of energy is required for the production of clay bricks. The CO₂ emissions during manufacturing of red clay brick have significantly contributed to the global warming [1]. For greener environment, it is recommended to replace the red clay brick with the equivalent sustainable material. The autoclaved aerated concrete (AAC) block or cellular lightweight concrete brick has emerged as the best alternative of clay brick as a sustainable building material. Further, the kilns meant for heat treatment source in clay brick create air pollution and are replaced by steam-based heat treatment, so-called autoclave, in the AAC production industries.

This work aims to review the manufacturing process of AAC blocks, particularly with regard to use of solid/industrial wastes as replacement of major raw material (sand). The use of solid/industrial wastes leads to sustainable manufacturing process for building bricks in place of burnt clay brick. In this paper, present status on manufacturing of AAC blocks as well as possible utilization of industrial wastes for its production is presented. A complete industrial manufacturing process of AAC blocks and its cost-benefit analysis have been studied. The detailed manufacturing stages of AAC along with the cost-benefit analysis for its application are not in open literature. It is a manufacturing process requiring urgent attention of manufacturing/industrial/production engineers, so that construction industry should be benefitted. A sample image of AAC block and AAC building has been shown in Fig. 69.1.

69.1.1 History and Present Status

The AAC block was invented in the mid-1920s by the Swedish Architect Dr. John Axel Eriksson and was patented in 1924 [2]. The first AAC block's plant in India was set up in the 1970s by Siporex at Pune. However, in the North-Eastern part of the country, it was produced in the year 2014 and opened for commercialization in 2015 [3]. The recent developments in AAC production have highlighted the need of

industrial wastes utilization in AAC manufacturing. In the past, several researchers have investigated the possibilities for the use of industrial wastes in the production of AAC, such as use of air-cooled slag in place of sand and lime. The use of coal bottom ash, efflorescence sand, copper trailing, carbide slag, and glassy waste has been also studied for the substitution of quartz sand in the production of AAC [3]. The use of industrial wastes for the production of AAC materials leads to the sustainable development, which is environmentally friendly, economical, and also energy efficient. Rozycka and Pichor [4] utilized the perlite wastes as a replacement of sand for the production of AAC materials and investigated the main properties of respective AAC products. Presently, due to ban on use of top soil for manufacturing clay bricks, AAC blocks are in great demand. At present, 10% of total requirement of bricks in construction industries is satisfied with AAC blocks [2].

69.2 Advantages and Applications of AAC

AAC material is a special kind of concrete, which uses fly ash or sand, lime, cement, water, and a foaming agent (aluminum powder) for its manufacturing. The AAC material is an eco-friendly material with almost negligible environmental impacts. The porosity property due to expansion process imparts lots of interesting features/advantages in this type of material to be well fitted for building application. The present AAC materials help in establishing a green environment due to its environmentally friendly nature. In the AAC production process, no toxic by-products or pollutants are evolved and diffused in air. Since all the ingredients used in the production of AAC blocks are inorganic in nature, it does not contain any toxic substance. The environmental pollution from clay brick-making operations is injurious to human health, animals, and plant life. A building constructed from AAC block emits only 2.2 kg of CO₂ compared to 17.6 kg of CO₂ emitted due to use of clay brick for a carpet area of 1 Sq Ft [1]. At a global level, the air pollution from clay brick-making operations contributes to the phenomena of climate change and global warming. Moreover, the AAC materials are also helpful for consuming the power plant by-products (fly ash).

The gas formed by the reaction of aluminum with a soluble alkali in the lime/cement slurry makes the aerated concrete lightweight [2]. The lightweight cellular structure of AAC makes ease of handling, transportation, and wall installation for the construction activities [5]. The lightweight property of AAC material also imparts high strength to weight ratio. AAC is typically a low density, porous, and lightweight, by which it reduces the dead load and seismic inertial force acting on the structure as compared to the conventional clay brick. With high porosity, thereby low density ($\sim 500 \text{ kg/m}^3$) and thermal conductivity (0.1 W/mK); the AAC material can serve as a sound-proofing and thermal insulation material [2]. The porosity value for AAC material was found to be 82.1–91.5% for a density range of 250–500 kg/m³ [6]. The porosity (H) of AAC is determined by helium pycnometry given by

$$H = \left(1 - \frac{\rho_m}{\rho_f}\right) \times 100, \quad (69.1)$$

where ρ_m and ρ_f are the apparent and skeleton density (kg/m^3), respectively. The apparent density is normally measured by immersion in mercury and the skeleton density can be measured by determining the difference in helium pressure between an empty chamber and the chamber containing a sample [6]. The total porosity decreases with increase in apparent density. The thermal conductivity of any cellular concrete is due to air-filled pores. The value of thermal conductivity, k (0.08–0.21 W/mK) of aerated concrete primarily depends on its density. Further, AAC is non-combustible (fire resistant), and hence in case of fire, it can help to prevent the fire from transfer to other rooms. Autoclaved aerated concrete (AAC) can be used in the form of block and panel for masonry wall constructions, floors, roof insulation, trench fills, and for other insulating purposes [5]. There is an increasing use of this material for non-structural applications in the past decade such as infill panels and cladding. AAC has a wide range of application for building construction with residential, commercial, and industrial buildings. AAC block, being lightweight, is suitable for the realization of masonry bearing wall of low to medium story building in the seismic zones for good seismic resistances [2]. As AAC is a prefabricated product, it is of high precision and drillable. AAC panels are used directly in internal wall of concrete structures and steel structures, such as schools, in hotels, offices, houses, and market places, etc. The internal walls such as separation walls, household walls, partition walls in bathroom, and kitchen also find the application of AAC panel. The AAC blocks have application in both external and internal walls. Several authors reported the application of AAC material in the structure of wall system and roof. Researchers [5] studied the utilization of AAC in green roof structure.

69.3 Manufacturing Phases of Autoclaved Aerated Concrete

There are mainly five phases involved in the production of AAC block and panel units. They include mixing of raw materials, casting, expansion or rising, wire cutting, and autoclaving. The present study focuses on detailed manufacturing steps of AAC blocks. All the observations and visualizations in manufacturing of AAC blocks have been made in a local industry (KD Infra, Guwahati, India). The stages for producing the AAC block are summarized in Fig. 69.2. The detailed manufacturing stages of AAC block have been described in the following subsections.

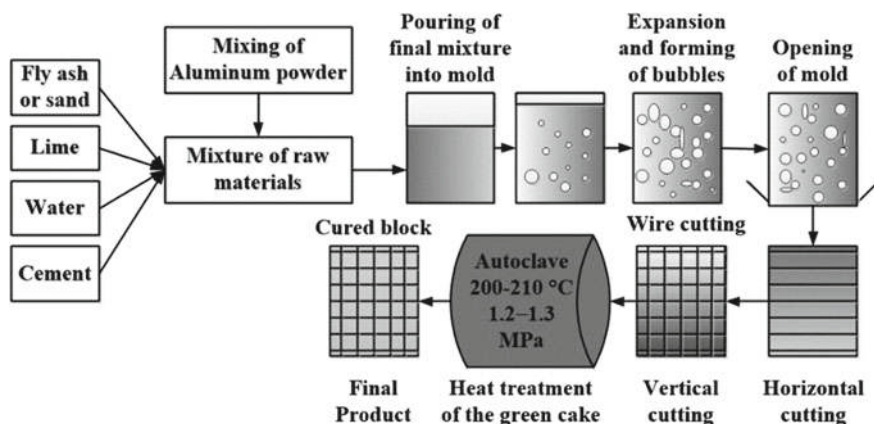


Fig. 69.2 Block diagram representing the manufacturing stages of AAC block

69.3.1 Mixing of Raw Materials

Raw materials used for producing autoclaved aerated concrete are fine grade materials. Fly ash or sand, lime, cement, water, and aluminum powder are the main raw materials for producing AAC. A total of 3150 kg of mixture (43–45% fly ash or sand, 36–38% water, 13–14% cement, 6–7% lime, and 0.025–0.03% aluminum powder by weight) is prepared for filling a single mold. The prepared raw material mixture is weighed by an electronic scale controlled by programmable logic controller (PLC) system automatically.

69.3.2 Pouring and Expansion of Mix Slurry

After the preparation of the final mix with all its ingredients, the final mix slurry is poured into an empty oil-treated mold through up-and-down pouring device. Presently, the mold used in the industry (KD Infra, Guwahati) is of size 4200 (length) $\times$ 1200 (thickness) $\times$ 600 (width) mm³. The internal surface of the mold is treated with oil for avoiding any sticking of mix on the surface of mold. The volume of raw material mixes allowed to occupy the mold is fixed to 60–65% of mold volume. The remaining 35–40% volume of mold is kept unfilled for the expansion process to save the material spillage. After the pouring process is completed, the filled mold is transferred to the pre-curing room by a ferry cart for pre-curing under constant temperature. The pre-curing of about 4–5 h is done in order to get the required hardness for wire cutting. The pre-curing also allows the cake for its expansion and gaining initial hardness. The aluminum powder reacts with calcium hydroxide, which is the product of reaction between cement and water. This reaction causes forming of microscopic air bubbles which results in increasing

of paste-volume. The formed hydrogen, which is a lighter gas rises and is replaced by air which is a denser gas. As a result, the hydrogen gas escapes out of the material leaving the small air bubble in it. The volume increase is dependent upon the amount of aluminum powder added. The volume increases with increase in the amount of aluminum powder.

69.3.3 Wire Cutting of Green Cake

The mold is transported to the locating position under tilting hoister by friction wheels, which take the mold to the cutting line. The tilting hoister turns the mold by 90° and puts on the cart for demolding. After the demolding process, the large green cake is transferred to the wire cutting line. The wire used in this cutting process is highly tensioned by air pressure with the help of a pneumatic cylinder. The size of wire used is of 1–2 mm in diameter. A group of wires in both horizontal and vertical directions are adjusted according to the variable desired dimensions of AAC block.

The size of AAC block generally produced in the industry are $600 \times 250 \times 200$, $600 \times 200 \times 200$, $600 \times 150 \times 200$, $600 \times 100 \times 200$, and $600 \times 75 \times 200 \text{ mm}^3$ corresponding to length, thickness, and width dimensions. The variable sizes of the AAC block produced are as per the need of customer. The wire cutting operations are carried out in two separate stages, horizontal cutting and then vertical cutting. After the horizontal cutting, the large green cake gets divided into six slices of dimension $4200 \times 200 \times 600 \text{ mm}^3$. Subsequently, the exchange device lifts the green cake and puts on the second cutting cart for vertical cutting. During the vertical wire cutting, the cutting machine frame swings back and forth like a saw.

In the vertical cutting, a 4200 mm length side large cake is divided into variable 43 parts. Finally, a stack of 258 (43×6) AAC blocks of same length and width but of variable thickness ($600 \times 250 \times 200$, $600 \times 200 \times 200$, $600 \times 150 \times 200$, $600 \times 100 \times 200$ and $600 \times 75 \times 200 \text{ mm}^3$) are obtained after the wire cutting. The total number of final AAC block obtained from a single mold may vary with desired dimension of individual AAC block; accordingly the arrangements of wires are adjusted in the cutting machine.

69.3.4 Autoclaving or Hydrothermal Treatment

After the final cutting operation, the stack of AAC block is transferred for heating or steam curing in an autoclaved chamber. This large steam-heated vessel is in fact a large pressure cooker by which the autoclaved aerated concrete is cured to gain strength. Autoclaving of AAC green cake is carried out with a steam at a high temperature ranging from 190 to 210°C and high pressure ranging from 1.0 to 1.3 MPa. The whole autoclaving process completes in 16–18 h. During the autoclaving or hydrothermal treatment, the formation of tobermorite ($5\text{CaO} \cdot 6\text{SiO}_2 \cdot 5\text{H}_2\text{O}$) occurs

[5]. The formation of tobermorite, during the cooking period is affected by the presence of aluminum powder. In other words, the mechanical properties of final AAC are influenced by the formation of tobermorite. After the autoclaving, the cured or hardened stack of AAC blocks is pulled out from the autoclave chamber with the help of the ferry cart. The final product is kept outside for cooling so as to attain the room temperature. Finally, the finished AAC blocks are taken to the storage yard with the help of a fork lift.

69.4 Cost-Benefit Analysis

The sustainability of any new product is governed by the cost-benefit ratio. In India, the demands and costs of building materials are increasing rapidly due to high growth of population. Both demands and costs can be met by adopting the AAC blocks as a primary building wall material. The building wall systems are comprised of mortar, brick, and reinforcements. In this section, the overall cost for AAC block wall system and clay brick wall system has been analyzed and compared. The cost analysis for both brick wall system is shown in Table 69.1. The analysis is carried out assuming same centering, shuttering, and scaffolding cost incurred for both AAC and clay brick wall of 10 m^3 . The market price of items for both of the wall system (AAC and clay brick) has been collected locally (Guwahati). The analysis of 10 m^3 of wall system has been carried out with the help of procedures followed in India [7]. A good margin of cost difference can be seen from the analysis values. It has been found

Table 69.1 The cost analysis of building wall of AAC and clay brick (for a 10 m^3 wall)

Particulars	Qty. or Nos.		Rate (Rs.)		Cost (Rs.)	
	Clay brick	AAC block	Clay brick	AAC block	Clay brick	AAC block
Materials						
(1) Brick	5000	330	8/unit	109/unit	40,000	35,970
(2) Cement	0.75 m^3	0.45 m^3	$11,524/\text{m}^3$	$11,524/\text{m}^3$	8643	5186
(3) Sand coarse	2.25 m^3	1.35 m^3	$2000/\text{m}^3$	$2000/\text{m}^3$	4500	2700
Labor						
(1) Head mason	$\frac{1}{2}$ nos.	$\frac{1}{2}$ nos.	612/day	612/day	306	306
(2) Mason	16 nos.	10 nos.	522/day	522/day	8352	5220
(3) Mazdoor	10 nos.	6 nos.	462/day	462/day	4620	2772
(4) Coolie	10 nos.	6 nos.	462/day	462/day	4620	2772
(5) Bhishti	2 nos.	1 nos.	462/day	462/day	924	462
Total					72,025	55,448

that total cost of wall system made of AAC block is approximately 22.73% less than that of wall system made up of ordinary burnt clay brick. Thus, AAC block is cost effective.

69.5 Challenging Issues and Future Direction

Porosity is one of the major challenging issues in the case of use of AAC blocks as 35–40% of total volume of block is occupied by air voids, which are seen on surfaces also. Thus, challenge is to prevent moisture inflow through a wall system made of AAC block in highly humid weather condition of our country. Intermolecular bonds in AAC blocks are weak as revealed by observed results of compressive strength that reaches only up to 3 MPa as compared to clay brick of 8.24 MPa [2], studied in India. Also, the tensile strength of AAC block in India has been reported as 0.26 MPa [8], whereas for clay brick it is 1.0 MPa [2]. The compressive and tensile strength of AAC block lie between 2.53–2.67 MPa and 0.54–0.64 MPa, respectively as reported by Ferretti et al. [9] in Italy. Low compressive and tensile strength of AAC do not play load-bearing roles in framed structures. Also, the seismic performance of AAC structure cannot be ignored. Further, AAC blocks are very recent building material in India. It is used in construction industries for less than thirty years. Thus, there is a lack of data on long term durability and threat to natural calamities.

Presently, China has over 1000 AAC block manufacturing units. However, only 50 AAC manufacturing units are there in India such as Aercon, Magicrete, Siporex, Xtralite, Brikolite, Superlite, Greentech, Biltech, etc. [10]. More manufacturing industries should be established throughout the rural and urban areas to meet the demand.

69.6 Conclusions

AAC blocks are manufactured in the industries very scientifically, with proper supervisions and inspections. The mechanized systems used in the industries are responsible to produce precise AAC blocks with close tolerances and with desired variable sizes. The manufacturing process of AAC blocks is more scientific than any other brick production process. The weight of AAC blocks is actually one third that of burnt clay bricks. The strength to density ratio of AAC blocks is very encouraging. The total cost incurred for the preparation of AAC wall is 22.73% less than that for constructing the clay brick walls for same wall dimension. Overall, the AAC block is an economical, efficient, sustainable, and reliable building material.

Acknowledgements This work is the part of a Department of Science and Technology (DST) sponsored project entitled “Design and development of proper bonding mechanism for individual AAC block units in wall system of a structure” through grant number DST/TSG/AMT/2015/375. Authors wish to acknowledge DST and industry partner *M/s K D Infra*, Guwahati for their support.

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Chapter 70

Influence of Flow Domain Parameters on Hot Water Actuation of Shape-Memory Alloy Spring for Barrier Gate System



R Mithun , Tameshwer Nath , S. S. Mani Prabu and I. A. Palani

Abstract The study concerns the comparison of the experimental and analytical techniques using hot water actuated shape-memory alloy (SMA) spring for energy-efficient barrier gate. Specifically, this script studies the heat transfer analysis of the shape-memory alloy spring, using COMSOL Multiphysics. The actuation rate is one of the important factors for using SMA spring for rapid operation of barrier gates. In hot water actuation, this speed of actuation is influenced by two important parameters, the temperature of the inlet water and the velocity at which the water flows into the system. Parametric sweep for an inlet temperature of 65, 75, and 85 °C and velocity rates of 5 and 7 mm/s were studied. The conclusions from this parametric investigation can be used to choose the optimum values for the parameters to achieve the required actuation. The results from the finite element analysis were compared with the experimental results.

Keywords Shape-memory alloy · Nitinol · Hot water actuation · COMSOL

70.1 Introduction

In a mechatronics system, actuators and sensors form an essential part of operation and control. A barrier gate system, considered especially for security purposes, is usually powered by hydraulic, pneumatic, or servo motors. These actuators are bulk and complex, and require many additional components for their working. The barrier gate actuation system can be easily replaced by smart technology. Degeratu et al. [1]

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_70

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proposed that a shape-memory alloy (SMA) spring can be used to lift the barrier gate when it is actuated using Joule heating. The term Shape-memory alloy refers to that group of inter-metallic alloys which expresses the capacity to reorient their predefined structure after interacting with thermal stimuli. Uniqueness in the materials is due to two properties: shape-memory effect and pseudo-elasticity [2–5]. Shape-memory alloys can be of two types: one way or two-way shape-memory alloys depending on whether they change only to heating or for cooling also [6, 7]. The actuation of these smart materials can be done through contact (Joule heating and hot water actuation) [8–17] or non-contact (laser actuation) [18–23] heating. When the SMA spring is deformed from its original position with the help of load, it can be brought to its original shape with the help of thermal interaction. On heating the SMA spring, it begins to attain its original shape, and this temperature is called austenite start temperature. On further increasing the temperature, the spring completely attains its original shape, this temperature is called austenite finish temperature. This reorientation to the parent state due to the interaction of the heating process is called martensitic phase transformation. There is an array of advantages like it is eco-friendly because of no moving parts, it needs no lubrication and the process is silent throughout the operation [24].

In this study, an SMA spring-operated barrier gate with hot water as actuation medium is proposed for the application. The arrangement consists of a hot water sump, heated by the thermal collector, that delivers the hot water to the container where the SMA spring is already positioned. Figure 70.1 shows the schematic diagram of the proposed model. This SMA spring is trained to contract on martensitic phase transformation. When the hot water flows into the container, it transfers its heat energy to the SMA spring. Thus, the SMA spring actuates and contracts, which in turn lifts the barrier gate. In order to bring back the barrier gate, the hot water is drained from the container to the cold water sump. For a quick return of the barrier gate, an expansion spring is used in the counter direction, once the hot water is drained from the container. In this work, the influence of the inlet temperature and velocity of water on the hot water actuation of the SMA spring is deliberated.

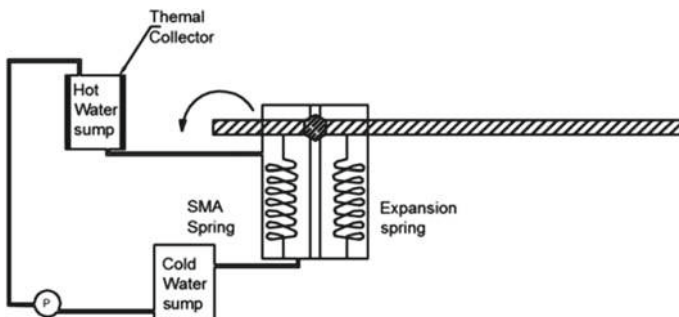
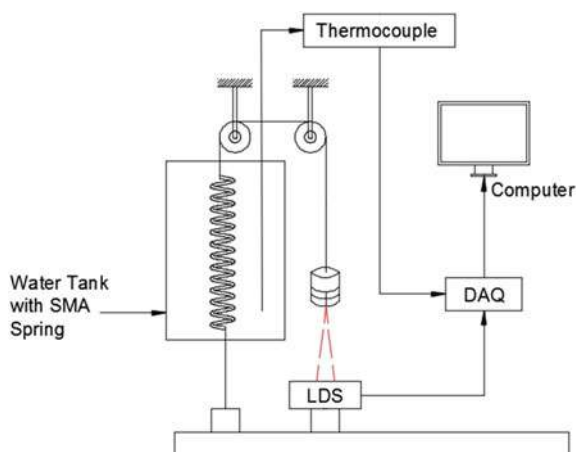


Fig. 70.1 A hot water actuated barrier gate arrangement

Fig. 70.2 Hot water actuation schematic diagram



Further, the parametric investigation of hot water actuation has been studied using the finite element method in comparison with the experimental results.

70.2 Experimental Test Bench

70.2.1 Components

A shape-memory alloy Ni–Ti (Nitinol) expansion spring, commonly used for micro actuations, is used in this study. This SMA spring is enclosed in the container, allowing the water to flow in and out. The spring has 18 turns with a wire diameter of 0.7 mm and mean diameter of 5.69 mm, and its solid length is 13.86 mm. One end of the spring is fixed, and the other end is loaded to extend the SMA spring. Figure 70.2 shows the schematic of the hot water actuation setup. A laser displacement sensor (LDS) was used to measure the displacement of the spring. A *K*-type thermocouple, with a measurement range of -270 to 1260 °C was used to measure the temperature change over the time. For recording the displacement and temperature readings, a data acquisition system (DAQ) was used (Agilent 34970A).

70.2.2 Working

When a deformed SMA spring is heated, it changes from a martensite state to austenite state, this is called the Martensitic phase transformation. The spring, deformed after the application of load, can be actuated by heating it to the austenite start temperature until reaching the austenite finish temperature (60 – 85 °C). The necessary

heat input for this transformation can be achieved by submerging the spring into the hot fluid. In this test setup, for actuation of the spring, hot water is poured inside the container. On contact with the hot water, the spring absorbs heat from the hot fluid and reaches austenite finish temperature. This makes the spring to contract and come to its original solid length along with the load [1].

The actuation percentage of the spring depends on the level of hot water in contact with the spring. Once the spring was fully immersed in the hot water, the spring was actuated fully to regain its solid length. After the actuation, the hot water in the container was drained and the spring was left to cool by natural convection.

K-type thermocouple, immersed in the water, was used to measure the temperature change during both the heating and cooling of the spring. A calibrated LDS was used to record the displacement of the spring. Temperature readings from the thermocouple and the displacement readings from the LDS were recorded in the DAQ system. From the recording, the time taken for heating and cooling can be found.

70.3 Mathematical Modeling

The actuation of the shape-memory alloy spring is achieved by introducing the hot water into the system. The heat energy required for the martensitic phase transition is supplied by the hot water to the spring. This heat transfer from the hot water to the spring is modeled as a conjugate heat transfer between a solid and a fluid. The conjugate heat transfer with the following assumptions was considered.

1. The flow inside the container is considered to be a laminar flow.
2. The heat lost by the fluid is the heat gained by the solid.

The motion of fluid inside the container transfers its heat energy to the shape-memory alloy spring. Conjugate heat transfer is a combination of conduction heat transfer in solids and convection heat transfer in fluids, exchanging heat energy at the interfaces. The heat transfer in solids is governed by Fourier's law of conduction.

$$q = -k\nabla T \quad (70.1)$$

where ' q ' is the heat flux density (W/m^2), ' k ' is the thermal conductivity ($\text{W/m } ^\circ\text{C}$), and ' ∇T ' is the temperature gradient ($^\circ\text{C}$).

The heat transfer in fluids, either free or forced convection, is governed by Newton's law of cooling.

$$q = h_c A dT \quad (70.2)$$

where ' h_c ' is the heat transfer coefficient, ' A ' is the heat transfer surface area and ' dT ' is the temperature difference between the surface and fluid.

The motion of fluid through or over the domain is considered to be a laminar flow. The laminar flow is governed by the Navier-Stokes equations and continuity

equation. Navier-Stokes equation stands for the conservation of momentum, where the inertial force is equal to the summation of the pressure, viscous and the external force applied to the fluid. For a time-dependent problem, the Navier-Stokes equation is given by,

$$\rho \frac{\partial u}{\partial t} + \rho(u \cdot \nabla)u = \nabla \cdot [-pI + \mu(\nabla u + (\nabla u)^T) - \frac{2}{3}\mu(\nabla \cdot u)I] + F \quad (70.3)$$

where ' u ' is the velocity of the fluid, ' p ' is the pressure applied to the fluid, ' ρ ' is the density of the fluid, ' I ' is the identity matrix, ' F ' is the external force and ' μ ' is the dynamic viscosity of the fluid.

The continuity equation represents the conservation of mass,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0 \quad (70.4)$$

In conjugate heat transfer, the heat lost due to the conduction compensates the heat gained due to the convection. Thus, for a time-dependent conjugate heat transfer, the governing equation is given by,

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p u \cdot \nabla T + \nabla \cdot q = Q \quad (70.5)$$

where ' C_p ' is the heat capacity at constant pressure.

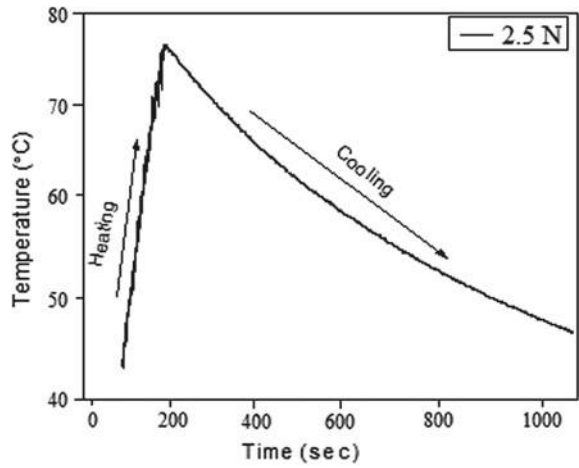
70.4 Results and Discussion

70.4.1 Experimental Results

Hanging a load of 2.5 N, causes the spring to deform up to a length of 4.5 mm. This deformation can be reduced to zero by heating the SMA spring. On heating the deformed SMA spring, due to the shape-memory effect, the SMA spring reaches its original parent solid length shape. The hot water poured into the container slowly rises up and starts to actuate the SMA spring. For experimental discussion, it is assumed that there is no heat loss between the hot water and the SMA spring.

The temperature with which the water is poured into the container influences the rate of actuation of the SMA spring. The velocity with which this water enters the container also plays a significant role in the rate of actuation. The inlet temperature and the velocity of the hot water were 75 °C and 5 mm/s, respectively. The time taken for the spring to reach 75 °C at a velocity of 5 mm/s was 25 s as shown in Fig. 70.3. After the complete actuation of the spring, the hot water was drained and allowed to cool by natural convection. The cooling time was measured as 1200 s.

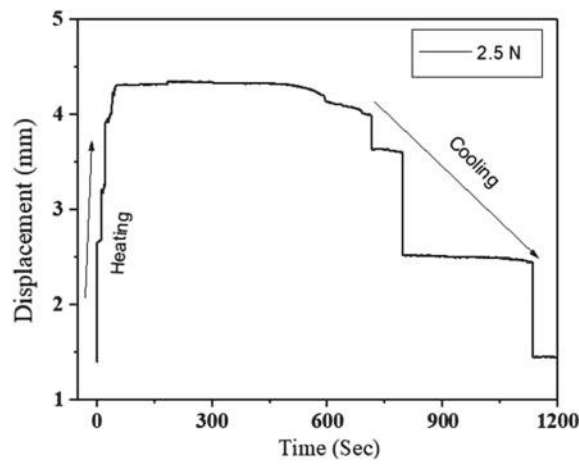
Fig. 70.3 Temperature versus time graph for a 2.5N load



The introduction of hot water causes the SMA spring to displace the load. The load displaced by the SMA spring on heating with the hot water is expressed as a graph with respect to time in displacement versus time graph. The displacement versus time graph for the actuation is shown in Fig. 70.4. This displacement versus time graph has a stepped curve in the heating phase, and this is due to the flow of hot water into the container. As the water flows into the container, the level of water slowly rises. The SMA spring starts to actuate only where the hot water is in contact with the spring. Since the load restrains the SMA spring, a stepped heating curve is observed.

Similarly, due to cooling, there is a sudden drop in the load. Hence a stepped cooling curve is observed.

Fig. 70.4 Displacement versus time graph for a 2.5 N load



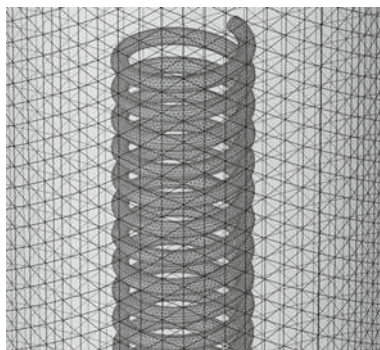
70.4.2 FEM Results

COMSOL Multiphysics 5.3a, a finite element tool for analysis and simulation of various engineering problems especially coupled physics, is used to analyze the parametric study of heat transfer in the hot water actuation. The conjugate heat transfer equation is incorporated into the COMSOL using a multiphysics interface between the general heat transfer and the laminar flow. COMSOL, in a conjugate heat transfer problem, enforces continuity of temperature at the fluid–structure interfaces.

The spring is defined as the solid domain, over which the water flows. So an enclosure of container is created as a fluid domain. The user-controlled mesh of 'Finer' size for the spring and the 'Fine' mesh size for the enclosure was done. Figure 70.5 shows the finely meshed geometry. The inlet and the outlet of the flow were determined. The velocity and temperature, at which the water enters the fluid domain are initialized.

The analysis is done based on the mathematical modeling and assumptions considered in this study. A time-dependent analysis of this conjugate heat transfer problem is considered, as the temperature of the spring varies with the time of flow of water into the container. Figure 70.6 shows the temperature change of the spring over the time as the water rises inside the container. In the conjugate heat transfer model, the fluid domain is provided with a water flow from bottom to top. This is because, during the experiment, when the water valve is opened, it falls to the bottom and rises to the top. As this is a time-dependent problem, the inlet water at a temperature of 75 °C is allowed to flow at 5 mm/s velocity until covers the whole spring. Only after 3 s, the hot water meets the SMA spring. After 15 s, approximately more than half of the spring is covered with hot water. The whole spring reaches an average probe temperature of 75 °C after 25 s. These results are satisfactory with the experimental results obtained with a water flow of 5 mm/s. Even though the time taken for the actuation (25 s) is less than the time taken for cooling (1200 s), this approach cannot be used in a real-time application. Hence, to optimize the parameters, a parametric sweep for the inlet temperature of the water (65, 75 and 85 °C) and the velocity of water flow (5 and 7 mm/s) is done.

Fig. 70.5 Finely meshed geometry



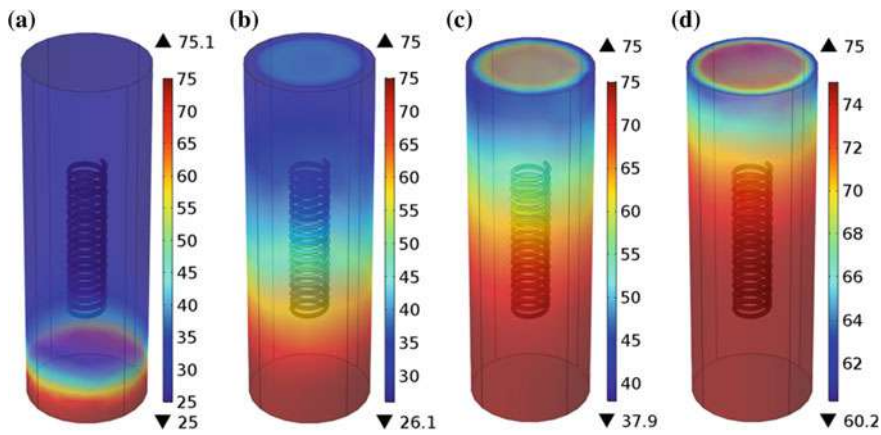


Fig. 70.6 a Flow after 3 s. b Flow after 9 s. c Flow after 15 s. d Flow after 25 s

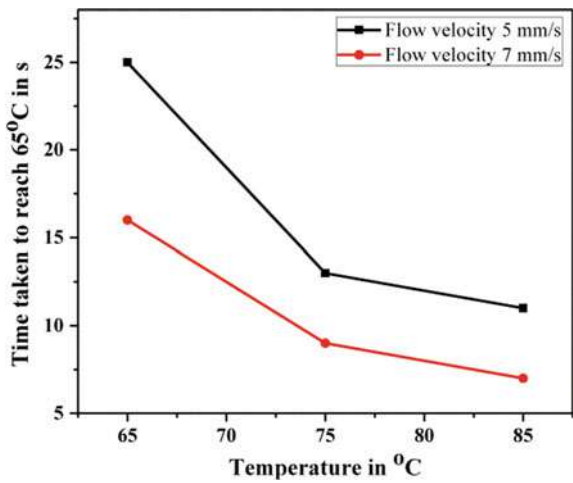
70.4.2.1 A Parametric Study

Figure. 70.7 summarizes the results of the parametric study.

SMA spring starts to actuate slowly at 65 °C itself. So, it is taken as the reference data. For a water flow velocity of 5 mm/s, when the inlet water temperature was increased from 65 to 75 °C, the time taken by the spring to reach 65 °C was reduced from 25 to 13 s. Further increasing the temperature of the water to 85 °C, the time gets reduced to 11 s.

When the flow velocity was increased to 7 mm/s, the time required to reached 65 °C were 16, 9 and 7 s for an inlet temperature of 65 °C, 75 °C, and 85 °C, respectively.

Fig. 70.7 Time taken versus temperature graph



70.5 Conclusions

The hot water actuation of the spring was experimentally performed and the parametric study for the crucial parameters was implemented. From the FEM results, the time taken for the actuation is inversely proportional to both the inlet temperature of the water and the flow velocity of the water. When the inlet temperature was increased from 65 to 75 °C, the time taken for the spring to reach 65 °C reduces from 25 to 13 s for a flow velocity of 5 mm/s. When the flow velocity was increased to 7 mm/s, the same time taken further reduces to 9 s. This analysis assists in preferring the optimum actuation parameters.

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Chapter 71

A New Approach to Control the Position of Joint Arm Robot Using Image Background Subtraction Technique



Pramod Kumar Thotapalli , CH R Vikram Kumar
and B Chandra Mohana Reddy

Abstract Vision-based systems enhance the degree of autonomy of the robot in manufacturing applications and help to increase productivity. Computer vision systems are used to perform multiple tasks by using various algorithms in the view field. In this paper, a new approach based on image subtraction technique using Gaussian mixture model (GMM) to control the position of joint arm robot for pick and place operations to sort the objects is presented. In this work, a simple vision system is used to capture the images of the robot and objects placed within the work volume of the robot, and these images are processed continuously using GMM background subtraction algorithm to find the coordinates of the objects with reference to robot base. These coordinate points are used to pick the object and place in the desired location. In this work, a prototype of 3R servo robot linkage system is fabricated to evaluate the algorithm using web camera, and a program is developed in MATLAB.

Keywords Robot · Vision system · Pick and place operation · GMM

71.1 Introduction

The robots are used for a numerous number of ways in manufacturing applications and changed the scenario in production systems. To accelerate the production as per the demands of the customers, sophisticated technological development is required in the manufacturing system. Industrial robots with artificial intelligence lead a vital role in technological development. Artificial intelligence is used to enhance the autonomy of robotic applications. Robots with the vision-based artificial intelligence provide more reliable information about the dynamic environments. Robots are commonly

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M. S. Shunmugam and M. Kanthababu (eds.), *Advances in Simulation, Product Design and Development*, Lecture Notes on Multidisciplinary Industrial Engineering, https://doi.org/10.1007/978-981-32-9487-5_71

used for repetitive tasks such as pick and place operations, for which it is essential to identify the location and orientation of the object of interest, and these intern help the robot to pick the desired object of interest. Artificial intelligence through the vision system provides the most reliable information to control the robot. Vision-based artificial intelligence uses the stored statistical information of each pixel in the image frame.

Digital image processing methodologies use the basic concepts and mathematical functions for the development of an algorithm for vision applications [1], and the features of the object in the image frame provide information about location and orientation. A static camera needs to capture the image with the entire scene in the visual field. These captured images are needed for real-time analysis in background subtraction. By processing those image frames, the changes in the scene with the reference image frame to the current image frame can be estimated [2]. The object detection, human activity, vehicle, etc., entering into the scene will be tracked by subtracting background in many video surveillance systems [3, 4]. This is the common methodology used in many vision applications, to extract foreground pixels of the object of interest [5]. For pick and place operations, background subtraction technique precisely provides information about a location of the object of interest with respect to the robot. The coordinates of the object can be determined based on the features of the object after separating the foreground from the background at that time.

Many researchers proposed numerous background subtraction methods. Frame difference subtraction technique is used the previous frame for subtraction rather than the initial frame. But it fails to update the background with reference to initial position when the object is in motion. Long and Yang [6] proposed stationary background generation based on flicker of a pixel using stable filling methods. Ridder et al. [7] developed the frame difference method based on the threshold by using the arithmetic mean of pixel values over few temporal sequences, which made their system more robust to lighting changes in the scene, that each pixel-wise automatic threshold with a Kalman filter. Lai and Yung [8] developed running mode and running average algorithms for better noise suppression and higher accuracy. Zhan et al. [9] proposed an improved algorithm based on frame difference and edge detection techniques. This methodology has great recognition rate at high speed, but it fails under a more complicated background. Stauffer and Grimson [10] proposed a “pixel process,” each pixel as a mixture of Gaussians and on-line approximation to update the background. It is based on statistical methods and commonly used in background subtraction algorithms due to a good compromise between their performance and their computation cost [5]. Many approaches have been developed using background subtraction methods for different applications mainly surveillance using constant background [10]. Background subtraction using a Gaussian mixture model (GMM) is a widely used approach to separate the foreground from the background. GMM is a most efficient method to separate the background when the background is static for a significant length of time and objects are moving continuously with a consistent speed. GMM is also a robust method for dynamic backgrounds.

The objective of the present work is to develop an algorithm to control the base rotation of jointed arm robot to find the location of the object dynamically using image background subtraction technique. In this work, a new approach is developed to extract the location information of the object within the work volume at any particular instant by subtracting background from initial reference image frame using GMM. The subtracted image can be used to extract the features of the objects like center coordinates, color, shape, and size. The center coordinates of the object are used to control the base rotation of the robot. This approach is used to track the objects dynamically, i.e., with respect to the present location of the robot, for pick and place operation. The complete methodology and construction details of the robot linkage system used in this work are presented in this paper.

71.2 Methodology

The schematic layout of the robotic work cell is shown in Fig. 71.1. A webcam is fixed at the top of the work cell in such a way that center coordinate of the captured image frame is collinear with the robot base center. A webcam is connected to the computer to store and processes images. Microcontroller-ATmega328P (UNO Rev3) is used as an interfacing device to process the output signals from the computer and the 3R servo robot linkage system. A program is developed in MATLAB software for image processing and controlling the interface hardware. Various stages involved in this work are shown in Fig. 71.1.

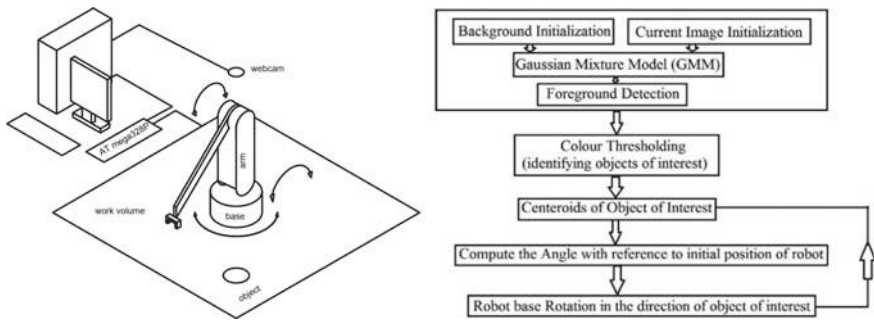


Fig. 71.1 Schematic layout of computer-controlled 3R servo robot work cell and flowchart for controlling the position of robot

71.2.1 Background Subtraction Technique

In the Gaussian mixture model, each pixel in the image frame is considered a mixture of Gaussians because the color image is a combination of RGB intensities of respective vectors [11]. Using probability density function (PDF) of pixels, the foreground from the background can be separated. A Gaussian mixture model (GMM) is the pixel-based parametric modeling method using multimodal distribution for dynamic background. The pixels of a frame $\{x_1, \dots, x_t\}$ is modeled by a mixture of K Gaussian distributions [10]. The predefined number of Gaussian component density is suggested to be $K = 3$ [10]. The pixel intensity is represented in form of the mean and covariance based on respective densities of each pixel to find the illumination changes of respective pixel at the time “ t .” If the mean of pixel densities is within 2.5 standard deviation of distributions, then it is considered as background and it is mainly due to illumination changes [12]. Some surface regions have inconstant lighting to model a background, and in these cases, GMM is an excellent method to identifying the irregular movement of objects effectively. Thus, the probability density function of the current pixel, $P(X_{x,t})$, from the current frame at the time which indicates occurrence of a color of a pixel x at time t is being represented as [10].

$$P(X_{x,t}) = \sum_{i=1}^K w_{i,x,t} \eta(X_{x,t}, \mu_{i,x,t}, \Sigma_{i,x,t}) \quad (71.1)$$

where $P(X_{x,t})$ is the probability density function of $X_{x,t}$ in a parametric form and composed of a mixture of Gaussians of each pixel x at time t . K is the number of Gaussian distributions. $\eta(X_{x,t}, \mu_{i,x,t}, \Sigma_{i,x,t})$, is the i th Gaussian probability density function, $w_{i,x,t}$ is the estimate of the weight of the i th Gaussian at time t , $\mu_{i,x,t}$ is the mean value of the i th Gaussian at time t , $\Sigma_{i,x,t}$ is the covariance matrix of the i th Gaussian probability density function. The covariance matrix is represented by [10].

$$\Sigma_{i,t} = \sigma_{i,t}^2 I \quad (71.2)$$

Background (B) Gaussian distributions, which exceed certain threshold T , are retained for a background distribution [4]:

$$B = \arg \min_b \left(\sum_{k=1}^b W_k > T \right) \quad (71.3)$$

where “ T ” is an assigned threshold.

For case, to account model each background pixel with a Gaussian distribution $\eta(\mu_{x,t}, \Sigma_{x,t})$ where $\mu_{x,t}$ and $\Sigma_{x,t}$ stand for the average background color and covariance matrix over pixel x at time t . In this context, the threshold value in terms of the Mahalanobis distance metric can be the likelihood [11].

$$\text{mahalanobis} = \sqrt{(|X_{x,t} - \mu_{x,t}|^T)(\Sigma_{x,t}^{-1})(|X_{x,t} - \mu_{x,t}|)} \quad (71.4)$$

A match is found with one of the K Gaussians. In this case, if the Gaussian distribution is identified as a background one, the pixel is classified as background else the pixel is classified as foreground [4].

To interpret the illumination variations in the scene, the mean and covariance of each pixel can be iteratively updated as follows [10]:

$$\mu_{x,t+1} = (1 - \alpha) \mu_{x,t} + \alpha X_{x,t} \quad (71.5)$$

$$\Sigma_{x,t+1} = (1 - \alpha) \Sigma_{x,t} + \alpha (X_{x,t} - \mu_{x,t})(X_{x,t} - \mu_{x,t})^T \quad (71.6)$$

The prior weights of the K distributions at time t , $w_{i,t}$ are updated as [10],

$$w_{i,t} = (1 - \alpha) w_{i,t} + \alpha M_{k,t} \quad (71.7)$$

where α is a learning rate and $M_{k,t}$ is 1 for the matched, otherwise, $M_{k,t}$ is 0.

When they match the new observation, they are updated as follows [10].

$$\mu_{i,t} = (1 - \rho) \mu_{i,t} + \rho \cdot x_{i,t} \quad (71.8)$$

$$\sigma_t^2 = (1 - \rho) \sigma_{t-1}^2 + \rho (x_t - \mu_t)^T (x_t - \mu_t) \quad (71.9)$$

where ρ is the second learning rate defined as $\rho = \alpha \eta (x_t | \mu_k, \sigma_k)$.

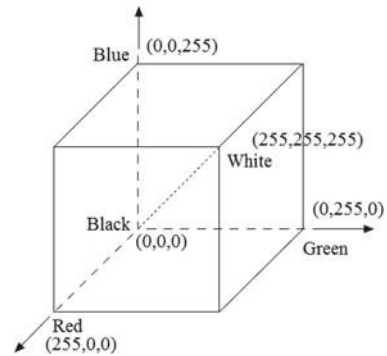
The parameters $\mu_{i,t}$ and standard deviation $\sigma_{i,t}$ for unmatched distributions remain the same and only the weight is replaced by Stauffer and Grimson [10].

$$w_{i,t+1} = (1 - \alpha) w_{i,t} \quad (71.10)$$

If none of the K distributions match the current pixel value, the least probable distribution is replaced with the current value as its mean value, an initially high variance, and low prior weight. Foreground detection can be made and so on.

71.2.2 Identifying Objects of Interests

RGB color space is widely used for color thresholding in computer vision applications. Red (R), Green (G), and Blue(B) are three primary colors that are represented by a 3 dimensional Cartesian coordinate system. For example the indicated diagonal of the cube as shown in Fig. 71.2 represents the percentage of intensities of RGB to represent the color.

Fig. 71.2 RGB color space

A color mask is a filter and is created based on the range (maximum limit and minimum Limit) of intensity values (RGB array) in each pixel in the region of interest. To describe any color of interest in an image frame, the respective pixel level persistence is required. That persistence contains respective RGB intensities of the particular pixel. For creating the RGB color mask, the following color threshold equation is used for the respective pixel in color array frame, and the general equation for creating a color mask is shown below

$$I_{r,g,b}(i, j) = \begin{cases} 0, & P(i, j) < T_{\min} \\ 1, & T_{\min} \leq P(i, j) \leq T_{\max} \\ 0, & P(i, j) > T_{\max} \end{cases} \quad (71.11)$$

The mask filtered the intensity values of a pixel in respective array frames. The output image frame contains the only blue color object ($1, T_{\min} \leq P(i, j) \leq T_{\max}$) in Fig. 71.2 of interest “1” and ($0, P(i, j) < T_{\min}$) ($0, P(i, j) > T_{\max}$) remaining pixel in frame turn dark “0.” Color thresholding is the preprocessing step to find the center coordinates of the object of interest in the work volume.

71.2.3 Center Coordinates of Object

Center coordinates of the object of interest are the primary task for identifying the location of the object in the work volume of the robot. Let us consider the centroid of the object, where $(x_i, y_i), i = (1, 2, \dots, n)$ are the boundary points (a cell array of boundary pixel locations) of the object of interest [13]. To get the boundary points, the threshold is applied to the binary image frame. The data are statistically extracted to compute the centroid (X_1, Y_1) of the object binary image frame [13].

$$X_1 = \frac{1}{n} \sum_{i=1}^n x_i, \quad Y_1 = \frac{1}{n} \sum_{i=1}^n y_i. \quad (71.12)$$

71.2.4 *Compute the Angle Between Object Center and Unloading Position of Robot*

To compute the angle θ between object center and unloading position of robot is used to rotate the base of the 3R servo robot. To calculate the angle, three points are identified in the image frame. In three points, two reference points are fixed and third point is variable. Let us consider first reference point $b = (X_n/2, Y_n/2)$ represents the center of the image frame that represents the center of the robot base (i.e., center coordinate image frame and robot base are same). The second reference point is the unloading position of robot. Let us consider 20 pixel distance in Y-direction from the center of robot base. The third reference point variable $c = (X_1, Y_1)$ is the center coordinate of the dynamic object position of interest in the image frame. One way to calculate angle is by using vector.

$$\theta = \cos^{-1} \left(\frac{\vec{ab} \cdot \vec{bc}}{\|\vec{ab}\| \|\vec{bc}\|} \right) \quad (71.13)$$

Hence, the calculated angle is used to rotate the base of robot such that all joint arms of robot are in collinear and in between the centers of object and base of robot.

71.2.5 *Directional Rotation of Robot Base*

The intelligent task is to control the rotation of the robot base in direction of the object of interest. The user defines the unloading position of robot. The coordinates of the objects of interest are (X_1, Y_1) in the image frame and the image center coordinates are $(X_n/2, Y_n/2)$. As mentioned earlier, center of robot base and the image frame coordinate are same. For this, a logic is developed such that image frame is divided into four quadrants as shown in Fig. 71.3.

Based on position of the object, robot can move in two possible rotational directions i.e., counterclockwise (C.C.W) and clockwise (C.W). The following conditional equations are used to judge the rotational direction.

Case 1: Counter clockwise (C.C.W) Rotation II-Quadrant

$$\text{if } X_1 < (X_n/2) \text{ and } Y_1 < (Y_n/2) \quad (71.14)$$

Case 2: Clockwise (C.W) Rotation I-Quadrant

$$\text{if } X_1 > (X_n/2) \text{ and } Y_1 < (Y_n/2) \quad (71.15)$$

The location of the object of interest coordinates is less than the robot base or image center coordinates and it is considered to lie in II-quadrant. So, the direction of rotation is CCW. In the same way, the other direction of rotation is decided.

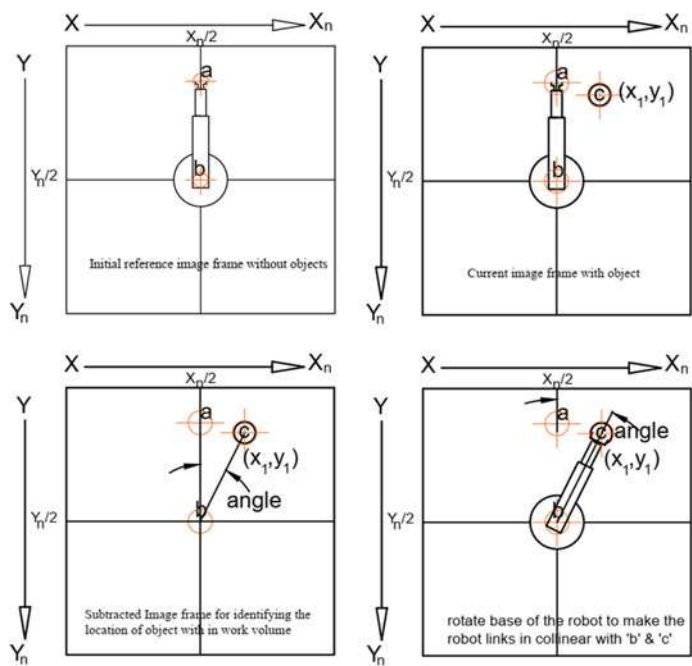


Fig. 71.3 Illustration for identifying the object location using background subtraction technique

71.3 Results and Discussion

A MATLAB program is developed using the above algorithm and it is evaluated on fabricated prototype 3R servo robot linkage system successfully, and the output image frame with three-pixel coordinates from two vectors is shown in Fig. 71.4 and the extracted three coordinates are tabulated in Table 71.1.

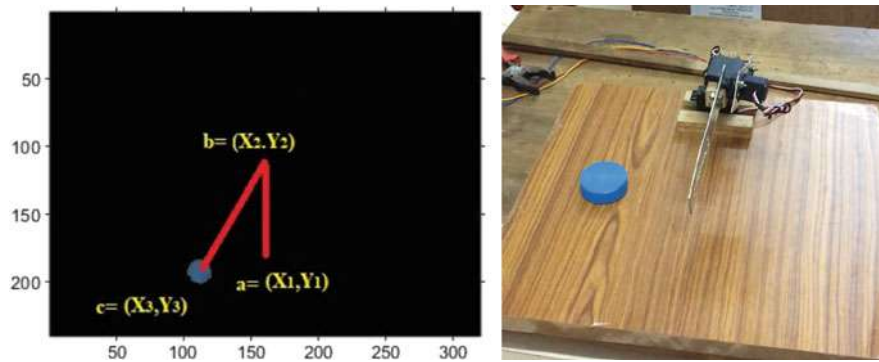


Fig. 71.4 Compute the angle in the image to jog the robot and webcam capture the video

Table 71.1 Extracted coordinate values of the output image frame

	The position of pixel in the 'X-direction'	The position of pixel in the 'Y-direction'
Size of image frame "I"	320	240
Unloading position of robot "a"	160	180
Centroid of image frame "b"	160	120
Centroid of the object "c"	125	190

The $\overrightarrow{ab}$ is the initial position of the joint arm robot and $\overrightarrow{bc}$ is the final position of joint arm robot. The angle between the $\overrightarrow{ab}$ and $\overrightarrow{bc}$ is 26.56° .

71.4 Conclusions

In this paper, a new approach is developed to control the position of jointed arm robot using GMM image subtraction technique. The location of the object with respect to the robot base is determined by subtracting the current image frame from reference image frame. In this work, a simple image system is used to rotate base of the robot to make the robot links, the center coordinate of the object, and robot base point in collinear. This approach reduces kinematic analysis of 3D joint angles to 2D. The position of the object from the image processing is used to find the joint angles for end effector using planer inverse kinematics for pick and place operation. The vision algorithm provides more reliable and real-time information about the position and orientation of the objects for accurate control of the position of joint arm robot.

Acknowledgements This project is funded by AICTE New Delhi under Research Promotion Scheme.

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